Lecture Notes in Civil Engineering

Claudio Cameselle · Susana Gouveia · Santiago Urréjola · Krishna R. Reddy · Arvind Kumar Agnihotri *Editors*

Sustainable Environmental Geotechnology and Pollution Control

Select Proceedings of EGRWSE 2025, Volume 2

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Conference Highlights

The sixth edition of the EGRWSE-2025 International Conference was held at the University of Vigo from June 11 to 14, 2025. The official opening ceremony took place on June 11 at the facilities of the Vigo Free Trade Zone Consortium (Consorcio Zona Franca de Vigo), which generously hosted both the inauguration and a short course on sustainable and resilient engineering.

The inaugural session featured several distinguished representatives, including Mr. David Regades, Delegate of the Vigo Free Trade Zone Consortium; Mrs. Nuria Rodríguez, City Councilor for the Environment of the Vigo City Council; Dr. Xanel Vecino, Deputy Director of the School of Industrial Engineering; and Prof. Manuel Joaquín Reigosa, Rector of the University of Vigo. The conference coordinators, Prof. Krishna R. Reddy (University of Illinois Chicago) and Prof. Arvind Kumar Agnihotri (NIT Jalandhar, India), participated alongside local organizers Dr. Susana Gouveia and Prof. Claudio Cameselle. This session underscored the significant interest demonstrated by both the scientific community and the business sector in advancing research and translating its outcomes into practical applications that address contemporary societal challenges through innovative solutions.

Following the opening ceremony, a 4-hour *short course* titled *Sustainable and Resilient Engineering: Drivers*, *Metrics*, *Tools*, *and Applications* was delivered by Krishna R. Reddy, Claudio Cameselle, Jeffrey A. Adams, and Jagadeesh Kumar Janga. The course covered key concepts in the field and aimed to equip participants with foundational knowledge and practical tools to develop university-level courses on sustainable and resilient engineering, as well as to integrate sustainability and resilience into engineering projects.

The scientific sessions were held on June 12 and 13 at the School of Industrial Engineering, University of Vigo. During the opening session on June 12, Prof. Claudio Cameselle welcomed all attendees, expressing his hopes for a productive scientific and social meeting. Professor Félix Quinteiro, Vice-Rector for Planning at the University of Vigo, extended a warm welcome and highlighted the university's strong commitment to hosting international scientific events of this nature. Dr. Cristina Trillo, Deputy Director for International Affairs at the School of Industrial Engineering, welcomed participants on behalf of the School, emphasizing its international reach and active engagement in fostering student and faculty exchanges with numerous universities across Europe and worldwide. Professors Krishna Reddy and Arvind Kumar Agnihotri underscored the growing global interest in sustainable engineering and resilience, as reflected by increased conference participation.

Dr. Susana Gouveia provided an overview of the event's participation statistics, noting that both the number of contributions and attendees nearly doubled compared to previous editions—a proof of the conference's expanding relevance. The scientific program comprised 24 parallel sessions, covering a broad range of topics, and featured 144 oral presentations alongside 40 poster displays.

During the opening session, special recognition was given to Prof. Madhavi Latha for her significant contribution to the Chenab Bridge project—the world's highest

railway arch bridge. Constructed in a challenging seismic and mountainous region, this project required innovative geotechnical solutions to ensure its stability and safety.

The closing session of the conference offered a comprehensive review of the event's academic and social activities. Participants expressed their satisfaction with the warm welcome, the excellent organization, and the high scientific quality of the presentations—delivered by keynote speakers, invited experts, and contributing participants, including both faculty and students. Awards were presented to the three best poster presentations and the three best oral communications, with the support of Springer Nature. The session also confirmed the strong interest from various universities and research centers in hosting future editions of the conference. It was officially announced that the next edition will be held in Sardar Vallabhbhai National Institute of Technology, Surat, India, in 2026.

On Saturday, June 14, the last day of the conference, participants enjoyed a guided visit to the Cíes Islands Natural Park—an outstanding example of sustainable management of a natural area with high ecological value.

Keynote Lectures

Opening Session: Urban Flooding and Stormwater Pollution Problems Under Changing Climate: Sustainable and Resilient Solutions. Krishna R. Reddy, Banuchandra Nagaraja. Professor Reddy presented a detailed analysis of sustainable systems for managing stormwater runoff and pollution in urban environments under the increasing challenges posed by climate change, using Chicago as a case study.

Session 1B. Biotechnological valorization of underutilized biomasses to produce sustainable animal feeds. Helena Fernandes, Nuno Muñoz-Seijas, Francisco Manuel Soto Beltrán, José Manuel Salgado, Borja Fernández, Manuel Domínguez. Dr. Fernandes explored biotechnological processes for transforming underutilized biomass into sustainable animal feed ingredients.

Session 2A. Advancing Circular Economy Solutions for the Olive Oil Sector: Integrating Biotechnological and Engineering Advances for Olive Pomace Utilization. Alicia Paz, Evangelos Topakas, Attilio Converti, Manuel Domínguez. Dr. Paz discussed circular economy strategies for the olive oil sector, focusing on the conversion of olive pomace into high-value products using biotech and engineering solutions.

Session 3A. Sustainable deployment of Industrial wastes to improve the load carrying capacity of granular soil. Arvind Kumar Agnihotri. This presentation focused on reusing industrial waste materials to enhance the mechanical performance of granular soils.

Session 4A. Transforming Waste to Solution: Moringa Seed Cake Biochar for Lead-Contaminated Soil Remediation. Chandresh Solanki, Bhoomi Kamdar. The

authors presented an innovative approach to soil remediation using biochar derived from moringa seed cake to immobilize heavy metals.

Session 5C. From Brownfields to Brightfields: The Benefits of Solar landfills on Closed Sites. Eugeniusz Koda, Magdalena Daria Vaverková. This talk highlighted the transformation of inactive landfills into solar energy sites, promoting the reuse of potentially idle land for clean energy production.

Invited Lectures

Opening Session. Quantifying PFAS in Water According to the Latest EU Regulations. Angel Antelo. This lecture provided a comprehensive overview of methods for quantifying PFAS in water in line with current European Union regulations.

Session 1A. Breakthrough Sustainable Energy: Energy Geo-structures. Yeliz Yukselen-Aksoy. Prof. Yukselen-Aksoy introduced cutting-edge developments in energy geo-structures for sustainable energy generation.

Session 1A. Conversion of calcium bentonite to sodium bentonite through dialysis. Nicolò Guarena, Francesco Mazzieri, Andrea Dominijanni, Evelina Fratalocchi, Mario Manassero. The team presented a method for converting calcium bentonite into sodium bentonite using a dialysis process.

Session 2A. Valorisation of food chain waste through solid-state fermentation for biostimulant production for a sustainable agriculture. Jana Font, Antoni Sánchez, Adriana Artola, Esther Molina-Peñate, Óscar Prado. This presentation discussed the valorization of food waste through solid-state fermentation to produce biostimulants for sustainable agriculture.

Session 2B. Micro-meso-macro interactions in bio-cemented sands. Madhavi Latha Gali, Geethu Thomas. The authors explored how interactions at various scales influence the bio-cementation process in sands.

Session 2B. Sustainable remediation of sediment using furnace slag mixture. Ahmed Benamar, Mohamed-Tahar Ammami, Ahmad Zein-eddin, Zulfiqar Ahmad. The study focused on using furnace slag mixtures for the sustainable remediation of contaminated sediments.

Session 3A. Exploring Sustainability Assessment of Geosynthetic-based solutions in Civil Engineering. Ivan P. Damians. This lecture examined sustainability evaluation frameworks for geosynthetic applications in civil engineering.

Session 3B. Contaminated Site Management and Sustainability Analysis Platform. Antonio Thome, Adeli Braun, Caroline Visentin. The team presented tools and methodologies for sustainable management and assessment of contaminated sites.

Session 4A. Air-water interface adsorption impact on PFAS transport through vadose zones. Arvin Farid, MD. Khorshed Alam, Krishna R. Reddy. The researchers analyzed how PFAS movement in unsaturated soils is affected by adsorption at the air-water interface.

Session 4B. Construction of unbound granular layers using steel slag—Performance compared to natural reference material on a road in use. António José

Roque, *Antonio Gomes Correia*, *Eduardo Fortunato.* This presentation assessed the feasibility of using steel slag in road construction compared to conventional materials.

Session 4B. An Alternative Approach to Reusing Mine Waste in Underground Mines: CPB. Okan Yüksel, *Oznur Karaca*, *Tugay Aktaş.* The authors explored the use of cemented paste backfill (CPB) as a method for reusing mining waste in underground operations.

Session 5A. Lab-Scale Assessment of Effect of Sodium Polyacrylate on the Shear Strength of Wet Coal Combustion Residuals. Milind Khire. Professor Khire presented lab results on how sodium polyacrylate can influence the shear strength of wet coal combustion residues.

Session 5A. Innovation Projects in Agroforestry Waste Valorization. Ana Magdalena Pérez-Lafuente, *Raquel Ledo.* The speakers discussed innovative projects for turning agroforestry waste into valuable products.

Session 5B. Assessment of Rainfall Influence on the Stability of Ash Dykes. Jaswanth Reddy Banka, *Goushya Begum*, *Kummari Sekhar*, *Hanumantha Rao Bendadi.* This study evaluated how rainfall impacts the structural stability of ash dykes.

Session 6A. Hexavalent Chromium (Cr(VI)) Immobilization in Cement via *Chemically Driven CO_2 Mineralization. Junboum Park, Kian Cho.* The presentation addressed a novel method of Cr(VI) immobilization in cement through CO_2 mineralization processes.

Session 6B. Workshop on Scientific Publishing. Swati Meherishi. In this workshop, Springer Nature's editor discussed best practices and challenges in using AI tools for scientific publishing.

Short course of sustainable and resilient engineering

Welcome reception at Consorcio Zona Franca de Vigo

Opening session at the school of industrial engineering

Auditorium of the school of industrial engineering

Group photo of EGRWSE-2025 participants

Group photo in the gala dinner

Technical trip to the Cies Island Natural Park

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Preface

As we move further into the twenty-first century, humanity faces an unprecedented convergence of global challenges that threaten both human well-being and the ecological balance of our planet. Among these challenges are the impact of population growth, increased resource consumption, the escalating generation of waste, and the profound effects of climate change on rising sea levels, biodiversity loss, and intensified threats to coastal regions. These interlinked issues demand urgent and coordinated responses. Compounding these challenges are the United Nations Sustainable Development Goals (SDGs), which emphasize the need for equitable access to food, housing, education, clean energy, and economic opportunity, while ensuring environmental stewardship for future generations.

In this context, Environmental Geotechnology has emerged as a vital and dynamic interdisciplinary field, offering innovative and science-based solutions to some of the most pressing environmental and geotechnical issues of our time. Expanding beyond the traditional boundaries of geotechnical and environmental engineering, the discipline now incorporates a wide range of knowledge areas, including hydrogeology, geochemistry, biotechnology, ecology, sustainable construction, and water resources engineering. The integration of these diverse domains allows for the development of holistic approaches to solid waste management, site remediation, infrastructure design, climate adaptation, and pollution control, among many others. Furthermore, sustainability and resilience criteria are becoming increasingly important in engineering projects to address current challenges and prevent future damage, enabling more suitable sustainable development that safeguards the interests of future generations.

Recognizing the need for collaboration across disciplines and sectors, the 6th International Conference on Environmental Geotechnology, Recycled Waste Materials and Sustainable Engineering (EGRWSE-2025) was held from June 11 to 14, 2025, at the University of Vigo, Spain. The conference provided a global platform for researchers, educators, practitioners, students, and policymakers to present their work, exchange ideas, and explore emerging trends and technologies aimed at building a more sustainable and resilient future.

EGRWSE-2025 took place across two complementary venues in the city of Vigo: the Vigo Free Trade Zone Consortium (Consorcio Zona Franca de Vigo), a centrally located and architecturally significant facility that hosted the opening activities and the short course on sustainable and resilient engineering, and the School of Industrial Engineering of the University of Vigo, located on the serene and scenic university campus. Together, these venues created an inspiring and practical environment for academic exchange, professional networking, and international collaboration. To ensure seamless participation, the organizers provided daily transportation between the city center and campus venues.

The conference welcomed approximately 230 delegates, including members of the local organizing committee, from 31 countries across five continents. During the event, 144 oral presentations were delivered across 24 technical sessions, along with 40 poster presentations, covering a wide array of topics. The themes of the conference included: solid waste and circular economy, environmental pollution control, renewable energy geotechnics, nature-based and biotechnological solutions, climate-responsive geoengineering, and sustainable infrastructure development, although participation was not restricted to these themes. One of the conference highlights was the Short Course on Sustainable and Resilient Engineering, taught by internationally recognized experts and focused on key principles, assessment tools, and applications for advancing sustainability in engineering practice. In addition, a series of keynote and invited lectures delivered by leading figures in academia and industry enriched the program and stimulated in-depth discussion.

Since its inception in 2018, the EGRWSE conference series has steadily grown into a highly regarded international forum dedicated to promoting sustainability and resilience in environmental geotechnology. Previous editions were held in India (2018, 2023), the USA (2019), Türkiye (2022), and Poland (2024), hosted by prestigious academic institutions. The 2025 edition, hosted in Vigo (Spain), continues this tradition of fostering interdisciplinary dialogue, promoting innovative research, and reinforcing the practical implementation of environmentally responsible engineering.

This volume presents a carefully selected collection of peer-reviewed papers from EGRWSE-2025, reflecting the most recent developments in theory, practice, and innovation across multiple domains of geoenvironmental engineering and sustainable technology. The breadth and quality of the contributions underscore the critical role of scientific and engineering research in addressing contemporary global issues and advancing public health, environmental protection, and sustainable development.

It is our sincere hope that this publication will serve as a lasting and meaningful resource for students, researchers, professionals, and decision-makers who share a commitment to engineering to have a more sustainable and equitable future for all.

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Vigo, Spain
June 2025

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This conference has been made possible thanks to the invaluable support of our partner institutions and sponsors. We would like to especially acknowledge the University of Vigo, the Vigo Free Trade Zone Consortium (Consorcio Zona Franca de Vigo), and the City Council of Vigo. Their logistical, material, and financial support has been essential in organizing an event of this scale, enabling us to provide a comprehensive and enriching experience to all our attendees while keeping registration fees as low as possible.

We also extend our deepest thanks to the authors for their valuable research and commitment to advancing the field, and to the reviewers for their rigorous assessments and constructive feedback. Our heartfelt appreciation goes as well to the invited speakers, session chairs, and all those who contributed through their participation and support.

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Innovative and Sustainable Environmental Geotechnology and Pollution Control

Micro-Meso-Macro Interactions in Bio-Cemented Sands

Geethu Thomas and Gali Madhavi Latha

Abstract The concept of sustainability is widely being implemented in various engineering applications and the use of eco-friendly alternatives is encouraged to replace traditional materials to gain significant, often overlooked positive environmental impacts. This study examines the feasibility of using natural alternatives such as bioenzyme (TerraZyme) and biopolymer (Xanthan gum) to enhance the strength of sands by bio-cementation. Although sand is chemically inert, it often encounters problems like stiffness reduction, settlement, erosion, and earthquake-induced liquefaction under various loading conditions. The study investigates the stress–strain behavior of sand influenced by TerraZyme and Xanthan gum and discovers the key role of a minimal Kaolin content in the treatment. Results from a set of static triaxial tests on treated soil samples showed that both additives significantly improved the strength of sand within 30 days of curing. TerraZyme modifies the elemental structure of sand–Kaolin mixtures, whereas Xanthan gum acts mainly through weak binding by altering the soil matrix. Further, comprehensive chemical and microstructural analyses including Environmental Scanning Electron Microscopy (ESEM), Energy-Dispersive X-ray Spectroscopy (EDS), high-resolution powder X-ray diffraction (XRD) analysis, Fourier Transform Infrared Spectroscopy (FTIR), and Solid-state Nuclear Magnetic Resonance (NMR) spectroscopy were carried out to understand the microstructural and chemical interactions at different scales that are responsible for the improvement. These analyses revealed that the alumina in Kaolin is the primary reason for better bonding through chemical interactions, while Xanthan gum serves just as a physical binder. The findings indicate that besides the additive type, the presence of Kaolin and its reactivity play a critical role in strength improvement. This study provides significant confidence on the use of these eco-friendly substitutes as a replacement for traditional additives like cement and lime in ground improvement applications, thereby lowering the environmental impact.

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1 Introduction

The stability of a structure is mainly related to the design and underlying soil conditions. Both natural and human impacts can lead to severe failures by pushing the overlooked subsurface geotechnical issues. The natural reasons for instability are due to soil erosion, high fluctuations in groundwater levels, weak or fractured rock formations, seismic occurrences, etc. Severe changes in climatic conditions and thereby floods and landslides also become a frequent cause of devastating damages globally which necessitates effective mitigation strategies. The soil instability is mainly caused by construction-related activities, such as excessive excavation without adequate support, which can be addressed by effective ground improvement techniques.

Traditional soil stabilization techniques, such as mechanical reinforcement and chemical treatments with cement or lime, enhance soil strength but vary significantly in their environmental impact. Mechanical methods, including geosynthetics and compaction, generally have a lower carbon footprint compared to cement-based stabilization, which is responsible for approximately 7–8% of global CO_2 emissions [1]. Each ton of manufactured cement releases nearly 1 ton of CO_2 into the atmosphere, contributing to climate change. Additionally, cement stabilization poses risks of chemical leaching, which can contaminate groundwater and lead to long-term degradation of soil quality. The extensive mining of raw materials for cement production further exacerbates habitat destruction and resource depletion. To address these pressing environmental issues, the field of geotechnical engineering is increasingly exploring sustainable alternatives that utilize biological processes for soil enhancement, underscoring the urgent need for greener solutions. Bioenzyme and biopolymer stabilization are more cost-effective than cement-based techniques due to reduced transportation and application expenditures. Since bioenzymes are concentrated liquids, they can be transported in concentrated form and diluted at the site for application, unlike cement and lime, thus minimizing logistics expenses. Further, cement-based additives pose high adverse impacts to the environment, increasing the cost of carbon control. While the initial manufacturing costs may be greater for bio-based stabilizers, life-cycle evaluations reveal that their low resource consumption and sustainable benefits make them cost-effective. Traditional stabilizers like cement and lime provide rapid strength gain, whereas eco-friendly additives achieve strength gradually through mineralogical alterations and physical bonding via microbial, enzymatic, or polymerization effects. While early strength gain is higher with cement, the sustainable and gradual improvement from bio-cementation enhances cohesion, friction, and shear strength, contributing to long-term soil stability.

Bio-geotechnical methods offer innovative pathways for soil stabilization, leveraging microbial activity, biopolymers, and bioenzymes to enhance soil properties. Among these, Microbially Induced Calcite Precipitation (MICP) relies on bacteria

such as *Sporosarcina pasteurii* [2] and Enzyme-Induced Carbonate Precipitation (EICP) utilizes urease enzymes [3] to induce mineral precipitation which demonstrated the potential to improve soil strength through calcite bonding. Despite their effectiveness, challenges such as non-uniform calcite distribution and enzymatic stability in EICP require further refinement. Additional research is carried out on its economic viability for large-scale field applications. Biopolymers, including Guar gum, Gellan gum Xanthan gum (XG) and Agar, have emerged as promising soil binders due to interaction with soil particles. Among which XG exhibits pseudo-plastic behavior and stability under varying environmental conditions, making it suitable for geotechnical applications. Similarly, bioenzymes such as Permazyme and TerraZyme derived from fermented plant extracts have shown potential in modifying fine-grained soils by altering the adsorbed water layer, thereby improving mechanical properties. Among many available bioenzymes, TerraZyme (TD) is capable of enhancing soil performance with minimal dosage and improving soil durability. These biological alternatives not only improve soil properties but also minimize the carbon footprint of conventional stabilization techniques. Since biopolymers are organic additives, they are vulnerable to microbial attack. Hence, their durability under dynamic and extreme stresses, and their long-term efficiency need additional investigations. These limitations are more relevant to their application in grouting.

Beyond biological treatments, geotechnical research has explored alternative strategies such as the reuse of industrial by-products for soil improvement. This aligns with circular economy principles and long-term sustainability. Further advancements in nanotechnology may also contribute to enhancing soil stabilization techniques by improving material interaction at the microstructural level. Comprehensive chemical and microstructural investigations are required to optimize the effectiveness of bio-geotechnical techniques. Advanced spectroscopic and microscopic analyses provide insights into mineralogical changes, cementation mechanisms, and the durability of biocemented soils. The formation and distribution of calcite in sands, along with polymer interactions in clay-rich soils, necessitate high-resolution imaging and chemical characterization to evaluate the in-depth variations in soil fabric.

This study focuses on the role of biopolymer and bioenzyme treatments in improving strength of sand and the cementation mechanisms involved. The role of Kaolin, a hydrous aluminum silicate as fines content, in sand is also explored. By integrating state-of-the-art spectroscopic and microstructural analyses, this research aims to bridge existing knowledge gaps and contribute to the advancement of sustainable soil stabilization techniques in geotechnical engineering. Ultimately, the adoption of bio-geotechnical methods can pave the way for resilient infrastructure and environmentally responsible construction practices.

2 Materials and Methods

2.1 Materials

The base soil is river sand with a specific gravity (G) of 2.63, consisting of subrounded particles and no fines. It is classified as poorly graded sand (SP) according to ASTM D2487-17e1 [4], with a coefficient of uniformity (C_u) of 1.82 and a gradation coefficient (C_c) of 0.85. Kaolin (denoted as K) powder (as fines) was sourced from Purenso Global Pvt. Ltd. The bioenzyme (TerraZyme with $G = 1.0$–1.1) was procured from Avjeet Agencies, Chennai, while the biopolymer (Xanthan gum with $G = 1.6$) was obtained from Urban Platter Pvt. Ltd. Both biomaterials are water soluble.

2.2 Method

Material Characterization. The samples were characterized using Environmental Scanning Electron Microscopy (ESEM) and Energy-Dispersive X-ray Spectroscopy (EDS) with Thermo Fisher XL-30 and JEOL SEM IT 300 for imaging and elemental analysis, employing gold coating for conductivity and real-time observation of soil stabilization mechanisms. Additionally, advanced techniques such as high-resolution X-ray diffraction (XRD) for mineral composition, Fourier Transform Infrared Spectroscopy (FTIR) for functional group identification, and Solid-state Nuclear Magnetic Resonance (NMR) for atomic-level structural insights were employed to analyze soil-additive interactions and stabilization effects.

Sample Preparation. For untreated pure sand (S) samples, air-dried river sand was prepared at a maximum dry density (MDD) of 1.78 g/cc and optimum moisture content (OMC) of 10%. For sand samples containing 15% Kaolin (S + 15%K) MDD was kept at 1.97 g/cc and OMC = 10%. For TerraZyme (TD)-treated samples, TD dosages ranging from 0.02 to 0.1 ml/kg were pre-mixed with water, stirred for five minutes, and added to the dry mix of S + 15%K while maintaining OMC, for S + 15% K + XG, sand and Kaolin were first dry-mixed, followed by dry mixing of Xanthan gum (XG: 0.2%–1.5% by dry weight) to ensure uniform distribution. OMC was then added, and the combination (S + 15%K + XG) was hand-mixed and stored for 1 h before the reconstitution of samples. This method is more effective than wet XG mixing, ensuring consistency and uniformity [5].

Static Triaxial Compression Testing Details. Soil samples were compacted in three layers in cylindrical molds, then removed, sealed, and cured in a desiccator for 30 days. After the inspection for water content and specimen dimensions (3.8 cm diameter, 7.6 cm height), static triaxial compression tests under unconsolidated undrained (UU) conditions were performed to assess strength with and without additives, under IS 2720 (Part 11):1993 [6]. This test replicates field stress conditions by using controlled confining pressure and axial loading to simulate the stresses that

soils face during foundation construction and rapid loading cases such as seismic events. The cured samples were placed in rubber membranes attached with O-rings, tested under 100 kPa confining pressure, and sheared at 1.25 mm/min until failure or 20% axial strain, with load and deformation monitoring via load cells and Linear Variable Differential Transformers (LVDTs).

3 Results

3.1 Bioenzyme Effect on Sand and Sand with Kaolin

The stress–strain behavior of untreated and biomaterial-treated sand was investigated through a series of static triaxial compression tests under unconsolidated undrained (UU) conditions. The optimum dosages of the biomaterials were determined based on peak deviatoric stress, with supporting chemical and microscopy analyses. Since bioenzymes are known to react effectively with soil only when a minimum fines content (7–12%) is available [7], trials were also conducted on pure sand to evaluate its potential without fines. Kaolin content was fixed at 15% to ensure adequate fines for bioenzyme interaction.

For untreated sand, the peak deviatoric stress was 176.58 kPa, which increased to 250.58 kPa with the addition of 15% K, representing a 41.9% improvement in strength. The maximum strength enhancement for pure sand with TD was observed at 0.06 ml/kg, achieving 266.01 kPa (50.7% increase compared to untreated sand). In contrast, for S + 15% K, the highest peak deviatoric stress of 429.88 kPa was attained at 0.075 ml/kg TD, showing a 144.6% increase relative to untreated sand and 71.5% compared to sand with Kaolin alone (Fig. 1).

The stress–strain response indicates that untreated sand exhibits strain softening with ductile failure characteristics. In contrast, the S + 15% + TD-treated samples

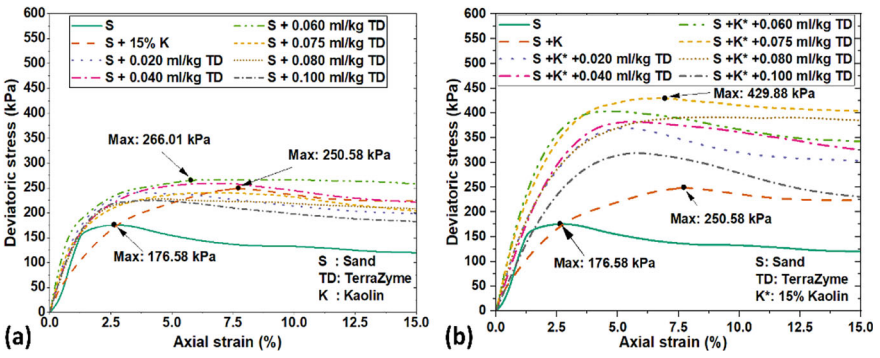

Fig. 1 Stress–strain response of TD-treated sand with and without Kaolin (effective confining pressure: 100 kPa, curing period: 30 days, static triaxial test in UU condition)

display strain-hardening behavior, signifying enhanced resistance to deformation and a more brittle failure mode at higher TD dosages. The observed brittleness, attributed to cementitious bonding induced by TD, suggests applicability in foundations and embankments requiring higher stiffness and reduced settlement. However, excessive TD dosage leads to reduced strength due to particle slippage caused by an excess of cementitious gel, highlighting the importance of optimizing treatment levels [8]. These results confirm that TD has limited impact on pure sand but significantly enhances strength in fines-rich sand, making it a viable option for sustainable soil improvement in geotechnical applications. Bioenzymes are plant-based materials and hence they are environmentally safe. Many of these bioenzymes are already being used in agricultural applications and wastewater treatment, where they aid in breaking down organic matter without introducing harmful contaminants. Unlike traditional stabilizers, bioenzymes do not produce toxic byproducts, hence they do not pose any risk of groundwater pollution.

3.2 Biopolymer Effect on Sand with Kaolin

The influence of Xanthan gum (XG) on the stress–strain behavior of sand with 15% Kaolin was evaluated through static triaxial compression tests (UU) after 30 days of curing and XG dosages ranged from 0.2% to 1.5% by dry weight. As seen in Fig. 2a, untreated sand shows the lowest peak deviatoric stress, exhibiting rapid strain softening, while adding 15% Kaolin enhances strength by 41.9%, demonstrating its role in soil improvement. With XG treatment, the peak deviatoric stress increases further, reaching a maximum of 448.46 kPa at 1.0% XG, which corresponds to a 153.9% increase in comparison with untreated sand and 78.9% over S + 15%K. Beyond 1.0% XG, a marginal reduction in peak strength is observed and possibly due to excess polymerized gel resulted in localized weak zones and increased water retention.

The addition of XG induces strain-hardening behavior, enhancing both the strength and ductility of the treated sand. This improvement aligns with previous studies, highlighting the potential of XG in geotechnical applications requiring enhanced soil stability [9]. The combination of increased strength and ductility makes XG-treated soil suitable for slope stabilization, road subgrade improvement, and other ground reinforcement applications. Xanthan gum can degrade over time due to microbial activity, which may result in gradual strength reduction. However, the rate of degradation is influenced by environmental factors such as temperature, moisture, and microbial growth. Its longevity can be improved through suitable remedial measures to enhance resistance to bacterial attack and environmental changes, or periodic reapplication may be suggested.

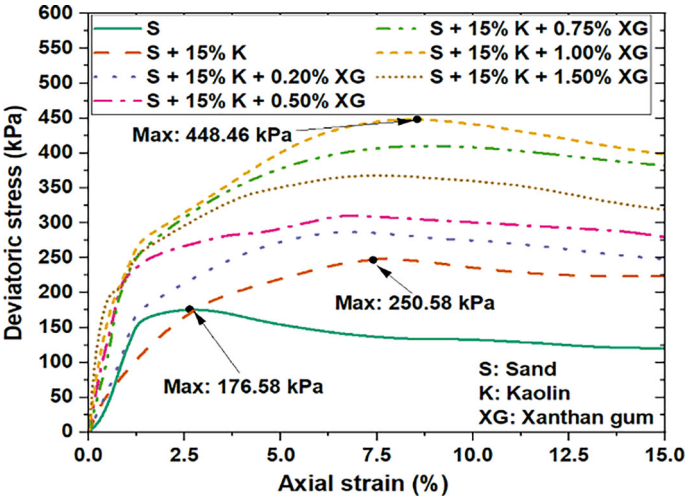

Fig. 2 Stress–strain response of XG-treated sand with Kaolin (effective confining pressure: 100 kPa, curing period: 30 days, static triaxial test in UU condition)

3.3 Environmental Scanning Electron Microscopy (ESEM)

The ESEM images (Fig. 3) display the distinct microstructural differences between the treated and untreated samples. Untreated sand (Fig. 3a) exhibits a loosely packed structure with minimal particle bonding and a high void ratio. With the addition of 15% Kaolin (Fig. 3b), the fine particles fill some of the voids, spread over the sand surface, and display a denser arrangement with improved cohesion. S + 15%K + TD (Fig. 3c) shows a flake-like structure with closely packed particles, indicating enhanced interparticle bonding and reduced porosity. In contrast, S + 15%K + XG (Fig. 3d) forms a fibrous, gel-like matrix that binds the grains, creating a more cohesive and compact structure. These differences suggest that while TD enhances particle packing, XG provides a stronger binding effect, improving structural integrity [10, 11].

3.4 Energy-Dispersive Spectroscopy

The EDS data (Fig. 4) indicate that untreated sand is primarily composed of Si and O, typical of quartz, with moderate Al and minor trace elements (Fig. 4a). The addition of 15% Kaolin increases Al and O levels while slightly reducing Si, confirming the presence of Kaolinite (Fig. 4b) [12]. Treatment with TD further enhances Al and O content while slightly reducing Si, suggesting stronger interaction between Kaolin and sand (Fig. 4c). Additionally, XG-treated samples exhibit higher C content, indicating the presence of organic matter (Fig. 4d).

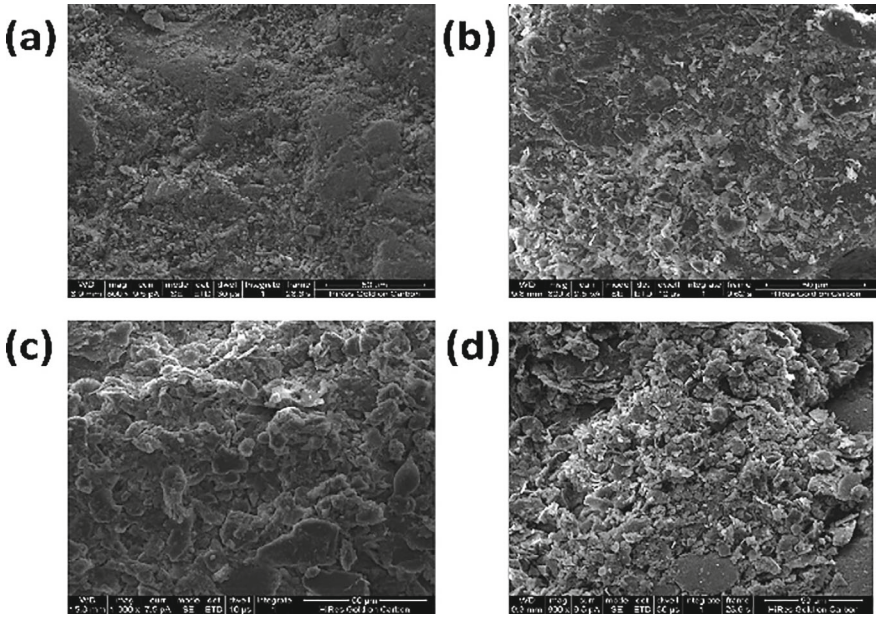

Fig. 3 ESEM images **a** Untreated sand **b** S + 15% K **c** S + 15% K + 0.075 ml/kg TD **d** S + 15% K + 1% XG (at 800 × magnification and scale bar = 50 μm)

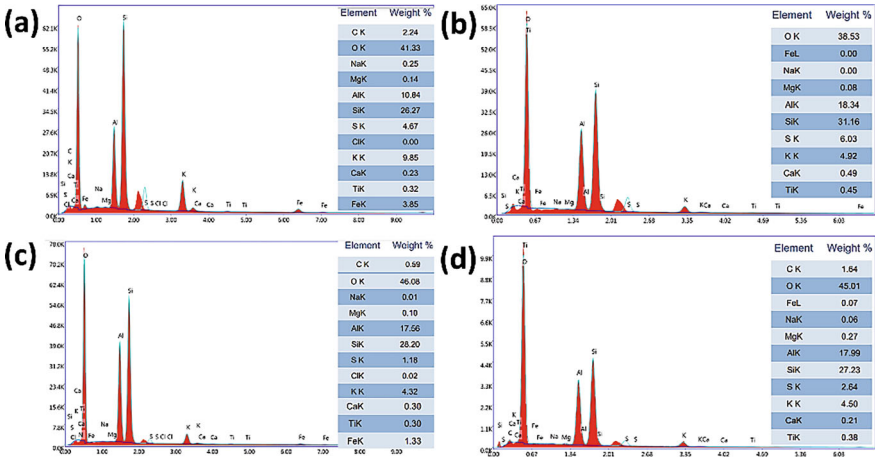

Fig. 4 Chemical composition of **a** Untreated sand **b** S + 15% K **c** S + 15% K + 0.075 ml/kg TD **d** S + 15% K + 1% XG from EDS

3.5 X-Ray Diffraction Spectroscopy

The XRD data in Fig. 5a confirm the presence of quartz in untreated sand, with primary peaks at 20.8°, 26.6°, and 42.5° (2θ).

In Fig. 5b, kaolinite is identified by major peaks at 35.0°, 41.2°, and 44.6° (2θ), while the absence of the characteristic 12.3° peak suggests possible structural modifications or reduced crystallinity in the Kaolin-sand mixture. In Fig. 5c, TD-treated S + 15% K exhibits peaks at 28.3° (C/C_3S), 34.9° (C), 36.8° (CH/CSH), 54.8° (C_2S), 60.9° (CSH/CASH), and 66.8° (G/CASH), indicating the formation of cementitious phases that enhance particle bonding, reduce porosity, and improve mechanical stability. The presence of CSH and CASH suggests ongoing pozzolanic reactions contributing to bio-cementation. In Fig. 5d, XG-treated S + 15% K shows peaks at 55.2°, 60.3°, 67.5–68.5°, and 80.1°, likely corresponding to newly formed or enhanced phases due to XG–soil interactions [13]. These peaks suggest the presence of amorphous compounds or weakly crystalline phases, contributing to improved cohesion and binding within the treated soil.

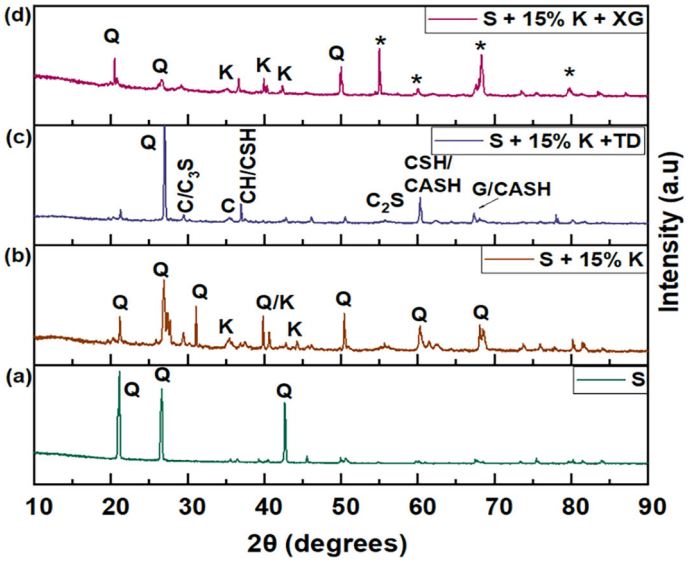

Fig. 5 XRD data of **a** Untreated sand, **b** S + 15% K, **c** S + 15% K + TD, **d** S + 15% K + XG (Q: quartz, K: kaolinite, C: calcite, CH: calcium hydroxide, CSH: calcium silicate hydrate, C_2S: dicalcium silicate, C_3S: tricalcium silicate, CASH: calcium aluminate silicate hydrate, G: gypsum, *: amorphous compound or a phase formed due to the interaction with XG)

3.6 Fourier Transform Infrared Spectroscopy

FTIR analysis reveals distinct chemical interactions in untreated and treated S + 15%K mixtures (Fig. 6).

Untreated sand exhibits stable calcite peaks at 450–500 cm^{-1}, with Si–O stretching confirming quartz and silicate minerals. The addition of 15% Kaolin maintains stable calcite peaks, with no significant spectral shifts. However, incorporating TD into the S + 15%K mixture introduces new peaks at 503.28 cm^{-1}, 517.85 cm^{-1}, 527.91 cm^{-1}, 547.08 cm^{-1}, and 567.10 cm^{-1}, indicating enhanced chemical interactions. Peaks at 1431 cm^{-1} and 1471 cm^{-1} confirm calcium carbonate formation, while C–S–H gel formation is evidenced by peaks in the 500–800 cm^{-1} (Ca–O stretching), 900–1100 cm^{-1} (Si–O stretching), and 3000–3700 cm^{-1} (O–H stretching) regions [14], with key peaks at 503.28 cm^{-1}, 907.93 cm^{-1}, and 3525.97 cm^{-1}. In contrast, XG-treated samples show FTIR shifts from 450–500 cm^{-1} to 1400–1600 cm^{-1}, confirming calcite formation, but exhibit weak or absent peaks in 500–800 cm^{-1} and 1000–1100 cm^{-1}, suggesting minimal C–S–H gel formation. The absence of peaks in the 1000–1200 cm^{-1} region rules out ettringite formation, while the removal of organic matter is indicated by missing peaks in the 1450–1500 cm^{-1} and 2800–3000 cm^{-1} ranges. Additionally, new peaks at 3500–3600 cm^{-1} and 1600–1700 cm^{-1} suggest hydration-driven changes and C–S–H gel formation. Key peaks at 503.49 cm^{-1}, 517.79 cm^{-1}, and 458.52 cm^{-1} confirm presence of Kaolin. While both TD and XG enhance soil stability, TD significantly promotes C–S–H gel formation, whereas XG primarily influences hydration interactions.

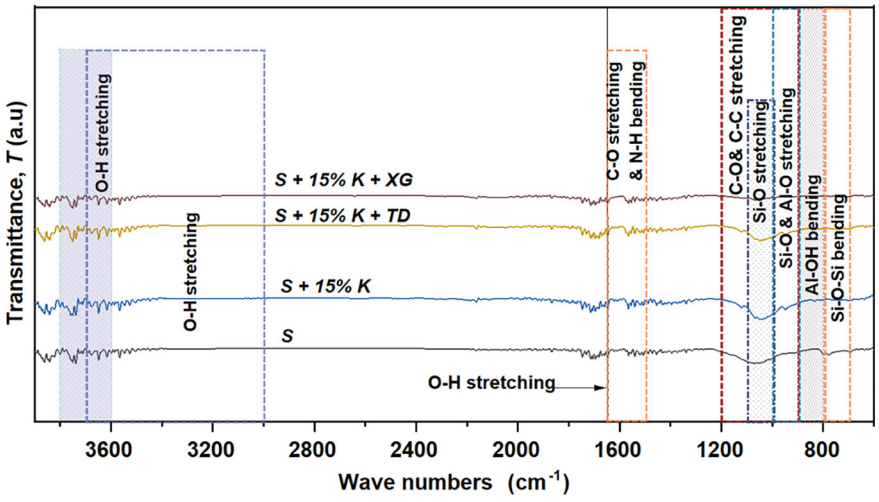

Fig. 6 FTIR data of TD and XG-treated sand

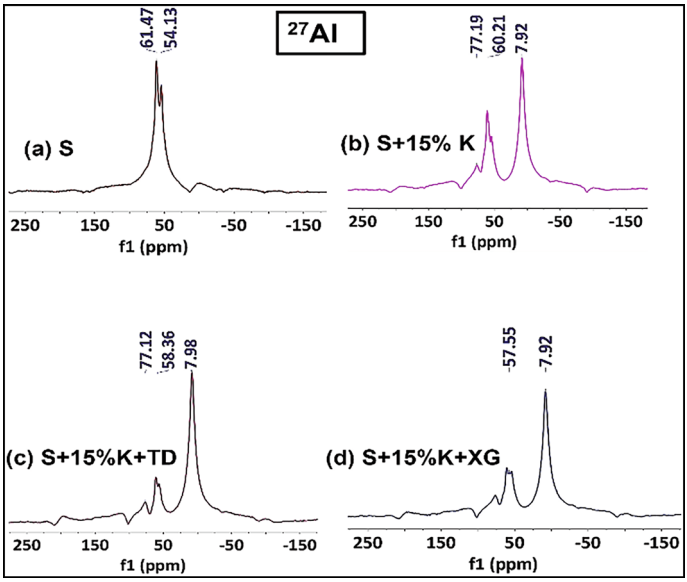

Fig. 7 Solid-state NMR spectra of TD- and XG-treated sand

3.7 Nuclear Magnetic Resonance

Solid-state ^{27}Al NMR analysis provides insights into the chemical environment and structural stability of aluminum in biomaterial-treated soils. In untreated sand, a peak at 61.47 ppm corresponds to tetrahedral coordination in quartz sand, consistent with the expected 60–80 ppm range (Fig. 7a).

Upon TD treatment in S + 15%K, this peak shifts to 60.21 ppm [15], with an additional peak emerging at 77.19 ppm, indicating modifications in the alumina coordination environment (Fig. 7b). The peak at 7.92 ppm, representing octahedral coordination in Kaolin, remains unchanged in untreated S + 15%K but slightly shifts to 7.98 ppm in TD-treated samples, suggesting minor alterations in Kaolin (Fig. 7c). Additionally, TD treatment results in new peaks at 77.12 ppm and 58.36 ppm, signifying changes in the alumina phase. In XG-treated S + 15% K, peaks at 57.55 ppm and 7.92 ppm indicate notable shifts in the quartz sand structure, while the Kaolin component remains unaltered (Fig. 7d) [16]. These spectral modifications suggest enhanced alumina interactions with biomaterials, leading to improved particle cohesion, frictional resistance, bonding strength, and overall soil stability, contributing to better mechanical performance in treated soils.

4 Conclusions

The addition of biomaterials significantly enhances the mechanical behavior of sand, particularly in the presence of Kaolin, which plays a crucial role in improving soil bonding. The alumina content in Kaolin enhances reactivity, stability, and binding sites, as confirmed by spectroscopic analyses, including EDS and NMR, along with its correlation to peak deviatoric stress. TerraZyme, a surfactant-based bioenzyme, interacts with Kaolin through its hydrophilic head, facilitating improved cohesion, while its hydrophobic tail exhibits minimal interaction with silica-dominant sand particles. Conversely, Xanthan gum, being entirely hydrophilic, strengthens soil primarily through physical bonding, with Kaolin further amplifying its effectiveness. Experimental results demonstrate that small amounts of Kaolin significantly enhance the chemical interactions and bonding characteristics of sandy soils. The combined use of TerraZyme and Xanthan gum has led to substantial improvements in soil strength, with a notable increase within 30 days of curing. The effectiveness of TerraZyme in soil stabilization has been validated in infrastructure projects, including road construction under the Pradhan Mantri Gram Sadak Yojana (PMGSY) scheme, showcasing its reliability under diverse loading conditions. Future research should focus on evaluating alternative bioenzymes, assessing their performance across varied environmental conditions, and conducting large-scale field studies to optimize bioenzyme-based soil stabilization techniques for broader geotechnical engineering applications.

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Modelling Agricultural Soils: Agro-Geotechniques Perspective

George Appiah-Kubi, Kai Wang, Ana Heitor, and Xiaohui Chen

Abstract Soil is viewed as a three-phase system consisting of solids, liquid and gas. This perspective does not seem to account for all the inherent variability of the soil, in particular, for residual soils and from agricultural perspective. It has been acknowledged that natural soils are heterogeneous consisting of biotic and abiotic factors. These factors are central to terrestrial ecosystem dynamics. In geotechnical engineering, the abiotic factors are accounted for mathematical formulations for soil properties derived from the three-phase conceptual model. However, because the biotic factors are not considered in the three-phase system, its contribution to soil properties are not mathematically represented. Consequently, the associated physics are not adequate to describe all physics needed to adequately model the natural heterogeneity in agricultural soils. This paper idealised soil as four-phase system (i.e. consisting of solid, soil biota, liquid and gas) allowing biotic factors to be adequately accounted for, in particular, for agro-geotechniques applications. It intends to extend the existing mathematical formulations for the basic properties of the soil derived from the three-phase model to include those of the biotic phase, thereby providing framework for modelling agricultural soils. This framework can be applied to develop hydro-soil-biota coupling model to analyse biological, chemical and mechanical stability of soils to address global issues facing agricultural lands.

Keywords Biotic · Abiotic · Heterogeneity · Four-phase soil model · Soil biota

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1 Introduction

Soils are naturally heterogeneous, multiphase and is a porous medium. Natural soils consist of inorganic mineral, organic matter, liquid and gas, making it a four-phase system. The fourth phase is living composed of organisms that are crucial to soil metabolism and soil formation [1]. There are two main factors that contribute to the soil's inherent heterogeneity: biotic and abiotic factors. These two factors' interaction contributes to terrestrial ecosystem dynamics. The biotic factors are the living things within the terrestrial ecosystem, such as plants roots and organisms, which are crucial for forming and stabilising soil aggregates both physically, chemically, biologically and mechanically, while abiotic factors are the non-living such as soil mineral, water, air, moisture content, temperature, rainfall, humidity, pressure and pH [2] which controls the biotic factors of an environment. The abiotic factors which are mainly the biological, physical and chemical conditions of the terrestrial ecosystem define the type of biotic suitable for a given terrestrial environment.

The interaction between the biotic and abiotic leads to changes in physical, biological, chemical and mechanical properties which are extremely variable that challenges our current understanding of time-dependent stability in soils and spatial distribution of soils [1, 3]. DeJong [1] noted that such changes can either improve or degrade soil properties from the as-constructed condition over the service life of a geotechnical infrastructure, prompting the importance of considering the inherent heterogeneity of soils from biotic and abiotic interaction perspective in current design philosophy.

The main objectives of this study are to (1) review and evaluate some of the naturally occurring soil types; (2) use the information gathered to extend three-phase conceptual model to four phase by including biotic phase, for applications in agro-geotechnology; (3) derive mathematical equations based on the four-phase conceptual model; and (4) analyse the equations by comparing to published data and/or experimental result.

2 Agro-Geotechnology Perspective of Soil

2.1 Soils

Natural soil is defined as biologically active, porous medium that has been developed in the uppermost layer of Earth's crust. Historical formation of soils makes it naturally heterogeneous with horizontal distinct layers in which micro-organisms (e.g. bacteria, fungi and algae) and macro-organisms (e.g. protozoa, nematodes, mites, springtails, insects and earthworms) and plant roots live, making the soil life. The biota activity in soils mostly takes place in the topsoil, which varies from a few mm to 30 cm of soil depth. It has been acknowledged that the biotic component occupies approximately 0.5% of total soil volume and makes up less than 10% of the total organic matter in soil. The biotic component significantly impacts on soil structure,

Fig. 1 Soil profiles from different parts of the world (Parikh & James, 2012; USDA-NRCS) images in **a** Mollisols, **b** Andosols, **c** Podzols and **d** Plinthosols

hence its impact on biological, physical, chemical and mechanical stability of soils. Figure 1a–d shows some of different types of naturally occurring soils from different regions of the world.

Mollisols are mostly found in the middle latitudes, common to grassland ecosystems, characterised by thick accumulation of humus in surface horizon and contain high organic matter. These are poorly drained due to the presence of swelling clay. They are known to be saturated with magnesium, calcium and nitrate and suitable for fruits, maize and tobacco.

Andosols (Fig. 1b) are found in ice-free areas, which is ~120 million hectares of the earth and contains ~5% of organic matter (OM) [4] and has mineralogical properties that are not common in other soil types. They are very fertile and support intensive sustainable agriculture practice, such as wet rice, fruit, maize, tea, coffee and tobacco farming, and have very low bulk density (<0.9 g/cm^3), thixotropy, high cation exchange and excellent water-holding capacity due to the high porosity and variability of pore size distributions found in them. Andosols have little swell-shrink properties, but have a strong hysteresis effect, extreme drying will irreversibly deteriorate water-holding properties, ion exchange capacity and ultimately the cohesion of soil particles. Andosols have high phosphate fixation capacity in steep topography.

Podzols (Fig. 1c) have grey or light-coloured horizon which resulted from severe leaching, or eluviation washing out everything but quartz grains. They are common to temperate regions, where the temperature varies between –40 °C and 20 °C with precipitation varying between 200 mm/year and over 1000 mm/year and evapotranspiration is limited. Most are infertile and are physically limiting soils for productive use due to high sandy portion resulting in extremely high permeability and low water-holding capacity. In addition, they are reported to have high C/N ratio and low pH**.**

Plinthosols (Fig. 1d) are soils commonly found in moist tropical ecosystems, mostly in West Africa, Central Congo and Brazil, which occupy ~0.5% of the earth surface. They are firm but soft, iron-rich, red-mottled clay made up of a humus-poor mixture of kaolinite, quartz and other constituents that irreversibly hardens to petroplinthite or petroferric contact when exposed to repeated wetting and drying. They have poor cation exchange properties and low base saturation.

From above review, it is not hard to infer that naturally occurring soils are inherently heterogeneous made up of horizontal layers with topsoil mostly OM where most soil biota thrives. Plinthosols are least productive owing to their lack to provide adequate abiotic factors, to support soil lives. Mollisols and Andosols are very productive because they provide suitable abiotic factors to promote biotic factors.

2.2 Soil Biota

Soil biota (e.g. plants and earthworms) are the sources of organic matter and lifeblood of fertile and productive soil without which plant growth will be hindered and crop production will be not sustainable. All soil biota eventually decompose into organic matter which return to soil and become valuable nutrient source for the biota, affecting soil structure, moderating soil pH, improving water-holding capacity and aeration, and increasing porosity and void ratio of the soil. Increase in OM by 1% results in increase in water-holding capacity of a soil by 3.7% [5]. This demonstrates that soil formation is dynamic, evolving over time and biota plays key role. The decomposed organic matter gives the topsoil dark colour containing ~1–6% OM. This decreases with soil depth. The dark colour increases absorption of heat thereby increasing the soil temperature, which decreases with depth. Through ingestion and egestion, earthworms pull OM below the soil surface and bring soil particles from deeper soil layers to the surface. This activity changes the soil texture in the affected soil horizons. Their burrowing creates soil pores, which improve soil aeration and drainage and reduce soil compaction. Hence, soil biota influences swelling and shrinkage of soils.

2.3 Abiotic

These are non-living and physical factors in the environment that directly controls biopotential of the soil biota, for example, soil, water, temperature, sunlight and soil pH. These factors control the biopotential of soil biota. Soil has a direct effect on the environmental conditions, habitat and nutrient sources available to the soil biota. Environmental changes alter soil biota and result in a shift in biopotential. These shifts can also result from climatic change, drought, extensive wildfires or changing rainfall patterns. Biota activities and shifts in microbial communities are site- and context-specific soil health indicators.

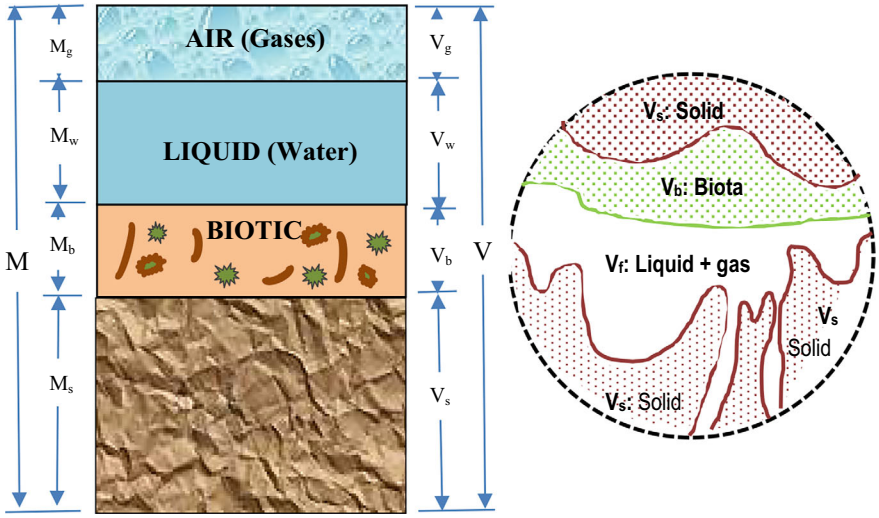

Fig. 2 Idealised soil profile and conceptual model for a typical agriculture soil

2.4 Conceptualised Model Agricultural Soil

To capture and fully understand all inherent factors which contribute to behaviour of agriculture soils, it is important to conceptualise the soil as multiphase material consisting of four phases, which are solid, organic matter (soil biota), water and air. Figure 2 shows conceptualised model which is an extension of the three-phase model to include the biotic phase.

2.5 Derivation of Mechanical and Hydraulic State Variable for Four Phase

In a four-phase soil consisting of solid, soil biota, liquid and gas, the properties depend on the relative percentages of these components, their spatial arrangements and other factors. For derivation purposes, it is convenient to show these components occupying separate spaces, as shown in Fig. 2. There are intrinsic relationships between the volume and mass of each phase which are prerequisite for complex constitutive modelling of soil. This relationship helps to develop state variables, i.e. mechanical and hydraulic, for constitutive modelling.

2.6 Mechanical State Variable

Bulk Density (ρ). Assume a Representative Elemental Volume (REV) V with a mass of M, then bulk density of the soil, ρ, which is the mass of solids in the given volume of soil can be defined mathematically by

$$\text{Bulk density of soil, } \rho = \frac{Mass of soil(M)}{total Volume of soil(V)} = \frac{M_s + M_w + M_b}{\mathrm{V}_s + \mathrm{V}_b + \mathrm{V}_w + \mathrm{V}_g} \quad (1)$$

where M is the total mass of the soil including biota, which includes V is the total volume, M_s is the mass of the solid grains, M_b is the solid mass of soil biota, V_s is the volume of the solid, V_b is the volume of the soil biota, V_w is the volume of water and V_g is the volume of the gas. In Eq. (1), the mass of the biota includes mass of the roots and earthworm, which is given by sum of the masses of the root and earthworm:

$$\mathrm{M}_b = M_r + M_e \quad (2)$$

Similarly, the volume is given by

$$V_b = V_r + V_e \quad (3)$$

where M_r and V_r are the mass and volume of the root, respectively, and M_e and V_e are the mass and volume of the earthworm. Worth noting dynamic nature of the biota is considered.

Equation (1) can be expressed in terms of the mixture density described by total (bulk) densities of the all the phases as

$$\rho = \frac{M}{V} = \frac{\rho_s^s V_s + \rho_s^w V_w + \rho_s^g V_g + \rho_s^b V_b}{V} = \rho_s + \rho_b \quad (4)$$

where ρ_s^s is the partial density solid in the soil phase, ρ_s^b is the partial density of the biota in the soil, ρ_s is the density of the soil phase excluding that of biota phase, ρ_b is the density in the biota phase, ρ_s^w is the density of water in the soil, ρ_s^g is the partial density of gas in the soil. Equation (4) represents extension of mixture bulk density for three-phase soil conceptual model to include the fourth phase (biotic phase).

Bulk density is extrinsic physical property which is influenced by pore space in the soil. Table 1 shows bulk densities of different soil textures and their effect on plant growth. The more the pore spaces, the lower the value for bulk density. From the table, sandy soil has the highest bulk density compared to silty and clayey soils. This is because sandy soils have larger pores spaces which results in fewer pore spaces compared to clayey soils which has small pore sizes resulting in high number of pore spaces.

Table 1 Relationship between soil bulk density and root growth based on soil texture [8]

Soil texture	Bulk densities suitable for plant growth (g/cm^3)	Bulk densities that restrict root growth (g/cm^3)
Sandy soil	< 1.60	> 1.80
Silty	< 1.40	> 1.65
Clayey	< 1.10	> 1.47

In agriculture fields, to improve soil structure, to boost aeration, increase nutrients assimilation, and increase water-holding capacity, soil compactness is disturbed mechanically or biologically. However, seasonally variations in temperature and precipitation, and other disturbance activities recompact the soil leading to increase in bulk density. [7] confirms earthworms are sensitive to compactness of soils, they tend to avoid area of high compactness as they have to spend more energy in burrowing. This leads to the conclusion that bulk density is one of the key mechanical properties of soil that can influence biopotential of soils. As the plant root growth dynamics and decomposition of dead roots increases, the bulk density decreases [8].

Void Ratio (*e*). e derivation for the four-phase agriculture soil is complex as it involves considering biota contributes to the volume of void in the soil. This involves understanding the plant's root growth dynamics and earthworm activities. These dynamics and activities evolve over time. By definition e is given by

$$e = \frac{V_v}{V_s} = \frac{(1-\eta)\mathrm{V}_b^v + \mathrm{V}_w^v + \mathrm{V}_g^v}{V_s} \tag{5}$$

where $\{e \in \mathbb{R} | e \geq 0 or e \leq 1\}$ and $\{\eta \in \mathbb{R} | \eta \geq 0 or \eta \geq 1\}$; V_v is the volume of void due to air, water exchanges and biota activities; $(1-\eta)\mathrm{V}_b^v$ indicates the volume of voids created as a result of plant growth and earthworm activities and η is the biota factor. It describes population of biota and extent of their activities that influences the evolution of void ratio in the soil. The value depends on biotic potential of the species under consideration, and on the abiotic factors. When $\eta = 0$,, it can be interpreted in two ways: (1) no soil biota present in the soil and (2) the plant roots have completely decayed, and the earthworm's egestion takes on surface of soil and there is no earthworm presence in their barrows. Alternatively, when $\eta = 1$, it indicates that due to the roots and earthworm presence in the soil, the soil is compacted, hence $1-\eta)\mathrm{V}_b^v = 0$. Biota phase does not contribute to void ratio. V_w^v and V_g^v are the voids created by water and gas, respectively.

From Eqs. (3) and (5) the void ratio can be expressed as

$$e = \frac{V_v}{V_s} = \frac{(1-\eta)\left(V_r^v + V_e^v\right) + V_w^v + V_g^v}{V_s} \tag{6}$$

From Eq. (6), the void ratio of the mixture of the REV in relation to the soil phase can be derived as

$$e = (1 - \eta)(e_r^s + e_e^s) + e_w^s + e_g^s \tag{7}$$

where e_r^s and e_e^s represent voids associated with the roots and earthworm, which mainly occupy the biota phase. The root's radial expansion and elongation compact the soil around its circumference and axially in front of the root leads to decrease in void ratio. However, earthworm burrowing activities increase the void ratio of the soil. e_w^s and e_g^s are the void ratio due to the presence of water and gas respectively. Equation (7) is the extension of three-phase void ratio to four phase by including the biota phase void ratio.

η can be partition into components depending on the number of species under consideration. In this study, two species are considered, which are plant and earthworm:

$$\eta = \eta_r + \eta_e \tag{8}$$

where $\{\eta_r \in \mathbb{R} | \eta_r \geq 0 or \eta_r \geq 1\}$, $\{\eta_e \in \mathbb{R} | \eta_e \geq 0 or \eta_e \geq 1\}$ and $\eta_r + \eta_e \leq 1$; η_r is the biota factor of the plant which is also influenced by the dynamics of root growth (i.e. growth and decay) and η_e represents earthworm barrowing activities (i.e. ingestion and egestion activities).

Substituting Eqs. (8)–(7) and rearranging lead to

$$e = (1 - \eta_r - \eta_e)e_r^s + (1 - \eta_r - \eta_e)e_e^s + e_w^s + e_g^s \tag{9}$$

Equation (9) can be simplified as

$$e = e_b + e_f \tag{10}$$

where $\{e_b \in \mathbb{R} | e_b \geq 0 or e_b \geq 1\}$, $\left\{e_f \in \mathbb{R} | e_f \geq 0 or e_f \geq 1\right\}$ and $e_b + e_f \leq 1$; $e_b = (1 - \eta_r - \eta_e)e_r^s + (1 - \eta_r - \eta_e)e_e^s$ is the void ratio due to the activities of soil biota at biotic phase and $e_f = e_w^s + e_g^s$ (i.e. due to fluid–soil–biota interaction).

Porosity (Ø). Porosity influences fertility and many physical processes such as soil erosion, infiltration, water-holding capacity, mechanical properties and evaporation. Due to the presence of biota in the soil, the porosity is not only dependent on the soil's natural arrangement and particle size but also the dynamics of root growth and earthworm activities. The soil biota promotes aggregation of soil particles, which increases porosity and permeability, and reduces bulk density (i.e. compactness). This influences the plant available water. Soil porosity is defined as the ratio of volume of voids (nonsolid) to the total volume of soil. Mathematically Ø can be expressed as

$$Ø = \frac{(1 - \eta)\left(V_r^v + V_e^v\right) + V_w^v + V_g^v}{V} = Ø_b + Ø_f = \frac{e_b + e_f}{1 + e_b + e_f} \tag{11}$$

where $\{\emptyset \in \mathbb{R} | \emptyset \geq 0 or \emptyset \geq 1\}$, $\{\emptyset_b \in \mathbb{R} | \emptyset_b \geq 0 or \emptyset_b \geq 1\}$ and $\left\{\emptyset_f \in \mathbb{R} | \emptyset_f \geq 0 or \emptyset_f \geq 1\right\}$. Equation (11) introduces the porosity contribution of the biotic phase ($\emptyset_b$), which is not considered in the three-phase classical geotechnical engineering modelling. Note that $\emptyset_b + \emptyset_f \leq 1$. Eq. (11) shows that the presence of soil biota increases porosity due to the soil faunal activities. The $\emptyset_b$ can boost the infiltration capacity of soil, reduce runoff and increase the amount of water that can be held in the soil.

2.7 Comparison of Results and Analysis

Figure 3a shows the relationship between porosity and organic matter (soil biota). Samples were collected from different terrestrial ecosystems from topsoil (0–15 cm deep) with different vegetation across Great Britain and analysed. Comparing the computed results shown in Fig. 3b to the experimental result in Fig. 3a, both graphs follow same trend. Both the experiment and computation assumed that there was mixing of the SOM by earthworms and plants in the soil. The four-phase model predicted higher values of $\emptyset$ than the three-phase mode, indicating that considering the biotic phase can increase the $\emptyset$ of the soil by ~2.5% at SOM > 0.10. Figure 3c and 3d shows the variation of initial e with SOM. Figure 3c shows the result and validation of an experiment carried out by [9] to study the effect of SOM on e. The linear relationship is because SOM bind helps the soil particles into aggregates which creates large voids in the soil. As SOM increases, more pores are created hence the linear relationship. Figure 3d shows the same trend with different initial e. Furthermore, Fig. 3d shows that considering the biotic phase in soil model can impact on the values of e. These findings confirm Eqs. (10) and (11). However, further investigations may be required to confirm these findings.

3 Conclusion

Three-phase conceptual model which forms the basis of classical geotechnical design has been extended to include biotic phase. This resulted in a new four-phase conceptual model. The integration of the biotic phase can help solve most of geo-environmental problems which challenges our currently understanding. Equations for the four-phase model have been derived and compared with field data and experimental data from literature. The findings were that the experiment results and field data mirror the equations derived. The study shows that inclusion of biotic phase can have significant effect on the physical, mechanical and hydraulic properties of soil. The biotic phase is dynamic due to the cycle of growth, death and decomposition which continuously contribute to SOM content of the soil. This enhances

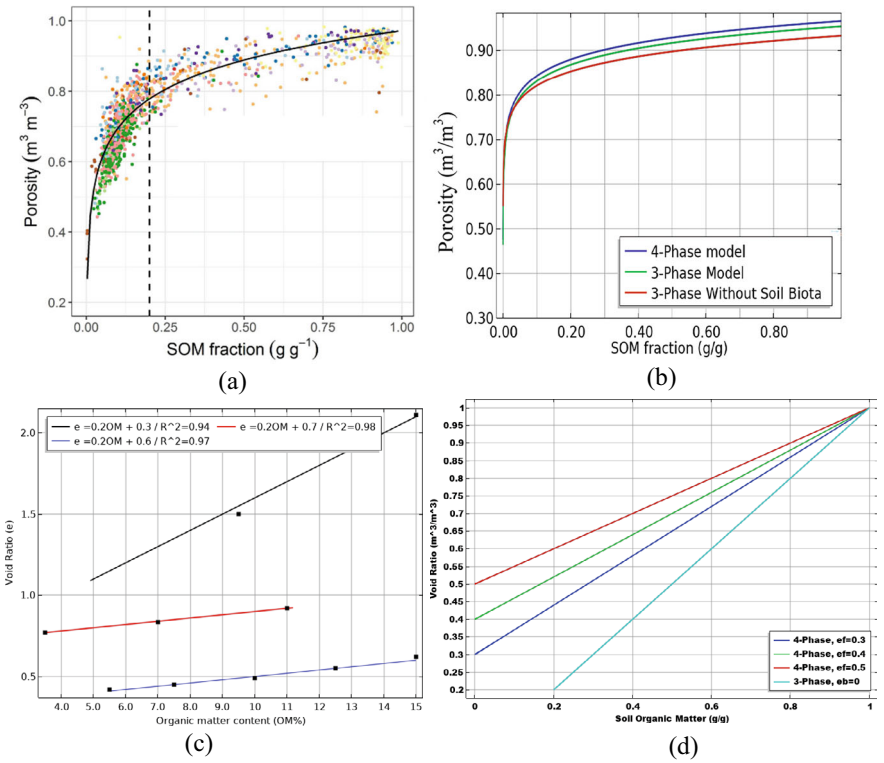

Fig. 3 Effect of SOM on Ø and *e* **a** Experiment by [6] to study effect of SOM on Ø; **b** Modelled result to study SOM and Ø in three and four phases **c** Experiment to study effect of SOM on initial *e* [9] and **d** Modelled results to study relationship between SOM and initial *e* in three- and four-phase soil models

porosity, moisture retention, aggregation (i.e. resulting in large voids) and biopotential leading to characteristics lower bulk density compared with the three-phase soils models where the biotic phase is not considered.

Hence, the incorporation of biotic phase captures the inherent heterogeneity in the soil and provides a good foundation for enhancing our understanding of geo-environmental problems. In addition, it could provide framework for modelling and analysing bio-inspired, bio-geotechniques and agro-geotechniques problems, in particular, those involving interaction between geoecological and hydrological cycle's feedback on land use and climate change, which challenges our current understanding.

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Evaluation of Modified Culture Media Obtained from Liquid and Solid Residual Fractions of Corn Steep Liquor, After Biosurfactant and Gramicidin Extraction, for *Lactobacillus Pentosus* Fermentation

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Abstract Corn steep liquor (CSL), an aqueous stream from the corn industry, is used as a nutritional supplement in animal feed and biotechnological processes like the production of Penicillin. CSL was also identified as a source of biosurfactants, patent WO 2020/234501, and Gramicidin S. This raises the possibility of evaluating the residual supernatant and biomass from CSL, obtained after biosurfactant and Gramicidin S extraction, as potential nutritional supplements for culture media. Hence CSL was subjected to cell-bound biosurfactant and Gramicidin S extraction with phosphate buffered saline (PBS) or ethanol, respectively. Subsequently, fermentations with *Lactobacillus pentosus* were ran at 31ºC for 30 h, using culture media based on extracted CSL containing a mixture of liquid and solid residual fractions and compared with commercial CSL. *L. pentosus* was chosen due to its high nutritional requirements, and the colony-forming units (CFU)/ mL of *L. pentosus* on the different CSL-based media were considered as dependent variable.

The results revealed that CSL subjected to extraction with either PBS or ethanol promoted similar growth of *L. pentosus*. Therefore, it can be concluded that extracted CSL can be used as a nutritional supplement in fermentative processes, increasing the added value of this secondary waste stream.

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Keywords Circular economy · Upcycling · Biosurfactants · Gramicidin S · Corn steep liquor

1 Introduction

Corn steep liquor (CSL) has garnered attention as a versatile and eco-friendly resource in the bioindustry, offering a sustainable alternative to traditional chemical feedstocks, as highlighted in the literature [1]. In microbial fermentation, the culture medium constitutes around 30% of the overall cost. CSL is a by-product obtained during the wet fractionation of corn. Usually, it is commercialized as viscous concentrate containing 50% in solids. CSL is considered an economical choice as a source of nitrogen, carbon, or vitamins in the production of lactic acid and other secondary metabolites obtained by biotechnological processes.

The Ministers of the Environment and Health of Canada have conducted a screening assessment of CSL studying different reports and research works from literature, concluding that CSL is a mixture of water-soluble corn extracts suspended in water, comprising approximately 50% water. Additionally, it contains lactic acid-producing bacteria and residual amounts of SO_2 from the manufacturing process (<0.01%) [2]. It is reported also that CSL contains apart from water, crude protein (47%), lactic acid (26%), ash (17%), phytic acid (7.8%), reducing sugars (2.5%), and fat (0.4%) as the primary components [3]. Although the amino acid and polypeptide class are the most highly represented in CSL, its predominant individual component, arginine, comprises only 7.2% of CSL (on a dry weight basis). Conversely, the individual chemical component with the highest concentration in CSL on a dry weight basis is lactic acid [2]. The ratio of D to L isomer found in CSL is influenced by the type of *Lactobacillus* bacteria present in the corn steep, and this ratio undergoes changes as the steeping process progresses [4]. It is important to mention that recently it was discovered that CSL not only contains lactic acid bacteria but also other bacillus strains with capacity to survive at high temperatures like *Aneurinibacillus aneurinilyticus* [5].

CSL is widely utilized in the fermentation industry due to its unique characteristics, cost-effectiveness, and ease of availability of raw materials. Its richness in key fermentation media components, such as nitrogen, carbon sources, and vitamins (see Table 1), makes it highly valuable. Additionally, the extraction of important compounds such as biosurfactants and the antibiotic Gramicidin S from CSL further enhances its applications. This approach not only aligns with the growing demand for renewable sources but also addresses environmental concerns associated with waste management.

Gudiña et al. [6] have used CSL as a nutritional supplement to enhance the production of Surfactin by *Bacillus subtilis,* whereas Lvova et al. [7] have evaluated the production and isolation of novel Gramicidin S analogues and biosurfactants from *A. aneurinilyticus* through a combination of fermentations involving corn steep liquor as the substrate and green chemistry extraction processes using HCl-ethanol

Table 1 Summary of the use of CSL to produce different metabolites

CSL	Microorganism	Metabolite	Ref.
100 g/L	*Bacillus subtilis*	Surfactin	[6]
55.0 g/L	*Lactobacillus rhamnosus*	Lactic acid	[10]
500 mg/L	*Lactobacillus rhamnosus*	Lactic acid	[9]
42.5 g/L	*Lactobacillus rhamnosus*	Lactic acid	[11]
100–400 g/L	*Aneurinibacillus aneurinilyticus*	Gramicidin and biosurfactants	[7]

or phosphate buffered saline (PBS) as the solubilizing agent. Moreover, it has been demonstrated that it is possible to extract biosurfactants from the microbial biomass naturally present in CSL using PBS as the extraction solvent (WO2020/234501) [8]. The maximum yields and productivities were achieved by Crivelari da Cunha et al. [9] that obtained 183.8 g/L of lactic acid after 72 h of fermentation using 220 g/L of potato residue, 55 g/L CSL and 45 g/L $CaCO_3$, presenting a productivity during the fermentation process of 2.55 g/L.h, 96.1% sugar consumption and 94.2% yield. Table 1 includes information about other biotechnological processes where CSL has been used as supplemental nutritional media.

Therefore, this straightforward green extraction technology enhances the environmental sustainability of chemical practices while aligning with current trends in chemical technology that prioritize ecological considerations. This approach holds significant implications in several areas related to discovering new sources of antibiotics and bioactive compounds.

1. Pharmaceutical innovation: Developing novel antibiotics and biosurfactants has the potential to lead to more effective treatments, especially in the face of growing antibiotic resistance. By creating new variations, researchers can explore enhanced efficacy, reduced side effects, and broader spectrum activity, which are crucial for modern healthcare challenges.
2. Sustainable practices: Adopting green chemistry methods in the production of these novel compounds is a proactive approach that supports global sustainability goals. Traditional chemical synthesis often involves hazardous substances and generates substantial waste. In contrast, green chemistry emphasizes reducing hazardous substances, energy efficiency, and waste minimization. This not only reduces the environmental impact but also agrees with regulatory trends toward more sustainable industrial practices.
3. Regulatory and market advantage: Companies that embrace green chemistry and sustainable practices are often perceived more favorably by both regulatory bodies and consumers. This can lead to faster regulatory approvals and a stronger market position, as consumers and stakeholders increasingly favor environmentally responsible products.

It is evident that the utilization of secondary raw materials as nutritional supplements to obtain bioactive compounds of commercial interest is crucial for achieving sustainability and a circular economy in conjunction with industrial symbiosis.

However, it is imperative to consider the streams remaining after fermentation, which are still rich in organic matter and other nutrients that should be reevaluated for potential future uses. Therefore, the purpose of this work was to subject CSL that is naturally fermented, to the extraction of biosurfactants and Gramicidin S with PBS and acidic ethanol, respectively, following the protocol established by Lvova et al. [7]. Subsequently, the sterilized streams will be evaluated as nutritional media for *Lactobacillus pentosus*, with the colony-forming units (CFU) as the dependent variable.

2 Materials and Methods

2.1 Materials

Corn steep liquor. Commercial CSL from FISH FeedStimulants, with a composition of 32% protein and 20% ash, was used. The CSL was diluted to a concentration of 400 g/L and subjected to an extraction process to obtain biosurfactants and Gramicidin S [7, 8].

2.2 Methods

Extraction of Gramicidin S and Biosurfactants from Corn Steep Liquor. Spontaneously fermented CSL containing lactic acid bacteria and *A. aneurinilyticus* [5] underwent various extraction processes for obtaining biosurfactants and Gramicidin extracts. The extraction of biosurfactants was performed using PBS in accordance with the protocol outlined in the patent WO 2020/234501 [8]. For the extraction of Gramicidin S, acidic ethanol was used following the method established by Lvova et al. [7]. Figure 1 illustrates the extraction scheme for biosurfactants and Gramicidin S from CSL, extracted with PSB (CSL-PBS) or extracted with ethanol (CSL-Ethanol) which has not been previously tested as a nutritional supplement in biotechnological processes.

Evaluation of corn steep liquor as a nutritional supplement in lactic acid fermentation. Before fermentation with lactic acid bacteria, the CSL was sterilized for 15 min at 121 ºC and employed as a nutritional supplement in a fermentation process conducted with *L. pentosus*. This species of lactic acid bacteria is known for its high nutritional requirements, making it an ideal candidate for assessing the efficacy of CSL as a nutritional supplement. Fermentation with *L. pentosus* was carried out at 31 °C for 30 h and the pH was adjusted at the beginning using $CaCO_3$ at the same concentration as the glucose used as the carbon source (30 g/L). This amount of $CaCo_3$ was added in excess, ensuring pH neutralization as lactic acid was produced. The fermentation medium also contained 10 g/L of CSL and 10 g/L of yeast extract.

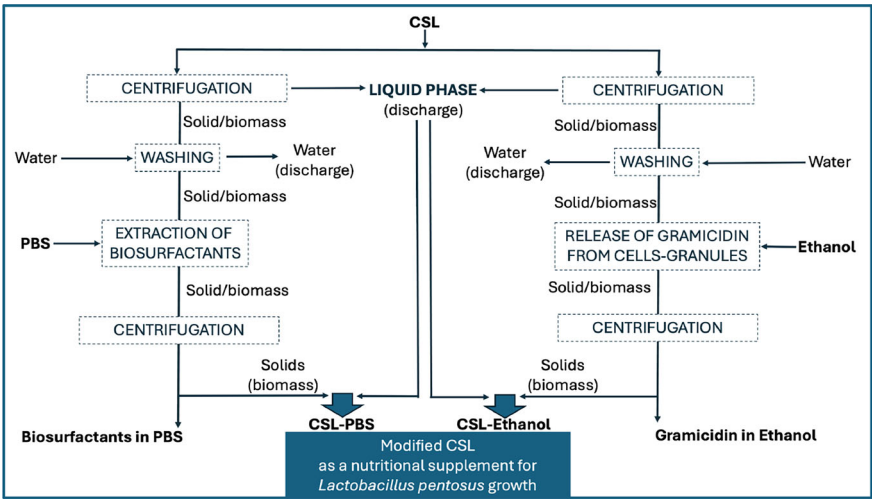

Fig. 1 Flowchart illustrating the extraction processes applied to fermented CSL

L. pentosus growth was monitored throughout fermentation by sampling, performing dilution series, plating 0.1 mL in duplicate, and counting CFUs on Man, Rogosa and Sharpe (MRS) agar. As a positive control, unmodified commercial CSL was used.

3 Results and Discussion

With the aim of zero waste and circular economy goals, solid and liquid fractions of extracted CSL were mixed, reestablishing the initial CSL composition. *L. pentosus* was subjected to fermentation in the presence of commercial CSL (CSL-Control), and two extracted CSLs: CSL-PBS and CSL-Ethanol. All fermentations with *L. pentosus* began with a 10% inoculum containing 5.58×10^9 CFU/mL. Figure 2 illustrates the kinetic profile of the logarithmic growth of *L. pentosus* under various CSL-based fermented media.

Remarkably, the extraction of Gramicidin S and biosurfactants from CSL had no impact on *L. pentosus* growth, as no significant differences were observed between the counts obtained for the three media, according to Tukey, Tukey B, and Duncan post hoc tests with 95% confidence. The resulting CFU/mL values were comparable for all three media after 30 h of fermentation: $4.45 \times 10^9 \pm 1.91 \times 10^9$ CFU/mL for CSL-Control, $6.45 \times 10^9 \pm 1.77 \times 10^9$ CFU/mL for CSL-Ethanol, and $3.75 \times 10^9 \pm 1.06 \times 10^9$ CFU/mL for CSL-PBS. In comparison with the literature Wu et al. (Wu et al. 2020) investigated the cultivation of *Lactobacillus rhamnosus* using corn starch liquid (CSL) to enhance the cost-effectiveness of probiotic production. Through continuous domestication, they developed a strain capable of growth in the presence of 12 g/L of glucose, 4 g/L of peptone, 10 g/L yeast extract, and 500 mg/

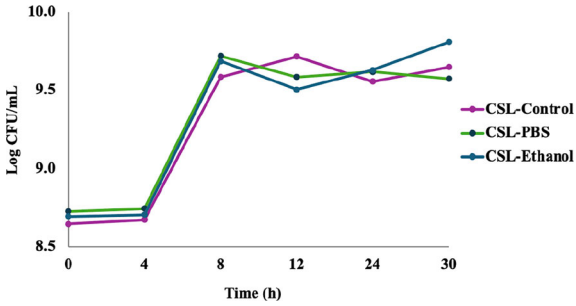

Fig. 2 Growth kinetics of *L. pentosus* (Log CFU/mL) in the presence of different CSL treatments

L of CSL. This strain retained the high antioxidant and antibacterial activity, as well as antibiotic resistance, of the original strain. Moreover, it exhibited enhanced probiotic potential. Using response surface methodology, the authors determined that the domesticated strain achieved a maximum cell yield of 9.78×10^{10} CFU/mL under optimal conditions of 10.96 g/L glucose, 0.86 g/L peptone, 12.17 g/L yeast extract, and 435.4 mL/L CSL. The values observed here are slightly higher than those seen with *L. pentosus.* However, it's worth noting that *L. pentosus* was not domesticated and its fermentation medium was less nutrient-dense, consisting only of CSL, glucose, and yeast extract. Other researchers [9] working with *L. rhamnosus* and using CSL as a nutritional supplement along with potato residue as a carbon source reported growth ranging from 4.4–6 $\times$ 10^7 CFU/mL to 2.3 $\times$ 10^9 CFU/mL, which falls within the range of the values obtained in this study. Additionally, Garcia-Hernández et al. [12] assessed an affordable culture medium for producing a probiotic additive with *Lactobacillus pentosus* LB-31 using a completely randomized design to track bacterial growth over 24 h. The study compared MRS broth (control) with a medium containing sugarcane honey and urea. The viable count varied based on the medium, with higher counts observed in the MRS broth. The highest concentrations were recorded at 8 h of fermentation, reaching 8.85 and 8.08 log CFU/mL, respectively, which are lower values than those achieved in the current work with *L. pentosus* and modified CSL.

On the other hand, Fig. 3 shows comparative images of colonies of *L. pentosus* growing in presence of CSL-Control, CSL-Ethanol, and CSL-PBS.

Based on the results presented here and compared to existing literature, this study demonstrates that CSL subjected to extraction processes to obtain naturally produced Gramicidin S or biosurfactants positively enhances its nutritional quality for the growth of *L. pentosus*, a lactic acid bacterium with high nutritional requirements, like *L. rhamnosus.* Therefore, modified CSL continues with its own function and effectively serves as a nutritional supplement in lactic acid fermentation, promoting the growth of lactic acid bacteria, which is the primary focus of this study.

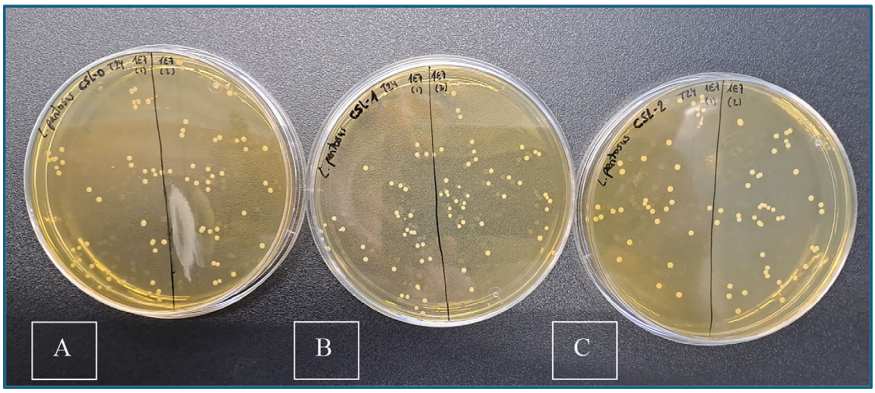

Fig. 3 Colonies of *L. pentosus* in **A** CSL-Control, **B** CSL-Ethanol, and **C** CSL-PBS at the same dilution factor (1E7) after 24 h growth

4 Conclusion

The data presented in this chapter align with the principles of promoting a circular economy and industrial symbiosis. These concepts involve resource optimization, waste reduction, and sustainability practices. The extraction processes of Gramicidin S and biosurfactants from CSL do not negatively impact the growth of lactic acid bacteria such as *L. pentosus*; instead, it enhances it. The findings suggest that exhausted CSL retains its nutritional value, ensuring that residual biomass or byproducts are effectively repurposed rather than wasted. The use of exhausted CSL in fermentation processes as a supplement reduces waste and minimizes the need for new raw materials, keeping resources in use for as long as possible and reducing the costs associated with purchasing additional nutrients. This is a key advantage for industries seeking to adopt more sustainable and financially viable practices. This study supports renewable industrial practices by potentially reducing the carbon footprint, which could contribute to reducing environmental pollution and resource depletion. Furthermore, the protocol promotes collaboration between different industrial sectors, where waste or byproducts from one industry become raw materials for another. This integration can lead to more efficient use of resources and innovation in biotechnological applications. However, further studies are needed that include the consumption of carbon sources and the production of metabolites, as well as the characterization of exhausted CSL.

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Sustainable Soil Stabilization Using Fly Ash, Biochar, and Polypropylene Fibers

Nitin Sharma and S. Rupali

Abstract This study explores the sustainable stabilization of clayey soils using fly ash, biochar, and polypropylene fibers to enhance geotechnical properties and mitigate heavy metal leaching. Laboratory experiments were conducted to evaluate the effects of varying proportions of biochar and fly ash on the strength and contaminant transport behavior of chemically stabilized clayey soils. Tests included Unconfined Compression Strength (UCS) after a 7-day curing period, and Scanning Electron Microscopy (SEM) to analyze microstructural interactions. Results indicated that fly ash significantly improved soil strength but introduced heavy metal leaching risks (e.g., Ba, B, Cu, Zn, Pb, Cd), potentially contaminating groundwater. Coconut husk biochar effectively reduced metal leaching but compromised soil strength. To overcome this limitation, polypropylene fibers were added, achieving a balance between strength enhancement and environmental safety. The study demonstrates the potential of combining fly ash, biochar, and fibers to develop stabilized soils with superior strength and reduced contaminant transport.

Keywords Sustainable soil stabilization · Fly ash · Biochar · Heavy metal leaching · Polypropylene fibers

1 Introduction

Soil stabilization is a reinforcement technique that ensures physical and mechanical modification of soil resulting in reinforced and stable soil structures. Stabilizers like lime, cement, and fly ash are effective but poses sustainability challenges. It has been observed that these materials pose the potential to release heavy metals when interacting with aqueous solutions [1]. Biochar is derived from the pyrolysis of biomass,

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and has demonstrated remarkable potential in contaminant immobilization and soil improvement [2]. This study explores the combined use of fly ash, biochar, and polypropylene fibers to enhance soil strength while mitigating contaminant transport. Several studies have examined the geotechnical and environmental benefits of fly ash, biochar, and polypropylene fibers in reinforcing the soil through stabilization technique [3–6]. On the environmental front, Leelarungroj et al. [7] performed leaching tests on cement and fly-ash-stabilized soils and arsenic leaching was found to be dominant. Xiao and Meng [8] highlighted the efficacy of biochar in reducing the mobility of contaminants through adsorption and ion exchange mechanisms. Similarly, He et al. [9] found that biochar effectively reduces the heavy metal contamination. Ahmad et al. [10] examined the potential of biochar for copper, cadmium, and lead from aqueous solutions. These studies underline the potential of integrating biochar into fly-ash-stabilized soil to address both geotechnical and environmental challenges.

2 Materials and Experimental Methods

2.1 Materials

The materials used in this study are clay, fly ash, biochar, and polypropylene fiber. The fly ash (class F) was procured from Guru Gobind Singh thermal power plant situated at Ropar in Punjab. Fly ash passing from 425 micron sieve was used in this study. Coconut biochar was procured from Ebee Nuporonix Agro Industry New Delhi India. Coconut biochar is a form of charcoal produced through the pyrolysis of coconut shells, and exhibits several properties that make it beneficial for various applications. Its diverse properties like high porosity, large surface area, carbon-rich composition, nutrient-retention capability, microbial habitat, soil structure improvement, and environmental sustainability prove beneficial for soil amendment for agriculture, stabilization, and environmental application. The polypropylene fibers were procured from Nina Concrete Systems Pvt. Ltd. These fibers are fibrillated type having length of 6 mm, diameter of 0.02 mm, and specific gravity of 0.9.

2.2 Material Characterization

The clay was air-dried for performing experiments. The general properties of soil are determined by conducting various basic experiments as presented in Table 1 following the Indian Standard Code of practice [11–14].

Column Setup and Microstructural Analysis. For the evaluation of the leaching properties, column test is conducted with the different compositions of biochar (0, 3, 5%) with standard fly ash–clay mixture shall be conducted. Vertical acrylic columns

Table 1 Properties of clay used in the study

Physical properties	Values	Physical properties	Values
Liquid limit	42%	MDD	16.75%
Plastic limit	18.9%	OMC	19.2%
Plasticity index	24	Color	White
Specific gravity	2.68		

(Fig. 1) with a diameter of 10 cm and a range of sample length 10 cm have been used in several laboratory tests for one-dimensional solute transport in homogeneous, 75% kaolin clay, and 25% fly ash with varying percentages of biochar in columns with low permeability. These samples were placed in the homogeneous columns. Hydraulic conductivity for the clay media employed in this investigation was discovered to be around 10^{-5} cm/s. The clay particles with grain sizes less than 0.3 mm were utilized as the testing medium after all contaminants, including gravel particles, residual leaves, and roots, had been eliminated. After the tests, the collected soil samples were air-dried, sieved, and kept at room temperature. The leachate obtained from the column set up was analyzed by atomic adsorption spectroscopy (AAS) and the results are presented in Fig. 2.

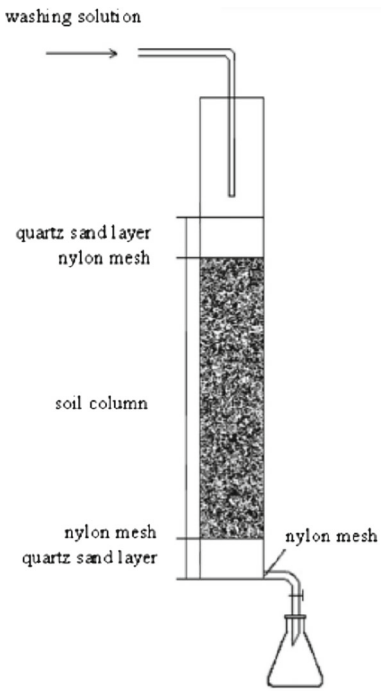


Fig. 1 Leaching column set up

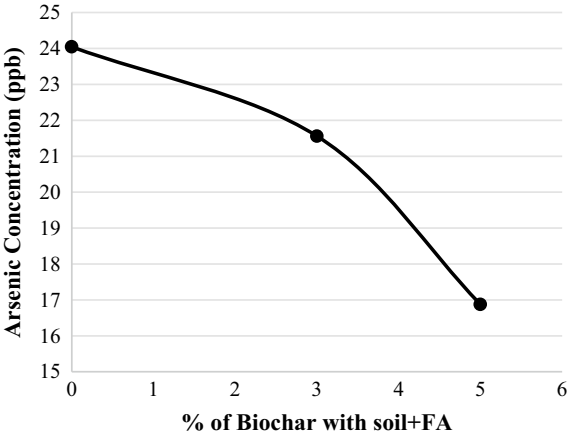


Fig. 2 Arsenic (ppb) at different percentages of biochar

Compaction and Strength Characteristics. Standard Proctor tests determined the maximum dry density (MDD) and optimum moisture content (OMC) of the soil–biochar–fly ash and PP fiber mixtures [14]. Further the unconfined compressive strength was determined for different mixes like soil and fly ash; soil, FA, and 3% biochar; soil, FA, and 5% biochar; soil, FA, and 7% biochar; soil, FA, 3% biochar, and 0.5% PP; soil, FA, 5% biochar, and 0.5% PP; soil, FA, 7% biochar, and 0.5% PP; soil, FA, 3% biochar, and 1.0% PP; soil, FA, 5% biochar, and 1.0% PP; and soil, FA, 7% biochar, and 1.0% PP [15]. The samples were cured for 3, 7, and 28 days.

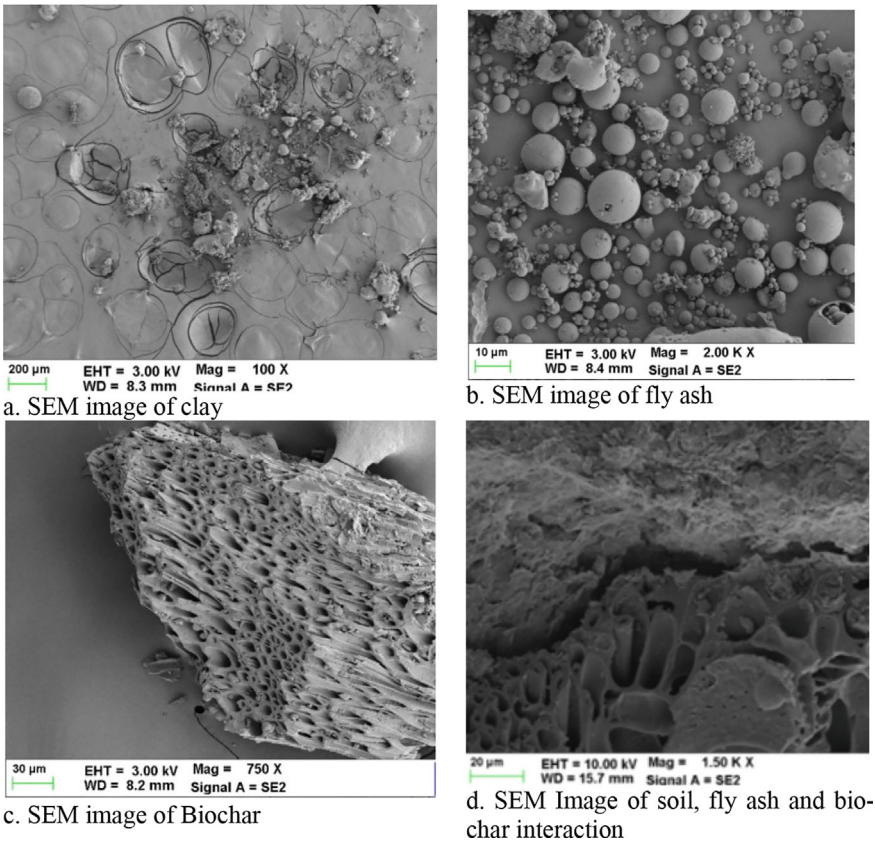
3 Results

3.1 Leaching Behavior

The Atomic Absorption Spectroscopy (AAS) was performed for the leachate obtained from the column experiment. Arsenic concentration was focused in this test as arsenic concentration is dominant in fly-ash-stabilized soils due to the generally alkaline pH of fly ash [7, 16]. For soil and fly ash mix, arsenic content was found to be 24.05 ppb, on addition of biochar (3%) the arsenic concentration reduced to 21.56 ppb and at 5% biochar addition the concentration lowered to 16.88 ppb, this highlights that biochar addition helps in mitigating heavy metal contamination.

3.2 Microstructural Analysis

Scanning electron microscopy tests (SEM) were performed on clay sample; fly ash sample; biochar sample; and soil, fly ash, and biochar samples and the same are presented in Fig. 3. The images reveal that clay has small pores, compact structure which affects its permeability, fly ash is spherical in shape which helps in improving the workability. Biochar has a porous, sponge-like structure, visible macropores help in adsorption capacity and the rough texture helps in adsorbing the heavy metals. The combined structure of soil, fly ash, and biochar demonstrates fly ash particles were effectively trapped within biochar's porous structure, confirming its role in contaminant immobilization.



a. SEM image of clay

b. SEM image of fly ash

c. SEM image of Biochar

d. SEM Image of soil, fly ash and biochar interaction

Fig. 3 SEM images for soil, fly ash, biochar and soil, fly ash and biochar interaction

Table 2 Compaction and strength characteristics of various samples

Mix composition	MDD (g/cc)	OMC (%)	0-day UCS (kN/m^2)	3-day UCS (kN/m^2)	7-day UCS (kN/m^2)	28-day UCS (kN/m^2)
75% soil + 25% FA	2.54	15.38	945	960	1010	1085
75% soil + 25% FA replaced with 3% biochar	1.58	18.5	925	940	1007	1045
75% soil + 25% FA replaced with 5% biochar	1.56	19	901	920	998	1039
75% soil + 25% FA replaced with 7% biochar	1.54	21.3	887	905	985	1025
75% soil + 25% FA replaced with 3% biochar + 0.5% PP	1.55	18.8	920	950	1025	1100
75% soil + 25% FA replaced with 5% biochar + 0.5% PP	**1.51**	**19.6**	**940**	**970**	**1070**	**1150**
75% soil + 25% FA replaced with 7% biochar + 0.5% PP	1.5	21	890	910	995	1010
75% soil + 25% FA replaced with 3% biochar + 1.0% PP	1.48	19.2	930	945	1015	1090
75% soil + 25% FA replaced with 5% biochar + 1.0% PP	1.43	20.5	960	965	1060	1142
75% soil + 25% FA replaced with 7% biochar + 1.0% PP	1.4	21.5	900	920	990	1005

3.3 Compaction and Strength Characteristics

The clay soil was stabilized using 25% fly ash as the optimal proportion for improving the mechanical properties [17–20]. The results of the different mixes are presented in Table 2. From the table, it is evident that as the biochar percentage increases the strength decreases and hence the polypropylene fibers were added to improve the unconfined compressive strength. The best mix is for soil + FA + 5% biochar + 0.5% polypropylene fibers.

3.4 ANOVA Analysis

ANOVA (Analysis of Variance) was employed to assess the significance of various factors on soil properties, which helps to understand whether different treatments or additives have a statistically significant effect on soil stabilization outcomes [21, 22]. A multi-way ANOVA at 5% level of significance was performed on MATLAB. Multiple factors like biochar percentage, PP fiber percentage, and curing time were fixed and the results are presented in Table 3. The *p*-value of biochar is 0.00 which

Table 3 Multi-way ANOVA analysis

Source	Sum of squares (SS)	(Degrees of freedom) D.f.	Mean square (MS)	F-value	p-value
Biochar%	20,857.7	2	10,428.9	61.72	0
PP%	0	0	0	0	NaN
Curing time	106,224.4	3	35,408.1	209.54	0
Biochar × PP	6306.4	4	1576.6	9.33	0.0011
Biochar × curing	4092.5	9	454.7	2.69	0.0564
PP × curing	1376.4	6	229.4	1.36	0.3064
Error (residuals)	2027.8	12	169	–	–
Total variance	186,582	39	–	–	–

indicates statistically strong effect and increasing biochar percentage reduces UCS, curing time significantly increases UCS, interaction of biochar and PP fiber is significant from its *p*-value of 0.0011 and that helps understand presence of PP fiber affects biochar effects on the UCS, and lastly the effect of biochar and PP fibers does not change significantly with curing time.

4 Conclusion

The study confirms that fly-ash-stabilized soils leach heavy metals especially arsenic. Addition of biochar reduces the arsenic concentration in the leachate but the strength is compromised on the addition of biochar, hence polypropylene fibers are added to compensate for the lost strength. It is observed that the optimum mix is obtained at 5% biochar and 0.5% polypropylene fiber added to 75% clay and 25% FA.

SEM analysis confirms biochar's interaction with fly ash but highlights potential bonding disruption at higher percentages. Multi-way ANOVA analysis confirms that biochar, PP fiber, and curing time have a significant effect on the UCS of the soil.

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Biological Degradation of High-Density Polyethylene Microplastics by *Comamonas Testosteroni*

Bess A. Newrick, Amanda Laca, and Adriana Laca

Abstract One of the most concerning issues related to global plastic production is the widespread presence of microplastics (MPs) in natural ecosystems. These recalcitrant micropollutants pose ecological and health risks. Biodegradation offers a promising approach to mitigating MP pollution. In aquatic environments, the most common microplastic type is polyethylene. This work investigates the ability of the bacterial strain *Comamonas testosteroni* CECT 326 to degrade high-density polyethylene (HDPE) MPs evaluating different process parameters. The results showed a reduction in polymer mass within 8 days, along with chemical and structural changes, indicating that the HDPE served as a carbon source for the bacteria. Temperature, MP size, and concentration influenced the degradation efficiency. ImageJ software analysis revealed a decrease in MP particle diameter and an increase in particle count, likely due to microbial activity fragmenting the MPs. High-Performance Liquid Chromatography-Mass Spectrometry (HPLC–MS) analysis of the culture medium before and after degradation identified the appearance of new compounds, some of which may be hazardous. This study confirms the potential of *C. testosteroni* to degrade HDPE MPs, contributing to mitigate the environmental impact of MP accumulation. However, it also underscores the need to address the release of toxic byproducts during biodegradation, which could lead to long-term adverse effects on ecosystems and human health.

Keywords Microplastics · Plastic pollution · Biodegradation · Bacteria · *Comamonas testosteroni*

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1 Introduction

Microplastics (MPs), defined as synthetic polymeric particles ranging from 1 μm to 5 mm in size, have become a significant pressing concern on a global level [1]. These micropollutants originate from various sources, including the fragmentation of larger plastics, the textile industry, fishing gear, tire wear, different industrial processes, and the direct release of MPs from certain products like cosmetics or personal care items [2]. MPs have been recognized as a potential threat to human health and wildlife, due to their bioaccumulation properties and persistence in natural environments [3]. Studies have linked MPs to health effects such as oxidative stress, immune response issues, endocrine and metabolic disorders, neurotoxicity, and reproductive disorders [4]. In addition, some plastic products are known to contain various toxic compounds that are added during the manufacturing process of the material, such as flame retardants, pigments, or plasticizers [5]. These additives are gradually released into the environment during fragmentation and aging. Due to their nature, MPs are also able to accumulate toxins from surrounding environments, such as heavy metals or polycyclic aromatic hydrocarbons, as well as microbial pathogens that attach to the plastic surface, leading to their distribution and concentration [6].

Plastics are designed to be durable, and therefore once they enter the environment they require extended time periods to be degraded and were initially considered an irreversible threat and environmental disruptor. A promising solution to address this issue is biodegradation, which involves the use of microorganisms, such as bacteria or fungi, or even higher organisms to degrade microplastics and minimize their presence in the environment [7]. Microorganisms possess many beneficial functions such as their adaptability, allowing them to survive in places with particularly harsh environments. Various species, including those belonging to the phyla *Proteobacteria*, *Firmicutes,* and *Actinobacteria,* have been reported for their ability to degrade polymeric particles, even especially recalcitrant polymers such as polyethylene (PE) [8].

In our previous study, we conducted an in-depth evaluation of the high-density polyethylene (HDPE) degrading potential of three bacterial strains, identifying *Comamonas testosteroni* CECT 326 as the most efficient [9]. A systematic analysis examined various process parameters, including bacterial species, temperature, MP size, and concentration on HDPE MP biodegradation. Additionally, an assay was performed adjusting all conditions to achieve maximum degradation efficiency.

This study further explores the biological degradation of MPs by examining reductions in particle diameter and the number of particles in the culture medium. It also examines the final compounds present in the culture medium, such as degradation intermediates, metabolites, or additives released from the HDPE, to assess potential environmental risks, such as the release of hazardous byproducts during biodegradation. Therefore, this research contributes to the understanding of biological processes for removing non-biodegradable fossil-based polymers, offering insights into bacterial MP degradation as a sustainable approach to mitigating plastic pollution, while also evaluating potential adverse effects, such as the release of toxic compounds.

2 Materials and Methods

2.1 Materials

High-density polyethylene (HDPE) commercial bottle caps used in food packaging were selected for the preparation of the model MPs employed in this study. The plastic was ground with an electric stainless-steel grinder (high-speed multifunction grinder, HC-150). The particles obtained were then sieved and separated by size, obtaining MPs with sizes between 20–100 μm and 100–250 μm, which were disinfected with 70% ethanol and dried completely before being employed in the biodegradation experiments.

2.2 Methods

Biodegradation Experiments. The biodegradation assays were carried out using the bacterial strain *Comamonas testosteroni* CECT 326, provided by the Spanish Type Culture Collection (CECT). Nutrient Broth (NB) medium was prepared for the pre-cultures and degradation assays, containing beef extract (1 g/L), yeast extract (2 g/L), peptone (5 g/L), and NaCl (5 g/L), according to the CECT Media Catalogue. All culture media were sterilized in an autoclave (Raypa AES-75) at 121 ºC for 20 min. For the pre-culture, the bacteria were plated on nutrient agar and then incubated at 30 ºC for 24 h. Afterwards, single colonies were transferred to the NB medium and kept at 30 ºC and 200 rpm for 24 h with a shaker incubator (New Brunswick G25 incubator shaker).

All biodegradation experiments were performed in 1 L Erlenmeyer flasks, which contained 300 mL of NB medium, prepared following the previous indications. Each flask was inoculated with 600 μL of the inoculum to achieve an initial concentration of $10^6 - 10^7$cells per mL. Various assays were performed to evaluate the effect of different parameters, including MP size (20–100 μm and 100–250 μm), temperature (30 ºC and 35 ºC), and concentration (200 mg MP/L and 500 mg MP/L). All experiments were conducted in duplicate at a minimum. The 20 mL samples of medium taken during the fermentation process were treated with 20 mL of 30% hydrogen peroxide and placed in a water bath at 60 °C for 45 min to eliminate the biomass. After cooling, the samples were filtered through grade 1 Whatman™ filter paper and dried overnight in an oven at 60 °C to recover the cleaned MPs. Further details on the experimental set-up can be found in our previous study [9].

Analysis of the Particle Diameter and Number. The mean particle diameter and the number of particles per milliliter in the culture media before and after degradation were measured using an LEICA M205FA stereomicroscope. Images were captured and processed with a high-resolution digital camera (LEICA DFC310FX) and analyzed using ImageJ software. For imaging, 1 mL aliquots were treated with

hydrogen peroxide as mentioned previously, and filtered using grade 1 Whatman™ filter paper and dried overnight at 60 °C.

Analysis of Compounds Formed After Biodegradation. To identify possible metabolites, degradation products or intermediates, and additives released from the HDPE during the biodegradation, a high-resolution quadrupole and time-of-flight mass spectrometry system equipped with an ion mobility unit (Agilent 6550) and coupled with an HPLC liquid chromatograph (Agilent 1290 Infinity II) (LC-IM-Q-TOF) was used. Culture medium samples, taken before and after degradation, were analyzed to detect compounds present only after treatment.

Statistical Analysis. IMB SPSS 27.01 was used for all statistical analyses performed, where the data was presented as the mean ± standard deviation (SD). The normality of the data from the experiments was determined using the Shapiro–Wilk test and Q–Q plots, and for assuming equal variances between groups, Levene's test was performed. The t-Student test was used to determine statistical differences. All the statistical analyses were conducted with $p < 0.05$ as the criterion for statistical difference.

3 Results

3.1 Measurement of MP Diameter and Count

The reduction of the MP diameter after biodegradation was evaluated to build upon the research presented in our previous article, where biodegradation assays focused on polymer weight loss, surface morphology changes, and functional group modifications [9]. Samples were selected for further analysis from experiments with the highest degradation efficiencies, where temperature, MP size, and concentration were evaluated. Consequently, the following results pertain to biodegradation under the following conditions: 30 ºC, MP size of 100–250 μm, 500 mg/L concentration during 8 days (A0); 35 ºC, MP size of 100–250 μm, 500 mg/L concentration during 8 days (A1); and 30 ºC, MP size of 20–100 μm, 500 mg/L concentration during 8 days (A2). Finally, a sample from the experiment conducted under the conditions that yielded the best results was analyzed: 35 ºC, MP size of 20–100 μm, 200 mg/L concentration during 10 days (A3). A lower MP concentration was used, as this led to higher degradation efficiencies, possibly due to microbial inhibition caused by environmental stress at higher MP concentrations.

Figure 1 presents the mean diameter of the MP particles before and after biodegradation assays within each size range. The figure shows that biodegradation led to significant changes in the particle diameter, and in all cases, a decrease of the mean particle size was found. As observed, increasing the temperature from 30 ºC to 35 ºC results in a greater reduction in particle diameter from 18.38% to 20.04% for MPs with a particle size between 100 and 250 μm, suggesting enhanced HDPE degradation at higher temperatures. In addition to temperature, MP size also affected

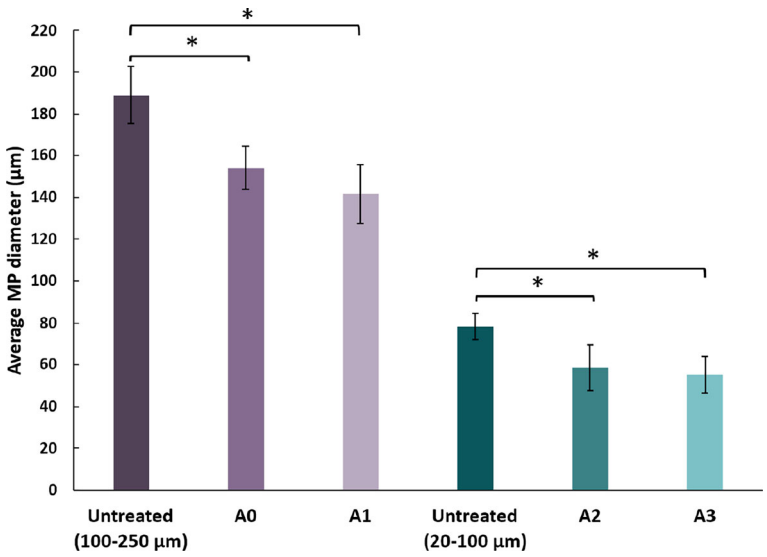

Fig. 1 Reduction in MP diameter after treatments with *C. testosteroni* 326. Error bars indicate standard deviation, and asterisks denote significant differences from the controls

degradation efficiency. The mean MP diameter decreased as particle size reduced, with the greatest diameter reductions observed in treatments A2 and A3, compared to those in treatments with larger particle sizes. This may be explained by the increase in the specific surface area of the MPs, which allows for higher interaction between the bacteria and the surface of the polymers, favoring biodegradation. The highest reduction of 29.02% was observed in treatment A3, which was conducted under the most favorable conditions.

Previous investigations have also reported a decrease in the mean particle size after exposure to different microorganisms. In a study of the degradation of polyethylene (PE) microplastics by mesophilic mixed bacterial cultures isolated from municipal landfill sediment, it was observed that *Bacillus* sp. and *Paenibacillus* sp. were the most dominant species, reducing the dry weight of PE MP by 14.7% and the diameter by 22.8% in 60 days [10]. In a different study by Gong et al. [11], a reduction in the particle diameter from 7.3 µm to 2.63 µm and 1.58 µm was observed with neutral bacteria-treated and alkali-resistant bacteria-treated polyethylene terephthalate (PET) particles, respectively, revealing that there is surface erosion during the biodegradation process of PET.

As shown in Fig. 2, the number of MP particles increased after treatments A0, A1, and A2 with *C. testosteroni* 326. This suggests that the bacterial activity leads to the breakdown of the HDPE MPs into smaller particles, thereby increasing the total number of detectable MPs while reducing the mean diameter. This phenomenon has been observed in previous biodegradation studies, where microbial enzymatic activity targets larger plastic particles, leading to their progressive fragmentation

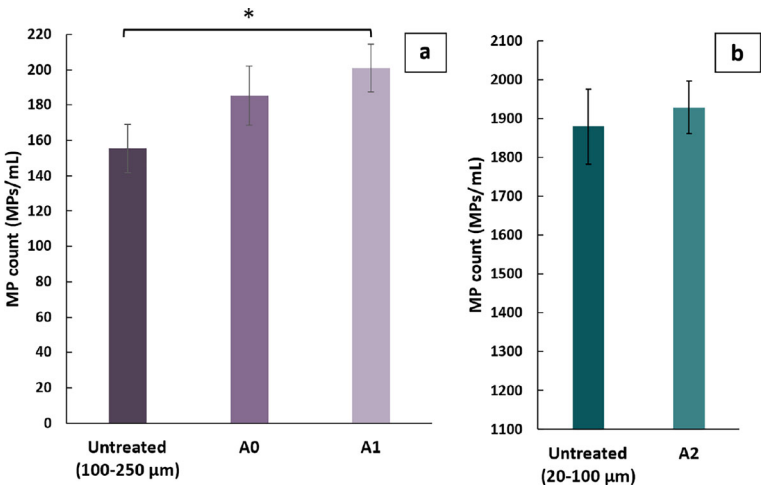

Fig. 2 MP counts in untreated and treated samples for different size ranges. Error bars indicate standard deviation, and asterisks denote significant differences from the controls

before achieving complete mineralization [12]. These results, along with the observed polymer weight loss and the morphological and chemical changes previously reported after bacterial exposure [9], indicate MP biodegradation.

In the case of treatment A3, the results were not included in Fig. 2, since a lower total number of particles is due to the concentration employed during the experiment in comparison to the control (200 mg/L).

3.2 Analysis of Degradation Products by HPLC–MS

Table 1 presents a list of possible organic compounds detected in the culture medium after the biodegradation process, according to HPLC–MS results. As shown, a range of compounds were identified following the treatment under conditions yielding the most favorable results, specifically at a higher concentration of MPs (A2). These compounds are likely to be bacterial metabolites, intermediates from the oxidation of polyethylene, or additives present in the polymer, which are released during its breakdown. As shown in Table 1, a total of 17 possible compounds were detected.

During biological degradation, the polyethylene chains undergo oxidation, leading to the incorporation of hydroxyl, carboxyl, or carbonyl groups. This oxidation facilitates the breakdown of the polymers, producing functionalized hydrocarbons, which may explain the presence of compounds such as $C_{16}H_{33}O_{12}$, and $C_{25}H_{41}O_{12}$. Additionally, $C_{15}H_{31}N_2O_6S$ was detected in the culture medium, which may correspond to Capryloamphohydroxypropylsulfonic acid, a known emulsifier and surfactant [13]. Furthermore, $C_{23}H_{42}N$ could correspond to a bioactive compound with antibacterial

Table 1 Possible organic compounds detected by HPLC–MS in the culture medium after degradation. The measured m/z and possible ion formulas with corresponding m/z are shown

Measured m/z	Ion formula	Theoretical m/z	Score
152.1064	$C_9H_{14}NO$	152.1070	100.00
397.1711	$C_{27}H_{23}O_3$	395.1642	100.00
417.1969	$C_{15}H_{31}N_2O_{11}$	415.1922	100.00
	$C_{14}H_{21}N_{14}O_2$	417.1966	97.81
	$C_{16}H_{33}O_{12}$	417.1967	81.91
620.2578	$C_{16}H_{42}N_7O_{18}$	620.2581	100.00
	$C_{14}H_{30}N_{21}O_8$	620.2581	93.22
367.1894	$C_{15}H_{31}N_2O_6S$	367.1897	100.00
	$C_8H_{23}N_{12}O_5$	367.1909	86.00
	$C_{12}H_{23}N_{12}S$	367.1884	80.24
279.1327	$C_{17}H_{16}N_3O$	278.1288	100.00
444.2265	$C_{10}H_{28}N_{13}O_7$	442.2229	100.00
	$C_{22}H_{24}N_{11}$	442.2211	81.70
536.2679	$C_{25}H_{41}O_{12}$	533.2593	100.00
	$C_{38}H_{33}N_2O$	533.2587	87.61
405.0906	$C_{14}H_7N_{14}O_2$	403.0871	100.00
332.3303	$C_{23}H_{42}N$	332.3312	100.00

activity which can be produced by microorganisms, such as *Dysidea herbacea* [14]. *C. testosteroni* has been found to produce secondary metabolite with antimicrobial properties [15], as well as surfactants [16]. Similarly, $C_9H_{14}NO$ may correspond to 2-Azaadamantane-N-oxyl (AZADO), an aliphatic hydrocarbon known to catalyze oxidation reactions, which may be related to oxidative processes occurring during the breakdown of the polymer [17]. While the exact origins of the detected compounds remain uncertain, they may either be byproducts from the degradation process or metabolites produced by *C. testosteroni* as part of its metabolic activity. Further in-depth analysis is required to confirm the detected compounds and evaluate their origin and potential environmental toxicity.

4 Conclusion

Due to the growing threat of microplastic contamination, there is an urgent need for effective removal methods, with biodegradation emerging as a promising solution. This study demonstrates the ability of *Comamonas testosteroni* CECT 326 to break down HDPE MPs, through the reduction of the mean particle diameter and changes in the MP count due to the fragmentation of the polymer. These findings build on previous observations of weight loss, as well as morphological and functional group changes, further indicating MP biodegradation. During the process,

several compounds were detected in the culture medium, which could correspond to metabolites produced during degradation, such as surfactants or antimicrobial agents, as well as intermediates or byproducts released from polymer degradation. Additionally, some compounds may have originated from additives used in the manufacturing of plastics, which may lead to the inhibition of bacterial activity. The findings from this investigation highlight the complex nature of MP degradation, as well as the need for further in-depth research to determine the fate of these particles in the environment and enhance microbial degradation processes.

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The Influence of Failure Patterns on the Tensile Strength of Plant Roots from *Salix Sp*

G. I. Otim, A. Zhelezova, G. Sorrentino, S. Trapp, and I. Rocchi

Abstract Vegetation can be used to stabilize slopes and for erosion control along water bodies, or for improvement of soil health and biodiversity, mainly through the functionality of their roots. Plant roots exhibit a segmented and non-uniform structure, with their diameter typically decreasing as they extend deeper into soil. This study explores the impact of evaluating possible failure types in roots on the tensile strength using Salix sp. (willows). The tensile strength of roots grown in pot experiments was determined on single root specimens at 30, 60, and 90 days, respectively. The failure pattern for each measurement was categorized as breakage failure from the top third, middle third, bottom third, and slippage after each tensile test. One-way ANOVA and post hoc analyses to establish pairwise comparisons were used to evaluate these failure types with tensile strength for each given age. This evaluation suggested that all failure types recorded along the root yielded a reliable measure of tensile strength, free from underestimation or overestimation until 2 months of age. Therefore, it would be beneficial to explore tensile strength beyond the middle third considering the root morphology is non-uniform.

Keywords Tensile strength · Failure pattern · Failure type · Slippage failure · Reinforced soil · Willows

1 Introduction

Plant roots are multicellular structures typically found underground. They play important roles in anchoring the plant, absorbing solutes and nutrients, and storing carbohydrates produced during photosynthesis [1]. Biogeotechnics and bioengineering leverage these functions to provide reinforcement to soil by integrating

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mechanical, biological, and ecological principles. This approach is used for sustainable upland slope protection, erosion control, fill slope stabilization, earth embankment protection, dune stabilization, reservoir drawdown area management, and the stabilization/restoration of shorelines and stream banks [2–5]. In thes systems mentioned, the structural integrity of plant roots significantly strengthens over time as vegetation becomes established [6]. Various types of monocotyledonous and dicotyledonous vegetation contribute to the development of rooted soils, including herbaceous plants (i.e., grasses) and woody plants (i.e., shrubs, and trees) [6]. Root systems can vary with respect to the monocotyledonous or dicotyledonous vegetation [1, 7]; however, in woody plants, the root morphology can be broadly categorized into three primary types: plate, heart, and tap [8]. Recent studies highlight a link between root architecture and slope stability and subsequently the influence of root systems can be based on stabilizing slopes and reducing erosion [9]. These can be classified as Horizontal (H), Vertical-Horizontal (VH), and Mixed (M) root types, respectively [10, 11].

Root strength exhibits significant variation, both between different species and within individual root systems, influenced by the mechanical functions they serve. Root strength can be characterized by parameters including tensile strength, elasticity, pullout strength, uprooting strength, and root cohesion. Among these, tensile strength is particularly important in rooted soils [6, 12, 13] considering that soils are strong in compression and their combined effect with the tensile strength from roots yields a reinforced soil. This also makes root tensile strength a basic parameter for root reinforcement evaluation [14].

The strength of roots is determined by various species- and site-specific factors, including terrestrial environment, season (winter, summer, spring, autumn), altitude, orientation, lignocellulosic biomass [15], root demography [16–19], and method of planting [20–25]. Recognizing these factors can deepen our understanding into the variation of root strength across diverse ecological settings. The tensile strength of roots tends to correlate with their diameter, and this relationship can be effectively described using a power law equation [6].

Root tensile strengths are typically assessed using Universal Testing Machines. In this process, the roots are cut to a specified length, depending on their diameter size as either fine ($\leq$ 2 mm) or coarse roots ($>$ 2 mm), their ends are appropriately prepared for gripping, and they are securely clamped into the machine. The roots are then subjected to tensile stress by increasing the distance between the two grips at a constant rate of displacement or constant strain rate, until they reach the point of failure where they break into two [6]. This yields valuable data on the ultimate tensile strength and the roots that exhibit failure within the middle third of the evaluated specified length are presented in conjunction with the ultimate tensile strength while those that show breakage failure out of this range are discarded. Additionally, a curve fit using the power law is then used to correlate all the root diameters and their respective tensile strengths.

This study focuses on understanding the influence of various failure types in *Salix sp.* (willows) roots, particularly those occurring outside the middle third of their length like top breakage, bottom breakage, or slippage owing to very limited

studies [26]. Specifically, breakage failures can occur at the top or bottom third of a given specified length or may result from slippage at either end of the clamp. Given that roots vary in diameter and are segmented along their length, our aim is to compare these failure types with those associated with the middle third, to verify the impact of including them and potentially reduce the number of testing required. At the same time, this would improve measurement of tensile strength and reliability with much fewer number of roots compared to the current practice which does not take into consideration their non-uniform structure and root morphology into consideration during this test and thus requires a large number of roots for laboratory and/or field evaluation. This will aid the design of effective conservation strategies and sustainable land management practices where using this approach much fewer roots will need to be tested to evaluate their tensile strength.

2 Materials and Methods

2.1 Pot Experiment

Pots made of Poly Vinyl Chloride (PVC) pipes were used measuring 100 mm inside diameter and 200 mm height. Loose sand of 0.3–0.5 mm grain size was used as substrate, which was placed using the pluviation method up to a height of 180 mm. Cuttings of *Salix sp.* with diameters ranging from 10 to 14 mm measured using a Vernier caliper and a length of 180 mm were carefully placed, ensuring that half of each cutting was positioned above the sand surface and the other half below, allowing for optimal rooting and shoot development.

The pots were transferred to a cellar, where they were watered daily with 100 ml of water and grown under the same light and nutrient conditions as described in Trapp et al. [27]. This light source was provided continuously for 16 h a day with the nutrients being only applied every 2 weeks during the initial 60 days. Additionally, separate pots were designated for evaluating tensile strength at 30, 60, and 90 days.

2.2 Tensile Strength

At the specified age, the shoot system was trimmed off the cutting, and the pot was disassembled to facilitate the manual washing of loose sand off the roots using tap water. Following this process, the root architecture was analyzed, and the primary roots were trimmed to a uniform length of 100 mm, while the second and higher order roots were discarded. To enhance grip and reduce slippage during testing, hot plastic glue with a setting time of about three minutes was molded at each end of the roots, resulting in an effective test length of 50 mm [28] along the root. A minimum of 20 diameter readings were taken along the length of each root using

a Nikon digital microscope, and the average diameter was calculated from these measurements. More readings were taken for roots with highly variable diameters along the effective test length.

The ends of the root were clamped in both the upper and lower jaws of the Instron 5965 universal tensile strength equipment, ensuring minimal or no slack. The equipment was then operated at a rate of 10 mm/min [28] until failure occurred, defined as the point at which the root broke into two parts. The data from each test was recorded by the in-built data logger, capturing the tensile curve to failure, and this was subsequently saved for each tensile test.

The failure type associated with each root was documented and illustrated through sketches, indicating whether it occurred in the middle third (M), top third (T), bottom third (B), or as slippage (S).

3 Results

The relative number of root specimens that failed with respect to the middle third (M), top third (T), bottom third (B), and slippage are presented in Fig. 1. This shows that failure within the middle third (usually defined as a successful test) was 49–62% at the three testing ages. Considering all ages this accounted for 192/330 root specimens in this study having this pattern. This left the remainder susceptible to either the bottom or top failure pattern. The slippage pattern was the least likely to occur with less than 4% in the test specimens. Some scholars have reported failure within the middle third as 7% [26], 33% [15], and 45–55% [14, 29] of the total specimens tested.

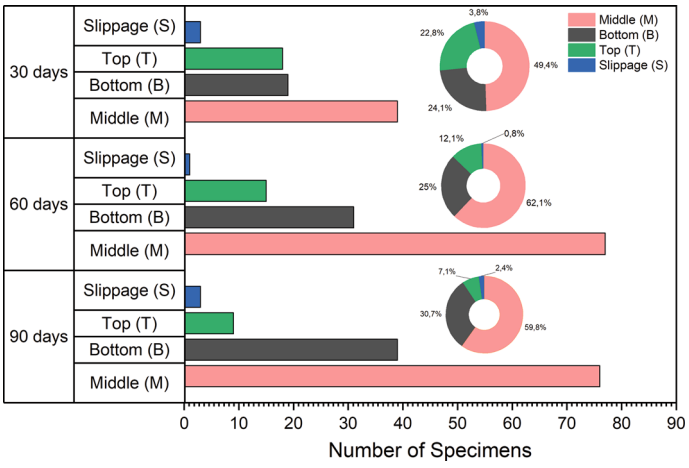

Fig. 1 Failure pattern for each age

The evaluation of the primary root diameters at 30, 60, and 90 days ranged from 0.18 to 1.08 mm, 0.25 to 1.04 mm, and 0.23 to 1.84 mm, respectively, as measured by optical microscopy. This indicated an increase in root diameter with age during plant growth as shown in Fig. 1. In addition, the tensile strength generally increased with the age of the plant roots as shown in Figs. 1, 2, and 3. The tensile strength measurements for 30, 60, and 90 days were subjected to normality and distribution tests to ensure their validity for statistical analyses. Following these assessments, a one-way ANOVA was conducted, complemented by mean pair comparisons using the Fisher Least Significant Difference (LSD) method at 30, 60, and 90 days. The one-way ANOVA identified that the failure type did not show a statistically significant influence on tensile strength, with p-values of 0.38 (F-value = 1.04) and 0.29 (F-value = 1.27), for the 30- and 60-day assessments, respectively. At 90 days, the failure type was shown to be statistically significant with p-values of 0,002 (F-value = 5.4).

Post hoc analyses using the pairwise Fisher LSD method further indicated that there was no statistically significant difference ($p > 0.05$) for the tensile strength within the failure modes at both 30 and 60 days as shown in Fig. 2 (values reported only when $p \leq 0.05$). However, at 90 days there were statistically significant differences in the tensile strength metrics between the middle breakage and the other modes $p < 0.0001$, $p = 0.028$, and $p = 0.040$, with respect to bottom breakage, slippage breakage, top breakage, respectively. This statistical analysis showed that, during the initial 2 months of *Salix sp.* root growth, the failure patterns did not significantly affect the tensile strength, as the results from the top, middle, bottom, and slippage

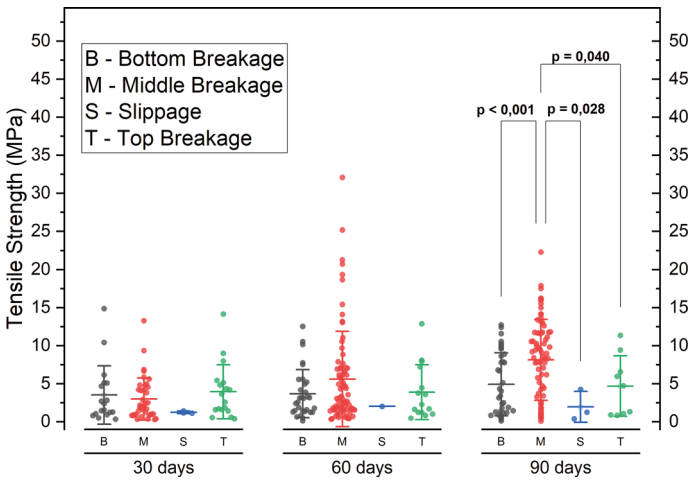

Fig. 2 Pairwise comparison utilizing Fisher tests with respect to tensile strength and failure type at the same ages

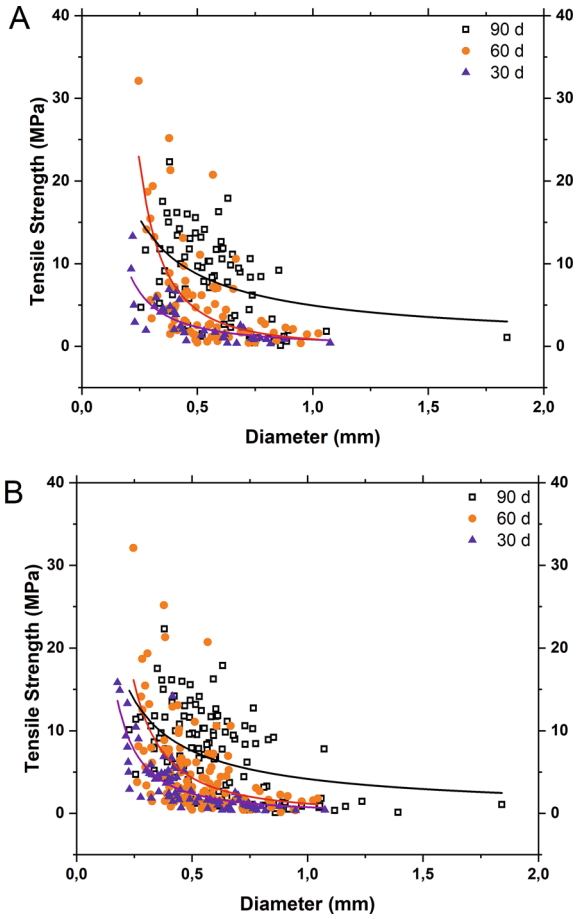

Fig. 3 Root tensile strength for middle failure **a** and combined failure types (**b**)

were quite statistically comparable. However, after 90 days, the morphological development of the roots began to influence tensile strength, indicating that the types of breakage become increasingly important at this stage and possibly beyond.

Slippage failure has been discussed above; however, this failure type is premature as it slips out of the plastic glue during the test either from the top or bottom of the equipment jaws. Therefore, slippage was not subsequently analyzed further with respect to the power law curve fit. In addition, this failure yielded a limited number of data points as fewer than three roots exhibited this failure type at any given age.

In the analysis of the relationship between tensile strength and diameter using the power law curve at different ages, the power law curve holds for each failure type as presented in Table 1. This was because the tensile strength increases with a decrease in root diameter [29, 30]. The correlation coefficients from the power curve fit for the middle third were observed to be 0.56, 0.42, and 0.21 for the 30-, 60-, and 90-day periods, respectively, as presented in Table 1 with $p < 0.0001$. This p-value indicated

Table 1 Correlation coefficients from the power fitting curve

	Coefficient of Determination, R^2			P-value
Failure type	30 days	60 days	90 days	
Middle third (M)	0.56	0.42	0.20	< 0.0001
Bottom third (B)	0.91	0.31	0.21	
Top third (T)	0.28	0.15	0.54	
combined	0.64	0.32	0.22	

statistical significance of the root diameter on tensile strength. When combining data from the middle third (M), top third (T), and bottom third (B), the overall correlation coefficients improved to 0.64, and marginally for 0.22 at 30 and 90 days as shown in Fig. 3. Therefore, it may be beneficial to re-evaluate the criteria for excluding samples that fall outside the middle range of root lengths [26]. This consideration is important when determining tensile test results due to the given age and tapered tubular structure of the roots.

4 Conclusion

The estimation of tensile strength contribution from plant roots, due to root reinforcement, could benefit from a more nuanced approach considering the current conservative practice. Given that plant roots have a segmented and non-uniform structure, with diameters typically decreasing as they penetrate deeper into the soil, there is an opportunity to improve the current practice for better assessments in geo-environmental engineering. Furthermore, the breakage occurring within the middle third of the effective length of the root specimen presents the need for conducting several destructive tests to measure tensile strength. In this study with *Salix sp.*, the success rate for these tests ranged from 49 to 62%, based on an analysis of 192 out of 330 root specimens over 30-, 60-, and 90-day period evaluated. The analysis using ANOVA and the pairwise Fisher LSD method suggested that the different failure patterns did not significantly ($p > 0.05$) influence tensile strength when compared to the middle breakage at 30 and 60 days as they neither over-estimated nor under-estimated the tensile strength. However, the failure patterns showed statistically significant ($p \leq 0.05$) differences at 90 days. These observations indicate that it might be worthwhile reconsidering the criteria for excluding samples that fall outside the middle third when evaluating the tensile strength.

5 Next Steps

This study will continue to build upon the results of root growth at 150 and 180 days for the same plant type *(Salix sp.)* by conducting a detailed evaluation. Furthermore, we plan to perform more systematic analyses to better understand the effective length extending beyond the middle third.

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Biodegradation of Rhodamine B Dye Via Vermicomposting: Advancing Nature-Based Solutions for Textile Waste Management

Bikash Dhadumia, Satya Sundar Bhattacharya, and Biswajit Paul

Abstract The textile sector generates a significant volume of waste, with the majority being included in municipal solid waste. Innovative management is crucial for improving sustainability in this sector. The biodegradation of Rhodamine B dye via vermicomposting methods is a novel concept. This study investigated the biodegradation capability of the epigeic earthworm species *Eudrilus eugeniae* in various combinations of dye, rice husk (RH), and cow dung (CD) relative to composting. The pH, electrical conductivity, and organic carbon availability substantially reduced following vermicomposting in comparison with aerobic composting. The dynamics of Ca-K availability stabilized effectively due to vermicomposting. In the experiment, *Eudrilus eugeniae* showed significant adaptation to the toxic dye. The metal accumulation capacity was notable, resulting in a 50–90% reduction of Cd, Cr, Zn, Mn, Co, and Pb concentrations in the vermicomposting system. The earthworm species similarly enhanced the proliferation of beneficial microorganisms, specifically nitrogen-fixing and sulfate-reducing types, within the unit. The degradation of dye was primarily examined using HPLC, revealing that vermicomposting achieved a 76–98% degradation of Rhodamine B. This study shows that the biodegradation of Rhodamine B via vermicomposting is the most simple and cost-effective method for dye degradation, contributing to textile waste management.

Keywords Biodegradation · Rhodamine B dye · Textile waste · Vermicomposting · Nature-based solutions

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1 Introduction

Synthetic dyes in industrial effluents pose a significant threat to the adjacent environments when discharged untreated due to their toxic properties. The textile industry is a primary consumer of such dyes. Rhodamine B is a prevalent xanthene dye that serves as a fluorescent water tracer [1]. It is extensively utilized in the textile sector for dyeing, and it also serves as a biological stain and food additive. It may induce carcinogenic and teratogenic consequences on health [2]. Dyes in effluents (~20%) from textile industries constitute a significant source of environmental pollution and require increased scrutiny due to the complex structures of most dye molecules and their photolytic stability, rendering them resistant to conventional biochemical degradation [3, 4].

The management of dye waste poses a significant challenge for industries. Indigenous approaches such as vermicomposting and aerobic composting provide significant benefits for biodegradation. Vermicomposting effectively transforms waste materials into excellent compost and enhances the diversity of microbes in vermicasts [5]. This technology is swift, economical, and efficient, employing earthworms to convert waste. The fragmented and decomposed final product (vermicompost) serves as an excellent soil conditioner and organic fertilizer for uses in agriculture [6]. In these last few years, numerous investigations have shown the efficacy of remediating industrial solid waste via vermicomposting [7].

Based on our research understanding, no research has been done on bioremediation of Rhodamine B dye with vermicomposting. The objective of this study is to examine the potential bioremediation of various concentrations of Rhodamine B dye absorbed in a substrate (rice husk) using vermicomposting with the help of an indigenous earthworm species *Eudrillus euginae*. Therefore, the study concentrated on the following elements to achieve this goal: (1) availability of organic components and range of important parameters like pH, EC, organic carbon, total Kjeldahl nitrogen, Ca and K availability in vermicompost; (2) assess the heavy metal accumulation and degradation capacity of the vermicomposting system; and (3) the assessment of dye degradation using HPLC.

2 Materials and Methods

2.1 Experimental Setup for Vermicomposting Experiment

This study employed industrial fluorescence grade dye Rhodamine B (Sigma-Aldrich). CD and RH were procured from a local source to function as the organic substrate and to serve as an organic medium for color retention, respectively, to enhance the biological conversion process [8]. The experiment was conducted in the Department of Environmental Science at Tezpur Central University, with analysis

Table 1 Substrate preparation with different treatment names

Substrate composition [11–13]	Normal composting	Vermicomposting
Only compost	C1	V1
Compost + Rhodamine B (0.3 gm/kg)	C2	V2
Compost + Rhodamine B (0.9 gm/kg)	C3	V3
Only cow dung	C4	V4

performed at the Indian Institute of Technology (ISM) Dhanbad, India. The compost for the experiment was prepared by combining RH and CD in a 1:2 ratio [8, 9].

Two separate sets for each treatment combination were set up (Table 1), one designated for aerobic composting (C) and the other for vermicomposting (V). In vermicompost treatments, the adult earthworm species *Eudrilus eugeniae* was introduced at a ratio of 10 worms per kilogram of substrate [10]. The experiment was conducted in earthen reactors made like circular truncated cones, with a capacity of 5 liters [0.32 ft. (lower diameter) × 0.66 ft. (upper diameter) × 0.9 ft. (height)]. The experimental treatment was incubated for 60 days, with samples collected at 30-day intervals (0, 30, and 60 days). The collected sample was stored to dry. Following the removal of moisture, samples were ground and evaluated for different parameters.

2.2 Parameters Analyzed and Statistics

The health of earthworms was assessed and parameters such as earthworm count, juvenile count, and cocoon count were recorded at 15-day intervals. The pH, electrical conductivity (EC), and total organic carbon (TOC) were assessed using the standard methods [14]. Total nitrogen and sulfur availability were assessed using the Kjeldahl and spectrophotometric technique, respectively. Potassium and calcium availability was assessed using the flame photometric approach [15]. The estimation of heavy metals was conducted using the DTPA method of extraction as outlined by Lindsay and Norvell [16]. The quantifications of these elements were done using ICP-OES (PerkinElmer, USA).

HPLC analysis was conducted utilizing an HPLC system (Thermofisher Scientific, Model: HPG-32002). The mobile phase consisted of 65% HPLC-grade methanol and 35% HPLC-grade water. The rate of flow was fixed at 1.1 ml/min at 35 °C. A 20 µl sample volume was injected, followed by excitation at 550 nm and emission at 50 nm.

For statistical analysis, multivariate ANOVA was applied to evaluate physico-chemical and biological factors. Furthermore, the LSD post hoc test was used to evaluate the efficacy of the different treatments.

3 Results and Discussion

3.1 Earthworm Health Study

During the health study (Fig. 1), rapid mortality of earthworms was observed up to 15 days. Substrate adjustment may be a major factor for initial mortality. The inherent availability of heavy metals in the substrate could also be a factor behind the sudden decline in population. The earthworm counts gradually increased after 15 days in all treatments, except for treatment V2, where the count remained stable and subsequently declined. This increase could be because of degradation of dye, which probably changed the substrate's physicochemical characteristics [17]. Additionally, higher cocoon count was encountered in the cow dung substituted treatments apart from the control (only CD). The cause may be the incorporation of cow manure, which significantly altered their habitat providing better porosity and moisture content resulting in a significant increase in number and offspring production [18].

3.2 Changes in Physicochemical Properties During Bio-Composting Experiment

A notable decrease in substrate pH and EC was noticed toward the conclusion of the incubation time (Fig. 2a, b). The decrease was mostly observed in the vermicompost treatments compared to aerobic composting. Earthworms decompose the substrate and enhance the environment by producing numerous organic acids, which significantly lower the pH and EC [19]. A substantial decrease in total organic carbon (TOC) was noted during the vermicomposting process (Fig. 2c). TOC declined significantly during the bio-composting process, especially in vermicompost pots. The loss of TOC is generally ascribed to the fast mineralization of organically bound carbon facilitated by the coactive activity of earthworms and microorganisms [20].

The total nitrogen concentration in the vermicomposting series increased considerably over time (Fig. 2d). An increase was observed in V1, then followed by V2 and V3. Vermicomposting accelerates the mineralization of nutrients into accessible forms, leading to an increase in nitrogen [21]. Conversely, the availability of S decreased across all treatments over time, with a significant reduction observed in the vermi-treatments (Fig. 2e) due to the production of several enzymes and by supporting sulfate-reducing bacteria [22]. K and Ca are vital elements for sustaining physiological equilibrium in living organisms [23]. The substrate possessed a high intrinsic K concentration. Nonetheless, advancements in incubation have effectively controlled the potassium concentration in the substrate using vermi-treatments (mostly V2 and V3) (Fig. 2f). The calcium content similarly declined in both types of treatments throughout time (Fig. 2g). Earlier studies also reference K and Ca stabilization during the vermicomposting of different wastes [23, 24].

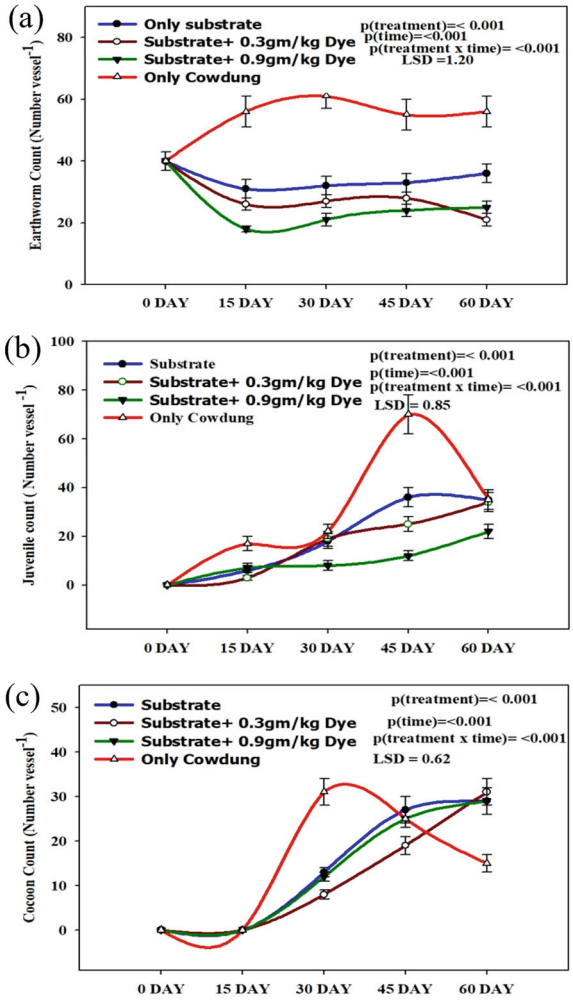

Fig. 1 Earthworm health study **a** Earthworm count **b** Juvenile count, and **c** Cocoon count per interval

3.3 Heavy Metal Content in the Bio-Composted Dye Waste

This study observed a significant reduction in the concentration of toxic heavy metals (Cr, Mn, Cu, Zn, Cd, and Pb) over time in the vermi-treated pots (Table 2). The study reported a general trend: V1 > V2 ≥ V3 > C3 = C2 = C1. Earthworms primarily accumulate non-essential micro-elements rather than essential trace elements [25]. Research also demonstrates the role of metallothionein isoforms in heavy metal-stressed earthworms in mitigating metal ion toxicity. Metallothionein possesses the capacity to bind metal ions via organo-metallic ligands, thereby limiting the release of these metal species from waste [26].

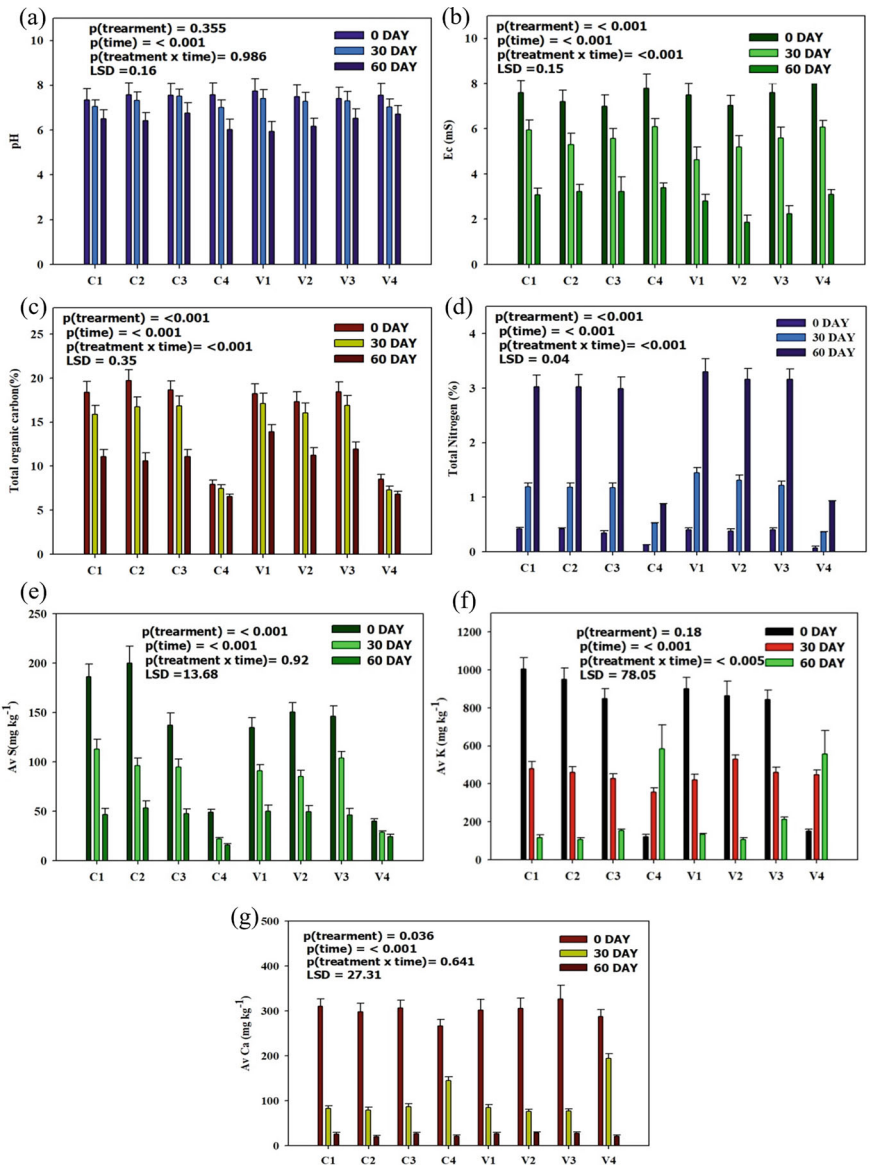

Fig. 2 Temporal variations in **a** pH, **b** EC, **c** Total organic C content, **d** Total nitrogen, **e** Available sulfur, **f** Exchangeable potassium, and **g** Available calcium during the bio-composting

Table 2 Reduction percentage in heavy metal concentrations in the dye under various treatments after 60-day period

Treatments	Cr (%)	Mn (%)	Cu (%)	Zn (%)	Cd (%)	Pb (%)
C1	49.20	60.77	48.83	52.18	20.00	35.98
C2	49.60	60.39	49.08	52.86	15.79	29.51
C3	49.41	60.30	49.75	51.03	14.29	35.21
C4	–	–	48.34	41.22	–	25.52
V1	74.47	90.22	87.27	80.49	59.15	89.45
V2	71.25	85.82	83.72	78.33	52.84	79.42
V3	70.78	84.65	82.15	74.73	49.84	77.26
V4	–	–	50.14	52.12	–	53.11

As previously illustrated in Fig. 1, earthworm growth was inadequate in treatment V3 (high dye concentration), and the majority of earthworms vacated, probably due to the substrate's intrinsic toxicity. Concerning the employed therapies, V2 proved to be the most effective combination for mitigating substrate toxicity. The earthworm, *E. fetida*, is an adept accumulator of harmful heavy metals from various industrial wastes [11].

3.4 Dye Degradation in HPLC

Dye degradation in this experiment primarily occurs through the activity of earthworms. The subsequent table (Table 3) illustrates the degradation of Rhodamine B. The HPLC measurement is recorded as Area [Retention time (min) X Height (mAU)] for the specified time frame. The results indicated a strong and swift decline in V2, succeeded by V3. The degradation of V2 exceeds that of V3 due to a lower dye concentration in V2 (0.3 gm/kg) compared to V3 (0.9 gm/kg). The microbiological and geological systems facilitate dye breakdown by reducing COD, whereas earthworms release substantial amounts of enzymes. These enzymes and microorganisms enhance the vermi-remediation process and contribute to the reduction of dye concentration [27, 28]. Concerning the employed treatments, V2 was the most effective combination for mitigating dye toxicity. Multiple authors have documented the efficacy of a vermifiltration system, achieving a removal rate of 90–95% for both suspended solids and dissolved solids from effluents, attributed to earthworms [29].

Table 3 Temporal change and reduction Rhodamine B dye under various treatments

Treatments	0 day	30 days	60 days	Reduction %
C1	BDL	BDL	BDL	–
C2	5.05	4.36	3.35	33.62
C3	0.86	0.76	0.62	27.96
C4	BDL	BDL	BDL	–
V1	BDL	BDL	BDL	–
V2	4.96	0.95	0.13	97.40
V3	0.89	0.47	0.21	76.37
V4	BDL	BDL	BDL	–

Note BDL = Below detection limit

4 Conclusion

During the bio-composting experiment earthworms were acclimating to the substrate in the initial days, followed by a significant increase in their population count. There were significant decreases in pH, EC, and TOC in all treatments. The N-K content in the vermicomposting series exhibited a significant increase over time, supporting good decomposition. Sulfur availability has diminished across all treatments over time. The depletion of toxic heavy metals from the substrate occurred progressively, particularly within vermicompost units. The removal of Rhodamine B dye reached up to 98% at the required concentration through vermicomposting, facilitated by microbial activity in the earthworm gut. Since this process is both an effective and cost-efficient method for dye degradation, it contributes to a nature-based approach to textile waste management.

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Applications of Microbial Cellulose: A Byproduct of the Kombucha Production Industry

Susana Gouveia and Claudio Cameselle

Abstract Kombucha is a beverage obtained by the fermentation of tea with sugar (*Camellia sinensis*) by the SCOBY, an acronym for Symbiotic Consortium of Bacteria and Yeast. It is believed that this beverage was already consumed in the Manchuria region (China) since the third-century BC. Recently, the increased demand for functional beverages has driven the consumption and industrialization of kombucha worldwide. This probiotic drink, rich in polyphenols, with antioxidant, and antibacterial properties, among others, has a pleasant and slightly acidic taste, which has likely contributed to its popularity compared to other probiotics, such as Kefir. During fermentation by the SCOBY, a cellulose biofilm forms on the surface. In this study, we will explore the enormous potential and applications of microbial cellulose obtained as a byproduct of the growing kombucha production industry, as a means to reduce industrial waste.

Keywords Bacterial cellulose · Byproduct valorization · Fermentation · *Camelia sinensis* · Kombucha tea

1 Introduction

Kombucha is a beverage traditionally produced through the fermentation of sugared tea (*Camellia sinensis*) by a symbiotic culture of bacteria and yeast (SCOBY). Typically, brewed using black or green tea, kombucha undergoes an aerobic fermentation process lasting from one to several weeks, depending on environmental conditions and desired acidity levels. During fermentation, the consortium of microorganisms consumes sugars, producing organic acids, carbon dioxide, and small amounts of ethanol. This process also generates bioactive compounds such as polyphenols and probiotics. The final product holds a diverse microbial community, with yeasts (primarily *Saccharomyces* spp., *Zygosaccharomyces* spp., *Brettanomyces* spp., and

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Candida spp.), acetic acid bacteria (mainly *Komagataeibacter* spp., *Gluconobacter* spp., *Acetobacter* spp., and *Gluconacetobacter* spp.), and sometimes also lactic acid bacteria (*Lactobacillus* spp., *Lactococcus* spp.) playing key roles in its fermentation dynamics [1]. In a complex biochemical transformation, kombucha not only enhances its sensory attributes but also acquires potential functional properties, contributing to its growing popularity as a beverage worldwide.

Microbial cellulose (MC), also known as bacterial cellulose, is a biopolymer produced by certain bacteria, including those found in the SCOBY used in kombucha fermentation. In this process, MC is produced, forming a floating gelatinous matrix on the surface of the liquid, which provides support for the bacteria and yeast to grow. Contrasting to plant cellulose, MC is highly pure [2], lignin free [3], with high water retention [4], mechanical strength [5], and biocompatibility [6]. Therefore, due to its unique physical and chemical characteristics, the resulting MC has numerous applications, from food and beverages to biomedical and industrial uses.

In recent years, there has been a substantial increase in kombucha consumption, driven not only by its purported health benefits but also by its pleasant taste, leading to a growing variety of products on the market. This beverage is now produced on both small scales—sometimes even through artisanal methods—and on an industrial scale [7]. However, current production is primarily focused on the commercialization of kombucha itself, and as a result, the MC generated during fermentation is often underutilized or not utilized at all. In many cases, it becomes a waste product, despite being a valuable byproduct with numerous potential applications. Most current research has primarily focused on the beverage itself, sometimes neglecting the unique properties and applications of the MC byproduct. Addressing this research gap could open new opportunities for sustainable and innovative uses of MC that comes from kombucha fermentation.

This short review aims to explore the various applications of MC, highlighting its sustainability and potential to bring changes to many industries. Even though the focus is on MC in general, this study aims to create a basis for using the MC that comes from kombucha fermentation as a byproduct in the food industry.

2 Cellulose Structure

Cellulose, a high-molecular-weight homopolysaccharide, is the most abundant biomolecule on Earth. Its structure comprises D-glucose ring units linked by β-1, 4-O-glycosidic bonds [8], resulting in a linear chain with molecular formula $(C_6H_{10}O_5)_n$. The repeating unit in the polymer synthesis consists of two glucose molecules bonded together, with one molecule rotated 180 degrees relative to the other (Fig. 1). For each D-glucose ring in the cellulose structure, two hydroxyl groups are involved in intramolecular linkage while three remain reactive hydroxyl groups (Fig. 1). These three reactive hydroxyl groups (attached to carbons 2, 3, and 6) enable cellulose chains to link together via intermolecular hydrogen bonds, forming microfibrils when the chains align in parallel and bundle [8, 9]. This organization

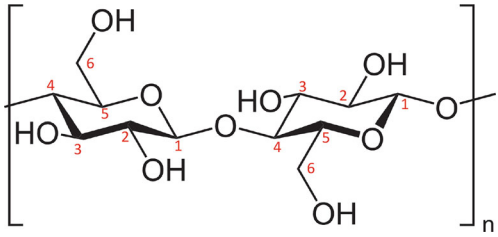

Fig. 1 Cellulose molecule chemical representation (n = degree of polymerization)

into microfibrils contributes to the high strength and fibrous structure of cellulose. In other words, while the individual cellulose chains are linear, they interact with each other in a way that creates a 3D porous arrangement when the chains pack together to form microfibrils [10]. The polymerization degree ranges 1000–5000 in native cellulose [11] or even reaching 10,000 in cotton plant [12] and its total hydrolysis yields exclusively glucose.

Lignocellulosic biomass, the primary source of cellulose, consists of cellulose (30–50% of dry weight), hemicellulose, and lignin, which are strongly bonded by covalent and non-covalent linkages [13, 14]. Cellulose isolation from biomass is efficiently achieved through various methods, including chemical pulping (e.g., Kraft, alkaline, or sulfite processes), thermal treatments, mechanical processing, or combinations thereof. The physicochemical properties of the resulting cellulose fibers will strongly depend on the specific isolation method employed as well as the severity of the processing conditions [15]. Furthermore, enzymatic treatments offer a versatile approach to modifying cellulose, impacting both traditional applications in the pulp and paper industry (such as bleaching and refining) and newer applications like nanocellulose production [16].

2.1 Microbial Cellulose

Cellulose is not exclusive to plants; certain bacteria, such as those from the genera *Acetobacter* spp., *Komagataeibacter*, *Agrobacterium*, and *Gluconacetobacter* spp. [10, 17–19], can also synthesize cellulose. These bacteria use a variety of carbon sources, including glucose and other organic molecules, to produce MC.

MC, despite sharing the same molecular formula as plant-derived cellulose, has several distinct characteristics. It is free of lignin and hemicellulose, which simplifies its purification process and makes it more cost-effective in terms of energy, reagents, and infrastructure compared to plant cellulose, which requires more complex and aggressive methods for isolation and purification [20]. Furthermore, MC generally has a higher crystallinity index and a lower degree of polymerization than plant cellulose [21]. It also exhibits superior properties such as greater tensile strength, higher water absorption and retention capacity, and better moldability. Additionally, MC fibrils are 100 times thinner than those of plant cellulose, contributing to its high

surface area and excellent structural integrity. These properties, combined with its biodegradability and insolubility in many solvents, make MC particularly valuable for a range of technological applications, including food, packaging, and medical fields [20–22].

As mentioned, one of the main advantages of MC over plant-derived cellulose is its high purity making MC more reliable and easier to process for various uses. Moreover, its production does not require agricultural land or natural resources, making it a more sustainable option compared to plant cellulose. However, after fermentation, MC is initially impure due to the presence of cells and their surrounding medium. To purify it, the pellicles are first extracted from the fermented broth and washed with purified water. The remaining MC cells are then removed by treatment with organic acids and alkaline solutions such as potassium hydroxide and sodium hydroxide. Finally, the solution is filtered and washed until it reaches a neutral pH, ensuring the purity of the MC, ready for use in various applications, as demonstrated by previous studies [23].

3 Production of Microbial Cellulose from Kombucha

3.1 The Kombucha Process: Fermentation and Production in Brief

Kombucha is produced by fermenting sweetened tea with a symbiotic culture of bacteria and yeast. This fermentation, typically lasting one to several weeks, begins when a sweetened tea infusion (5–10 g/L of tea leaves, 60–100 g/L sucrose) is inoculated with the SCOBY (approximately 3% w/v), along with 10–20% of a previously fermented kombucha infusion, commonly referred to as the "starter" (Fig. 2). The starter introduces a sufficient population of beneficial microorganisms, both bacteria and yeasts, which establish a symbiotic relationship, crucial to the stability of the microbial ecosystem and inhibition of potential contaminants growth. Additionally, the starter culture reduces the pH, creating an acidic environment that prevents the proliferation of undesirable microorganisms. Thus, the starter culture plays a key role insuring the quality of kombucha fermentation acting as both chemical and biological barriers [24, 25].

During this complex process, microorganisms metabolize sugars and tea components, generating a range of bioactive compounds, including acetic acid, lactic acid, and gluconic acid. These metabolites contribute to the unique fruity and slightly acidic flavor of kombucha. The yeasts present in the SCOBY facilitate the hydrolysis of sucrose into the monosaccharides glucose and fructose, providing carbon sources for bacterial metabolism, particularly in the synthesis of MC [23]. A characteristic feature of kombucha fermentation is the formation of a gelatinous cellulose-based pellicle on the liquid surface, which serves as a visual indicator of ongoing fermentation. This cellulose layer, newly formed in each batch, provides structural support

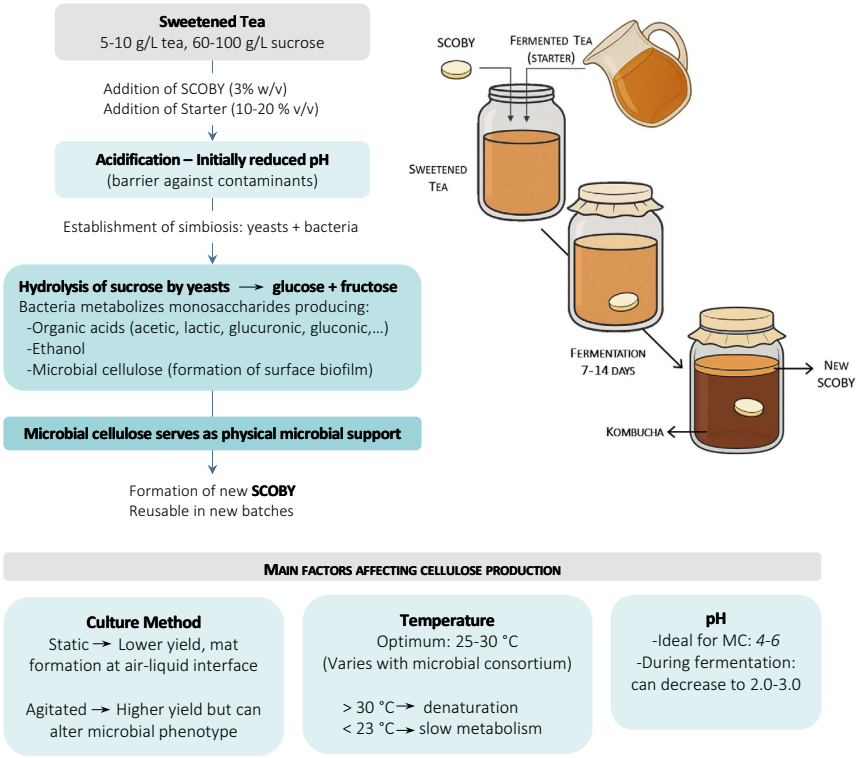

Fig. 2 Kombucha fermentation scheme and factors affecting MC cellulose production

for bacterial and yeast colonies, collectively forming the SCOBY. In addition to cellulose production, fermentation also results in small amounts of ethanol and other secondary metabolites.

Several studies have attempted to comprehensively characterize the microbial communities within kombucha. Research conducted in different geographical regions, analyzing samples of varying origins, has shown that there is no single universal microbial consortium but rather multiple distinct consortia. These microbial populations may include different genera of bacteria and/or yeasts. However, one consistent finding across all studies is the simultaneous presence of yeasts and acetic acid bacteria, while lactic acid bacteria may or may not also be present [26, 27].

One of the main differences between MC from kombucha and MC produced in a traditional way is the type of microorganisms involved. MC production usually depends on individual bacteria or small groups of them, while kombucha fermentation involves a stable symbiotic culture of bacteria and yeast, which keeps cellulose production ongoing. Another important aspect is cost: kombucha-based MC is produced using a simple sweetened tea mixture, while MC requires more expensive culture media. The way carbon sources are used is also different—MC mainly

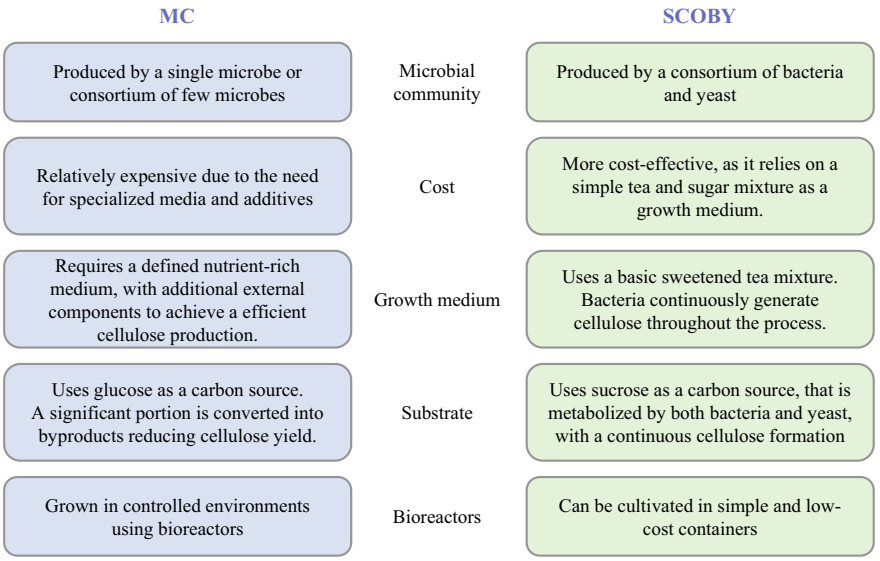

Fig. 3 Main differences between microbial cellulose and SCOBY cellulose. Adapted from [28]

relies on glucose, but a large part of it turns into gluconic acid and other byproducts, which can reduce cellulose production. On the other hand, in kombucha fermentation, sucrose is broken down by yeasts and then metabolized by both bacteria and yeast, keeping cellulose formation steady. Finally, while MC is usually produced in specialized bioreactors like rotary disk reactors or biofilm contactors, kombucha fermentation takes place in simpler and more affordable containers, such as glass jars or plastic vessels, making it a more practical and scalable option. These differences are summarized in Fig. 3.

3.2 Factors Affecting Microbial Cellulose Production

MC production typically occurs via two methods: static and shaking cultures. The static culture method requires long periods of culture, usually resulting in reduced cellulose yields. Under static conditions, bacterial cells produce cellulose mats at the air–liquid interface, where oxygen availability is high. On the contrary, the shaking culture method enhances cellulose production more efficiently but under these conditions variations in bacteria phenotype may lead to a reduction of cellulose synthesis [21]. Agitation rate also affects MC production. Low agitation rates ($\leq$100 rpm) lead to the production of uniform spheres with diameters ranging from 0.5 to 1.0 cm, while higher agitation rates tend to produce aggregates with an irregular form and smaller size [18].

Temperature is a vital parameter that affects fermentation and MC production. Although some *Acetobacter* spp. have their optimum cellulose production in temperatures as high as 33.5 °C most MC microorganism producers are able to adjust their normal homeostatic physiology in a temperature range of 23–30 °C [18, 29]. In the specific case of cellulose production during Kombucha fermentation, temperature must be adjusted according to the microbial consortium used, as each species involved has a specific optimal temperature for its growth and metabolic activity. Therefore, it is essential to find a balance between the metabolic activities of bacteria and yeasts that maximizes cellulose biosynthesis. Studies indicate that the optimal temperature range for this process is between 25 and 30 °C [28]. Higher temperatures may lead to the denaturation of microbial community, compromising their viability and efficiency in cellulose production, while lower temperatures may reduce microbial metabolic rates.

pH is another crucial factor, with an optimal range of 4–6 being considered ideal for MC production in most strains [18]. However, during fermentation, various metabolites are generated, leading to changes in the pH of the growth medium. In the specific case of Kombucha, acetic, glucuronic, and lactic acids are the predominant organic acids, progressively lowering the pH as fermentation advances, ultimately reaching values as low as 2.5–3.0 [1, 27].

As in any other fermentation process, the carbon source and its concentration significantly influence microbial growth patterns. In the case of Kombucha, the yeasts present, mainly *Saccharomyces spp.*, *Zygosaccharomyces spp.*, *Brettanomyces spp.*, and *Candida spp.*, hydrolyze sucrose into its monomers, glucose, and fructose. These sugars are subsequently metabolized by acetic acid bacteria (primarily *Komagataeibacter spp.*, *Gluconobacter spp.*, and *Acetobacter spp.*), lactic acid bacteria (*Lactobacillus spp.* and *Lactococcus spp.*), and the same yeasts that performed hydrolysis, leading to the production of various metabolites, including acetic, lactic, glucuronic, and gluconic acids, as well as ethanol. Variations in sugar type and concentration during fermentation influence microbial metabolic behavior, and affect growth dynamics and progression of the process. During this oxygen-dependent process, a cellulose biofilm forms on the surface of the medium. This gelatinous material thickens as fermentation progresses, serving as a physical support for bacterial and yeast adhesion, enhancing the stability of the microbial consortium [17, 24, 26].

4 Applications of Microbial Cellulose

MC widely used in diverse areas, ranging from the food industry and personal care products to household chemicals, biomedicine, textiles, and composite materials. A brief description of its main applications within each of these areas is presented in the subsequent sections.

4.1 Biomedical Applications

Although sharing the identical molecular formula ($C_6H_{10}O_5$) to plant-derived cellulose, MC shows exceptional purity and biocompatibility, making it highly suitable for various biomedical applications. The high water-retention capacity, oxygen permeability, and an adequate mechanical strength of MC-based films used as temporary skin in patients with extensive tissue loss, such as burns or skin ulcers, have demonstrated a positive effect in the healing, regeneration, and regrowth of skin tissue while preventing infections [30]. This leads to a reduction in hospitalization time for both pediatric and adult patients, as well as a decrease in the number of procedures requiring anesthesia [31]. These biofilms are also available as antimicrobial varieties, containing silver particles or other antimicrobial agents. Moreover, there are several other commercial MC-based products for medical applications, including drug delivery systems with selective and controlled release of pharmaceuticals, contact lenses, vascular grafts [10], bone tissue engineering, tumor cell culture, and cancer treatment [32]. The use of MC in biomedical applications needs the highest level of purity and is the most regulated and restricted among all MC applications followed closely by its use in the food industry, which usually does not require such a high level of purity.

4.2 Food Industry

MC has long been used in the food industry for packaging and edible coatings, due to its biocompatibility and ability to form films. The possibility of functionalizing these MC-based films to enhance properties such as mechanical strength and moisture resistance makes this biomaterial an effective choice for food packaging. For example, MC can be impregnated with antimicrobial compounds like chitosan or nisin to prevent bacterial growth, or with active agents such as antioxidants or oxygen scavengers that are slowly released to prevent food degradation, thus extending shelf life [20].

Another common use of cellulose in the food industry is as a multifunctional additive, following the safety regulations set by the European Union and the U.S. Food and Drug Administration. The microcrystalline structure of MC makes it a versatile and valuable ingredient for developing innovative and health-conscious food products. It is used as an emulsifier, stabilizer, and thickener in many products, such as minced meats, sauces, ice creams, and other processed foods [20, 33]. These functions improve the texture, consistency, and overall quality of the products, which is why cellulose is widely used in food production. Since humans lack the enzyme cellulase, cellulose cannot be digested, making it a good option as a low-calorie dietary fiber [21]. It helps with intestinal health, supports digestion, and also improves the texture and taste of foods. Besides its crystalline form, cellulose is often converted into carboxymethyl cellulose, a water-soluble polysaccharide which acts as a thickening

agent or emulsifier. However, some recent studies suggest that MCM has strong pro-inflammatory activity and argue that its use as a food industry additive should be reconsidered [34].

4.3 Environmental Applications

Cellulose-based materials have been widely studied in liquids filtration and adsorption. For instance, modified nanocellulose—owing hydrophobic structures—can adsorb oils and organic solvents from water in a highly efficient way. Due to their large specific surface areas surfaces, modified nanocellulose often achieve adsorption capacities of oil exceeding 150 g per gram of cellulose base material. Furthermore, their functionalized surface makes them highly promising for the removal of water pollutants, including heavy metals, organic dyes, and pharmaceutical waste [35].

Another commercially widespread application is the use of hydrophilic cellulose-based membranes, such as cellulose acetate or cellulose esters, for ultrafiltration. They are commonly used in wastewater treatment facilities, water treatment plants, as well as in sterilization processes, as the pore diameter achievable with these filters is extremely small effectively blocking passage of viruses and bacteria [36].

4.4 Other Applications

Electronics: MC is widely recognized for being lightweight and rigid, qualities that make it particularly suitable for acoustic applications. Studies on MC membranes have demonstrated excellent dielectric and acoustic responses [37]. One of the earliest examples of MC's commercial application dates back to the late 1980s when SONY incorporated this material into the acoustic diaphragm of one of its iconic models, the MDR-R10 [38]. More recently, manufacturers like Creative have highlighted that their MC diaphragms exhibit sound propagation speeds comparable to metallic materials such as aluminum and titanium, while retaining the warm and delicate sound characteristics typical of paper diaphragms.

Cosmetics: The high purity, biocompatibility, and remarkable water-retention capacity turn MC in a perfect candidate for several cosmetic applications. For instance, MC-based materials have been efficiently used as facial masks. Its high water-holding capacity ensures prolonged hydration, while the malleability of this biomaterial allows adequate adherence to facial contours, enhancing the delivery of active ingredients to the skin [39]. Another example of MC in cosmetics is its use in creams, lotions, and serums, where its emulsifying properties and compatibility with active compounds enhance formulation stability. Furthermore, the natural origin of MC matches with the growing consumer demand for eco-friendly and sustainable cosmetic products, making it more interesting for commercial and marketing purposes [40].

Textile: MC's high mechanical strength, flexibility, and biodegradability makes it a promising alternative to traditional textile materials. Its nanofibrous structure provides excellent tensile strength and durability, both essential for creating strong fabrics. However, some challenges still limit the broader use of MC in textiles. One of the main issues is its natural hydrophilicity that reduces its suitability for water-resistant applications. To overcome this, researchers have explored chemical modifications and the addition of hydrophobic compounds to improve the water resistance of MC-based textiles [41]. A similar strategy has been applied to Lyocell fibers made from wood pulp cellulose that are establishing a strong presence in the textile industry [41, 42]. Still, a major disadvantage of MC is its low production capacity compared to cellulose from wood sources, which remains a barrier to its large-scale industrial use.

5 Summary and Future Perspectives

MC, particularly as a byproduct of kombucha fermentation, has emerged as a highly versatile and sustainable biomaterial. Its nanofibrillar structure, high purity, and biodegradability makes it an attractive alternative to plant-derived cellulose in applications ranging from textiles and food packaging to biomedical and environmental technologies. Unlike dedicated MC production processes, kombucha-derived cellulose benefits from the growing market for kombucha as a probiotic beverage. Since its production costs are primarily linked to the beverage itself, rather than to cellulose extraction, this could help offset the traditionally high costs associated with MC commercialization. Additionally, strategies such as using industrial byproducts, fruit waste, and alternative nutrient sources could further reduce costs while promoting circular economy principles. Further research into microbial strain selection, fermentation optimization, and bioreactor design will be crucial for improving production efficiency and ensuring consistent material quality, making kombucha-derived MC an increasingly viable option for large-scale applications.

Material properties also require further improvement. For instance, MC high water absorbency limits its use in textiles unless modifications are made to improve hydrophobicity and mechanical durability. Advances in functionalization techniques, hybrid composites, and bio-based coatings may offer solutions to enhance its performance without compromising sustainability.

Looking ahead, the increasing demand for eco-friendly materials presents a major opportunity for kombucha-derived MC. With constant technological improvements and interdisciplinary research, this sustainable byproduct could become a key player in the transition to bio-based materials, providing a scalable and environmentally responsible alternative to traditional cellulose sources.

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A Comparison of the Stripping Capabilities of Water-Biobased Solvent Microemulsions as Paint Strippers with Commercial Solvents

Santiago Urréjola-Madriñán, Ángel Sáchez-Bermúdez, Clara Rodríguez-García, Mercedes Pardo-Clavijo, and Ana-María González-Sanchez

Abstract This paper explores the potential of microemulsions derived from biobased solvents, such as alkyl esters of natural oils (dibasic esters), in comparison to traditional industrial oil-based solvents. The study compares their capacity to clean, de-grease, and strip paint with that of traditional industrial oil-based solvents. These microemulsions, which can be classified as O/W (oil in water) or W/O (water in oil) based on their composition, exhibit noteworthy thermodynamic stability and the capacity to solubilise both hydrophilic and hydrophobic substances due to their dual nature. Consequently, they have become the focus of extensive research, which suggests that they could offer a promising alternative to traditional solvent-based systems. Moreover, the incorporation of water in their composition has the added benefit of reducing the cost of bi-based solvents, which are generally more expensive than traditional ones. To evaluate the microemulsions' stripping potential, a test by confrontation was designed and performed against a commercial solvent mixture that is utilised in industry. The formulations assessed in this study are subject to specific criteria, namely, the biodegradability of all their components, and the absence of any requirement for classification as environmentally hazardous substances according to the European Chemicals Agency (EC) regulations. The ensuing results demonstrate the efficacy of these mixtures.

Keywords Microemulsions · Biobased solvents · Clean · Dibasic esters

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1 Introduction

Microemulsions are defined as liquid mixtures that are dispersed, isotropic, transparent, and thermodynamically stable systems with extremely low surface tension. They are constituted of two immiscible liquids (water–oil) and the appropriate amounts of surfactant and/or co-surfactant.

The term "microemulsion" was first coined by Schulman in 1959 to denote emulsions composed of minute droplets. While emulsions and microemulsions are related, there is a difference between them. While emulsions are typically formed through the application of mechanical and/or heat energy, microemulsions are capable of thermodynamic stability without the need for agitation [1].

The International Union of Pure and Applied Chemistry (IUPAC—International Union of Pure and Applied Chemistry), founded in 1919, is recognised worldwide as the highest authority on chemistry-related decisions. It is the IUPAC that defines microemulsions: "Water-, oil- and surfactant-based dispersion that is an isotropic and thermodynamically stable system with the diameter of the dispersed domain between 1 and 100 nm, usually from 10 to 50 nm" [2].

Paint stripping is a pivotal process in the marine industry for the purpose of maintaining the structural integrity of vessels and ensuring the proper adhesion of new coatings. Historically, this process has relied on mechanical and chemical methods that utilise organic solvents. However, growing environmental concerns and regulations on volatile organic compound (VOC) emissions have prompted the search for safer and more sustainable alternatives, with microemulsions standing out [3].

In the late twentieth century, research began to be conducted on the use of microemulsions as a coating solution to replace organic solvents. Initial applications of microemulsions were in the domain of industrial cleaning and contaminant removal on metal and plastic surfaces. The utilisation of microemulsions has been demonstrated to offer superior performance, flexibility, and environmental and safety benefits in water-based cleaning and degreasing applications, where only organic solvents could previously be used. The replacement of petroleum solvents with microemulsions is facilitated by the use of near pH-neutral water. The composition of microemulsions can be tailored to meet specific criteria, including effectiveness, environmental and health profile, and biodegradability. The resultant microemulsions are biodegradable, non-flammable, and do not contribute to ozone depletion. When diluted with water, they enhance performance and reduce the amount of solvents transported. The compatibility of these compounds with common solvents, their stability over an extended period, ease of water dilution, the possibility of maintaining a pH close to neutral, and their low odour are some of the advantages of these compounds [5]. The development of this research aims to establish a new method to improve the performance of these compounds, which will be used in the development of a new product.

The objective of this research is to obtain biodegradable, non-flammable microemulsions, devoid of VOCs, with a view to replacing the conventional solvents utilised by industries. The continued utilisation of these conventional solvents is

primarily driven by their cost-effectiveness. However, their employment is associated with deleterious consequences for human health and the natural environment, including their adverse effect on the ozone layer [6].

2 Materials and Methods

2.1 Materials

The microemulsion used was made up of the following components:

- Surfactant: Sodium lauryl ester sulphate.
- Cosurfactant: Ethyl alcohol (96%).
- Aqueous phase: Deionized water.
- Oily phase: Dibasic ester.

The microemulsions have been formulated with varying concentrations of surfactant and co-surfactant, thus establishing the following theoretical proportions: 5, 10, 15, and 20%. In a similar manner, the water/oil ratios (W/O) are determined, with these being adjusted from the amount remaining after defining the surfactant and co-surfactant fraction. As an illustration, Table 1 presents the amounts of water and oil for 5–5% of surfactant and co-surfactant, and similarly, another 5 microemulsions would be prepared for each of the theoretical ratios, resulting in a total of 20 microemulsions.

A paint frequently utilised in the shipbuilding industry was employed to assess its efficacy. For each microemulsion, three test tubes with a paint thickness of 30 µm were prepared, resulting in a total of 60 tests. The paint selected for this study was a COMERCIAL NAVAL PAINTING®, which is based on alkyd resins, a synthetic resin that hardens when exposed to oxygen in the air, forming a hard, protective film of high quality and resistance to fading. Its characteristics include fast drying (approximately 2 h) and a glossy finish. It is considered ideal for naval silver-forms exposed to severe weather conditions [7].

Table 1 Quantities of water (W) and oil (O) with 5% surfactant and 5% co-surfactant

W/O	W	O	Surfactant	Cosurfactant	Total
(%)	(g)	(g)	(g)	(g)	(g)
20/80	3.6	14.4	1.0	1.0	20.0
35/65	6.3	11.7	1.0	1.0	20.0
50/50	9.0	9.0	1.0	1.0	20.0
65/35	11.7	6.3	1.0	1.0	20.0
80/20	14.4	3.6	1.0	1.0	20.0

2.2 Methods

Figure 1 shows schematically the development of the experimental phase.

In order to approach the research successfully, a four-stage strategy has been designed (Fig. 2).

The first stage consists of the application of paint on the metal plates and test tubes (which simulate the material of the naval platforms). For this purpose, a paint, commonly used on naval platforms, Naval Painting® (fast-drying, high-strength, and glossy naval industrial enamel) has been selected. After its application with a quadrangular applicator shown in Fig. 3, the plates were left to dry for subsequent use in the following stages. Once dry, they are checked with a thickness gauge to ensure that they are 30 microns thick.

In the second stage, the microemulsions are prepared, 20 mixtures with different concentrations of water, oil, surfactant and co-surfactant. For this purpose, tables are projected in which the data of the measurements are shown, that is to say, measurements are taken of the weight of the mixtures in order to know the necessary quantities once the optimum one is determined.

The third stage consists of pouring the microemulsions into the enamel paint with a thickness of 30 µm and between 10 and 15 µm in the spray paint, leaving them to act for a sufficient time, which is recorded to compare the results. This stage

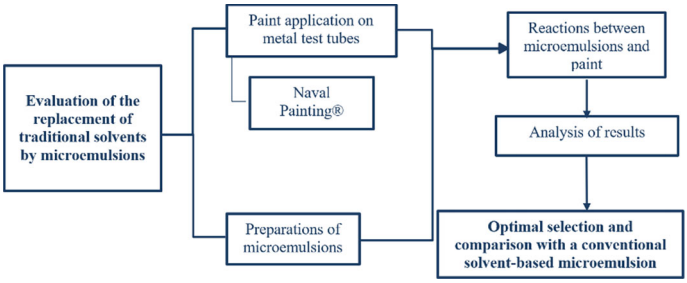

Fig. 1 Schematic experimental phases

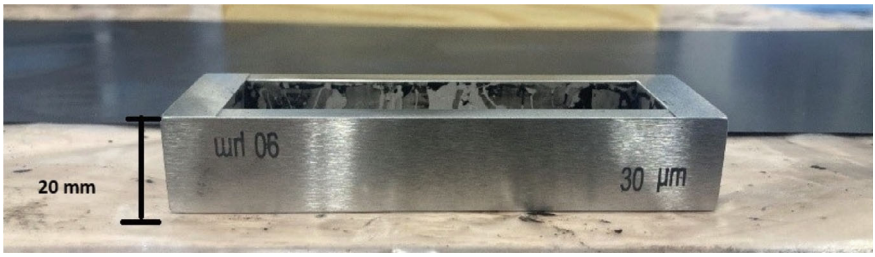

Fig. 2 Quadrangular applicator

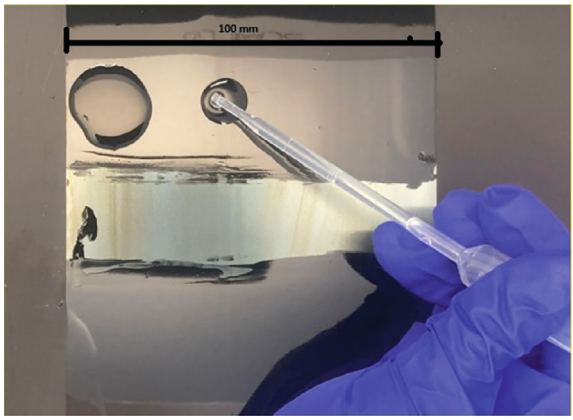

Fig. 3 Application of 10 drops of microemulsion on the paint

allows to evaluate the effectiveness of the coating removal process, ensuring that the paint is completely removed without leaving residues on the metal surface.

The fourth stage consists of the comparison between microemulsions. For this purpose, those that are optimal are selected and another microemulsion is prepared whose composition is different but only differs in one component from the rest.

3 Results

3.1 Material Characterisation

Subsequent to the execution of Stages 1 and 2, the third stage is initiated. The purpose of this stage is to observe and examine the reactions that occur after the application of the microemulsions on the two types of paint selected. In order to guarantee the pre-accuracy of the experiment and to avoid contamination of the samples, an individual pipette is used for each microemulsion, ensuring that there are no interferences that could affect its effectiveness. The application of ten drops of microemulsion onto the painted surface is undertaken, allowing adequate reaction time for the process to be completed and for the paint to be entirely removed.

During the interaction, it can be observed how the microemulsions produce a visible reaction on the paint, causing a crackling effect on the treated surface, as shown in Fig. 4. This phenomenon can be attributed to a chemical reaction between the di-solvent and the paint, resulting in the weakening of the paint's chemical bonds and the subsequent loss of its original structure. The reaction is complete when the applied area exhibits uniform wrinkling along its entire length. The paint is then removed using paper or a cloth.

This procedure is then repeated systematically with the 20 microemulsions in three experiments, i.e. the same sample is applied three times to evaluate its performance

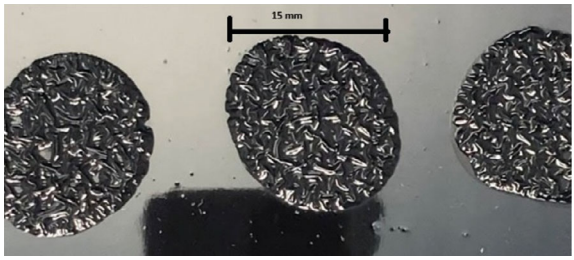

Fig. 4 Cracking reaction between the microemulsion and the paint

and determine which of the two types of paint is more effective in removing the coating.

3.2 Results

The criterion for selecting the best microemulsions is the one that takes the shortest time to react with the paint. As illustrated in Fig. 5, the results for various microemulsions are presented, with those coloured red indicating the most rapid reactions. These are then selected for further analysis.

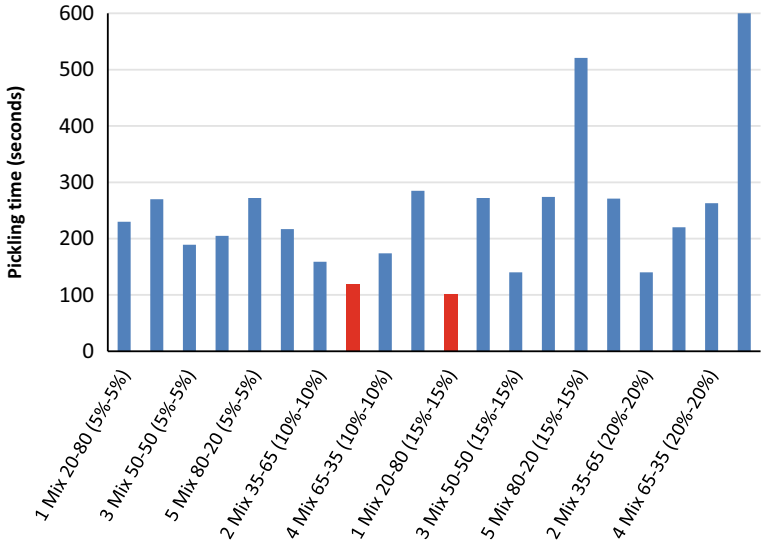

Fig. 5 Reaction time for different microemulsions

Table 2 Comparison between microemulsions according to the mean cracking starting time

Microemulsion	Crackle start time (s)	
	Dibasic ester	Conventional solvent
50–50 (10–10%)	234	241
20–80 (15–15%)	214	216

3.3 Results

This section will reflect the data obtained for samples 20–80 (15%–15%) and 50–50 (10%–10%). As these samples were identified as the optimum ones for the enamel paint, the experiment was repeated, but on this occasion substituting the dibasic ester with a conventional bilayer solvent. The objective of creating the microemulsions selected as optimal, but substituting the dibasic ester by a conventional solvent, is to enable a comparison between them and determine the advantages offered by those whose composition contains dibasic ester. Table 2 presents a comparison of the reaction time for painting the emulsion prepared with dibasic ester and the reaction time for the emulsion prepared with a conventional solvent.

- Microemulsion 20–80 (15–15%): It has very similar times among the three experiments carried out and comparing it with the same sample based on dibasic ester, it shows practically equal action times, so its effectiveness is similar.
- Microemulsion 50–50 (10–10%): Like the previous one, it has similar times between the three times taken and shows practically no difference with the optimum microemulsion formulated with dibasic ester.

With regard to the concentration of co-surfactant, it is important to note that a considerably higher amount is required to stabilise the microemulsion, at least twice as much as that composed of dibasic ester. This discrepancy can be attributed to the molecular interactions between the compounds present within the system, which exhibit variation in accordance with the chemical structure of the solvent, its polarity, and the type of surfactant employed. The necessity for a greater proportion of co-surfactant is driven by the need to attain equilibrium in order to form stable and transparent microemulsions, a process that is contingent on the physicochemical properties of the constituent components of the mixture.

The observation that both demonstrate these action times indicates that those with a higher solvent concentration are more effective than those with a higher water content. This is due to the propensity of oily organic compounds to dissolve in paint, formed by resins and polymers, allowing solvent molecules to penetrate the polymer chains, weakening them, breaking the internal interaction, and consequently dissolving the paint.

It is thus concluded that microemulsions formulated with dibasic ester and those formulated with conventional solvent act in similar times. However, the differences lie in toxicity and environmental consequences, which makes these microemulsions a viable alternative to traditional formulations.

4 Conclusion

The primary objective of this study was to develop and evaluate the efficacy of microemulsions with ecological formulations in paint stripping. The experimental phase of the study entailed the execution of a series of tests, which were then meticulously analysed and compared with the existing data from previous studies. The findings of this analysis have provided substantial evidence that validates the efficacy of the microemulsions.

It was determined that a greater amount of co-surfactant was required to stabilise the mixture. This finding underscores the notion that the type of paint is not the sole determining factor in identifying the optimal microemulsion. It is evident that the compounds employed play a pivotal role in determining the requisite proportions of surfactant and co-surfactant.

In all cases, the action time is similar to that of the dibasic ester pre-prepared microemulsions, which are more environmentally friendly, allowing the substitution of conventional reagents, such as the bilayer solvent, facilitating their use in open spaces, without the need to work with microemulsions in closed spaces. It can be concluded that the use of microemulsions ensures the transition towards a more sustainable industry. The present study aims to establish a foundation for future research endeavours that seek to enhance the utilisation of microemulsions within the naval industry.

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Efficacy of Vermicompost Tea and Xanthan Gum Solutions as Flushing Agents in Biodegrading and Removing Spent Lubricant Oil from Sandy Soils

Mohammad Rakhodaei, Ikrema Hassan, Kyle Rogers, and Laura Romero-Zerón

Abstract Globally, vermiremediation is gaining significant research interest. This study evaluated the effectiveness of vermicompost tea (VCT) and solutions of Xanthan Gum (XG) prepared in VCT (e.g., VCT/XG solutions) as flushing agents in conjunction with air injection to bioremediate sandy soil contaminated with spent lubricant oil (SLO). Batch biodegradation tests were conducted to determine the biodegradation of SLO by the bacterial consortia contained in VCT. Sand-packed experiments were conducted to evaluate the effect of two injection sequences: injection sequence # 1 (IS#1) with alternating injections of the flushing agents VCT-VCT/XG-VCT-VCT/XG with air and injection sequence # 2 (IS#2) with alternating injections of VCT-VCT/XG-VCT under continuous airflow. Baseline experiments were conducted using tap water rather than VCT. The remediated sandy soil was subjected to phytotoxicity tests. FTIR analysis demonstrated that the SLO was biotransformed by the microorganisms present in VCT. The experimental results demonstrated that the IS#2 was more effective in removing SLO (e.g., 98% removal) compared to the baseline experiments (e.g., 80% removal). However, IS#1 produced a better quality remediated sand for seed growth and development as per phytotoxicity test results. Overall, the combination of VCT-VCT/XG solutions as flushing agents and continuous air injection shows effectiveness for bioremediation of polluted soils for potential application at industrial sites contaminated with hydrocarbon derivatives.

Keywords Bioremediation · Vermicomposting · Vermicompost tea · Spent lubricant oil contamination · Xanthan gum

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1 Introduction

Petrochemical hydrocarbons (PHCs) constitute major environmental pollutants due to their resistance to natural degradation, non-uniform nature, and high toxicity. Numerous PHCs are classified as carcinogens, posing serious risks to ecosystems and human health [1–3]. PHCs are released into the soil due to accidental spills; leaking underground storage tanks; and during oil production, exploration, and refining processes. These PHCs alter the properties and functionality of soils (e.g., soil fertility), impact native microorganisms, and percolate through the soil, potentially contaminating groundwater [4]. One of the most promising approaches for soil remediation is bioremediation, due to its cost-effectiveness, in situ applicability, and lack of harmful byproducts. In recent years, vermiremediation has attracted significant research interest [5–8]. A study by [6] showed that *Eisenia fetida* earthworms effectively biotransformed polycyclic aromatic hydrocarbons by up to 95% [6]. Similarly, another research [7] demonstrated 86.8–95.4% removal of diesel from contaminated soil via vermiremediation. Several studies confirmed that biotransformation of hydrocarbon derivatives (e.g., phenanthrene, pyrene, crude oil, diesel, and engine oil, among others) from contaminated soils via vermiremediation occurred in time frames ranging from 35 to 160 days [8–12]. In these studies, the optimum environmental conditions for bioremediation included temperatures ranging from 15 to 30 °C and humidities varying from 50 to 80%. Both vermicomposting and vermiremediation processes use earthworms to degrade organic matter. However, vermicomposting deals with the management of waste organic matter such as food waste, and manure, while vermiremediation is applied for the biotransformation of pollutants. Vermicompost, also known as earthworm's casts or vermicast, is a nutrient-rich organic product that contains a substantial number of bacteria that produce biosurfactant-like components [13–15]. During vermicomposting, earthworms consume organic matter that is biotransformed as it passes through the earthworm's digestive system in conjunction with microbial enzymatic activity. VCT is a solution made from the dilution of vermicast in 5 or 10 parts of tap water, and therefore VCT contains a high concentration of micronutrients, enzymes, biosurfactant-like components, and microorganism consortia [16].

In this study, it was hypothesized that due to the unique composition of VCT, it could be applied as a bio-flushing agent to bioremediate soil contaminated with SLO via biotransformation and flushing action [17, 18]. Furthermore, it was also anticipated that the presence of biosurfactants-like components in VCT would aid the detachment, mobilization, and microbial bio-accessibility of SLO aiding its biotransformation and removal from the soil [19]. This study evaluated the effectiveness of VCT and solutions of XG prepared in VCT (VCT/XG) as flushing agents in biotransforming and removing SLO from sandy soils. The specific research objectives were first to evaluate the capacity of the microbial consortia contained in VCT to biotransform SLO. Secondly, to assess the effectiveness of using VCT and XG solutions as flushing agents through the application of two injection sequences to displace SLO

from sandy soil. Finally, to determine the phytotoxicity of the remediated soil via seed germination tests.

2 Materials and Methods

2.1 Materials

VCT was prepared from a vermicast obtained from a vermicompost station located in our research group (Chemical Engineering Department, University of New Brunswick, Canada). The vermicast was obtained from the vermicomposting of organic matter (e.g., vegetable peels, spent coffee, spent tea bags, eggshells, etc.) using the earthworm species *Eisenia fetida* acquired from Mother Worms (ON, Canada). VCT was prepared by mixing one part of vermicast with five parts of tap water. XG (Red Mill Co., USA) was used as a flushing agent at a concentration of 0.4 wt% prepared in VCT and/or tap water as the solvent. XG was selected as a flushing agent because it has been demonstrated that XG–soil mixtures improve soil properties [20]. SLO was used as a proxy of PHCs. Engine lubricant oils contain up to 80% of low and high molecular weight aliphatic and aromatic hydrocarbons. The SLO was provided by GPS Auto Repair Ltd. (Fredericton, NB, Canada). Pre-washed White Play sand distributed by Shaw Resources (NS, Canada) was selected as the soil media instead of topsoil to avoid the inclusion of additional bacterial consortia and organic matter.

2.2 Methods

Materials Characterization. The grain size distribution of the sand was characterized by the weight average diameter using a shaker Retsch, model AS200 (Retsch, USA). Canadian Standard Sieve Series sieves (Tyler Ltd., Canada) mesh sizes 20, 40, 60, and 80 were employed. The elemental composition of the VCT was determined using an Inductively Coupled Plasma Optical Emission Spectrometer model Vista MPX CCD (Varian, USA). Bacteria count in VCT was conducted following a routine procedure [21] using a hemocytometer using a compact inverted microscope, Olympus model GX41, an objective MPlan N 10X/0.25 ∞/0/FN22 magnification (Olympus, Japan), and a camera Infinity model 2 (Teledyne Lumenera, Canada). The surface tensions of VCT and SLO, and the interfacial tension of the VCT/SLO system were determined at 20 °C using a Fisher Surface-Tensiomat model 21 (Fisher Scientific, USA). The rheological characterization of the VCT/XG and tap water/XG solutions at 0.4 wt% concentration was conducted using a Gemini HR Rheometer model Nano 150 (Malvern Rheology Instruments, UK) under oscillatory analysis at 25 °C.

Bulk SLO Biodegradation Tests. Bulk SLO biodegradation tests were conducted by placing 100 ml of VCT in a beaker. Afterward, 100 ml of SLO was added on top of the VCT, and the beakers were covered but not sealed allowing air to contact the samples. These tests were conducted in triplicates. The samples were mounted on a Wrist Action shaker (Burrel Corp., USA) at 200 rpm. The samples were shaken for 35 days at 21 °C. Afterward, the SLO was separated and subjected to FTIR analysis using a Nicolet 6700 FTIR spectrometer (Thermo Fisher Scientific, USA). The FTIR specimens were prepared by placing 20 µl of the SOL on a KBr pellet to make a thin film consistent among samples.

Experimental Set-Up/Sand-Pack Cell. An in-house made sand-pack cell was used as a proxy for a shallow underground contaminated soil site. The dimensions of the rectangular metal model are length of 19.2 cm, width of 14.2 cm, and depth of 5.3 cm. The bulk volume of the empty cell is 1445 cm^3 requiring 2.49 kg of play sand to fill it ($\rho_{sand} = 2.65$ kg/L, $Porosity_{average} = 35\%$). The inlet of the model is located at the bottom of the rectangular cell and includes nine aligned holes following a pattern of 3 × 3 simulating three shallow horizontal wells. Figure 1 presents a simplified schematic of the experimental set-up. The flushing agents were injected through the bottom (base) of the sand-pack and the displaced fluids (e.g., a flushing agent and pollutant) were produced from the top of the cell, which was inclined at an angle of ~10° to allow the produced fluids to drain from the top of the cell by gravity. An Eldex Optos metering pump (Eldex Laboratories Inc., USA) was used to pump the corresponding flushing agents to the sand-pack cell. The sand-pack was characterized by dead volume, pore volume, porosity, and permeability using routine procedures [22].

In Situ Bioremediation. The first stage of each sand-pack experiment was the injection of the SLO as the pollutant into the water-saturated sand. For each experiment, 2.2 pore volumes (PVs) of SLO were injected until the production of water ceased. Afterward, the bioremediation process was initiated by injecting the corresponding flushing agent using an average flow rate of 1.33 ml/min ± 0.01. Breathable air was co-injected at low pressure (0.25 psi). Two injection sequences were evaluated: Injection Sequence # 1 (IS#1) and Injection Sequence # 2 (IS#2). The flushing agents were VCT and XG solution prepared in VCT at a 0.4 wt% concentration. Table 1 summarizes the injection sequences evaluated.

In IS#1, breathable air was co-injected with the flushing agents and as soon as the injection of the flushing agent(s) was completed, the flow of breathable air to

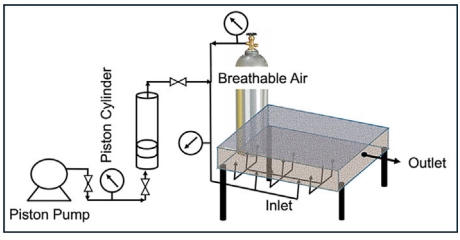

Fig. 1 Schematic of the experimental apparatus set-up

Table 1 Summary of flushing agent injection sequences

Injection steps #	Type of flushing agent	IS#1. PVs injected	IS#2. PVs injected
1	VCT	2	2
2	VCT/XG	4.3	4.3
3	VCT	3	2
4	VCT/XG	1	–
5	VCT	2	–

the sand-pack was stopped. For IS#2, on the other hand, breathable air was injected continuously during the entire bioremediation process (e.g., about 3–4 weeks per experiment). Injection of breathable air is necessary because the biotransformation of hydrocarbon pollutants requires the presence of oxygen [23]. For each injection sequence, baseline tests were conducted following the same injection sequence, except with tap water and tap water/XG solution at a concentration of 0.4 wt%. The volume of fluids injected, fluids produced, and differential pressure (ΔP) across the sand-pack were continuously monitored. Material balance was applied to calculate the saturation of SLO remaining in the sand-pack after each injection step. Each experimental run was conducted using a fresh sand-pack (e.g., a newly prepared sand-pack). Therefore, each sand-pack showed variations in permeability and porosity. Hence, for comparison purposes, it was necessary to normalize the sand-packs for permeability and porosity. The capillary bundle parameter model was used to normalize the different sand-packs for porosity and permeability [24].

Phytotoxicity Tests. Phytotoxicity tests were conducted using Jalapeno pepper seeds (*Capsicum annuum*) SPK299-PPJAPKT (Halifax Seed Co. Inc. Canada). Jalapeno pepper seeds (JP seeds) were selected in this study due to their low tolerance to hydrocarbon pollutants [25–27]. 40 g of the corresponding bioremediated sand were placed on large Petri dishes. Then, 10 JP seeds were placed on the sand and 40 ml of distilled water was added to the Petri dishes, which were covered and placed in a humidity control chamber of 70% at 25 °C for 14 days. These tests were conducted in duplicate (e.g., 20 JP seeds were tested per bioremediation treatment). The JP seeds in the Petri dishes were watered using a fixed amount of distilled water as needed. Table 2 displays the experimental matrix of the phytotoxicity test conducted in this study.

Table 2 Phytotoxicity test. Experimental matrix

Sand sample	Description	Sand sample	Description
Clean control	Clean sand*	IS#1. Baseline	Remediated sand
Polluted control	Sand** ($S_{SLO} = 0.98$)	IS#2. Exp. #1	Remediated sand
IS#1. Exp. #1	Remediate sand	IS#2. Exp. # 2	Remediated sand
		IS#2. Baseline	Remediated sand

*Clean sand rinsed with XG at a 0.4 wt% concentration, **Sand contaminated with 98% SLO

Table 3 Traits and corresponding equations [28–30]

Trait	Equations	Variable definition
Trait # 1	$T\#1 = \#gs/\#ts$	***#gs***: number of germinated seeds, ***#ts***: total number of seeds,
Trait # 2	$T\#2 = rl_{av}/rl_{avcc}$	$\boldsymbol{rl_{av}}$: average root length, $\boldsymbol{rl_{avcc}}$: average root length of clean control
Trait # 3	$T\#3 = sl_{av}/sl_{avcc}$	$\boldsymbol{sl_{av}}$: average steam length, $\boldsymbol{sl_{avcc}}$: average stem length clean control

The results of the phytotoxicity tests were subjected to the t-*statistic* test for related samples. The objective was to determine the effect of the soil bioremediation treatments under different injection schemes (IS#1 and IS#2) on the germination performance of JP seeds (e.g., root and stem development). For this statistical analysis, the hypotheses stated were H_O: $\mu_D = 0$ (there is no change in the germination performance of the JP seeds), H_1: $\mu_D \neq 0$ (there is change in the germination performance of the JP seeds). The level of significance was set at $\alpha = 0.05$ for a two-tailed test. Furthermore, several commonly used Traits for the analysis of phytotoxicity tests/germination tests as discussed by [28–30] were determined: Trait # 1, Trait # 2, and Trait #3. Table 3 summarizes the corresponding equations for each of the Traits calculated.

3 Results

3.1 Materials Characterization

The weight average diameter of the sand was 393 ± 6 μm that corresponds to a "medium sand" classification within particle size ranging from 250 to 500 μm [31].

Table 4 summarizes the elemental composition of VCT.

VCT contains a high concentration of K and Ca, which are necessary in soil for seed germination. This composition agrees with previous research [32–39]. The average bacteria count in the VCT was 26.3×10^6 cells/ml ± 2.4×10^6. Some of

Table 4 Elemental composition of VCT

Element	Concentration [mg/L]	Element	Concentration [mg/L]
B	0.12 ± 0.01	Mn	1.4 ± 0.02
Ca	270 ± 4.0	Na	81.1 ± 1.1
Fe	1.5 ± 0.14	P	5.4 ± 0.72
K	541 ± 7.2	Si	27.0 ± 0.36
Mg	81.1 ± 1.1	Sr	0.3 ± 0.003
Zn	0.14 ± 0.002		

Table 5 Surface and interfacial tension

Sample	Surface and/or interfacial tension [dynes/cm]
VCT	54.3 ± 0.8
SLO	23.9 ± 1.3
VCT/SLO system	4.5 ± 0.2

the bacteria species in VCT include *Kluyvera ascorbate, Rhizobium sp., Baus sp., and Serratia sp.* [40], which are also hydrocarbon-degrading bacteria. *Enterobacteriaceae* improves nutrient availability and promotes root development, *Pseudomonadaceae* reduces disease incidence by production of antibiotics, and *Flavobacteriaceae* breaks down organic matter and contaminants in the soil, enhancing the bioremediation process.

The average dynamic viscosities of the VCT/XG and tap water/XG were 290 ± 14 cP and 284 ± 6 cP at an angular frequency, $\omega = 7.055$ rad/s and 25 °C, respectively. Table 5 outlines the surface tension of VCT and SLO, and the interfacial tension of the system VCT/SLO. The surface and interfacial tension data indicates that VCT contains biosurfactant-like components that drastically reduce the interfacial tension of the VCT/SLO system from 23.9 to 4.5 dynes/cm. Likewise, the presence of organic acids in VCT can act similarly to surfactants and reduce the interfacial tension between the VCT/SLO. Hence, the presence of both (e.g., biosurfactants and organic acids) promoted the formation of inverted emulsions in our system, in which fine droplets of water were dispersed in the SLO, as observed under the microscope (micro-pictures not reported in this paper).

3.2 Bulk SLO Biodegradation Tests

Figure 2 displays the FTIR spectra of the original SLO and three samples of SLO that were subjected to biodegradation in VTC. Figure 2a displays the spectra of wavenumber ranging from 3200 to 2700 cm^{-1} and Fig. 2b shows the spectra of wavenumbers ranging from 1600 to 1300 cm^{-1}. The spectra of Fig. 2a show significantly broader peaks for samples 1 to 3 compared to the narrow peaks of the original SLO at the wavenumbers 2850, 2920, and 2960 cm^{-1} attributed to methyl C-H asym./sym.stretch. These changes indicate that biotransformation of the SLO took place in the presence of VCT. The spikes at 1370, 1380, and 1460 cm^{-1} of the spectra of Fig. 2b also show changes compared to the original SLO.

The inherent capacity of microorganisms to biodegrade and/or biotransform organic pollutants via metabolic or enzymatic processes is well established [41]. Bioremediation processes take place aerobically or anaerobically. However, under aerobic conditions the biodegradation of hydrocarbon pollutants is much faster [42], in which the organic pollutant is oxidized [41]. Vermicompost and by extension VCT is rich in enzymatic and microbial activity [43, 44]. VCT contains glutinous mucus

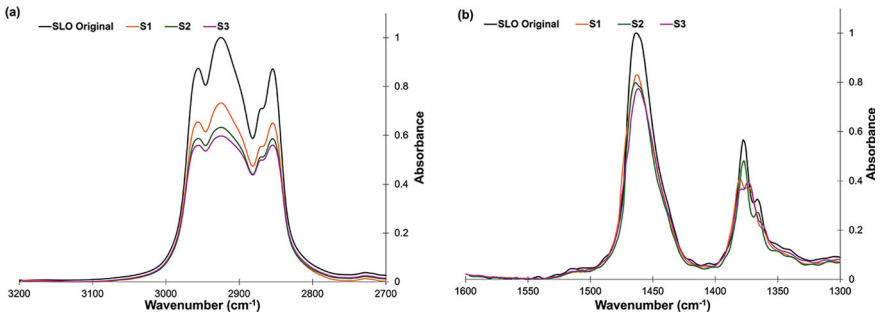

Fig. 2 FTIR spectra. **a** FTIR spectra in the wavelength range from 2700 to 3200 cm^{-1} and **b** FTIR spectra in the wavelength range from 1300 to 1600 cm^{-1}

and casts secreted from the earthworm's body walls, which contain dissolved organic carbon, amino acids, glycerolipids, glycoproteins, mucopolysaccharides, lectins, and hemocyanin that stimulate the growth of eukaryotic grazer populations (e.g., protozoa and nematodes). Some of the secreted earthworm's compounds display surfactant-like properties, facilitating the bioavailability of hydrophobic organic pollutants. The increased bioavailability of pollutants activates an array of microbial processes that break down, mineralize, and degrade organic pollutants [44, 45]. A large variety of hydrocarbon decomposer bacteria is associated to vermicomposting including *Acaligenes*, *Acidobacteria*, *Actinomycetota*, *Aeromonas*, *Bacillota*, *Bacteroidetes*, *Burkholderia* sp., *Chloroflexi*, *Clostridium*, *Firmicutes*, *Fusobacteria*, *Gemmatimonadetes*, *Micrococcus Paenibacillus*, *Planctomycetes*, *Propionibacterium*, *Proteobacteria*, *Pseudomonadota*, *Sphingomonas* sp., *Staphylococcus*, *Stenotrophomonas*, *Verrucomicrobia*, etc. [45–47]. From these bacterial phyla *Bacillus, Paenibacillus, Verrucomicrobia, and Fusobacteria* are dominant [45, 46]. As well as hydrocarbon-degrading fungal genera including *Aspergillus*, *Penicillium*, and *Mucor* [45]. Figure 3 displays biodegradation pathways of n-alkanes proposed by [48] for the *Geobacillus* genus originally classified within the *Bacillus* genus [49].

The biodegradation pathway of long-chain alkanes as explained by [48] (Fig. 3a) is initiated via oxidation by alkane oxygenases producing primary or secondary alcohols, which are further oxidized into aldehydes by dehydrogenases, and into fatty acids. The fatty acids are further degraded into acetyl Co-A by the classical β oxidation pathway. Finally, the released carbon enters the tricarboxylic acid cycle, where it decomposes into CO_2 and H_2O to release energy. The metabolic pathway of hexadecane degradation (Fig. 3b) is initiated by the oxidation of the alkanes by monooxygenases. Followed by further enzymatic oxidation by alkyl alcohol dehydrogenase, aldehyde dehydrogenase, and subsequent enzymes involved in fatty acid metabolism pathways as presented by [48].

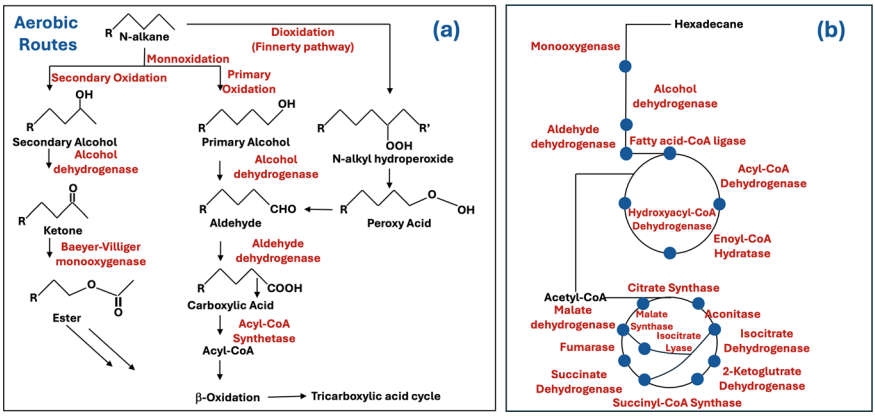

Fig. 3 Biodegradation pathways of n-alkanes by the *Geobacillus* genus under aerobic conditions: **a** Biodegradation of long-chain alkanes and **b** Hexadecane biodegradation [adapted from 48]

3.3 In Situ Bioremediation

Figure 4 displays the removal of SLO from the sand as a function of volume of flushing agents injected. Figure 4a shows the results for IS#1, in which the injection of VCT and VCT/XG removed 96% of the SLO from the sand. While the baseline (tap water and tap water/XG) flushed out a maximum of 81%. Therefore, the VCT-based flushing agent removed 15% more SLO than the baseline. Figure 4b displays the results for IS#2. The VCT and VCT/XG removed, on average, 98% of the SLO from the sand, while the baseline run removed a maximum of 79%. IS#2 achieved higher SLO removal, emphasizing the role of continuous airflow in bioremediation.

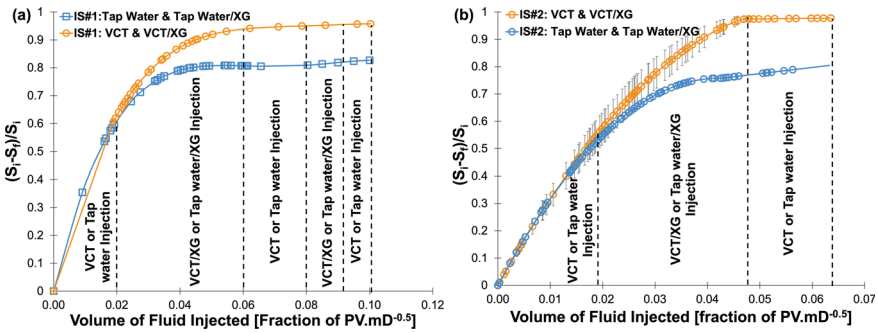

Fig. 4 Removal of SLO from sandy soil: **a** IS#1 and **b** IS#2

3.4 Phytotoxicity Tests

Table 6 summarizes the *t*-statistical analysis of the phytotoxicity test results, which demonstrated that the polluted sand significantly restricts the germination and development of JP seeds. On the contrary, all the bioremediation treatments significantly improved the soil properties, which supported seed germination, root, and stem development. The exception of this trend was treatment IS#2 baseline, which statistically did not improve root and steam development. These statistical results agree with the information obtained from the Traits analysis provided in Fig. 5.

Figure 5a and b summarizes the results of Traits #1, #2, and #3 obtained from the phytotoxicity tests for IS#1 and IS#2, respectively. The experimental values obtained for Traits #1 to #3 indicate that the in situ bioremediation processes render non-toxic sandy soils. The trait values also reveal that a much better quality bioremediated soil for seed germination and development of root and stem was obtained from the application of IS#1. This is noteworthy because the IS#2 removes more SLO from the contaminated sand (e.g., $\cong 2\%$) than the IS#1. This could be attributed to the injection of a larger volume of VCT during the application of IS#1, producing better soil conditions due to the significant larger addition of essential nutrients contained in

Table 6 *t*-statistic for related samples of the phytotoxicity test results. H_O: $\mu_D = 0$ (no change in the germination performance of the jalapeno pepper seeds), H_1: $\mu_D \neq 0$ (there is a significant change in the germination performance of the jalapeno pepper seeds). Critical values: + 2.093 to −2.093. Level of significance $\alpha = 0.05$ for a two-tailed test, df = 19

Reference	Treatment	Root development	Stem development
Clean sand	Polluted control	$t(19) = -4.39$, H_o rejected	$t(19) = -4.44$, H_o rejected
Polluted control	IS#1 Exp#1	$t(19) = 6.04$, H_o rejected	$t(19) = 7.88$, H_o rejected
Polluted control	IS#1 Baseline	$t(19) = 3.52$, H_o rejected	$t(19) = 4.54$, H_o rejected
Polluted control	IS#2 Exp#1	$t(19) = 5.73$, H_o rejected	$t(19) = 2.85$, H_o rejected
Polluted control	IS#2 Exp#2	$t(19) = 2.43$, H_o rejected	$t(19) = 4.48$, H_o rejected
Polluted control	IS#2 baseline	$t(19) = 2.07$, H_o supported	$t(19) = 1.71$, H_o supported

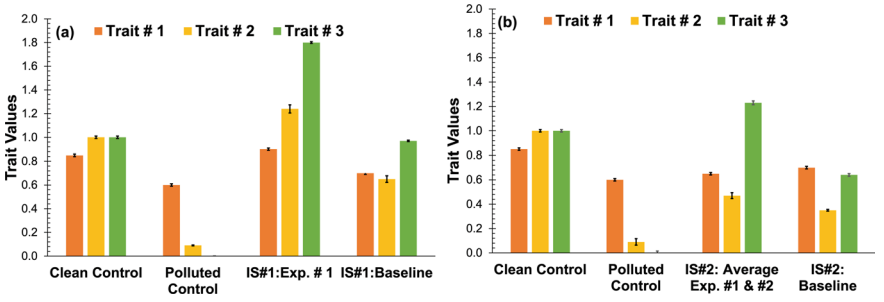

Fig. 5 Phytotoxicity test results: **a** IS#1 and **b** IS#2

the VCT (e.g., N, K, P, and micronutrients such as Fe and Mn) and the augmentation of the microbial consortia within the sand.

4 Conclusions

The elemental characterization of VCT verified its rich composition in nutrients and micronutrients. Likewise, the bacterial count revealed the presence of a large bacteria consortia in VCT. FTIR analysis demonstrated modifications of the chemical structure of the SLO subjected to bioremediation. Therefore, the VCT's microorganism consortia are capable of degrading hydrocarbons. The flushing agent's injection sequence # 2 (IS#2) removes approximately 98% of the SLO from the contaminated soil. Phytotoxicity tests established that the injection scheme # 1(IS#1) produces better quality remediated sandy soil for seed germination and root and stem development than IS#2. Overall, using alternating injections of VCT and VCT/XG as flushing agents in conjunction with the infusion of air combines three bioremediation techniques: biostimulation, bioaugmentation, and bioventing. The synergism of these techniques is effective for the bioremediation of sandy soil contaminated with SLO. A potential application of this co-operative bioremediation process is the treatment of industrial sites polluted with hydrocarbon derivatives, which include crude oil production facilities, gasoline distribution stations, and petrochemical plants, among others.

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Variations in Mechanical Strength of Enzymatically Induced Calcite

Preity Yadav, Amit Prasad, and Kala Venkata Uday

Abstract Enzymatically induced calcite precipitation (EICP) is a bio-inspired method that has gained increasing attention over the past years for its wide applications in ground improvement. The concentration of non-fat milk (NFM) powder, urease enzyme (UE), and cementation media (CM) plays a major role in calcium carbonate's crystal structure and precipitation, which can influence its interaction with soil particles. $CaCO_3$ is a polymorph consisting of vaterite, aragonite, and calcite. Among which, calcite is the most stable form, providing the highest mechanical strength and resistance to soil. UE concentration was varied with CM for synthesizing $CaCO_3$ polymorph in the laboratory. Various characterization techniques, such as SEM, XRD and nanoindentation, were involved in the analysis of mechanical properties of calcite. A detailed microstructure examination of the precipitated $CaCO_3$ crystals recommended that the hardness, elasticity, and texture are possibly the major strengths influenced by UE concentration. The current study unravels the possible reasons for the distinct responses of UE at constant concentrations of CM which can further assist in optimizing the EICP processes based on the targeted field application, resource consumption, economics, and time.

Keywords Enzymatically induced calcite precipitation · Cementation media · Urease enzyme · Ground improvement · Mechanical strength

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1 Introduction

Enzyme-induced carbonate precipitation (EICP) has been used to bind soil particles together through calcium carbonate ($CaCO_3$) precipitation to improve their strength and other mechanical properties [1, 2]. EICP has emerged as a sustainable method to improve soil strength by binding soil particles through calcite formation. This process not only increases unconfined compressive strength (UCS) but also enhances other mechanical properties, making it a valuable technique in soil stabilization [3, 4].

EICP process involves the breakdown of the urea through the urease enzyme (UE). The cementation media (CM) provides calcium and urea, whereas the source for UE is plants. Carbonate ions and ammonium ions are released in the surrounding area through urea hydrolysis [5, 6]. The role of ammonium ions is to maintain the pH in the reaction for the formation of carbonate ions. Calcium ions react with carbonate ions to form the $CaCO_3$ precipitate [7]. Calcium carbonate formed is a polymorph consisting of vaterite, aragonite, and calcite [8]. Among which, calcite is the most stable form, providing the highest mechanical strength and resistance to soil. The mechanical strength of calcite in soil is crucial for enhancing soil stability and performance, particularly in geotechnical applications.

The hardness of calcite is influenced by various factors, including its composition, crystal orientation, and the presence of organic molecules. Recent studies have shown that the incorporation of amino acids can significantly enhance the hardness of calcite [9]. UE and NFM powders are composed of various amino acids, which can impact the mechanical properties of calcite. Multiple studies have been carried out to optimize cementing and enzyme solution concentrations. However, studies done on optimizing the mechanical strength of calcite under the influence of UE-CM are limited in literature, which is the major objective of the current study. This study evaluates the effect of calcite crystal size and UE concentration on the mechanical properties of calcite by performing different characterization techniques on the $CaCO_3$ precipitate. Mineralogical and morphological analysis of the crystal was done through XRD and SEM. Nanoindentation was performed to analyze the mechanical strength of the sample. Also, the $CaCO_3$ precipitation efficiency with respect to the change in UE concentration was obtained.

2 Materials

2.1 Urease Enzyme (UE) and Non-Fat Milk Powder (NFM)

Urease is a nickel-dependent metalloenzyme that breaks down urea into ammonium and carbonate ions. It can be extracted from both plants and microorganisms. UE used in our study is purchased from Sigma Aldrich extracted from jack beans, having a specific activity of 25,920 units per gram. It was stored at freezing temperature, − 20 °C, as enzymes lose their ability to function and get denatured at low temperature.

NFM powder is known to enhance the precipitation process. UE and NFM powders were mixed in a ratio of 1:0.75 to form an enzyme solution [10].

2.2 Cementation Media (CM)

Calcium chloride and urea are components of CM. Calcium chloride and urea were obtained from Thermo Fischer Scientific Pvt. Ltd., with molecular weights of 147.01 g/mol and 60.06 g/mol, respectively. Previous studies have indicated that calcium chloride serves as the most effective source of calcium ions for optimizing carbonate production [11, 12]. Equimolar concentrations of calcium chloride and urea were combined to prepare the CM. Earlier studies have reported that an equimolar ratio of urea–calcium chloride may produce an equal proportion of CO_3^{2-} and Ca^{2+} ions that are required to form $CaCO_3$ precipitation. Furthermore, if non-equimolar ratios are used, these ions may become inadequate [13].

3 Methodology

3.1 Enzymatically Induced Calcite Precipitation (EICP)

Equimolar concentrations of CM were combined with UE and NFM powders. 10 ml of cementation solution and 10 ml of enzyme solution were combined in a 50 ml Falcon tube to prepare a 20 ml EICP solution. 1 M of CM was added individually to each test tube, followed by the UE solution of 8kU/L and 10kU/L concentrations. To enhance clarity, the concentrations of CM and UE used in the test tube precipitation experiment were designated as C1E8 and C1E10. Control tubes were prepared separately for each set. C1E8, C1E10, and control tubes were kept on a shaker at 30 °C with a rotation speed of 200 rpm. The sample in the test tube was characterized at the end of the 28th day.

3.2 Precipitation Efficiency

After the drying process of the precipitate, final precipitate is checked with acid leaching for any impurities. The samples were weighed, and the precipitate formed was further used for its characterization. Efficiency of the precipitate formed at the bottom of the test tube was calculated by Eq. (1) which was proposed by Ahenkorah et al. 2021 [13].

$$P.E = \frac{W1}{W2} \times 100 \quad (1)$$

where P.E is precipitation efficiency.

W1 is the weight of $CaCO_3$ precipitated.

W2 is theoretical weight of $CaCO_3$.

3.3 Nanoindentation

Mechanical strength analysis was evaluated through nanoindentation. $CaCO_3$ samples C1E8 and C1E10 were prepared in the form of pellets to ensure a flat surface. Pellets were prepared from 0.250 mg of samples, after which pressure was applied with a force of 40 kN for 30 s and then pressed for the same time with a force of 80 kN by means of a hydraulic press for each sample. A total of six indents were done on each sample. Maximum load (P_{max}) was 0.5 mN. Load was increased linearly for a time period of 5 s to reach P_{max} and then again held at P_{max} for 2 s. The sample was unloaded linearly for 5 s [8].

3.4 Mineralogical and Morphological Analysis

Quantitative analyses of different $CaCO_3$ polymorphs were performed through the XRD characterization technique. Before performing characterization, a sterile mortar and pestle was used for grounding the samples. With a step size of 0.02°, an angle of diffraction (2θ) between the range of 10–80° of the X-rays was set. Using copper (Cu) Ka, the scanning speed was set at 4°/min. Morphology of the samples was analyzed through SEM via drop-cast on a silicon wafer. The medium used for drop-cast was ethanol as calcium carbonate does not dissolve in ethanol [14, 15].

4 Results and Discussion

4.1 Precipitation Efficiency

The precipitation efficiency of C1E8 and C1E10 at the 28th day reaction time obtained 100% $CaCO_3$ precipitation efficiency. It can be observed that both concentrations achieved their maximum precipitation at the end of the experiment. No precipitate was obtained in control tubes. This may be due to the absence of UE, which initiates urea hydrolysis reaction.

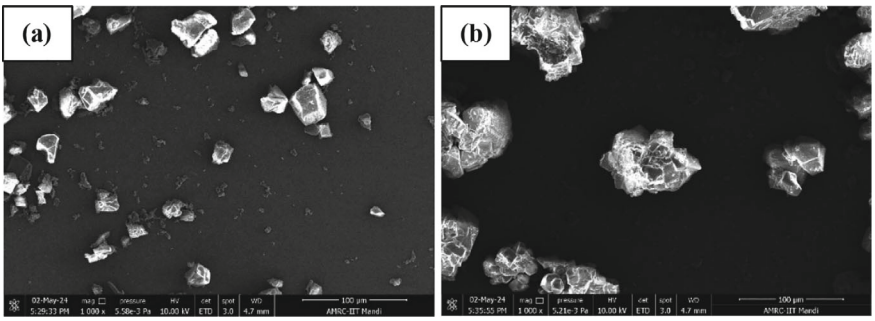

Fig. 1 **a** 28th day crystals of C1E8, **b** 28th day crystals of C1E10

4.2 Scanning Electron Microscopy (SEM) Images

The SEM images of C1E8 and C1E10 for the 28 day are represented in Fig. 1a and b. In contrast with C1E8 $CaCO_3$ crystals, the crystals formed in C1E10 appeared to be larger in size when measured at the same scale.

4.3 X-Ray Diffractometer (XRD) Analysis

Figure 2a and b represents XRD analysis of C1E8 and C1E10 done on the 28th day. It is observed that calcite, vaterite, and aragonite peaks are present in both samples representing all the polymorphs of $CaCO_3$ are present in the sample. The peaks represent the crystalline state of the sample.

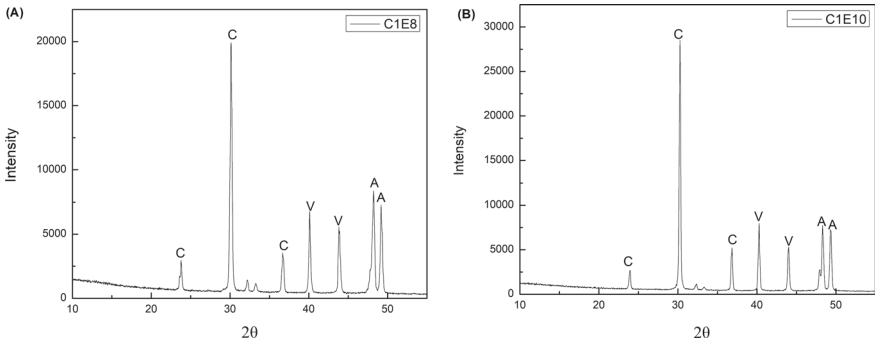

Fig. 2 XRD analysis of $CaCO_3$ for **a** C1E8, **b** C1E10

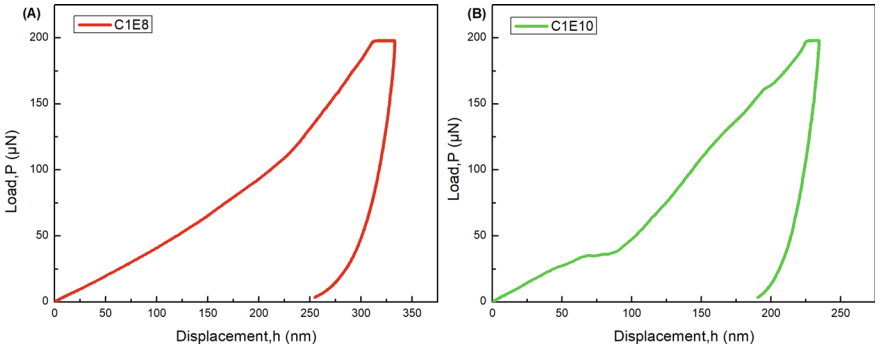

Fig. 3 Nanoindentation plot for **a** C1E8 and **b** C1E10

4.4 Crystal Strength

Nanoindentation was performed to analyze the mechanical strength of the crystals obtained for C1E10 and C1E8 on the 28th day as shown in Fig. 3a and b. Table 1 represents the elasticity, hardness, and stiffness of C1E10 and C1E8 precipitates. It was observed that the elasticity, hardness, and stiffness of C1E10 are higher than C1E8.

An increase in hardness can make calcite stiffer and can resist deformation better. An increase in elasticity can resist deformation and make it more difficult to scratch or break, essentially making it stronger and more durable. This can be beneficial in applications where resistance to wear and tear is important. When added to other materials, calcium carbonate with high mechanical hardness can increase the overall strength, stiffness, and resistance to abrasion of the composite material.

Biogenic calcite is generally harder than geological calcite. Comparative studies using nanoindentation have shown that biogenic calcite exhibits increased hardness due to its unique structural properties, while geological calcite displays mechanical anisotropy under different loading conditions [16].

The addition of amino acids, such as glycine and aspartic acid, into calcite crystals has been shown to increase hardness. Jack bean urease and NFM powder used in the current experiment consist of aspartic acid. C1E10 has a higher UE concentration than C1E8 and exhibits more hardness. This can be due to the increases in UE concentration and the crystal orientation. In literature, nanoindentation tests revealed that hardness values reached those comparable to biogenic calcite, indicating a direct correlation between amino acid content and hardness [17]. The hardness of calcite

Table 1 Elasticity, hardness, and stiffness evaluated for C1E8 and C1E10

Combination	Elasticity (GPa)	Hardness (GPa)	Stiffness (μN/nm)
C1E8	3.65	0.056	7.75
C1E10	7.26	0.105	11.25

also varies with crystal orientation [18]. Crystals obtained for C1E10 are larger in comparison with C1E8.

5 Conclusions

Characterization techniques such as IC, SEM, XRD, and nanoindentation were performed on C1E8 and C1E10 samples to analyze the effect of UE on the strength of calcite. Both samples exhibited crystalline form through XRD evaluation. The following inferences were obtained from the experimental work conducted:

- C1E8 calcite crystals were smaller in comparison with C1E10.
- Nanoindentation revealed that C1E10 exhibited more hardness and stiffness that is 0.105 GPa and 7.75 μN/nm in comparison with C1E8. This may be attributed to the difference in the UE concentration. The presence of amino acids in the UE increases the hardness of calcite crystals.
- The role of UE is crucial in EICP affecting the mechanical properties of calcite crystals. Thus, for field applications, an optimized amount of UE can be added with CM for an increase in the strength of calcite crystals leading to effective binding properties.

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Biochar as a Promoter of the Circular Economy in Environmental Remediation

Antón Puga, Ana Rita Alves, Cristina Soares, and Cristina Delerue-Matos

Abstract The micropollutants detected in waters and their control are serious problems affecting ecosystems and, consequently, human beings. The primary issue in the inefficiency of the wastewater treatment plants (WWTPs) is their inability to eliminate contaminants. For this reason, the European Community legislates this problem, forcing WWTPs to correct these inefficiencies by 2045. However, there is currently no technology available that can eliminate this necessity. Biochar derived from waste can serve as an ecological solution to this problem, while providing new life to undervalued products. This work aims to respond to the environmental problem of water pollution, through a two-stage process, characterized by the principles of the circular economy. The first stage involves producing biochar from agricultural and forest waste and use it as a micropollutant adsorbent in wastewater. In the second stage, once the biochar is saturated and without greater adsorption capacity, it will be regenerated using electrochemical techniques.

Keywords Wastewater treatment · Biochar · Adsorption · Advanced oxidation processes · Electro-Fenton

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1 Introduction

Water pollution by organic and inorganic compounds is a major environmental issue worldwide. The uncontrolled discharge of industrial and agricultural wastewater has led to the accumulation of persistent pollutants, such as heavy metals, dyes, and pharmaceuticals [1], which pose a threat to aquatic ecosystems and human health. In this context, the development of sustainable and efficient water remediation technologies has become a priority for the scientific community [2, 3].

Wastewater treatment plants (WWTPs) play a crucial role in protecting the environment by removing organic matter, nutrients, and pathogens from wastewater. However, WWTPs were not originally designed to eliminate micropollutants, such as pharmaceuticals, pesticides, personal care products, and industrial chemicals. As a result, these substances can bypass conventional treatment processes and enter natural water bodies, potentially harming ecosystems and human health [4].

To reduce harmful substances in water sources, the European Union has introduced new legislation that requires member states to upgrade their wastewater treatment facilities. This initiative aims to enhance overall water quality and promote long-term environmental sustainability. By 2045, all WWTPs must implement advanced treatment technologies capable of effectively removing micropollutants [5].

Investing in these upgrades will be essential to meeting regulatory requirements and protecting aquatic ecosystems. Technologies such as advanced oxidation processes, activated carbon filtration, and membrane treatments are among the promising solutions that will help WWTPs comply with the new standards [6].

Among the various treatment strategies, the use of carbon-based adsorbent materials has proven to be an effective solution for removing contaminants from water [7]. Biochar, a carbonaceous material obtained from biomass pyrolysis, has gained increasing interest due to its low cost, high chemical stability, and excellent adsorption capacity [8]. Producing biochar from agroforestry waste not only provides an efficient means of contaminant removal but also contributes to waste valorization and the mitigation of the environmental impact associated with waste disposal [9].

This study proposes the use of biochar derived from the quince tree (*Cydonia oblonga*) as an innovative adsorbent for the remediation of contaminated water. The quince tree is a widely cultivated fruit tree in various regions of the world, whose agroforestry residues represent an abundant and underutilized biomass source [10]. Transforming these residues into biochar through controlled pyrolysis processes enables the production of a material with physicochemical properties suitable for the adsorption of emerging contaminants in wastewater [11].

Beyond the adsorption process, biochar can play a key role in advanced oxidation processes, such as the electro-Fenton method, where it acts as a catalyst and supports the generation of highly reactive hydroxyl radicals. This synergistic combination of adsorption and advanced oxidation enhances the efficiency of organic pollutant degradation, opening new perspectives for the treatment of hard-to-remediate wastewater [12].

The present study focuses on the characterization of biochar obtained from quince residues, evaluating its structural and chemical properties as well as its performance in contaminant adsorption. Additionally, its application in the electro-Fenton process is investigated to enhance the degradation of persistent organic compounds. The findings of this research are expected to contribute to the development of sustainable materials and innovative technologies for water decontamination, promoting the utilization of agroforestry waste for environmental protection.

2 Materials and Methods

2.1 Biochar Preparation

Biomass wastes from different vegetal species, namely, pruning from *Cydonia oblonga* (quince tree) were used to produce biochar. The biomass wastes were pyrolyzed in an industrial oven from Ibero Massa Florestal (Portugal) for 8 h at 500 °C and cooling for 18 h until room temperature. Due to the design of this pyrolytic oven, the presence of any gas was not required (nitrogen or other), and only the oxygen from the biomass itself was present. Before use, the biochar was ground to a particle size of less than 75 μm.

2.2 Biochar Characterization

The biochar characterization was presented in Table 1. The elemental classification of biochar, as well as its quantification, was carried out using an Elemental Analyzer Thermo Finnigan—CE Instruments, model Flash EA 1112 CHNS series, based on sample combustion dynamics. The quantification of the moisture, volatile matter, and ashes were studied following the ASTM D1762 standard.

The pH at the point of zero charge (pHpzc) for the quince biochar was determined according to Carabineiro et al. [13]: 0.1 g of biochar was added to 20 mL of 0.1 M NaCl solutions with pH values between 2.0 and 12.0 (pH adjustment: NaOH or HCl

Table 1 Characteristics of quince biochar

pHpzc	9.75
Surface area (m^2/g)	230
Moisture (%)	6.2
Volatile matter (%)	15.8
Ashes (%)	5.0
Elemental analysis: C (%)	79.9
Mineral analysis: Fe (mg/kg)	1296.2

1 M). After 24 h, the final pH was subtracted from the initial pH and the pzc value was obtained through the curve (pH_{final} vs. $pH_{initial}$).

The determination of the Fe content was performed according to the European Standard EN 15,290 [14] with digested samples, in an Inductively Coupled Plasma-Atomic Emission Spectroscopy (ICP-AES) (Horiba Jobin–Yvon equipment).

Textural parameters: Surface area (SBET) and total pore volume (Vtotal) were evaluated by nitrogen adsorption kinetics 77 K (ASAP 2010 Micromeritics equipment) by using the single point method at the relative pressure of P/Po = 0.3; all samples were previously outgassed overnight, under vacuum, at 150 °C before analysis.

2.3 Kinetic Assays

For the kinetic experiments, 75 mg of biochar in 150 mL of solution containing 10 mg/L of sulfamethoxazole (SMX) was used and stirred at 150 rpm (VMS–C7 advanced, VWR) during 45 min at room temperature. An aliquot of 1 mL was collected over time, centrifuged (8,000 rpm/5 min), and analyzed, to determine the remaining concentration of SMX. Furthermore, adsorption studies were carried out at different pHs (natural pH and pH adjusted to 3) to evaluate the influence of pH (Consort pH Meter model C861, Belgium) on both adsorption and degradation assays (next section).

The mass of SMX adsorbed per gram of adsorbent (uptake: qt, mg/g) was calculated according to Eq. 1:

$$qt = C_0 - C_e \times Vm \tag{1}$$

where C_0 and C_e (mg/L) are the initial and equilibrium concentrations of SMX, V (L) is the volume of the SMX solution, and m (g) is the mass of quince biochar.

2.4 Degradation Assays

The electro-Fenton treatment in a solid matrix was carried out with the same setup as described in Sect. 2.3. In addition, the selected electrodes (metal mix oxide: MMO in the anode and stainless steel: STS in the cathode) played a crucial role, together with the intensity (0.1 A). Once the biochar (25, 50, 75, 100 mg) was added to the electrochemical cell, the pH was adjusted to 3, in a volume of 150 mL and with 150 rpm of agitation. The time of the experiment was the time required for the total removal (60–120 min). The electro-Fenton was carried out through a heterogeneous process (with biochar), and through a homogeneous treatment (without biochar, but with Fe_2SO_4 dissolved in the solution at a concentration of 0.29 mM). In both types of assays, a concentration of 0.1 M of Na_2SO_4 (electrolyte) was added. An aliquot

of 1 mL was collected over time, centrifuged (8,000 rpm/5 min) and analyzed, to determine the remaining concentration of SMX.

3 Results

3.1 Adsorption

As mentioned throughout the introduction, one of the water remediation strategies is adsorption through carbonaceous materials [8]. In this case, a pyrolyzed carbonaceous material (biochar) has been produced, specifically from the quince tree. This biochar has been tested as an adsorbent material to remove the antibiotic sulfamethoxazole (SMX) from wastewater. As can be seen in Fig. 1, the biochar has been successful in the adsorption of SMX, with a removal rate of 100% in 45 min.

Going deeper into this adsorption process, it can be observed that SMX has two pka: 1.6 and 5.7. These pKa values dictate its ionization, below pH 1.6, biochar is positively charged, between 1.6 and 5.7, it remains neutral, and above 5.7 it becomes negatively charged, influencing its adsorption capacity [15]. Therefore, it is necessary to know the zero charge point of the adsorbent, pH_{pzc}, and the pH of the solution. The first has resulted in a pH_{pzc} of 9.75, which indicates that below this value it is positively charged, and above, negatively charged [16]. The second, the natural pH of the solution has been between 6.5 and 7.8 (initial and final, respectively). For this reason, at the working pH, the adsorbent is positively charged, and the contaminant negatively charged, so electrostatic adsorption can be deduced.

Biochar exhibits outstanding potential as an adsorbent, primarily due to its unique physicochemical properties, which are critical for the efficient removal of various

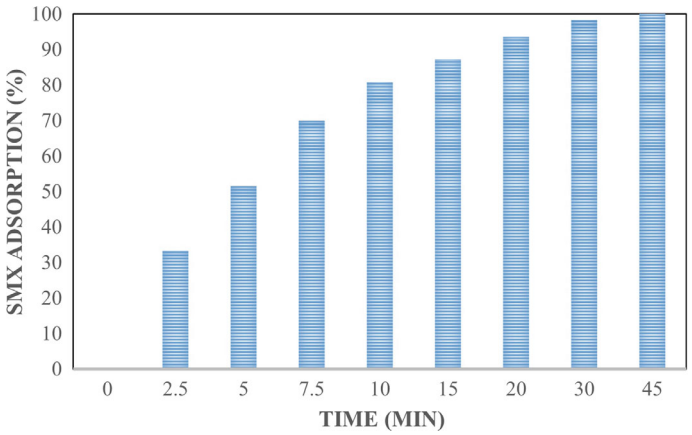

Fig. 1 Bar chart showing SMX adsorption percentage over time in minutes

emerging contaminants from aqueous systems. As shown in Table 1, the material displays a high specific surface area (230 m2/g), low moisture content (6.2%), moderate volatile matter (15.8%), low ash content (5.0%), and a notably high carbon content (79.9%). Among these, the high BET surface area plays a pivotal role, as it directly correlates with the number of available adsorption sites, enhancing the material's capacity to immobilize a broad range of organic micropollutants [8, 15].

3.2 Electrodegradation

The quince biochar produced was tested in the elimination of the known antibiotic sulfamethoxazole (SMX) in a liquid medium. To do this, its efficiency as a catalyst was tested, once its elemental composition was known, among which its iron content (1296 mg/kg) stands out, a transition metal known for its role in the Fenton reaction [17]. Therefore, quince biochar has been proposed as a catalyst for the Fenton reaction, in its variant with electric current (electro-Fenton). The procedure carried out was to apply two electrodes known for their good performance in electro-Fenton, namely, MMO in the anode and STS in the cathode [18].

The experiment was conducted under acidic pH conditions (ideal pH for Fenton-like processes [17]), where SMX exhibits negligible adsorption (less than 2%), confirming that the biochar primarily acts as a catalyst. The electro-Fenton has been compared with biochar: heterogeneous electro-Fenton (with the solid metal that the biochar itself contains), and without biochar: homogeneous electro-Fenton (0.29 mM of dissolved iron). In addition, different amounts of biochar were provided to check which was the most appropriate (between 25 and 100 mg). As can be seen in Fig. 2, the biochar with the lowest presence in the reactor (25 mg) was the slowest in degrading the contaminant under study (SMX), achieving 83.2% in 120 min. On the other hand, the test with 50 mg reached 97% at the same time. As this was a significant difference (more than 15%), it was decided to increase the quince biochar to 75 mg (red circles, Fig. 2) achieving total removal in 90 min (30 min less than with 50 mg of biochar). With these results, a final test was attempted, with 100 mg, waiting, again, for another considerable reduction in time.

However, as can be seen in Fig. 2, the electro-Fenton with 100 mg of quince biochar (purple circles) was slower than that with 75 mg, which leads to think that it is a saturation of the solution, which is too much mass, and produces a rebound effect: that is, instead of providing a greater speed between the electrons, the reaction is slowed down by an excess of biochar.

Therefore, and after observing thse differences between the different electro-Fenton and the catalyst masses used, a comparison has been made with the homogeneous electro-Fenton, at an optimal iron concentration previously studied (0.29 mM) [19]. In this case, the difference, as expected, is large. The homogeneous electro-Fenton (blue circles, Fig. 2) reaches total elimination at 60 min, and therefore 30 min before the optimal electro-Fenton (with 75 mg of quince biochar). To make a better decision on which is the most appropriate electro-Fenton, two studies have been

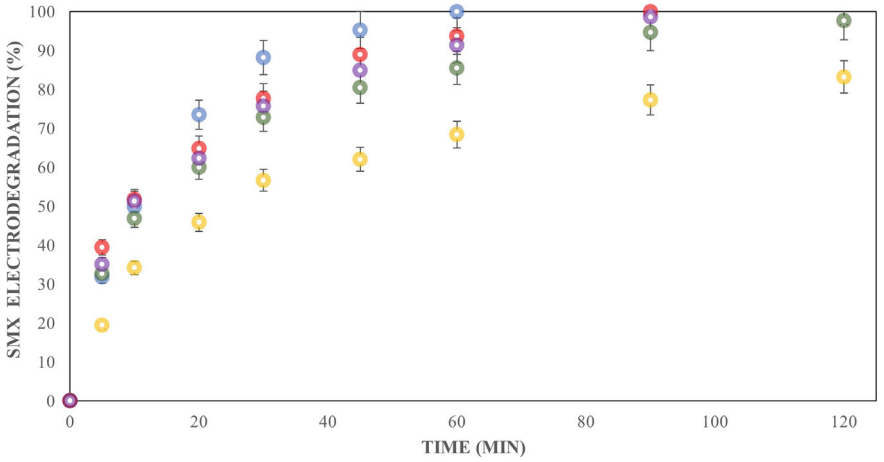

Fig. 2 Scatter plot showing SMX electrodegradation percentage over time in minutes: in heterogeneous (25 mg: yellow; 50 mg: green; 75 mg: red, 100 mg: purple) and homogeneous (blue) treatment

carried out on the system: on the one hand, the leaching study of the optimal heterogeneous electro-Fenton, and, on the other hand, the study of electrical consumption. Starting with the first one, the heterogeneous electro-Fenton has leached less than 1% of the total iron content of the biochar between the beginning and the end of the process, which makes the process can be called ecological, and does not require extra processes to remove the dissolved iron before discharging the treatment into public waterways. Regarding the study of electrical consumption [20], considering that the intensity has been fixed (0.1 A, as reflected in Sect. 2.4), the difference has been relatively large, since the electrochemical reactor with the biochar has obtained higher voltage values, due to having the biochar in the liquid medium and making a little more resistance. On average throughout the tests, the heterogeneous electro-Fenton consumed 21.5 V, while the homogeneous one has not reached 20 V (specifically, 18.8 V). For the heterogeneous electro-Fenton, consumption reached 3.23 kWh, while for the homogeneous, 1.88 kWh. This is a difference of almost 42% more consumption for the reactor with biochar [18]. However, the fact that the process is catalyzed by a residue, thus giving it a new useful life, and that it can be recovered in a simple way and that it has a very low leachate, makes the electro-Fenton catalyzed by quince biochar the best option. The overall analysis of the process shows that the choice of biochar as a catalyst turns the process greener, despite the increase in energy consumption. Moreover, it supports the potential for large-scale application, as it involves the reuse of waste materials, giving them a new useful life to tackle another residual problem.

4 Conclusion

The study demonstrated the effectiveness of quince-derived biochar in wastewater treatment through two distinct approaches: adsorption and the Electro-Fenton process. Both techniques achieved complete (100%) pollutant removal, highlighting the versatility and efficiency of biochar as a sustainable material for water purification.

Adsorption provided a straightforward, cost-effective, and energy-efficient solution, making it suitable for passive treatment systems or pre-treatment stages. On the other hand, the electro-Fenton process leveraged biochar's catalytic properties to enhance the degradation of contaminants through advanced oxidation, offering a powerful alternative for treating highly persistent pollutants.

These findings emphasize the potential of biochar from quince residues as a multifunctional material for wastewater treatment, addressing environmental concerns while promoting resource valorization. Future research should focus on optimizing operational parameters and evaluating the long-term stability of biochar in real wastewater conditions.

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Comparison of Antibiotic Efficacy of Gramicidin S-Biosurfactant Combined Formulations Against *Staphylococcus Aureus*

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Abstract Biosurfactants (BS) are amphiphilic molecules, more biocompatible and biodegradable than their chemical homologous, used to reduce water surface tension and stabilize non-aqueous soluble active principles. BS can improve drug solubility reducing phase separation and aggregation. Thus, BS can be good candidates for combination with poorly water-soluble Gramicidin S (GR-S) enhancing its antimicrobial activity in aqueous environment against *Staphylococcus aureus* (*S. aureus*) and reducing the potential toxicity for human skin natural microbiome like *Staphylococcus epidermidis* (*S. epidermidis*), as ethanol is commonly used solvent in pharmaceutical formulations. In this study, two formulations containing GR-S dissolved in the aqueous solutions (1 g/L) of cell-bound or extracellular BS, coming from corn steep water (CSW), were prepared. Minimum inhibitory concentrations (MIC)

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against *S. aureus* and *S. epidermidis* were assayed after 24 h using an ethanolic solution of commercial GR-S as a reference. Results showed that GR-S formulations in BS solutions presented similar MIC values against *S. aureus* compared to the reference GR-S in ethanol. Moreover, GR-S formulation in extracellular BS keeps the selectivity toward *S. aureus* in comparison with *S. epidermidis*. In conclusion, aqueous solutions of cell-bound and extracellular biosurfactants derived from CSW can replace ethanol and/or synthetic surfactants in pharmaceutical formulations containing GR-S as an active ingredient.

Keywords Antimicrobials · Pathogen · Water solubility · MIC · Corn steep water

1 Introduction

Staphylococcus aureus is a Gram-positive bacterium commonly found on the skin and mucous membranes of humans. While it can be a harmless commensal organism, it is also a leading cause of various infections, including skin and soft tissue infections, pneumonia, and bloodstream infections [1, 2]. Additionally, MRSA strains are resistant to methicillin and other beta-lactam antibiotics that complicate treatment options and leading to higher morbidity and mortality rates [3, 4]. The World Health Organization (WHO) in 2024 report underscores the global threat posed by MRSA [5]. For example, in some regions of Africa, up to 80% of *S. aureus* infections are reported to be methicillin resistant, rendering standard antibiotic treatments ineffective [6].

Among the actions WHO emphasizes are (1) the need for enhancing surveillance by improving monitoring of *S. aureus* infections and resistance patterns to inform treatment strategies; (2) the research and development by investing in the development of new antibiotics and alternative therapies to combat resistant strains; and (3) infection prevention and control by implementing stringent infection control measures in healthcare settings to prevent the spread of MRSA [5].

Considering the growing resistance to conventional antibiotics, Gramicidin S (GR-S), a cyclic peptide antibiotic composed of ten amino acids, cyclo[L-Leu-D-Phe-L-Pro-L-Val-L-Orn-L-Leu-D-Phe-L-Pro-L-Val-Orn] (IUPAC condensed description), has emerged as a promising candidate in the fight against *Staphylococcus aureus*, including its resistant forms [7]. GR-S is known for its ability to disrupt the bacterial cell membrane, leading to the lysis (breaking apart) of bacterial cells, effectively killing the bacteria [8]. This mode of action, which targets the bacterial cell membrane, makes GR-S a strong candidate as an alternative treatment for resistant strains like MRSA [9].

However, while Gramicidin S shows significant potential, there are challenges to overcome. The peptide's potent bactericidal activity can be toxic to mammalian cells as well, which raises concerns about its use in humans. Although GS exhibits potent bactericidal activity, its high hemolytic and nephrotoxic effects limit its clinical use to topical applications [10]. Hence, further research into more biocompatible

formulations of GR-S is needed. The development of novel drug delivery systems or formulations that can target bacterial cells more specifically, without harming human cells, is crucial for maximizing the therapeutic potential of GR-S.

Moreover, because of the poor solubility of Gramicidin S in water, it is used for skin or eyes application dissolved in ethanol where potential damage caused by this solvent to the organs cannot be ignored. In this context, incorporating an aqueous solution of biosurfactants in Gramicidin S formulations could enhance the solubility and stability of this antibiotic, thereby facilitating its incorporation into various pharmaceutical forms. The amphiphilic nature of biosurfactants allows for the formation of micelles, which can encapsulate hydrophobic compounds like Gramicidin S, improving their delivery and efficacy. Additionally, the biosurfactant's antimicrobial properties could provide a synergistic effect, enhancing the overall therapeutic action against *S. aureus*. For example, López-Prieto and coworkers [11] have developed a multifunctional biosurfactant extract from corn steep water (CSW), a byproduct of the corn wet milling industry, containing lipopeptides and other bioactive compounds, which contribute to its antimicrobial and surface-active properties. This extract is obtained through a liquid–liquid extraction using ethyl acetate as the organic solvent. The biosurfactant extract demonstrated significant surface-active properties, being able to reduce the surface tension of water in more than 20 units with a critical micellar concentration lower than 0.5 g/L. Moreover, the resulting biosurfactant at a concentration of 1 g/L exhibits significant antimicrobial activity, inhibiting 100% of *S. aureus* growth.

In this work, it is evaluated by the first time the use of two sustainable and biocompatible formulations based on Gramicidin S combined with natural biosurfactants. Some of them exhibited antimicrobial activity against *S. aureus* comparable to the less biocompatible formulation composed of Gramicidin S and ethanol. Additionally, the study includes an antimicrobial test against *S. epidermidis*.

2 Materials and Methods

2.1 Materials

Antibiotic. Commercial antibiotic, Gramicidin S TFA salt, ≥95% was purchased at Biosynth (China) in dry powder form. For making the Gramicidin S-based formulation, the antibiotic was dissolved in ethanol or two aqueous solutions of the biosurfactants.

2.2 Methods

Preparation of Gramicidin S-biosurfactant formulations. For assay, both extracellular (BS-1) and cell-bound (BS-2) biosurfactants were tested. Biosurfactant 1 (BS-1) was obtained following the protocol stablished by López-Prieto et al. [11] where an aqueous CSW solution (50 g/L) was previously prepared and extraction process of biosurfactant occurred using ethyl acetate. Whereas Biosurfactant 2 (BS-2) was obtained from CSW through a multi-step extraction process with PBS according to the protocol established in patent WO2020234501A1 [12]. To prepare Gramicidin S-biosurfactant-based formulations, first, the two aqueous solutions of BS-1 and BS-2 (1 g/L) were prepared and then used to solubilize 1 g/L and 0.7 g/L of the antibiotic, respectively, with following filtration (0.2 microns). Table 1 includes composition of the antibiotic formulations containing GR-S in ethanol (F1), BS-1 (F2), and BS-2 (F3).

Bacteria and inoculum preparation. The strains of *Staphylococcus aureus* (ATCC 6538) and *Staphylococcus epidermidis* (ATCC 1457) from Microbial Culture Collection of Centre for Biological Engineering (Portugal) were stored on Mueller–Hinton agar at 4 °C and used in all assays. The working stocks of cultures for inoculum preparation were incubated in Mueller–Hinton broth (MHB) (15 mL) at 37 °C under shaking conditions (180 rpm) overnight. For antibacterial assay, the test culture suspension was diluted down to OD_{600} of 0.1 in sterile MHB obtaining the total volume of 100 mL.

Antibacterial assay. Minimum inhibitory concentration was determined according to the Clinical Laboratory Standards Institute (CLSI) M07-A10, 2015, Wayne, PA in a 96-well cell plate. It was prepared in the range of dilutions for the formulations (F1, F2, F3) of 0.3125, 0.625, 1.25, 2.5, 5, 10, 20, 40, 80, 160 mg/L in order to obtain a final range of 0.15625, 0.3125, 0.625, 1.25, 2.5, 5, 10, 20, 40, 80 mg/L making a serial two-fold dilution with sterilized MHB. Moreover, two controls of sterility and vitality of the bacteria were included using uninoculated and inoculated wells of antibiotic-free MHB, respectively. Plates were then incubated at 37 ºC under

Table 1 Nomenclature, composition, and MIC values for the formulations tested in this chapter

Formulation	Composition	MIC, μg/mL	
		S. aureus	*S. epidermidis*
F1	Gramicidin S	2.5–5	5–10
	Ethanol		
F2	Gramicidin S	2.5–5	5–10
	Biosurfactant (BS-1: water)		
F3	Gramicidin S	2.5–5	2.5–5
	Biosurfactant (BS-2: water)		

shaking conditions (140 rpm) overnight and the bacterial susceptibility to the antibiotic was interpreted as the visible growth inhibition. Additionally, for the concentrations where visible growth was detected, the growth inhibition percentage was determined by OD_{600} measurement. These values were elaborated using Microsoft Excel 2020 (version 16.43), and the results are presented as an average ± standard deviation. Each concentration was tested in six repeats. When complete growth inhibition was observed, the corresponding values were fixed at 100% for all replicates, which prevented the calculation of standard deviation.

3 Results

The development of antibiotic formulations free from conventional organic solvents like ethanol and DMSO is of critical importance, both for environmental sustainability and biocompatibility. The minimum inhibitory concentrations (MIC) of the GR-S formulations (F1, F2, and F3) against *S. aureus* and *S. epidermidis* were evaluated after 24-h incubation. F1 was based on ethanol whereas F2 and F3 were formulated using aqueous solution containing biosurfactants (see Table 1).

Results indicated that the GR-S formulations in biosurfactant (F2 and F3) exhibited MIC values against *S. aureus* similar to those of the reference GR-S in ethanol (F1) (Fig. 1, Table 1).

Additionally, the GR-S formulation in extracellular BS maintained its selectivity toward *S. aureus*, showing less activity against *S. epidermidis* (Table 1, Fig. 2).

The results obtained suggest a viable pathway toward greener pharmaceutical formulations without compromising performance, offering a safer alternative to traditional ethanol-dissolved GR-S systems or DMSO-based delivery methods reported

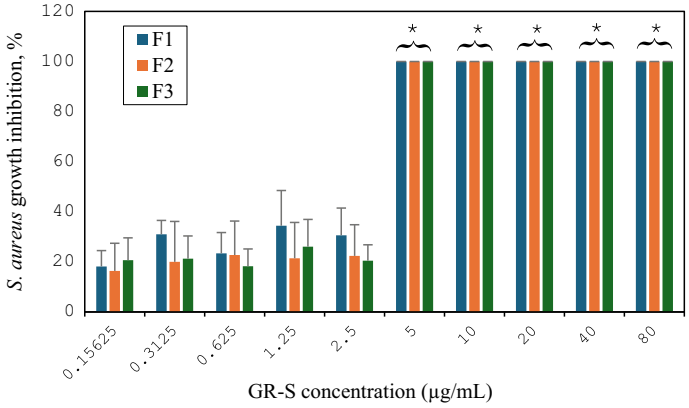

Fig. 1 Antimicrobial activities of gramicidin S in ethanol (F1), BS-1 (F2) and BS-2 (F3) against *S. aureus*; values not labeled with * represent the average of six replicates and corresponding standard deviations; values labeled with * were fixed at 100% for all replicates

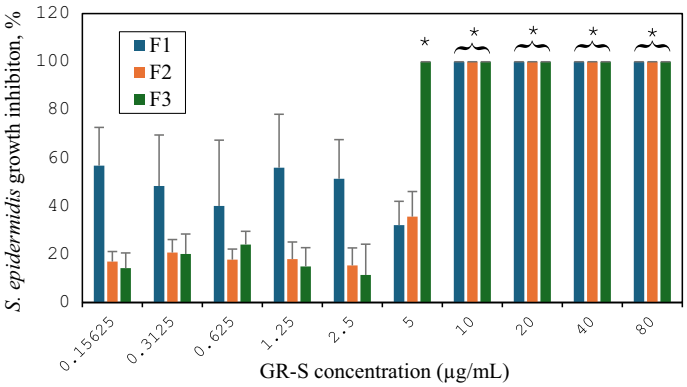

Fig. 2 Antimicrobial activities of gramicidin S in ethanol (F1), BS-1 (F2), and BS-2 (F3) against *S. epidermidis*; values not labeled with * represent the average of six replicates and corresponding standard deviations; values labeled with * were fixed at 100% for all replicates

in the literature. In comparison with the data presented in this chapter, Wang et al. [10] developed a series of GR-S analogues to improve its antibacterial activity while reducing its toxicity, addressing the rise of multidrug-resistant bacteria like MRSA. Among these, they produced a GR-S analogue with a two-fold increased in antibacterial activity against MRSA (MIC: 2 µg/mL) compared to a standard of GR-S (MIC: 4 µg/mL), though with increased cytotoxicity. The standard MIC value observed by these authors is in the range of the MIC obtained against *S. aureus* with the formulation F2 and F3 based on GR-S dissolved in natural solvents (see Table 1). It is important to point out that Wang et al. [10] determined the antibacterial activity using the broth microdilution method recommended by the CLSI, where GR-S was dissolved in 25% dimethyl sulfoxide (DMSO) and sequentially in cation-adjusted Mueller–Hinton (CAMH) broth following by a serial two-fold dilution, although DMSO already has been recognized as an antimicrobial agent which works against *S. aureus*. Hence, Kligman [13] determined the bacteriostatic concentration in nutrient broth for two *S. aureus* varieties among other pathogenic strains showing that 20% DMSO was bacteriostatic to all microorganisms tested. Hence, the data presented in the current work are more favorable than those presented by Wang et al. [10], as F2 and F3 are extent of organic solvents. Moreover, Summer et al. [14] have provided evidence that DMSO at concentrations between 0.03 and 25% significantly inhibits biofilm formation of *P. aeruginosa* and acts synergistically with standard antibiotics at very low concentrations (<1%), hence DMSO can increase the antimicrobial activity of GR-S analogues against *S. aureus*. The European Chemicals Agency (ECHA) identifies DMSO as a combustible liquid that can cause serious eye and skin irritation [15]. On the other hand, ECHA is communicated by companies in REACH registrations and reports about potential dangers of ethanol, including damage to organs, skin irritation, harmful effects to the skin and internal organs, and possible carcinogenicity [16]. Moreover, ethanol is a highly flammable liquid

and vapor. In the current study, formulations F2 and F2 demonstrated competitive efficacy while relying solely on natural, biodegradable solvents, eliminating the toxicity risks and ecological concerns associated with synthetic solvents like DMSO or ethanol.

4 Conclusion

The combination of GR-S with biosurfactants extracted from corn steep water (CSW) presents a promising approach for developing more biocompatible and sustainable treatments against *S. aureus*, offering an alternative to harmful and unsustainable solvents like ethanol or DMSO. However, further research is crucial to optimize formulation parameters and evaluate the safety and efficacy of aqueous formulations containing GR-S and biosurfactants with minimal use of organic solvents. It is essential to conduct additional cytotoxicity tests and clinical trials to ensure the viability of these formulations for therapeutic use.

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Merging Tuneability and Toxicity of Ionic Liquids for the Removal of Hydrophobic Aromatic Pollutants

Fatima Moscoso, Marta Markiewicz, and Stefan Stolte

Abstract Polycyclic Aromatic Hydrocarbons (PAHs) are pollutants of concern with potential adverse health impacts. The very low solubility of PAHs in water derails the possibility of using microorganisms to biodegrade them. In the present work, different Ionic Liquids (ILs) possessing surface active properties have been selected to shed light on the implementation of a true remediation strategy. An outstanding enhancement of solubility for phenanthrene (used as a model PAHs) was found by $[IM12]^+$ cation when is combined with docecylbenzosulfonate anion [DDBS]. The ammonium-based ILs [N1182OH][Cl] have been satisfactorily able to solubilize aromatic compounds by means of mixture strategies. A complete metabolization for this cation was detected in less than 10 days classifying it as a readily biodegradable compound. The metabolic pathway was also reported.

Keywords PAHs · Hydrophobicity · Ionic liquids · Biodegradability · Toxicity

1 Introduction

Polycyclic Aromatic Hydrocarbons (PAHs) are pollutants of concern with potential adverse health impacts that are present in oil, coal, or tar and are formed in several industrial processes. It has been long known that these substances are characterized by low aqueous solubility which implies a high tendency to strongly adsorb onto particulate matter. Consequently, the removal of these pollutants by means of eco-friendly techniques can be assumed as a difficult task [1, 2]. PAHs have been

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the focus of tremendous number of scientific publications since the 80 s regarding not only their persistence, but also their acute toxicity, mutagenicity, genotoxicity and carcinogenicity to living organisms as recently have been summarized in the review reported by Louvado et al. [3]. Attends based on physical and chemical methodologies had been traditionally approached in many works to remediate PHA-polluted areas. However, these conventional technologies present some operational and economical disadvantages [4]. From an environmentally friendly perspective, biotechnological methodology seems more attractive to remove PAHs from soils and groundwater [5–9]. Thus, since the very low solubility of PAHs in water derails the possibility of using microorganisms to biodegrade them, Surfactant-Enhanced Remediation (SER) must be considered. SER has been already used as a promising strategy for enhancing solubility of hydrophobic pollutants such as dodecane from soils [10–12].

Ionic Liquids (ILs) can be defined as new types of substances solely compound by ions which are liquids at room temperature. Over the last years, ILs have garnered much attention from widespread scientific and industrial communities, not only due to their green potential (due to their "negligible" volatility) but also to their unique combination of properties. A particularly interesting result is their tuning capability, as the right combination of chemical and physical properties of these cationic and anionic monomers allows them to design ILs specific to each task [13].

Literature shows examples of different applications of ILs based on extraction processes. Hence, some authors reported the use of imidazolium-based ILs for the extraction of organic contaminants from soil as well as for the removal of dyes and pesticides from water. Moreover, it had been already reported that ILs composed of cation or anions with long alkyl chain possess active surface properties and exhibit behaviors like conventional surfactants [17–19]. Accordingly, it seems clear that active surface ILs can act as solubility enhancers for PAHs due to their ability to form micelles/aggregates in water. This surfactant-like behavior will be the solution for the implementation of a bioremediation strategy for PAHs by means of SER strategies.

Imidazolium-based ILs are, by far, the most studied ILs in biotechnological applications although their true greenness character is still questionable because of their high stability (toward abiotic and biotic processes) [19, 20]. On the other hand, by contrast, pyridinium-based ILs have shown an elevated biodegradation potential as some authors have already reported [13, 21]. According to this, from toxicity and biodegradability considerations, ammonium-based ILs could be considered for SER, as promising candidates.

Therefore, properties such as biodegradability and toxicity must be mandatorily considered to achieve a sustainable SER strategy by ILs. During the last years, a great effort has been made to provide insight into the IL structure-based properties. Thus, as it was reviewed by Stolte and co-workers, biodegradability ranges widely from very good to very poor. Namely, good primary biodegradability has been observed in ILs with longer alkyl side chains at the cationic core whereas the same head group with shorter side chains is classified only as poorly biodegradable. In terms of toxicity toward living organism, similar behavior was found [21–23].

To date some studies are focused on solubility enhancement of PAHs by different surfactants, but from the best of our knowledge, this is the first work mainly focused on the use of ILs as solubility enhancers of PAHs with biodegradability and toxicity considerations. This study aims to assess PAHs solubility enhancers in relation to their likely toxicity and to gain a structure-based understanding of their biodegradability and toxicity.

2 Materials and Methods

2.1 Materials

Ionic liquids from different families (imidazolium, pyridinium, ammonium, and phosphonium) combined with different anions such as sulfonates or chlorides were selected based on their structural properties. Acronym scheme proposed by UFT-Merck Ionic Liquids database was used in this work: N-ethyl-N,N-dimethyl-1-butanaminium chloride ([N1124][Cl]), 2-hydroxy-N,N-dimethyl-1-octylethanaminium chloride ([N1182OH][Cl]), N,N,N-trimethyl-1-butanaminium methanesulfonate ([N1114][1SO3]), 2-hydroxy-N,N,N-trimethylethanaminum methanesulfonate ([N1112OH][1SO3]), N,N-dibutyl-N-methyl-1-butanaminium methanesulfonate ([N1444][1SO3]), 1-butyl-3-methyl-1H-imidazolium 4-methylbenzenesulfonate ([IM14][4MePhSO3]), 1-hexyl-3-methyl-1H-imidazolium 6-methyl-2,2-dioxo-1,2,3-oxathiazin-4(3H)-onate ([IM16][AC]), 1-ethyl-3-methyH-imidazolium dodecylbenzenesulfonate ([IM12][DDBS]), trihexyltetradecylphosphonium decanoate ([P666-14][9COO]), N-decyl-N,N-dimethylbenzenemethanaminium chloride ([N11-10-1Ph] [Cl]), 1-methyl-3-octadecyl-1H-imidazolium chloride ([IM1-18][Cl]), 4-(dimethylamino)-1-hexylpyridinium chloride ([Py6-4NMe2][Cl]), and 1,3-didecyl-2-methyl-1H-imidazolium chloride ([IM-10-10-2Me][Cl]).

Aromatic compound, Phenanthrene (PHE), was supplied by Sigma Aldrich at 99% of purity. Millipore-grade water was used for the preparation of all aqueous solutions. Acetonitrile (HPLC grade) for chromatography was obtained from VWR International GmbH (Darmstadt, Germany).

2.2 Methods

Solubility assays. Stock solutions of ILs were prepared at 100 mM final concentrations. Vials of nominal volume of 5 mL containing 1.5 mL of various concentrations of the aqueous solution were saturated by addition of PHE and were maintained under vigorous agitation for 72 h. Teflon-coated caps were used to avoid any contamination of the samples. After the tubes were equilibrated at room temperature, undissolved

PHE particles were removed by centrifugation at 5000 rpm for 10 min. Samples were filtered by 0.45-μm syringe-filter prior their injection into chromatographic device. PHE concentration in aqueous solutions was monitored by HPLC technique. A control sample (water saturated with PHE) was also prepared.

Chromatographic methodology. Specific chromatographic methodology, namely, ion chromatography (IC system) and high-performance liquid chromatography (HPLC) were developed for the quantification of both ILs and aromatic compound *Metrohm 881 Compact IC system* (Metrohm, Herisau, Switzerland) equipped with a 20 μL injection loop and a conductometric detector was used for the ILs determination. For cation separations, a Metrosep C4 ion exchange column (50 × 4.0 mm ID and 5 μm mean particle size) coupled with its corresponding guard column was used (Metrohm, Herisau, Switzerland). The isocratic mobile phase was composed of 30% of an acid organic solution (2 mM of nitric acid) of an organic modifier, acetonitrile. The mobile phase was pumped at a constant flow rate of 0.9 mL min^{-1} for 12 min at 30 ºC. The volume of injection was set at 10 μL. For the quantification of PHE, isocratic reversed phase HPLC was carried out on *Agilent 1100 Series system* equipped with a Multosphere waters PAH C18 column (150 × 4.6 mm i.d., 5 μm). PHE concentration was calculated from a standard curve based on peak area using the absorbance at 252.4 nm (photodiode array UV − vis detector). The injection volume was set at 5 μL. Isocratic eluent, 60:40 acetonitrile/water, was pumped at a flow rate of 1 mL min^{-1} for 12 min. *HPLC 1100 coupled to a Bruker Esquire ESI–MS ion trap detector* was used for the identification of metabolites. The samples were diluted in methanol and deionized water [13, 21].

Primary biodegradation test. The biodegradation test was conducted according to OECD guideline 301 D (closed bottle test). Primary biodegradation of the cation was monitored by ion chromatography for 28 days. Samples were taken at regular time intervals. The inoculum used was derived from the wastewater treatment plant Bremen (Germany). Briefly, 5 g sludge flocs were suspended in 1 L mineral medium and pre-conditioned for 5 days under aerobic conditions. Mineral medium was composed of (mg L^{-1}): 8.5 KH_2PO_4, 21.75 K_2HPO_4, 22.13 $Na_2HPO_4 \bullet 2H_2O$, 1.7 NH_4Cl, 36.4 $CaCl_2 \bullet 2H_2O$, 22.5 $MgSO_4 \bullet 7H_2O$, and 0.25 $FeCl_3$. pH was adjusted at 7.2. Solutions of ILs were prepared at a concentration of 100 μM (corresponding to 17–44 mg L^{-1} depending on each IL) in inoculated test media (100 mL total volume). Blank samples (inoculated media without tested substance) and positive controls (inoculated media with 10 mg L^{-1} benzoic acid) were also prepared. The sampling was done every 5 days. 5 mL of all samples were taken, centrifuged (16,400 rpm, 30 s), and frozen for further determinations [13].

Toxicity test: *Daphnia magna* Acute immobilization test using crustacean *Daphnia magna* was performed following OECD guideline 202 (MicroBioTest). Briefly, tests with neonates less than 24 h old, obtained by hatching of ephippia, were performed at 20 °C in dark. Five pre-fed animals have been incubated with test substances in 10 mL of mineral medium. The number of immobilized or dead organisms has been checked after 24 and 48 h. The relative toxicity of the samples was expressed as percentage of not effected organisms compared to the controls. All experiments were conducted at least twice. Full dose–response curve parameters and

plots were obtained using the drift package (version 0.05–92) for the R language and environment for statistical computing.

3 Results

3.1 Enhancement of Solubility

Solubilization tests of PHE were performed using an aqueous solution of 13 ILs. Experiments were carried out at concentrations of 10 and 100 mM, as these are considered harmless from an environmental perspective. Furthermore, these concentrations are within the usual range used in SER [24]. Control tests in water are also carried out. Results are presented considering the amount of PHE detected in the aqueous solution per concentration of ILs (Fig. 1). To facilitate interpretation, these results are presented in successive figures classified by families.

As the results show, of all the ILs tested, only one, namely, [N1112OH][1SO3], was unable to increase PHE solubility, even at the highest concentration tested. All the other ILs were able to enhance aromatic compound solubility by at least 0.5-fold than water and can, therefore, be considered good candidates for SER applications. Figure 1a shows that the five selected imidazolium-based ILs are effective solubility

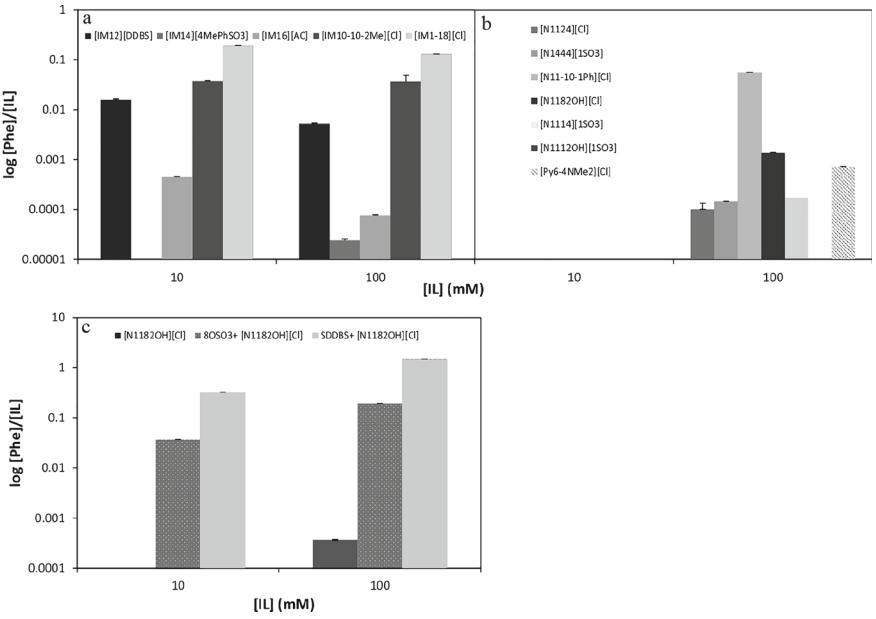

Fig. 1 Ability of different families of ILs to enhance the solubility of phenanthrene: **a** imidazolium-based ILs, **b** ammonium- and pyrrolidinium-based ILs, **c** Mixtures

enhancers at a concentration of 10 mM, except for the one with the butyl alkyl chain, [IM14][4MePhSO$_3$]. Moreover, comparing [IM1-18][Cl] and [IM-10–10-2Me][Cl], it seems that solubilization is better with one long substituent than two shorter ones, even if the overall amount of carbon is higher ([IM1-18]$^+$ *vs.* [IM-10–10-2Me]$^+$). This could be explained in terms of the feasibility of access in relation to the structure. On the other hand, short imidazolium cations, such as [IM12] and [IM16], exhibit significantly different solubility-promoting abilities when combined with anions such as 6-methyl-2,2-dioxo-1,2,3-oxathiazin-4(3H)-onate ([AC]$^-$) or dodecylbenzenesulfonate ([DDBS]$^-$) compared to halide anions, such as [Cl]$^-$. The behavior of imidazolium-based ILs in aqueous solutions has been thoroughly investigated by Jungnickel and colleagues [17]. They concluded that 1-alkyl-3-methylimidazolium chloride ILs can form micelles into an aqueous solution with alkyl chains longer than hexyl. However, the exceptional solubility enhancement observed in this study with the [IM12]$^+$ cation is likely due to the interaction with the dodecylbenzosulfonate anion ([DDBS]$^-$). There are some examples in the literature of this anion being used in combination with sodium as the cation (SDDBS) as an anionic surfactant [25, 26]. However, the use of this anion as a counterion in ILs to enhance PAHs solubility has not been reported before.

Regarding the ability of ammonium-based ILs to promote solubility, it was found that this family cannot enhance PHE solubility at low aqueous concentrations (10 mM), as shown in Fig. 1b. However, four of the five selected ammonium-based ILs successfully increased PHE solubility up to 100 mM aqueous concentrations. By comparing different substituted cations, such as [N1114]$^+$, [N1124]$^+$, and [N1444]$^+$, it can be concluded that an increase in the length of the alkyl chain enhances the enhancer behavior by up to 0.5 times. Furthermore, the cation [N11-10-1Ph]$^+$ exhibits exceptional solubilizing properties, dissolving 556 mg of PHE at a concentration of 100 mM. Additionally, the levels of PHE detected in the aqueous solution are comparable to those detected in the presence of [IM-10–10-2Me]$^+$ and [IM1-18]$^+$ from the imidazolium family of ILs, despite the fact that these are known to be more toxic and difficult to biodegrade than ammonium compounds, which is the main focus of this study. It can be concluded that, since the ILs were selected based on biodegradability and toxicity properties, most of them are unable to enhance the solubility of the hydrophobic pollutant PHE at concentrations higher than 100 mM.

The ability of pyridinium- and phosphonium-based ILs has also been examined in this work. Thus, [P666-14][COO] was selected due to the long alkyl chains present in the cation and the anion's apparent eco-friendly nature. This IL's ability to enhance solubility was satisfactory in a low aqueous solution since it is highly hydrophobic. Conversely, the results obtained for the pyridinium-based IL [Py6-4NMe2][Cl] solution in water allow us to conclude that this IL is suitable for use in the SER process (its solubilization capacity is up to 2.5 times higher than that of water).

Regarding the alkyl-substituted ammonium ILs, namely, [N1182OH][Cl], it is worthy to highlight that the introduction of the octyl alkyl chain enhances its ability to solubilize hydrophobic pollutants compared to the unsubstituted analogue, [N1112OH][Cl]. Moreover, this IL is relevant in terms of environmental concerns,

since it is readily biodegradable and non-toxic to standard bio-indicators such as *Vibrio fisheri* as is detailed in the next section [23, 27].

Despite most of the work on micellar solubilization focused on individual compounds, the use of mixtures of active surfaces ILs could be considered as useful strategy [26]. To date some works have been published based on this strategy using common surfactants [10, 28, 29]. Nevertheless, from the best of our knowledge, no one has reported yet using mixtures of ILs as an efficient tool for solubility enhancement. Based on the preliminary solubility data obtained in this work regarding cation and anion behaviors, alkyl-substituted ammonium-based IL [N1182OH][Cl] has been selected for combining SDDBS and NaC8OSO3. Thus, aqueous solutions of these mixtures were tested at final concentrations from 1 to 20 mM in the first case and from 5 to 100 in the second, respectively, for each component. In both mixtures, outstanding behavior was detected reaching values like the ones obtained for the IM1-18 cation and up to three orders of magnitude higher than the values obtained for functionalized ammonium, by itself (Fig. 1c).

In summary, this work satisfactorily approached an improvement in the solubility behavior of PAHs in an aqueous solution of ILs. Thus, the surfactant-like behavior exhibited by some of these novel solvents could be the solution for implementing a SER strategy to remove PAHs from polluted areas. Furthermore, it was demonstrated that mixtures of active surface ILs could be a useful strategy.

3.2 Biodegradability of Selected ILs

According to the main principle of green chemistry, biodegradation is one of the most important properties for the environmental hazard assessment of a substance [21]. Hence, biodegradability of ILs must be assessed prior to developing any new biotechnological process based on SER strategies. The aerobic biodegradability of the ILs used in this work has been disclosed in this section. It is well known that ammonium-based ILs present good biodegradability behaviors in comparison with pyridinium- and imidazolium-based ones [19, 23, 28]. In this work, the concentration of three imidazolium-based and three ammonium-based ILs was monitored for 28 days following the guidelines given by OECD test 301D (closed bottle test). In this context and to compare the obtained data with the data reported before, primary biodegradability data are summarized in Table 1 and the trend of the concentration along the time is depicted in Fig. 2a. The results obtained show that common cations such $[IM12]^+$, $[IM14]^+$, and $[IM16]^+$ are classified as not readily biodegradable (NRB). These results are supported by data previously reported by Stolte et al. [21]. They concluded that imidazolium cores substituted with short alkyl side chains are highly resistant to biological degradation by the microbial community in activated sludge. These results also agree with the general trend described by Markiewicz and co-workers [28] when they evaluated the toxicity toward activated sludge communities with alkyl chain lengthening.

Table 1 Primary degradation rates for different families of ILs

Cation	Anion	Biodegradation (%)	Classification	Reference
[IM14]	[4MePhSO3]	17	NRB	This work
[IM16]	[AC]	27	NRB	This work
[IM12]	[DDBS]	22	NRB	This work
[IM1-18]	[Cl]	nd	NRB	This work
[IM-10-10-2Me]	[Cl]	nd	NRB	This work
[P666-14]	[9COO]	nd	NRB	21
[Py6-4NMe_2]	[Cl]	100	RB	20
[N1444]	[1SO3]	13	NRB	This work
[N1124]	[Cl]	100	RB	This work
[N1182OH]	[Cl]	100	RB	This work
[N11-10-1Ph]	[Cl]	nd	NRB	This work

Note NRB: Not readily biodegradable; RB: Readily biodegradable, nd: not detected

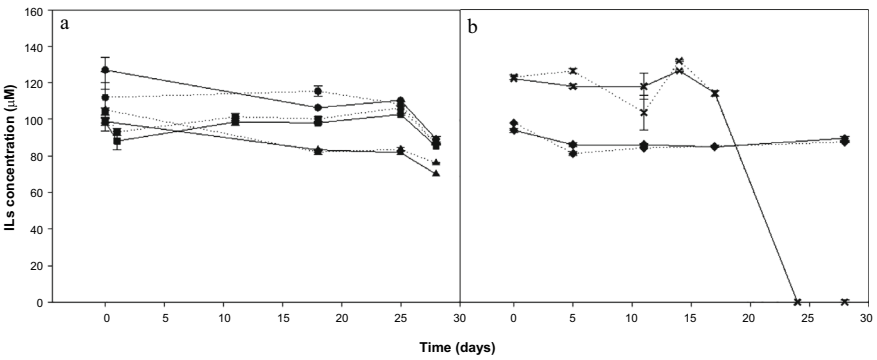

Fig. 2 **a** Degradation kinetics of imidazolium-based ILs: () [IM12][DDBS], () [IM14] [4MePhSO3] and () [IM16][AC] **b** Degradation kinetics of ammonium-based ILs () [N1124][Cl] and () [N1444][1SO3]. Two independent cultures are presented by solid and dotted lines

Although little information exists regarding the biodegradability of phosphonium-based ILs, $[P666\text{-}14]^+$ was reported to promote inoculum inhibition and is therefore classified as not readily biodegradable [21]. In contrast, complete primary degradation of the pyridinium-based IL was detected after 31 days, allowing it to be classified as readily biodegradable IL [20]. Although ammonium-based ILs are more environmentally friendly and readily biodegradable than imidazolium ones, their biodegradation rates vary greatly depending on the structure and active moiety of their substituents. Thus, it was found that the introduction of long alkyl chains makes these ILs not as readily biodegradable as their shorter alkyl chain counterparts. This is the case for cations such as $[N1124]^+$ and $[N1444]^+$. While the former is completely

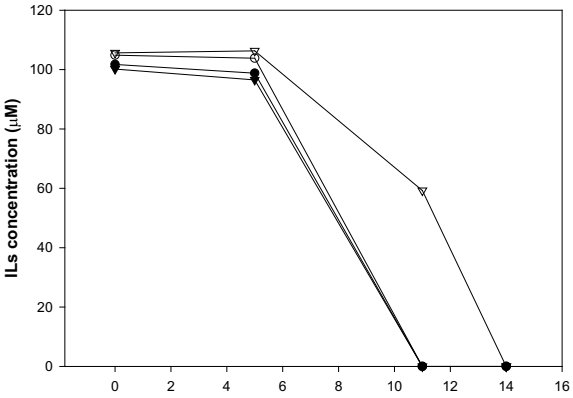

Fig. 3 Degradation kinetics of [N1182OH][Cl]: (,) consortium from wastewaters and (,) consortium + *Pseudomonas stutzeri*. Two independent cultures are presented by full and void symbols

metabolized in 20 days, the latter only reduces its concentration to 15% after 28 days (Fig. 2b).

Furthermore, it is noteworthy that the introduction of an alkyl-octyl chain to the choline cation, $[N1182OH]^+$, turns it as readily biodegradable as the choline cation, $[N1112OH]^+$, while this capacity as solubility enhancers of hydrophobic compounds is also achieved. Figure 3 shows that this cation was completely metabolized in less than 10 days.

Moreover, the trend detected in this work is agreement with the behavior described by Neuman et al. [13]. They concluded that hydroxyl-functionalized alkyl chains (side chain and core) allow complete mineralization to be achieved. Furthermore, they reported that an increase in side-chain elongation is responsible for the classification of some ILs as "not readily biodegradable" (ethyl) to "readily biodegradable" (octyl).

As a result of the experiments, it can be concluded that [N1182OH][Cl] seems to be a good candidate, from a biodegradation point of view, to be used in SER strategies. Therefore, this IL was selected to study its biodegradability. Two different strategies have been considered: one by consortium from wastewater and the one by consortium supplemented with a common bacterium present in polluted areas, such as *Pseudomonas stutzeri* [24]. It was found that the biodegradation rate in the presence of both microorganisms did not improve the biodegradation rate at all. In this case, synergism is not a good strategy for biodegradation of this ammonium-based IL, as the removal rate, in the presence and in the absence of the additional microorganisms, is completely similar. The rate and trend of biodegradation in both cases (presence and absence of *P. stutzeri*) are depicted in Fig. 3. In addition, a metabolic pathway was also reported. Initial intermediates with m/z^+ relation of 218.3 (which belong to hydroxyl group), 216.3 (aldehydes), and 234 (carboxylic acid) were detected within a short time, namely, 11 days. These types of transformation products have been already identified [21]. Figure 4 shows the intensity of the signals within the mass-to-charge ratios recorded in positive mode and degradation pathway. Based on the identified metabolites, a degradation pathway for alkyl choline is proposed, being,

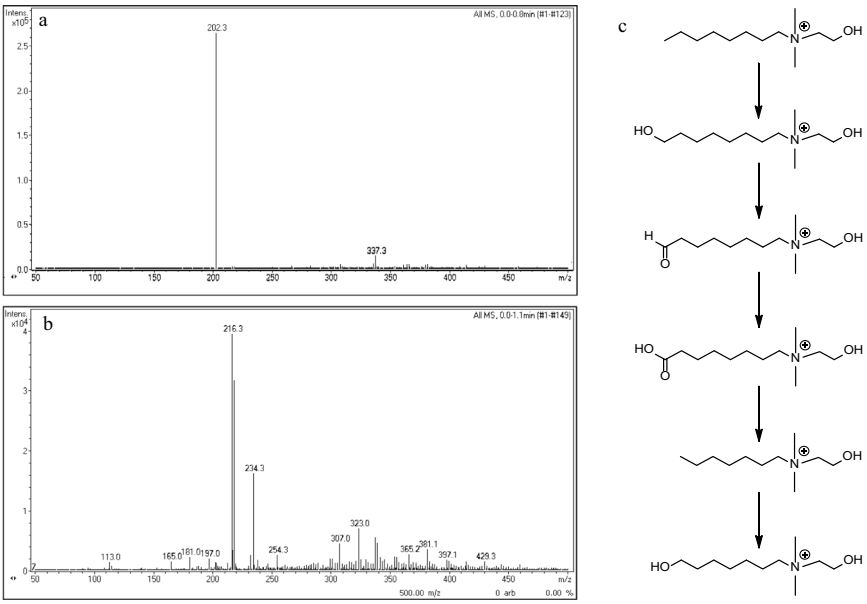

Fig. 4 A **a** Mass spectra of initial biotic [N1182OH][Cl] sample; **b** Mass spectra after 11 days of biotic [N1182OH][Cl] sample; and **c** [N1182OH][Cl] biodegradation pathway

to our knowledge, the first time that a degradation pathway for alkylcholine has been presented.

3.3 Ecotoxicity of Selected ILs

Although ILs are well known because of their almost null volatility, which makes them attractive for industrial processes, their high solubility in water must be considered before large-scale industrial applications. In this work, an ecotoxicity evaluation of selected ILs has been carried out by *Daphnia magna* in vivo test. The use of this invertebrate in bioassays test has been reported in the literature in a plethora of papers focused on the environmental and ecotoxicological impact of ILs [28–30]. Therefore, this freshwater crustacean is considered of extreme relevance as this invertebrate links the higher microbial and trophic levels [30]. Hence, from the determined values of EC50 (mg L^{-1}) it has been observed that there is a well-established relationship between toxicity and alkyl chain length of ILs. Therefore, the most toxic ILs toward *D. magna* were the imidazolium-based ones with long alkyl chains as substituent, [IM-18][Cl] and [IM-10–10-2Me][Cl], and the ammonium-based one that also presents an aromatic ring, [N11-10-1Ph][Cl]. Consequently, these three ILs have been considered toxic to aquatic life according to CE 1772/2008 regulation, as the EC50 values are below 1 mg L^{-1}. In contrast, the alkyl-substituted

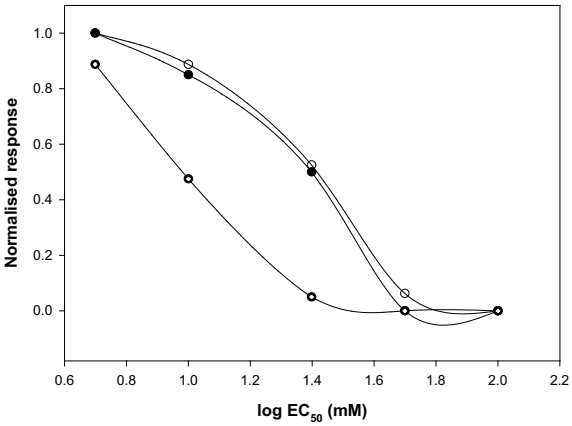

Fig. 5 Dose–response curves of ()[N1182OH][Cl], ()[N1182OH][Cl] + NaC8OSO3, and ()[N1182OH][Cl] + SDDBS

ammonium-based IL, [N1182OH][Cl], has been classified as harmless to aquatic organisms (EC50 value in the range 10–100 mg L^{-1}). Some authors, such as Santos et al. and Stolte et al., had already reported the safety of cholinium-based ILs when they evaluated environmental toxicity of cholinium-based salts and the ecotoxicity and biodegradability of ammonium ILs, respectively [31, 23]. The biological effect of pyridinium-based IL [Py6-4NMe2][Cl] to *D. magna* was found lower than for the ammonium-based one (EC50 80 mg L^{-1}, value reported by UFT-Merck database) and consequently it can be classified as harmless for aquatic living microorganisms. In contrast for the phosphonium-based [P666-14][COO] any data was reported yet although it is supposed that it will be classified as toxic to aquatic life since this IL results toxic for the inoculum in the primary biodegradation tests [30]. Interestingly, it was noted that the toxicity evaluation of mixtures containing ammonium-based IL ([N1182OH][Cl] + SDDBS or [N1182OH][Cl] + S8OSO3) can be classified as harmless for aquatic microorganisms (EC50 of 15 and 25 mgL^{-1}, respectively) (Fig. 5).

The experimental results obtained can be considered extremely relevance since opening completely new windows for further investigations related to the enhancement of solubility of hydrophobic compounds in water by means of SER strategies based on mixed solutions of ILs. Accordingly, further investigations should address questions like the study of more binary systems and the degradation of the mixtures and toxicity of the formed intermediates. Furthermore, it can be concluded that enhanced biodegradability can be designed into ILs by tuning the functional groups present in the cation core.

4 Conclusion

In this work, the improvement of the solubility behavior of PAHs by aqueous dissolution of ILs was successfully addressed. Based on the results obtained, it is now possible to use ILs to improve the solubility of hydrophobic pollutants. Thus, the surfactant-like behavior of some of these neoteric solvents will be the solution for the implementation of an SER strategy. To shed light on the implementation of a real remediation strategy, ILs possessing surfactant properties have been selected. Thus, imidazolium-based ILs have revealed higher efficiency than ammonium-based ones. However, by comparison between different substituted cations such as $[N1114]^+$, $[N1124]^+$, and $[N1444]^+$ it can be concluded that increasing the alkyl chain increases the potential behavior. Furthermore, it was shown that the use of mixtures of active surfaces ILs could be considered as a useful strategy.

The heuristic rule defining biodegradability and ecotoxicity of ILs was successfully addressed. The results obtained show that common cations like $[IM12]^+$, $[IM14]^+$, and $[IM16]^+$ are classified as not readily biodegradable, whereas most of the ammonium-based ones can be classified as readily biodegradable depending on the structure and substituents groups. Furthermore, it was concluded that synergism is not a good strategy for the biodegradation of $[N1182OH]^+$ ammonium-based IL. In terms of ecotoxicity, the most toxic ILs were the imidazolium-based ones with long alkyl chains as substituent, [IM-18][Cl] and [IM-10–10-2Me][Cl], and the ammonium-based one with the presence of an aromatic ring, [N11-10-1Ph][Cl].

In summary, it can be concluded that enhanced biodegradability can be designed into ILs by tuning the functional groups present in the cation core.

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Influence of Bacterial Encapsulation on Urease Activity During Microbially Induced Calcite Precipitation

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Abstract Microbially induced calcite precipitation (MICP) is a sustainable biomineralization process that utilizes naturally abundant bacteria for soil stabilization. However, the bacterial encapsulation in MICP significantly affects urease activity, urea hydrolysis, and reaction fluid's pH. While past studies have acknowledged bacterial encapsulation, its direct impact on enzymatic activity, crystal growth, and mineralogy remains insufficiently quantified. This study aims to assess the effect of bacterial encapsulation on urease activity, pH, urea hydrolysis, and MICP efficiency. Flask-scale experiments and reactive kinetic modeling studies were conducted under two scenarios: bacterial culture with urea alone to prevent encapsulation and bacterial culture with urea and cementation reagent (calcium chloride, $CaCl_2$) to induce encapsulation. Both experimental and reactive modeling studies corroborated and indicated that urea hydrolysis occurs rapidly without time lag in the absence of cementation reagent. In contrast, adding cementation reagent leads to the bacterial surface being covered by $CaCO_3$, which restricts metabolic activity and decelerates urea hydrolysis and urease activity. However, the effect of encapsulation diminishes with increased bacterial concentrations. The encapsulation significantly lowers the pH due to the minimal availability of free carbonate ions. Overall, the study successfully captures the biomass encapsulation and its impact on urease activity during microbially induced calcite precipitation.

Keyword Microbially induced calcite precipitation (MICP) · Bacterial encapsulation · Reaction kinetic modeling

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1 Introduction

Microbially induced calcite precipitation (MICP) is a biochemical process that employs urease-producing bacteria to aid urea hydrolysis and converts urea into carbonate ions, increasing the pH of the reaction fluid and, in the presence of calcium source, calcium carbonate ($CaCO_3$) is precipitated. This process has gained significant interest due to its potential in soil stabilization, concrete self-healing, aqua pond sealing, leak mitigation from a CO_2-sequestrated aquifer, and liquefaction resistance [1–3]. The ultimate strength of biocemented soils depends on various factors, such as the amount, type, and relative size of $CaCO_3$ crystals. The $CaCO_3$ crystal characteristics are, in turn, a function of biochemical parameters such as biomass, cementation reagent concentration, temperature, and pH [4]. Among these factors, the biomass concentration is the most significant parameter and affects the MICP process in two ways. On the one hand, the bacteria release the urease enzyme necessary for urea hydrolysis; on the other hand, it acts as the nucleation site for crystal precipitation [5, 6]. As the urea hydrolysis progresses, the $CaCO_3$ matrix grows further, leading to bacterial encapsulation within the precipitated crystal matrix. The precipitation starts around bacterial cell walls and eventually surrounds them with crystal deposits, leading to entrapment (Fig. 1). The biochemical factors, such as biomass, cementation reagent concentration, and environmental conditions, affect the biomass encapsulation [4]. Multiple mechanisms arise during the biomass encapsulation. In the early stages of biomass encapsulation, bacteria can still be metabolically active and produce urease enzymes. However, with the intensification of $CaCO_3$ precipitation, complete encapsulation occurs with full bacterial embedment within the mineral structure, ultimately restricting urease production and exchange of nutrients. This bacterial inactivation, post-encapsulation, leads to bacterial cell death with the drop-in urease activity of the system [4].

Past research has emphasized the impact of bacterial encapsulation on ureolysis rates and the overall efficiency of MICP. For instance, Cuthbert et al. [7] showed

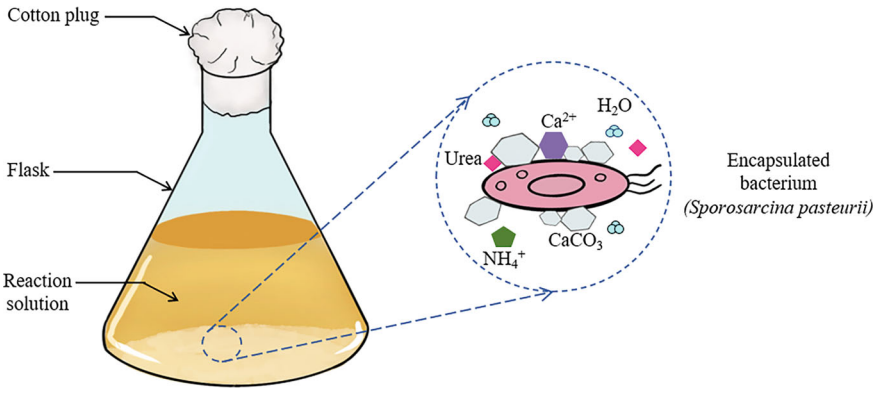

Fig. 1 Cementation flask featuring bacterial encapsulation

the influence of encapsulation by proposing a micro-scale semi-empirical relation to account for encapsulation through experimentally derived maximum calcite crystal size. In contrast, Minto et al. [8] performed batch-scale experiments to quantify the encapsulation rate for the reactive transport modeling. These limited previous studies primarily focused on qualitative observations of encapsulation and its indirect impact on microbial activity through the amount of calcium carbonate precipitated. Therefore, there remains a lack of direct experimental validation quantifying how encapsulation affects urease activity and urea hydrolysis rates over time. Despite various modeling attempts to simulate coupled processes in MICP, a predictive framework specifically for urea hydrolysis trends under encapsulated versus non-encapsulated conditions has been minimal [4]. Since bacterial encapsulation by $CaCO_3$ can impair their metabolic functions and limit urease activity, ultimately affecting the efficiency of the precipitation process. Therefore, understanding the extent to which encapsulation impacts urease activity is crucial for optimizing MICP-based applications.

Given these research gaps, the current study aims to investigate the effect of bacterial encapsulation on urease activity, urea hydrolysis rates, and pH dynamics. For this purpose, two scenarios were considered: (1) bacterial culture with urea alone, where $CaCO_3$ formation does not occur and (2) bacterial culture with urea and $CaCl_2$, where encapsulation is triggered. Additionally, a reactive kinetic model was developed to describe urea hydrolysis trends with a clear distinction between the rapid initial reaction phase and slower far-along phases, which has not been explored in previous studies. By bridging this knowledge gap, the present study provides new insights into optimizing MICP efficiency and offers a more precise understanding of how encapsulation alters microbial functionality in biomineralization processes.

2 Materials and Methods

2.1 Bacterial Culture

The present study utilized *Sporosarcina pasteurii* of ATCC 11859 culture as a urease-producing bacterium. The chosen bacterial culture is the most commonly adopted with high urease activity [6]. The bacteria were grown in a growth media comprising 20 g/l yeast extract and 10 g/l ammonium sulfate in 0.13 M tris base solution at a pH of 9. Before the bacterial inoculation, the growth media was autoclaved at 121 °C and 15 psi pressure to ensure no contamination. All the experiments were performed in a sterilized laminar airflow chamber. Turbidity in the media confirmed the growth of the bacteria. A UV–visible spectroscope was used for quantitative analysis of bacterial growth, and the optical density was measured at 600 nm wavelength (OD_{600}).

Table 1 Experimental design for flask-scale MICP studies

Sl. No.	Case	Description	Biomass concentration, B_0 (OD_{600})	Cementation solution concentration, C_0	
				Urea (M)	$CaCl_2$ (M)
1	Case-1	Inhibiting bacterial encapsulation	0.25	0.50	0.00
2			0.50		
3			1.00		
4	Case-2	Promoting bacterial encapsulation	0.25	0.50	0.50
5			0.50		
6			1.00		

2.2 Cementation Reagent Preparation

The bacterial culture from the growth media was taken and centrifuged to separate the pellet from the liquid, and the bacterial pellet was accordingly mixed in cementation media at various concentrations. Cementation media consisted of urea and $CaCl_2$ at various concentrations (Table 1), with a minimal amount of yeast extract (2 g/l). Two types of experiments were performed by employing urea with $CaCl_2$ (Case-1) and without $CaCl_2$ (Case-2) in cementation media to restrict and allow bacterial encapsulation, respectively (Table 1). Three different bacterial concentrations of B_0 (0.25, 0.50, 1.00 OD_{600}) were chosen to assess the encapsulation effect. However, the cementation solution concentration (C_0) was maintained constant with 0.5 M of urea and varying $CaCl_2$ concentration based on the considered scenario (Case-1 and Case-2). The bacterial concentration ranging from 0.25 OD_{600} to 1 OD_{600} was selected to ensure optimal balance, so sufficient urease activity is present to capture bacterial encapsulation effectively. For instance, a lower bacterial concentration (< 0.25 OD_{600}) can be too low for the precipitation to commence. However, at higher bacterial concentrations (> 1 OD_{600}), having bacterium in large numbers, the effect of biomass encapsulation on urea hydrolysis and urease activity is negligible [9]. Similarly, the 0.5 M cementation reagent concentration was chosen based on prior studies demonstrating its effectiveness in controlled $CaCO_3$ precipitation [1, 9]. This concentration allows sufficient carbonate availability for crystal growth while preventing urease enzymatic activity inhibition observed at higher cementation reagent concentrations.

2.3 Quantitative Analysis of Urea Hydrolyzed and Urease Activity

The cementation media was sampled at regular intervals for the quantitative analysis of urea hydrolyzed. The media was centrifuged, and the supernatant was used for

pH measurement and available urea quantification. The amount of urea consumed was analyzed by quantifying the ammonium ions in the solution using the phenol hypochlorite method [10]. The collected supernatant was diluted 5000 times, and 500 µl of phenol nitroprusside and 500 µl alkaline hypochlorite reagent were mixed within it and stranded for incubation. After half an hour, the optical density was measured at a wavelength of 626 nm (OD_{626}). A known concentration of ammonium ions samples ranging from 10 µM to 100 µM using ammonium chloride was prepared similarly to establish the correlation between the optical density and ammonia present in the sample. From the chemical stoichiometry, the amount of urea consumed was evaluated as 1 M of urea converted to 2 M of ammonia. Similarly, 1 unit of urease activity was calculated as the enzyme required to catalyze the hydrolysis of 1 µM urea per minute [11].

2.4 Reaction Kinetic Modeling

A reaction kinetic model was developed following the urea hydrolysis rate as per Michaelis–Menten (MM) kinetics to understand the urea hydrolysis dynamics over time [4]. For the first time, a dual-parameterized MM equation was developed to capture faster hydrolysis of urea in the initial lag time (t_{lag}) and creeping hydrolysis rates in the later stages. This modification is imperative as a quicker rate of hydrolysis in the early stages has several implications on precipitated biocement crystal mineralogy if not on the overall quantity of the precipitated mineral. The following is the governing mass balance equation for simulating the enzymatic urea hydrolysis reaction within the flask using MM reaction kinetics (Eq. 1).

$$\frac{\partial C_u}{\partial t} = u_{sp} \cdot B_0 \cdot \left(\frac{C_u}{K_m + C_u}\right) \cdot \left(1 - \frac{A_c}{A_T}\right) \tag{1}$$

$$u_{sp} = \begin{cases} u_{sp1} & \text{if } t < t_{lag} \\ u_{sp2} & \text{if } t > t_{lag} \end{cases} \quad \text{and} \quad K_m = \begin{cases} K_{m1} & \text{if } t < t_{lag} \\ K_{m2} & \text{if } t > t_{lag} \end{cases} \tag{2}$$

where C_u is the concentration of urea (kmol·m^{-3}), u_{sp} is the maximum urease rate constant (kmol·m^{-3}·s^{-1}·OD_{600}^{-1}), B_0 is the initial concentration of bacteria in the flask (OD_{600}), K_m is the half-saturation constant (kmol·m^{-3}), A_c is the area of precipitated calcite (m^2), and A_T is the total area available for precipitation (m^2). The evaluation of A_c is based on the biocement precipitation amount and A_T is based on the total surface area of bacterial cells available for precipitation in the flask fluid. The term ($1–A_c/A_T$) is the fractionalized area, which fractionates the urea hydrolysis rate to account for the lowered reaction rate as the encapsulation initiates. The two major MM parameters (u_{sp} and K_m) were determined using urease activity tests at different urea concentrations [4]. From experimental urease activity tests, two distinct slopes from urea hydrolysis curves over time for each urea concentration were determined to capture initial faster hydrolysis rates and slower well-along hydrolysis rates. Based on

the MM data points determined from experiments, the calibration of MM parameters for each faster and creeping hydrolysis rate was performed and two distinct maximum urease rate constants ($u_{sp1} = 3 \times 10^{-3}$ kmol·m^{-3}·s^{-1}·OD$_{600}$$^{-1}$ and $u_{sp2} = 6 \times 10^{-5}$ kmol·m^{-3}·s^{-1}·OD$_{600}$$^{-1}$) and half-saturation constants ($K_{m1} = 2.46$ kmol·m^{-3} and $K_{m2} = 0.4$ kmol·m^{-3}) were obtained (Eq. 2), respectively. The reaction kinetic simulations were performed for biochemical concentrations reported in Table 1, utilizing the above-proposed framework to understand the bacterial encapsulation effect on urea hydrolysis.

3 Results and Discussion

The presence of bacteria and its concentration in the MICP process is of prime importance as it dictates the rate of MICP reaction and the size of the final precipitated crystals and morphology. Bacteria primarily have two roles in MICP. On the one hand, the bacteria catalyze the ureolysis reaction, converting urea into carbonate ions. On the other hand, bacteria act as a nucleation site for crystal precipitation owing to their negatively charged cell wall [6, 12]. Consequently, the calcium cations (Ca^{2+}) attach to the bacterial cell, and the bacteria becomes the nucleus for the crystal precipitation, which finally leads to bacterial encapsulation within calcium carbonate minerals and with complete encapsulation, bacterial death or loss of activity occurs. Therefore, in the present study, various flask-scale experiments were conducted to evaluate the effect of bacterial encapsulation on the MICP process. For the following purpose, the experiments were performed without and with $CaCl_2$ addition (Case-1 and Case-2) to capture the bacterial encapsulation.

The SEM analysis was categorically performed for Case-2 with the crystals precipitated at the bottom of flasks in Case-2 (Fig. 2). It is clearly visible that bacteria (length, $L \sim 2.41$ μm and diameter, $\Phi \sim 0.67$ μm) were getting encapsulated within calcium carbonate crystals (Fig. 2a, b). Residual bacterial imprints are also visible on the surface of some precipitated crystals (Fig. 2b), indicating strong evidence of bacterial encapsulation due to $CaCO_3$ crystals. In fact, the surface area of bacterial cells in reactive kinetic modeling was evaluated based on the dimensions of bacterial cells reported (Fig. 2b).

Analyzing the urea degradation over time demonstrates that bacterial encapsulation induces a lag time (t_{lag}) for the reaction and decreases the rate of ureolysis (Fig. 3). This time lag effect is observed to be highest in the case of bacterial concentration of 0.25 OD$_{600}$ and reduces as the bacterial concentration increases. This is because as the bacterial concentration increases (from 0.25 OD$_{600}$ to 1.00 OD$_{600}$), there are numerous bacteria to compensate for the decrement in the rate of reaction caused by bacterial encapsulation (significant to moderate and from moderate to negligible), as shown in Fig. 3. Lag time is of prime importance in MICP as it influences the ultimate crystal size. A higher lag time to the MICP reaction decreases the reaction rate, thereby decreasing the saturation state of the solution. A reduction in saturation is beneficial for crystal growth as, at a critical supersaturation state,

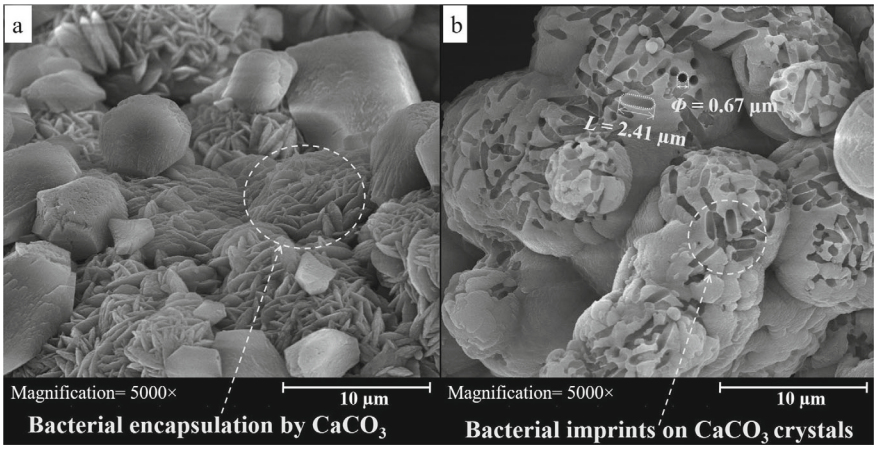

Fig. 2 **a** Bacterial encapsulation by $CaCO_3$ crystals, **b** Bacterial imprint over some of the $CaCO_3$ crystals

new crystal nucleation facilitates the existing crystal growth [1, 8]. Encapsulation also reduces the number of bacteria in the system, resulting in lower nucleation sites, which is desirable for big-sized crystal precipitation [8]. The above-discussed observation of urea hydrolysis was corroborated using the reactive kinetic modeling simulation results (Fig. 3), with a reasonable agreement by attaining minimal root mean square error (RMSE) between 0.047 and 0.022 M of urea. The time lag concept for reaction with dual-parameterized MM model parameters facilitated the prediction of realistic urea hydrolysis phenomena during MICP considering without and with encapsulation scenarios (Case-1 and Case-2).

The effect of encapsulation is also visible in terms of pH changes as the reaction proceeds. As seen in Fig. 4, the trend of pH is entirely different for both cases (Case-1 and Case-2). With encapsulation, the maximum pH of the solution is limited to 8.5 (Case-2). However, in the case of no bacterial encapsulation, the maximum pH of the solution was around 9.5 (Case-1). This is because of the reduction in the urease activity due to bacterial encapsulation (Fig. 5). For instance, in Case-1, the absence of $CaCl_2$ and the presence of free carbonate ions (CO_3^{2-}) can contribute to an increase in pH due to their buffering effect. During urease-driven hydrolysis, urea converts to ammonia (NH_3) and carbonic acid (H_2CO_3), which further dissociates into bicarbonate and hydrogen ions, converting into carbonate ions and water. Without a calcium source, where no encapsulation is allowed (Case-1), the free carbonate ions consume the hydrogen ions in the system, making the solution more basic. However, in the presence of a calcium source (Case-2) contributing encapsulation, the carbonate ions combine with calcium and get captured in calcium carbonate, leaving residual hydrogen ions in the solution, making it slightly acidic compared to Case-1 (Fig. 4).

While the precipitation occurred (Case-2), a significant reduction in urease activity was observed due to bacterial encapsulation (Fig. 5) compared to Case-1. For the

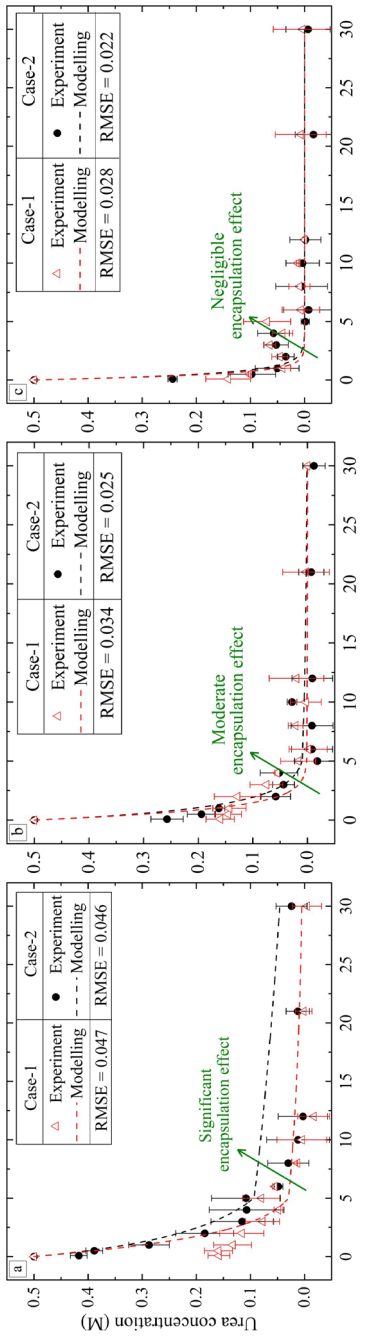

Fig. 3 Urea hydrolysis with encapsulation (with $CaCl_2$) and without encapsulation (without $CaCl_2$) for different bacterial concentrations for a time period of 30 h having biomass concentration as **a** 0.25 OD_{600}, **b** 0.5 OD_{600}, **c** 1 OD_{600}

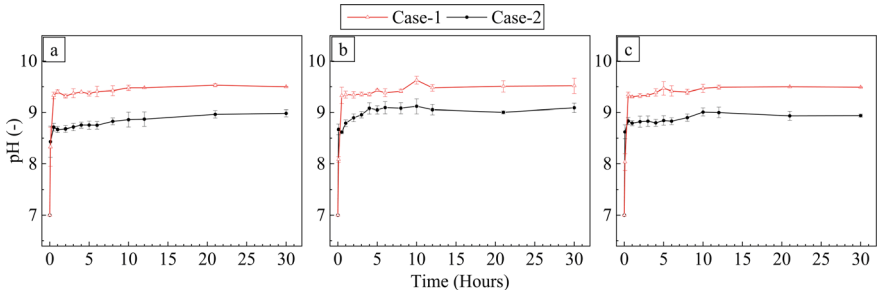

Fig. 4 pH variation of MICP without encapsulation (Case-1) and with encapsulation (Case-2) for different bacterial concentrations for a period of 30 h having biomass concentration as **a** 0.25 OD_{600}, **b** 0.5 OD_{600}, **c** 1 OD_{600}

bacterial concentration of 0.25 OD_{600}, the urease activity reduces from 68 μM/min (Case-1) to 16 μM/min with encapsulation (Case-2). However, the reduction is lower with increased bacterial concentration, from 0.50 OD_{600} to 1.00 OD_{600}. This reduction in urease activity is generally advantageous for the slow growth of crystals. It has been observed in the literature that the size of the crystal decreases as the biomass concentration increases [1, 8]. The following observation can be explained from Fig. 5 as urease activity is very high in case of higher bacterial concentrations, as the effect of encapsulation diminishes. Changes in urease activity also impact the mineralogy of precipitated crystals. When urease activity decreases, it precipitates the most stable mineral form of calcium carbonate, calcite. However, at elevated urease activity, vaterite is a more prominent mineral [13]. Encapsulation decreases the urease activity in the system and leads to more stable calcite precipitation in MICP.

Overall, bacterial encapsulation plays a key role in impacting MICP efficiency by reducing urease activity and slowing urea hydrolysis, thereby influencing biocement characteristics like amount, size, morphology, and mineralogy. While encapsulation limits prolonged urease activity, it promotes a more stable mineralogy of calcium carbonate, i.e., calcite over less stable vaterite, enhancing soil strength. Unlike previous studies that primarily attributed the dependency of the size of precipitated crystal to the availability of bacteria as nucleation sites [1, 8], this study highlights that in MICP, both nucleation sites and reduced urease activity due to encapsulation play a crucial role in shaping precipitated crystal's size alongside mineralogy. Reduced urease activity leads to slow ureolysis and results in lower saturation levels, favoring the growth of fewer, larger crystals over numerous smaller ones. The encapsulation of bacteria due to calcite precipitation influences the biochemical reaction rate and can significantly influence the amount of calcite precipitation in field-scale problems. Thus, these findings highlight the need to balance reaction rates, bacterial viability, and precipitation patterns due to bacterial encapsulation for optimizing MICP in applications such as soil stabilization, dust suppression, erosion control, and contaminant immobilization [1, 3, 11, 14].

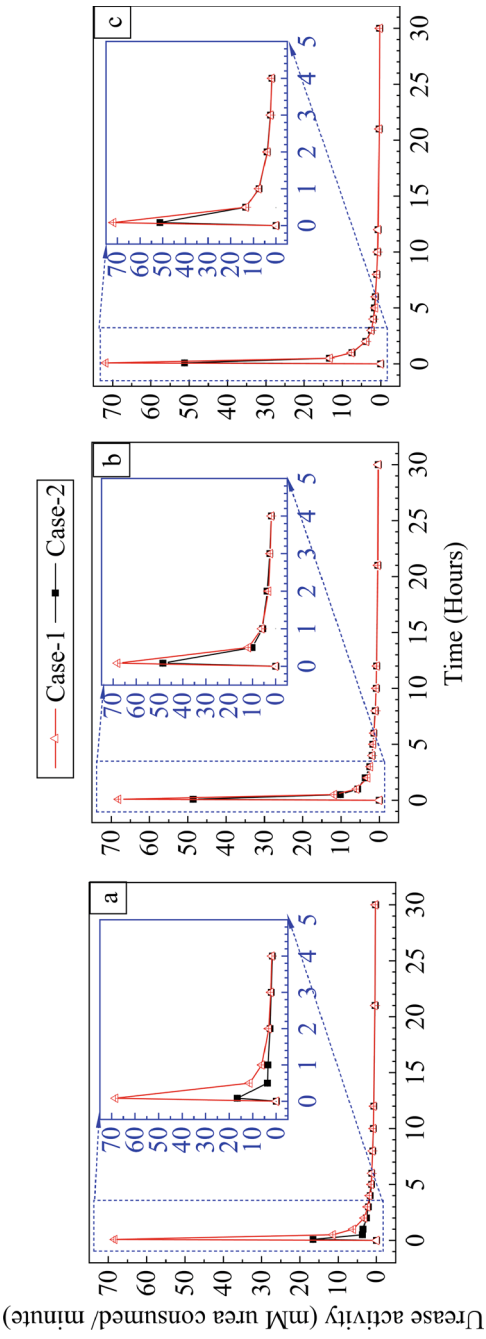

Fig. 5 Urease activity variation of MICP without encapsulation (Case-1) and with encapsulation (Case-2) for different bacterial concentrations for a period of 30 h as **a** 0.25 OD_{600}, **b** 0.5 OD_{600}, **c** 1 OD_{600}

4 Summary and Conclusions

The present study quantified the effect of bacterial encapsulation on urea hydrolysis, pH, and urease activity during the soil biocementation treatment. Various flask-scale experiments and reactive kinetic modeling study were conducted with and without $CaCl_2$ to induce and restrict bacterial encapsulation by $CaCO_3$, respectively. For the first time, a dual-parameterized Michaelis–Menten equation was proposed to capture faster hydrolysis of urea in the initial lag time and creeping hydrolysis rates in the later stages for reaction kinetic modeling studies. Experimental observations indicated that, in the case of no encapsulation, the bacteria remained uncovered and rapidly hydrolyzed the urea without lag time, ensuring high metabolic activity. However, introducing $CaCl_2$ resulted in $CaCO_3$ precipitation, leading to bacterial encapsulation and limiting enzymatic activity. Both experimental and modeling studies comprehensively suggested encapsulation is a limiting factor in prolonged MICP activity and should be considered when designing applications requiring sustained urease function. The controlled reduction in urease activity due to bacterial encapsulation is advantageous for crystal growth, leading to large-sized crystals that are thereby advantageous for soil strengthening. Decreased urease activity due to bacterial encapsulation is also beneficial for crystal mineralogy, leading to more stable calcite mineralogy precipitation over vaterite. Experimental results demonstrate that encapsulation decreases the pH of the reaction fluid. This is attributed to capturing free carbonate ions in $CaCO_3$ crystals, which otherwise were combined with free hydrogen ions, making the solution alkaline in case of no encapsulation. Overall, this study highlights the role of bacterial encapsulation in urea hydrolysis rate and shaping precipitated crystals characteristics in MICP for its effective use in dust suppression, erosion control, and other surface-strengthening applications.

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Solid Waste and Circular Economy

Sustainable Remediation of Sediment Using Furnace Slag

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Abstract Ports and river waterways dredging generates large volumes of sediment, and its dumping creates an environmental risk to the marine environment due to its potential contamination. An alternative to intensive dredging is to deposit the material on land for treatment before possible recovery, but regulations governing dredged sediments remain an obstacle to the development of recovery pathways. Similarly, the iron industry generates a huge amount of by-products (furnace slag), which are deposited in the production area. Furnace slag is generated from the smelting or refining of metals in a furnace. The slag consists primarily of silica, alumina, lime, and iron oxides, which are by-products of the reduction process. Furnace slag has significant potential as a resource in various sectors. Proper recycling and reuse can not only reduce the environmental impact of mining raw materials but also contribute to sustainable development practices. The concept of combining sediment with blast furnace slag to enhance electrokinetic remediation comes from the idea that the properties of slag can improve the electrokinetic process and increase its effectiveness. In addition, mixing two different waste materials to make a remediation process more efficient contributes to reducing by-products deposits and the beneficial use of dredged sediments. This article details the use of a sediment-slag mixture to enhance the electrokinetic remediation of a river sediment. Because of the low electrical properties of the river sediment, electrokinetic remediation was not efficient in comparison with marine sediment. A light proportion (5%) of mixed slags with sediment was tested using friendly environmental electrolytes. The results showed a large removal of metals (up to 70% and 90% for Zn and Cd, respectively) from the mixture, and a performance improved 4 to 5 times, advocating the advantage provided by furnace slag in electrokinetic remediation of low conductive sediments. The results support the concept of combining two waste materials to degrade pollutants in soil materials.

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Keywords River sediment · Furnace slag · Trace metals · Electrokinetic remediation · Sustainable reuse

1 Introduction

In France, dredging operations are essential for maintaining navigability, preventing flooding, and supporting economic activities. Each year, approximately 50 million cubic meters of marine sediments are dredged to sustain harbor activities and the economy of coastal regions. Regarding river sediments, between 2015 and 2017, around 413,000 m^3 were dredged from rivers, canals, and tributaries in the Paris Region and its surroundings. Specific regions, such as the Deûle Canal in northern France, have been identified as heavily polluted, containing large volumes of contaminated sediments [1]. These sediments require careful characterization and management before disposal or reuse. For example, the Compagnie Nationale du Rhône (CNR) conducts regular maintenance dredging on the Rhône River, removing approximately 536,000 m^3 of sediment annually. This represents about 1 to 3% of the river's annual sediment transport [2]. River sediment remediation presents several challenges due to the complexities involved in dealing with contaminated, eroded, or excessive sediment [3]. These challenges span technical, environmental, economic, and regulatory aspects. Sediments need to be remediated before any reuse. Sediments in rivers can accumulate a variety of harmful substances, including heavy metals, persistent organic pollutants, and petroleum hydrocarbons. These contaminants can pose significant risks to aquatic life, water quality, and human health. Once contaminated sediments are dredged, finding an appropriate disposal method is a significant challenge. They may require treatment or need to be placed in special landfills designed to contain hazardous materials, which can be costly and environmentally challenging. With rising concerns about pollution and land contamination, there are fewer suitable sites for the disposal of dredged sediments. Securing approval for disposal can be difficult, and the cost of transporting sediment to distant disposal sites can be high. Finding sustainable and beneficial reuse options for dredged sediments is an ongoing challenge. In some cases, sediments can be used for land reclamation or as construction material, but the contaminated nature of some river sediments makes this unfeasible or unsafe. There is a push for "green" or ecological remediation techniques that help restore sediments to their natural state, but these are still under research and development in many cases.

On the other hand, slags, which are primarily composed of silicates, lime, alumina, and iron oxide, are by-products generally produced in metals refining and smelting processes [4]. The physico-chemical properties of these materials vary greatly depending upon their source, processing conditions, and end products. Steel slag, which is generally produced in the steelmaking process, contains a rich quantity of iron minerals like magnetite (Fe_3O_4) and wustite (FeO). Additionally, it contains magnesium oxide (MgO) and free lime (CaO). These varying compositions have a significant impact on reactivity, stability, and their potential future environmental

application [5]. Slags can interact with water and other substances to produce cement-like compounds due to their pozzolanic characteristics. Additionally, the alkaline nature due to high levels of CaO and MgO leads to high pH, which is useful for neutralizing acidic environments. Furthermore, due to their extensive surface area and silicate and oxide fractions, slags are well suitable choice for the adsorption of heavy metals. Leaching of different elements like lead, chromium, and vanadium under specified environmental conditions is one of the limitations in the successful application of slags in waste reuse [6]. Appropriate weathering and stabilization of slags by mixing them with fine materials can result in the development of secondary mineral forms and stabilized products, thus reducing their leachability and environmental concerns [7]. The distinct properties to immobilize pollutants in various environmental samples and stabilization, make slags a viable choice for electrokinetic remediation. Recent remediation studies focus on the use of electrokinetics, using different additives like activated carbon to increase the surface area and conductivity of the environmental samples [8]. Therefore, more state-of-the-art studies are needed with the aim of investigating the feasibility of slags in remediation and reuse of contaminated dredged sediments through electrokinetic, thus promoting the reuse and circular economy concept for sustainable sediment management.

2 Materials Characterization and Testing Procedure

2.1 Sediment

The river sediment was collected from channels in northern France and provided by the company *Baudelet Environment*. The collected sediment was reserved in sealed buckets to perform the experiments. The initial sediment had 17% water content, a pH of 7.8, and low electrical conductivity (1.2 mS/cm) compared to estuarine sediments (13.2 and 16.2 mS/cm). The measured physical–chemical parameters (pH and conductivity) were measured using similar probes used for previous sediment samples (Table 1). Similarly, the same technique was applied to measure the physical–chemical parameters of the pore water in the sediments as described by Du Laing et al. (2009). The river sediment is classified as Silty Soil according to USCS. The large sand content enables a significant permeability and easy material implementation.

2.2 Slags

Slag particles were shipped from an industrial company for analysis and pollutant remediation. The slags used in this study were obtained from the iron industry. These by-products result from the smelting process in blast furnaces during the extraction of iron from ore. The mineral composition was determined using XRF analysis,

Table 1 Initial sediment properties (average ± standard deviation) of the sediment

Parameter	Value		Standard deviation
pH	7.8	±	0.1
Electrical conductivity (mS/cm)	1.2	±	0.3
D50 (μm)	45.1	±	0.5
Clay (%)	2.7	±	0.7
Silt (%)	55.7	±	1.4
Sand (%)	41.6	±	0.4
Water content (%)	18.4	±	1.7
Loss in ignition 550 or OM (%)	19.8	±	0.6
Carbonates (%)	4.2	±	0.2

as appears in Table 2. The pH, electric conductivity, and oxidoreduction potential were also measured for the pore fluid leached from slags after being centrifuged for 20min (4640 g). For the physical–chemical parameters measurements of slags, Ultra-pure water was added to each slag sample in a proportion of 1:5 according to the norm NF X 31–103. The values of pH, electric conductivity, and oxidoreduction potential are respectively 12.3 ± 0.1, 6.1 ± 0.3, and 35 ± 5. The analysis of contained elements shows a large proportion of Molybdenum (Mo) and Barium (Ba), but also interesting amounts of Ca and Fe. This last element contributes to making these by-products interesting to be used for EK enhancement. Slags are by-products from the steel industry that are iron-rich materials. A significant amount of Mo and salts is also present and can be detrimental to the reuse of such materials in construction applications.

2.3 Experimental Setup and Electrokinetic (EK) Tests

The electrokinetic (EK) system, adapted from Ammami et al. (2015) [9], consists of a Teflon PTFE cylindrical chamber serving as the main cell, flanked by two electrode compartments (Fig. 1). Paper-Durapore membranes with a pore size of 0.45 μm were positioned between the sediment compartment and each electrode chamber. Electrolyte solutions were supplied to the anode and cathode chambers using two pumps operating at a flow rate of 5 mL/h. Electrical power was delivered by a generator capable of providing 0–50 V, connected to two graphite electrodes measuring 10.4 cm in length, 2.75 cm in width, and 0.5 cm in thickness. The power source could be operated in either constant-voltage or constant-current mode.

Following homogenization, collected sediments were stored in sealed containers at <5 °C to preserve their initial properties and water content. Approximately 360 g of homogenized dredged sediment were loaded into the chamber and allowed to stabilize for 24 h prior to experimentation. Citric acid (0.1 M) and ultrapure water

Table 2 Major and trace elements composition of the initial slag sample

Major elements	%	Trace elements	mg/kg
Ca	19.75	Ba	1748
Fe	16.64	Mo	312
Si	5.52	Sr	300
Zn	3.61	V	225
Cu	2.85	Pb	181
Mn	2.62	K	134
Cl	2.31	Ce	97
S	2.04	As	72
Mg	1.99	Ga	69
Al	1.34	Hf	55
Cr	1.02	Nb	51
Na	0.65	Zr	50
P	0.19	Y	5
Ti	0.16		
Ni	0.12		

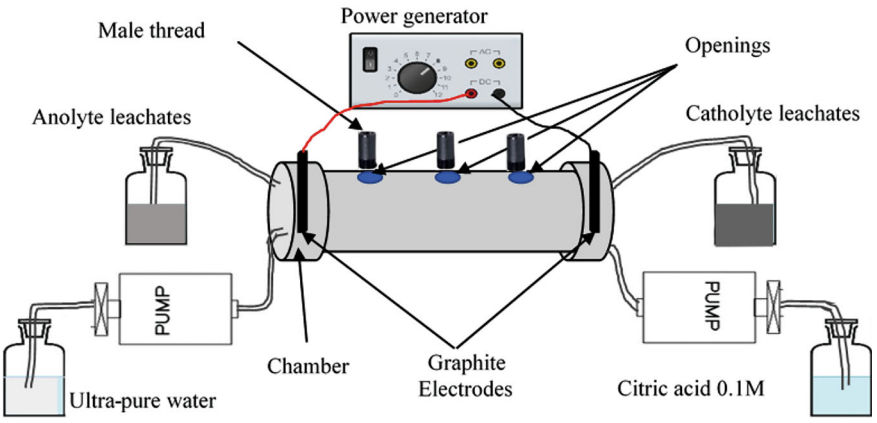

Fig. 1 Schematic draw of the experimental setup

were used as electrolytes in the cathode and anode compartments, respectively. Three electrokinetic remediation (EKR) tests were conducted under comparable electrolyte conditions: (i) continuous voltage applied to fluvial sediment, (ii) intermittent voltage applied to fluvial sediment, and (iii) intermittent current applied to a mixture of 95% fluvial sediment and 5% slag by volume (Fig. 2).

For the mixture preparation, a steel rod was positioned at the center of the chamber during sediment loading, then removed and replaced with ground slag (18 g) to

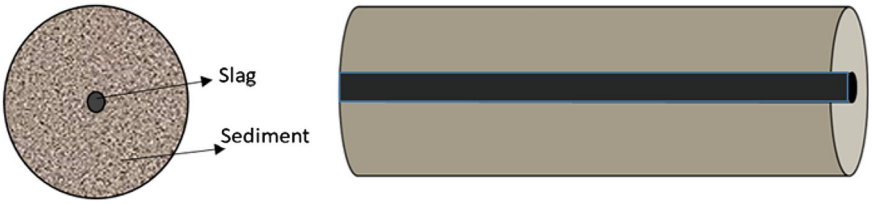

Fig. 2 The pouring of slag inside the center of the chamber and surrounded by sediment

Table 3 Experimental conditions

Test	Voltage	Anolyte/Catholyte	Power source	Duration	Material
Exp1	1V/cm	UP/ 0.1M CA	Continuous 24 h ON	21 d	Sediment
Exp2	1V/cm	UP/ 0.1M CA	8 h On/ 16 h OFF	21 d	Sediment
Exp3	1V/cm	UP/ 0.1M CA	8 h On/ 16 h OFF	21 d	Sediment + 5% slag

occupy the corresponding cylindrical volume. Experimental conditions are summarized in Table 3. Initial sediment characterization indicated low salinity and, consequently, low electrical conductivity, which may hinder EKR efficiency. To address this, the third test incorporated slag to enhance conductivity and promote electro-Fenton processes. The slag, rich in Fe^{2+}, facilitates hydroxyl radical generation, thereby improving the degradation of organic contaminants. Throughout the experiments, electrical and chemical parameters were continuously monitored. Metal speciation was determined by ICP analysis due to its superior sensitivity and accuracy for this application.

Following each experiment, the total sediment mass was measured to determine mass loss. The sediment was then sectioned into six layers along the anode–cathode axis (S1–S6). Layers adjacent to the electrodes (S1, S2, S5, S6) were cut to a thickness of 1.75 cm, while the central layers (S3, S4) were 3.5 cm thick, allowing higher resolution near the electrodes where greater gradients were expected. Each section was centrifuged for 20 min at 4640 g, and the supernatants were analyzed for pH and electrical conductivity. The remaining sediments were dried, homogenized, and subjected to metal content analysis.

3 Results

3.1 Temporal Evolution of Current Intensity

In preliminary tests, low current intensity with no significant temporal variation was observed. The addition of 5% slag was therefore examined as a means of improving current flow and treatment efficiency. The presence of iron salts (Fe2+) is expected

to lead to the generation of hydroxyl radicals, which are highly reactive and can break down a wide range of pollutants into simpler and less harmful compounds. A constant-current density with an initial voltage gradient of 1 V/cm was applied in the three experiments. As shown in Fig. 3, the initial electric current in Exp. 1, 2 had a low value, around 1 mA, which remained low throughout the 21 days of treatment. This low current is due to the low conductive species in fluvial sediments, where the conductivity was 1.2 mS/cm. The conductivity of metals, salts, and electrolyzed ions represents the main factor of variation in electric current. Thereby, no variation of current occurred in the case of application of permanent voltage (Exp1). The application of intermittent potential voltage in Exp2 did not enhance the current density in the sediment. This stable current density can be attributed to the medium becoming balanced, where the release of species from sediment is restricted, and the mobility has no significant impact on the system. The current was too low in both Exp1 and 2 while the voltage gradient was kept constant. However, due to the presence of conductive slag mixed with the sediment in Exp3, the initial current value reached a value of 12 mA, which was higher than in the previous two experiments. The electric current started to increase, reaching the highest value of 19 mA after 6 days. Overall, the mix of slag with the sediments increased the electric current passing through the medium. However, the current significantly started to drop at that time until reaching 9 mA at the end of the experiment. The decrease in current indicates high material resistivity, or low conductivity, suggesting that ionic species were removed from the sediment and/or pore fluid [10]. This trend of current evolution (initial increase of current followed by a decrease in the current trend) was reported in previous EKR studies [11]. Previous studies addressed by Kanbar et al. (2023) [12] showed that after 14–16 days of treatment, the current achieves a stable value, which was lower than the duration needed in Exp3. Considering the average current values in Exp2 (I2) and Exp3 (I3), the average ratio I3/I2 is approximately equal to 5. Similarly, the electrical conductivity ratio between slag and fluvial sediment, ECslag/ECRiverl-sediment, is also close to 5. The choice of adding the slag in the center of the sediment chamber instead of homogenizing it is to maintain high electrical continuity in the sample in order to enhance the EK process. Due to the electro-fenton process, which occurs due to the presence of slags, the hydroxyl radicals and other soluble ions contribute to the sediment, leading to better electrical conductivity through the sediment compared to other experiments.

3.2 Variation of Cumulative Electroosmotic Flow

The mass of leachates was monitored during the EKR process in each electrode compartment. The cumulative EOF in Exp1 and Exp2 was increasing linearly (Fig. 4). The intermittent application of a voltage gradient in Exp2 boosts the EOF compared to Exp1 where a continuous voltage gradient was applied. The main significance of EOF appears between the comparison of Exp2 and 3, where intermittent current is applied in both experiments. Due to the higher electric current in Exp3 and the

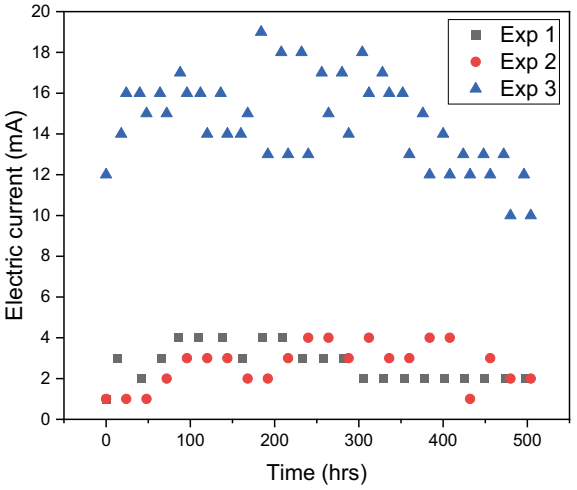

Fig. 3 Evolution of current intensity during test

presence of conductive species, the EOF was higher than Exp2. Previous studies have shown that the highest EOF is achieved when a higher electrical current occurs in the sample [13]. Even though the EOF is not significant in slag coarse particles alone, the mixture of 5% slag with the sediment enhances the EOF. The presence of sediment (charged fine particles) with a low proportion of slag contributes to increasing the EOF. The high EOF in Exp3 can enhance the removal of elements in the fluid phase.

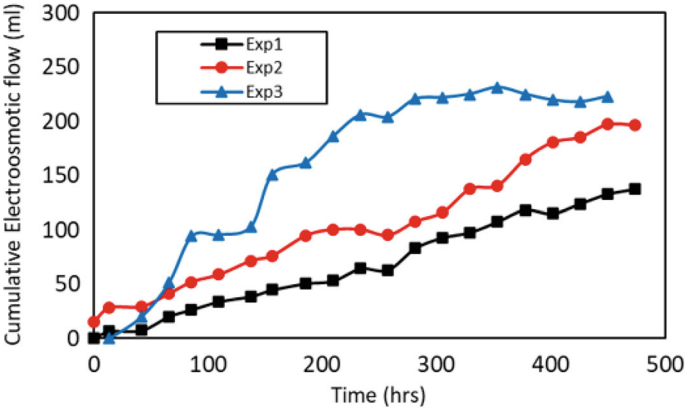

Fig. 4 Cumulative Electroosmotic flow (toward the cathode)

3.3 *Evolution of pH and OM in Sediment After EK Remediation Test*

Figure 5 displays the pH value within the different slices from the sample after the EK treatment. The evolution trend shows that pH did not vary extensively between the sections from anode to cathode, and the initial pH in both Exp1 and 2. The pH near the cathode was a bit lower due to the low pH exerted in cathode leachates. In previous studies, three main parts were distinguished, which are the anode side, the middle side, and the cathode side. Similar patterns in the pH of sediments were observed along the anode–cathode, as demonstrated in previous studies [14]. The pH values at the end of each experiment were lower than the initial pH values, especially in Exp3, indicating that the electrolysis had an acidifying effect on the anode side, while the addition of citric acid led ultimately led to acidifying the cathode side. The highest variation of pH between the initial and final state clearly appears in Exp3 due to the higher efficiency of the EK process in the presence of slag. The pH in Exp1 and 2 had a higher value than Exp3, which can be attributed to the low efficiency of EKR in both experiments. This meant that there was less exchange of H + ions, resulting in a lower acidic medium near the anode compared to Exp3. In summary, the variation in pH observed in each experiment can be explained by the differing electric current. Similar behavior was revealed by (Nasiri et al. 2020) [13]. The lower pH impacts elements to become more exchangeable and mobile toward the electrode compartments.

The organic matter near the anode (Fig. 6) showed a slight decrease compared to the initial sample, which can be attributed to electrolysis and oxidation processes that tend to break down organic compounds near the anode. This indicates that the EKR process is effective in mobilizing and altering the composition of sediment components near the anode. In the cathode side (S6), the organic matter content was

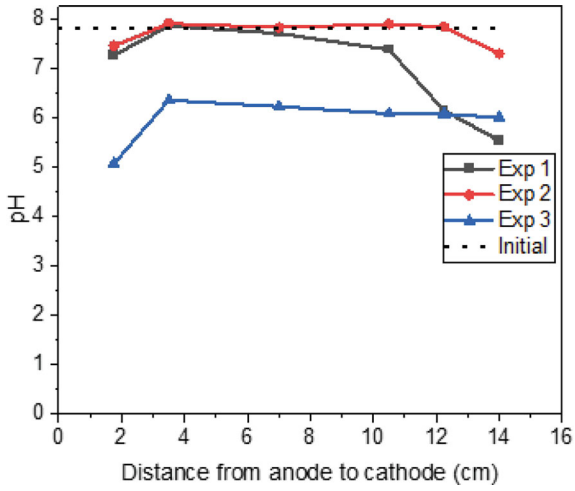

Fig. 5 Variation of pH after treatment from anode to cathode

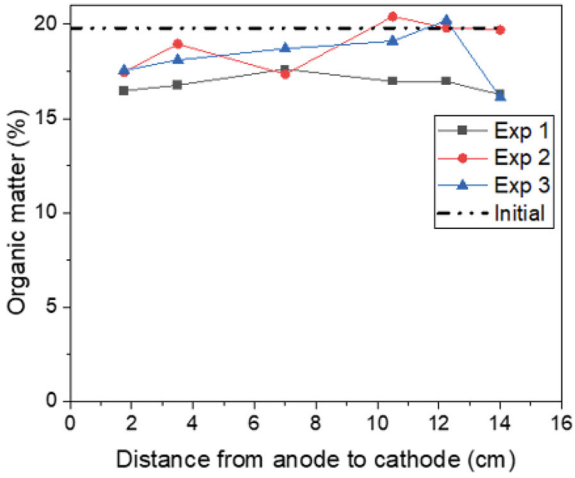

Fig. 6 Organic matter distribution along the sample (six slices) after the test

higher in Exp2 compared to Exp1 and Exp3. The overall trend indicates a decrease in organic matter within the matrix.

3.4 Metals Removal

The speciation of target elements was performed using ICP analysis. On purpose and for focusing on comparison with similar experimental conditions, the analysis was done only on Exp2 and Exp3. In Exp3, the addition of 5% slags to the sediments, which are conductive species, appears to have a significant impact on the removal of various metals. Arsenic (As), Chromium (Cr), Zinc (Zn), and Cadmium (Cd) removal shows notable differences between Exp2 and Exp3 (Fig. 7). For Arsenic, the removal percentage in Exp3 is higher across all sections (S1, S3, and S6) compared to Exp2. Because the conductive slags enhanced the electric current (Fig. 3), they improved the conductivity of the sediments and facilitated the migration of arsenic ions toward the electrodes. Similarly, for Zinc and Cadmium, the removal percentages in Exp3 are generally higher across all sections compared to Exp2. Cadmium was nearly totally removed in Exp3. The higher removal of As (28%), Cu (32%), Zn (55%), and Cd (85%) was achieved in Exp3. This overall improvement in metal removal efficiency can be attributed to the enhanced conductivity and combined electro-fenton process provided by the slags, which likely facilitated better ion migration and electrochemical reactions throughout the sediment matrix. By improving the overall conductivity of the sediments, the slags enhance the efficiency of metal removal by making the EKR process more effective. This finding is significant for the field of environmental remediation, as it suggests a practical approach to improving the performance of EKR systems in treating low-conductivity sediments.

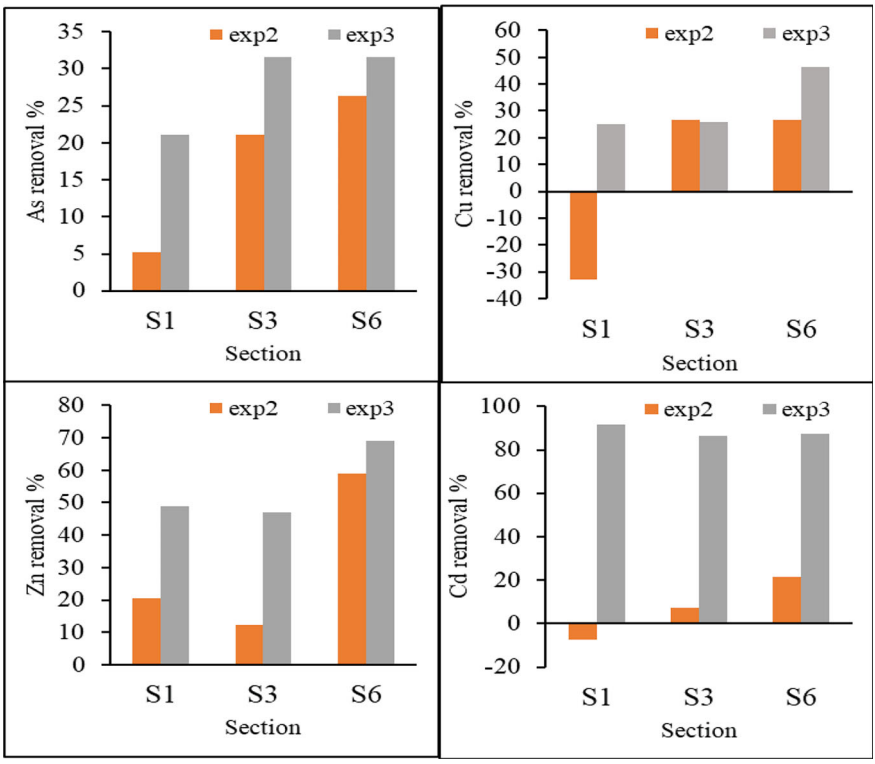

Fig. 7 Removal of As, Cu, Zn, and Cd from the sample sections

The addition of slag, a by-product of steelmaking, into these sediments can influence the physical–chemical properties of the matrix, affecting the overall behavior of EKR processes. Slag, which is often rich in iron, can alter the pH, electrical conductivity, and buffering capacity of the sediment, potentially enhancing or inhibiting the migration of contaminants under an applied electric field. The use of slag mixed with sediment (5/100) enhances the mobilization of heavy metals. The sediments with added industrial by-products like slag have shown that while some metals may become more mobile due to the increased ionic strength of the pore water, others may precipitate out of solution, reducing their availability for removal. The findings of this study underscore the significant influence of sediment properties, such as conductivity, on EKR behavior in river sediments. In particular, the low initial current observed in the fluvial sediments (Exp1 and 2) highlights the role of sediment conductivity in EKR processes. As expected, the fluvial sediments exhibited minimal variation in current, consistent with their low conductivity (1.2 mS/cm), which limited the mobility of ions and thus the overall electrokinetic efficiency.

This behavior is in line with previous studies such [15], which reported that low-conductivity sediments typically result in reduced ion mobility, leading to a stabilization of current over time. On the other hand, the mixing of slag in Exp3

led to an increase in current, peaking at 19 mA before stabilizing at 9 mA, indicating that the slag enhanced ion migration through its higher conductivity, where the introduction of conductive materials (slag) significantly boosted the electrokinetic efficiency by promoting the migration of contaminants. However, the electro-Fenton (EF) process is a powerful advanced oxidation process [16]. The Fenton reaction led to the generation of hydroxyl radical in an acidic medium. Due to the presence of oxidation reactions, hydrogen peroxide is generated. The hydroxyl radical ($^{\cdot}OH$) is a highly reactive molecule, formed through various processes, including electrochemical reactions. The hydroxyl radicals produced are powerful oxidizing agents capable of reacting and degrading unsaturated compounds and aromatic pollutants present in aqueous solutions [17].

4 Conclusion

Electrokinetic remediation (EKR) can be used for different kinds of materials, even though its efficiency was widely demonstrated for fine materials. This study aimed to evaluate the behavior of various materials submitted to EKR during and after treatment, an area that remains less investigated in the literature. The mixture of marine sediment with slags enhanced the electric current and the electroosmotic flow, and so the mobility of salts and other heavy metals. The mixture of slags with fluvial sediment can enhance the EK efficiency. The results showed that the addition of 5% slags increased the EOF and the electric current during the experiment. The highest removal of As (28%), Cu (32%), Zn (55%), and Cd (85%) was achieved in slag/fluvial sediment mixture. The findings from these studies have significant implications for optimizing EKR processes for the removal of metals and salts from different contaminated materials. The use of slag mixed with sediment (5/100) enhances the mobilization of heavy metals. The sediments with added industrial by-products like slag have shown that while some metals may become more mobile due to the increased ionic strength of the pore water, others may precipitate out of solution, reducing their availability for removal. The findings of this study underscore the significant influence of sediment properties, such as conductivity, on EKR behavior in river sediments. But it highlights the interesting idea for combining different waste materials to make the remediation process more effective and to develop sustainable methods for by-products materials. Given the huge volumes of deposited sediments and slags, the use of these materials in a mixture with more slag content can lead to a better performance of the EK remediation technique.

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Phosphorus Removal from Wastewater Using Carbonated Biochar/Steel Slag Composites

Zongqiang Ren, Devin Sapsford, and Fei Jin

Abstract Phosphorus (P) pollution is a major contributor to eutrophication, while the accumulation of solid wastes such as steel slag and biomass without proper disposal poses significant environmental risks. To address these challenges, biochar/steel slag (BC/SS) composites were synthesized via the carbonation of BC and SS as a novel adsorbent to remove P from simulated wastewater. In this study, the carbonated BC/SS effectively captured P and reduced the pH and conductivity of the treated solutions. Different proportions of BC and SS were tested to optimize the adsorptive performance of the composites. Results show that BC10 (containing10 wt.% of BC) exhibited the highest adsorption capacity for P in the batch experiments. The adsorption process followed the pseudo-second-order kinetics and conformed closely to the Langmuir isotherm, with a maximum adsorption capacity (q_m) of 5.36 (mg/g) in 48 h. Column experiments further confirmed the strong P removal efficiency of the composite. Material characterisations revealed that carbonation primarily led to the formation of calcium carbonate from Ca-species, and the addition of BC enhanced the carbonation reaction. This study highlights the potential of carbonated BC/SS composites as an effective adsorbent for P removal from wastewater, offering a sustainable method for the utilization of solid wastes.

Keywords Biochar · Steel slag · Phosphorus · Carbonation · Adsorption

1 Introduction

Excess phosphorus (P) contamination from agricultural runoff has become a critical environmental concern, triggering eutrophication of watersheds [1]. Elevated P loading also promotes the growth of aquatic bacteria, which can adversely

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affect ecosystem functions and pose microbiological risks to human health [2]. Concurrently, evolving regulatory frameworks on waste management regulations and circular economy principles have intensified the exploration of industrial byproducts such as basic oxygen furnace steel slag (SS) as sustainable remediation materials [3].

SS has emerged as a promising P removal candidate due to its inherent chemical/mineral composition [4, 5]. SS contains Ca-, Fe-, and Al- rich mineral phases, making it a cost-effective and efficient material for wastewater treatment, especially for P removal through synergistic chemisorption-precipitation mechanisms over a wide range of water characteristics [6]. However, the high pH and conductivity of the outflow solution attributed to the alkaline leaching pose significant challenges for the practical implementation, such as accelerated corrosion of the equipment [7]. Furthermore, the high density and low porosity of SS limit its adsorption efficiency by diminishing hydraulic permeability, and restricting contaminant diffusion and active site accessibility [8].

Carbonation technology involves the reaction of calcium-bearing minerals with CO_2 to form stable calcium carbonate, which can reduce the leaching of constituents and lower the pH and conductivity [9, 10]. In order to further improve its properties, researchers have found that the incorporation of porous materials such as biochar (BC) can enhance the carbonation efficiency and hydraulic performance of the material [11, 12]. BC is a carbon-rich material derived from the pyrolysis of biomass in an oxygen-limited environment. BC is a promising sorbent for removing organic and inorganic contaminants from wastewater due to the excellent porous structure and various functional groups [13]. The modification with BC, including impregnating with metal-containing minerals to optimize the physical and chemical properties, has been proven to meet practical environmental application requirements [14].

BC/SS has been designed to remove P from aqueous solution [15], but the composites are usually manufactured by co-pyrolysis of biomass and raw slag, which is associated with energy consumption and potentially harmful gas emissions [16, 17]. This study proposed the combination of SS and BC fabricated by carbonation as a porous sorbent in P removal from the aqueous solution. Carbonation-induced calcite precipitation binds the powders together while simultaneously storing CO_2 in the composite. It also reduces the potential risks of applying raw SS for water treatment, and the addition of BC would enhance carbonation efficiency, improve hydraulic performance, and adjust the microstructure of the composites. This work aims to: (1) evaluate the P sequestration performance of the composites with different proportions of BC and SS; (2) determine the P removal kinetics and isotherms with the batch adsorption test; and (3) investigate the long-term P removal performance using the column adsorption test.

Table 1 Mixing ratios of SS and BC

Samples	Steel slag	Biochar	Carbonation
BC0	100	0	Yes
BC10	90	10	Yes
BC20	80	20	Yes
BC50	50	50	Yes
SS	100	0	No
BS	90	10	No

2 Materials and Methods

2.1 Fabrication of BC/SS Composites

BC used in this study was purchased from Oxford Biochar Ltd. It was produced from the pyrolysis of mixed hardwoods at 400 °C. SS is a basic oxygen furnace slag supplied by Tarmac Ltd. The raw BC and SS were crushed and sieved to obtain particles with sizes of 75–150 μm and then stored in sealed plastic bags before use. The BC/SS composites were prepared by direct mixing and carbonation curing. Briefly, different proportions of SS and BC (Table 1) were firstly mixed in a blender at 800 rpm for 5 min while adding 30 wt.% deionised water during mixing to achieve homogeneity. The BC content varies from 0 to 50% by mass, and the samples were denoted as BC0, BC10, BC20, and BC50, respectively. The obtained mixtures were then placed into polystyrene petri dishes (100 mm in diameter and 15 mm in height) and then moved into a carbonation chamber for CO_2 exposure for a duration of 7 days. The inner temperature and CO_2 concentration were maintained at 50 °C and 20%, and the relative humidity was kept at around 70% by saturated sodium chloride solution. After carbonation, the specimens were dried in an oven at 75 °C for 24 h to remove the excess water and then crushed into less than 0.25 mm for the batch adsorption test and between 0.25 and 2 mm for the column test. The smaller particles in batch tests maximize the adsorbent's reactivity due to the larger surface area, but they do not perform well in column tests due to their poor permeability and potential clogging issues. Thus, for the column test, larger particles are preferred to ensure proper flow and better simulate what really happened in the field for contaminated groundwater remediation.

2.2 Batch Adsorption Test

The phosphorus removal performance of the composite was investigated by batch tests. For comparison, pristine BC, SS, and non-carbonated BC (10 wt.%) + SS (90 wt.%) mixture (BS) were also included in the experiments. Phosphorus stock solution (1 g/L) was prepared by dissolving dipotassium hydrogen phosphate (K_2HPO_4) in

deionised water. Subsequent solutions with desired concentrations were obtained by diluting the stock solution. All batch tests were carried out in triplicate, and the average values were reported here (with a coefficient of variation <8.12%).

Based on preliminary trials and literature, approximately 0.16 g of solid samples was added into 50 mL centrifuge tubes containing 40 mL of P solution (solid/liquid ratio of 0.04 mg/mL) with an initial P concentration of 10 mg/L. This ratio allows effective interaction between the adsorbent and the solution. The P concentration of 10 mg/L is similar to that in the agricultural runoffs and some industrial water. The mixture was shaken at 200 rpm for 48 h at room temperature. After reaching equilibrium, 2 mL of solution was extracted, filtered through a 0.45 μm filter, and analyzed for P concentration using ICP-OES.

BC10 was selected to evaluate the adsorption kinetics and isotherms of P removal. The adsorption kinetics were assessed by measuring the P concentration at different time intervals (5 min, 30 min, 2 h, 5 h, 10 h, 24 h, and 48 h) using an initial P concentration of 10 mg/L. The kinetic data were fitted with nonlinear pseudo-first-order (1) and pseudo-second-order (2) models. Adsorption isotherms were determined by equilibrating the prepared samples with initial P concentration solutions ranging from 1 to 100 mg/L for 48 h. The equilibrium results were analyzed using the Langmuir (3) and the Freundlich (4) models.

$$y = q_e\left(1 - e^{-k_1 x}\right) \tag{1}$$

$$y = \frac{k_2 q_e^2 x}{1 + k_2 q_e x} \tag{2}$$

$$q_e = \frac{q_m k_l c_e}{1 + k_l c_e} \tag{3}$$

$$q_e = k_f c_e^{\frac{1}{n}} \tag{4}$$

where y represents the amount of adsorbate at time x, q_e is the equilibrium adsorption capacity, k_1, k_2 are the rate constants, q_m is the maximum adsorption capacity, c_e is the equilibrium concentration of the adsorbate, k_l, k_f are the Langmuir and Freundlich constants, respectively, and n is the heterogeneity factor.

2.3 Column Test

In order to compare the effects of carbonation and BC modification on the wastewater treatment performance, a column study was conducted to evaluate the long-term P removal performance [18]. Columns (300 × ø40 mm) were packed with 150 mm of B0, B10, SS, and BS with a 30 mm layer of glass beads placed at both top and bottom. The choice of the size of the column is to ensure the diameter of the column (40 mm)

is more than 10 times the maximum size of the composite (2 mm) to minimize the boundary effect. A peristaltic pump was employed to inject a synthesized P solution (12.5 mg/L) at a steady flow rate of 1 mL/min through the column, which simulates the slow groundwater flow in a low-gradient sandy aquifer. Prior to the experiment, columns were equilibrated in a deionised water bath for approximately 18 h to eliminate impurities in the samples. At specific time intervals, a certain volume of the filtered solution was collected and stored in a plastic vial in a fridge for subsequent analysis.

2.4 Analytical and characterization Methods

The crystalline phases were identified by X-ray diffraction (XRD) analysis using a Siemens D5000 Diffractometer operating at 40 kV and 30 mA. During the adsorption experiments, the pH and conductivity of the solutions were detected by a Multiparameter (Mettler Toledo, S479-K). P concentrations were measured by Inductively Coupled Plasma Optical Emission spectroscopy (ICP-OES) on a Perkin Avio 220 Max instrument.

3 Results

3.1 Characterization of Adsorbents

According to X-ray diffraction (XRD) analysis (Fig. 1), the primary mineral phases in SS consist of dicalcium silicate (C_2S), $Ca_2Fe_2O_5$, and the RO phase (continuous solid solution formed by iron oxide, magnesium oxide, and manganese oxide). After carbonation, $CaCO_3$ was generated through the reaction between Ca ions and dissolved CO_2. The diffraction peak intensity of C_2S decreased after the addition of BC, and as the BC content increased, the intensity of $CaCO_3$ peaks became more pronounced, indicating that BC promoted the carbonation of Ca-containing phases. This can be attributed to its hydrophilic property and porous structure. When combined with SS, BC functions as an internal curing agent, supplying absorbed free and bound water, as well as CO_2, thereby enhancing the carbonation reaction [12].

3.2 Batch Results

The P removal performance among different adsorbents is shown in Fig. 2. Results indicated that P removal by pristine BC was almost negligible (BC = 0). This is because the surface of BC is often negatively charged and repels negatively charged

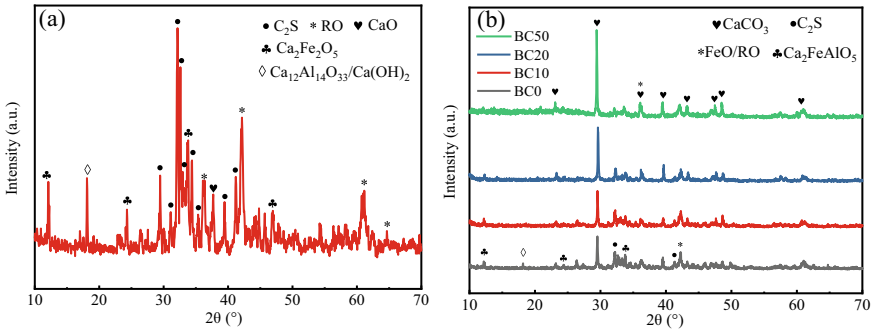

Fig. 1 XRD spectra of the: **a** raw SS and **b** composites after carbonation

ions such as phosphate [19]. The uncarbonated samples exhibited a strong affinity for P adsorption; however, the pH and the conductivity of the solution after treatment were significantly elevated, with pH exceeding 11 and conductivity surpassing 1000 μS/cm. This poses a major challenge for the application of pristine SS as a substrate for wastewater treatment directly [4, 7]. The P removal efficiency of SS decreased after carbonation, primarily due to the precipitate of carbonate compounds on the surface, blocking or altering the active sites for P adsorption. Additionally, the less dissolved Ca^{2+} in the solution reduced the P removal efficiency via precipitation. Compared to BC0, the addition of 10 wt.% of BC greatly improved the adsorption efficiency, reaching approximately 99%, similar to that observed with uncarbonated samples. The incorporation of BC may have improved the pore structure and provided additional adsorption sites. Nevertheless, when BC was added in higher amounts (20 wt.% and 50 wt.%), it reduced the reactive components present in SS, resulting in a lower overall P removal efficiency. Therefore, the combination of carbonation and BC modification can effectively lower the pH and conductivity, while maintaining a high P removal efficiency, ultimately leading to improved water quality after treatment.

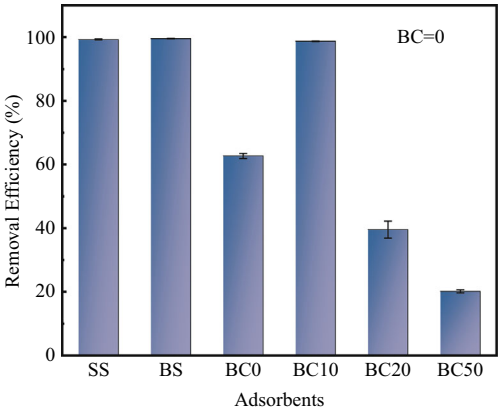

Fig. 2 P removal efficiency of different adsorbents

Since BC10 exhibited the best P removal performance compared to other carbonated composites, it was chosen for further evaluation of kinetics and isotherms, as shown in Fig. 3. This was consistent with the decision to choose BC10 for the column test as well. Unlike other adsorbents that exhibited rapid P removal efficiency in the initial stages [7, 20], the P adsorption performance of the carbonated material was much slower, with adsorption equilibrium being reached after 40 h. Although the addition of BC may enhance the physical adsorption of P, the carbonation process lowered the availability of Ca^{2+} in SS, thereby limiting the chemical precipitation rate of P. While SS primarily removes P through chemical adsorption [3], the slow dissolution process of calcium carbonate formed after carbonation results in an overall sluggish adsorption behavior. Figure 3b shows that the P removal capacity of BC10 increased with the rise in initial P concentration, stabilizing at around 5.3 mg/g beyond 60 mg/L, as adsorption saturation had been reached.

To better understand the P sorption performance of BC10, two adsorption kinetic models were used for evaluation. The pseudo-first-order model fit resulted in $R^2 = 0.94$, while the pseudo-second-order model had $R^2 = 0.98$, suggesting that the pseudo-second-order model better describes the P removal process. Figure 3b compares two isotherm models to further fit the P adsorption data on BC10. Results indicate that the Langmuir isotherm showed a better fit ($R^2 = 0.93$) than the Freundlich model ($R^2 = 0.85$). The maximum adsorption capacity $q_m = 5.36$ (mg/g) reflects the P adsorption potential of BC10 under optimal conditions (Table 2).

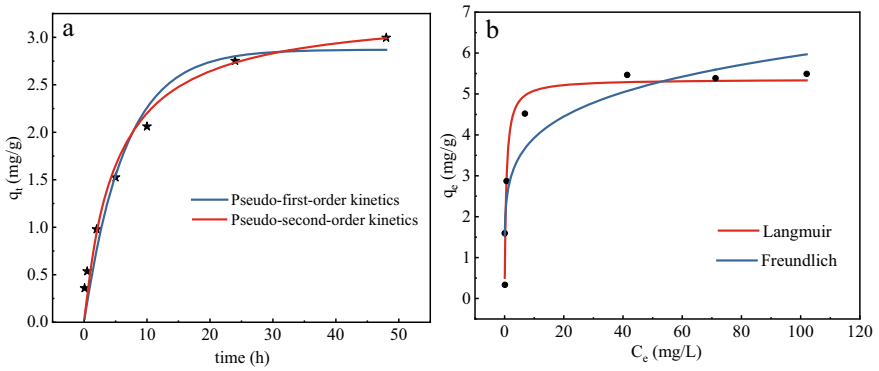

Fig. 3 The P adsorption kinetics and isotherms for sample BC10

Table 2 Fitting parameters for adsorption kinetics and isotherms

Adsorption model	Fitted parameters		R^2
Pseudo-first-order kinetics	$q_e = 2.88$ (mg/g)	$k_1 = 0.15$ (h^{-1})	0.94
Pseudo-second-order kinetics	$q_e = 3.30$ (mg/g)	$k_2 = 0.0604$ (g/(mg•h))	0.98
Langmuir	$q_m = 5.36$ (mg/g)	$k_l = 1.82$ (L/mg)	0.93
Freundlich	$1/n = 0.18$	$k_f = 2.59$ (mg/g)$(L/mg)^{1/n}$	0.85

3.3 Column Results

Figure 4 presents the changes in P removal efficiency, pH, and conductivity over time for different adsorbents (BC0, BC10, SS, and BS) to compare the effects of carbonation (SS vs. B0 & BS vs. BC10) and the modification of steel slag with biochar (B0 vs. B10) on the wastewater treatment performance. As shown in Fig. 4a, all adsorbents exhibited high P removal efficiency (~100%) and remained stable during the column experiments. Figure 4b illustrates that the carbonation of SS lowered the pH of the solution, and the addition of BC further intensified this reduction. As for the conductivity, the SS and BS samples showed higher initial conductivity, suggesting greater ion release. The B10 sample showed a smaller fluctuation and lower values in both pH and conductivity, implying better chemical stability. This is due to the more developed pore structure and increased specific surface area of the modified composite, which limited the diffusion of conductive ions. During the course of the experiment, white precipitates were observed in the effluent water collected in the SS sample column, suggesting that more $Ca2^{+}$ dissolved and subsequently formed $CaCO_3$ upon exposure to air. These precipitates may interfere with the filtration process by clogging the pores and reducing the permeability of the column.

4 Conclusion

The results of this study indicate that the carbonated BC/SS composite is a promising adsorbent for P removal. XRD analysis revealed that BC facilitated the carbonation of calcium-containing phases. Batch experiments demonstrated that carbonation lowered the pH and conductivity, while BC addition improved P sequestration efficiency. The superior performance of BC10 made it the most suitable candidate for further evaluation and modeling. The adsorption process of BC10 followed pseudo-second-order kinetics and fitted well with the Langmuir isotherm. Column study further confirmed that BC10 not only exhibited excellent P removal efficiency, but also effectively reduced and stabilized the effluent's pH and conductivity, which may be advantageous for long-term water treatment applications. Future studies will focus on the characterization of column media after adsorption to investigate the mechanism of P removal. In addition, more parametric studies will be conducted by varying the flow rate, pH value, etc., to achieve the breakthrough points and model the dynamic adsorption behavior of the materials.

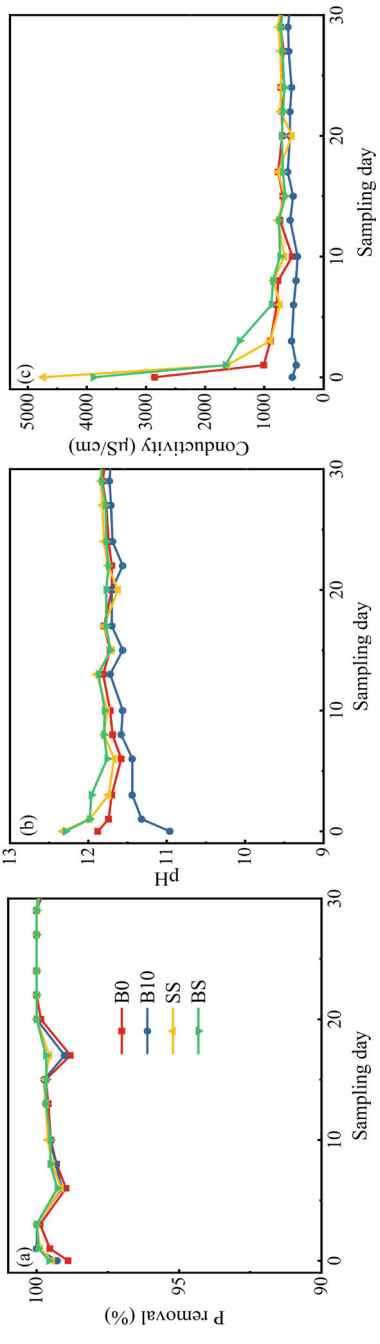

Fig. 4 Effect of adsorbents on P removal, pH, and conductivity

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Use of Residual Materials in Adsorption and Tertiary Treatment Processes for Industrial Wastewater

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Abstract This study aimed to repurpose two solid industrial wastes as adsorbents, replacing raw materials and promoting an effective circular economy. The investigated wastes included blast furnace slag and coke fines sourced from an integrated steel plant. Their adsorption performance was assessed for the removal of fluoride and COD from the effluent of an industrial wastewater treatment facility. The initial COD concentration was 216.49 mg/L, while the fluoride concentration was 36.92 mg/L. Blast furnace slag underwent activation using a commercial aluminum product containing 10% Al_2O_3, with solution concentrations ranging from 0.1 M to 0.5 M. Thermal activation was carried out at temperatures between 550 °C and 800 °C. The highest adsorption efficiency was achieved with activation at 800 °C and a 0.5 M aluminum solution, resulting in a reduction of COD to 123.02 mg/L (43% removal efficiency) and fluoride concentration to 3.4 mg/L (90.70% removal efficiency). Similarly, coke fines were chemically activated using the same commercial aluminum product at concentrations between 0.1 M and 0.5 M. The optimal results were obtained with a 0.2 M aluminum solution, reducing COD to 97.22 mg/L (55% removal efficiency) and fluoride concentration in the final effluent to 2.56 mg/L (93.1% removal efficiency).

Keywords Fluoride removal · Blast furnace slag · Adsorption · Industrial waste reuse · Circular economy

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1 Introduction

Slag is one of the waste materials obtained from the blast furnace in integrated steel plants, where steel is produced from iron ore. In this process, iron ore—mainly composed of iron oxides and other substances—is introduced into the furnace along with coke and a fluxing material, typically limestone. The process occurring inside the blast furnace involves the reduction of iron to its elemental state, facilitated by a countercurrent exchange between the charged materials that descend and the ascending reducing gas. This reducing gas is generated through the combustion of coke and coal in the lower part of the furnace, using oxygen-enriched air. The reduction occurs because the gas contains carbon monoxide (CO), which has a higher affinity for oxygen than iron does for forming iron oxides, leading to the formation of carbon dioxide (CO_2). The resulting slag is characterized by its porous structure and a chemical composition rich in calcium, silicon, aluminum, and magnesium oxides.

Currently, blast furnace slag is primarily used as an additive in cement production or for asphalt generation, as it enhances strength and durability. Additionally, it is also utilized as a fertilizer or soil amendment in agriculture, as it provides beneficial nutrients for plant growth.

Another waste product associated with integrated steelmaking is coke fines. Coke is introduced into the blast furnace, where it acts as a reducing agent; however, when its particle size is extremely small, it is discarded. On the other hand, coke manufacturing involves several water-intensive operations, including quenching hot coke, washing the ammonia still, cooling and washing coke oven gases, and isolating coke by-products [1]. Worldwide, 750 million cubic meters of coke wastewater are generated annually (equivalent to 0.6–1.6 cubic meters of wastewater per ton of coke produced), with 92 million cubic meters generated in Europe [2].

The treatment of wastewater from industrial processes has become a highly relevant environmental concern, aiming to return used water to the environment in a state that does not alter its natural characteristics. The conditions under which it is discharged can have significant environmental consequences due to the pollutant load present, as some contaminants cannot be neutralized and may modify the physical, chemical, and biological parameters of the receiving environment [3]. Due to its large volume and complex pollutant load, coke wastewater is among the most problematic industrial wastewaters. It contains toxic organic (organic matter, phenols, polycyclic aromatic hydrocarbons) and inorganic (cyanides, thiocyanates, ammonia, fluorides, sulfides) compounds [4, 5].

The increasing restrictions on the concentrations of certain pollutants in wastewater discharged into public water bodies have led to the continuous adaptation of industrial wastewater treatment plants to ensure compliance with regulatory limits. Naturally, these treatment processes entail costs that companies must bear, making it essential to optimize and assess them to identify the most effective solution from both a technical and economic perspective. In this regard, the aim is to explore new alternatives that meet the established requirements [6].

Adsorption processes for contaminant removal in wastewater have already been implemented in water treatment, particularly in tertiary treatment processes. These involve the use of columns where water comes into contact with a solid material—typically activated carbon—to retain and eliminate certain pollutants. In such cases, the primary objective is to reduce organic compounds and heavy metals.

This study presents the preliminary results of the investigation into the adsorption capacity of blast furnace slag and coke fines for the treatment of wastewater from Coke oven battery operations.

In this context, blast furnace slag has emerged as a promising adsorbent due to its favorable physical and chemical properties. This slag is characterized by its porous structure and a chemical composition rich in calcium, silicon, aluminum, and magnesium oxides. Similarly, steel coke is also a highly porous material.

The use of blast furnace slag and coke fines as adsorbents offers several significant advantages. Firstly, they are abundant and low-cost industrial waste materials, making them an economical and sustainable option for wastewater treatment. Additionally, their porous structure provides a large surface area and a wide pore size distribution, which enhances the adsorption of various contaminants present in wastewater. Moreover, the presence of certain components within their composition facilitates chemical interactions between the contaminants and the adsorbent [7].

2 Materials and Methods

2.1 Materials

Blast Furnace Slag. The blast furnace slag, used as a precursor for the adsorbent material, comes from a local integrated steel plant with an approximate production of 5 million tons of steel per year.

Coke Fines. The coke fines used are generated during the coke quenching process after being extracted from the furnace. Subsequently, these fines are separated from the aqueous phase through sedimentation. Coke fines are classified as residual waste or by-products, as they cannot be directly reutilized, and the company must manage them as industrial residues.

Wastewater. The wastewater utilized in this study was the effluent from the aerobic biological treatment process, which includes integrated nitrification–denitrification stages, designed for treating wastewater generated in the coke oven plant (production capacity: 1 million t coke/year) of an integrated steel mill (ECWT). ECWT samples were stored in a refrigerator at 4.0 ± 0.1 °C.

2.2 Methods

Preparation of the Adsorbent. The residues are washed with distilled water through an agitation process, followed by drying at 105 °C to remove dirt, dust, and other substances present on the solid that could interfere with the adsorption process. Subsequently, the material is sieved using a vibratory sieve to obtain the desired particle size ranging from 1 to 2 mm.

Filtration of the Effluent from the Coke Wastewater Treatment Plant. A filtration process was conducted to remove solids from the ECWT using a 0.45-μm filter prior to the adsorption process. The pH, total suspended solids (TSS), chemical oxygen demand (COD), and five anions (fluoride, chloride, nitrate, nitrite, and sulfate) were analyzed.

Adsorbent Activation. The thermal activation of blast furnace slag is performed using a muffle furnace (HOBERSAL, Mod. 8B 12 PR/400) at the required activation temperature (550 °C or 800 °C) for a duration of 90 min. After this period, the material is allowed to cool and is subsequently stored in a desiccator. Chemical activation is carried out using a commercial aluminum solution containing 10% Al_2O_3. The concentration of this solution varies across activation tests to determine the optimal activation conditions based on the results. This process involves contacting and agitating the activation solution with the blast furnace slag at a ratio of 10 mL of solution per gram of slag. The mixture is left for 24 h, after which the adsorbent material is separated and dried at 80 °C for 48 h.

Adsorption Batch Tests. Equilibrium experiments are conducted to determine the adsorption capacity of the materials after activation. The procedure consists of adding 250 mL of wastewater into an Erlenmeyer flask, along with a variable amount of slag, depending on the desired concentration, ranging from 10 to 60 g/L. The mixture is kept under agitation for 24 h and subsequently filtered using a 0.45-μm membrane filter. The resulting sample is then analyzed to determine the adsorption performance.

Analytical Methods. The parameters analyzed to characterize the wastewater and effluents from the laboratory treatment included pH, conductivity, TSS, COD, fluoride, chloride, nitrate, nitrite, sulfate, and heavy metals.

The pH, TSS, and COD were measured following standard analytical methods [8]. COD was determined using Method 5220 D (closed reflux, colorimetric method) with a PerkinElmer Lambda 35 UV–Visible spectrophotometer. The determination of F^-, Cl^-, $NO_3{}^-$, $NO_2{}^-$, and SO_42^- was performed by ion chromatography (861 Advanced Compact IC 2.861.0010) after filtering the wastewater through a 0.45-μm pore size membrane filter. Heavy metal concentrations were analyzed by inductively coupled plasma mass spectrometry (ICP-MS) following acid digestion.

Table 1 Physicochemical parameters of wastewater from coke oven batteries

Parameter	Value
pH	7.62
TSS ($mg.L^{-1}$)	13.33 ± 8.33
COD ($mg.L^{-1}$)	216.49 ± 4.06
Iron ($mg.L^{-1}$)	14.92 ± 4.40
Copper ($mg.L^{-1}$)	0.90 ± 0.44
Nickel ($mg.L^{-1}$)	21.12 ± 9.17
Zinc ($mg.L^{-1}$)	3.14 ± 0.15
Chromium ($mg.L^{-1}$)	<0.044
Manganese ($mg.L^{-1}$)	<0.,24
Lead ($mg.L^{-1}$)	$5.58 \pm 0.,62$
Cadmium ($mg.L^{-1}$)	2.21 ± 0.11
Fluoride ($mg.L^{-1}$)	36.92 ± 1.91
Chloride ($mg.L^{-1}$)	1506.45 ± 39.10
Nitrite ($mg.L^{-1}$)	13.94 ± 0.25
Nitrate ($mg.L^{-1}$)	70.18 ± 1.10
$Sulfate^{-}$ ($mg.L^{-1}$)	1093.58 ± 21.52
Bromide ($mg.L^{-1}$)	15.60 ± 0.98

3 Results

3.1 Wastewater Characterization

The results are presented in Table 1. This wastewater exhibits a COD of 216.49 mg/L in the filtered water and a total suspended solids concentration of 13.33 mg/L. The low solids concentration corresponds to the slight decrease in COD observed after filtration compared to the raw wastewater. Additionally, among the heavy metals, iron and nickel are present at higher concentrations, with values of 14.92 mg/L and 21.12 mg/L, respectively. Regarding the analyzed anions, chloride and sulfate concentrations are notably high, along with a fluoride concentration of 36.92 mg/L, a value that will be of particular relevance throughout this study.

3.2 Adsorption Preliminary Assays

These assays are conducted to determine the optimal conditions for the adsorption process. Based on the results, the adsorbent concentration, slag particle size, and operating pH for subsequent experiments are established.

Table 2 The test with slag and coke fines at a concentration of 50 g·L-1, particle sizes between 2–5 mm and 1–2 mm

	COD ($mg.L^{-1}$)	F^- ($mg.L^{-1}$)	Br^- ($mg.L^{-1}$)	$NO_3{}^-$ ($mg.L^{-1}$)
BFS (50) S (2–5)	192.79 ± 5.11	40.90 ± 0.58	11.96 ± 9.88	382.14 ± 7.22
BFS (50) S (1–2)	238.46 ± 36.86	33.48 ± 1.34	13.40 ± 0.30	84.16 ± 0.67
CF (50) S (1–2)	161.57 ± 2.80	24.72 ± 4.34	10.36 ± 0.92	15.71 ± 0.62

Note BFS = Blast Furnace Slag ($g.L^{-1}$), S = Particle Size (mm), CF = Coke Fines ($g.L^{-1}$)

As observed in Table 2, in the case of blast furnace slag with a particle size between 2 and 5 mm, no reduction in fluoride concentration is detected. However, COD decreases, indicating that the adsorption process may be effective under appropriate conditions. A particle size between 1 and 2 mm shows a reduction in fluoride concentration, as well as in other components such as chlorides, nitrites, and nitrates. Comparing all the results, it is concluded that the adsorption process is more effective when the adsorbent has a particle size between 1 and 2 mm. This is due to the increase in specific surface area, which enhances adsorption capacity. For this reason, it was decided to use this particle size range in subsequent experiments. In the case of coke fines, COD decreases from 216.49 mg/L initially to 161.57 mg/L, while fluoride concentration is reduced from 36.92 mg/L to 24.72 mg/L. This indicates favorable adsorption processes, even before thermal and/or chemical activation is applied.

3.3 Activation Assays

Blast Furnace Slag. Two types of activation were performed on the blast furnace slag: thermal activation and chemical activation. Thermal activation was carried out at two temperature ranges: 550 °C and 800 °C. Chemical activation was conducted using a commercial aluminum solution containing 10% Al_2O_3, with concentrations ranging from 0.1 M to 0.5 M. The results are presented below, identifying the conditions that achieve the highest adsorption capacity. The outcomes of the assays are summarized in Table 3. As a result of thermal activation, increased porosity was observed, leading to enhanced adsorption capacity. This improvement resulted in a lower equilibrium concentration of both COD and fluoride in the treated wastewater. Regarding chemical activation, a significant reduction in fluoride concentration was noted, highlighting the specificity of the adsorbent for fluoride removal.

As shown in Table 3, the key parameters considered (COD and fluorides) exhibit a significant reduction compared to their initial concentrations in the coke battery wastewater used in this study, as well as in relation to previous trials. The results indicate that reductions exceed 85% for fluorides and 43% for COD, demonstrating that the activated adsorbent prepared in this manner possesses good adsorption capacity.

Table 3 Results of the equilibrium tests on blast furnace slag activated thermally and chemically

	COD (ppm)	F^- (ppm)	Br^- (ppm)	NO_3^- (ppm)
TA (550) CA (–)	147.35 ± 9.58	26.71 ± 0.49	10.62 ± 0.12	70.55 ± 1.07
TA (–) CA (0.1)	183.84 ± 5.42	20.72 ± 1.48	10.05 ± 0.68	74.60 ± 11.41
TA (550) CA (0.1)	173.50 ± 6.50	20.35 ± 0.41	10.20 ± 0.28	67.42 ± 0.97
TA (800) CA (0.1)	137.37 ± 4.81	24.34 ± 0.24	13.07 ± 1.28	187.73 ± 1.94
TA (800) CA (0.2)	172.37 ± 5.03	11.83 ± 0.16	10.51 ± 0.70	210.70 ± 1.50
TA (800) CA (0.3)	122.49 ± 6.57	5.21 ± 0.46	12.51 ± 0.10	188.62 ± 0.49
TA (800) CA (0.4)	113.69 ± 8.03	5.78 ± 0.16	12.68 ± 0.07	186.50 ± 0.55
TA (800) CA (0.5)	123.03 ± 6.01	3.40 ± 0.20	12.33 ± 0.33	185.81 ± 2.08

Note A = Thermal Activation (ºC), CA = Chemical Activation (Al_2O_3 molar concentration)

In Fig. 1, the adsorption capacities resulting from the adsorption experiments with blast furnace slag can be observed. The highest adsorption capacity was achieved at 800 °C and 0.5M Al_2O_3, with an adsorption capacity of 0.67 mg·g-1.

Coke Fines. The coke fines are not subjected to thermal activation, as they have already been exposed to 900 °C for 16 h in an inert atmosphere during their production process. Chemical activation is performed using a commercial aluminum solution containing 10% Al_2O_3, with concentrations ranging between 0.1 M and 0.5 M. After conducting six experimental trials, it was concluded that the optimal result was achieved with coke activation using a commercial aluminum solution containing 10% Al_2O_3 at a concentration of 0.2 M. Under these conditions, the highest fluoride adsorption capacity was 0.69 mg·g-1. In this case, the COD value was reduced to 97.22 mg·L-1, corresponding to a removal efficiency of approximately 55% (Table 4 and Fig. 2).

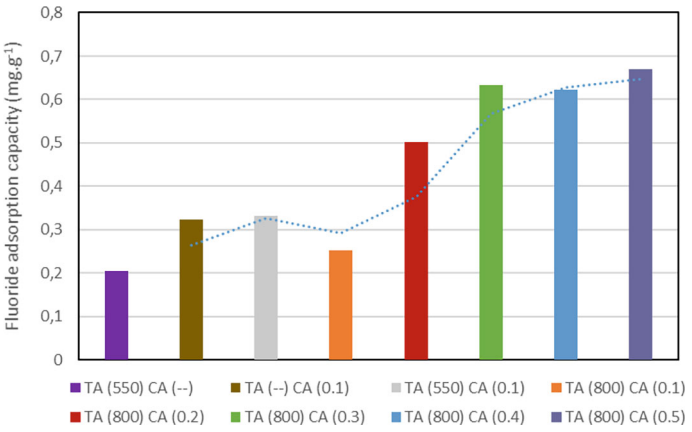

Fig. 1 Fluoride adsorption capacity in blast furnace slag assays

Table 4 Results of the equilibrium tests for chemically activated coke fines

	COD (ppm)	F^- (ppm)	Br^- (ppm)	NO_3^- (ppm)
TA (–) CA (–)	161.57 ± 9.58	24.72 ± 4.34	10.36 ± 0.92	15.71 ± 0.62
TA (–) CA (0.1)	94.11 ± 0.92	4.55 ± 0.16	14.18 ± 2.19	41.38 ± 2.03
TA (–) CA (0.2)	97.22 ± 9.96	2.56 ± 0.02	12.26 ± 0.20	39.01 ± 2.21
TA (–) CA (0.3)	100.60 ± 7.89	5.46 ± 0.15	11.89 ± 0.06	39.98 ± 1.17
TA (–) CA (0.4)	124.44 ± 5.02	7.45 ± 1.31	13.95 ± 3.44	10.28 ± 8.64
TA (–) CA (0.5)	199.87 ± 6.21	4.53 ± 2.71	10.67 ± 1.34	9.59 ± 0.55

Note TA = Thermal Activation (ºC), CA = Chemical Activation (Al_2O_3 molar concentration)

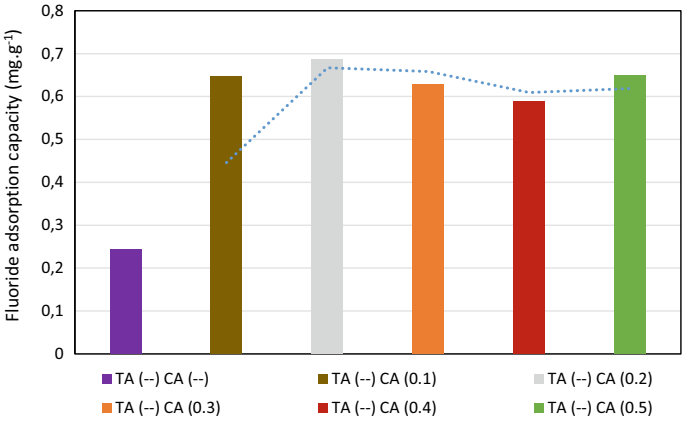

Fig. 2 Fluoride adsorption capacity in coke fines tests

4 Conclusions

Based on the results, it is concluded that both blast furnace slag, after physical activation followed by chemical activation, and coke fines, after chemical activation, exhibit effective adsorption behavior, significantly reducing the concentration of specific contaminants such as COD and fluorides. For blast furnace slag, the highest adsorption efficiency was achieved at an activation temperature of 800 °C and a 0.5M commercial aluminum solution, resulting in a reduction of COD to 123.02 mg/L (43% removal efficiency) and fluoride concentration to 3.4 mg/L (90.70% removal efficiency). In the case of coke fines, the best performance was observed using a 0.2M aluminum solution, leading to a 55% COD reduction and a final fluoride concentration of 2.56 mg/L (93.1% removal efficiency).

Acknowledgements This work was partially supported by the Gijón Council through the University Institute of Industrial Technology of Asturias (Ref. IUTA-23-GIJON-1-11).

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Geoenvironmental Pollution Control

Integrated Electrochemical Technologies for Wastewater Treatment and Soil Remediation: Tackling Contaminants of Emerging Concern and Pathogens

Nazaré Couto, Paula Guedes, Ana R. Ferreira, Vanessa G. Correia, Raquel Carvalho, João Brinco, Eduardo P. Mateus, and Alexandra B. Ribeiro

Abstract Treated wastewater can help address water scarcity, but it's crucial to test technologies that remove emerging contaminants of emerging concern (CECs) from both the effluent and potentially contaminated soils. This document combines findings from two works investigating the use of electrochemical oxidation (EO) and electrokinetic remediation (EKR) technologies to tackle CECs, including pharmaceuticals and *Escherichia coli*. For the treatment of effluent, EO showed removal

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efficiencies exceeding 90% for compounds such as sulfamethoxazole (SMX) and diclofenac (DCF) when using MMO/Ti circular mesh electrodes. EKR was applied to remove CECs, including SMX, DCF, and oxybenzone, from soils irrigated with effluent. It was shown that after being introduced into the soil by effluent irrigation, 20–100% of the CECs remained in the soil after 6 days. EKR improved the elimination of CECs by up to 20% compared to the control (without current), which depends on the contaminants' distance to the electrodes. The simultaneous application of reversed electrode polarization and On/Off currents enabled a more uniform removal while maintaining soil characteristics, validating EKR as a viable in-situ remediation method. These findings highlight that integrated EO-EKR systems offer scalable, eco-friendly solutions for wastewater and soil remediation. On this topic, results from stakeholders' perceptions of wastewater reuse are also presented to understand the needs and challenges of promoting an effective application. Subsequent studies should refine operating parameters, explore electrode configurations, and evaluate long-term environmental impacts to improve acceptance and effectiveness.

Keyword Electro-assisted technologies · Contaminants of emerging concern · Water reuse · Soil remediation

1 Introduction

Water scarcity and contamination are critical global challenges exacerbated by population growth, industrial activities, and climate change. Effluent from wastewater treatment plants (WWTPs) is anticipated as a potential strategy to alleviate pressure on water resources [1]. Effluent reuse is underpinned by specific legislation and policies, reinforced by the principles of the Circular Economy, Sustainable Development Goals (SDG6), and the EU Green Deal. The European Commission implemented quality requirements within an integrated water management framework to guarantee the safe reuse of water from urban wastewater treatment plants [2]. Reusing treated wastewater for irrigation has been proposed as a solution to mitigate this pressure; however, contaminants of emerging concern (CECs), such as pharmaceuticals and *Escherichia coli*, remain a significant risk. The Regulation (EU) 2020/741 establishes minimum requirements for water reuse to ensure the safety of reclaimed water for agricultural irrigation. *E. coli* is included in the quality requirements for reclaimed water. Annex II, B) outlines the conditions associated with the additional requirements, contingent upon risk assessment, which examines pharmaceuticals and other substances.

Advanced water treatment technologies encompass activated carbon adsorption, membrane filtration, advanced oxidation processes (AOPs), and the integration of multiple technologies. The results obtained to date regarding the complete elimination of CECs indicate that their susceptibility to various treatment processes is variable. Consequently, additional research on advanced wastewater treatment

technologies for removing emerging and currently unregulated contaminants is necessary.

Electrochemical oxidation (EO) has earned significant attention among various AOPs, owing to its effective performance in removing persistent and recalcitrant organic contaminants from wastewater [1, 3]. EO is recognized as an environmentally friendly technology due to its requirement of no additional chemicals, with the electron acting as a controllable reagent that can be classified as "green". Several factors, such as the type of electrode, the material of the electrocatalyst, the composition of the wastewater, the pH, and the applied current and voltage levels influence the mechanism of the EO process.

On the same note, soil may be contaminated by CECs resulting from previous irrigation procedures [4]. Once in the soil, the CECs may bioaccumulate in vegetables [5]. Although electrochemical remediation (EKR) is a promising technique to remove CECs from soil [6–8], its efficiency depends on parameters like the type of electrodes used and the control of soil pH, which are pivotal to guarantee an effective and sustainable remediation process.

Another critical aspect toward increasing the applicability of these environmental technologies is to involve stakeholders and understand their needs and challenges, aiming to find common ground and move closer to their requirements. Thus, this document combines the technical and social/strategic aspects of improving sustainable water reuse systems, addressing (Fig. 1):

- The use of electrochemical technologies for water treatment [1];
- Soil irrigation and electrokinetic treatment [4];
- Stakeholder analysis based on strengths, opportunities, aspirations, and results of using wastewater as an alternative water source.

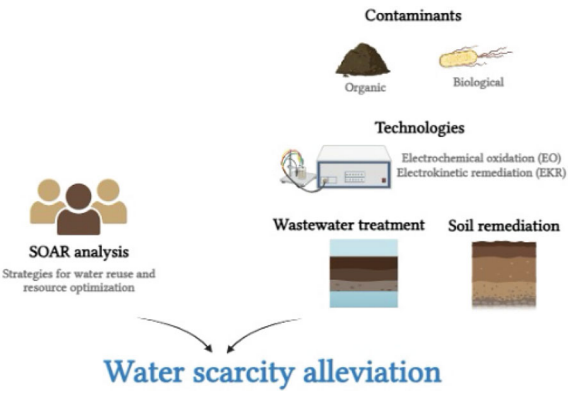

Fig. 1 Overview of the methodological approach. Created in BioRender

2 Materials and Methods

2.1 Applied Studies

Wastewater Treatment Using Electrochemical Technologies [1] with a focus on removing emerging organic contaminants and *E. coli* using electrooxidation. **Experimental setup**: Effluent collected from the secondary settling tank in a WWPT and spiked with a concentration of 2 mg/L of sulfamethoxazole (SMX), carbamazepine (CBZ), estradiol (E2), ethinylestradiol (EE2), and diclofenac (DCF). Reactor with length of 10 cm and diameter of 8 cm, with a capacity to treat 0.45 L of effluent; experiments carried out for 2 h; optimization of the process (*Phase I*: fixed cathode—activated titanium bar coated with MMO, IrO_2: RuO_2 and anode selection—electrode material (MMO/Ti or Pt/Ti) and shape (bar, flat-plate or circular mesh); direct current of 100 mA; *Phase II* three current intensities (100, 125, 175 mA) applied to the best conditions; *Phase III*—cathode and anode with the same shape/material (MMO/Ti circular mesh shape); 100 mA) and *Phase IV—scale up feasibility* Reactor with a length of 20 cm and diameter of 8 cm), with a capacity to treat 0.9 L of effluent; 9 CECs in total: SMX, CBZ, E2, EE2, DCF, caffeine (CAF), oxybenzone (OXY), triclosan (TCS), bisphenol A (BPA). MMO/Ti material with circular mesh shape as anode and cathode; two pairs of electrodes with an equally split current of 175 mA; kinetics evolution—sample collected every 10 min during 2 h from the start of the experiment.

Soil Irrigation with Treated Effluent and Electrokinetic Remediation [4] Application of EKR to remediate CECs present in soils irrigated with effluent. **Experimental Setup:** Effluent collected from the secondary settling tank and spiked with a concentration of 5 mg/kg of each compound: SMX, DCF, ibuprofen (IBF), CBZ, EE2, and OXY. Parallelepiped-shaped glass reactor (14 × 14 × 5 cm), covered to prevent light exposure. For each experiment, the microcosm was filled with 300 g of soil and irrigated using the spiked effluent (3:1 m:V), simulating a flood irrigation system; MMO electrodes with mesh shape (IrO_2/RuO_2-Ti). Experiments carried out for 6 days, 2.5 mA and different current strategies: Continuous Current (CC): Direct current (DC) was applied continuously throughout the experiment; Switched Current (On/Off): DC was switched off on day 3 for 24 h and turned back on from day 4; Reversed Electrode Polarization (REP): Electrode polarization was reversed on day 3 for 24 h; Combined On/Off + REP: DC was switched off on day 3 for 24 h, and from day 4 onward, the electrode polarization was reversed.

2.2 Value perception—Stakeholders' View

The dynamic "SOAR Analysis for Territorial Strategies to Address Water Scarcity" involved the formation of working groups to collaboratively conduct a SOAR analysis

(Strengths, Opportunities, Aspirations, and Results) focused on territorial strategies for addressing water scarcity. Groups discussed their perspectives on each topic within the SOAR framework and then voted for the ones that best aligned with their vision. The SOAR framework allows for the identification of: (i) Strengths: Distinctive qualities and unique attributes; (ii) Opportunities: Emerging capabilities, processes, and services; (iii) Aspirations: Strategic initiatives (projects, programs, processes); (iv) Results: Measurable outcomes that can assess the success of the envisioned future. This collaborative approach fosters a shared vision for strategic development and actionable insights to address water scarcity and the driver for the technical research for wastewater reuse and soil irrigation.

3 Results

3.1 Wastewater Treatment Using Electrochemical Technologies

The article [1] shows that the removal rate of CECs through electrooxidation is influenced not only by the structure and physicochemical properties of CECs but also by the material and form of the electrode applied for the treatment process. In Phase I it was reported that MMO/Ti mesh with a circular mesh shape presented a higher CECs removal than the other tested anodes with removal efficiencies reaching, e.g., 91% for SMX, 90% for DCF, and 85% for E2 and EE2. In contrast, results from Phase II showed that even using MMO/Ti with circular mesh shape as anode and MMO/Ti bar as cathode, the increase in current intensity did not improve CECs removal, except for one compound (CBZ that enhanced oxidation rate by 30%). In Phase III, with cathode and anode with a circular shape, the removal efficiency for CECs improved, e.g., CBZ removal increased by 50%), adding to lower energy consumption compared with electrode combinations in Phase I and II. In Phase IV (scale up), with double the reactor capacity and electrodes /current intensity adapted to the scale up, more than 90% of all compounds were degraded in 2 h. The rate constants (k) for the 9 tested compounds, according to their pseudo first-order kinetics rate constant, categorized the compounds as follows: rapidly degradable ($k > 0.0103$, e.g., EE2), moderately degradable ($0.0076 < k < 0.0103$, e.g., DCF), or persistent ($k < 0.0076$, e.g., CBZ) (Fig. 2) [1], with differences explained by chemical structure and/or physicochemical properties, thus influencing the susceptibility to undergo EO. In parallel, the applied treatment was effective in inactivating *E. coli* [(NMP/100 mL) $2.0 \text{ X } 10^5$ before treatment and < 1 after treatment], showing the capacity of EO to be used as an effective technique for wastewater treatment for further reuse.

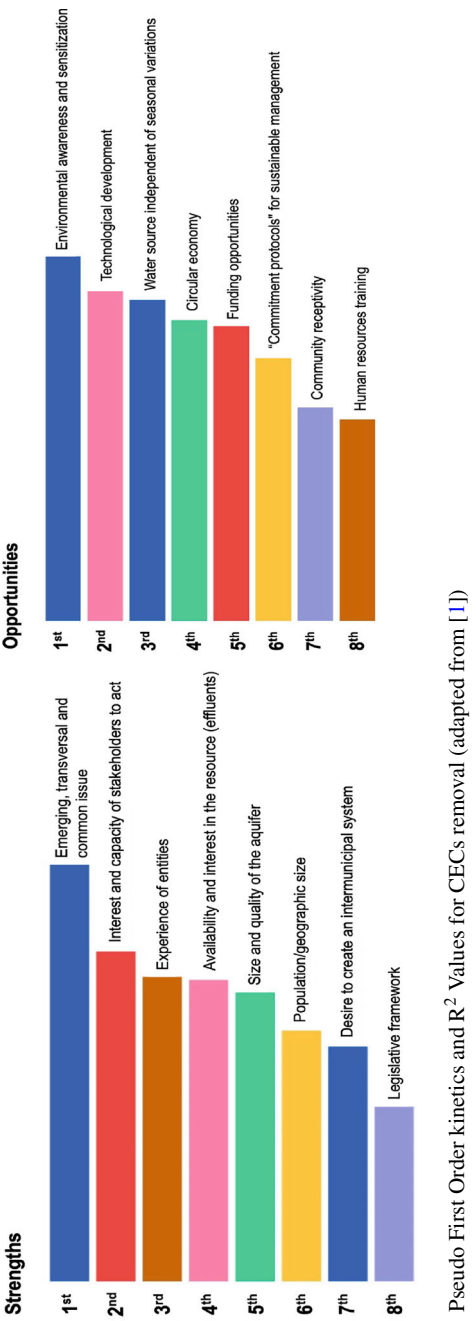

Fig. 2 Pseudo First Order kinetics and R^2 Values for CECs removal (adapted from [1])

3.2 Soil Irrigation with Treated Effluents and Electrokinetic Remediation

The article [4] showed that while soil pH remained more homogeneous in the On/Off + REP strategy, mitigating pH shifts across the anode, central, and cathode sections (approx. 6), soil conductivity showed higher variability. Natural attenuation (control) presented remediation above >30% for some CECs (e.g., SFM 57%, DCF 36%). EKR enhanced CECs removal by up to 20% compared to the control. The removals were inhomogeneous among the soil's sections (anode, central, cathode sides) with a tendency for higher CECs removal near the electrodes (>20%). This proves that the removal of CECs was associated with the microcosm location (anode, central, or cathode sections) and dependent on the CECs properties. The remediation of IBF and DCF was significantly influenced by the direction of the applied current—their migration toward the anode led to their accumulation in the central compartment. The combination of On/Off and REP proved to be the most effective strategy, as it preserved soil pH characteristics and enabled a more uniform removal of the studied EOCs under the tested conditions.

3.3 Stakeholders' Perception

The SOAR analysis (Fig. 3) allowed for the expression of stakeholder perception about wastewater reuse. For **strengths**, the stakeholders refer to the existence of diverse strategies to combat water scarcity as both an emerging and shared concern. They emphasized that there is adequate interest, expertise, and capacity within relevant entities to act effectively in this area. Key strategies and opportunities identified include raising environmental awareness, advancing technological development, and utilizing water sources independent of seasonal variations. To achieve these goals, participants highlighted the importance of improving water distribution systems, enhancing wastewater treatment infrastructure, and optimizing the quality-to-cost ratio of water services.

These actions are expected to lead to efficient and integrated water management, a reduction in the impacts of climate change, and the preservation of aquifers. The analysis highlights key differentiators, with a significant emphasis on innovation and technology development as the primary strengths. This is followed by effective stakeholder collaboration and existing expertise and experience in water management strategies, which are critical for addressing water scarcity. Additionally, environmental awareness and established governance frameworks are also identified as contributing factors. The participants identified **opportunities** centered around technological advancements and the promotion of environmental awareness and education. Other prominent opportunities include the optimization of resource allocation through improved water distribution systems and the development of infrastructure

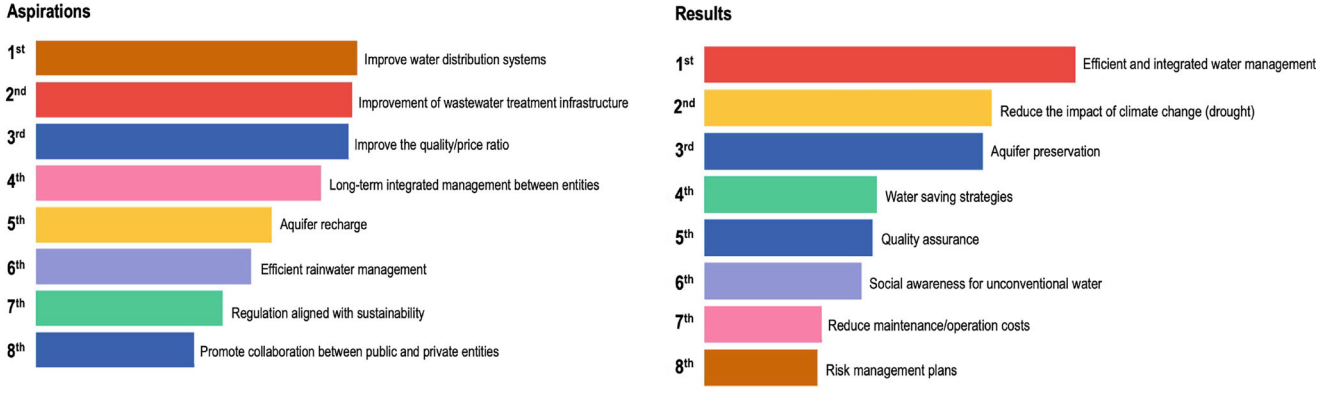

Fig. 3 Summary of SOAR analysis outcomes: stakeholder-identified strengths, opportunities, aspirations, and expected results related to wastewater reuse and water scarcity strategies

resilient to seasonal variations. These aspects collectively point toward a significant potential for leveraging innovation to enhance water security. The **aspirations** focus on achieving integrated and sustainable water management systems, alongside strategic initiatives to strengthen the resilience of water supply and treatment infrastructure. There is also a strong emphasis on creating a balance between cost-efficiency and quality of water services, as well as implementing proactive measures to mitigate the impacts of climate change. The expected **results** include efficient water resource management, reduced environmental impact, and preservation of aquifers. Other measurable results include enhanced stakeholder engagement and improved alignment with sustainable development goals, which collectively ensure long-term success in addressing water scarcity challenges. The stakeholders involved were representatives from local authorities, environmental agencies, research institutions, and water management companies. Responses were gathered during group workshops using a voting and prioritization process within the SOAR framework. The SOAR analysis underscores the importance of leveraging existing strengths in innovation and collaboration while capitalizing on opportunities to implement resilient and sustainable water management strategies. The aspirations and results highlight a shared vision of efficient, integrated systems that ensure water security, environmental preservation, and climate change mitigation.

4 Conclusion

Treated wastewater reuse for soil irrigation offers a promising solution to water scarcity but requires effective removal of contaminants of emerging concern (CECs). This study explored the complementary application of electrochemical oxidation (EO) and electrokinetic remediation (EKR) as integrated approaches for effluent treatment and in-situ soil remediation. EO achieved >90% removal of selected CECs using MMO/Ti circular mesh electrodes, with degradation following pseudo-first-order kinetics. EKR enhanced soil remediation by up to 20%, with effectiveness influenced by the contaminant's properties and location relative to the electrodes. The combined application of reversed electrode polarization and On/Off current strategies promoted a more uniform contaminant removal while preserving soil pH, validating the feasibility of EKR for sustainable soil management. Moving forward, the implementation of EO and EKR systems at a larger scale requires addressing several key challenges: (a) Optimization of operational parameters for different environmental conditions and contaminant profiles; (b) Long-term monitoring to assess stability, effectiveness, and environmental impacts post-treatment; (c) Development of modular, field-deployable systems to enhance scalability and cost-efficiency; (d) Integration of real-time monitoring tools to support adaptive management of treatment processes. In parallel, stakeholder engagement remains essential to ensure alignment with societal needs and regulatory expectations. The SOAR analysis emphasized the importance of innovation, collaboration, and knowledge sharing to build resilient and integrated water management strategies. Future

studies should further investigate how stakeholder-driven frameworks can guide technology implementation, particularly in regions facing increasing water stress and soil contamination.

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Transforming Waste to Solution: Moringa Seed Cake Biochar for Lead-Contaminated Soil Remediation

Chandresh H. Solanki and Bhoomi A. Kamdar

Abstract This study synthesized a novel biochar through the slow pyrolysis of moringa seed cake, a de-oiled cake obtained as a by-product of oil extraction, and analyzed its physicochemical properties. The results revealed that the biochar exhibits a high pH and abundant surface functional groups, making it a promising material for lead adsorption. The research further investigated the impact of biochar content (0% to 10% w/w) and curing time on stabilizing soil contaminated with lead at concentrations of 5,000 mg/kg and 10,000 mg/kg. The toxicity characteristic leaching test (TCLP) demonstrated that treatment with 10% w/w biochar combined with 28 days of curing reduced lead leachability to within regulatory limits, achieving over 89% immobilization efficiency. Additionally, soil strength and pH progressively increased with higher biochar content and longer curing durations, maintaining stability after 56 to 90 days. Microstructural analysis revealed the mechanisms responsible for effective lead stabilization, including lead precipitation, surface complexation with functional groups (C=O, O=C–O), and encapsulation in calcium silicate hydrates (C–S–H).

Keywords Biochar · Surface complexation · Lead contamination

1 Introduction

Lead (Pb) contamination in soil is a critical environmental issue caused by industrial activities such as smelting, mining, and improper waste disposal. Unlike organic pollutants, lead persists indefinitely in the environment, posing severe risks to human

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health, agriculture, and water quality [1]. Exposure to lead can lead to neurological disorders, developmental impairments in children, and cardiovascular diseases, making its remediation a priority.

Traditional remediation methods, such as excavation and chemical washing, are often costly and environmentally disruptive. Stabilization/solidification (S/S) techniques using binders like Portland cement are widely used but have high carbon footprints and sustainability concerns [2]. As an alternative, biochar produced through the pyrolysis of organic biomass has emerged as an eco-friendly and effective solution for immobilizing heavy metals in soil.

Several studies have demonstrated the effectiveness of biochar in heavy metal stabilization [3–5]. Current research reported that biochar derived from bagasse and orange peel increased soil pH and organic matter content, facilitating lead stabilization in contaminated soils [6]. Similarly, a study found that biochar produced from spent coffee grounds effectively immobilized lead in mine-contaminated soils, with a 10% biochar application achieving regulatory limits for leachability [7].

Moringa seed waste biochar (BC), derived from de-oiled moringa seed cake, is a promising material for lead stabilization due to its high alkalinity, porous structure, and functionalized surfaces [8]. It reduces lead bioavailability and leaching potential, improving soil stability and fertility. Through a series of laboratory experiments, this research investigates the physicochemical interactions between biochar and lead-contaminated soils, evaluates the immobilization efficiency, and assesses the long-term stability of biochar-treated soils. The findings will contribute to sustainable, cost-effective remediation strategies, promoting greener environmental practices for soil pollution management.

2 Materials and Methods

2.1 Materials

Uncontaminated soil was collected from an open site in Surat, India, and characterized according to ASTM standards. The soil had a specific gravity of 2.55, with 79% silt and clay-sized particles, classifying it as low-plasticity silt (ML). Biochar was produced by slow pyrolysis (350 °C and 6 h residence time) of de-oiled moringa seed cake. The biochar exhibited high alkalinity (pH 9.69), significant carbon content (68.03%), and a porous structure suitable for lead stabilization.

2.2 Methods

Lead-contaminated soil samples were prepared by artificially spiking soil with lead nitrate solution to achieve contamination levels of 5,000 mg/kg and 10,000 mg/

kg. The samples were stored for 14 days to ensure uniform contamination. The remediation process involved mechanically mixing contaminated soil with biochar in proportions of 5% and 10% by dry weight. The treated samples were molded into cylindrical specimens and subjected to curing for up to 56 days under controlled humidity and temperature.

Experimental analysis included:

- Unconfined Compressive Strength (UCS) to assess mechanical improvement.
- Soil pH measurement to evaluate alkalinity effects.
- Toxicity Characteristic Leaching Procedure (TCLP) to determine metal leaching potential.
- Sequential Extraction Procedure (BCR-SEP) to analyze metal fractionation.
- Fourier Transform Infrared Spectroscopy (FTIR), and Scanning Electron Microscopy with Energy Dispersive Spectroscopy (SEM–EDS) for microstructural analysis.

3 Results

3.1 Effect of Biochar Dosage on Soil pH Over Time

Figure 1 illustrates the variation in soil pH with different dosages of biochar over curing durations of 0, 7, 14, 28, and 56 days. The initial pH of untreated contaminated soil was 7.51 (5000 mg/kg) and 7.39 (10,000 mg/kg), indicating moderate alkalinity. However, biochar addition significantly increased pH at all curing stages due to its high alkalinity, attributed to inorganic compounds such as CaO, MgO, K_2O, and $CaCO_3$, along with basic functional groups like C-O and C=O [9]. After 28 days of curing, soil pH for Pb0.5% soil increased from 7.94 (untreated) to 8.55 (5% BC) and 8.6 (10% BC). A significant pH rise was observed during the first 14 days, particularly between 5 and 10% biochar amendments, after which the differences were minimal at 56 days. Similarly, for Pb1% soil, pH increased by 7%–8.42% within 7 days with 5% and 10% BC, while a 6.05%–7% increase was noted at 28 days compared to untreated soil. Regardless of curing duration, 10% biochar content consistently maintained the highest pH, confirming its superior alkalinity and stabilization potential. The findings highlight that biochar amendments provide sustained alkalinity, ensuring long-term stabilization of lead-contaminated soils.

3.2 Influence of Biochar on Unconfined Compressive Strength of Lead-Contaminated Soil

The unconfined compressive strength (UCS) of soil samples treated with varying concentrations of biochar (BC) and heavy metal contamination levels was analyzed

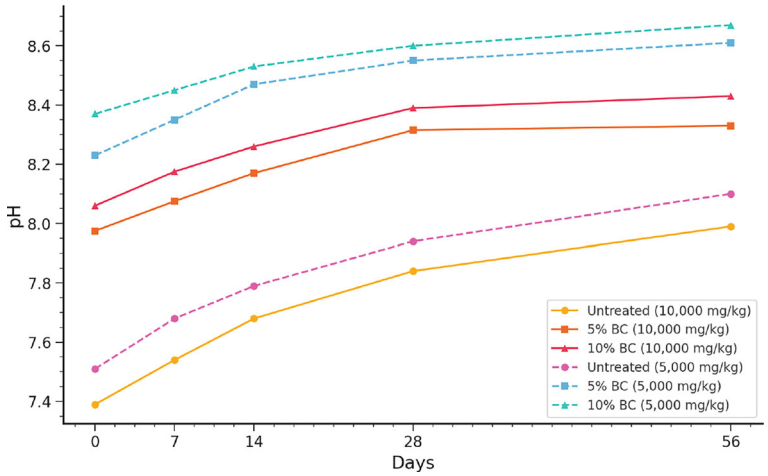

Fig. 1 pH of contaminated and treated lead soils

over different curing periods, as depicted in Fig. 2. It is evident that for both contamination levels (10,000 mg/kg and 5,000 mg/kg of heavy metals), untreated soil exhibits the lowest UCS values throughout the curing period. This suggests that heavy metal contamination impairs soil strength development, likely due to the disruption of soil particle bonding and hindered cementation reactions. At 56 days of curing, untreated soil with 10,000 mg/kg contamination exhibits a UCS of approximately 700 kPa, while the untreated soil with 5,000 mg/kg contamination slightly surpasses it, reaching around 850 kPa.

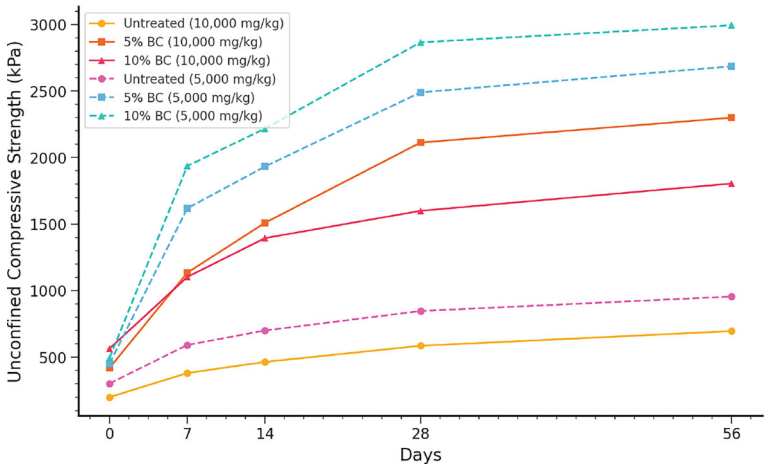

Fig. 2 Unconfined compressive strength of contaminated and treated lead soils

The addition of biochar significantly enhances the UCS values, with higher biochar dosages yielding better results. The 5% BC-treated samples show noticeable improvements in compressive strength compared to untreated soil at both contamination levels. For the 10,000 mg/kg contaminated samples, the UCS progressively increases, reaching approximately 2,000 kPa after 56 days. Similarly, for the 5,000 mg/kg contaminated samples, the UCS reaches nearly 2,500 kPa by the final curing period. The enhancement in strength can be attributed to the improved soil structure and binding capacity induced by biochar particles, which promote the agglomeration of soil particles and improve load-bearing capacity. A further increase in biochar content to 10% yields even more pronounced strength improvements. For the 10,000 mg/kg contamination level, the UCS steadily increases and reaches around 2,500 kPa at 56 days, whereas for the 5,000 mg/kg contamination level, the UCS reaches its peak at approximately 3,000 kPa. Biochar contributes to strength gain through its high specific surface area and porous nature, which facilitates particle interaction, reduces void ratios, and enhances cementitious bonding. Additionally, the ability of biochar to adsorb heavy metals may contribute to a reduction in their adverse effects on soil strength, thus explaining the observed improvements.

The rate of strength gain over time follows a similar trend across all treatments, with a rapid increase in UCS during the initial 7 to 14 days of curing, followed by a slower rate of strength gain up to 56 days. This is consistent with the hydration and pozzolanic reactions that contribute to strength development in treated soils [10]. The initial rapid gain is likely due to early-stage particle bonding facilitated by biochar, while the gradual increase over time suggests progressive cementation and consolidation effects.

3.3 Effect of Biochar on Lead (Pb) Leachability in Contaminated Soil Over Time

The reduction in lead (Pb) concentration in soil leachate over time, as shown in Fig. 3, highlights the effectiveness of biochar (BC) amendments in mitigating Pb leachability in contaminated soils. The graph presents Pb concentration trends for untreated and biochar-treated soils under two different contamination levels (5,000 mg/kg and 10,000 mg/kg). It is evident that biochar application significantly reduces Pb leaching, with higher biochar dosages yielding better immobilization effects. Untreated soils exhibit the highest Pb concentrations in the leachate throughout the curing period. The untreated sample with 10,000 mg/kg contamination starts at approximately 120 mg/L and progressively decreases over time, but remains substantially higher than other treatments, reaching around 30 mg/L at 56 days. Similarly, the untreated soil with 5,000 mg/kg contamination starts at approximately 60 mg/L and decreases gradually to about 20 mg/L after 56 days. The relatively slow reduction in Pb concentration in untreated samples suggests minimal natural attenuation or adsorption in the absence of stabilizing agents.

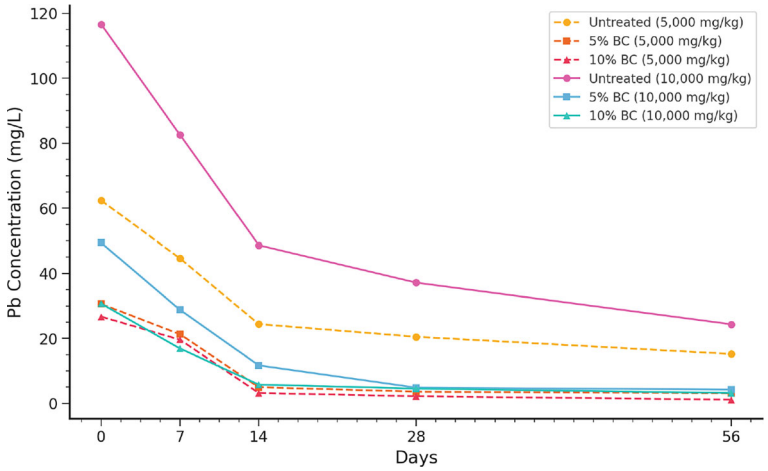

Fig. 3 TCLP test results of contaminated and treated lead soils

The inclusion of 5% BC significantly reduces Pb leaching across both contamination levels. At the highest contamination level (10,000 mg/kg), the Pb concentration initially drops from around 50 mg/L to nearly 5 mg/L by 28 days, maintaining low levels thereafter. Similarly, for the 5,000 mg/kg contaminated soil, Pb leaching decreases sharply within the first two weeks and stabilizes below 5 mg/L. The 10% BC amendment demonstrates the highest Pb immobilization efficiency. In both contamination levels, Pb concentration in the leachate reduces rapidly, reaching nearly undetectable levels by 28 days and remaining stable through 56 days.

3.4 Effect of Biochar on Pb Fractionation in Contaminated Soil

The distribution of lead (Pb) in different geochemical fractions of contaminated soil with and without biochar (BC) treatment is illustrated in Fig. 4. The Pb fractionation is categorized into four fractions: acid soluble (F1), reducible (F2), oxidizable (F3), and residual (F4) [11]. The results indicate a significant shift in Pb distribution upon biochar application, demonstrating its ability to immobilize Pb in more stable fractions.

In untreated soils, the majority of Pb is present in the exchangeable fraction (F1), with values exceeding 70% in highly contaminated samples (10,000 mg/kg). This indicates that Pb remains in a highly mobile and bioavailable form, posing serious environmental risks. The application of biochar reduces the proportion of Pb in the F1 fraction, with 10% BC treatment showing the most significant reduction. Correspondingly, there is an increase in Pb content in more stable fractions, particularly

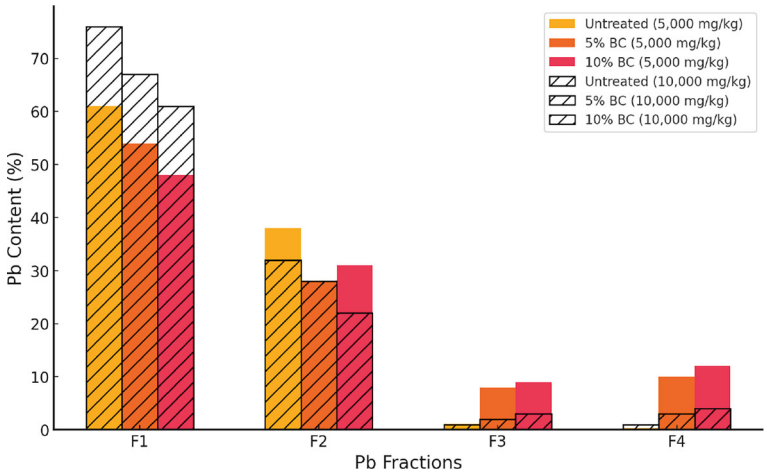

Fig. 4 BCR-SEP fractionation of contaminated and treated lead soils

F3 and F4, which represent less bioavailable forms. The reduction in exchangeable Pb suggests that biochar enhances Pb stabilization, likely through adsorption, complexation, and precipitation mechanisms.

3.5 Microstructural Studies

Figure 5 illustrates the FE-SEM-EDS analysis of 10% BC stabilized Pb-contaminated soil for both concentrations. After biochar amendment, the micrographs displayed a more flocculated structure, indicating improved soil stabilization. The integration of calcium silicate hydrate (C-S-H) gel in biochar-treated soil led to a more compact structure with fewer pores compared to untreated soil, suggesting enhanced encapsulation of soil particles [9]. EDS microanalysis further validated the SEM findings, showing an increase in calcium content to 10.10% in Pb0.5% contaminated soil when treated with 10% biochar. Additionally, lead content in the soil was reduced by 41.15% for 10% biochar amendment compared to the untreated sample, demonstrating the effectiveness of biochar-based stabilization/solidification. A similar trend was observed in Pb1% stabilized soil, further supporting the role of biochar in reducing lead mobility and enhancing soil stabilization.

The FTIR spectra (Table 1) highlight the main functional groups in untreated and biochar-stabilized Pb-contaminated soils after 28 days of curing. In untreated Pb0.5% and Pb1% soils, the O-H stretching and bending peaks appeared at 3618 cm^{-1} and 3622 cm^{-1}, respectively [12]. After biochar stabilization, these peaks shifted to lower wavenumbers (3610–3614 cm^{-1}), indicating interactions between hydroxyl groups and biochar. A strong peak around 1028 cm^{-1} in untreated soil, shifting to

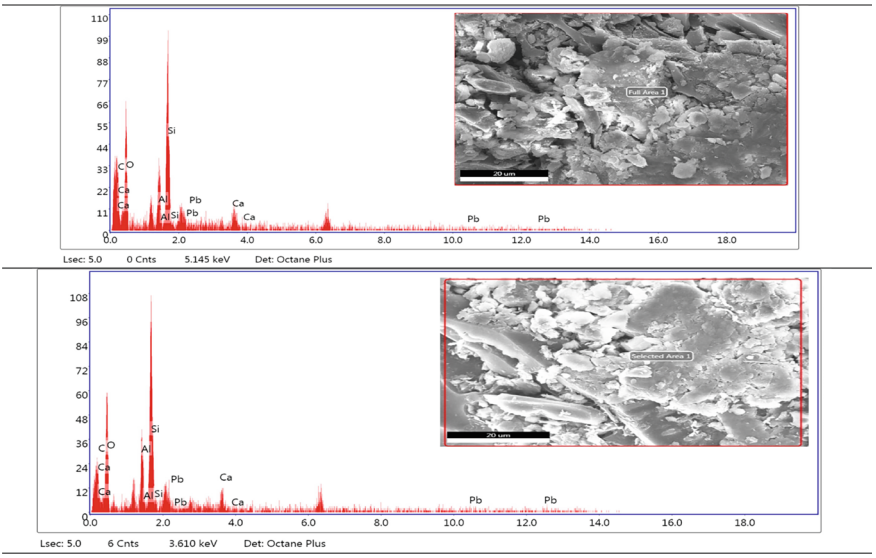

Fig. 5 SEM–EDS of contaminated and 10% BC-treated lead soils

1002–1010 cm^{-1} in treated samples, corresponds to asymmetric Si-O-Si/Al vibrations, suggesting C-S-H gel formation due to hydration reactions. Similarly, the C-O stretching band associated with carbonate formation shifted from 1512–1516 cm^{-1} in untreated soil to 1508–1520 cm^{-1} in biochar-treated samples, signaling the precipitation of lead carbonate minerals through surface complexation [13]. Additional peaks between 2874–2881 cm^{-1} (C-H stretching), 650–655 cm^{-1} (Si-O-Si/Al symmetric stretching), and 459–470 cm^{-1} (O-Si-O bending) further confirm the presence of hydration products.

Table 1 FTIR spectra of untreated and BC-treated soils

Functional Group	Wave Number (cm^{-1}) Untreated Soil	Wave Number (cm^{-1}) 5%BC Treated Soil	Wave Number (cm^{-1}) 10% BC Treated Soil
O-H stretching (Hydroxyl group)	3622	3610–3618	3606–3614
O-H bending (Hydroxyl group)	1690	1688	1685
Si-O-Si/Al asymmetric stretching	1028	1010	1002
C-O stretching (Carbonates)	1516	1511–1512	1508–1520
C-H stretching	2874–2881	2874–2881	2874–2881
Si-O-Si/Al symmetric stretching	650–655	650–655	650–655
O-Si-O bending	459–470	459–470	459–470

4 Conclusion

The research findings confirm that MSBC, produced through slow pyrolysis, possesses significant physicochemical properties, including high alkalinity, porous structure, and abundant surface functional groups, which enhance its capacity to immobilize lead (Pb) in contaminated soils. The study's experimental results revealed that the addition of biochar to lead-contaminated soils significantly increased soil pH, reducing Pb leachability through a combination of adsorption, precipitation, and encapsulation mechanisms. The toxicity characteristic leaching procedure (TCLP) showed that soils treated with 10% biochar achieved an immobilization efficiency of over 89%, reducing lead leaching to within regulatory limits. Furthermore, biochar-amended soils exhibited sustained pH elevation over time, providing long-term stabilization benefits. The strength gain was attributed to biochar's role in promoting soil particle agglomeration and enhancing cementitious bonding through pozzolanic reactions. The progressive increase in UCS over 56 days demonstrated the durability of biochar-treated soils, making them suitable for long-term stabilization and potential reuse in construction applications. Microstructural investigations using scanning electron microscopy with energy-dispersive spectroscopy (SEM–EDS) and Fourier transform infrared spectroscopy (FTIR) further confirmed the effectiveness of biochar in lead stabilization. The formation of calcium silicate hydrate (C-S-H) gels, along with the shift in FTIR functional group peaks, provided strong evidence of lead entrapment within stable mineral phases. Overall, this study provides strong evidence supporting the use of moringa seed cake biochar as a low-cost, environmentally friendly, and efficient remediation material for lead-contaminated soils. By enhancing soil stability, reducing heavy metal mobility, and improving mechanical properties, MSBC offers a viable alternative to conventional remediation methods.

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Triaxial Performance of Geopolymer-Stabilized Industrial Waste for Pavement Sub-base Applications

Kuldeep Sharma and Arvind Kumar

Abstract Using industrial waste as a construction material reduces the consumption of natural materials and saves the environment. Copper slag (CS) and rice husk ash (RHA) treated with geopolymer have been developed as composite materials. Geopolymer is a good alternative binding material compared to traditional binders (lime and cement). It may be cost-effective and have low carbon emissions. It may contribute to environmental conservation. In this paper, triaxial (unconsolidated undrained) tests have been performed on a developed composite material. Deviator stress at failure (kPa), elastic modulus (MPa), and modified failure envelopes have been examined. Triaxial tests were performed at three different confining pressures, i.e., 50 kPa, 100 kPa, and 150 kPa. Specimens have been tested at different periods of curing, i.e., 3, 7, 14, and 28 days. Results indicate that deviator stress at failure (kPa), elastic modulus (MPa), and shear strength parameters (c and ϕ) increased with increasing the curing period and confining pressure. It is concluded that geopolymer is a good binder for CS and RHA. Developed composite materials may be used as construction materials for flexible pavement (sub-base) layers.

Keywords Copper slag · Geopolymer · Rice husk ash · Shear strength parameters · Triaxial test

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1 Introduction

The increasing environmental concerns associated with conventional cement-based materials have prompted the exploration of alternative binders that can reduce carbon emissions without compromising mechanical performance and durability. Geopolymer technology, which involves the alkali activation of aluminosilicate materials, presents a promising alternative. This binder, first introduced by Davidovits [1], has garnered significant attention due to its low carbon footprint, high strength, and excellent durability, making it a superior option compared to ordinary Portland cement [2].

In recent years, the incorporation of industrial by-products, such as CS and RHA, into geopolymer composites has gained extensive interest. These by-products not only enhance the mechanical properties of geopolymers but also contribute to sustainable waste management practices [3]. Copper slag, a waste material generated during the pyro-metallurgical refining of copper, is rich in silica (SiO_2), alumina (Al_2O_3), and iron oxide (Fe_2O_3), making it a promising precursor for geopolymerization. Similarly, RHA, a by-product of rice milling, is abundant in amorphous silica and exhibits high pozzolanic activity, which is beneficial for the formation of a strong geopolymer matrix [4]. By integrating industrial by-products, this approach not only reduces reliance on natural resources but also significantly lowers the carbon footprint of construction activities [5]. When these materials (CS and RHA) are activated using an alkaline solution (sodium silicate and sodium hydroxide), they undergo a chemical transformation that results in the formation of a three-dimensional polymeric network, imparting superior strength and stability. This combination of CS and RHA in geopolymer matrices has been shown to enhance the mechanical properties of the composite, making it suitable for geotechnical applications such as pavement sub-bases, soil stabilization, and embankment construction. One of the key properties influencing the effectiveness of geopolymer-stabilized CS-RHA composites in geotechnical applications is shear strength. The shear strength parameters, including cohesion (c in kPa) and the internal friction angle (ϕ in degrees), play a critical role in determining the material's stability and load-bearing capacity. This study investigates the shear strength properties of CS-RHA geopolymer composites through triaxial tests at varying confining pressures (50, 100, and 150 kPa) and curing durations (3, 7, 14, and 28 days).

The findings of this study also highlight the superior shear strength of CS-RHA geopolymer composites compared to traditional cement and lime-stabilized materials. This demonstrates the potential of CS-RHA mix with an AA as a promising alternative for ground improvement, pavement construction, and other geotechnical applications.

2 Materials and Methods

2.1 Materials

Copper Slag: It was procured from Hindustan Copper Limited, Gujarat, India. It is a cohesionless material with a high internal friction angle, making it suitable for use in alternative materials and geotechnical applications. The physico-chemical properties of CS are presented in Tables 1 and 2. The raw CS used in this study is shown in Fig. 1a.

Rice Husk Ash: It was collected from Jackpot Rice Mill, Jalandhar, India. It is a cohesionless material with low specific gravity and contains a high amount of amorphous silica, which enhances its pozzolanic activity. The physico-chemical properties of RHA are listed in Tables 1 and 2. A raw image of RHA is depicted in Fig. 1b.

Table 1 Physical properties of CS and RHA [7]

Major properties	Materials	
	CS	RHA
Color	Black, Blackish gray	Black
Plasticity	Non-plastic	Non-plastic
Swelling	Non-swelling	NT
Particle size >425 μm >300 μm >150 μm >75 μm <75 μm	– – 4.16% 91.07% 3.65% 1.12%	– 0.23% 0.32% 47.35% 5.760% 46.34%
Cu	1.534 (Poorly graded)	7.08
Cc	0.960	0.44
Specific gravity	3.50	1.45
OMC	11%	53%
MDD	2.232 g/cc	0.861 g/cc
Fineness modulus	2.94	1.43

Table 2 Major chemical compounds of CS and RHA [7]

Major compounds	Percentage (%)	
	CS	RHA
Fe_2O_3	40.77	1.70
SiO_2	35.22	91.90
CaO	6.79	0.688
Al_2O_3	2.80	1.38
MgO	1.75	0.683

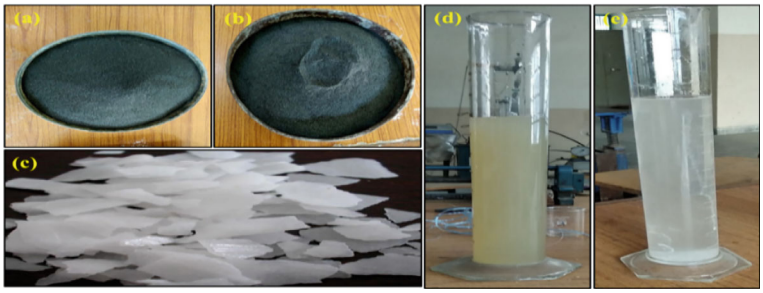

Fig. 1 **a** CS, **b** RHA, **c** Flack SH, **d** SS solution, **e** SH solution

Geopolymer (Alkali Activator): A geopolymer binder was prepared using sodium hydroxide (SH) and sodium silicate (SS) as alkali activators. The NaOH solution was prepared with a 10 M molarity concentration, and the SS to SH ratio was maintained at 2.5:1 to ensure optimal geopolymerization. The images of SH and SS are shown in Fig. 1c–e. The preparation procedure of the AA solution has been detailed in a previous study [6]. A comprehensive discussion on the properties and behavior of CS, RHA, and the AA (geopolymer solution) has been provided in previous studies [6, 7].

2.2 Methods

A triaxial test under Unconsolidated Undrained (UU) conditions was conducted in compliance with the ASTM D2850-07 standard to analyze the shear strength and stress–strain response of the composite material [8]. UCS specimens, measuring 38 mm in diameter and 76 mm in height, were prepared while maintaining an aspect ratio of 1:2 to ensure consistency in testing and accurate strength assessment. The sample preparation process, including compaction and curing methods, has been previously documented by the same authors [7]. The triaxial test apparatus and samples are depicted in Fig. 2a, b.

3 Results and Discussion

The results of the triaxial test (unconsolidated undrained) for the composite mix (CS, RHA, and alkali activator) are discussed in the following subsections. The unconsolidated undrained triaxial test was selected for this study due to its suitability for assessing the short-term strength behavior of the developed composite material under rapid loading conditions. The discussion includes the composite mix's stress–strain

Fig. 2 **a** Digital Triaxial test setup; **b** Triaxial samples

behavior, deviator stress at failure, elastic modulus, and shear strength parameters, providing insights into its mechanical performance under varying conditions.

3.1 Stress–Strain Behavior at Various Curing Periods and Confining Pressures

Figure 3 presents the stress–strain behavior of a composite mix containing 90% CS, 10% RHA, and 0.25% B (total base material: CS and RHA) under different confining pressures (50, 100, and 150 kPa) and varying curing durations of 3, 7, 14, and 28 days. The results indicate that there is an increase in strength with increasing curing period, as the 28-day cured samples exhibit the highest deviator stress at failure. This can be attributed to the ongoing pozzolanic reactions and binder hydration, which enhance the overall strength and stiffness of the composite material. Additionally, higher confining pressures contribute to increased peak deviator stress, with the samples tested at 150 kPa displaying significantly greater strength compared to those at 50 kPa, indicating improved load-bearing capacity under higher lateral confinement.

The stress–strain curves demonstrate an initial elastic phase followed by strain hardening and eventual softening beyond the peak stress. The post-peak reduction in deviator stress is more pronounced for specimens with shorter curing durations, suggesting that prolonged curing improves the material's ductility and resistance to failure. Comparing these results with the lime-stabilized composite mix from the previous study [7, 9], it is evident that the geopolymer-stabilized mix exhibits superior strength and deformation capacity. The peak stress values are significantly higher, highlighting the enhanced bonding effects of the geopolymer binder. These findings confirm that geopolymer-based stabilization effectively improves the mechanical performance of the composite mix, making it a viable and sustainable alternative to traditional cement and lime stabilization in geotechnical applications.

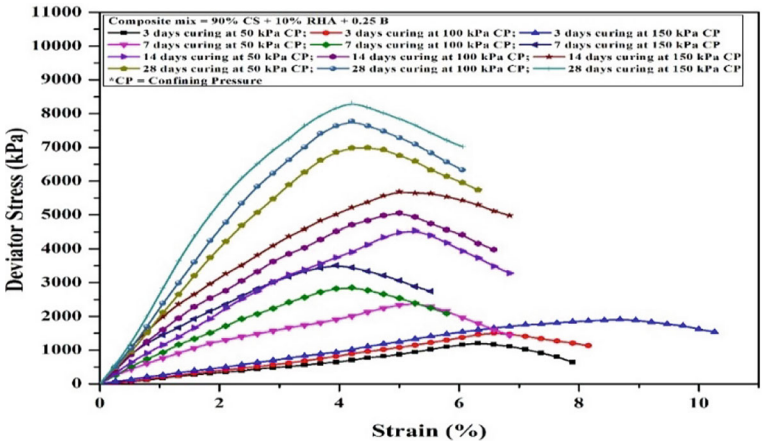

Fig. 3 Deviator stress (kPa) vs. strain (%) for composite mix (90% copper slag, 10% rice husk ash, 0.25B) at various periods of curing and cell pressures

3.2 Deviator Stress at Failure for Different Curing Periods and Confining Pressures

Figure 4 illustrates the variation in deviator stress at failure concerning confining pressure for a composite mix consisting of 90% CS, 10% RHA, and 0.25% B (total base material: CS and RHA) under various periods of curing (3, 7, 14, and 28 days). The results indicate a consistent increase in deviator stress with prolonged curing time and higher cell pressure. Specimens cured for 28 days exhibit the highest deviator stress, demonstrating a significant strength enhancement due to prolonged pozzolanic reactions and improved binder activation. For the 90CS-10RHA-0.25B composite mix, the percentage increase in deviator stress at failure at 50 kPa cell pressure is 97.40% (7 days), 276.01% (14 days), and 478.83% (28 days) compared to the 3-day curing period. Similarly, at 100 kPa cell pressure, the increase is 88.94% (7 days), 235.06% (14 days), and 415.42% (28 days), while at 150 kPa cell pressure, the values are 83.33% (7 days), 196.89% (14 days), and 334.44% (28 days). These findings confirm that curing duration significantly influences strength development, with longer curing periods resulting in higher deviator stress. The observed trend highlights that higher confinement pressure enhances strength, as lateral support minimizes excessive deformation and improves load resistance. This effect is particularly evident in specimens cured for only 3 days, which exhibit the lowest strength, whereas those cured for 7, 14, and 28 days show a gradual increase in deviator stress, reinforcing the time-dependent nature of strength development in geopolymer-stabilized composites. The geopolymerization process is inherently slow, and as the curing period extends, a greater quantity of N-A-S-H and C-A-S-H gels is formed. These gels effectively bind CS and RHA particles, leading to increased stiffness and failure stress. A similar linear correlation between deviator stress at failure and elastic

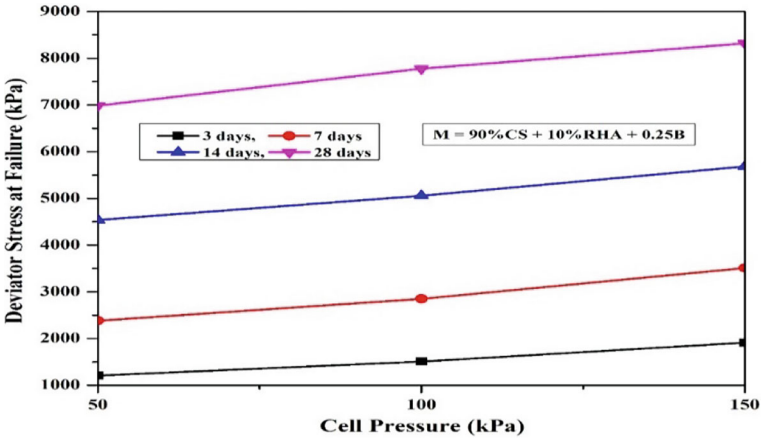

Fig. 4 Deviator stress (kPa) at failure vs. cell pressure (%) for composite mix (90% copper slag, 10% rice husk ash, 0.25B) at various curing periods

modulus with confining pressure has been observed in various pavement materials [10, 11].

A comparison with lime-stabilized samples further demonstrates that the geopolymer-treated composite consistently achieves higher deviator stress at failure across all curing durations and confining pressures. This suggests that geopolymerization significantly enhances particle bonding, leading to superior mechanical performance. These findings underscore the potential of geopolymer-based stabilization as a sustainable [6, 12] and high-performance alternative to traditional cement and lime treatments, particularly for applications requiring enhanced strength and durability in geotechnical engineering.

3.3 Elastic Modulus Variation with Curing Period and Confining Pressure

Figure 5 illustrates the variation in elastic modulus (MPa) with cell pressure (kPa) for a composite mix comprising 90% CS, 10% RHA, and 0.25% B (total base material: CS and RHA) at different periods of curing (3, 7, 14, and 28 days). The results indicate that the elastic modulus increases with both curing time and confining pressure, signifying improved stiffness and deformation resistance over time.

For all curing durations, the modulus of elasticity increases linearly with cell pressure. Increased specimen confinement reduces lateral strain, allowing the sample to withstand axial stress at a lower strain level, thereby increasing the elastic modulus. For the 90CS-10RHA-0.25B composite mix, the percentage increase in elastic modulus at 50 kPa cell pressure is 137.17% (7 days), 351.83% (14 days), and 718.43% (28 days) compared to the 3-day curing period. Similarly, at 100 kPa cell pressure,

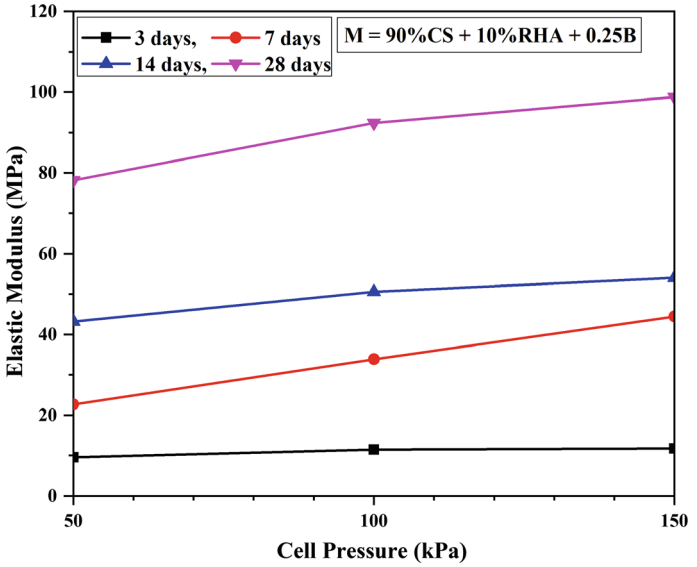

Fig. 5 Elastic modulus (MPa) versus strain (%) for composite mix (90% copper slag, 10% rice husk ash, 0.25B) at various periods of curing and confining pressures

the increase is 195.20% (7 days), 340.80% (14 days), and 705.32% (28 days), while at 150 kPa cell pressure, the values are 278.53% (7 days), 360.39% (14 days), and 741.74% (28 days). These findings confirm that curing duration and confinement pressure significantly impact material stiffness and load-bearing capacity.

As shown in Fig. 5, the elastic modulus increases with the curing period, which aligns with the slow nature of geopolymerization reactions. As curing progresses, a greater quantity of N-A-S-H and C-A-S-H gels is produced, effectively binding the CS and RHA particles, leading to enhanced stiffness and failure stress. Similar linear relationships between deviator stress at failure and elastic modulus with cell pressure have been reported for various pavement materials [10, 11].

At 3 days of curing, the material exhibits the lowest elastic modulus, suggesting incomplete geopolymerization and inadequate bonding between particles. However, as the curing period extends to 7, 14, and 28 days, a significant increase in stiffness is observed, with the 28-day cured specimens displaying the highest modulus values. This confirms the effectiveness of prolonged geopolymerization in enhancing material performance. Furthermore, as cell pressure increases from 50 to 150 kPa, the elastic modulus rises across all curing durations, reinforcing the role of confinement in improving load-bearing capacity.

Comparing these findings with lime-stabilized samples, the geopolymer-based composite exhibits substantially higher elastic modulus values, underscoring its superior stiffness and durability. The enhanced performance of geopolymer-treated samples is attributed to stronger gel formation and improved particle interlocking,

making it a viable alternative to traditional stabilization techniques in geotechnical applications.

3.4 Modified Failure Envelopes for Different Curing Periods and Confining Pressures

The Mohr–Coulomb failure envelope for the geopolymer-stabilized composite mix (90 CS + 10RHA + 0.25B) at different curing durations (3, 7, 14, and 28 days) is presented in Fig. 6, with cohesion (c) and the angle of internal friction (ϕ) for each curing period provided in the inset table. The results indicate that both cohesion and the angle of internal friction increase with a longer curing period, enhancing shear strength and stability. For the 90CS-10RHA-0.25B composite mix, the percentage increase in cohesion compared to 3-day curing is 105.86% (7 days), 277.31% (14 days), and 430.19% (28 days). Similarly, the angle of internal friction increases by 2.74% (7 days), 4.54% (14 days), and 6.09% (28 days). This increase in cohesion is primarily attributed to enhanced particle bonding due to the geopolymerization process, which plays a more significant role in shear strength development than frictional resistance. The data further demonstrate that cohesion increases significantly with curing time, rising from 76.16 kPa at 3 days to 403.79 kPa at 28 days, indicating improved load-bearing capacity and resistance to shear failure. In contrast, the angle of internal friction (ϕ) shows only a marginal increase, from 38.75° at 3 days to 40.51° at 14 days, stabilizing at 41.11° at 28 days. This trend confirms that cohesion is the dominant factor contributing to shear strength enhancement. The linearity of the failure envelopes across different curing periods confirms that the geopolymer-stabilized composite follows Mohr–Coulomb behavior, making it highly suitable for geotechnical applications requiring long-term strength gain. Compared to traditional lime stabilization, the higher cohesion and sustained strength development of geopolymer-treated samples highlight their potential as a durable and sustainable alternative for soil stabilization and ground improvement projects.

4 Conclusion

- Deviator stress increases with curing time and confining pressure, indicating enhanced strength and ductility. The highest deviator stress is observed at 28 days due to geopolymerization. Improved strain at peak stress enhances deformation capacity and energy absorption, improving overall mechanical performance.

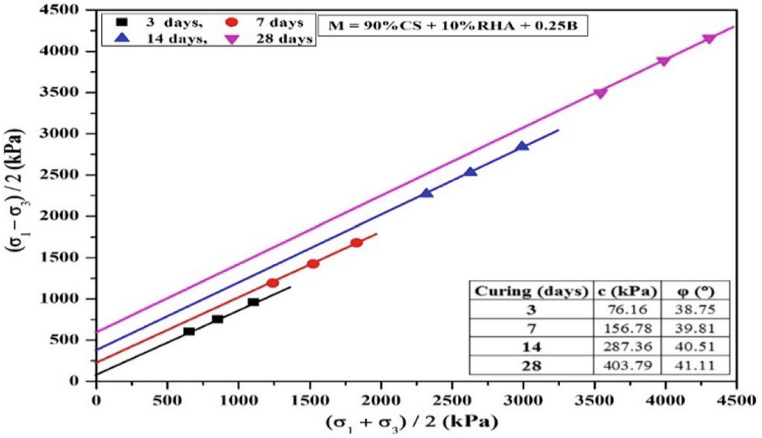

Fig. 6 Modified failure envelope for composite mix (90% copper slag, 10% rice husk ash, 0.25B) at various curing periods and confining pressures

- Strength increases with confining pressure and curing duration, indicating improved particle bonding. Moderate strength gain occurs at 3 and 7 days, while a significant increase at 14 and 28 days highlights progressive strength development. Continuous geopolymerization enhances mechanical properties, making the composite more suitable for load-bearing applications.
- Deviator stress at failure and elastic modulus increase with curing time and confining pressure, enhancing stiffness and resistance to deformation. The formation of N-A-S-H and C-A-S-H gels strengthens particle bonding, improving strength and load-bearing capacity over time.
- The geopolymer-stabilized composite shows a significant increase in cohesion with curing time, rising from 76.16 kPa at 3 days to 403.79 kPa at 28 days, while the angle of internal friction stabilizes at 41.11°. The Mohr–Coulomb failure envelope confirms its suitability for geotechnical applications, making it a sustainable alternative to lime stabilization for soil improvement.
- Geopolymer-stabilized CS-RHA composites show potential as sustainable, high-performance materials for geotechnical applications such as flexible pavements, embankments, and soil stabilization. Optimizing these formulations can contribute to greener and more resilient infrastructure solutions.

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Radon and CO_2 Content in Puerto Naos After the Volcanic Eruption on the Island of La Palma: Recalculation with Artificial Intelligence

Wolfgang Wehr and Roger P. Frey

Abstract Radon and CO_2 measurements on the ground floor of a building were combined with meteorological data and analyzed using artificial intelligence. Neural networks from the field of machine learning were used. After training the neural networks, a high level of accuracy was achieved in both the training and the prediction. A correlation matrix was used to identify the most important parameters influencing the radon and CO_2 contents. Wind speed, wind gusts, and wind direction have the greatest influence on the radon and CO_2 values: the more wind, the lower the gas values. An evaluation over the course of the day revealed maximum gas levels at night, when air temperatures are at their lowest. Northern winds lead to the highest gas values due to suction effects.

Keywords Radon · CO_2 · Artificial neural networks

1 Introduction

Details on the topic of artificial intelligence in geotechnics and how neural networks work can be found in Wehr [1].

Soil gas measurements of radon play an important role as a basis for measurements in buildings to predict building radon concentrations. Two examples have already shown how important it is to take soil parameters such as water content and grain size into account [1, 2]. However, if a building is situated directly on the bedrock, there is no need for soil gas measurements.

Using an example from the Canary Island of La Palma, indoor measurements in a non-ventilated room are to be evaluated with neural networks. Following the

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volcanic eruption from September to December 2021, elevated radon and CO_2 levels were measured in the towns of Puerto Naos and La Bombilla. The reason for this is a new fault zone in the rock with increased gas permeability. The main problem is the excessively high CO_2 values of up to 4% or 40000 ppm. Locations were evacuated to avoid endangering people. CO_2-values of up to 100% are quite possible directly when the gases escape.

2 Setup and Data Preparation

The radon and CO_2 values were measured in the paragliding school at Juana Tabares road N°3 on a table approximately 80 cm above the floor and 1.5 m from the shop window. The room is not ventilated, but the sliding door and windows are not very tight. The ground floor of the house is approx. 1 m below the street level. There is an underground garage under the ground floor, which has a permanent connection to the outside air.

The street slopes down from the house to the beach 100 m away. There is no groundwater, but there is seawater in the rock fissures at high tide. This also explains the parallel course of the CO_2 with the tidal range [3].

The radon measurements were taken every hour from 10th November 2023 using a Radon Eye Plus. The data can be viewed online at https://radon-net.com/auth/radonmap.html, see Fig. 1.

A Dioxcare CX700 with NDIR sensor and a measuring range of up to 10,000 ppm was used as the reference device, and an Imbomsi CO_2 measuring device with NDIR sensor and measuring range of up to 5,000 ppm was used as the control device (Fig. 2). The Dioxcare was periodically calibrated in fresh air to correct for any sensor drift.

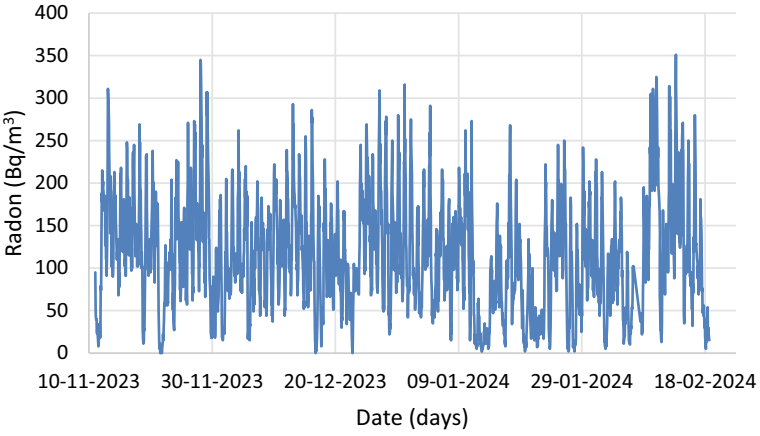

Fig. 1 Measured Radon over time

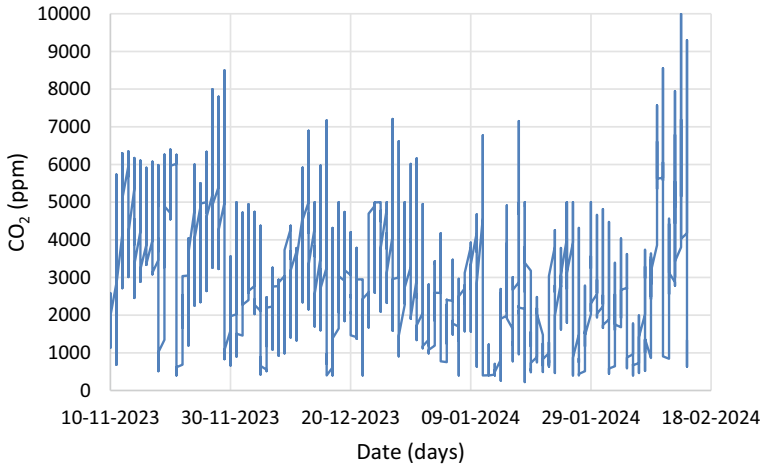

Fig. 2 Measured CO_2 over time

To obtain additional meteorological data, the Davis Vantage Pro2 Plus weather station was installed on the neighboring building (Martín apartment building) in Puerto Naos. It is freely exposed to all wind directions and has been providing data via Wunderground (https://www.wunderground.com/dashboard/pws/ILOSLL12) to the internet since mid-2020. The recorded parameters are: temperature Temp (C), dew point Dew (C), humidity Hum (%), wind direction Wdir, wind speed Wspeed (km/h), wind gust Wgust (km/h), air pressure Pres (hPa), precipitation Rain (mm), UV radiation UV and solar radiation Solar (W/m^2).

In previous measurement data evaluations [4], it has proven useful to use additionally derived measured variables:

(1) Temperature difference day/night = maxT - minT = delta_L
(2) Temperature difference air/ground = aveT - ground temperature = delta_LB
(3) Pressure difference high pressure/low pressure = delta_p.

3 Calculations with Artificial Neural Networks

The calculations were carried out using the Anaconda program package and the Spyder program for entering the programming code in the Python language. The measurement data was imported using Pandas.

The hyperparameters are described here as an example:

```
n_in = 22 # Input neurons
n_hid1 = 440 # Neurons hidden layer 1
n_hid2 = 440 # Neurons hidden layer 2
n_hid3 = 22 # Neurons hidden layer 3
n_out = 1 #neurons
```

bat_size = 1 # Batch size
n_epochs = 300

There are therefore 22 input values (Input neurons), 3 hidden neuron layers (Neurons hidden layer) with different numbers of neurons, and 1 output value (Radon). Each line of the measured values is taken into account (batch size), and everything is calculated 300 times (n_epochs). It is very important that all hyperparameters are optimized according to a specific strategy.

4 Results of the Calculations

Hourly calculations for the entire time resulted in a coefficient of determination R^2 of 0.98 for training, but a maximum of 0.90 for testing (forecast) despite all optimizations. Compared to other measurement data evaluations, this is very good, which speaks for the quality of the measurement data. 90% of all values were used as training data. It was crucial to use not only the radon values and the meteorological parameters, but also the CO_2 values. It is scientifically proven that CO_2 is the carrier gas for radon and carries radon out of the soil.

Figures 3 and 4 compare the measured and predicted CO_2 values during training and testing. The solid line corresponds to 0% deviation, and the dashed lines to + 30%/−30% deviation. Of particular interest is which influencing factors determine the results. A correlation matrix (heatmap) was used for this purpose, whereby the correlation between the input and output values was examined. Value from minimum 0 no correlation to 1/−1 results in 100% correlation.

Important influencing variables on the CO_2 concentration are (Fig. 5):

> 0.7: Radon,

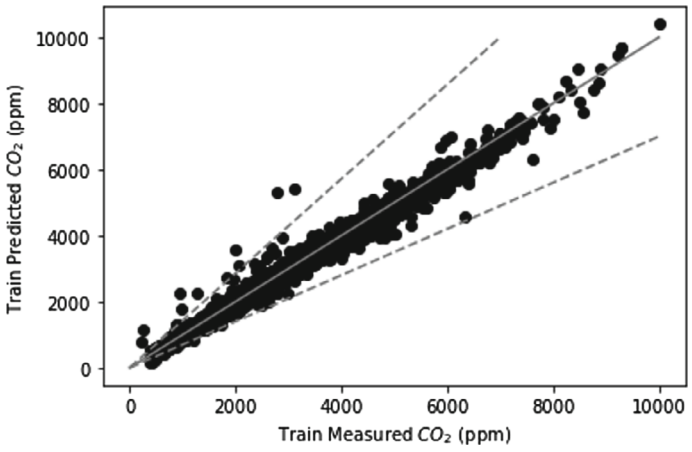

Fig. 3 Comparison of the measured and calculated CO_2 values in the training

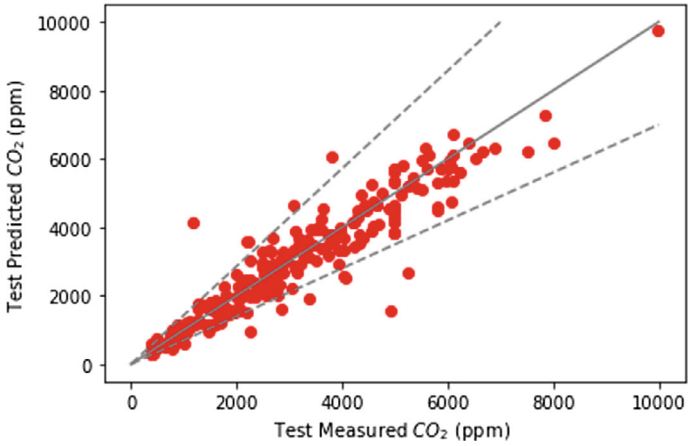

Fig. 4 Comparison of the measured and calculated CO_2 values in the test

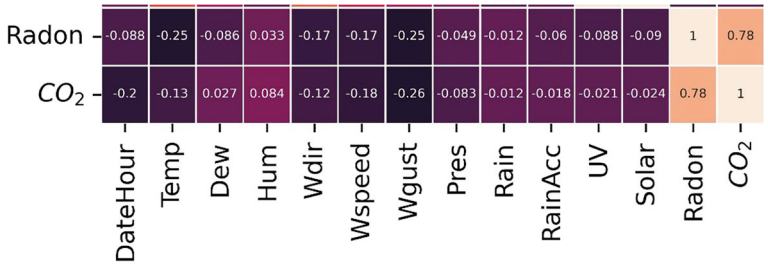

Fig. 5 Heatmap of the input values and radon or CO_2

> 0.2: Wind gust speed Wgust, date/time of measurement DateHour,
> 0.1: temperature Temp, wind speed Wspeed, wind direction Wdir

5 Interpretation

The distribution of CO_2/radon values over time is analyzed. The highest gas values are generally caused in winter by the air pressure difference between the high and low pressure areas of 3.5 hPa, and also by the temperature difference between the atmospheric air and the ground air of −0.7°. This leads to an increased rise of gas-containing soil air in winter.

However, the evaluation of the heatmap (Fig. 5) of the neural networks shows that the above-mentioned parameters do not have a decisive influence in this series of measurements, but rather the wind. The wind gusts reach speeds of up to 73 km/h. The stronger the wind, the lower the CO_2 values and the lower the radon values.

The major influence of wind has also been observed in measurements in Germany [1, 2]. The correlation between radon values and CO_2 values is evident in Fig. 5. The higher the radon values, the higher the CO_2 values.

Figures 6 and 7 show that the CO_2 and radon values depend on the time of day. The average radon values are 115 Bq/m^3. The maximum is between 4 and 7 am, and the minimum is between 6 and 8 pm. The reason for the high night-time values could be that the ground temperature in Puerto Naos is around 21° (night-time average) instead of 10° in Germany. This would lead to an increased rise of warm gases into the cooler air at night (between 1 a.m. and 8 a.m.), see Fig. 8.

In Gran Canaria, 2 deep boreholes measured approx. 20° at the ground surface. The temperature increased at 65° per km depth (ITC2020). Temperature measurements in Tenerife showed 25° at a depth of 1 m and 18° at a depth of 20 m [3]. These data are presented for comparison reasons.

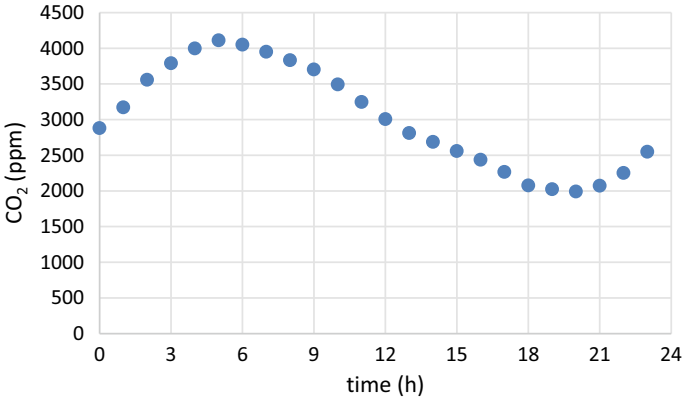

Fig. 6 Average CO_2 values depending on the time of day

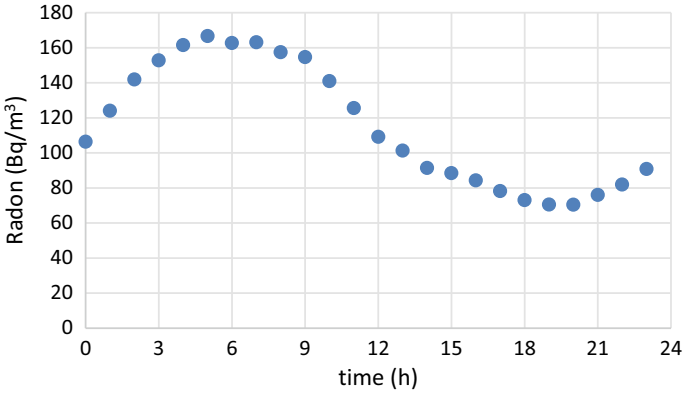

Fig.7 Average radon values as a function of the time of day

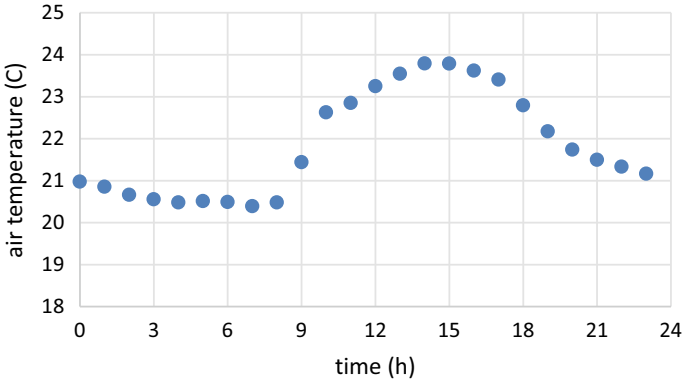

Fig. 8 Mean air temperature as a function of the time of day

The influence of wind direction can be seen in the CO_2 values at Juana Tabares (Fig. 9), with southerly winds lowering the values and northerly winds tending to raise them. This can be explained by the CO_2 source, which mainly outgasses in the region of Playa Chica, north of Juana Tabares. There is an old lava tunnel there.

Similar data was analyzed by Pezzutti [4]. The temperature, humidity, air pressure, and CO_2/radon were measured in a building in Puerto Naos. The measurement period was from August 2022 to January 2023. The measurements, therefore, took place approximately 1 year before the measurements presented here. The temperature was between 27 and 33° compared to the 20–24° measured here. The house has therefore heated up by the sun. The humidity of 40–60% is significantly lower than our 60–70%, and the air pressure fluctuated between 1005 and 1025 hPa compared to our 1020–1022 hPa. Wind was not measured, as it was probably not expected to have this effect with leaky windows and doors.

6 Conclusions

After training artificial neural networks, a coefficient of determination of approx. 0.9 was achieved by taking hourly radon and CO_2 values into account in both the training and the prediction. The gas values can therefore be predicted well, and the radon and CO_2 values correlate best.

Wind speed, wind gusts, and wind direction have the greatest influence on the radon and CO_2 values: the more wind, the lower the gas values. An evaluation of the time of day revealed maximum gas levels at night, when air temperatures are at their lowest. North winds lead to the lowest gas values.

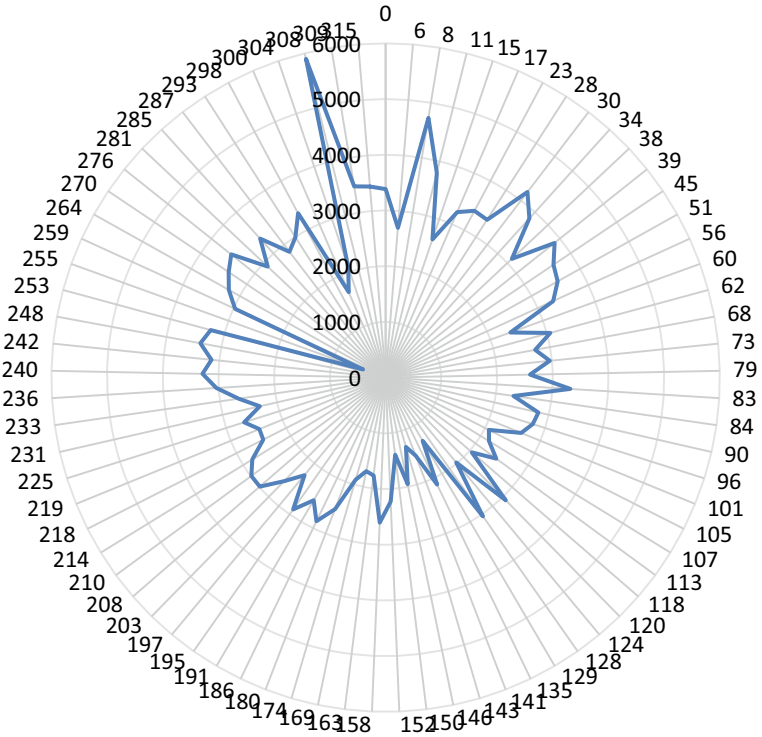

Fig. 9 CO_2 as a function of the wind direction

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Effects of Alternative Electrolytes on the Removal of Hexavalent Chromium from Paint Sludge by Electrokinetic Remediation

A. Metali, M. Ziati, C. Cameselle, and S. Gouveia

Abstract Heavy metal content in the waste sludge from a paint manufacturing plant showed hexavalent chromium concentrations above the regulatory limit according to European standards. The objective of this study was the removal of Cr using the electrokinetic method. To achieve this, a comparative evaluation of two electrolytes was conducted to identify the most effective option for Cr(VI) removal. The electrolytes tested were sulfuric acid and citric acid (1 M). Electrokinetic experiments were performed under potential differences of 5 and 15 V for 10 days. The results showed that citric acid was more effective than sulfuric acid in removing Cr(VI) due to citric acid capacity to extract and dissolve Cr from the solid matrix and its contribution to the electromigration and electroosmosis. Specifically, 76% of Cr(VI) was removed with citric acid at 5 V, with an energy consumption of 3.27 Wh, compared to a removal of 70% with sulfuric acid at the same voltage, requiring 4.75 Wh of energy. Increasing the potential difference to 15 V resulted in lower Cr(VI) removal efficiency for both electrolytes. At 15 V, the removal ratio dropped to 68.93% for citric acid and 58% for sulfuric acid.

Keywords Electrokinetic · Remediation · Paint sludge · Hexavalent chromium · Citric acid

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1 Introduction

Heavy metals, particularly chromium, are commonly utilized as pigments in various industrial applications, including the manufacturing of paints. These processes, especially those involving solvent-based paints, generate hazardous waste materials like paint sludge that contain elevated concentrations of toxic metals—such as lead, chromium, zinc, and copper—when compared to waste from water-based paints [1].

The most prevalent form of chromium found in the environment is hexavalent chromium (Cr(VI)), which is highly soluble and extremely toxic. Its adverse effects on human and environmental health include liver damage, skin conditions, asthma, gastrointestinal issues, and an increased risk of cancers affecting the lungs and digestive system [2, 3]. Given these serious health risks, it is critical to treat Cr(VI) before it is released into the environment.

Numerous studies have concentrated on methods to reduce Cr(VI) levels in soil. For example, Zhao et al. [4] applied a novel technique involving sonoporation-mediated gene transfer to introduce chromium-reducing microbes into contaminated soil. Gao et al. [5] investigated how soil pore structures respond to electric fields and how this affects Cr(VI) removal. Their findings indicate that an electric field strength between 1.00 and 1.50 V/cm and the presence of soil mesopores support the removal process. Cai et al. [6] developed an optimized biochar (HFBC), modified with phosphoric acid and ferric chloride, which demonstrated a high adsorption capacity of 55.5 mg/g for Cr(VI) at pH 6.0 and 25 °C in both soil and water environments. Liu et al. [7] used sulfidated nanoscale zero-valent iron to stabilize Cr(VI) in contaminated soils, with the process resulting in the formation of Fe–Cr complexes, confirming the reduction of Cr(VI) to its less harmful trivalent form (Cr(III)).

Electrochemical and electrokinetic methods have also shown promise for enhancing the extraction of heavy metals—particularly Cr(VI)—from soil. Ribeiro et al. [8] applied an electrokinetic remediation technique to chromium-contaminated soil using a reactive barrier made from eggshell inorganic fraction powder (EGGIF). They observed that lower initial Cr concentrations facilitated more effective removal, especially when citric acid was used as a buffer to maintain soil pH. Their experiments yielded maximum Cr removal rates of 55%, 59%, and 60% at a normalized distance of 0.8 from the cathode for initial Cr concentrations of 500, 250, and 100 mg/kg, respectively. In a separate study, Reddy and Chinthamreddy [9] explored various purging solutions to enhance the removal of Cr(VI), Ni(II), and Cd(II) from kaolin clay. Tap water proved ineffective, but the use of 1 M acetic acid significantly improved metal removal. Notably, 83% of Cr was extracted using 0.1 M EDTA, though Ni and Cd removal remained low. A sequential purging method combining water, acetic acid, and NaOH achieved removal efficiencies of 68–71% for Cr, 71–73% for Ni, and 87–94% for Cd.

Wang et al. [10] proposed a new electrokinetic system for soil remediation using graphite electrodes and auxiliary polypyrrole electrodes, achieving a Cr(VI) removal efficiency of 91.7%. Wu et al. [11] demonstrated that adding citric acid at specific locations within the treatment system enhanced Cr(VI) removal by 60.17% while

reducing energy consumption through improved current focusing. Similarly, Yu et al. [12] used biosurfactants, specifically rhamnolipids at 15 g/L, to facilitate electrokinetic treatment of Cr(VI) in chromite ore processing residues, attaining a removal efficiency of 90.04%.

Several operational factors—such as electrolyte composition, applied voltage, and duration of treatment—affect electrokinetic remediation effectiveness [13]. Studies by Ribeiro et al. [8], Wu et al. [11], and Kada et al. [14] have confirmed that using citric and sulfuric acids enhances the removal of chromium and lead.

This study aims to identify optimal conditions for the electrokinetic removal of hexavalent chromium from paint sludge. The experimental setup compares the effectiveness of 1 M citric acid and 1 M sulfuric acid as catholytes, under both low (0.5 V/cm) and high (1.5 V/cm) voltage conditions, over a treatment period of 10 days [15, 16].

2 Materials and Methods

2.1 Sampling and Characterization of Paint Sludge

A sample of solvent-based paint sludge was collected from a paint production plant in Algeria. The sample was oven-dried at 70 °C for 36 h, crushed, and then sieved through a standard sieve (CB/T6003.1–2012, 200 mesh) at less than 0.2 mm. The physicochemical characterization of the sludge was carried out by measuring the pH and electrical conductivity using the "Hanna HI 5522 instrument" device in the supernatant after mixing the sludge with water (ratio 1:10, sludge-water) and stirring for 1 h. Dry matter (DM) or non-volatile matter is measured by dry weight after drying 2 g of sludge at 150 °C for 4 h according to the method EN ISO 3251[17]. The content of heavy metals in the paint sludge was determined according to US EPA Method 3050B SW 846 [18]. The hexavalent chromium was determined according to the US EPA 3060A method by ICP-OES (Inductively Coupled Plasma—Optical Emission Spectroscopy).

2.2 Electrokinetic Decontamination of Hexavalent Chromium

The schematic diagram of the electrokinetic (EK) setup, used for the removal of hexavalent chromium from paint sludge, is shown in Fig. 1. The experiments were carried out in a parallelepipedal acrylic cell made of Plexiglas whose dimensions were 10 cm × 4 cm × 5 cm. The cell consisted of three compartments: anode, cathode (2.5 cm × 4 cm × 5 cm), and a central compartment for the sludge specimen.

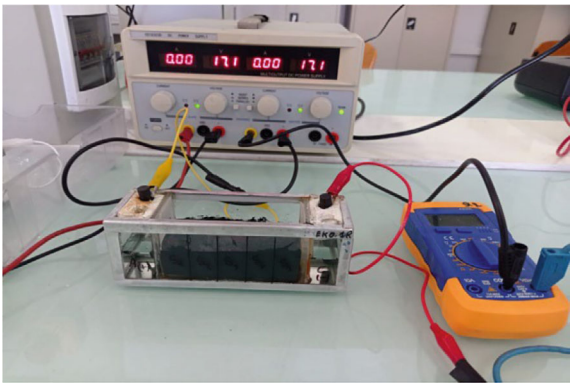

Fig. 1 Electrokinetic cell setup

A mass of 75 g of paint sludge, previously mixed with 50 mL of distilled water, was introduced into the central compartment. Before starting the EK experiments, each electrode compartment was filled with 20 mL of electrolyte (1 M sulfuric or citric acid as catholyte and 0.1 M sodium nitrate as anolyte) with a graphite electrode placed in each compartment. Between the central compartment and each electrode compartment, a polyamide fabric filter was installed to facilitate the transport of ions without the sludge particles crossing to the electrode compartments. The electric current intensity was measured periodically with a digital multimeter (UNI-T, UT61A), as well as the pH in the electrode compartments. The experimental conditions are summarized in Table 1.

At the end of the experiment, the treated sludge was immediately divided into five equal sections of 2 cm each. These sludge segments were dried at 105 °C for 2 h in a laboratory oven, grounded, and then analyzed for hexavalent chromium content as described above.

The selection of these experimental conditions (electrolyte concentration, applied potential difference, and treatment time) is based on previous work carried out at the laboratory.

Table 1 Experimental conditions

Test	Anolyte	Catholyte	Potential drop (V)	Treatment time (days)
Exp. 1	$NaNO_3$ (0.1 M)	H_2SO_4 (1 M)	5	10
Exp. 2	$NaNO_3$ (0.1 M)	H_2SO_4 (1 M)	15	10
Exp. 3	$NaNO_3$ (0.1 M)	Citric acid (1 M)	5	10
Exp. 4	$NaNO_3$ (0.1 M)	Citric acid (1 M)	15	10

2.3 Data Analysis

The removal efficiency from the sludge (R, %) was calculated using Eq. (1).

$$R = [(C_0 - C)/C_0] \times 100 \quad (1)$$

where $R(\%)$ is the removal ratio, C_0 and C (mg/L) are the initial and final concentration of Cr(VI) in the sludge, respectively.

The energy consumption E (Wh) was calculated using Eq. (2)

$$E = I \cdot U \cdot t \quad (2)$$

where I, U, and t are current intensity (mA), the potential drop (V), and treatment time (h), respectively[19].

3 Results and Discussion

3.1 Characterization of the Paint Sludge Sample

Table 2 shows the physicochemical properties of the studied sludge. It is noted that pH is close to neutral (7.35), the conductivity is 461.1 µS/cm, and the dry residue is 75%. The sludge is very rich in metals such as iron, zinc, nickel, and chromium; the concentration of the latter is 1251 mg/kg; this content exceeds the European standard value, which is 1000 mg/kg [20]. Furthermore, hexavalent chromium, which is the most toxic form, is found at a concentration of 208.8 mg/kg. It is used in paint production as pigments and additives. However, this element has harmful effects on the environment and on living species.

Table 2 Physicochemical properties of the paint sludge

Parameter	Value
pH	7.29 ± 0.002
Electrical conductivity (µS/cm)	461.10 ± 0.002
Dry matter (% wt.)	75.01 ± 1.356
Fe (mg/kg)	14,395
Cr (mg/kg)	1251
Cr(VI) (mg/kg)	208.8
Ni (mg/kg)	114.45
Cd (mg/kg)	106.15

3.2 *Removal of Hexavalent Chromium by the Electrokinetic Process*

Electrokinetic remediation has recently received significant attention for the treatment of soils and sludges due to its distinct advantages. These include its effectiveness in remediating fine-grained and low-permeability materials, as well as its ability to simultaneously achieve soil consolidation, dewatering, and removal of both organic and inorganic contaminants in a single stage.

In order to reduce the concentration of the hexavalent chromium in the sludge, a fourth electrokinetic test was designed. The residual Cr(VI) concentration was measured in the sludge at the end of the tests. Figure 2 shows the ratio of the residual concentration of Cr(VI) to the initial concentration in all sludge sections, from the anode to the cathode. In all experiments, a higher concentration of Cr(VI)was detected in the anodic compartment, compared to the cathodic compartment. This is due to the nature of the Cr soluble ions that move according to the experimental conditions (the pH of the medium and redox conditions), which migrate as oxyanions as hydrochromate ($HCrO_4^-$)under very low pH conditions, as dichromate ($Cr_2O_7^{2-}$) under acid to neutral conditions, and chromate (CrO_4^{2-}) in alkaline conditions [9, 21]. The low residual chromium rate is obtained in experiment 3, using citric acid as catholyte, which further dissolves Cr(VI) and improves the electroosmotic flow, due to the interaction between citric acid and sludge particles [21] compared to sulfuric acid in experiment 1 at low voltage and energy consumption.

In exp. 3 the residual chromium concentration decreases from cathode to anode. It is noted that the low residual concentration is present in soil sections S1 and S2 compared to soil sections S3, S4 and S5. In neutral to basic medium, the most predominant form of Cr(VI) is the oxyanion, which explains the migration of Cr(VI) toward the anode. Reddy and Chinthamreddy [9] observed the same phenomenon.

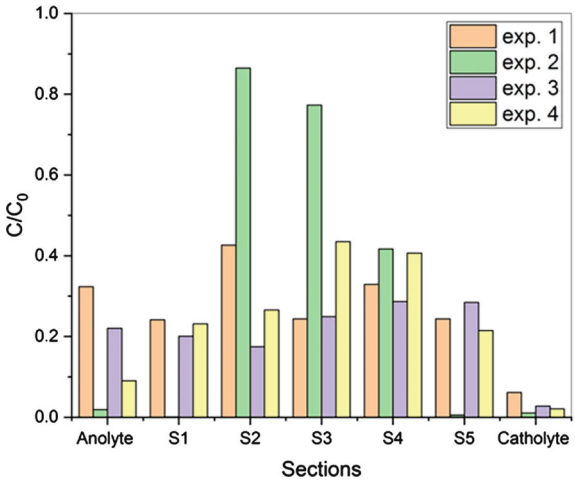

Fig. 2 Residual ratio of hexavalent chromium from the anode to the cathode after electrokinetic tests

In exp. 1 the use of sulfuric acid as a catholyte reveals the migration of Cr(VI) toward the anode. The same phenomenon is observed with citric acid. However, some Cr(VI) exists in the cathode with a low residual concentration in sections S1, S3, and S5. Hexavalent chromium migrates in the form of oxyanion and cations (Cr(VI)) [9, 11], compared to S2 and S4, where the residual concentration is high due to the accumulation of chromium in this section and its precipitation in the presence of sulfate and hydroxide anions.

At high voltage (15 V), a low residual concentration is observed in experiments 2 and 4 compared to the other experiments, and the migration of chromium occurs in both compartments (anolyte and catholyte), which exist in anionic and cationic forms.

Moreover, in sulfuric acid (exp. 2) and citric acid (exp. 4) medium, the residual concentration of Cr(VI) in soil sections S1 and S5 is lower than that of soil sections S2, S3 and S4, which present an accumulation or precipitation of this metal.

In this study, hexavalent chromium present in paint sludge has different forms depending on the type of electrolyte and applied voltage, noting that a low potential of 0.5 V cm^{-1} and using citric acid or sulfuric acid as electrolyte dissolve more Cr(VI) and favor its migration toward the anode compartment. On the other hand, Gao et al. [5] found that a potential lower than 1.5 V cm^{-1} facilitated Cr(VI) removal in the soil. In our experiments, we found that a 0.5 V cm^{-1} applied voltage could remove hexavalent chromium from the paint sludge.

3.3 Current and Energy Consumption Variation

Figure 3 shows the current intensity profiles in the electrokinetic tests. In the first 72 h, the current intensity increases from 13.25, 13.56, 3.9, and 2.46 mA to 18.6, 14.35, 4.82, and 3.50 mA for exp. 2, 4, 1, and 3, respectively, then decreases to stabilize at 6.81, 3.80, 3.73, and 2.32 mA, respectively.

The increase in electric current is due to the migration of mobile ions in the electrolytes and the dissolution of metals in the paint sludge compartment. Then, it decreased gradually with the extension of the treatment time, due to the continuous migration of cations and anions toward the cathode and anode sides, respectively [9, 12]. In addition, the products of water electrolysis can contribute to the current decrease by neutralizing the migrating ions.

For energy consumption (Fig. 4), it reaches the following values: 35.22, 23.74, 4.75, and 3.27 Wh for experiments 2, 4, 1, and 3, respectively. This is a function of the treatment time, the imposed potential, and the acid used as treatment fluid. Lower values were registered in experiments 1 and 3, where the potential difference is 5 V.

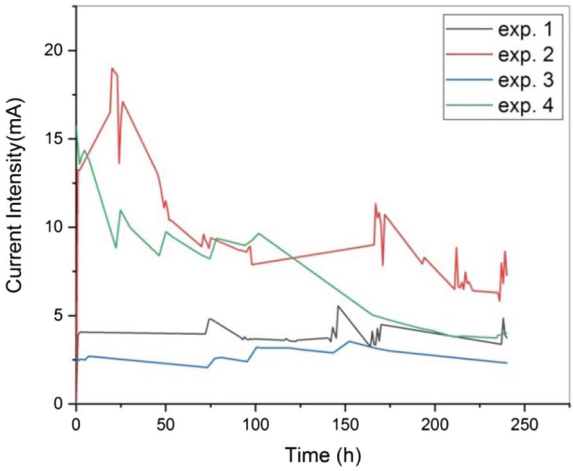

Fig. 3 Current intensity profiles in electrokinetic tests

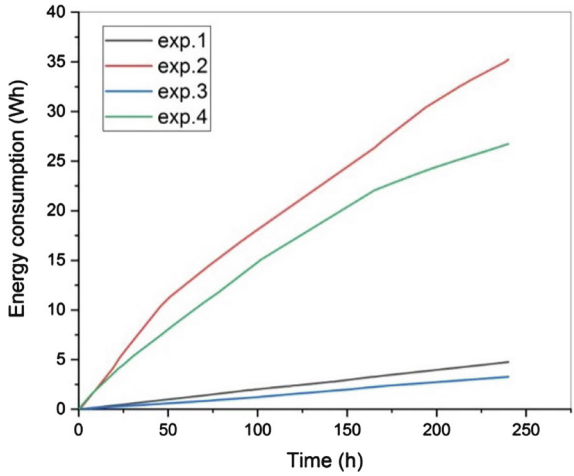

Fig. 4 Cumulative energy consumption profiles in electrokinetic tests

3.4 Variation of the pH of the Electrolytes During the Treatment

In all the experiments, the pH in the electrolytic compartments is shown in Fig. 5. Before the EK treatment, the pH of the anolyte for the four experiments is between 4 and 6 (pH exp. 1 = 6.84, pH exp. 2 = 5.32, pH exp. 3 = 5.84, pH exp. 4 = 4.66). After the electrokinetic treatment, the pH of the anolyte stabilizes below 2. At the cathode, the phenomenon is the opposite; the pH does not exceed 1.7 at the beginning of the experiments, then it increases up to 10 and 12 at the end of the experiments. This can be explained by the usual water electrolysis reactions, where hydroxyl ions (OH^-) and hydrogen (H_2) are produced at the cathode, while hydrogen ions (H^+)

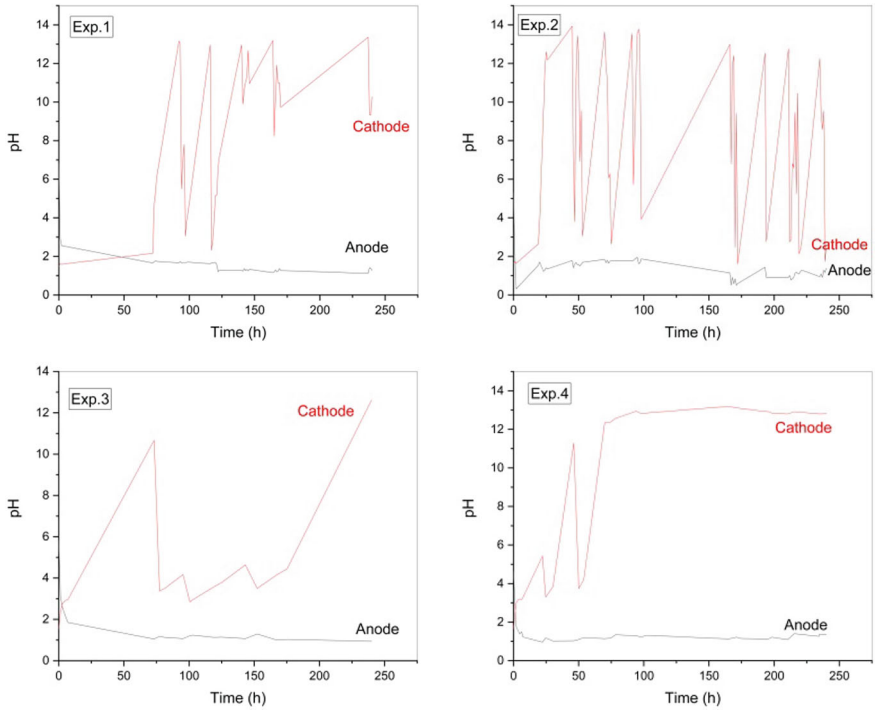

Fig. 5 pH evolution in the anolyte and catholyte during electrokinetic tests

and oxygen (O_2) are produced at the anode. Therefore, bubbles were continuously generated in both the anode and cathode electrodes [12, 22].

3.5 Variation of Sludge pH After Treatment

Figure 6 shows the pH profiles in paint sludge for tests 1, 2, 3, and 4 after the electrokinetic treatment, using citric acid and sulfuric acid as catholytes. The initial pH of the sludge is 7.29, because paint sludge has a high acid buffer capacity, associated with carbonate that causes a slightly basic pH [22]. After treatment, the pH of the first section, near the anode for all experiments, varies from 7 to 8 due to the presence of carbonate, which generates a basic environment. Then, the pH gradually increased from the anode to the cathode, with a pH value of 9–10 for all tests. On the other hand, H^+ ions generated at the anode do not readily migrate into high carbonate sludge, which has high acid buffering capacity and neutralizes H^+ near the anodic region, unlike OH^- ions, which migrate through the sludge [9].

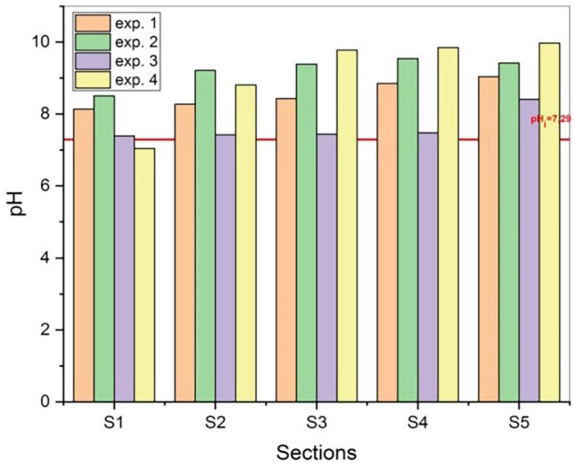

Fig. 6 Sludge pH at the end of electrokinetic tests

3.6 Variation of Electrical Conductivity After Treatment

Figure 7 shows the variation of electrical conductivity after electrokinetic treatment of sludge in different tests. Before treatment, the electrical conductivity was 461.1 μS/cm. After treatment, it increases near the anodic (S1) and cathodic (S5) compartments to 2236 μS/cm, 641.7 μS/cm, 780.6 μS/cm, and 1602 μS/cm, 535.9 μS/cm, 460.8 μS/cm, respectively, for experiments 2, 3, and 4. This is compared to the central sections: S2, S3, and S4 (S2: 589.9 μS/cm, S3: 399.4 μS/cm, and S4: 408.8 μS/cm), (S2: 423.9 μS/cm, S3: 441.8 μS/cm, and S4: 460.8 μS/cm), and (S2: 277.7 μS/cm, S3: 407.4 μS/cm, and S4: 445.4 μS/cm) for the same experiments. This increase in conductivity is linked to the presence of electrolytes in the electrode compartments (sodium nitrate as anolyte and citric acid or sulfuric acid as catholyte).

3.7 Removal Efficiency of Hexavalent Chromium from Paint Sludge in Electrokinetic Tests

Table 3 shows the Cr(VI) removal efficiency in all electrokinetic tests. At low voltage (5 V) and in the presence of citric acid (1 M) as electrolyte, the Cr(VI) removal ratio is 76% with a current intensity and energy consumption of 3 mA and 3.27 Wh, respectively. At the same voltage, and replacing citric acid with sulfuric acid (1 M), the removal rate is 70% with a electric current intensity and energy consumption of 4.85 mA and 4.75 Wh, respectively.

At a higher voltage (15 V), the Cr(VI) removal rate is lower in both cases: 68.93% with citric acid and 58% with sulfuric acid, with a current intensity and energy consumption of 4.01 mA and 26.73 Wh, respectively, for citric acid and 7.27 mA and 35.22 Wh, respectively, for sulfuric acid as electrolyte.

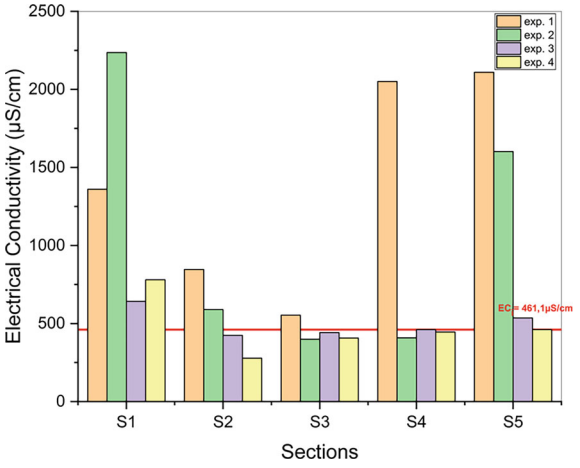

Fig. 7 Electrical conductivity of the sludge after electrokinetic tests

Table 3 Removal efficiency of Cr(VI)

Assay	Anolyte	Catholyte	Potential (V)	Treatment time (days)	Removal efficiency (%)
Exp. 1	$NaNO_3$ (0.1 M)	H_2SO_4 (1 M)	5	10	70.326
Exp. 2	$NaNO_3$ (0.1 M)	H_2SO_4 (1 M)	15	10	58.774
Exp. 3	$NaNO_3$ (0.1 M)	Citric acid (1 M)	5	10	76.098
Exp. 4	$NaNO_3$ (0.1 M)	Citric acid (1 M)	15	10	68.935

In our study, a low voltage of 5 V and the use of citric acid as the electrolyte resulted in a higher dissolution of hexavalent chromium, achieving a removal efficiency of 76%. This represents an improvement compared to previous studies. Citric acid is commonly used as an electrolyte due to its biodegradability and environmental friendliness [21, 23].

4 Conclusion

This study presents a comparison between two electrolytes, sulfuric acid and citric acid, for the removal of Cr(VI) from paint sludge using electrokinetic remediation, under both low and high applied potentials.

The results show that citric acid is more effective than sulfuric acid for extracting Cr(VI) from the studied sludge. The optimal conditions for dissolving hexavalent chromium from this matrix were determined as follows: an electrolyte concentration of 1 M citric acid, an applied voltage of 5 V, and a treatment duration of 10 days. Under these conditions, a removal efficiency of 76% was achieved.

The energy consumption of the process, which is directly related to the current, was calculated to be 3.27 Wh.

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Evaluation of the Efficiency of a Mixture of Moringa Seed and Banana Peel Powder as a Coagulant in Wastewater Treatment (Batik Effluent as a Case Study)

Adekunle Olorunlowo David, Julius Musyoka Ndambuki, Jacques Snyman, Williams Kehinde Kupolati, Mpho Muloiwa, Samuel Ayomikun Owolabi, and Toheeb Lakoja

Abstract The purpose of this research was to examine how well a mixture of Moringa oleifera (MO) seed extract and Banana peel powder performed as an alternative to traditional coagulants in improving Batik effluent wastewater. To treat wastewater, coagulation/flocculation and sedimentation are utilized. The seeds of Moringa oleifera were dried, crushed, and sieved, and this process was also done on the banana peel. Jar tests were performed, and the effectiveness of coagulation tests was compared to the use of aluminum sulfate (alum). In comparison to using the mixture of Moringa oleifera seeds extract and Banana peel powder at the prescribed dose of 15 g of moringa seed powder and 5 g of banana peel in a liter of the effluent and 20 g of moringa seed powder in a liter of the effluent provided outstanding results (though none of the mixture gives a clear water) and alum at 5 g/L. In reality, mixture of MO seeds extract plus powder peel and alum reduced turbidity by −39.72% and 77.50%, TSS by 82.44% and 47.98%, TDS by 69.45% and 41.26%, color by −99.21% and 32.67%, conductivity by −149.85% and −280.67%, COD by −49.11% and −101.32%, and BOD by −46.55% and 67.82%, respectively. In contrast to alum, increasing the dosage of MO seed extract and banana peel resulted in a minor pH change and a decrease in conductivity. The usage of a mixture of Moringa oleifera seed extracts and banana peel is recommended to avoid the numerous disadvantages associated with the use of chemical coagulants. This research highlights the potential of utilizing sustainable and locally available materials for effective and eco-friendly treatment of batik effluent, contributing to the development of cleaner and more sustainable practices in the textile industry.

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Keywords Moringa seed · Banana peel · Coagulant · Batik effluents

1 Introduction

Ensuring access to clean water and sanitation is a worldwide issue, specifically in developing nations. The United Nations Sustainable Development Goals aim to improve sanitation, minimize open defecation, treat wastewater, and promote global recycling [1]. Water is known to be vital to human existence. Water pollution from farming practices, domestic, and commercial activities has become a crucial problem hindering the economic stability of developing countries and, by extension, causing major health challenges to people. However, due to a number of issues, such as population increase, industrialization, and pollution, many places around the world strive to guarantee the supply of potable water. Exploring recently developed and environmentally friendly water treatment methods that can efficiently remove pollutants and impurities without harming the environment is therefore urgently needed.

Traditional water treatment processes often rely on chemical coagulants, such as aluminum sulfate or ferric chloride, which can be costly and may pose health risks, such as Alzheimer, and environmental risks [2]. Consequently, researchers have turned their attention to natural coagulants derived from plant-based sources that offer potential as safer and more environmentally friendly alternatives. Coagulation is the physicochemical phenomenon that involves the breaking down of colloidal suspended particles upon the addition of a coagulant, which may be a chemical-based or naturally occurring coagulant. Moringa oleifera is a plant with a variety of uses in food, wastewater treatment, and medicine. This research project aims to evaluate the feasibility and efficiency of a novel coagulant mixture composed of moringa seed extract and banana peel in the context of water treatment. By harnessing the coagulation properties of both moringa seeds and banana peels, it is hypothesized that this eco-friendly blend may yield enhanced coagulation performance compared to traditional chemical coagulants. The significance of this research lies in its potential to provide a sustainable and cost-effective solution for water treatment, especially in resource-constrained regions where access to conventional coagulants may be limited. The effect of polluted water and water treated with chemically based coagulants on human health and other living organisms cannot be overemphasized. Some research results also revealed that using alum as the main coagulant in water treatment reflects a wide effective PH range. Treating water using alum is known to be very costly, which is a major economic challenge. The nonchalant discharge of BATIK effluents into stream water without proper treatment has led to water pollution. Many naturally existing plant extracts can be used as coagulants in water treatment, such as moringa seed extract, banana peel, cassava peel, and so on. Water is an inert substance with the chemical formula H_2O, which is covalently bonded to itself. Access to clean drinking water and proper sanitation is considered a fundamental human right. It is important for maintaining good health and preserving the well-being of every individual. Water has been going up globally by around 1% each year

since the 1980s. This is because of population growth, development, and changing consumption patterns. The demand for water is expected to keep rising at a similar rate until 2050, with a 20–30% increase from current levels. This is mainly due to the higher demand for water in industry and homes. Sadly, over 2 billion people live in countries with high water stress, and about 4 billion people face severe water scarcity for at least one month each year. As water demand and the effects of climate change grow, stress levels will keep elevating [1]. If things keep going the way they are, life on Earth could become impossible. We might face a situation where there's a huge need for water, but not enough of it to go around. The standard water treatment processes that are utilized in developing nations have become a challenge due to the increased costs of implementation, especially for chemicals. Coagulation is one of the most important unit processes used with hydrated aluminum sulfate (alum), a frequently used coagulant. Coagulation and flocculation processes are key in the removal of turbidity and color. Replacement of chemically based coagulants with plant-based coagulants is crucial to mitigate the consequences posed by the former.

Native to the Indian subcontinent, the Moringa oleifera tree is a swift-growing, water-efficient member of the Moringaceae family. This plant is also known as "moringa," "drumstick tree," "horseradish tree," "ben oil tree," or "benzolive tree." The nomenclature is obtained from the horseradish-like flavor of the roots.

Moringa is a tasty, pungent herb that has been used for centuries to improve and strengthen human health, enhance the flavor of food, and aid with digestion. In terms of the potential advantages to enhance human health, Moringa is a neglected and underutilized plant species (NUS). This plant's roots have a long history of utilization in traditional medicine and are recognized to be excellent remedies for a variety of human health issues. Moringa contains a variety of biologically active components, and in recent years, its use as a functional food has increased due to its abundance in phytochemicals. A single tree can yield between 15,000 and 25,000 seeds, each of which weighs about 0.3 g on average [3]. Moringa inflorescences are axillary loose panicles which range in length from 10 to 25 cm. The fragrant, up to 12 mm long, bisexual zygomorphic blooms are white or cream in color, contain five pale green sepals, five white petals, and five stamens with anthers and five without (staminode) [4]. As a decoction, raw or crushed M. oleifera seeds are used in conventional medicine to treat joint discomfort, poor vision, stomachache, ulcers, and ulcerative colitis [5]. It was discovered that the seed extract had effective antibacterial action against a variety of bacterial and fungal species. Numerous phytochemicals that have been extracted from seeds could stop the growth of some pathogenic bacteria that cause diseases in humans [6].

Textile manufacturing processes, from fiber production to fabric finishing, have a significant impact on the environment, including air, water, and soil. To address this, both public and private sectors are collaborating on initiatives to make textile processing more environmentally friendly. Dyeing, printing, and coating are techniques used to add vibrant colors and patterns to textiles. Printing and coating are primarily used for surface coloration and can be applied to various types of fibers, including natural fabrics and synthetics. There are approximately 10,000 different dyes and pigments used globally in industrial applications [7]. During the wet

processes like dyeing and printing, a large number of dyes and chemicals are used, resulting in effluents that are often released into water bodies without proper treatment. These effluents contain high levels of pollution, including colors and other pollutants. As a result, the receiving water bodies become heavily polluted. Colors are particularly noticeable in wastewater, even in small amounts [8]. Types of textile manufacturing processes include yarn manufacturing, fabric manufacturing, garment manufacturing, and technical textiles. Textile factories utilize both dry and wet techniques for fiber production. The wet process generates heavily polluted effluent and requires a substantial amount of drinking water. The steps involved in this process encompass sourcing, sizing, de-sizing, bleaching, mercerizing, dyeing, printing, and finishing [9, 10].

2 Materials and Methodology

2.1 Materials

Sample Collection

The ripped moringa oleifera seed pod sample was acquired in Gbadebo Street, Ojota, Lagos State. Also, in the Abeokuta (Itoku market) area of Ogun state, the ripped banana peel was obtained. The 10 L of batik effluent used in this research study were collected in Itoku Adire industry, Abeokuta, Ogun state.

Sample Preparation

The seeds were peeled to reveal the nuts, which were then air-dried for three days before being ground into a fine powder and sieved using a sieve with a number 0.85 mm. The banana peel was also air dried for one week before being ground into a fine powder and sieved using a sieve with a number 0.85 mm.

3 Results and Discussion

3.1 Jar Test

Using various chemical concentrations, a pilot-scale laboratory experiment called a "jar test" can be used to simulate coagulation or flocculation. The primary purpose of flask testing is to find the optimum amount of coagulant required to achieve a given water quality standard. The pot test utilizes a four- or six-position agitator which can be used to precisely simulate conditions related to mixing and settlements. Jar testing aids in the determination of the proper dosage of wastewater treatment because very little is required to treat waste, and it settles quite well. The jar test makes it feasible

to determine the right chemical and dosage to add to wastewater treatment plant water in order to eliminate pollutants and particles in suspension. The coagulation process was performed using a TT-TB centrifuge and a total of five one-liter beakers, each containing 500 ml of batik effluent for each test run. The amount of coagulant (mixture of granulated moringa seed and banana peel powder) added to the five one liter's beakers, each containing 500 ml of the tested sample, was 5 gM + 15 gB,10 gM + 10 gb, 15 gM + 5 gB, 20 gM, and 20 gB (all in grams per liter). The amount of alum added to the five one liter's beakers, each containing 500 ml of the tested sample, was 5, 10, 15, 20 and 25 g/l. The test was conducted in two stages under identical circumstances. Alum alone was used in the first stage, and a mixture of granulated moringa seed and banana peel powder was used for the second stage. After blending the mixture aggressively for two minutes at 180 rpm, slowly for twenty minutes at 40 rpm, and letting it settle for thirty minutes, the process was repeated. Samples of the supernatant (top layers) were collected after the settling process in order to examine the parameter under study.

In the analysis above; ml = milliliters, B = banana peel powder,

M = granulated moringa seed, g = gram, l = liter

Following the formula; % Removal efficiency = (ci − cf)/ci where (ci) is the initial concentration of raw water effluent and (cf) is the final concentration for each parameter, the removal efficiency of this process was based on the removal percentage of turbidity, TSS, COD, BOD, conductivity, color, and TDS between the batik effluent and the effluent after the treatment with alum and mixture of moringa seed and banana peel powder at the Forest Research Institute of Nigeria, PMB 5054, Jericho hill, Ibadan and at Bumlab Care Services Limited, Ring- Road, Ibadan, physio-chemical tests were done on the sample. The results are displayed in Table 1 (Figs. 1, 2 and 3).

Table 1 Comparison of removal efficiency of physicochemical properties

	Ci	Cf		100%	
Physicochemical properties	BEW	After adding the coagulant to the batik effluent		Removal efficiency (Ci − Cf/Ci) *100%	
Parameters		Mixture of MS and BPP	Alum	Mixture of MS and BPP	Alum
PH	7.55	4.8	3.41		
Temperature	26	27	28		
T.D.S (ppm)	1614	493	948	69.45	41.26
Conductivity (ns)	1.95	4.872	7.423	−149.85	−280.67
Turbidity (ntu)	56.9	79.5	12.8	−39.72	77.50
TSS (mg/l)	1115	197	580	82.33	47.98
C.O.D (mg/l)	1515	2259	3050	−49.11	−101.32
B.O.D (mg/l)	348	510	112	−46.55	67.82
Color	1010	2012	680	−99.21	32.67

Fig. 1 Batched samples after undergoing jar test

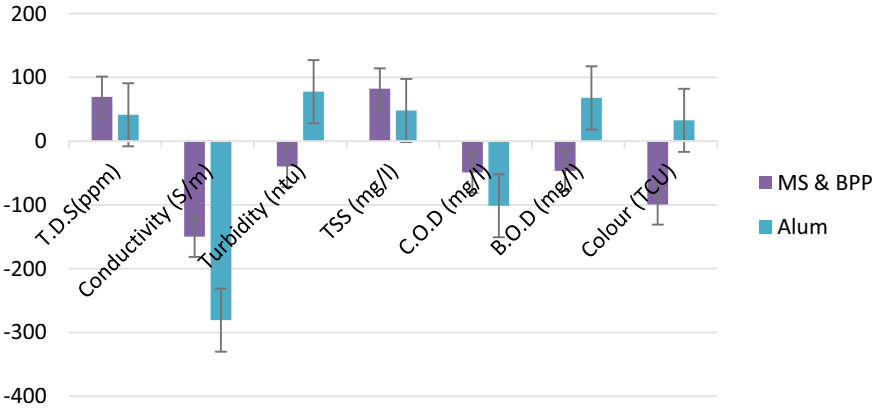

Fig. 2 Chart for comparison of removal efficiency of physicochemical properties

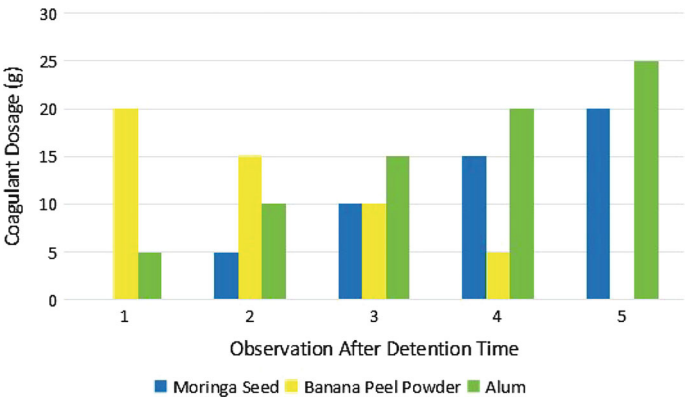

Fig. 3 Chart for jar test optimal dosage

The section above compares the efficiency of the two coagulants in removing TSS, COD, BOD, TDS, turbidity, conductivity, and color from effluent paint. Two coagulants were used with batik effluents that had an initial TSS of 1115 mg/l to achieve this. Turbidity of 56.9 Ntu, COD of 1515 mg/l, 348 mg/l BOD, 1614 mg/l TDS, 1.95 ns conductivity, and 1010 color. Alum exhibited a slightly higher removal efficiency for turbidity, BOD, and color than a mixture of granulated moringa seed and banana peel powder at 77.50%, 67.82% and 32.67%, respectively. A mixture of granulated moringa seed and banana peel powder outperformed alum in terms of TDS reduction, getting 69.45% against 41.26%. The best dosing conditions for all the measures under the plus 5 g concentration of banana peel powder in a liter of the effluent (though none of the mixture really gives clear water) and an alum concentration of 5 g/l.

4 Conclusions

The purpose of this study was to determine the efficiency of a mixture of Moringa oleifera seed extract and Banana peel powder as a natural coagulant for water treatment. The result of this research shows that a mixture of Moringa oleifera seed extract and banana peel powder is an effective and environmentally friendly alternative to traditional chemical coagulants in water treatment. The mixture dramatically reduced turbidity and removed a variety of pollutants, including bacteria and organic debris, with an efficiency comparable to or greater than chemical coagulants. There are various advantages of using a mixture of Moringa oleifera seed extract and banana peel as a coagulant, including sustainability and environmental friendliness. Moringa trees are a cheap and useful option for water filtration because they are widely disseminated and produced on a regular basis in developing countries. A mixture of Moringa oleifera seed extract and Banana peel has been used since the further advantage of using a mixture of Moringa oleifera seed extract and Banana peel as a coagulant is that it might improve the quality of the water in poorer countries and rural locations, where access to more traditional water treatment methods might be difficult or prohibitively expensive. It may be practicable and affordable to use a mixture of Moringa oleifera seed extract and Banana peel powder to enhance water quality and protect public health.

The mixture of granulated Moringa oleifera and Banana peel powder has great potential to successfully and practically replace conventional chemical coagulants in the water treatment process. In addition to improving dosage and administration techniques for different kinds of water sources, more research is required to fully comprehend the effects of the mixture of Moringa oleifera seed extract and Banana peel powder on water quality over time. People should be made aware of the situation through public awareness efforts, benefits, and potential uses of a mixture of Moringa oleifera seed extract and Banana peel in water treatment, which would encourage greater adoption and improve access to safe and clean drinking water in communities

around the world. Therefore, employing Moringa oleifera seed extract as a coagulant in water treatment warrants additional investigation, funding, and consideration.

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Modelling of Electrokinetic Soil Treatments: Verification Cases

Rubén López-Vizcaíno, Adrián Sánchez-Migallón, Laura Asensio, Xiaojuan Yang, Vicente Navarro, and Ángel Yustres

Abstract Electrokinetic (EK) soil remediation involves several processes, including transport, electrochemical reactions and thermal effects. These processes interact in a complex way, making it difficult to accurately predict the results. To address this problem, the authors are developing a project called 'Coupled Numerical Modelling of Electrokinetic Soil Treatments'. The aim of this project is to develop a model that integrates thermal, hydraulic, mechanical, chemical and electrical (THMCE) phenomena during EK treatment. The first phase consists of defining the conceptual model, implementing it in COMSOL Multiphysics and verifying it with reference parameters. The M4EKR model serves as the code base, using the Nernst-Planck equations for species transport and charge conservation together with the electroneutrality condition for electrical charge transport. For chemical reactivity, only the electrolysis of water is considered. Four verification exercises have been carried out to test: (1) Electromigration and electroosmosis with a constant electric potential gradient. (2) Multicomponent electromigration under steady state conditions. (3) Time-dependent multicomponent electromigration. (4) Electromigration with chemical reactivity. The model showed satisfactory agreement with the reference benchmarks in all cases.

Keywords Multicomponent transport · Multiphysics modelling · Electrokinetics · Nernst-Planck-Poisson equations

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1 Introduction

The world's growing population requires more land to be used for a variety of purposes, the most direct of which is the expansion of existing urban areas or the creation of new ones [1]. In line with this, many of the possibilities for city growth involve the reuse of land previously used for industrial or even agricultural activities. There is a high probability that these soils are not in optimal condition and require treatment aimed at both improving their structural properties and removing possible contaminants. In this context, the development of soil remediation and soil improvement techniques is of great interest. Among the wide range of options currently available, electrokinetic (EK) treatments can play an important role. This technology can be applied in situ, as the energy requirements can be supported by renewable energy sources [2], and is a promising candidate to stimulate species and liquid phase transport by various mechanisms in low hydraulic permeability media.

When an electrical potential difference is applied to the soil, various transport processes take place. Mass and energy may be mobilised in these cases [3]. Electromigration, which corresponds to the movement of ions towards the oppositely charged electrode, is the most intuitive phenomenon [4]. Electroosmosis means the movement of the liquid phase driven by an electrical gradient. This process is very useful for soil remediation by electrokinetic flushing [5] or for soil stabilisation by electroconsolidation. Electrophoresis refers to large (colloidal) particles or molecules migrating when subjected to an electric field. In biological remediation processes or electrokinetic biocementation [6], this phenomenon can be important for the mobilisation of micro-organisms [7].

The simultaneous presence of these physical phenomena leads to a series of coupled interactions which make the application of electric fields in the soil a very complex phenomenon. For instance, accelerated drainage of soil water due to electroosmosis provokes an increase in effective stresses and, consequently, changes in soil porosity due to enhanced consolidation. These events involve changes in the effective diffusion coefficients, in soil deformability and in hydraulic conductivity and electroosmotic permeability. The initial chemical equilibrium between soil and pore water is also altered by electromigration and electroosmotic transport. This can lead to mineral dissolution and/or precipitation, resulting in changes in the internal structure of the soil and cementation of solid particles. In addition, the whole geochemical system and therefore the mechanical behaviour of the soils can be altered by the electrochemical reactions that take place on the surface of the electrodes (mainly water hydrolysis). Finally, if the electric current is high and the resistance of the soil is also high, heating of the soil can occur due to the Joule effect. High temperatures have been observed to cause evaporation of the porewater in the proximity of the electrodes and consequently alter the saturation conditions of the soil and modify the salt concentration level in that zone ('sauna' effect), which favours the precipitation of minerals.

This wide variety of interacting thermal, hydraulic, mechanical, chemical and electrical/electrokinetic phenomena means that the availability of a numerical model

to understand the coupled phenomena in soils during electrokinetic treatment is essential for the design, construction and monitoring of these systems [8, 9]. It has been observed that in recent years the models continue to evolve more rapidly, with the following being the most widely used: DIFFUSE [10], MODFLOW-MT3D [11], EKGEO-CHEM [12], M4EKR [13], NP-Phreeqc-EK [14], SE-EK Model [15], SEKRET Model [16]. Thus, most of the models are oriented towards solving very specific problems. In general, the treatment of soil as a porous medium is often poor, especially in clayey soils. There is no mechanical interaction and the role of temperature in the generation of heat by the Joule effect and the coupled heat transport with chemical reactivity and hydraulic behaviour are not usually considered. Therefore, only electro-hydro-geochemical models can be considered.

To solve the gap, the authors of this work have started a project entitled 'Coupled numerical modelling of electrokinetic soil treatments (THMC-EKsoil)', which is focuses on the design, implementation, verification and validation of a numerical code with these capabilities (See Fig. 1). The work presented in this article is part of the first activities of this project, specifically the design and implementation of the electrokinetic transport model and its verification against analytical solutions and other codes. COMSOL Multiphysics has been used as the implementation platform where all the equations have been programmed following the constitutive models presented in the next section. The previously published model, M4EKR [13], without pre-programmed COMSOL modules, has been used as the code base. The code solves all the balance equations simultaneously. It uses a monolithic, fully coupled approach. COMSOL handles pre-processing (discretisation, initial and boundary conditions), solving the system of partial differential equations (PDE), and post-processing the results.

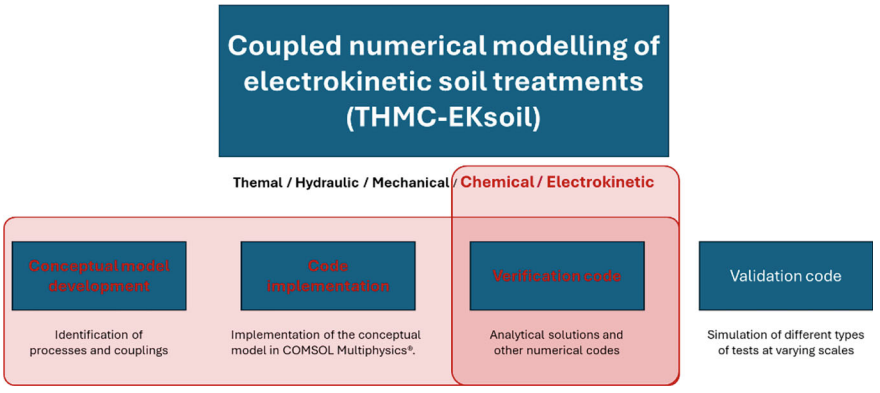

Fig. 1 Schematic of the activities presented in the paper and their location within the different actions envisaged in the project

2 Mathematical Model

The Nernst-Planck-Poisson equations, which describe the transport of charged species and electric current in porous media considering the effect of electrostatic interactions in natural transport [17] and under the application of an external electric gradient [14, 18], are used as the basis for the conceptual transport model, as are more advanced codes for the transport of chemical species.

2.1 Mass Transport

The conservation of mass is defined by a general balance equation for each of the chemical components. The equation takes the form

$$\frac{\partial m_{\mathrm{i}}}{\partial t} + \nabla \cdot \mathbf{j}_{\mathrm{i}} = f_{\mathrm{i}} \tag{1}$$

where m_{i} is the total mass per unit of volume, $\mathbf{j}_{\mathrm{i}}$ is the flux and f_{i} is the sink/source term of the i-th component, respectively. The calculation of the total mass depends on whether the soil is structured as a double porous medium and its saturation state [19]. On the other hand, the Nernst-Planck equations were used to calculate the total flux. These equations consider different transport mechanisms. The following contributions have been distinguished: (i) diffusive flux, $\mathbf{j}_{\mathrm{dif,i}}$, caused by concentration gradients (the effect of the activity coefficients is included), (ii) Electromigration flux, $\mathbf{j}_{\mathrm{em,i}}$, resulting from the action of an external electrical potential gradient (ψ_{ext}), (iii) flux induced by coulombic effects, $\mathbf{j}_{\mathrm{ce,i}}$, resulting from local charge neutrality constraint during multicomponent transport, and (iv) advective flux, $\mathbf{j}_{\mathrm{adv,i}}$, which considers the contributions of hydraulic and electroosmotic gradients. The diffusive flux is defined as

$$\mathbf{j}_{\mathrm{dif,i}} = -D_{e,\mathrm{i}} c_{\mathrm{i}} \nabla \left(\ln a_{\mathrm{i}} \right) \tag{2}$$

where c_{i} and a_{i} are the concentration and chemical activity of i-th component, respectively, and $D_{e,\mathrm{i}}$ its effective diffusion coefficient. The electromigration flux is

$$\mathbf{j}_{\mathrm{em,i}} = -u_{e,\mathrm{i}} c_{\mathrm{i}} \nabla \psi_{\mathrm{ext}} \tag{3}$$

where $u_{e,\mathrm{i}}$ is the effective ionic mobility [19]. The flux induced by coulombic effects

$$\mathbf{j}_{\mathrm{ce,i}} = -u_{e,\mathrm{i}} c_{\mathrm{i}} \nabla \psi_{\mathrm{ce}} \tag{4}$$

where ψ_{ce}^{L} is the electrical potential that includes the electrostatic effects generated by multicomponent diffusion that can be calculated as [17]

$$\nabla\psi_{ce} = -\frac{\sum_{i=1}^{N} z_i D_{e,i} c_i \nabla\left(\ln a_i\right)}{\sum_{i=1}^{N} z_i u_{e,i} c_i} \tag{5}$$

The advective flux is calculated considering the contribution of hydraulic and electroosmotic water fluxes ($\mathbf{q}_h$ and $\mathbf{q}_{eo}$)

$$\mathbf{j}_{adv,i} = c_i\left(\mathbf{q}_h + \mathbf{q}_{eo}\right) \tag{6}$$

2.2 Electric Current Transport

A formulation based on electric charge conservation and the electroneutrality condition is implemented in the proposed model. This approach is equivalent to the use of Gauss's law to obtain Poisson's equation [19]. The electrical charge balance can be formulated in a simple way, assuming that the medium is initially electrically balanced and considering that during the electrokinetic process, no charge accumulation process is enabled [20].

$$\nabla \cdot \mathbf{i} = 0 \tag{7}$$

In this model, the electric current $\mathbf{i}$ is calculated assuming only the contributions from the transport of ionic species in the liquid phase (the conduction in the soil matrix is not assumed in the verification cases performed) according to this expression

$$\mathbf{i} = F\sum_{i=1}^{N} z_i \mathbf{j}_i \tag{8}$$

where z_i is the electric charge of i-th component.

2.3 Additional Considerations in the Mathematical Formulation

From a chemical reactivity point of view, only water electrolysis has been considered. This process involves two semi-reactions: (i) the oxidation of the water at the anode

and (ii) its reduction at the cathode. The mass of protons and hydroxyl ions generated in this process is calculated using Faraday's law. The benchmark in which this process has been verified presents an experimental setup in which the electrodes are in electrolyte compartments. In this case, it is necessary to implement a mass balance using ODEs to calculate the variation of proton and hydroxyl ion concentration in the anolyte and catholyte compartments. The state variable of these ODEs will be the concentration of these species in the compartments and will act as a boundary condition in the mass balance equations formulated in the soil domain [13].

3 Description of Verification Exercises

This section describes the four verification exercises (VE) simulated in this work. The main parametrization can be found in Table 1.

The first VE (VE1) aims to verify the electrokinetic transport of a synthetic cation (A+) in a one-dimensional domain through the processes of electromigration and electroosmosis under a constant electric potential gradient. The concentration of the soil porewater is much higher (1000 times higher) than in the anodic contour where A+ infiltrates, which allows to use as a benchmark the time-dependent analytical solution for 1-D advection–diffusion presented in [21] and extensively detailed in [14].

The purpose of VE2 is to check the transport by electromigration of four species under a multicomponent diffusion scenario. As a benchmark, the stationary analytical solution developed in [22] has been used to verify both the spatial distribution of the electric potential and the concentration of the different species involved. The

Table 1 Main parameterisation of the verification exercises

Parameter				VE1	VE2	VE3	VE4
Length [m]				0.2	0.01	1	0.03
Electric potential gradient [$V \cdot m^{-1}$]				100	300	100	100
Electroosmotic permeability [$m^2 \cdot V^{-1} \cdot s^{-1}$]				$2.00 \cdot 10^{-9}$	0	0	0
Tortuosity [−]				0.1	1	0.4	0.1
Porosity [−]				1	1	0.4	0.1
Multicomponent diffusion				No	Yes	Yes	No
Diffusion coefficient [$m^2 \cdot s^{-1}$]							
VE1		VE2		VE3		VE4	
A+	$9.31 \cdot 10^{-9}$	Na+	$1.33 \cdot 10^{-9}$	K+	$1.96 \cdot 10^{-9}$	Na+	$1.00 \cdot 10^{-9}$
B−	$9.31 \cdot 10^{-9}$	Cl−	$2.03 \cdot 10^{-9}$	$C_3H_5O_3$–MnO_4	$1.03 \cdot 10^{-9}$–$1.63 \cdot 10^{-9}$	Cl−	$1.00 \cdot 10^{-9}$
C+	$9.31 \cdot 10^{-9}$	K+	$1.96 \cdot 10^{-9}$	Na+	$1.33 \cdot 10^{-9}$	H+	$1.00 \cdot 10^{-9}$
D−	$9.31 \cdot 10^{-9}$	OH−	$5.27 \cdot 10^{-9}$	HCO_{3-}	$1.18 \cdot 10^{-9}$	OH	$1.00 \cdot 10^{-9}$

concentration values of the boundaries and the domain can be found in Table 3 of [14].

The new feature of VE3 is that it evaluates electromigratory transport in a multicomponent diffusion problem, but in this case from a time-dependent perspective. To demonstrate the capabilities of the model, the spatial concentration profiles at different times of two species with different diffusion coefficients have been analysed, using as a benchmark a simulation exercise performed with the NP-Phreeqc-EK model [23], where the necessary information to define the boundary conditions can be found.

Finally, the VE4 aims to verify the electromigratory transport in a problem that assumes chemical reactivity, specifically the water electrolysis process and, consequently, the effects of the pH front generated, which is so important in electrokinetic treatments. For this purpose, a synthetic case simulated with Phreeqc has been used as a benchmark in which the spatial profiles of pH, electric potential gradient, and concentration of species have been analysed, as well as the temporal evolution of the electric current in the process. Details of the initial and boundary conditions can be found at [24].

4 Results of Verification Exercises

This section presents the results of the four verification exercises performed and their direct comparison with the reference benchmarks described in the previous section.

Figure 2 shows the results of VE1, specifically the propagation front of the A+ species evaluated at different observation times. In this case, the advective term included in the analytical solution considers the contribution of electromigration and electroosmosis generated by a constant electric potential gradient. The fit with the analytical reference solution is very good.

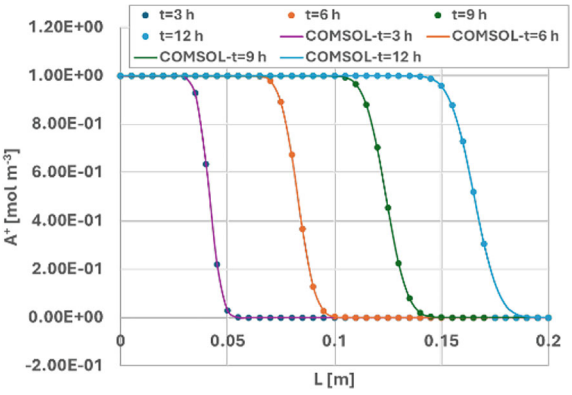

Fig. 2 Results of VE1. Spatial distribution of A+ concentration at different times. Comsol model (lines), Benchmark (markers)

The results of VE2 are illustrated in Fig. 3. Figure 3a shows the spatial distribution of the species concentration, and Fig. 3b, that of the electric potential, both evaluated at steady state. In this exercise it can be seen that the concentration profiles obtained in the domain show different trends, which are largely influenced by the application of individualised diffusion coefficients for each species. This in turn results in a non-linear electrical potential distribution. In both cases, the agreement between the model and the benchmark is satisfactory.

Figure 4 shows the results obtained in VE3. Unlike the previous case, in this exercise, the propagation fronts of two species (with different diffusive properties) are analysed at different observation times. It can be seen how the "multicomponent effect" influences both the trends of the fronts and their interaction with the species present in the domain, being able to reach concentrations higher than the background (10 mol m-3) in the case of MnO_4^- (Fig. 4a) and lower in the case of $C_3H_5O_3^-$ (Fig. 4b). In both cases, they show identical behaviour to that observed in the benchmark.

The results obtained in the simulations carried out on VE4 are presented in Fig. 5.

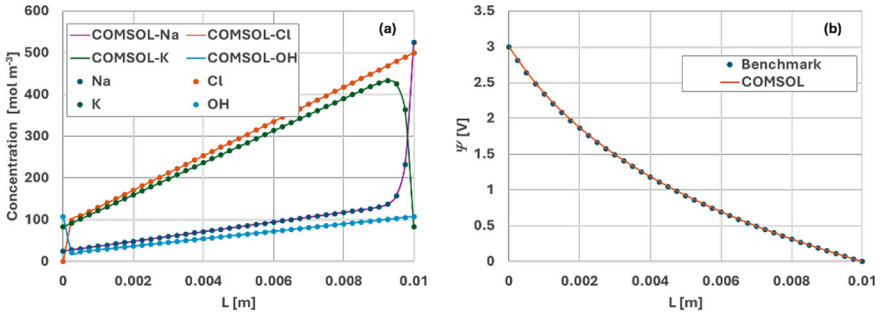

Fig. 3 Results of VE2. Spatial distribution of (**a**) concentration and (**b**) electric potential evaluated at the stationary state. Comsol model (lines), Benchmark (markers)

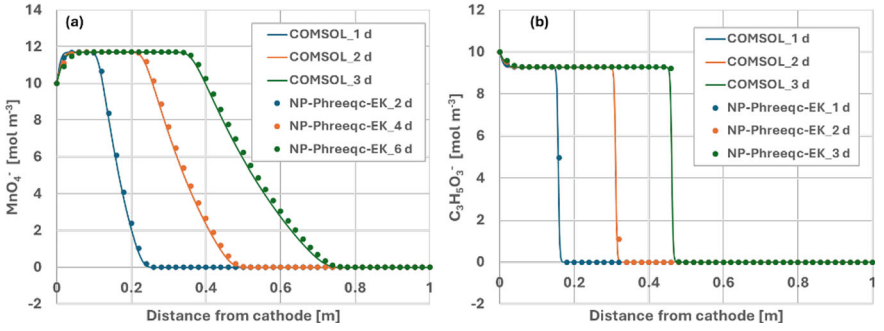

Fig. 4 Results of VE3. Spatial distribution of (**a**) MnO_4^- and (**b**) $C_3H_5O_3^-$ concentration evaluated at different times. Comsol model (lines), Benchmark (markers)

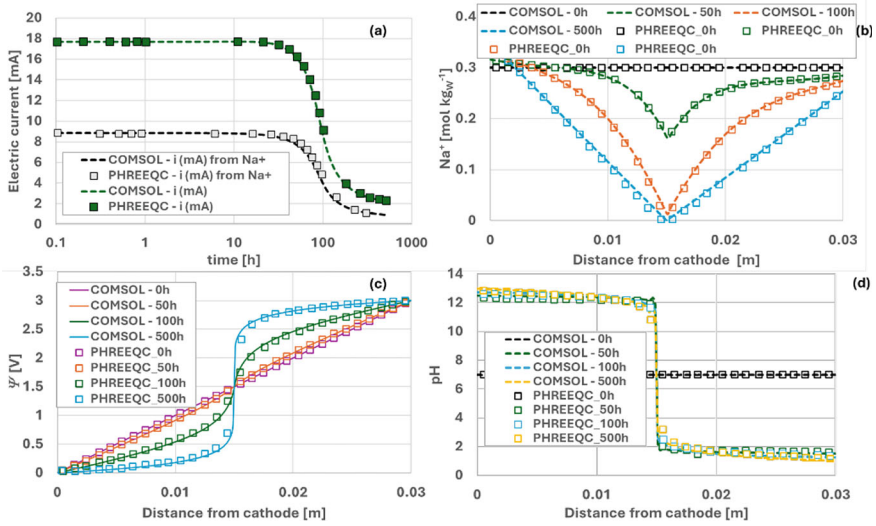

Fig. 5 Results of VE4. (**a**) Electric current time evolution. Spatial distribution of (**b**) Na concentration, (**c**) electric potential, and (**d**) pH evaluated at different times. Comsol model (lines), Benchmark (markers)

Figure 5a shows the effect of ionic mobility on the calculation of the electric current and how it decreases with time as the ion concentration in the domain decreases due to its mobility. This is illustrated in Fig. 5b, where there is an increase in Na concentration in the cathodic zone (attraction to oppositely charged electrodes) and a decrease in the anodic zone (repulsion from electrodes of the same charge). This in turn has a response in the electrical potential (Fig. 5c). Initially, the electrical gradient is linear when the concentrations in the region are constant. However, as the concentrations change, the gradient also changes, becoming higher where concentrations are lower, thus keeping the current constant across the domain. Finally, Fig. 5d shows the pH profiles characteristic of a water electrolysis process. In general, there is good agreement between the model and the benchmark results.

5 Conclusions and Future Work

In this work, the following phenomena have been successfully verified: advective transport by electroosmosis, transport of ionic species by electromigration, multicomponent diffusive transport, charge balance and electric potential distribution, simple chemical speciation, and coupling between liquid phase ionic transport and electric charge transport. From a chemo-electrokinetic point of view, the next developments will focus on the implementation of a real geochemical system with heterogeneous chemical reactivity (minerals and gases) and the inclusion of the solid phase

contribution to electric charge transport. Next, according to the project schedule, is the implementation and verification of the thermal and mechanical models.

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DAT File for Geochemical Modeling (DAT4GM): A Python Tool to Provide Reactive Transport Models for Multiphysics Platforms

Adrián Sánchez-Migallón, Rubén López-Vizcaíno, Xiaojuan Yang, Vicente Navarro, and Ángel Yustres

Abstract Geochemical modeling plays a crucial role in understanding natural processes in soils and other porous media. However, the implementation of reactive transport models in multiphysics platforms often encounters difficulties, especially when the use of extensive geochemical databases is needed, as it requires the manual creation of libraries containing the necessary formulation of each chemical species. The present work addresses this limitation by means of a tool created in Python capable of efficiently managing large geochemical databases in Phreeqc format. DAT4GM software (DAT file for Geochemical Modeling) allows the generation of the necessary code to model reactive transport in multiphysics platforms automatically, regardless of the number of species composing the selected geochemical system. Additionally, it can define the initial and boundary conditions of the transport problem. For the verification of this tool, different cases have been implemented in COMSOL. These include geochemical speciation and reactive transport with redox species in a one-dimensional configuration. In all cases, the results are satisfactory, even with geochemical configurations involving a large number of components and secondary species.

Keywords COMSOL · Python · Multiphysics modeling · Chemical speciation · Reactive transport

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1 Introduction

In recent years, computational models for simulating complex physical, chemical, and biological processes have advanced significantly across disciplines like environmental engineering, materials science, and geoscience. Key processes such as reactive transport are crucial for predicting environmental impacts and developing sustainable technologies [1]. Accurate modeling often requires integrating multiple interacting phenomena, including thermal, hydraulic, mechanical, and chemical (THMC) processes, to capture the diverse behavior of porous media [2].

Efforts to develop coupled THMC models have led to two primary approaches: integrating independent modules (operator splitting strategy) or adopting monolithic frameworks for seamless process coupling [3]. These models are essential for applications like carbon capture and storage or radioactive waste disposal [4]. However, their complexity often restricts use to academia, requiring expertise across disciplines.

The demand for accessible multiphysics platforms has encouraged the rise of tools like COMSOL Multiphysics, known for solving coupled partial differential equations. Yet, reactive transport models in such platforms often rely on operator splitting, which can hinder computational efficiency and accuracy in geochemical equilibrium systems.

The aim of this work is to provide an enhancement that facilitates the development of reactive transport models, considering a monolithic computational strategy. The DAT4GM tool automates the implementation of complex geochemical systems, thereby simplifying workflows and enhancing computational efficiency. It features a user-friendly graphical interface that enables the generation of libraries containing numerous geochemical formulations with just a few steps and with a very reduced consumption of time resources since this action occurs in a fraction of a second. The computational model developed by the tool and implemented in COMSOL has been rigorously verified with Phreeqc [5], using porewater references from nuclear waste repositories [6].

2 Calculational Model

2.1 Chemical Speciation

The chemical speciation model employed distinguishes between two types of elements: the components or master species (M), which represent the primary families of species, and the secondary species (S), which are derived from the master species. The total number of species in the system is thus the sum of these two groups ($I = M + S$).

Master species are defined by reactions that depend solely on themselves (e.g., $\mathrm{Cl} = \mathrm{Cl}$), whereas secondary species are formed through reactions involving only

master species (e.g., CH_4—H=CH_3, where CH_4 and H are master species and CH_3 is a secondary species).

For this model, the number of unknown variables to solve is M + 1 (the logarithm of the activity of the master species plus the ionic strength, I_S). However, pH, pE, and the I_S must be provided as inputs (that will be adjusted in each iteration), concentrations of protons and electrons are not required, meaning that the number of unknowns is reduced to M—1.

In order to compensate for these number of unknowns, total concentration equations (Eq. 1) of M—4 components—the master species, with the exception of H^+, e^-, H_2O, and a species that will ensure electroneutrality—plus three additional equations—electroneutrality (Eq. 2), water activity (Eq. 3) and I_S (Eq. 4)—will be used.

$$C_m = \sum_{i=1}^{I} (\alpha_{i\ m}\ c_i) \tag{1}$$

where α_{im} is the value of the stoichiometric coefficient given by the formula of the chemical reaction considered from the database, and c_i is the molal concentration of the i-th species.

$$c_m = -\frac{1}{z_m} \sum_{i=1}^{I-m-e} c_i\ z_i \tag{2}$$

$$a_{H_2O} = 1 - 0.017 \sum_{i=1}^{I-H_2O-e} c_i \tag{3}$$

$$I_S = \frac{1}{2} \sum_{i=1}^{I-e} c_i\ z_i^2 \tag{4}$$

The molal concentration of the i-th secondary species is determined from the chemical activity definition, where the logarithm of the activity is obtained from each equilibrium reaction and the equilibrium constant from the databases:

$$c_i = \frac{a_i}{\gamma_i} \tag{5}$$

$$\log_{10}(a_i) = \log_{10} K_{eq,i} + \sum_{m=1}^{M} \alpha_{im} \log_{10}(a_m) \tag{6}$$

A critical component of ensuring an optimal model fit is the calculation of the activity coefficient, γ_i. This varies depending on whether a Lawrence Livermore National Laboratory (LLNL) aqueous model database or a common Phreeqc database is utilized, as illustrated in Fig. 1 and the corresponding equations Eq. 7–12 [7–10].

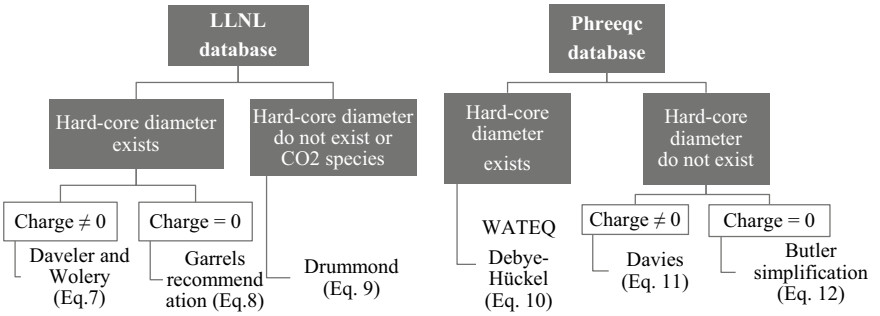

Fig. 1 Computational model used in each case

$$\log_{10}\gamma_i = -\frac{Az_i^2\sqrt{I_S}}{1 + a_i^0 B\sqrt{I_S}} + B^0 I_S \tag{7}$$

$$\log_{10}\gamma_i = 0 \tag{8}$$

$$\ln\gamma_i = \left(C + FT + \frac{G}{T}\right)I_S - (E + HT)\left(\frac{I_S}{I_S + 1}\right) \tag{9}$$

$$\log_{10}\gamma_i = -\frac{Az_i^2\sqrt{I_S}}{1 + a_i^0 B\sqrt{I_S}} + b_i I_S \tag{10}$$

$$\log_{10}\gamma_i = -Az^2\left(\frac{\sqrt{I_S}}{1 + \sqrt{I_S}} - 0.3\, I_S\right) \tag{11}$$

$$\log_{10}\gamma_i = 0.1 I_S \tag{12}$$

2.2 Reactive Transport

Once the speciation is solved, the tool is also designed to generate libraries with transport equations to be implemented in COMSOL. It should be noted that the approach given to the electron is the same as for the other components, except that, since it does not exist as an aqueous species (otherwise there would be an electric current), only its concentration bound to other species was considered in calculating its diffusive flux. These M-2 mass balance equations (referring to the master species except H_2O and the species ensuring the electroneutrality) are formulated as follows:

$$\frac{\partial m_m}{\partial t} + \nabla \cdot \boldsymbol{l}_m = R_m \tag{13}$$

where m_m is the total mass of the m-th master species per unit total volume and $\boldsymbol{l}_m$ is the total molal flux.

$$m_m = \varphi Sr C_m \tag{14}$$

$$\boldsymbol{l}_m = \boldsymbol{l}_m^{\text{adv}} + \boldsymbol{l}_m^{\text{dif}} \tag{15}$$

Here, φ is the porosity, Sr is the degree of saturation, and C_m is the total molal concentration. And in the second equation $\boldsymbol{l}^{\text{adv}}$ and $\boldsymbol{l}^{\text{dif}}$ are the advective and diffusive flux, respectively, defined as:

$$\boldsymbol{l}_m^{\text{adv}} = C_m \boldsymbol{q}^{\text{adv}} \tag{16}$$

$$\boldsymbol{l}_m^{\text{dif}} = -\sum_{i=1}^{I} D_i^e \alpha_i \nabla c_i \tag{17}$$

where $\boldsymbol{q}^{\text{adv}}$ is the advective velocity in Eq. 16. And, in Eq. 17, c_i is the concentration of the i-th secondary species, and ${D_i}^e$ is the effective diffusion coefficient, which depends on the tortuosity and ${D_i}^0$, the binary diffusion coefficient:

$$D_i^e = \varphi \; Sr \; \tau \; D_i^0 \tag{18}$$

3 Tool Description

The interface of the DAT4GM tool is composed of four panels (Fig. 2). The Component Panel comprises the available master species (automatically added after selecting a database), temperature, and the species employed to ensure electroneutrality in the calculations. The selection of the libraries of code generated is made in the Library Panel. Each library is displayed in the Results Panel and can be exported in TXT format. Finally, the Speciation Panel contains the inputs to perform a chemical speciation, the results of which serve as initial and boundary conditions for the multiphysics software.

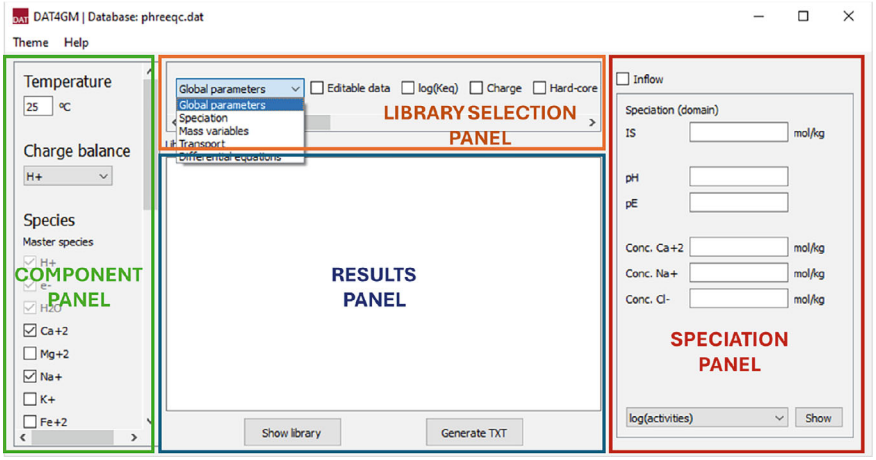

Fig. 2 DAT4GM interface

4 Results

To verify the correct functioning of the tool, a brackish water (with pH 7.12 and pE–3.3611) and a diluted water (with pH 7.49 and pE–3.7942) were used, both taken from reference waters for deep geological storage of radioactive waste.

4.1 Chemical Speciation Verification

For the first verification, a speciation of two waters with 17 components each, a brackish water and a diluted water at 25 °C each, was performed (Table 1).

Chemical speciation results were verified using Phreeqc, one of the most widely recognized and validated geochemical modeling software. Satisfactory results were obtained in the logarithms of the activities of the master species (Table 2) and in the rest of the speciation variables.

4.2 Reactive Transport Verification

The final verification exercise examines the application of the libraries automatically generated by DAT4GM and implemented in COMSOL to simulate a one-dimensional transport case. This simulation spans two days and involves advective-diffusive transport within a 0.5-m domain. The initial condition throughout the domain corresponds to the brackish water described in the preceding section, while a Dirichlet boundary condition is applied at $x = 0$ m, representing the diluted solution (Table 1). At the

Table 1 Total concentrations for brackish and diluted pore waters [7]

Total concentrations [Mol/kg]	Brackish	Diluted
Na	$7.71 \cdot 10^{-2}$	$1.32 \cdot 10^{-2}$
K	$4.70 \cdot 10^{-4}$	$2.50 \cdot 10^{-4}$
Mg	$7.45 \cdot 10^{-3}$	$7.40 \cdot 10^{-4}$
Ca	$1.62 \cdot 10^{-2}$	$1.19 \cdot 10^{-3}$
Cl	$1.14 \cdot 10^{-1}$	$1.10 \cdot 10^{-2}$
S	$4.82 \cdot 10^{-3}$	$9.60 \cdot 10^{-4}$
C	$1.74 \cdot 10^{-3}$	$4.52 \cdot 10^{-3}$
Sr	$9.30 \cdot 10^{-5}$	$5.70 \cdot 10^{-6}$
Si	$1.80 \cdot 10^{-4}$	$1.80 \cdot 10^{-4}$
Mn	$2.00 \cdot 10^{-5}$	0.00
Fe	$5.70 \cdot 10^{-6}$	$8.10 \cdot 10^{-6}$
F	$1.60 \cdot 10^{-5}$	$3.20 \cdot 10^{-5}$
Br	$1.70 \cdot 10^{-4}$	$1.80 \cdot 10^{-5}$
B	$5.70 \cdot 10^{-5}$	$2.70 \cdot 10^{-5}$

opposite boundary ($x = 0.5$ m), a Cauchy flux-type boundary condition is imposed, equivalent to the sum of the advective flux.

The libraries for this particular case include the calculations of 78 secondary species (apart from the 17 components), which have been taken fully automatically from the database based on the components present in the solution. Furthermore, the tool has produced the formulation considering the strong couplings in the model.

As demonstrated by the concentration profiles obtained in COMSOL with the libraries generated by DAT4GM, the model exhibits a high degree of compatibility for all species, including redox species, characterized by their high reactivity (Fig. 3).

Furthermore, the pH and pE plots, which are known to have slightly higher sensitivity to minor alterations, demonstrate a perfect fit to those obtained by Phreeqc, as can be seen in Fig. 4.

The physical parameters employed for this simulation of brackish and diluted pore water transport are listed in Table 3. The degree of saturation, porosity, and tortuosity has been set equal to one.

And, as parameters for the model built in Phreeqc and the multiphysics software, a mesh of 200 elements and an element ratio of 20 have been used.

Table 2 Speciation results of Phreeqc versus DAT4GM

	Brackish (domain)			Diluted (boundary)		
	Phreeqc	DAT4GM	Rel. error [%]	Phreeqc	DAT4GM	Rel. error [%]
Log (activity)						
H_2O	$-1.63{\cdot}10^{-3}$	$-1.63{\cdot}10^{-3}$	$3.69{\cdot}10^{-10}$	$-2.36{\cdot}10^{-4}$	$-2.36{\cdot}10^{-4}$	$7.26{\cdot}10^{-9}$
e^-	3.36	3.36	0.00	3.79	3.79	0.00
H^+	−7.13	−7.13	$3.95{\cdot}10^{-8}$	−7.49	−7.49	$1.36{\cdot}10^{-9}$
Ca^{+2}	−2.27	−2.27	$1.76{\cdot}10^{-10}$	−3.19	−3.19	0.00
Mg^{+2}	−2.60	−2.60	$1.16{\cdot}10^{-10}$	−3.40	−3.40	$2.94{\cdot}10^{-11}$
Na^+	−1.23	−1.23	0.00	−1.94	−1.94	0.00
K^+	−3.46	−3.46	$2.89{\cdot}10^{-11}$	−3.66	−3.66	$1.09{\cdot}10^{-10}$
Fe^{+2}	−5.76	−5.76	$6.25{\cdot}10^{-10}$	−5.46	−5.46	$1.83{\cdot}10^{-10}$
Mn^{+2}	−5.25	−5.25	$1.24{\cdot}10^{-9}$	−999	−999	0.00
Sr^{+2}	−4.51	−4.51	$6.64{\cdot}10^{-11}$	−5.51	−5.51	$7.26{\cdot}10^{-11}$
H_4SiO_4	−3.73	−3.73	$2.68{\cdot}10^{-10}$	−3.75	−3.75	$2.67{\cdot}10^{-11}$
Cl^-	−1.08	−1.08	0.00	−2.02	−2.02	$4.96{\cdot}10^{-11}$
$CO_3{}^{-2}$	−6.17	−6.17	$6.00{\cdot}10^{-8}$	−5.27	−5.27	$2.09{\cdot}10^{-9}$
$SO_4{}^{-2}$	−3.01	−3.01	$1.33{\cdot}10^{-10}$	−3.32	−3.32	$1.81{\cdot}10^{-10}$
H_3BO_3	−4.23	−4.23	$8.50{\cdot}10^{-10}$	−4.58	−4.58	$2.19{\cdot}10^{-11}$
F^-	−5.01	−5.01	$2.00{\cdot}10^{-11}$	−4.57	−4.57	$3.50{\cdot}10^{-10}$
Br^-	−3.91	−3.91	0.00	−4.81	−4.81	$2.70{\cdot}10^{-10}$
Ionic strength [Mol/kg]						
I_S	$1.47{\cdot}10^{-1}$	$1.47{\cdot}10^{-1}$	$6.81{\cdot}10^{-11}$	$1.96{\cdot}10^{-2}$	$1.96{\cdot}10^{-2}$	$3.58{\cdot}10^{-9}$

Note Thirteen significant figures were used in the error calculations

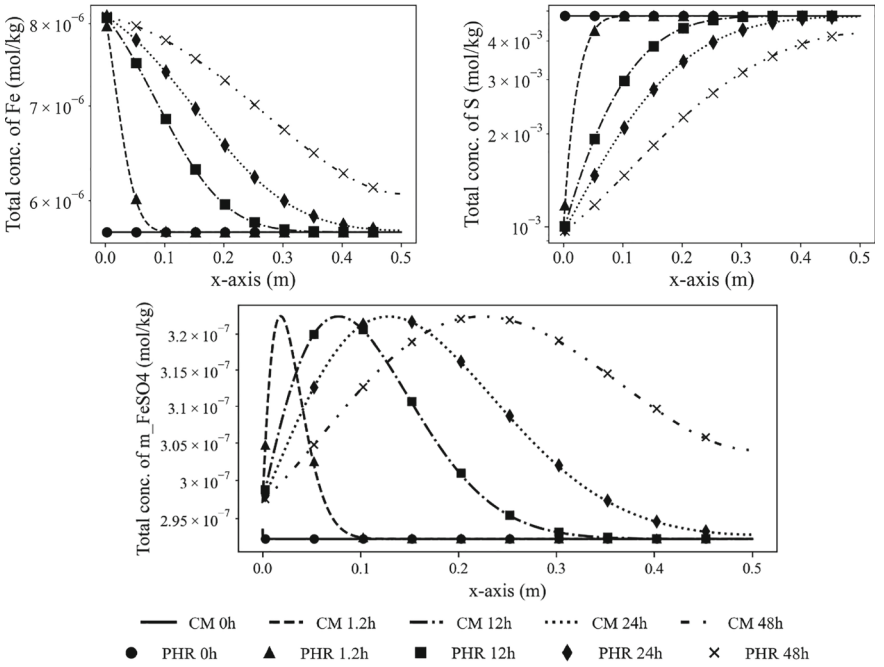

Fig. 3 Representation of $FeSO_4$ concentration and its components Fe and S

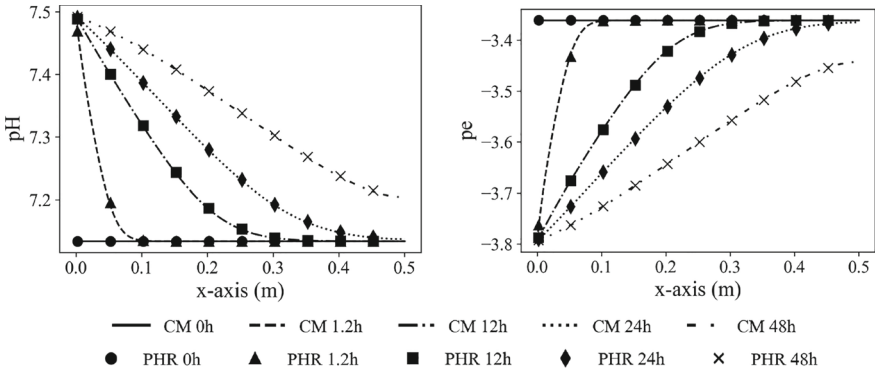

Fig. 4 Representation of pH and pE curves

Table 3 Physical parameters

Parameter	Value
Temperature (T)	25°C
Advective velocity (q^{adv})	10 cm d^{-1}
Binary diffusion coefficient ($D^0{}_i$)	$1.157 \cdot 10^{-7}$ m^2 s^{-1}

5 Conclusion

This study presents a novel Python-based tool designed to streamline the creation of geochemical modeling libraries. By offering an intuitive graphical user interface, the tool significantly reduces the complexity of developing libraries for speciation and transport models. It achieves this by automatically extracting accurate information from geochemical databases, enabling users to generate libraries with just a few interactions. The tool's flexibility allows for the creation of models of any size, tailored to the specific components selected by the user.

Verification results demonstrate that the speciation graphs produced by the tool align perfectly with those generated by established geochemical modeling software, confirming the accuracy and reliability of the libraries.

Moreover, as an open-source software, the tool is freely available for download and modification, empowering users to adapt it to their specific needs and contribute to its ongoing development, supporting practical applications in geochemical modeling.

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Comparison of SiteWise and SimaPro for Sustainability Assessment of Remediation Alternatives

Banuchandra Nagaraja and Krishna R. Reddy

Abstract Contaminated sites significantly threaten public health and the environment, underscoring the urgent need for effective remediation efforts. The regulations to address these sites require the contamination to be reduced to acceptable levels based on the risk of exposure. Numerous remediation technologies have been developed for cleanup; however, selecting an appropriate technology is contingent upon the site's geology, hydrogeology, contamination characteristics, and associated environmental risks. Abandoned or uncontrolled contaminated sites in the United States are regulated under Superfund (CERCLA), which employs nine-point criteria tailored to the site-specific characteristics for remediation method selection. However, these approaches do not account for the broader negative environmental impacts and the social and economic sustainability considerations. USEPA's Green Remediation promotes consideration of the environmental impacts to maximize the net environmental benefits of adopting remediation methods. This study focuses on these environmental aspects of sustainability. It employs SiteWise™ and SimaPro LCA, two standard tools for assessing the environmental sustainability of remediation technologies, to evaluate and select technologies to remediate contaminated soil and groundwater at an Oil refinery facility in Lincoln, Nebraska, USA. The study provides an overview of the site's background, contamination, and potential remediation alternatives. Each tool provided several unique sets of indicators that were not quantified in the other. SiteWise offers a more tailored approach to assessing remediation alternatives. In contrast, SimaPro's complexity and broader scope make it more suitable for detailed environmental sustainability assessment. Common indicators between the tools are identified and analyzed, while the differences and additional indicators unique to each tool are discussed. The study further evaluates the strengths and limitations of these tools, emphasizing their practical application to quantify the broader environmental sustainability of remediation technologies. This study highlights the benefits of adopting sustainability assessment tools in selecting

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remediation technologies, demonstrating their critical role in advancing sustainable environmental remediation practices.

Keywords Contaminated site · Remediation technologies · LCA · SimaPro · SiteWise

1 Introduction

Environmental contamination is a significant consequence of anthropogenic activities, impacting soil, air, and water. Contaminated sites pose severe risks to human health, ecosystems, and wildlife. The traditional approaches to handling contaminated sites focus on risk-based remediation [1]. The regulatory requirements address the contamination risks by characterizing the contaminated site to find the risk to human health and the environment. Remedial goals are defined based on the risk, and the remediation methods are chosen to reduce the contamination to acceptable levels. Further, the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA), also known as Superfund, as a part of the United States regulatory framework, was enacted to address the cleanup of abandoned waste and hazardous sites [2]. This regulatory framework addresses remedial action funding, liability, and prioritization of such sites [3]. Moreover, CERCLA's nine criteria guide remediation based on effectiveness, compliance, cost, and state and community acceptance [2].

However, this practice does not account for the broader negative environmental impacts of adopting remediation methods. Several agencies around the world, including the US Environmental Protection Agency (USEPA), Sustainable Remediation Forum (SURF), Interstate Technology and Regulatory Council (ITRC), and American Society for Testing and Materials (ASTM), have been driving for the incorporation of broader environmental impacts as well as social acceptance and economic considerations to promote sustainability [3]. The current study focuses on the environmental impacts of remediating a contaminated site. USEPA's Green Remediation is one such initiative that promotes considering all environmental effects of remedy implementation and incorporating options to maximize the net environmental benefits of cleanup action [4].

Environmental sustainability can be quantified using various tools, ranging from qualitative yes/no evaluations to quantitative methods such as carbon footprint analysis and life cycle assessment (LCA). Tools such as SiteWise and Spreadsheets for Environmental Footprint Analysis (SEFA) are specifically designed to evaluate the environmental impacts of remedial methods [5, 6]. Furthermore, LCA is one of the most widely recognized tools due to its standardized methodology, as defined by ISO 14040 [7]. Various LCA tools have been developed to quantify a project's environmental impacts, which include SimaPro, GaBi, OpenLCA, and Umberto [8]. SimaPro is widely used for various projects [9, 10], while SiteWise is primarily applied to remediation. This study describes both of these tools, highlights their differences, and explains their impact assessment methods to evaluate and select technologies for

addressing contaminated soil and groundwater at an oil refinery facility in Lincoln, Nebraska, USA.

2 SiteWise and SimaPro

The environmental sustainability assessment tools evaluate a project or process's potential broader environmental impacts throughout its life cycle. These tools help quantify various environmental metrics. This section discusses the two tools, SiteWise and SimaPro, for environmental sustainability assessment of remediation projects.

2.1 SiteWise

SiteWise is a Microsoft Excel-based tool initially developed by Battelle Memorial Institute. The United States Army Corps of Engineers and the US Navy further developed the tool to quantify the energy requirements, emissions, and other metrics associated with environmental cleanup projects.

SiteWise 3.2 was used in this study; it consists of a series of Excel sheets and provides a detailed baseline assessment of several environmental sustainability metrics, as listed in Table 1. The Excel sheets include an input sheet, which is the primary data entry point; a worksheet where users can calculate custom operations; separate calculation sheets for four components linked to the input sheet, where the environmental footprints of each component are computed; a summary, which compiles the results from the different calculation sheets; and final summary file compares the outputs of the remedial alternatives.

Methodology. The assessment in SiteWise is carried out by breaking down the remediation process into distinct phases: remedial investigation, remedial action construction, remedial action operation, and long-term monitoring. Designed for versatility, the tool can accommodate any remediation method. It includes setting up and decommissioning monitoring wells and extensive types of equipment that can be used for varied remedial applications. The input materials are considered as consumables, and the environmental impacts are calculated based on the material weight and pre-set emission factors. While SiteWise contains an extensive database of commonly used materials, users can select from five generic impact categories (very low, low, medium, high, and very high) when a specific material is not listed. However, the impact factors of these materials are collected from various sources, making the assessment less transparent. Furthermore, the transportation of personnel, materials, and equipment can be accounted for using road, rail, or air travel. The emission factors are derived from Argonne National Laboratory's Greenhouse Gases, Regulated Emissions, and Energy Use in Transportation (GREET) model [11].

Table 1 Output impact categories and their units from SiteWise and SimaPro

SiteWise		SimaPro (TRACI)	
Impact category	Unit	Impact category	Unit
GHG emissions	Metric ton	Global warming	Kg CO_2 eq
Onsite and total SO_x emissions	Metric ton	Acidification	Kg SO_2 eq
Onsite and total NO_x emissions	Metric ton	Eutrophication	Kg N eq
Onsite and total PM_{10} emissions	Metric ton	Respiratory effects	Kg $PM_{2.5}$ eq
Electricity usage	MWH*	Carcinogenic	CTUh*
Total energy used	MMBTU*	Noncarcinogenic	CTUh*
Water consumption	Gallons	Ecotoxicity	CTUe*
Accident risk fatality		Fossil fuel depletion	MJ surplus
Accident risk injury		Ozone depletion	Kg CFC-11 eq
Additional indicators:		Smog	Kg O_3 eq
Non-hazardous waste landfill space	Metric ton		
Hazardous waste landfill space	Metric ton		
Topsoil consumption	Cubic yards		
Costing	Dollars		
Lost hours- injury	Hours		
Electricity from renewable sources	Percentage		

Note: Abbreviations: GHG—Greenhouse gases; MWH—Megawatt hour; MMBTU—Million British thermal units; CTUh—Comparative toxic units-human; CTUe - Comparative toxic units-ecology; PM—Particulate Matter; CFC– Chlorofluorocarbons

SiteWise can quantify the environmental footprint associated with various equipment types. The tool utilizes established models such as GREET, the Non-Road model, and AP-42 [12], along with USEPA's Current Methodologies in Preparing Mobile Source Port-Related Emissions Inventories [13], to determine impact factors for different types of equipment. The tool's residual handling section also assesses the environmental impacts of transporting residual waste. SiteWise also accommodates the inclusion of laboratory analysis, onsite labor hours, and activities in the environmental assessment, highlighting its flexibility. SiteWise relies on the USEPA Climate Leaders Program [14], the GREET model by the National Renewable Energy Laboratory (NREL), and US EPA models such as Mobile 6, the Non-Road model, and AP-42 [12] for environmental impact calculations. Energy and water consumption are estimated based on equipment use and site operations. At the same time, accident risk, which can be used in social considerations for holistic sustainability assessment, is evaluated using statistical data on fatalities and injuries from sources such as Automobile Transport Statistics, Airplane Transport Statistics, Railroad Transport Statistics, and Labor Statistics.

2.2 SimaPro (Life Cycle Assessment)

SimaPro, developed by PRé Sustainability, is a widely used LCA software designed to quantify the environmental impacts of various projects and processes. This study utilized SimaPro V9.0 for analysis. The LCA, as defined by ISO 14040 [7] and its requirements and guidelines defined in ISO 14044 [15], comprises four fundamental steps: defining the goal and scope of the project, inventory analysis, impact assessment, and interpretation. The evaluation in SimaPro adheres to these steps and allows users to define the goal and scope of the project.

SimaPro contains Eco-invent and other databases that provide standard data covering over 10,000 processes and their impact factors. The user can use the database to build a life cycle inventory. Based on the inventory information, SimaPro produces a table containing elementary flow, representing the environmental emissions and extractions. These elementary flows are assigned to impact categories based on their potential to contribute to various environmental problems. While interpreting results, SimaPro analysis can incorporate factors such as uncertainty in the model due to data variations, accuracy and completeness of the model, sensitivity of the results to assumptions, and the influence of individual processes on the results.

Impact Assessment Methods. SimaPro incorporates multiple impact assessment methodologies and databases tailored to different regions worldwide. Methods developed for the European context include the Center of Environmental Science of Leiden University—Impact Assessment (CML-IA), Environmental Prices, Ecological Scarcity, the Environmental Footprint method (developed by the European Commission), and Environmental Product Declarations (EPDs). The Building for Environmental and Economic Sustainability (BEES) and Tool for the Reduction and Assessment of Chemical and other environmental Impacts (TRACI) have been developed for North America. Additionally, methodologies such as IMPACT, LC-IMPACT, and ReCiPe provide comprehensive life cycle impact assessment approaches with a global scope.

The USEPA developed the TRACI 2.1 model to assess environmental impacts for the US with input parameters consistent with US locations. The impact categories from the TRACI compared to those of SiteWise are presented in Table 1. The indicators include broader environmental impact categories such as carcinogenic and noncarcinogenic emissions, ozone layer depletion potential, ecotoxicity, smog, and fossil fuel depletion, which are not quantified in SiteWise.

3 Contaminated Site Remediation: Case Study

A case study was conducted to compare the two tools, SiteWise and SimaPro, to assess alternatives for the cleanup of soil and groundwater at a contaminated site. The contaminated site considered in this study is a soybean extraction, processing,

and oil refining facility located in Lincoln, Nebraska, USA. The site was investigated after discovering potential carbon tetrachloride contamination in nearby wells. Contamination was primarily found in the soil near the bean silos and a contaminant plume in groundwater, extending up to 0.75 miles northeast of the facility. Carbon tetrachloride, chloroform, methylene chloride, and dibromoethane (volatile organic compounds—VOCs) were the contaminants identified in soil and groundwater. The risks associated with the contaminants for human exposure were significant, necessitating remediation.

3.1 Selection of Remediation Technologies

A technical evaluation was conducted using the Treatment Technologies Screening Matrix [16] developed by the Federal Remediation Technologies Roundtable and USEPA-CERCLA nine-point criteria [2] for screening various remediation technologies. Following this, excavation, electrokinetic (EK) remediation, and emulsified zero-valent iron (EZVI) were selected as the potential soil remediation methods, and Pump and Treat (P & T) and EZVI were chosen as potential groundwater remediation methods.

Excavation involves physically removing contaminated soil, addressing immediate risks to human health or the environment. The excavated hazardous material is then safely disposed [17]. EK remediation uses vertical wells that dissipate electricity into the sub-surface. The heat generated by electricity propagation can aid VOCs' vaporization [18]. EZVI is a method that combines food-grade vegetable oil with a surfactant, elemental iron, and water to form an emulsified mixture. This mixture destroys dense non-aqueous phase liquids (DNAPLs) through abiotic and biotic processes [19]. In pump and treat remediation, groundwater is pumped from wells to an above-ground treatment system, where contaminants are removed [20]. This study considered tray air stripping as the above-ground treatment to remove VOCs efficiently.

3.2 Environmental Sustainability Assessment

The assessment used SiteWise 3.2 and SimaPro V9.0 with TRACI 2.1. The inventory used for the analysis is presented in Table 2. The results from the evaluation using the two tools for soil and groundwater remediation are presented in Figs. 1 and 2. The metrics were normalized by setting the remedial alternative with the highest impact to 1 for comparison.

SiteWise results for soil remediation indicated that EK remediation had the highest impact in all impact categories except fatality and injury accident risk, with excavation posing the highest risks in those categories due to extensive activity. EZVI had the lowest impacts in most categories for soil remediation. Similarly, P & T

Table 2 Materials inventory for environmental sustainability assessment

Materials	Soil remediation			Groundwater remediation	
	Excavation	EK	EZVI	P & T	EZVI
Excavating soil (m^3)	9908.5	509.40	–	–	–
PVC (metric ton)	10.63	0.12	–	1.17	–
Backfill sand (m^3)	9908.5	509.40	–	–	–
EDTA (kg)	–	410.5	–	–	–
NaOH (kg)	–	1132.0	–	–	–
Steel electrodes (metric ton)	–	7.44	–	–	–
Zero valent iron (metric ton)	–	–	19.89	–	15.95
Vegetable oil (metric ton)	–	–	8.95	–	7.18
Steel sheets-Tray air stripper (kg)	–	–	–	272	–

showed higher environmental impacts across all categories in groundwater remediation. However, EZVI posed higher accident risks in both fatality and injury. Further, SiteWise can help quantify additional sustainability indicators (Table 3). These indicators are calculated based on input information, such as cost, residual handling, and electricity usage based on the project's location.

Similarly, SimaPro results suggested that excavation and EK have higher environmental impacts than EZVI in soil remediation, and P & T had higher impacts in groundwater remediation. Excavation was found to have higher ozone layer depletion potential, smog, acidification, carcinogenicity, and fossil fuel depletion, while EK remediation had higher impacts in other categories.

The differences can be attributed to material selection and processes. SiteWise - designed explicitly for assessing remediation methods - includes materials such as zero-valent iron and employs a direct injection process, without the need to specifically estimate corresponding process details. In contrast, in SimaPro, the ZVI material was defined using user-specified energy usage, and the injection process was incorporated through calculated fuel consumption based on the number of injection points and the quantity of material to be injected, both of which were obtained from SiteWise documentation. Similarly, the chemical considered for electrokinetic remediation, Ethylenediaminetetraacetic acid (EDTA), was selected as a low-impact material in SiteWise, while it was available directly in SimaPro. However, both methods estimated that the environmental impacts were lower for EZVI in soil and groundwater remediation, suggesting it is a more environmentally sustainable alternative among the considered methods.

Table 4 presents the similar environmental impacts from the two tools, which estimate similar impacts for each remedial alternative. Despite different approaches, the metrics show that both tools estimate comparable impacts for all the methods. SimaPro quantifies eutrophication and acidification of nitrogen emissions equivalent and sulfur dioxide emissions equivalent, respectively. In contrast, SiteWise quantifies

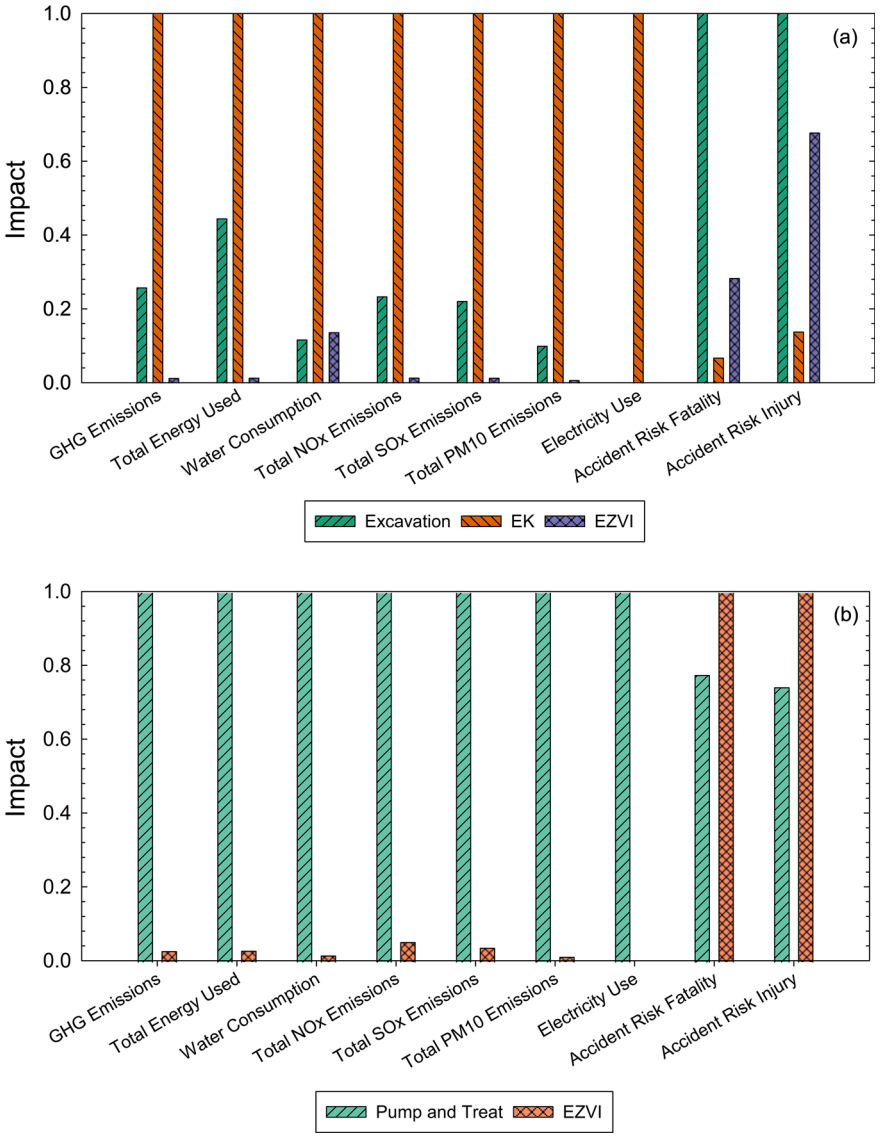

Fig. 1 SiteWise environmental impact results for (a) soil remediation using Excavation, EK (electrokinetic remediation), and EZVI (emulsified zero-valent iron), (b) groundwater remediation using pump and treat, and EZVI (emulsified zero-valent iron)

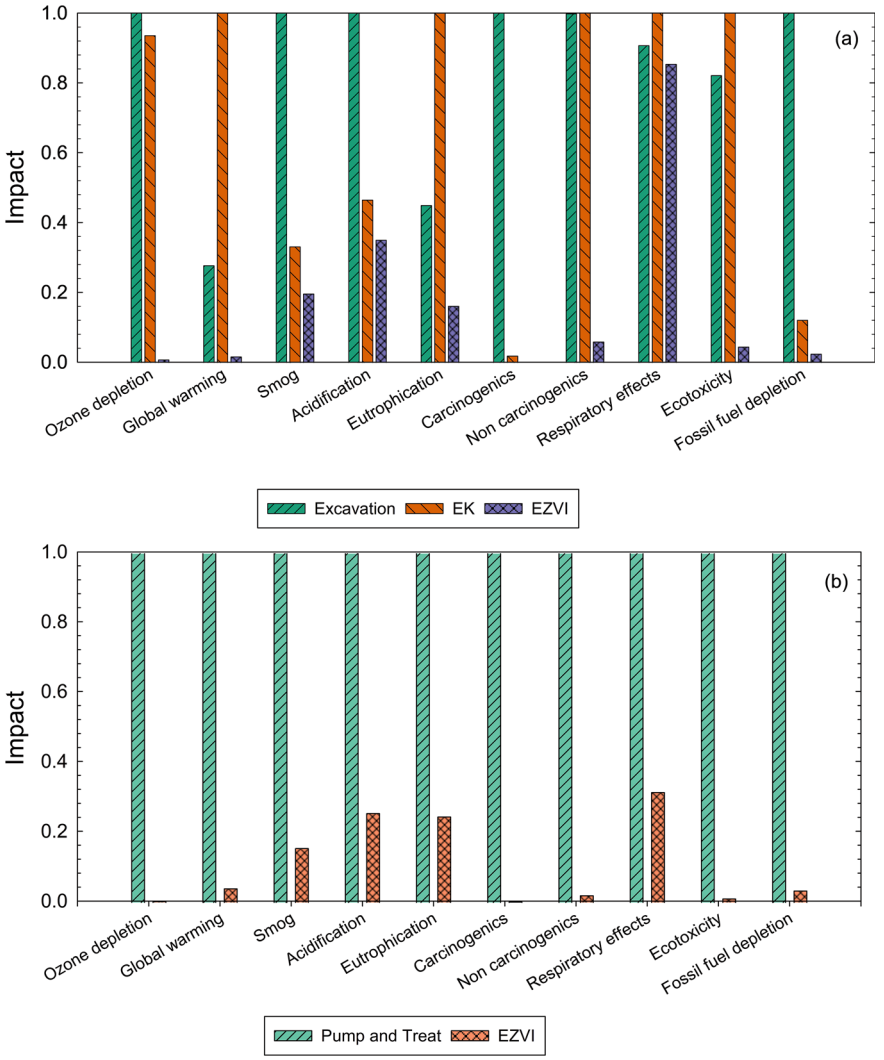

Fig. 2 SimaPro environmental impact results for (a) soil remediation using Excavation, EK (electrokinetic remediation), and EZVI (emulsified zero-valent iron), (b) groundwater remediation using pump and treat, and EZVI (emulsified zero-valent iron)

oxides of nitrogen and sulfur directly, which can be compared with eutrophication and acidification potential from SimaPro. Similarly, SiteWise quantifies PM_{10} emissions, and SimaPro quantifies $PM_{2.5}$ emissions, which are compared. Greenhouse gas emissions (global warming potential is the common indicator in both the tools) and the values were comparable for all the methods.

Table 3 Additional indicators from SiteWise

Indicators	Units	Soil remediation			Groundwater remediation	
		Excavation	EK	EZVI	P & T	EZVI
Non-hazardous waste landfill space	Metric ton	944.74	701.16	–	1.06	–
Hazardous Waste landfill space	Metric Ton	16174.20	831.80	–	–	–
Topsoil Consumption	m^3	–	–	–	–	–
Costing	Dollars	1,100,664	1,086,200	817,524	979,500	684,466
Lost hours-injury	Hours	0.04	0.0056	0.027	0.012	0.016
Electricity from Renewable sources	%	–	4.2%	–	4.2%	–

4 Conclusions

This study evaluates two environmental impact assessment tools: (1) SiteWise, designed for remedial alternatives, and (2) SimaPro, a comprehensive Life Cycle Assessment tool. These tools were used for a case study on the remediation of a contaminated site. The assessment showed that emulsified zero-valent iron had lower environmental impacts than excavation, electrokinetic remediation for soil remediation, and pump and treat in groundwater remediation. Despite differences in methodology, both tools effectively estimate the environmental impacts of remediation. However, they quantify only the environmental sustainability and require a comprehensive triple bottom line evaluation, including social and economic considerations, for holistic sustainability assessment. Other than the common indicators quantified by both the tools, SimaPro includes broader impacts like ecotoxicity, ozone depletion, and fossil fuel depletion, whereas SiteWise includes specific local impacts such as accident risks and fatality, and landfill space occupied. Both methods are limited by database accuracy for site-specific conditions. SiteWise accounts for region-specific electricity usage; however, broader impacts are generalized. Despite these differences, the choice between the two models depends on the scope and depth of assessment required. SiteWise is practical for straightforward assessments, while SimaPro is recommended for comprehensive environmental assessments with methods to verify the veracity of the interpreted results.

Table 4 Comparison of similar environmental impacts from SiteWise and SimaPro

SiteWise		SimaPro	
Impact category	Impact (ton)	Impact category	Impact (ton)
Excavation– Soil			
Greenhouse gas emissions	1019.22	Global warming (CO_2 eq)	1054.74
Total NO_x emissions	2.07	Eutrophication (N eq)	0.61
Total SO_x emissions	3.14	Acidification (SO_2 eq)	3.83
Total PM_{10} emissions	0.57	Respiratory effects ($PM_{2.5}$ eq)	0.28
Electrokinetic remediation- soil			
Greenhouse gas emissions	3942.18	Global warming (CO_2 eq)	3791.12
Total NO_x emissions	8.83	Eutrophication (N eq)	1.36
Total SO_x emissions	14.16	Acidification (SO_2 eq)	1.78
Total PM_{10} emissions	5.66	Respiratory effects ($PM_{2.5}$ eq)	0.31
Emulsified zero-Valent iron- soil			
Greenhouse gas emissions	53.23	Global warming (CO_2 eq)	66.20
Total NO_x emissions	0.13	Eutrophication (N eq)	2.2
Total SO_x emissions	0.2	Acidification (SO_2 eq)	1.34
Total PM_{10} emissions	0.044	Respiratory effects ($PM_{2.5}$ eq)	0.26
Pump and treat- groundwater			
Greenhouse gas emissions	1666.52	Global warming (CO_2 eq)	1316.80
Total NO_x emissions	2.97	Eutrophication (N eq)	7.21
Total SO_x emissions	4.44	Acidification (SO_2 eq)	4.22
Total PM_{10} emissions	2.88	Respiratory effects ($PM_{2.5}$ eq)	0.92
Emulsified zero-Valent iron- groundwater			
Greenhouse gas emissions	49.00	Global warming (CO_2 eq)	52.92
Total NO_x emissions	0.16	Eutrophication (N eq)	1.77
Total SO_x emissions	0.17	Acidification (SO_2 eq)	1.08
Total PM_{10} emissions	0.04	Respiratory effects ($PM_{2.5}$ eq)	0.29

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Mechanical and Durability Study of Fly Ash-Based Geopolymer Bricks Without Fillers

Sher Bahadur Singh, Pabitra Ranjan Maiti, and Supriya Mohanty

Abstract This experimental investigation explores geopolymer bricks made from fly ash of dimensions 230 mm × 110 mm × 70 mm. These ambient-cured bricks were manufactured without incorporating any filler materials. The activating solution used comprised sodium hydroxide (NaOH) and sodium silicate (Na_2SiO_3) blend. Sodium hydroxide solutions with molarities of 2 M, 4 M, and 6 M were employed. Twenty-eight days after ambient curing, bricks were evaluated for compressive strength and water absorption. Additional properties like bulk density, porosity, and initial rate of absorption were also assessed. The results revealed that bricks produced with 2 M NaOH and Na_2SiO_3, tested after 28 days of curing, met the requirements for Class 10 bricks. Further, it was noticed that the increment in NaOH molarity from 2 to 4 M and 6 M significantly enhanced the properties of bricks. Bricks developed in this investigation (compressive strength 39.82 MPa and water absorption 15.83%) demonstrated the potential to replace conventional burnt clay bricks. It promoted the extended utilization of fly ash that could solve its accumulation issue and contribute to sustainable development and a reduced carbon footprint.

Keywords Fly ash · Geopolymer bricks · Sustainability

1 Introduction

Bricks are a pivotal ancient building material used as a masonry unit. There has been a high demand for bricks nowadays due to the ever-increasing population. The conventional bricks made using clay are energy- and resource-intensive in nature. High amounts of carbon footprints are also associated with the clay brick manufacturing, contributing to global warming. On average, CO_2 emissions related to a single fired clay brick are approximately 0.41 kg [1]. It also engulfs a huge area of fertile land that can be used for farming purposes. Keeping these things in mind, research

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has been done on producing non-fired bricks using cement and lime as binding material with some industrial byproducts, e.g., pulverized coal ash, rice husk ash, etc. [2]. But these binders also have high carbon footprints associated with them. Precisely, cement production emits approximately 1 ton of CO_2 while producing 1 ton of cement [1, 3].

A number of research projects are also being done to produce building materials that neither require cement nor firing to achieve desirable strength and durability properties. One such approach was suggested by Davidovits in 1978, stating the utilization of silica and alumina-rich materials that can be activated by strong alkali to produce a material through three-dimensional networking of precursor ions, which has a binding property known as geopolymer [4, 5]. Fly ash, an industrial byproduct, is abundant in amorphous silica and alumina phases, and it is widely available worldwide and suitable for the manufacturing of geopolymer-based construction materials [6]. Alkali hydroxide facilitates the dissolution of silica and alumina, while soluble silicate ions from alkali silicate (typically sodium silicate (Na_2SiO_3)) play a key role in geopolymer formation [7]. Sodium hydroxide (NaOH) demonstrates greater effectiveness compared to potassium hydroxide (KOH) [8]. In this context, Abdullah et al. [9] formulated geopolymer masonry units using fly ash and sand. Hardjito et al. [10] developed geopolymer mortar from fly ash having a compressive strength of 20 MPa using alkaline activator (8 M-16 M NaOH with Na_2SiO_3). Lavanya et al. [11] attempted the production of geopolymer bricks by activating a blend of fly ash and GGBS. Shilar et al. [12] also manufactured masonry units exhibiting compressive strength of 10 MPa through geopolymerization using a mixture of fly ash and granite powder.

From an extensive literature survey, it is observed that fly ash-based geopolymer materials require temperature curing, and some kind of fillers are added to produce construction materials. A higher molar concentration of alkali solution in geopolymer production may make the material uneconomical. Keeping these things in mind, the primary goal is to produce ambient cured geopolymer bricks by activating fly ash without adding any fillers. The amorphous silica and alumina in fly ash are dissolved by NaOH and lead to the formation of SiO_4 and AlO_4^- tetrahedral precursor ions [13]. With the passage of time, these precursor ions come together to form three-dimensional amorphous networks of aluminosilicates. Sufficient presence of soluble silicate ions (from Na_2SiO_3) enhances the geopolymerization process [8, 10]. NaOH and Na_2SiO_3 are used as activating solutions because they are readily and cheaply available as compared to their potassium counterparts. Previous studies also suggest the utilization of NaOH and Na_2SiO_3 as activating solutions in geopolymer production [14–16]. It is aimed to use a low molar concentration of NaOH in alkali activator (combination of NaOH and Na_2SiO_3) in brick production of dimensions 230 mm $\times$ 110 mm $\times$ 70 mm. These masonry units are ambient cured and tested for various strength and durability characteristics after 28 days. Bricks developed in this study can serve as a replacement for conventional clay bricks. It promotes fly ash use (on an extended level) in brick making through geopolymerization, which may also solve the disposal problems related to fly ash.

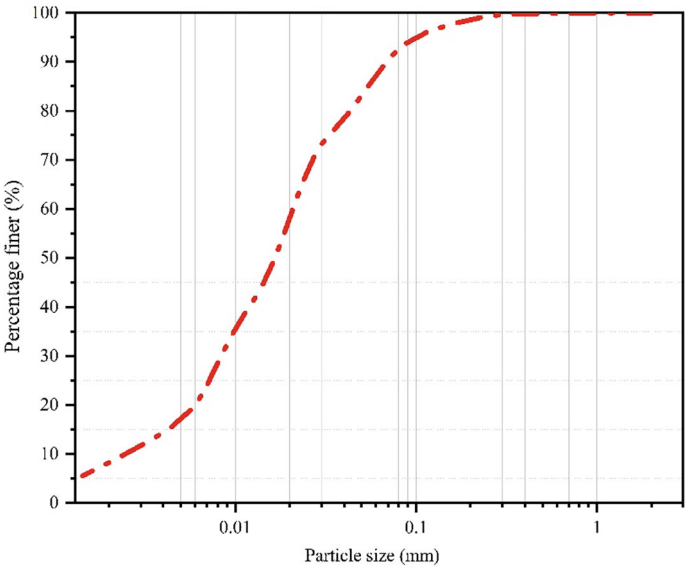

Fig. 1 Fly ash gradation curve

2 Methodology

2.1 Materials

Fly ash. This investigation employed fly ash sourced from Grasim Industries Ltd., Renukoot, (U.P.), India, and having an average size lower than 20 μm, as displayed in Fig. 1 [17]. It consisted of SiO_2: 63.9%, Al_2O_3: 23.70%, Fe_2O_3: 7.20%, TiO_2: 2.50, CaO: 0.60, SO_3: 0.03, loss on ignition: 0.30, suggesting that it belongs to class F [18].

Alkali activator. Alkali activator employed in this study consisted of a combination of NaOH solutions of molarity 2 M, 4 M, and 6 M and sodium silicate (Na_2SiO_3) solution of density 1.39 g/cc having Na_2O/SiO_2 ratio of 1:2.25. Fly ash/alkali activator ratio was 2.2, while Na_2SiO_3/NaOH ratio was 2.5. NaOH and Na_2SiO_3 solutions were mixed prior to mixing with the fly ash.

2.2 Mix Proportions Adopted and Sample Preparation

The materials required to produce 20 bricks in the laboratory are presented in Table 1. These mixed proportions were initially established on a mortar scale (cube of volume

351.9 cm^3), as documented in the literature by Singh et al. [17]. Then, the experimental work has been extended for actual brick production (230 mm × 110 mm × 70 mm) in the present study. Mixing of ingredients (fly ash and alkali activator) was done for a total of 10 minutes in a pan mixer, consisting of 2 cycles of 3 minutes of mixing with a 1 minute rest in between, and a final 4 minutes mixing phase. Thoroughly mixed ingredients were transferred to the brick-making machine, where hydraulic pressure of 14–17 MPa was applied for 3–4 s to form the bricks. Bricks were allowed to cure under a shed at ambient conditions for 28 days. Figure 2 shows the geopolymer bricks produced in this experimental investigation.

Table 1 Mix ratios adopted in the study

Sample	Fly ash (kg)	NaOH concentration (M)	NaOH (kg)	Na_2SiO_3 (kg)	Alkali activator (kg)	Extra water (kg)
M2SG	40.7	2	5.3	13.2	18.5	Nil
M2SWG	40.7	2	5.3	13.2	18.5	2.035
M4SG	40.7	4	5.3	13.2	18.5	Nil
M4SWG	40.7	4	5.3	13.2	18.5	2.035
M6SG	40.7	6	5.3	13.2	18.5	Nil
M6SWG	40.7	6	5.3	13.2	18.5	2.035

2 M (2, 4, 6) = NaOH Molarity, S = Sodium Silicate, W = Water, and G = Geopolymer, Nil = No extra water was used

Fig. 2 Ambient cured geopolymer bricks developed in the laboratory after 24 h

2.3 Testing

Compressive strength. Compressive strength of the bricks was computed in a compression testing machine of capacity 2000 kN with a loading rate of 14 MPa/min as per the specifications laid in IS 3495 (Part 1): 2002 (Part 1) [19].

Flexural strength. It was conducted to estimate the strength of bricks against bending forces following the guidelines provided in ASTM C67/C67M [20].

Water absorption. Bricks were oven-dried at 110°C for 24 h, followed by cooling to room temperature. The dry weight of the bricks was recorded. Bricks were then immersed in a water tank for 24 h, and the wet weight was recorded. Water absorption of the bricks was expressed as a percentage increase in the weight relative to the dry weight [19].

Porosity. Porosity of bricks was calculated according to the procedures laid down in the literature of Singh et al. [17].

Bulk density. Brick samples were oven-dried at 110°C for 24 h and allowed to cool at room temperature. Its dry weight and all the dimensions were measured, and bulk density was expressed as the ratio of weight and overall volume [21].

Initial rate of absorption (IRA). It was estimated for bricks following the procedures laid down in ASTM C67/C67M [20].

3 Results and Discussion

3.1 Strength Characteristics

Bricks should have the ability to sustain axial compression and bending forces during the service life. Strength characteristics of the bricks, i.e., compressive and flexural strength, were reduced when extra water was used in the alkali activator, as depicted in Fig. 3, and rightly so, as the overall morality of NaOH decreased with the addition of extra water, interfering with the extent of geopolymerization, also suggested by Patankar et al. [22]. However, it contradicts the observations noticed by Singh et al. [17] on the mortar scale. The possible reason could be the compaction method adopted (forming of bricks through hydraulic press) for geopolymer production in this study. Increasing the NaOH concentration in the alkali activator from 2 to 4 M and 6 M enhanced the strength properties of the bricks. This improvement occurred because higher molarity facilitated the leaching of amorphous contents in fly ash to an extended level, and consequently formed more geopolymer gel [13, 17, 23]. Failure patterns of the brick in compression and flexure test are displayed in Fig. 4.

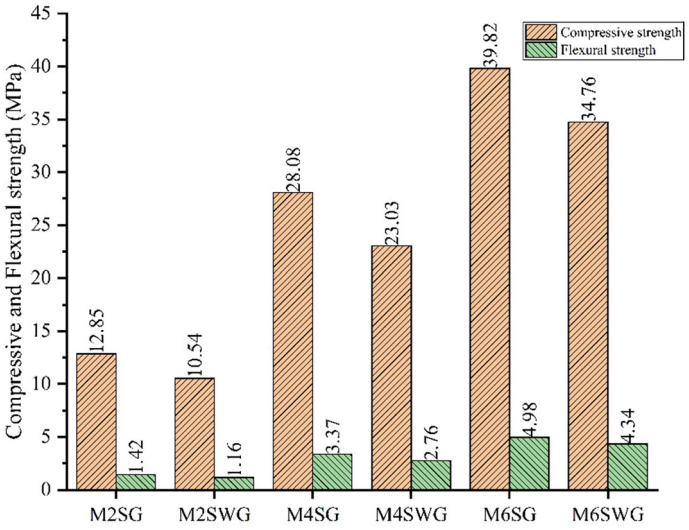

Fig. 3 Strength properties of geopolymer bricks

Fig. 4 Failure behavior of geopolymer bricks under compression (Left) and flexure test (Right)

3.2 Durability Characteristics

Estimation of water absorption and porosity is essential for the quality of bricks, as a porous brick may absorb water from the immediate environment and interfere with the overall performance of the structure. A balanced amount of voids is also necessary in order to impart lightweight property to the bricks. Observations suggested that as NaOH molarity increased, water absorption and porosity decreased, as presented in Fig. 5. Higher NaOH concentration resulted in a more compact N-A-S-H gel [13, 14]. Furthermore, water absorption was less than 20% in each case, satisfying the criteria for Class 10 bricks (water absorption < 20%) [24].

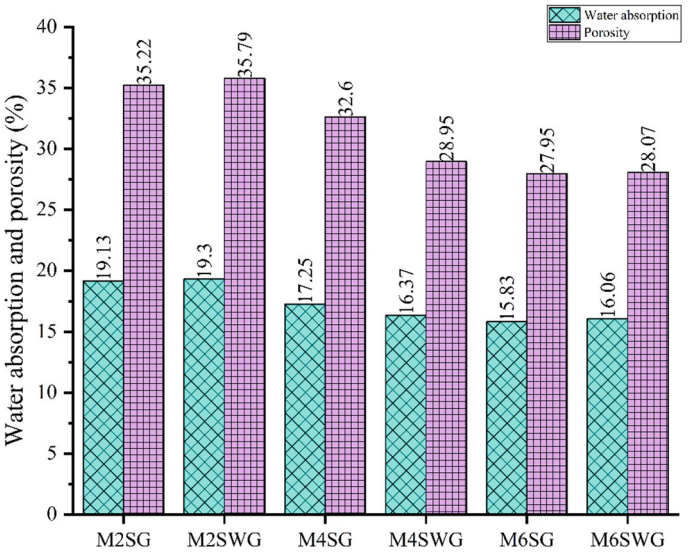

Fig. 5 Water absorption and porosity of geopolymer bricks

3.3 Bulk Density and IRA

Bulk density and IRA of geopolymer bricks are presented in Table 2. The bulk density of the bricks was found to range from 1.48 to 1.68 g/cc. Increasing the molarity of NaOH in the alkaline activator led to the formation of a dense gel. The IRAs of the bricks ranged from 0.103 to 0.296 g/cm^2/min, in close agreement with that of burnt clay bricks [25]. Decreasing IRA values with increasing molarity of NaOH in alkaline activator were validated by the trend of porosity.

Table 2 Bulk density and IRA of geopolymer bricks

Samples	Bulk density (g/cc)	IRA (g/cm^2/min)
M2SG	1.48	0.296
M2SWG	1.52	0.288
M4SG	1.48	0.234
M4SWG	1.57	0.127
M6SG	1.48	0.103
M6SWG	1.68	0.149

4 Conclusions

This study was an effort to produce fly ash-based geopolymer bricks without adding any fillers. The following conclusions were drawn from the above study:

- Activating solution having a 2 M concentration of NaOH yielded geopolymer bricks with a standard of class 10.
- Bricks with compressive strength 39.82 MPa and water absorption 15.83% were obtained when 6 M NaOH was used in alkaline solution.
- The addition of extra water along with the alkali activator reduced the strength of the bricks, contrary to that obtained on the mortar scale (which was using the vibration-compaction technique) for geopolymer production.
- The developed bricks were around 15–25% lighter in weight than traditional masonry units, with an IRA ranging from 0.103 to 0.296 g/cm^2/min.

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An Investigation of Suction Behavior of Zeolite-Bentonite Mixtures with Hyprop and Filter Paper Methods

Ceren Ercan, Sukran Gizem Alpaydin, Esra Güneri, and Yeliz Yukselen-Aksoy

Abstract Soil suction is a controlling parameter of the engineering behavior of unsaturated soils used in landfill liners. The combination of bentonite's swelling capacity and zeolite's ion-exchange properties makes these mixtures a good candidate for preventing long-term contaminant migration. Therefore, as landfill liner material, the suction characteristics of compacted zeolite–bentonite mixtures (ZBMs) should be known depending on different factors. The present study investigated the suction properties of compacted ZBMs for varying bentonite contents. The suction characteristics of powdered zeolite and 0, 10, and 20% bentonite in ZBMs were measured and compared with each other. The soil suction measurement techniques chosen were the Hyprop (Hydraulic Property Analyzer) and filter paper method. The findings obtained from these two methods were also compared. The results show that the suction capacity of zeolite increases with finer grain size. As bentonite content increases, both mixtures' matric and total suction increase. The suction values obtained using the filter paper method were generally higher than those obtained with Hyprop..

Keywords Suction · Zeolite-Bentonite mixture · Filter paper method · Hyprop method

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1 Introduction

Soil suction is a key parameter in unsaturated soil mechanics, defined as the negative pore water pressure that occurs in the soil matrix. It impacts many different properties, including strengths, volume changes, hydraulic conductivity, contaminant transport, and the control of soil behavior. Soil suction behavior is crucial for accurately characterizing, predicting, and controlling soil behavior [1, 2].

Matric suction refers to capillary pressure resulting from surface tension at the water–air interface in the soil's pores. As pore size decreases, capillary pressure increases, and matrix suction value increases. It is a reverse function for specific parameters such as soil structure, mineral composition, and water content. On the other hand, osmotic suction is caused by the difference in water potential between pore water and free water due to dissolved salts and other solutes in the pore water. The force is, therefore, created because of this difference that pulls the water into the soil. With an increase in the concentration of solutes, osmotic suction increases [3].

Zeolites are crystalline assemblies of aluminosilicate with an open network structure at the molecular size made up of interconnected channels and cavities. The complex framework, constituted by SiO_4 and AlO_4- tetrahedrons, is a negatively charged cage stabilized by the exchange cation within the channels [4]. These inherent features make zeolites suitable ion-exchanging materials in industrial, environmental, and agricultural applications. Their new architecture, with easily exchangeable cations, provides a location for highly selective and efficient ion-exchange.

Due to their structural and chemical properties, zeolites and bentonites are frequently used in various applications. Combining these two materials under zeolite-bentonite mixtures creates the opportunity to benefit from their complementary properties and achieve synergistic effects. The structural framework of zeolites can increase the mechanical stability of bentonite-based materials, while bentonite can act as a binder and improve the processability of zeolites by providing plasticity [5]. The resulting mixtures can exhibit improved adsorption capacity, increased selectivity for certain contaminants, and improved durability in various applications.

This study investigates the suction properties of compacted ZBMs for varying bentonite contents. The suction characteristics of powder and 0, 10, and 20% bentonite in ZBMs were measured and compared with each other. The soil suction measurement techniques chosen were the Hyprop and filter paper methods. The results of these two methods were also compared.

Table 1 Physico-chemical properties of all materials

Property	Zeolite	Bentonite
Specific gravity	2.36	2.54
Liquid limit (%)	60	295
Plastic limit (%)	NP	65
Plasticity index (%)	60	230
Inherent water content (%)	5.0	8.8

2 Materials and Methods

2.1 Materials

In this study, zeolite and zeolite-bentonite mixtures were used. Bentonite was gathered from Canbensan Export Co. While preparing the mixtures, 10 or 20% of the total weight of the dry zeolite was taken as bentonite. Physico-chemical properties of all materials used in this study are given in Table 1.

2.2 Methods

This study conducted liquid limit, plastic limit, and compaction tests on zeolite and zeolite-bentonite mixtures. In addition, suction pressure values were obtained using filter paper and the Hyprop methods.

Standard Proctor Test was performed to determine the optimum water content and maximum dry unit weight of the mixtures [6]. Mixtures with different bentonite contents (0%, 10%, and 20%) were thoroughly mixed in a dry state. Then, they were prepared in four different water contents by the water spraying method and placed in air-tight bags. The mixtures were kept for 24 h for homogeneous mixing. Following the tests conducted according to the standard, representative samples of each mixture were taken, and the water content was determined.

For the filter paper method, test samples were compacted at the pre-determined maximum dry unit weight and optimum water content values in the Harvard miniature compaction mold (R = 3.78 cm and V = 84.8 cm^3) (Fig. 1a). After the compaction, the samples were divided into two equal parts for a total height of 5 cm. Suction measurements due to drying were made using the filter paper method [7]. Matric and total suction values were determined. A total of three filter papers were placed between the samples and divided into equal pieces to measure the matric suction (measurements were made with the filter paper in the middle of these filter papers). A lid was placed on the upper sample to measure and realize total suction, and filter paper was placed on the lid. Then, the samples were kept in a closed container, the filter papers were weighed at regular intervals, and the tests were terminated when the suction values reached equilibrium (Fig. 1b).

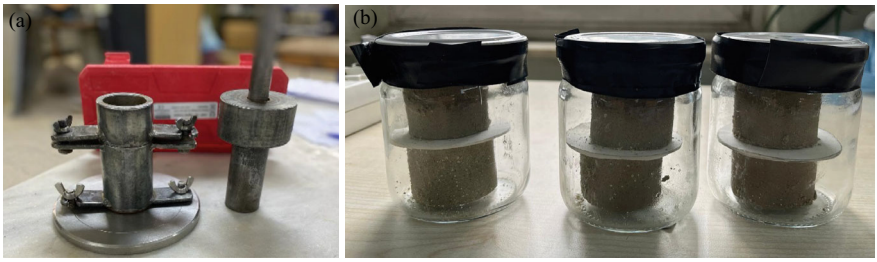

Fig. 1 Filter paper method procedure: **a** Harvard miniature compaction mold, **b** filter paper method test setup

The hyprop method is based on the evaporation technique, where two tensiometers, placed at different depths within a soil sample, measure the water potential as the soil dries. The soil samples were compacted in stainless-steel cylinders (8 cm diameter, 5 cm height) according to Standard Proctor parameters (Fig. 2a). After compaction at optimum water content and maximum dry unit weight, the samples were saturated and prepared for testing. Two tensiometers were installed vertically to measure water potential at different depths, ensuring precise data collection (Fig. 2b). During the test, the HYPROP system monitored changes in soil weight due to evaporation and corresponding variations in water potential. The collected data were analyzed using HYPROP-Fit software to construct SWCCs and estimate unsaturated hydraulic conductivity. The Van Genuchten model is well-known for effectively modeling water retention behavior and was used to interpret the results.

3 Results

According to the compaction test results presented in Fig. 3, it was observed that the dry unit weight increased with the addition of 10% bentonite and decreased with the addition of 20% bentonite. For example, with the addition of 10% bentonite, the dry unit weight increased from 11.20 to 11.50 kN/m^3, while with the addition of 20% bentonite, it decreased back to 11.20 kN/m^3 (Table 2). However, it should be noted that the change in these unit weight values was not significant. In addition, it was determined that the optimum water content increased with increasing bentonite content. This can be explained by the fact that bentonite has a higher liquid limit value than zeolite, thus increasing the optimum water content of the mixture.

Matric and total suction values of zeolite and zeolite-bentonite mixtures were determined by the filter paper method and presented in Table 3. Accordingly, as the bentonite content increased, both matric and total suction values increased. However, while matric suction more than doubled with the addition of bentonite, total suction values showed no significant change. The total suction value increased by approximately 4%, reaching 2043 kPa with a 20% bentonite addition.

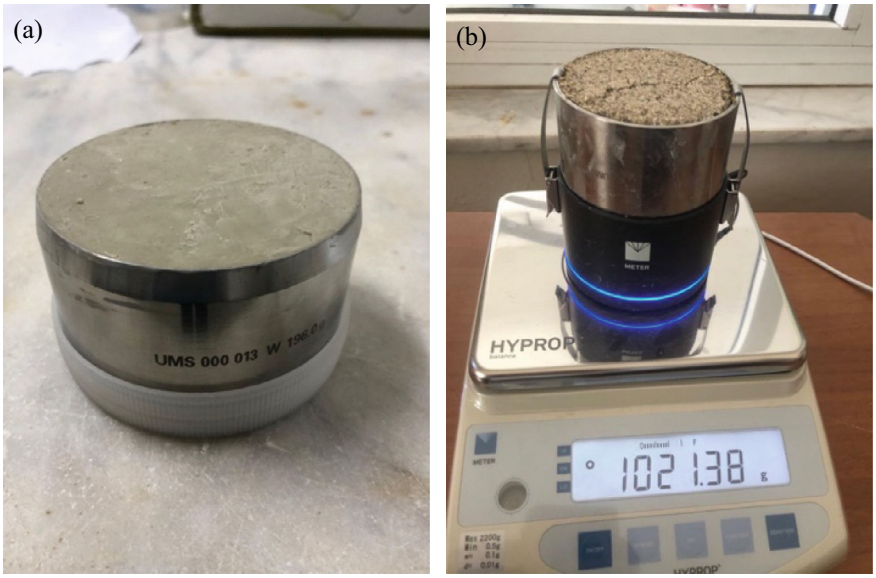

Fig. 2 HYPROP procedure **a** compacted sample in the mold **b** measuring sample weight

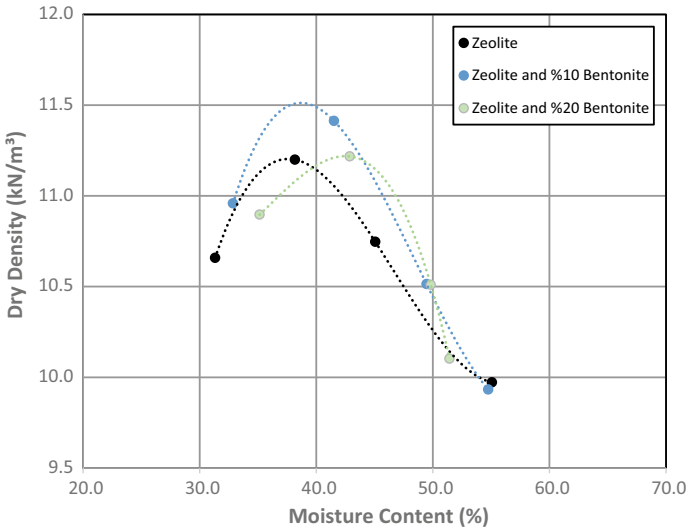

Fig. 3 Compaction curves of the mixtures

Table 2 Compaction parameters of the mixtures

	Fine Zeolite	Fine Zeolite and %10 Bentonite	Fine Zeolite and %20 Bentonite
Max dry unit weight (kN/m^3)	11.2	11.5	11.2
Optimum water content (%)	38	39	43

Table 3 Matric and suction values of the mixtures by using the filter paper method

	Matric suction (kPa)	Total suction (kPa)
Zeolite	70	1974
Zeolite-10% Bentonite	150	1978
Zeolite-20% Bentonite	182	2043

The soil water characteristic curves of the mixtures were also obtained using the Hyprop device (Fig. 4). At zero suction, the difference in water content among the mixtures was approximately 5%, but this variation remained relatively stable as suction increased. The air-entry value (AEV) of all mixtures ranged between 10 and 20 kPa. As suction increased to the AEV, the water content rapidly decreased due to the accelerated release of water from the samples.

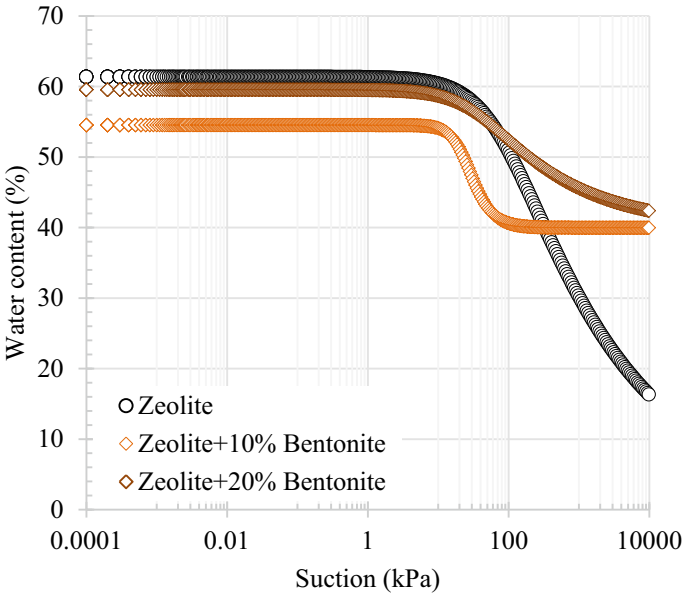

Fig. 4 Soil–water characteristic curves of the mixtures by HYPROP

The water content values at a suction of 2000 kPa were determined as 25% for zeolite, 40% for the 10% bentonite-zeolite mixture, and 45% for the 20% bentonite-zeolite mixture. This indicates that increasing the bentonite content improves the water retention capacity. As the bentonite content increased, suction values also increased. This effect was especially noticeable at lower water content values, where suction increased with a higher bentonite content.

Both methods had consistent results, confirming that suction values increased with the bentonite addition. Surface tension plays an important role in water retention capacity, and as bentonite content increases, suction values increase. Agus et al. [8] reported that the total suction of bentonite–sand mixtures mainly depends on bentonite content. Bentonite minerals have surface electrical charges, and as their content increases, the surface electric charge of the particles also increases due to the large specific surface area of bentonite. In other words, mixtures with higher bentonite content show higher osmotic and sorptive forces at the same water content [9].

4 Conclusion

Based on the experimental results, the following conclusions were drawn:

- The dry unit weight did not change significantly with the bentonite addition, while the optimum water content increased.
- Both matric and total suction values increased with bentonite content, though total suction showed only a minor increase.
- Increasing the bentonite content improved the water retention capacity of the mixtures.
- The filter paper method's suction values were generally higher than the Hyprop's results.

These findings highlight the role of bentonite in improving water retention in zeolite mixtures, with potential applications in geotechnical and environmental fields.

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A Sustainable Assessment of the Suitability of an Abandoned Quarry for Red Mud Disposal at Dalla Mines, Uttar Pradesh: A Feasibility Study

Ashish D. Gharpure, Prashant Navalakha, C. K. Aswathi, and Vaishali Surawar

Abstract Large-scale industrial activities are leading to a substantial rise in industrial waste production, posing challenges for its disposal and potential environmental risks. In parallel, there are abandoned open-cast mines with deep craters getting filled with rainwater and providing breeding areas for mosquitoes, posing a danger to surrounding habitats. This study assesses the feasibility of disposing of red mud, a by-product of alumina extraction in cement plants, into three abandoned quarry ponds at Dalla, Uttar Pradesh, India. The focus is on assessing the stability of the ponds and the environmental impacts of red mud disposal. Due to its alkaline nature, red mud causes risks to the surrounding soil, groundwater used for agriculture and daily consumption, and nearby human settlements. The sedimentary rock quarry ponds are separated by uncut rock barriers, including a larger pond and three smaller ponds. The red mud disposal must ensure the stability of the pond walls and barriers, as any failure could lead to contamination of the surrounding soil and water, creating a significant risk. To evaluate the feasibility, comprehensive investigations were conducted, including topographic and bathymetric surveys, geological mapping, geophysical, and geotechnical assessments, at and around the ponds. Geophysical methods such as Electrical Resistivity Tomography (ERT) and Seismic Refraction Tomography (SRT) helped to identify subsurface features, seepage zones, and potential anomalies. Kinematic analysis using geological data, and stability and seepage analyses for the pond barriers with Slide2 software were performed to ensure safe red mud filling. The paper also discusses additional measures for safe disposal and a cost–benefit analysis. Geotechnical and geophysical investigations indicated that there are no major anomalies except for a few seepage points in the study area. While the rock barrier between the ponds is structurally stable with a factor of safety higher than the

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recommended values, potential small rockfalls and seepage from the barrier could occur. Total seepage of $1.137 \times 105\ m^3/s$ is estimated through the pond barriers, from seepage analysis. It is concluded from geotechnical stability analysis and seepage analysis that ponds are safe to dispose of the red mud after implementing appropriate mitigation measures, including lining of ponds after dewatering.

Keywords Geotechnical stability · Sustainable waste disposal · Environmental impact assessment · Red mud disposal · Feasibility study

1 Introduction

The rapid growth of industrial activities has led to a significant increase in industrial waste, posing major management and disposal challenges. Red mud, a by-product of alumina extraction in cement plants, is notable for its high volume and strong alkalinity (pH 12–13), containing traces of heavy metals and radionuclides [1]. Classified as "High Volume Low Effect Waste," red mud generation varies from 55 to 65% of processed bauxite. Each ton of red mud solids is accompanied by up to 2 tons of highly alkaline liquid [2]. All over the world, disposal of red mud is being done either on land or in the nearby sea/ocean [3].

Red mud disposal presents significant environmental challenges due to potential alkali seepage into groundwater, storage instability, airborne alkaline dust affecting plant life, and the large land area required for storage [4]. Mora et al. [5] identified several human health hazards associated with red mud, including the release of fine-grained particles into the atmosphere by wind action. The proximity of caustic ponds to rural settlements increases the risk of resuspended alkaline particles posing health risks to local populations. Remisha et al. [6] reviewed the environmental impact, utilization methods, and treatment processes of red mud, highlighting the lack of treatment or utilization in India.

Improper disposal of red mud can contaminate soil and groundwater, affecting ecosystems, agriculture, and human health. Studies show red mud harms aquatic organisms, inhibiting phytoplankton growth and contaminating them with toxic substances [4]. Fine-grained red mud particles can become airborne, posing health risks to nearby populations [5]. There is a need for sustainable disposal solutions to mitigate these environmental risks.

Abandoned open-cast mines from extensive mining activities cause environmental and public health challenges. Over time, these mines, now quarry ponds, have filled with rainwater, creating stagnant water bodies that breed mosquitoes, increasing the risk of vector-borne diseases like malaria, dengue, and chikungunya, thereby endangering nearby communities. Open-cast mining activities have led to significant land use changes between 1971 and 2006, with forest land decreasing to nearly half its original size, open scrap land increasing by 2.5 times, agricultural land expanding by 1.5 times, and residential land growing 7.5 times compared to 1971 [7]. Abandoned quarry ponds contribute to the increase in open scrap land, which needs utilization.

These ponds hold potential for alternative uses, including serving as repositories for industrial waste like red mud, provided proper precautions are taken to ensure environmental safety and structural stability.

This study explores the feasibility of disposing of red mud in three abandoned quarry ponds in Dalla, Uttar Pradesh, India. The disposal process must prioritize the stability of pond walls and barriers to prevent contamination of nearby soil and water, posing significant environmental and health risks. Additionally, pond water must be diverted to remain suitable for community use and prevent mosquito breeding. Comprehensive investigations assessed the topographical and geotechnical characteristics of the ponds and the quality of the pond water. Based on these findings, detailed stability analyses were conducted to ensure the geotechnical integrity of the pond barriers. The study aims to propose a sustainable and safe disposal method for red mud, focusing on reducing its environmental impact.

2 Study Area

The study area is located at Dalla Mines, Sonbhadra, Uttar Pradesh, India, known for its rich mineral resources, including cement-grade limestone, dolomite, clay, calcite, sillimanite, and base metals. The study area includes three abandoned quarry ponds (see Fig. 1) referred to hereafter as Pond 1, Pond 2, and Pond 3, which remain water-filled throughout most of the year. Additionally, a larger pond (Pond 4) is present in the vicinity, separated from the three smaller ponds by uncut rock barriers.

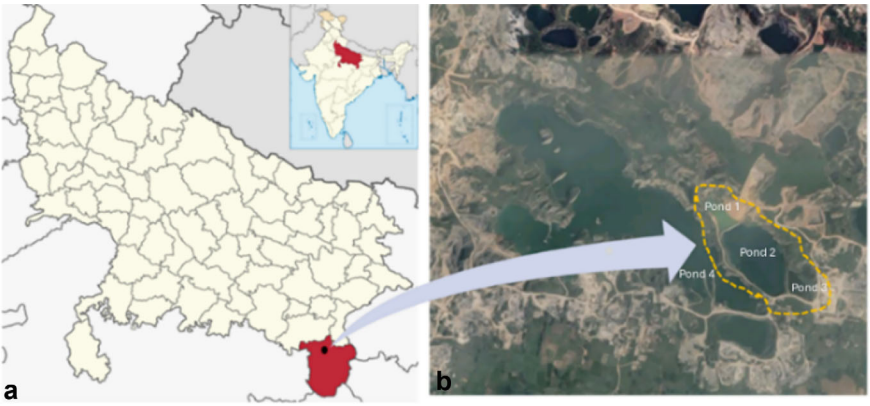

Fig. 1 Location of the Sonbhadra district and Google earth image of Dalla Mines

3 Survey and Investigation

3.1 Topography and Bathymetric Survey

A topographical and bathymetric survey was conducted to understand the terrain and features above and below the water level. An aerial survey using drones and a total station survey covered approximately 40 hectares. Drone images were processed to generate a 3D model of the study area. The total surface areas of Pond 1, Pond 2, and Pond 3 are approximately 0.55, 3.65, and 0.5 hectares, respectively. Depth measurements from the bathymetric survey revealed water depths of 8.77 m, 13.5 m, and 5.36 m in Pond 1, Pond 2, and Pond 3, respectively, with a maximum quarry depth of 45 m.

3.2 Geological Mapping

The area primarily consists of Porcellanite rock from the Kaimur Group, with bedding planes spaced 1.0–2.0 m apart. Generally, closed joints are observed. Continuous blasting and mining activities have caused surface cracks near the ground. Bushes and shrubs growing in the joints on cut sections indicate wet joint surfaces. Isolated block failures have formed rock outcrops at multiple locations in the pond barriers.

3.3 Geophysical Survey

A geophysical survey, using Electrical Resistivity Tomography (ERT) and Seismic Refraction Tomography (SRT), was conducted to study the subsurface strata, seepage zones, and potential anomalies. The surveys were conducted along the barrier between smaller ponds (Pond 1, Pond 2, and Pond 3) and Pond 4; and the barrier between Pond 1 and Pond 2. The survey revealed no major anomalies but showed a low-velocity zone indicating potential seepage between Pond 1 and Pond 2 in SRT. ERT profile along the pond barrier between smaller ponds and Pond 4 (See Fig. 2) and SRT profile along the barrier between Pond 1 and Pond 2 (See Fig. 3) are presented below.

3.4 Geotechnical Investigations

Two boreholes were drilled at the pond barriers to assess subsurface conditions. Geotechnical investigations revealed that the subsurface primarily consists of highly

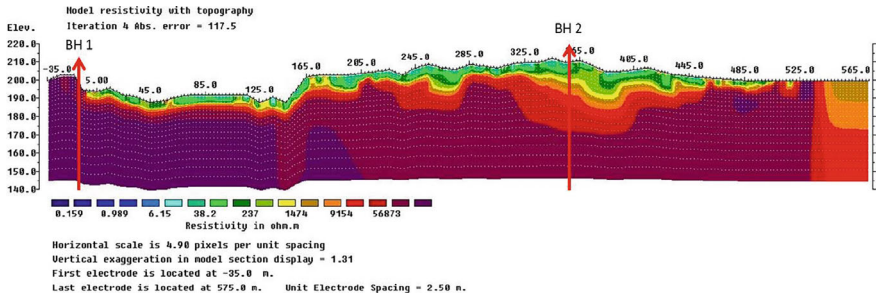

Fig. 2 ERT profile along the rock barrier between smaller ponds (Pond 1, Pond 2 and Pond 3) and Pond 4

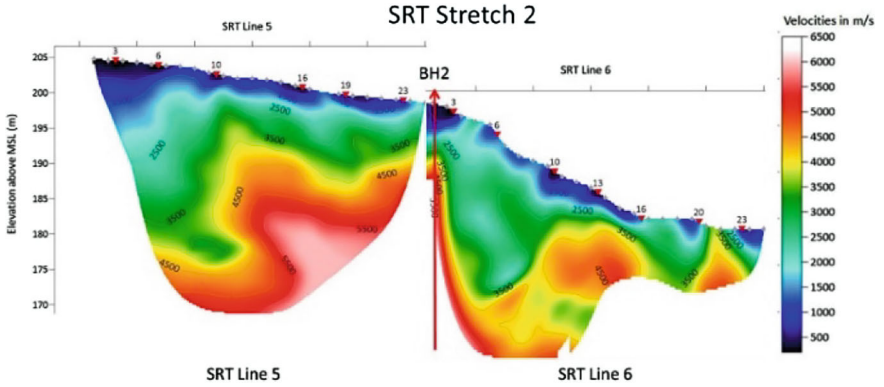

Fig. 3 SRT Profile along the barrier between Pond 1 and Pond 2

to slightly weathered rock with some overburden soil. Water seepage through fractures and joints was observed, aligning with seepage points noted during geological and geophysical surveys. Laboratory tests were conducted on samples from the boreholes, and field permeability tests were performed at both locations. The calculated equivalent permeability of the rock mass was $1.02 \times 10{-}4$ m/s. The total seepage expected from the barrier between Pond 4 and the other ponds is approximately 100,000 L per day (9.97×104 m^3/day).

3.5 Environmental Considerations

The survey and investigation findings highlight significant environmental risks due to high seepage rates from fractured rock. These seepage points may allow red mud to infiltrate the surrounding soil and water, leading to contamination. Additionally, the topographic and bathymetric survey indicates that the ponds can retain large volumes

of stagnant water, creating favorable conditions for mosquito breeding and increasing disease outbreak risks. Before filling the ponds with red mud, dewatering is essential to mitigate these risks. Redirecting stagnant water for alternative uses can prevent contamination and curb disease spread, ensuring effective water management.

Water quality was assessed to confirm its suitability for alternative uses. A comprehensive literature review identified necessary quality tests. Pawari et al. [8] found high total hardness in water linked to heart diseases and kidney problems. Hard water is unsuitable for domestic use due to its unpleasant taste and reduced soap lather. Tank et al. [9] examined groundwater hydrochemistry in Jaipur, evaluating its quality for drinking and agricultural purposes. Groundwater samples were analyzed for various physico-chemical parameters, finding water quality for irrigation purposes ranges from excellent. Ayub et al. [10] reviewed recent research on water quality, highlighting significant findings and advancements.

Parashar et al. [11] studied drinking water quality in Bhopal, concluding it is within standard values prescribed by various agencies, indicating generally good and safe water for consumption. Sarala et al. [12] analyzed groundwater quality in Jawaharnagar, Ranga Reddy district, Andhra Pradesh, finding acidic and hard water with some pollutants exceeding permissible limits, indicating pollution.

Physical analysis of pond water shows turbidity (0.5–1.3 NTU) and pH (8.3–8.4) within permissible limits as per IS 3025 [13]. Chemical analysis reveals total hardness in Pond 1 (210.4 mg/l) and Pond 3 (206.3 mg/l), and fluoride content in all ponds (1.9–2.0 mg/l) is slightly higher than the permissible limits for drinking water. Due to higher fluoride content, the water is unsuitable for drinking but can be used for irrigation.

4 Stability and Seepage Analysis

Based on the investigation and survey results, a feasibility study was conducted to evaluate the suitability of quarry ponds for red mud disposal. The findings and outcomes of this feasibility study are detailed in this chapter.

4.1 Kinematic Analysis

Various types of failures can occur in rock mass formations, such as planar, wedge, and toppling, depending on the orientation of joints and cut slopes. Before conducting strength-based analysis, it is essential to identify potential joint-based failures of rock slopes through kinematic analysis. Stereo net plots, using dip and dip direction of joints and slopes, will assess the possibility of such failures geometrically. Kinematic analysis using stereo net plots (see Fig. 4) of discontinuities at the rock barrier between the ponds indicates potential failure during red mud filling. The plot reveals

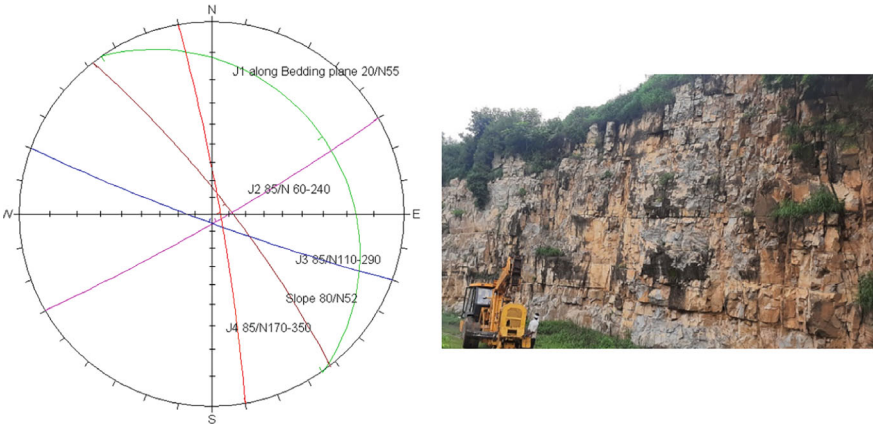

Fig. 4 Stereo net plot and discontinuities observed at the rock barrier between the ponds

the possibility of isolated planar sliding along the bedding plane and individual block failures along Joints 1, 2, and 3, posing risks to workers.

4.2 Slope Stability and Seepage Analysis

Slope stability and seepage analysis of the barrier between Ponds 1, 2, 3, and Pond 4 has been evaluated using the Limit Equilibrium Method (Bishop Simplified Method) with critical sections analyzed in Slide2 Software under both static and seismic loading conditions. The target factors of safety are 1.3 for static and 1.1 for seismic conditions. Two scenarios were analyzed:

Case 1: Complete Drawdown: Ponds 1, 2, and 3 will be dewatered before red mud dumping, leaving them empty while Pond 4 remains filled with water. The analysis considers the maximum possible water level in Pond 4 to simulate the worst-case scenario (See Figs. 5 and 6).

Case 2: After dewatering, Ponds 1, 2, and 3 will be filled with red mud, while Pond 4 may be dewatered for future use. This scenario accounts for the pressure exerted on the barrier by the red mud, ensuring its stability when the outer pond is empty. Red mud will be filled with a 4H:1V slope and a 3 m berm at 6 m height (See Figs. 7 and 8).

The rock barrier between the ponds is geotechnically safe under both static and seismic loading conditions. Seepage analysis at the critical section shows a total seepage rate of 275.98 m^3/s/m, spanning approximately 412 m, with an estimated total seepage of 1.137 × 105 m^3/s. While the barrier is structurally stable, potential rockfalls and seepage could occur. Therefore, appropriate mitigation measures should be implemented to ensure safe and sustainable red mud disposal.

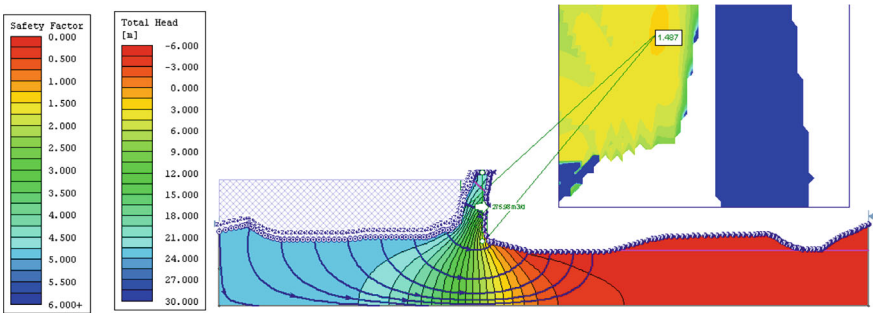

Fig. 5 Slope stability analysis: Case 1-Static loading

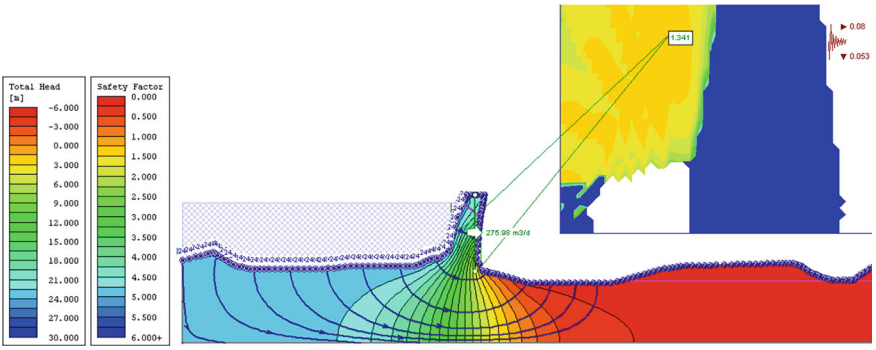

Fig. 6 Slope stability analysis: Case 1-Seismic loading

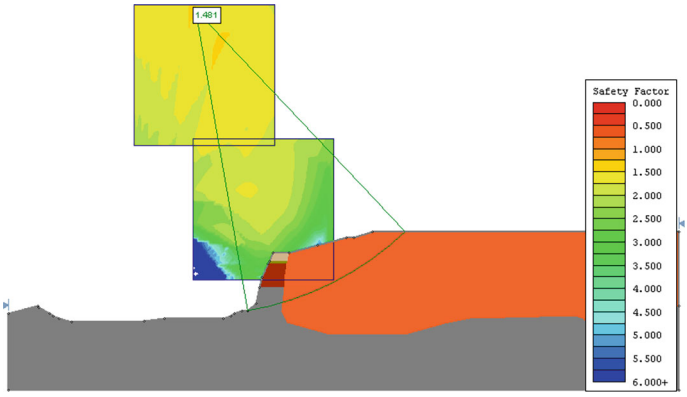

Fig. 7 Stability analysis: Case 2-Static loading

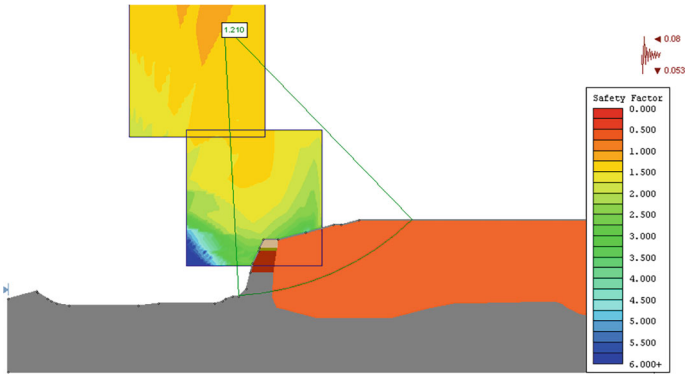

Fig. 8 Stability analysis: Case 2-Seismic loading

4.3 Cost–Benefit Analysis

Dumping red mud on flat land is considered an alternative to using an abandoned quarry. The calculated volume of the ponds is approximately 1.4 million m^3. To dump an equivalent amount of red mud on flat land, assuming a 4:1 slope and a height of 25 m, about 11.2 hectares would be required. This method would need a starter dam, a lining system, and proper drainage. Comparing the required mitigation measures and conducting a cost–benefit analysis shows that dumping red mud in the quarry pond is more cost-effective. Additionally, acquiring suitable flat land for dumping could be challenging. Therefore, using the quarry pond is the easier and more economical option.

5 Mitigation Measures

According to CPCB guidelines [1], red mud storage will be implemented through a Dry Stacking Facility (DSF) using filtered tailings placed mechanically via a conveyance system. Due to the alkaline nature of leachate, a lined facility with a drainage system is essential to collect and manage leachate, preventing groundwater contamination.

- **Rockfall safety**: To mitigate rockfall risks, overhanging rocks along cut slopes will be removed or reinforced with concrete buttresses. Fractured rock from quarry blasting will be covered with safety nets to prevent accidents during tailing placement.
- **Dewatering and grouting**: A dewatering system has been proposed to facilitate water removal. A sump will be placed at the pond's lowest point for continuous water pumping, aiding the grouting process. Adjacent to the DSF, a nearby pond contains water, and to minimize seepage through joints and fissures in the rock

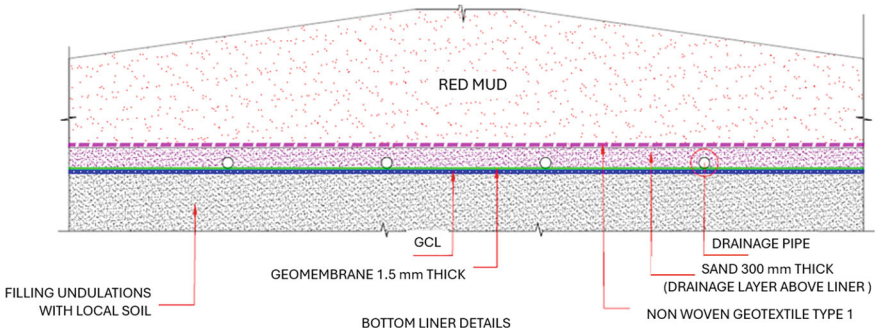

Fig. 9 Proposed lining system at the bottom of the Ponds

mass, grouting will be conducted. Additional surficial grouting may be carried out based on seepage observations post-dewatering.

- **Pond lining**: A liner system is proposed to prevent groundwater contamination from DSF leachate (see Fig. 9). The pond will be cleared of debris, and local soil will fill it up to 5.4 m above the lowest level, with a 1:200 slope toward the leachate collection well. A Geosynthetic Clay Liner (GCL) will replace a 600 mm thick Compacted Clay Liner (CCL). Over the GCL, a 1.5 mm thick High-Density Polyethylene (HDPE) geomembrane will be laid without gaps to ensure leak-proof containment. A non-woven geotextile will be placed above the geomembrane.
- **Leachate Collection System and Drainage System**: A leachate collection system is provided through a sump to manage leachate generated from rainwater infiltration. The sump base will be positioned 0.5 m below the lowest liner level in the pond. A 300 mm diameter HDPE pipe will extend from the sump to the top of the DSF, allowing a submersible pump to extract accumulated leachate over time.
- **Red mud filling**: The DSF will be constructed using an up-valley placement method, with red mud stacked in thin layers (typically 0.3 m) and compacted to 95% dry density using bulldozers and vibrating compactors.
- **Final cover system at the top of the DSF**: The cover system at the top of the DSF will consist of a 1.0 m thick layer of fly ash or pond ash. A layer of vegetative soil will be placed over the ash to minimize rainwater infiltration and support vegetation growth.
- **Surface and Drainage Systems:** A 0.5 m $\times$ 0.5 m garland drain will be constructed around the pond perimeter to channel surface runoff. Intermittent berm drains will direct water into the garland drain, which will discharge the runoff toward the northwest corner.

6 Conclusion

This study provides a comprehensive framework for the safe and sustainable disposal of red mud in abandoned quarry ponds at Dalla Mines, Uttar Pradesh. Integrating detailed geotechnical, geological, and environmental assessments, the proposed Dry Stacking Facility (DSF) design ensures structural stability of pond barriers and mitigates environmental risks like groundwater contamination and ecological damage. The estimated total seepage is approximately 1.137×105 m^3/s along the pond barrier. While the barrier is structurally stable, potential rockfalls and seepage could occur. Therefore, appropriate mitigation measures should be implemented for safe and sustainable red mud disposal. Key measures, including advanced liner systems, leachate collection systems, grouting, and slope stabilization, ensure environmentally sound and cost-effective disposal. This approach addresses industrial waste disposal challenges and repurposes abandoned quarry ponds for sustainable use, contributing to ecological restoration and public health improvement in the region.

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Use of AI-Powered Technologies for Review of Environmental Regulations

Manuel Joy and P. E. Alexander Abraham

Abstract The compliance with environmental regulations creates a major burden for manufacturing companies due to challenges such as limited skilled personnel, tight project timelines, complexity, changes in regulations, and development of new technologies. Artificial Intelligence (AI) presents an opportunity to address these barriers by automating repetitive tasks, enabling professionals to focus on critical thinking, intuition, and judgment. This research explores the integration of AI for review of environmental regulations using advanced Large Language Models (LLMs) combined with Retrieval-Augmented Generation (RAG) for domain-specific knowledge application. The proposed AI systems leverage historical data and machine learning to generate insights and support decision-making with minimal human intervention. A targeted approach to training AI on curated, facility-specific datasets ensures reliable outputs, fostering user trust and facilitating effective decision-making based on review of environmental regulations. Additionally, incorporating an AI agentic workflow empowers systems to operate autonomously, proactively addressing compliance challenges, adapting to evolving regulatory landscapes, and offering actionable recommendations with minimal reliance on human oversight.

Keywords Artificial intelligence · Environmental guidelines · Large language model · Agentic AI · Automation

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1 Introduction

Artificial Intelligence (AI)-based tools and techniques can be used to speed up the environmental compliance by manufacturing organizations and facilities. AI can be used to streamline the review of environmental regulations and compliance with changes in regulations in conjunction with the adoption of new techniques. An organization can develop a proprietary database of searchable applicable regulations. There is lot of variation in the structure and format of environmental documents. Utilizing AI will help organizations to quickly search, interpret, and synthesize information. This will assist organizations with more effective decision-making. The environmental compliance is a tedious, expensive, and time-consuming process for manufacturing companies. Reducing the cost and time for environmental compliance will improve the profitability of manufacturing companies. Data from previous compliance documents can be annotated to make it compatible with AI large language models. This will further contribute to time and cost savings. In this paper, we conduct deep research on AI-driven environmental compliance systems, focusing on integrating LLMs with Retrieval-Augmented Generation (RAG) techniques for automated regulation review. We explore methods for fine-tuning LLMs to domain-specific tasks, optimizations in RAG to improve information retrieval and answer accuracy, and an agent-based workflow for end-to-end compliance automation.

2 Technical Research

Environmental regulations are growing rapidly, making compliance management increasingly challenging. The volume of laws and regulatory updates grows exponentially, outpacing the capacity of compliance teams. Traditional manual review struggles to keep up, risking non-compliance. Artificial intelligence, especially Large Language Models (LLMs), offers a potential solution by automating regulation review and interpretation. According to a 2023 industry survey, nearly half of risk and compliance professionals see AI as a mechanism to boost efficiency, and 35% believe it helps organizations stay current with regulatory changes [1]. In this paper, we explore AI-driven environmental compliance systems, emphasizing the integration of LLMs with Retrieval-Augmented Generation (RAG) for automated regulation review. Our research examines methods for fine-tuning LLMs to domain-specific tasks, optimizing RAG to enhance information retrieval and answer accuracy, and incorporating an agent-based workflow for end-to-end compliance automation. We provide experimental evidence of the system's effectiveness and offer concluding insights on the future of AI in regulatory compliance.

2.1 Fine-Tuning Methods for LLMs

Domain Adaptation through Transfer Learning: General-purpose LLMs are pre-trained on large-scale, heterogeneous data with broad linguistic capabilities. However, when we applied to environmental compliance, these models lack the specialized vocabulary and legal reasoning required to interpret regulatory texts. Our approach begins with domain-adaptive pre-training, where the base LLM is further trained on a corpus of environmental laws, guidelines, and compliance reports. This step enables the model to capture domain-specific terminology and contextual nuances.

Parameter-Efficient Fine-Tuning (PEFT): Fine-tuning an entire LLM is computationally intensive. We leverage techniques such as Low-Rank Adaptation (LoRA) to introduce lightweight adapter modules that are trained on compliance data. This parameter-efficient fine-tuning (PEFT) approach reduces computational cost while ensuring that the model adapts to the domain without losing its general language understanding capabilities [2]. The resulting model is better able to interpret regulatory phrases, resolve ambiguities, and produce outputs that are aligned with compliance standards.

Instruction Tuning and Reinforcement Learning: Instruction tuning is used to train the model on specific prompt–response pairs typical of compliance scenarios. For example, the model learns to generate summaries of complex regulations and convert legal clauses into actionable items. Further, we incorporate Reinforcement Learning from Human Feedback (RLHF) to refine the model's responses. In this process, domain experts evaluate sample outputs, and their feedback is used to shape a reward model that guides the LLM toward more accurate and contextually grounded responses [3]. This multi-stage training pipeline ensures that the model not only understands the domain language but also follows the appropriate regulatory logic in its answers. Our fine-tuning work on a corpus of EPA guidelines and enforcement actions improved the LLM's accuracy in identifying violations from 70 to 85%.

While fine-tuning offers improved domain performance, it presents several challenges as well. Environmental compliance data (laws, standards, permits) may be relatively sparse or proprietary. Fine-tuning requires high-quality, representative text; obtaining and labeling enough diverse examples across different jurisdictions and contexts is important. Overall, fine-tuning LLMs for environmental compliance demands a balanced approach. When done well, a fine-tuned LLM can reliably interpret complex regulations and provide domain-accurate responses. But given the challenges of data, cost, and evolving knowledge, fine-tuning can be complemented by retrieval-based methods, as discussed in next section.

2.2 Optimizations with Retrieval-Augmented Generation (RAG)

RAG is a pattern that combines an LLM with a retrieval system to ground the model's outputs in relevant external knowledge. Instead of relying solely on the LLM's internal memory, RAG pipelines fetch documents (e.g., laws, clauses, prior guidance) from a knowledge base and provide them as context for the model during generation. This approach is especially valuable for compliance, where precise factual accuracy and up-to-date information are important. RAG is essential in addressing two major shortcomings of LLMs that are hallucination and token limitations.

Advanced Retrieval Techniques: To ensure that responses are grounded in factual regulatory documents, we segment long legal texts into coherent chunks based on their logical structure. Each chunk is embedded using a domain-tuned model and indexed with high-performance algorithms such as Hierarchical Navigable Small World (HNSW) graphs. This method enables rapid retrieval of relevant documents from a vast compliance knowledge base [4]. A purely semantic retrieval system sometimes misses critical keyword-specific information. We therefore integrate a hybrid retrieval approach that combines vector similarity search with traditional keyword matching using BM25 scoring. This dual strategy ensures that key legal citations and phrases are not overlooked, thus maintaining high recall and precision [5].

Our evaluations indicate that the RAG-enhanced LLM exhibited a reduction in hallucination incidents from 30% in the baseline model to approximately 10% after integration. Moreover, the hybrid retrieval approach consistently improves the accuracy of answers in questions that involve specific legal references, increasing precision by 15% for queries involving permit requirements. Overall, the RAG component supplied most relevant snippets of regulations to the LLM, and it resulted in more accurate, up-to-date, and context-rich responses from LLM.

3 AI Agentic Workflow

Adopting AI-driven agentic workflows in manufacturing can significantly improve environmental compliance management. By automating tasks such as data aggregation, regulatory monitoring, and document preparation, AI agents help manufacturers streamline administrative efforts and concentrate on core operations. Additionally, AI's capability to evaluate environmental impacts and customize reports to local community standards enables manufacturers to more effectively align their operations with sustainability objectives.

Data Aggregation Agent: One of the primary challenges in environmental compliance is the management of vast amounts of regulatory information and historical compliance records. The Data Aggregation Agent addresses this challenge by automatically collecting environmental regulations, historical compliance documents,

and impact data from both internal systems and external sources. By streamlining data management, manufacturing companies can reduce the time spent searching for documentation, thereby improving the efficiency of compliance efforts.

Regulatory Monitoring Agent: Environmental regulations are frequently updated, and it is crucial for manufacturing companies to stay informed about any changes that might impact their operations. The Regulatory Monitoring Agent continuously tracks updates in environmental laws and regulations across multiple jurisdictions relevant to the organization. The agent proactively alerts relevant teams to new mandates, ensuring that the company remains in compliance with the latest regulatory requirements.

Impact Analysis Agent: Assessing the environmental impact of manufacturing processes is central to ensuring compliance and promoting sustainability. The Impact Analysis Agent evaluates various factors such as energy consumption, emissions, waste production, and resource usage associated with manufacturing activities. This agent generates detailed reports that are tailored to local community standards and end-user concerns, enabling manufacturers to identify areas for improvement. By providing insights into the environmental consequences of their operations, manufacturers can take proactive steps to reduce their ecological footprint and meet regulatory expectations.

Documentation Agent: Documenting compliance efforts is a crucial aspect of regulatory adherence, but it is often a time-consuming and resource-intensive task. The Documentation Agent automates the preparation of compliance documents by reviewing historical compliance records and generating new reports based on current regulatory requirements. By automating this process, manufacturing companies can streamline their compliance reporting, minimize manual labor, and ensure that documentation is consistently up-to-date and accurate.

The Collaboration and Training Agent: Supports new employees by providing guided training on the company's compliance practices and linking them to relevant historical data. Additionally, the agent fosters collaboration across teams working on overlapping compliance tasks by facilitating knowledge-sharing and communication.

3.1 Agent Interaction

These agents do not operate in isolation; they work in together as a pipeline for compliance management. A typical sequence might proceed as follows:

Monitoring: The Regulatory Monitoring Agent detects a new or changed regulation and fetches the details. Aggregation: The Data Aggregation Agent ingests this new information, updates the regulatory database, and indexes it for retrieval. Analysis: Triggered by the new data, the Impact Analysis Agent connects to a LLM-RAG system to determine the effect of the regulatory change on the organization. It produces an impact report highlighting any compliance gaps. Documentation: If changes or actions are needed, the Documentation Agent prepares the necessary documents, e.g., updates to compliance manuals, draft of an implementation plan, or

a report to management about the needed changes. Collaboration: The Collaboration Agent notifies relevant staff about the findings and recommended actions. It presents the draft documents for review and facilitates any Q&A. For instance, compliance officers can ask follow-up questions through the agent, and the LLM will answer them using the updated knowledge base. Feedback and Training: Once the organization implements the changes, the Collaboration & Training Agent incorporate the scenario as a training module for others and schedule a review to ensure the changes are effective. If any part of the AI's analysis was corrected by humans, that feedback is logged to refine future AI performance.

This workflow operates continuously; even in the absence of new regulations, the agents can perform routine audits. For instance, the Data Aggregation Agent might periodically pull operational data (emissions logs, etc.), and the Impact Analysis Agent checks compliance status against existing rules on a schedule. The Documentation Agent could then log a periodic compliance status report, and the Collaboration Agent could ping a human if any anomalies are found. By delegating distinct tasks to dedicated agents, the system becomes modular, transparent, and easier to maintain. Each agent can be improved independently, and the overall workflow remains robust.

This agentic approach mirrors how human compliance teams operate with individuals or sub-teams focusing on tracking laws, analyzing impact, documentation, and training. However, here the AI agents tirelessly augment or automate each piece of the process. The AI-driven decision-making framework here aligns closely with real-time compliance needs. The framework allows many decisions to be automated (especially routine ones) while ensuring that critical judgments can be reviewed by humans. The real-time monitoring ensures the decision-making is always based on the latest information. In effect, the AI agents collectively function as an autonomous compliance co-pilot allowing human experts to focus on strategy and complex judgment calls.

3.2 Efficiency Gain Agentic Workflow

Integrating AI-driven automation into compliance processes has demonstrated measurable efficiency gains across various operational areas. Based on historic data collected from manufacturing industry, we analyzed data for different use cases. Below Table 1 summarizes specific improvements observed.

These data points underscore the tangible benefits of implementing AI-based automation systems for organizations, leading to enhanced efficiency, reduced costs, and improved adherence to regulatory standards.

Table 1 Agentic efficiency

Process	Details	Efficiency gain (%)
Regulatory review time	AI agent analyzed large amounts of data in real-time, identifying compliance issues that can lead to project delays or penalties. A proactive approach is invaluable in maintaining regulatory adherence	85
Compliance documentation preparation	Agent created new reports based on current regulatory requirements. By automating this process, organizations can streamline their compliance reporting, minimize manual labor	80
Manual regulatory searches	By analyzing historical data, regulatory guidelines and current project parameters, LLM optimized human effort. This can ensure resources are allocated effectively and potential delays are anticipated and mitigated proactively	75
Compliance training	Supports employee onboarding by providing guided information on the company's compliance practices linking them to relevant historical data and compliance guidelines	65

4 Model Evaluation and Effectiveness

In this framework, we evaluate the model's performance by comparing its outputs to baselines crafted by subject matter experts (SMEs). These SME-generated references serve as a gold standard for accuracy and completeness, providing a benchmark against which the RAG-based LLM's retrieval and generation capabilities. Specifically, SMEs develop reference outputs for a predefined set of queries or tasks, and the LLM processes the same queries.

To capture meaningful insights, we compared outputs at the sentence level and acknowledge that expert interpretations may vary, allowing for multiple valid baselines. The evaluation also rewards partially correct responses to reflect real-world scenarios where partial relevance can be valuable. Finally, user surveys measure clarity, relevance, and overall satisfaction, ensuring the model's practical utility and user acceptance are thoroughly assessed.

Table 2 summarizes the evaluation metrics observed when comparing the LLM's outputs against SME-created baselines for various retrieval queries. We also verify that the system meets its Key Performance Indicator (KPI) for real-time usability by maintaining a response time of under three seconds. This multifaceted evaluation approach provides a comprehensive measure of the RAG-based LLM's performance, effectiveness, and user acceptance, ensuring it aligns with both expert standards and real-world user expectations.

Table 2 Evaluation metrics

Metric	Definition	Range
Precision	Fraction of retrieved documents relevant to the query. Measures retrieval exactness	0.79–0.85
Recall	Fraction of relevant documents successfully retrieved. Measures retrieval completeness	0.78–0.84
F1 score	Harmonic mean of precision and recall, balancing both aspects. Balanced view of the model's overall performance	0.76–0.83

5 Challenges and Limitations

LLMs can misinterpret subtle context or policy nuances, especially if the rules are vague or conflicting. They lack true legal reasoning or understanding of complete intent, they operate on patterns in text. So, they might struggle with interpreting gray areas or balancing competing requirements, something human regulators and lawyers do with judgment. There is also the issue of accountability: if an AI gives a wrong recommendation that leads to a compliance failure, who is responsible? Organizations must treat AI outputs as advisory and ensure human oversight, especially in critical decisions.

6 Future Scope

The trajectory of AI in regulatory compliance is very promising. As models get better at understanding not just text but the intent behind laws, we can expect even more sophisticated analyses. For instance, predicting how proposed legislation might impact a company (scenario modeling), or optimizing operations to not only meet compliance but also improve sustainability metrics (using AI to suggest process tweaks that both save cost and stay green). The agent framework can be extended with new agents, such as a Risk Prediction Agent that uses historical data to foresee where compliance issues are likely to arise, or an Audit Agent that prepares the organization for external inspections by simulating an auditor's checks. Integration with IoT and real-time monitoring systems is another future potential, Compliance AI monitoring sensors in a factory flagging any reading that move toward non-compliance in a real-time manner, explaining which rule would be violated if not corrected. This addresses the gap between digital and physical compliance monitoring.

7 Conclusion

In this paper, we presented an AI-driven framework for environmental compliance that integrates fine-tuned LLMs, Retrieval-Augmented Generation, and a modular agentic workflow. Through adaptation of LLMs to the compliance domain with transfer learning, instruction tuning, and RLHF, and by leveraging RAG to ground responses in a continuously updated knowledge base, our system addresses key challenges in regulatory review. Experimental results confirm that our approach yields improvements in accuracy, response speed, and operational cost savings. The agentic architecture, with dedicated modules for data aggregation, monitoring, impact analysis, documentation, and collaboration, mirrors traditional human workflows while offering the scalability and speed of automated systems. However, several challenges remain. The continuous evolution of regulatory content demands ongoing updates to the knowledge base and periodic re-training.

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Numerical Modelling of Landfill Gas Emission Through Landfill Covers Based on Computational Fluid Dynamics

Prince Kumar, G. L. Sivakumar Babu, and Lakshminarayana Rao

Abstract Landfilling is the most common waste disposal method adopted all over the world. The degradation of biodegradable fractions of municipal solid waste (MSW) produces methane (CH_4) and carbon dioxide (CO_2). The migration of these gases into the atmosphere occurs through the cover layer in the landfill. In the present study, the 1D coupled multicomponent transport and the methane oxidation in the landfill covers by methanotrophs of landfill gas (LFG) in landfill covers were simulated using the computational fluid dynamics (CFD) numerical simulation technique. COMSOL Multiphysics_5.6 simulation tool has been used to solve the equations. The simulation has been carried out to simulate the concentration of different gases along the cover's vertical profile and quantify the methane emission from the cover at different parametric values. The numerical model was partially validated with the published data. The parametric study was carried out for different gas generation rates (F_{CH4}), the porosity of cover (e), the maximum oxidation rate (V_{max}) and different thicknesses of various cover materials (T) to understand their effects. The results show that methane oxidation efficiency becomes constant after a particular cover thickness. Methane oxidation efficiency decreases with the increase in the F_{CH4} and decrease in the V_{max}. It was also noted that as the porosity increases, the methane oxidation efficiency increases. The developed model can be used to predict methane emissions from landfills.

Keywords Multiphase · Computational fluid dynamics (CFD) · Landfill covers · Numerical simulation · Landfill gas (LFG)

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1 Introduction

The waste generation in the world has increased manifold due to the massive rise in population and the improvement in the standard of living. Most of the waste generated ends up in landfills. The waste that ends up in landfills contains a large quantity of biodegradable fractions. The degradation of biodegradable waste generates landfill gases (LFG), which primarily constitute methane (CH_4) and carbon dioxide (CO_2). Global CH_4 emissions from landfills accounted for approximately 11% of the total CH_4 emissions in 2010. In 2030, emissions from landfills are expected to represent 10% of the total global CH_4 emissions from all sources [1]. The LFG collection system is an effective measure for mitigating CH_4 emissions from landfills. However, very few landfills worldwide have active gas collection systems. Final covers are placed upon the placement of the last waste in landfills. The landfill final cover's fundamental purposes are to reduce water infiltration to reduce leachate volume, containment of LFG, minimise odour, promote vegetation growth, stabilise the landfill slope and regulate the LFG emissions [2]. One of the essential roles of landfill cover is to reduce the CH_4 emission into the atmosphere through CH_4 oxidation with the help of methanotrophs present in the cover soil [3]. In the absence of an LFG collection system, landfill cover plays an essential role in mitigating fugitive CH_4 emissions from landfills [4].

Naturally, methanotrophs present in the cover soil help in the oxidation of CH_4 and thus reduce the uncontrolled fugitive CH_4 emission from landfills. The experimental and field studies show a wide variation in oxidation efficiency of the cover soil from 0 to 100% [5, 6]. A comparison of various gas generation models for methane emission has shown very different values [7]. The lack of consideration of methane oxidation and transport in the cover has led to the inaccurate estimation of the LFG generation model. Few studies have reported using methane transport and oxidation to calculate the field methane emission [8–10]. Diffusion is the most used transport model for the gas flow in landfill covers. Most of the model uses Fick's law as the process of LFG transport [10–12]. However, due to the heterogeneous nature of the LFG, the application of Fick's law is unreliable to study the LFG transport in landfill cover. The multiple-component gas diffusive equation is more suitable for representing the LFG transport in landfill cover. The multicomponent diffusion model was applied to the Dusty Gas Model to analyse the LFG transport in the landfill cover soil [9]. However, the effect of various parameters like the gas generation rate, cover thickness, porosity and maximum oxidation rates on the oxidation efficiency, methane emission flux and O_2 concentration in the cover soil has not been studied.

In the current study, a model has been developed combining the multicomponent gas diffusion equation and Darcy's law coupled with the dual Monod kinetic equation to understand the mechanism of CH_4 transport, oxidation and emission through landfill cover soils. The model is partially validated against the literature data. The effect of four main parameters has been studied, i.e. Porosity of cover (e), Thickness of cover (T), CH_4 generation rate (F_{CH4}) and the maximum methane oxidation rate (V_{max}) on the CH_4 oxidation efficiency, CH_4 emission rate and O_2 concentration

have been studied. The model helps determine methane emission flux and methane oxidation efficiency for various scenarios without field or experimental study.

2 Modal Formulation

2.1 LFG Transport, Oxidation and Emissions in Cover Soil

Many processes govern LFG emission through the landfill covers, but LFG transport and CH_4 oxidation in the cover soil are the dominant processes [13]. The current model is based on a gas diffusion framework incorporating multicomponent gas migration, Darcy's equation for modelling convective flow behaviour and dual Monod kinetic equation for oxidation processes.

Multicomponent gas diffusion equation. Maxwell–Stefan diffusive equation developed for molecular diffusion of the multicomponent gaseous mixture in an unsaturated porous media is given by Eq. 1:

$$\left(\frac{-p}{RT}\frac{\partial y_i}{\partial z}\right) = \sum_{\substack{k \neq 1}}^{n} j = 1 \left(\frac{N_i y_k - N_k y_i}{Dsoil, ik}\right) \tag{1}$$

where $D_{soil,ik}$ = Binary diffusion coefficient of a mixture of gases *iandk* (m^2/s), z = Depth (m), y_i= Mole fraction, p = Absolute pressure (Pa), R = Gas constant (8.3145 J/mol/K) *and* N_i= Flux (mol/m^2. s).

Here, we solved diffusion using the Wilks' approximation approach to Fick's law with varying diffusivities. Diffusive flux is given by Eq. 2:

$$j_i = -\left(\rho y_i \sum_k \widehat{D}_{ik} d_k\right) \tag{2}$$

where j_i= Average flux of i (kg/m^2/s), ρ = Density of mixture (kg/m^3) and d_k= Driving force of i (1/m).

Darcy's equation. The movement of fluid through a porous media can be defined mathematically by the convection equation with Darcy's law.

$$u = -\frac{k}{\mu}\nabla p \tag{3}$$

where u= Velocity of soil gas transport (m/s), k = Permeability of cover (m^2) and μ = Kinematic coefficient of viscosity (Pa.s)

Total emission flux is the sum of diffusion flux and the convective flux. Equation 4 is used to calculate the total emission flux of methane.

$$N_i = j_i + \rho u \tag{4}$$

where N_i= total emission flux (mol/m^2/s) and ρ= dry density of soil(kg/m^3)

Oxidation rate. CH_4 oxidation can be described by Monod kinetics.

$$R_{CH4} = -\sigma t * \sigma m V_{max} \frac{[CH_4]}{K_{CH4} + [CH_4]} \times \frac{[O_2]}{K_{O2} + [O_2]} \tag{5}$$

where R_{CH4} = Oxidation rate (mol/kg/s), V_{max} = Maximum CH_4 oxidation rate (mol/kg/s). K_{O2} and K_{CH4} are the half-saturation rate constants of O_2 and CH_4, respectively.

Methanotrophic reaction. Methane undergoes partial conversion to biomass and partial oxidation to CO_2. The chemical reaction for methane oxidation in landfill cover soil can be given by Eq. 6 [6, 13]:

$$CH_4 + 1.5O_2 = 0.5CO_2 + 0.5CH_2O + 1.5H_2O \tag{6}$$

$$R_{O2} = 1.5R_{CH4} \tag{7}$$

$$R_{CO2} = 0.5R_{CH4} \tag{8}$$

where R_{O2} = O_2 consumption rate (mol/kg/s) and R_{CO2} = CO_2 production rate (mol/kg/s)

Continuity equation. For the conservation of mass of the system, the continuity equation is solved for the different gas components in the landfill cover soil. The continuity equation can be written as Eq. 9:

$$\rho \frac{\partial y_i}{\partial_t} + \nabla . j_i + \rho (u . \nabla) y_i = R_i \tag{9}$$

where R_i= The gas component i loss rate by biodegradation in cover soil (kg/m^3/s). A more detailed model overview can be found in the literature [13].

2.2 *Model Boundary Conditions, Assumption and Numerical Implementation*

The surface of the landfill cover is exposed to the atmosphere, so the surface is assumed to be atmospheric pressure, and the concentration of gas just above the surface is taken to be the atmospheric composition of 78% N_2, 21% O_2, 0.0039% CO_2 and 0.00017% CH_4. The composition of gases from the waste decomposition was assumed to be 50% CO_2 and 50% CH_4 and fed from the bottom boundary of the

Table 1 Values of parameters used for simulation

Properties	Values
Density (kg/m^3)	1500
Porosity (e)	0.5
Absolute permeability (m^2)	5.00E-13
Degree of saturation	0.5
F_{CH4} (kg/m^2/s)	8.00E-06
V_{max} (mol/kg/s)	1.00E-07
K_{CH4}	6.60E-03
K_{O2}	1.20E-03
L (m)	0.8

cover. Temperature and moisture were constant in the cover soil. The permeability of the cover soil is also assumed to be steady. COMSOL Multiphysics Version 5.6a was used to implement the equation governing the transport and the oxidation of the landfill gas to estimate the methane oxidation efficiency, methane oxidation rate and gas mass fraction.

3 Results

3.1 Validation of the Simulated Model with the Experimental Column Study

The developed numerical model is partially validated with the column study from the literature [14]. The model was applied to predict the gas concentration profile and CH_4 emission from the column study. The parameter used for simulation is shown in Table 1 [15, 16]. The experimental and simulated gas concentration along the vertical profile of the landfill cover soil is shown in Fig. 1. The model simulated the gas concentration well within the experimental values. The methane oxidation rate reported in the literature is 81.21% at a methane influx rate of 13.6 mL/min. The simulated methane oxidation rate is 80.30%, which agrees with the literature-reported value [15].

3.2 Parametric Analysis

The methane emission from landfill covers depends on various factors, including the cover soil and landfill gas production rate. This section selects four major parameters to evaluate their influence on methane oxidation and emission rates. Table 2 shows

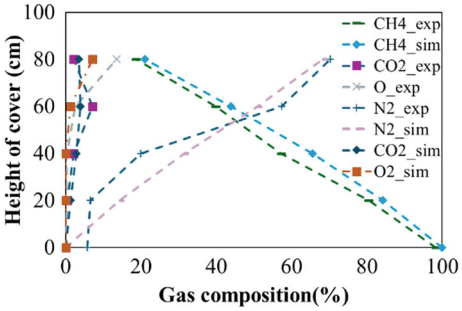

Fig. 1 Experimental and simulated gas concentration profile of the column study at $F_{CH4} = 13.6$ mL/min

Table 2 Properties used for parametric study

Parameters	Values
Thickness of cover (T, m)	0.1,0.2,0.3,0.5,1,1.5,2
F_{CH4} (kg/m^2/s)	1e-3,5e-4,1e-4,5e-5,1e-5
V_{max} (mol/kg/s)	1e-8,1e-7,5e-7,1e-6
Porosity (e)	0.3,0.5,0.7

the values of parameters chosen for parametric analysis. For the parametric study, the depth of landfill cover is taken as 0.6 m, except for the study of the thickness effect.

Effect of porosity (*e*). The porosity of the landfill cover plays a vital role in the reduction of methane emissions and methane oxidation. The lowest value of porosity is taken as 0.3, and the maximum value of porosity is taken as 0.7 to represent the dense and loose state of the cover material. Figure 2a shows the variation of O_2 concentration with depth for different porosity values. As the porosity decreases, the penetration of O_2 is limited to the upper surface of the landfill cover. As the porosity increases, the O_2 concentration increases in the lower zone of landfill cover.

Figure 2b shows the variation of the CH_4 oxidation efficiency with depth at different porosity values. The CH_4 oxidation efficiency increases with the increase in the porosity values. The zone of 100% efficiency increases from 0.05 m to 0.25 m as the porosity increases from 0.3 to 0.7. Figure 2c shows variation in CH_4 oxidation rate with depth at various porosity levels. The CH_4 oxidation rate decreases as the porosity increases. This can be attributed to the fact that at higher porosity, the gases spend very little time in the cover, reducing the chance of oxidation. At lower porosity values, the oxidation rate is higher than at higher porosity values [15, 17].

The thickness of the cover soil (*T*). The thickness of the cover defines the retention time of LFG. The higher the thickness, the greater the retention time and the more chances of oxidation of CH_4 [16]. Figure 3a shows the CH_4 emission flux for different thicknesses of the landfill cover at different F_{CH4}. The methane emission flux is calculated on the surface of the landfill cover for various thicknesses of the cover. As

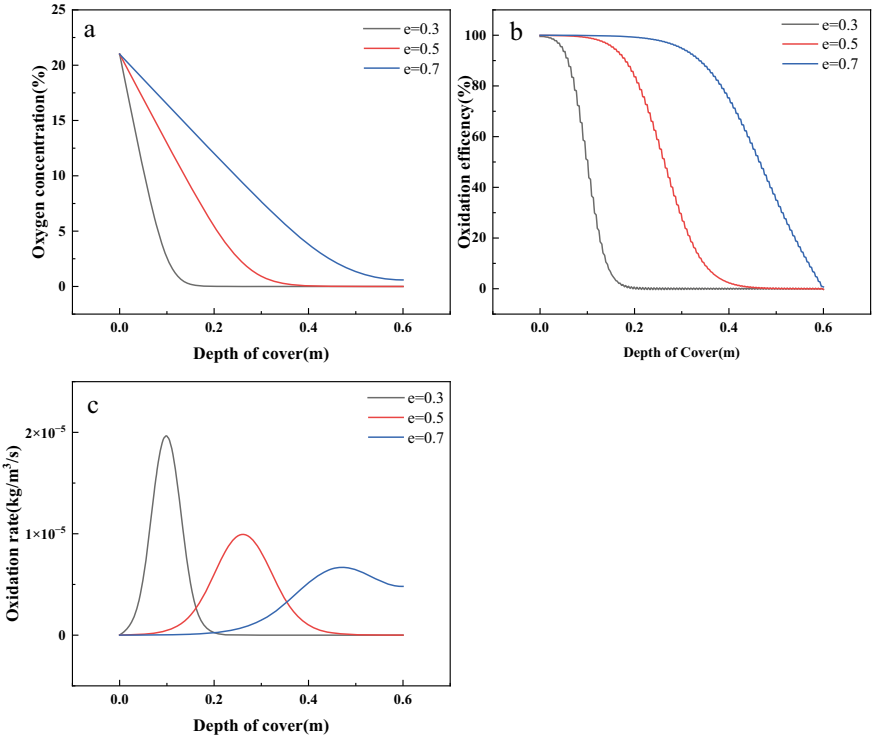

Fig. 2 Variation of **a** O_2 concentration, **b** CH_4 oxidation efficiency and **c** CH_4 oxidation rate with depth at different porosity

the thickness of the cover increases, the methane emission flux decreases to specific values and becomes constant thereafter at a given methane flux rate.

Figure 3b shows the CH_4 oxidation efficiency with the depth at various CH_4 flux rates. The CH_4 oxidation efficiency increases with the thickness of the landfill cover up to a certain extent, and then it becomes constant. Figure 3c shows the variation of CH_4 oxidation efficiency and CH_4 emission flux with the thickness of the landfill cover at a methane flux rate of 8e-5 mol/m^2/s. The CH_4 oxidation efficiency increases to a threshold value and remains constant even when the thickness of the cover increases [18]. The experimental results also show that oxidation primarily occurs in the shallower zone of the cover. Therefore, the suitable thickness of the cover is essential for achieving the maximum oxidation of methane.

Effect of CH_4generation rate (F_{CH4}). The CH_4 generation in landfills mainly influences the fugitive CH_4 emission from landfills. The CH_4 generation rate in landfills varies depending on the nature of the waste, the size of the landfill, the area of the landfill and weather conditions. The CH_4 oxidation capacity of landfill cover materials varies from 0 g/m^2/d to 500 g/m^2/d [17, 18]. In the present study, six CH_4

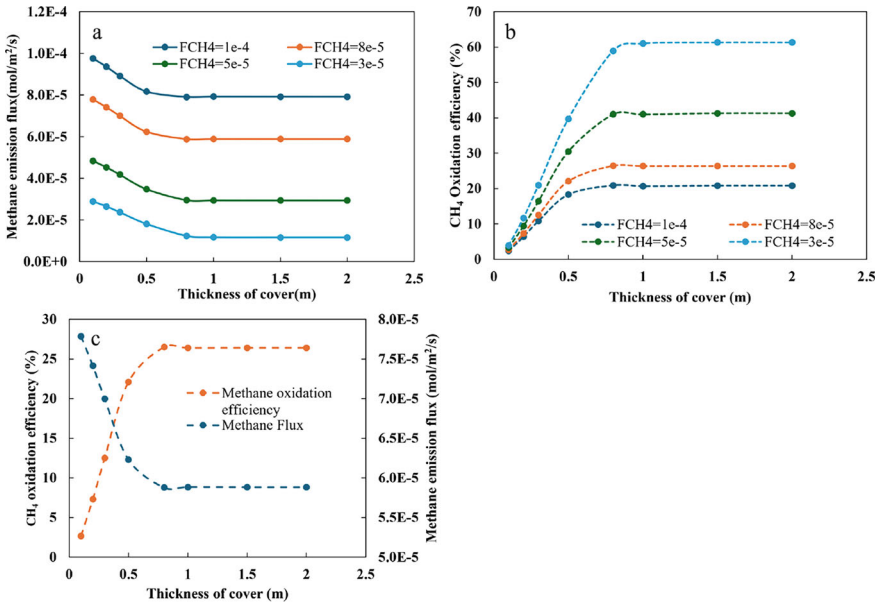

Fig. 3 Variation of **a** CH_4 emission flux and **b** CH_4 oxidation efficiency with cover thickness at different CH_4 flux rates **c** CH_4 oxidation efficiency and methane emission flux with cover thickness @ $F_{CH4} = 8\text{e-}5$ mol/m^2/s

generation rates range from 14 g/m^2/d to 1400 g/m^2/day. The thickness of the cover is assumed to be 0.6 m.

Figure 4a–c shows the influence of methane flux rate on O_2 concentration, oxidation efficiency and the methane oxidation rate with the cover thickness at various F_{CH4}. The experimental study showed that the CH_4 oxidation occurs 0–0.2 m below the cover layer [15, 17]. This is mainly because of the limiting availability of O_2. The simulation shows that at the lower value of CH_4 flux, oxidation can happen up to a depth of 0.4 m–0.6 m, as in the case of methane flux of 1e-5 mol/m^2/s. Also, the highest rate of CH_4 oxidation increases with the increase in the F_{CH4} (Fig. 4c). CH_4 oxidation efficiency increases with the decrease in the F_{CH4} (Fig. 4b). The CH_4 oxidation efficiency remains constant when the CH_4 flux rate is below 1e-5 mol/m^2/s. This is because of the greater penetration of O_2 at the low CH_4 generation rate compared to the higher CH_4 generation rate (Fig. 4a) [18].

Effect of maximum oxidation rate (V_{max}). The oxidation capacity of the cover mainly depends on the V_{max}. The influence of V_{max} on oxidation efficiency is shown in Fig. 5. The oxidation efficiency is calculated at the surface of the landfill cover. Figure 5a shows the variation of oxidation efficiency with the depth of the cover for different V_{max}. As the V_{max} increases, the oxidation efficiency increases, reaching up to 100% when the V_{max} is more than 5e-7 mol/kg/s [18]. Figure 5b shows the O_2 concentration for different V_{max}. It can be noted that as the V_{max} decreases, the O_2

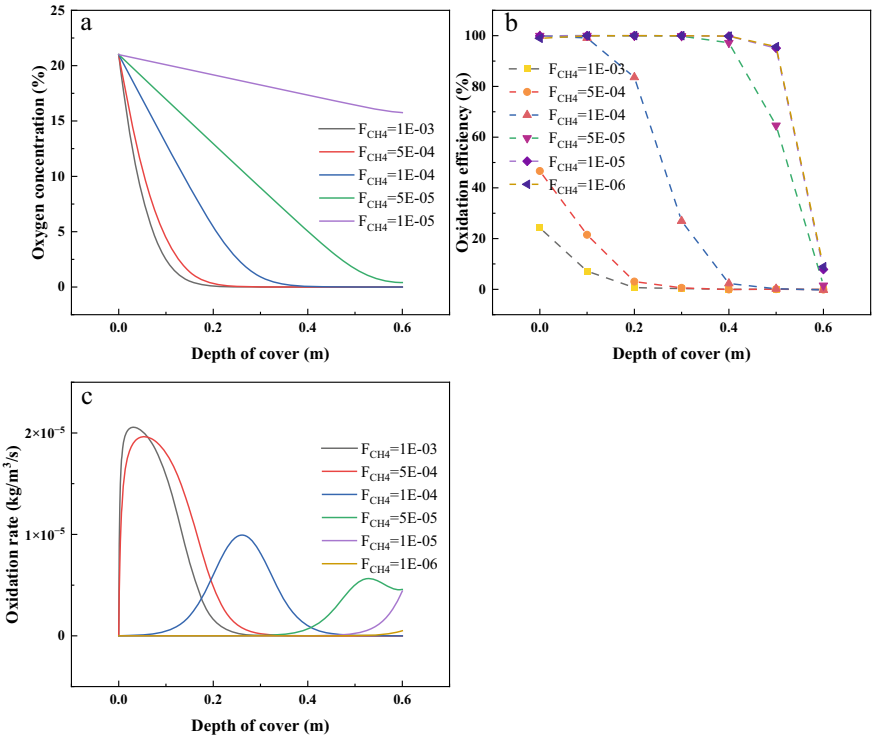

Fig. 4 Variation of **a** O_2 concentration, **b** CH_4 oxidation efficiency and **c** CH_4 oxidation rate with depth of cover at different FCH_4

concentration increases to the deeper cover layer. This can be attributed to the lower consumption of O_2 due to the low values of V_{max}, which makes more O_2 available in the deeper section. The higher V_{max} material is suitable for reducing the CH_4 emission flux. This will reduce the cover thickness required to achieve a certain level of CH_4 emission [17].

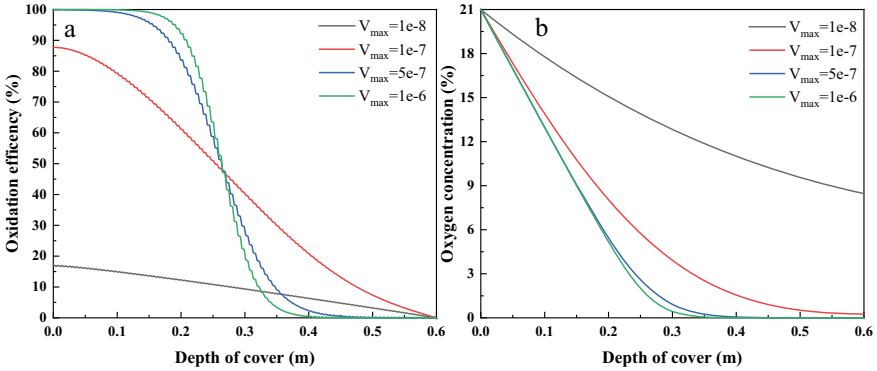

Fig. 5 Variation of **a** CH_4 oxidation efficiency and **b** O_2 concentration with depth of cover at different V_{max}

4 Conclusion

Landfill cover is essential in reducing fugitive methane emissions from landfills. Designing a suitable cover layer to mitigate methane emissions is very challenging. The computational approach to determine the various cover parameters to reduce methane emissions is a powerful tool. The developed model incorporates different physics of multicomponent gas transport coupled with Darcy's and Monod's dual oxidation kinetic equations to simulate the flow of gases in the landfill cover. Various parameters are essential in determining the methane flux, methane oxidation efficiency and the O_2 concentration in the landfill. The porosity of the cover is a critical parameter for selecting the cover material. The thickness of the cover has been found to influence methane emission and oxidation efficiency. The threshold thickness of the cover is essential to reduce the cost of construction and maximise the capacity of the landfill. The developed model fits well with the results of the literature. The proposed model can be used to determine the fugitive methane emission from the landfill provided reliable data related to landfill cover and the methane generation rate are provided in the modelling.

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Influence of Particle Deposition Morphology on Solute Transport in Porous Media: A CFD-DEM Study

Dong-Jiang Lv and Hong-Xin Chen

Abstract Accurate prediction of solute transport is crucial for environmental pollution forecasting and remediation. As the number of mobile particles, such as microplastics, in soils and groundwater increases, the risk of inaccurate solute transport predictions also rises. This study uses a coupled computational fluid dynamics-discrete element method (CFD-DEM) along with an advection–diffusion equation (ADE) to simulate the transport processes of particles and solute at the pore scale. We observed the migration and deposition of particles in porous media at various injection velocities. Further studies investigated the effects of different particle deposition statuses and Péclet numbers on solute transport. A continuous-time random walk (CTRW) model with a truncated power law (TPL) was used to describe solute transport behaviors under various conditions. The results show that particles form stable agglomerates under low injection velocity conditions, leading to a significant increase in the number of particles retained with time. In contrast, under high flow conditions, the variation in particle deposition at different times is minimal. The solute transport simulations revealed anomalous solute transport behavior, which is linked to increased particle deposition and higher Péclet numbers. The significant rise in particle deposition caused preferential flow in the porous media, resulting in early breakthroughs and tailing of solute transport. This study highlights the impact of particle-induced pore structure evolution on solute transport and emphasizes the importance of incorporating particle-structure interactions into solute transport models.

Keywords CFD-DEM · Pore-scale simulation · Particle deposition · Anomalous solute transport

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1 Introduction

The co-transport of particles and solute in porous media is a common phenomenon in environmental geotechnical engineering, with significant implications for pathogen and pollutant transport, groundwater resource management, and hydrocarbon extraction [1]. The deposition of particles in porous media leads to dynamic changes in pore structure, complicating the prediction of solute transport processes [2]. Therefore, a comprehensive study of particle deposition characteristics is crucial for accurately predicting pollutant transport patterns and optimizing hydrocarbon recovery strategies.

Continuum-scale models based on colloid filtration theory and the advection–dispersion equation have played a key role in explaining this phenomenon [3]. These models provide statistical explanations for colloidal particle deposition in porous media and describe the influence of particle adsorption on solute transport. However, these models typically assume static pore geometries, overlooking the dynamic evolution of pore structure and hydraulic properties due to particle deposition. This limitation stems from the complex and dynamic coupling between particle deposition, pore structure, and solute transport [4]. Traditional experimental methods are limited in capturing transient deposition patterns and quantifying solute transport dynamics in evolving porous structures over time. Recent advancements in numerical modeling have partially addressed these challenges. The computational fluid dynamics-discrete element method (CFD-DEM) has proven effective in elucidating particle transport, deposition behaviors, and the permeability reduction mechanisms caused by pore clogging [5]. Liu et al. [6] applied the CFD-DEM coupling method to study particle flow in porous media, revealing the influence of the Reynolds number on particle deposition patterns. Subsequent studies have also explored colloid electrochemical interactions and observed particle aggregation and clogging in porous media. However, few CFD-DEM studies on particle migration in porous media have addressed solute transport dynamics. Specifically, research on the impact of particle deposition-induced pore structure evolution on solute transport remains limited, leaving a gap in understanding the effects of particle deposition structures on solute transport.

In this study, we examine the transport of particles and solute in a porous media, aiming to reveal the influence of the deposition characteristics of particles on the solute transport behavior. A CFD-DEM numerical simulation method with coupling the advection–diffusion equations (ADE) is used. This study provides new perspectives and methodological support for understanding the co-transport processes of particles and solute in porous media.

2 Methodology

2.1 CFD-DEM Coupling

In this study, the transport of particles and solute was simulated in a porous media as shown in Fig. 1. The numerical model was developed using the open-source platform CFDEM, which integrates computational fluid dynamics (CFD) and discrete element method (DEM). The three key components are OpenFOAM 5.x for CFD, LIGGGHTS 3.8.1 for DEM, and CFDEM for coupling [7]. In OpenFOAM, the finite volume method (FVM) is employed to solve the Navier–Stokes (N-S) equations that governing fluid motion, and the continuity equation is

$$\frac{\partial \varphi \rho_w}{\partial t} + \nabla v(\varphi \rho_w u) = 0 \tag{1}$$

where φ is the fluid volume fraction, ρ_w is the density of the fluid, t is time, and u is the velocity of the fluid. The momentum equation is written as

$$\frac{\partial (\varphi \rho_w u)}{\partial t} + \nabla v(\varphi \rho_w u v u) = -\varphi \nabla p + \nabla v(\varphi \mu v u) + \varphi \rho_w g - \sum F_{drag} \tag{2}$$

where p is pressure, μ is the dynamic viscosity, g is the gravitational acceleration vector, and F_{drag} is the drag force exerted by the particle within the FVM cell. In this study, F_{drag} can be described as

$$F_{drag} = \frac{1}{2} C_d \rho_w A \varphi^{2-\chi} \left| u - v_p \right| \left(u - v_p \right) \tag{3}$$

where C_d is the drag coefficient, A is the projection area of the particle, χ is an empirical constant, and v_p is the velocity of particle. The exact expression of the drag coefficient varies in different drag force models, but they do not make much difference in terms of their effectiveness in describing the resistance to particle flow. In this study, the drag model proposed by Abraham [8] was selected for its simplicity:

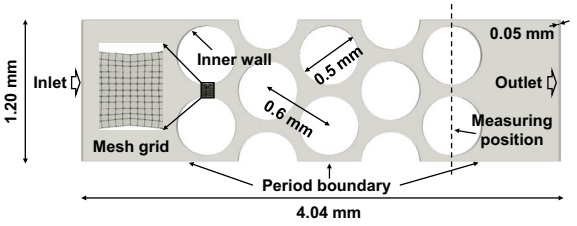

Fig. 1 Numerical simulation modeling of porous media

$$C_d = \frac{24}{(9.06)^2}\left(\frac{9.06}{\sqrt{\mathrm{Re}_p}} + 1\right)^2 \tag{4}$$

where Re_p is the particle Reynolds number:

$$\mathrm{Re}_p = \frac{\rho_w d_p (u - v_p)}{\mu} \tag{5}$$

where d_p is the particle diameter. This drag force model exhibits good correspondence with experimental results when $\mathrm{Re}_p < 5000$, covering the range of the fluid flow in soil media and later cases in this study. To balance computational accuracy with resource consumption, a semi-analytical CFD-DEM simulation code is used, which has been well proven [7]. In the semi-resolved CFD-DEM coupling method, the smoothed relative velocity between fluid and particle u_r can be written as

$$u_r = \frac{\sum_{j=1}^{N} \varphi_j V_{cell,j} u_{cell,j} \exp\left[-\left|p_i - c_j\right|^2 / \left(2\kappa^2 d_{p,i}^2\right)\right]}{\sum_{j=1}^{N} \varphi_j V_{cell,j} \exp\left[-\left|p_i - c_j\right|^2 / \left(2\kappa^2 d_{p,i}^2\right)\right]} \tag{6}$$

where N is the total number of fluid grids within the particle's smoothing domain, and j denote the index of a specific fluid grid. V_{cell} is the volume of the mesh grid, p_i-c_j is the distance between particle i and grid j, k is the ratio of the smoothing distance to the particle diameter and is typically ranges from 1.5 to 2. The motion of particle is calculated by Newton's law of motion. The linear velocity of the particle is given by

$$m_p \frac{dv_p}{dt} = m_p g + F_{buoyancy} + F_{drag} + F_{p-p} + F_{w-p} \tag{7}$$

where m_p is the mass of the particle, $F_{buoyancy}$ is the buoyance force caused by the pressure gradient, while $F_{p\text{-}p}$ and $F_{w\text{-}p}$ are the collision force between particle and particle and between wall and particle. In addition to using the Hertzian model to describe the particle–particle and particle–wall collisions, the simplified JKR (sJKR) model was utilized to incorporate additional normal adhesive forces between them [4], thereby promoting the expected phenomena of particles aggregation and inner wall's collection. The sJKR model calculates the normal force based on the contact area A_{sjkr} and the density of cohesion energy k_{sjkr}:

$$F_{\mathrm{sjkr}} = k_{\mathrm{sjkr}} A_{\mathrm{sjkr}} \tag{8}$$

The cohesion energy density was chosen as 1×10^5 J/m3, referencing experimental results [9]. The rotation of the particle is governed by

$$I_p \frac{d\omega}{dt} = \sum T_t + T_r \tag{9}$$

where I_p is the rotational inertia of the particle, ω is the angular velocity of the particle, and T_t, T_r are the moment of the tangential force exerted by the contacting particles and the rolling friction toque, respectively.

2.2 Transport of Solute and Evaluation of Breakthrough Curves

The ADE is utilized to simulate solute transport in porous media:

$$\frac{\partial C}{\partial t} + U v \nabla C = \nabla v (D \nabla C) \tag{10}$$

where C is the solute concentration, U is the advection velocity, and D is the effective diffusion coefficient $D = \varphi D_{m0}$, D_{m0} is the molecular diffusion coefficient. The breakthrough curves (BTCs) were used to evaluate the solute transport behavior, and the flux-averaged concentration can be calculated as

$$C_{flux}(t) = \frac{\int C(t)_i \varphi_i u_i dV_{cell}}{\int \varphi_i u_i dV_{cell}} \tag{11}$$

The BTCs were fitted with a continuous-time random walk (CTRW) model using the CTRW MATLAB toolbox [10]. The truncated power law (TPL) was used to describe the long tail of the anomalous solute transport:

$$\tilde{\psi}(u) = (1 + \tau_2 u t_1)^{\beta} \exp(t_1 u) \Gamma\left(-\beta, \tau_2^{-1} + t_1 u\right) / \Gamma\left(-\beta, \tau_2^{-1}\right) \tag{12}$$

where β is the degree of anomalous solute transport, t_1 is the characteristic transition time, and t_2 is the cutoff time to Fickian transport. The anomalous solute transport can be observed when $\beta < 2$.

2.3 Problem Formulation and Parameters

In the porous media illustrated in Fig. 1, inlet injection velocities of 0.002 and 0.01 m/s [2] were applied in the CFD simulations to obtain different particle deposition morphologies under varying Reynolds numbers, defined as $\mathrm{Re} = \rho_w u_f L/\mu$, where L is the characteristic length. The outlet was set to a constant pressure of 0 kPa, while no-slip conditions were applied to the inner walls. Periodic boundary conditions were imposed on the remaining boundaries in the CFD model, corresponding to the y and z

Table 1 Parameters used for simulation

Model parameter	Units	Value	Model parameter	Units	Value
Particle radius	μm	10	Coefficient of restitution	–	0.55
Particle mass density	kg/m^3	2600	Coefficient of friction	–	0.8
Shear modulus	MPa	10	Cohesion energy density	J/m^3	1×10^5
Poisson ratio	-	0.25	Gravity	m/s^2	9.81
Injection velocity	m/s	0.002, 0.01	DEM time step	s	1×10^{-7}
Fluid viscosity	Pa·s	0.001	CFD time step	s	1×10^{-5}
Fluid density	kg/m^3	1000	Coupling interval	s	1×10^{-5}
Molecular diffusion coefficient	m^2/s	1×10^{-11}-5×10^{-7}			

directions in the DEM model. The time step for the CFD and DEM simulations were set to 1×10^{-5} and 1×10^{-7} s, respectively, to ensure numerical stability based on the Courant number for CFD and the Rayleigh time criterion for DEM. The simulation parameters were listed in Table 1 [2].

3 Simulation Results and Discussion

3.1 Numerical Simulations Results

Particle retention status. Figure 2a, b display the evolution of particle deposition under injection velocities of 0.002 m/s and 0.01 m/s, respectively. It can be observed that at an injection velocity of 0.002 m/s, the quantity of retained particles is significantly higher compared to that at an injection velocity of 0.01 m/s. Additionally, particles tend to aggregate more easily and accumulate on the inner walls, leading to pore clogging and preferential flow. At an injection velocity of 0.01 m/s, the number of retained particles decreases. Furthermore, particles deposit in a triangular pattern on the inner walls, which is consistent with experimental observations shown in Fig. 2c.

Solute transport simulation results. For the structures shown in Fig. 2a, a solute with $D_{m0} = 1 \times 10^{-11}$ m^2/s and a dimensionless concentration of 1 was injected at a flow velocity of 0.002 m/s for 2 s. Following the injection, clean water was flushed through the system for 6 s to capture any potential concentration tail. For the structure presented in Fig. 2b, a solute with $D_{m0} = 5 \times 10^{-11}$ m^2/s was injected at a flow velocity of 0.01 m/s for 0.4 s to ensure the same Péclet number, defined as $\mathrm{Pe} = u_f L/D_{m0}$, followed by a 1.2 s flushing. The resulting BTCs are displayed in Fig. 3. Under an injection velocity of 0.002 m/s, the change in the form of BTCs

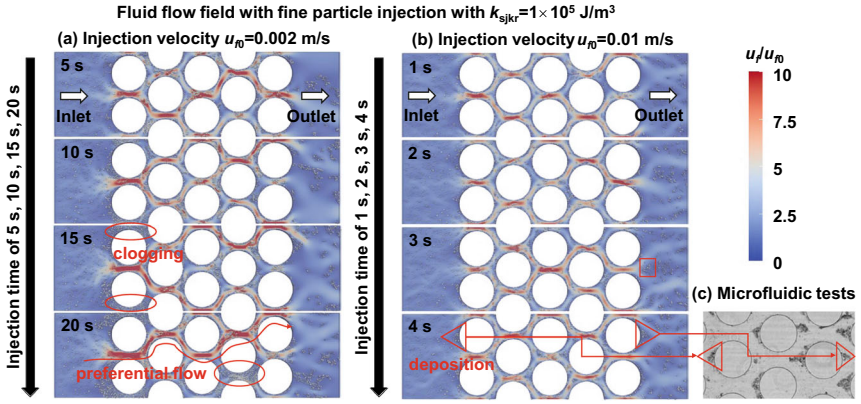

Fig. 2 Particle retention status within porous media: **a** $u_{f0} = 0.002$ m/s; **b** $u_{f0} = 0.01$ m/s and **c** particle deposition observed in microfluidic experiments [6]

is particularly evident, exhibiting early breakthrough and concentration tails. At an injection velocity of 0.01 m/s, the changing in BTCs are minimal, and the BTCs are nearly symmetric between the injection and flushing stages. Notably, BTCs with different structures show concentration tailing.

CTRW fitting results. Taking the final deposition status (i.e., the structure shown in Fig. 2a formed after injecting particles for 20 s) at an injection velocity of 0.002 m/s as an example, the CTRW provides a good fit for the BTCs shown in Fig. 4a. The fitting results of β for different deposition status and Pe numbers are shown in Fig. 4b. At an injection velocity of 0.002 m/s, β gradually decreases with particle deposition and almost all values are below 2, indicating non-Fickian transport of solute. However, at an injection velocity of 0.01 m/s, β does not decrease with particle deposition, and almost all values are greater than 2, indicating that the solute transport follows

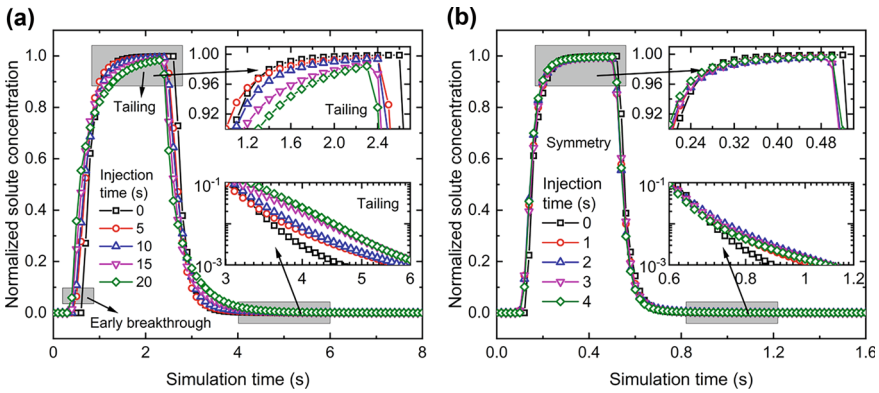

Fig. 3 BTCs at measuring position: **a** $u_{f0} = 0.002$ m/s and **b** $u_{f0} = 0.01$ m/s

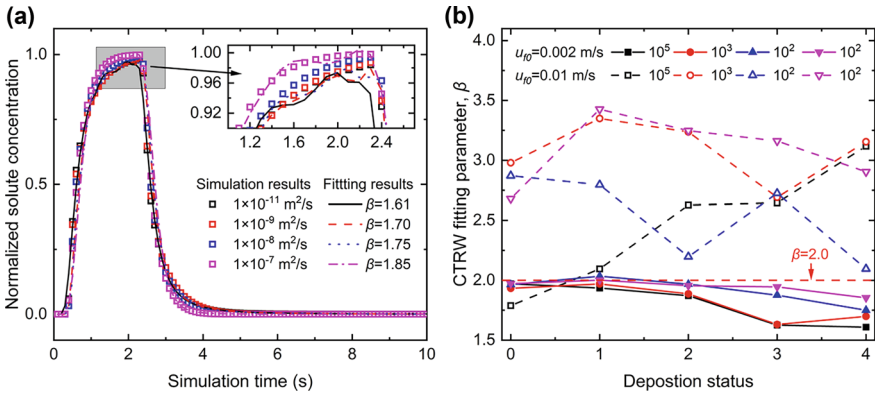

Fig. 4 Fitting results: **a** injection velocity of 0.002 m/s and molecular diffusion coefficient of 1×10^{-11}–1×10^{-7} m^2/s and **b** β with different deposition status and Pe numbers

Fickian behavior. An increase in the molecular diffusion coefficient results in a higher β value, and this trend is consistent across different injection velocities.

3.2 Discussion

The evolution of pore structure due to low injection velocity (0.002 m/s) particle deposition triggered anomalous solute transport. Analysis of deposition morphologies and flow paths attributes this phenomenon to preferential flow paths and solute retention in low-velocity zones created by particle deposition. In previous studies, the discussion on the effect of particle deposition state has often been limited. Moreover, this geometry-driven anomalous transport intensifies with increasing Pe numbers. However, high injection velocities result in a relatively homogeneous particle distribution, leading to Fickian transport. This study demonstrates that structural alterations in porous media due to particle deposition significantly affect solute transport. Long-term predictions of solute transport require advanced mathematical frameworks that account for particle-structure interactions.

4 Conclusion

This study investigated particle deposition morphologies in porous media using a CFD-DEM coupled approach and analyzed their impact on solute transport. The results show that a low injection velocity (Re = 1) leads to markedly higher particle retention and promotes the formation of preferential flow paths compared to a high injection velocity (Re = 5. Extensive particle deposition at low injection velocity

induces anomalous solute transport, while a higher injection velocity promotes uniform particle distribution, leading to Fickian transport. The CTRW model with a TPL effectively quantifies the anomalous solute transport behavior under clogging conditions. The anomalous solute transport behavior is correlated with elevated Péclet numbers and enhanced structural heterogeneity in porous media. These results emphasize the necessity of coupling particle-structure interactions in solute transport modeling and long-term prediction.

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Conflict of Interest There are no conflicts of interest to declare.

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An Integrated Approach to Groundwater Vulnerability Assessment and Monitoring at a Reclaimed Landfill Site

Anna Podlasek, Eugeniusz Koda, Piotr Streich, and Wojciech Pietruszewski

Abstract This paper presents research on groundwater vulnerability to pollution in the municipal solid waste (MSW) landfill area. The study focuses on a large-capacity MSW landfill in central Poland, established in 1977 and closed in 1991. Following its closure, technical reclamation was carried out between 1994 and 2001, and biological reclamation was completed in 2005. Groundwater monitoring has been carried out regularly since reclamation. The DRASTIC and GOD methods were used to assess groundwater vulnerability. The results show different levels of vulnerability influenced by several hydrogeological conditions. The results of the groundwater vulnerability assessment were compared with monitoring data to assess their accuracy and relevance in identifying real contamination risks. The analyses of contaminant migration paths indicated the effectiveness of reclamation efforts. Chloride and ammonium concentrations in most piezometers were within acceptable ranges for good chemical status. Adopted an integrated approach of vulnerability assessment and monitoring data analysis, made it possible to identify critical areas at risk of contamination, assess reclamation effectiveness and provide recommendations for improved environmental protection. The results highlighted the usefulness of parametric methods as valuable tools to complement groundwater monitoring, particularly in landfill areas.

Keywords Groundwater · Landfill · GOD · DRASTIC · Contamination · Reclamation

1 Introduction

Groundwater vulnerability refers to the susceptibility of groundwater quality to human activities that could negatively impact its present or future usefulness [1]. It is also defined as a natural property of the environment that determines the risk of

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migration of pollutants of anthropogenic origin from the ground surface to deeper aquifers [2].

Methods for assessing vulnerability to pollution include parametric methods that consider and prioritise the most important parameters influencing vulnerability, e.g. GOD [3], AVI [4] and ranking methods, which rank individual parameters according to their influence on vulnerability and aggregate the results into a vulnerability index, e.g. DRASTIC [5], DIVERSITY [6], CLASS [7] and SEEPAGE [8]. Vulnerability assessment analyses also use mathematical and hydrogeochemical modelling methods, which describe the system and its behaviour by mathematical models and allow vulnerability prediction, but require large amounts of data and advanced computational tools [9]. Hydrochemical methods are also used to assess vulnerability based on groundwater chemistry [10]. Statistical and geostatistical methods analyse relationships between different parameters, such as soil type, depth to water table and water quality. They are used to verify other methods and to identify key factors influencing vulnerability [11].

This study aimed to assess the vulnerability of groundwater to pollution in the area of a reclaimed landfill in central Poland, using the DRASTIC and GOD methods. The analysis aimed to identify areas with different contamination levels and assess the effectiveness of remediation measures. Based on the analysis of the technical documentation and the operational history of the landfill, it was hypothesised that the groundwater is exposed to a significant risk of contaminant migration due to the local hydrogeological conditions. This hypothesis was complemented by the assumption that the key factor influencing the vulnerability of the groundwater to contamination is the effectiveness of the landfill reclamation measures, particularly in terms of groundwater isolation.

2 Material and Methods

2.1 Study Area

The MSW landfill is located in Marki, about 20 km from Warsaw (central Poland). The landfill started operations in 1977 as a temporary facility for 3 years with a planned area of 6 ha. In practice, it operated for several years, covering an area of more than 10 ha and reaching a height of 24 m. The landfill was finally closed in 1991.

Since 1994, the MSW landfill has been subject to reclamation works, which have included covering the landfill with a layer of low permeable soil, drainage of rainwater and leachate from the landfill, ditches and pumping stations, introduction of an active degassing system with electricity conversion, seeding of the slopes with a grass mixture, introduction of a network of monitoring piezometers for environmental control, introduction of slope protection and levelling of the area adjacent to the landfill. The technical reclamation of the landfill was completed in December 2001.

Since 2002, work has been carried out on the landfill's leachate collection system, including replacing pumps and repairing the drainage system. Monitoring has also been carried out, including studies of surface water quality, groundwater quality, leachate quality and landfill gas control.

2.2 Groundwater Vulnerability Assessment

Two commonly used methods were applied to assess the vulnerability of groundwater to contamination: DRASTIC and GOD.

The choice of these methods was justified by their widespread use in practice and relatively low data availability requirements. Although both methods are based on a similar principle of assigning weights to individual parameters, they differ in the set of factors considered and how they are evaluated. A comparison of the results obtained with the two methods allowed for the verification of their suitability in the study area and the identification of the dominant factors for assessing groundwater vulnerability in a given case [12]. The analysis was based on data obtained from archive documentation and technical reports. These data were supplemented by monitoring reports containing long-term measurements of groundwater levels, groundwater quality and detailed hydrogeological parameters.

DRASTIC method. The groundwater vulnerability classification system of the DRASTIC method is a standard tool in water resources management in many countries and is based on the following assumptions [13]: the area analysed does not exceed 0.5 km^2, contaminants enter the aquifer directly from the ground surface, contaminant migration in the aquifer follows the direction of groundwater flow, groundwater flow is linear, i.e. it takes place in a pore environment, which is typical for aquifers with a permeable porous structure.

The DRASTIC method takes into account the following factors [14]: D—depth to water, R—recharge network, A—aquifer media, S—soil media, T—topography, I—influence of the vadose zone and C—hydraulic conductivity of the aquifer. The method uses a simple calculation algorithm that is a combination of two key elements: weights (w) and ranks (r) [5]. The vulnerability index determines the final vulnerability score, which is the sum of the products of the weights and ranks [13]. Each factor considered in the DRASTIC method is assigned a weight from 1 to 5. A weight of 5 means that the factor is very important for assessing the pollution risk, and a weight of 1 is the least important. Depending on the specific environment, each DRASTIC parameter is assigned a particular rank value that determines the potential for contamination. The DRASTIC index is calculated by summing the products of the weights and the ranks according to the following formula:

$$DRASTIC = D_r \times D_w + R_r \times R_w + A_r \times A_w + S_r \times S_w + T_r \times T_w + I_r \times I_w + C_r \times C_w \quad (1)$$

Table 1 Classifications of DRASTIC index

DRASTIC index	Class vulnerability
47–92	Low
93–136	Moderate
137–184	High
>184	Very high

Table 2 Classifications of GOD index

GOD index	Class vulnerability
0–0.1	Negligible
0.1–0.3	Low
0.3–0.5	Moderate
0.5–0.7	High
>0.7	Extreme

where: *D, R, A, S, T, I, C* are the seven parameters of the method, while subscripts *w* and *r* represent the weight of each parameter and the corresponding rating, respectively.

The DRASTIC index was classified according to Table 1 [15]:

GOD method. The GOD method allows a rapid assessment of groundwater vulnerability based on three parameters: *G*—groundwater conifer, *O*—overall aquifer class and *D*—depth to groundwater [16]. The GOD index is calculated according to the equation [12]:

$$GOD = G \times O \times D \tag{2}$$

The GOD index was classified according to the Table 2 [12].

2.3 Groundwater Quality Monitoring

In the landfill area, there is a groundwater table in sandy formations. The depth of the water table ranges from 1.0 to 3.8 m below the ground surface and varies with the precipitation rate.

The groundwater monitoring is based on a network of piezometers installed in 2002 in the landfill area. The location of the piezometers and the direction of groundwater flow are shown in Fig. 1.

In the present study, a comprehensive analysis of temporal variations in contaminant concentrations in groundwater was conducted, building upon the findings of Koda et al. [17]. The research framework encompassed an evaluation of long-term

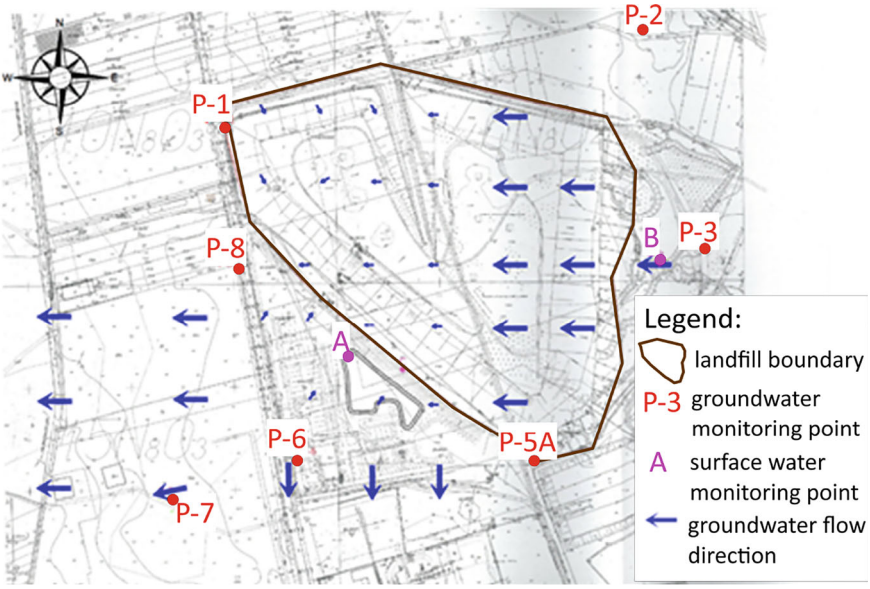

Fig. 1 Location of the groundwater monitoring network [17]

trends in concentration changes within piezometers, particularly on piezometer P-6, situated in the groundwater discharge zone. Furthermore, the study investigated the dynamics of chloride and ammonium concentration fluctuations throughout the monitoring period (2002–2009) across all observation points. This in-depth analysis facilitated the assessment of groundwater quality following established environmental standards [18].

3 Results and Discussion

3.1 Analysis of Groundwater Vulnerability

Analysis of the results showed that the differences in groundwater vulnerability to pollution are mainly due to the lithological diversity of the vadose zone and the aquifer, with the hydraulic conductivity of the geological formations playing a key role.

The vulnerability assessment results are presented in Table 3. The results of the DRASTIC index showed a moderate to high vulnerability of groundwater to pollution. In contrast, the vulnerability to pollution based on the GOD index ranged from negligible to high.

Table 3 The results of DRASTIC index assessment

Piezometer	DRASTIC index	Class vulnerability	GOD index	Class vulnerability
P-1	112	Moderate	0.189	Low
P-2	162	High	0.630	High
P-3	164	High	0.567	High
P-4	164	High	0.567	High
P-5A	112	Moderate	0.090	Negligible
P-6	128	Moderate	0.210	Low
P-7	167	High	0.630	High
P-8	106	Moderate	0.090	Negligible

Similar to the results obtained with the DRASTIC method, the GOD index values showed significant differences between piezometers, reflecting the different risks of groundwater pollution depending on the location of the monitoring point. High GOD index values occurred in 50% of the monitoring area. Of particular note is piezometer P-7, which is located in the western part, and is the lowest point of the site into which water from the entire eastern part flows. The differences in the vulnerability of the groundwater to contamination, as expressed by the GOD index value, are mainly determined by the lithology of the geological formations and the depth of the water table.

A groundwater vulnerability assessment based on the DRASTIC and GOD methods identified areas of increased vulnerability to contamination, allowing the identification of potential contaminant migration paths and their spatial extent [19]. The difference in interpreting the results obtained by the two methods is worth noting. In the GOD method, the values are so low that it is practically possible to consider the area around piezometers P-5A and P-8 entirely resistant for contamination. In contrast, the DRASTIC method classified this area as having moderate vulnerability to contamination. This difference is because the DRASTIC method is much more detailed and considers a broader range of hydrogeological parameters. Similarly, the area around piezometer P-6 was classified as low vulnerability to pollution by the GOD method, but moderate by the DRASTIC method. In addition, a study by Hosseini and Saremi [19] showed that the DRASTIC method is more precise than the GOD method due to the inclusion of a greater number of parameters, which allows for more accurate mapping of the location of potential groundwater contamination. Kirlas et al. [12] proposed that zones with increased pollution risk are mainly located in low-lying areas with high soil permeability and intensive anthropogenic activity. Similar conclusions were drawn by Gonçalves et al. [14], who analysed the vulnerability of groundwater to contamination in the area of a cemetery in Figueira da Foz, Portugal, and concluded that the high susceptibility of this area to contamination was due to the specific characteristics of the vadose zone. In addition, the area's low slope (0–2%) favoured the accumulation of rainwater and, combined with the

high infiltration capacity and hydraulic conductivity, allowed the rapid movement of contaminants to the lower aquifers.

Analysis of studies carried out in Greece [12], Portugal [14] and at the landfill site in this study clearly showed the key role of factors such as soil permeability, vadose zone characteristics, and slope in assessing the vulnerability of groundwater to contamination. Krogulec [13], in her study of the Kampinos National Park in Poland, indicated that the DRASTIC model is helpful for a preliminary, general assessment of groundwater vulnerability in a regional area. However, the author emphasised that more precise results require additional research tools, such as laboratory analyses. This approach allows a more precise determination of the hydrogeological properties of the pore medium, which in turn allows more detailed modelling of contaminant transport processes under local conditions. Despite the limitations of the DRASTIC and GOD methods, their use in hydrogeological analyses remains an invaluable tool in delineating groundwater protection zones in vulnerable environments, complementing the information obtained from other research methods [20]. The methodology provides essential data on the vulnerability of groundwater in the study region, which can serve as a basis for more restrictive land use regulations and water quality standards. The study results are particularly valuable for policymakers and local governments, who can use them to develop effective groundwater protection strategies [21].

3.2 Integration of Groundwater Monitoring Analysis and Vulnerability Assessment

Analysis of the groundwater monitoring results showed that the highest concentrations of pollutants occur in the western part of the area, which is consistent with the modelling results.

The selection of piezometer P-6 for detailed investigation (Table 4) was justified by its location in the groundwater outflow zone on the western side of the landfill. Analysis of the concentrations of key contaminants in the groundwater sampled from piezometer P-6, in the period 2002–2009, showed that the values of these concentrations did not exceed the standards for water of good chemical status (class III).

It was also found that between 2002 and 2009 the water quality in piezometer P-6 showed a gradual improvement, particularly evident in the decrease in chloride and ammonium levels, which can be linked to the effectiveness of applying landfill remediation. These results are consistent with the vulnerability assessment to pollution carried out by the DRASTIC and GOD methods. Both methods indicated a low (GOD) or moderate (DRASTIC) vulnerability of this area to pollution. This confirms that the parametric models adequately reflect the actual pollution status.

Different results were obtained for piezometer P-7. Despite a high vulnerability score using the DRASTIC and GOD methods, contaminant concentrations were

Table 4 Concentrations of selected groundwater pollution indicators in piezometer P-6 (at the landfill outflow) in 2002–2009 [17]

Parameter	Unit	Min	Max	Mean	Median	Standard deviation	Limit values*
pH	–	6.7	7.6	7.1	7.2	0.3	6.5–9.5
Electrical conductivity	μS/cm	893.0	1228.0	1020.3	1013.0	121.8	2500
Ammonium	mg/L	0.004	2.900	1.770	2.600	1.110	1.500
Chloride	mg/L	48.00	110.00	79.67	81.50	25.14	250

Notes * according to the Regulation of the Minister of Maritime Economy and Inland Navigation (for class III of groundwater quality) [18]

lowest at this site. The analysis of the spatial distribution of the chloride and ammonium levels confirmed this observation. This means that the hydrogeological vulnerability of the area is not directly reflected in current contaminant concentrations due to the distance from the source. The DRASTIC and GOD models, which do not directly consider the effect of distance, cannot fully reflect this situation. A similar situation occurred at piezometers P-1 and P-8. Ammonium concentrations at these sites were higher than at other sites, while chloride concentrations only slightly exceeded the class III water chemical status standard at piezometer P-1. Nevertheless, the DRASTIC and GOD methods indicated a low to moderate sensitivity of these areas to pollution.

4 Conclusion

The analyses indicate that, despite certain limitations inherent in the DRASTIC and GOD methods, such as their reliance on averaged parameters and subjective assessments, these approaches provide valuable insights into the spatial distribution of groundwater vulnerability to pollution.

Although neither method provides precise quantification, they effectively highlight areas where enhanced environmental monitoring would be prudent. Comparison of the contaminant concentration data with the vulnerability assessments derived from the DRASTIC and GOD methods suggests that the remediation measures implemented at the site have been effective. In particular, groundwater monitoring has shown good water quality in areas identified as highly vulnerable by both methods. This correlation demonstrates the usefulness of selected methods in validating the effectiveness of remediation strategies. In addition, groundwater quality was consistently better in piezometers located downstream of the landfill than in those located close to the landfill, further confirming the effectiveness of remediation measures.

In the context of the study area, DRASTIC and GOD methods provided a valuable complementary tool for environmental risk assessment. They have significant potential to guide the selection of suitable sites for landfills or other facilities requiring

rigorous environmental risk assessment. The vulnerability maps can support spatial planning and land use management by identifying areas where certain types of development should be restricted or carefully controlled. For example, zones classified as highly vulnerable should be excluded from activities posing a high risk of contamination, such as the construction of new facilities involving the storage of hazardous substances.

Furthermore, in the context of reclaimed landfill sites, the assessment outcomes can inform decisions regarding the need for additional protective measures. In zones where natural protection is limited, engineered solutions, such as impermeable liners, may be necessary to mitigate the risk of leachate infiltration into the aquifer.

From a policy perspective, the identification of vulnerable areas enables more efficient allocation of resources, focusing financial and administrative efforts on regions where groundwater is most at risk. Therefore, as presented in this study, the integration of parametric vulnerability assessment methods with traditional monitoring is recommended for comprehensive groundwater management and protection strategies.

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Removal of Two Azo Dyes with Different Ionic Natures by Adsorption onto a Hydrotalcite-Type Layered Double Hydroxide

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Abstract The textile industry is considered the second most polluting industry worldwide due to the huge generation of textile waste and highly polluted wastewater. This study arises from the crucial need to explore novel methods for enhancing wastewater treatment in the textile industry, specifically focusing on the removal of dyes, which are persistent compounds that pose a serious risk to the environment and human health. In this study, an Mg–Al-layered double hydroxide (LDH) was investigated as an adsorbent for two commonly used textile dyes with different ionic natures: methyl orange (anionic) and crystal violet (cationic). BET and XRD analyses were conducted to characterize the LDH. Three dosages of LDH were tested (5.0, 2.5, and 1.25 mg LDH/ml). Methyl orange showed fast adsorption in all cases; even at the lowest LDH dose, a 90% reduction was achieved within the first 10 min (1408 mg dye/g LDH·min). In contrast, crystal violet exhibited slower adsorption rates; thus, only 30% of dye removal was achieved after 20 min of exposure to 5 mg LDH/ml, and a complete color fade was not observed until 12 h. Therefore, LDH has shown itself to be a good alternative for removing anionic dyes; however, for cationic compounds, it should be combined with other technologies.

Keywords Removal · Adsorption · Hydrotalcite · Azo dyes

1 Introduction

The textile industry is currently in the spotlight because of its high water consumption, and consequently the wastewater it generates. Over the last few decades, the clothing market has grown exponentially, which has led the industry to move toward a style known as "fast fashion", manufacturing garments of generally lower quality and

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a shorter useful life, resulting in a higher rate of disposable clothing in a shorter period [1]. As a result, the industry has had to modernize its processes, leading to the increased use of synthetic fibers such as polyester, polyamide, nylon, and elastane. This shift has also driven the greater use of chemical substances, including acids, sulfates, and phenols, which are essential for transforming synthetic fibers into fabrics [2].

The main issue is not just the large amounts of wastewater generated but also the persistence and hazardous nature of the chemicals used in various production processes. This is the case of textile dyes, which pose serious health risks [3].

Given these concerns, it is crucial to develop new methods for improving the purification of wastewater from the textile industry, particularly in removing dyes [3]. Current conventional methods are costly and not efficient enough. Extensive research is being conducted to develop new treatment systems that are simpler to implement, more cost-effective, and more efficient. Some of these studies focus on novel adsorbents, such as layered double hydroxides (LDH), which offer a large surface area and high anion exchange capacity [4].

The objective of this study is to evaluate the effectiveness of Mg–Al-layered double hydroxide (LDH) as an adsorbent for removing textile dyes with different ionic charges from wastewater. Specifically, the study aims to investigate the adsorption performance of different LDH dosages for two commonly used dyes—methyl orange (anionic) and crystal violet (cationic).

2 Materials and Methods

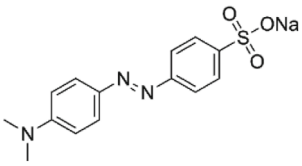
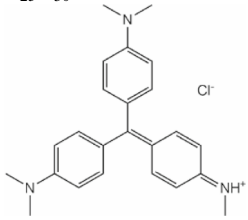
2.1 Selected Azo Dyes

The two dyes selected are Methyl Orange [5] and Crystal Violet [6], as both are primarily used in the textile industry. These dyes have significantly different chemical structures, as shown in Table 1, resulting in distinct wavelengths that can be easily identified using spectrophotometry.

2.2 Synthesis and Characterization of LDH

For the synthesis of hydrotalcite-type layered double hydroxides (LDHs), two separate solutions were prepared. The first was a 1 L solution (0.7 M) containing the interlayer anion, composed of aluminum nitrate (256.41 g) and magnesium nitrate (134.61 g). The second solution contained the metal cations and was prepared using sodium carbonate at a concentration of 1 mg/mL. The reagents used in this stage included sodium hydroxide (NaOH), magnesium nitrate [$Mg(NO_3)_2$], aluminum nitrate [$Al(NO_3)_3$], and sodium carbonate (Na_2CO_3) [7].

Table 1 Applications and chemical characteristics of the two azo dyes

	Methyl orange	Crystal violet
Uses	azo dye used in the textile, food, pulp, paper and leather industries	components of textile inks and disease treatment
Wavelength	465 nm	589–594 nm
MW (g/mol)	327.33	407.98
Structure	$C_{14}H_{14}N_3NaO_3S$	$C_{25}H_{30}ClN_3$
	$C_{14}H_{14}N_3NaO_3S$ 	$C_{25}H_{30}ClN_3$ 

To promote the formation of the characteristic three-dimensional layered structure, the cation solution was added gradually at a controlled rate of approximately 1 mL/min. During this addition, continuous agitation was maintained at 300 rpm using a magnetic stirrer in a 2 L glass beaker. The pH of the mixture was carefully regulated within the range of 9.8–10.2. After complete addition of the reagents, the resulting suspension was allowed to rest for 24 h to enable lamellar structure formation.

Following the resting period, the precipitate was collected by vacuum filtration and washed several times with distilled water until the pH of the filtrate reached 8.5, effectively removing excess hydroxide ions. The filtered solid was then dried in an oven at 100 °C for 24 h. After drying, the material was gently ground using a laboratory mortar to break up agglomerates and ensure a homogeneous powder.

A total of 49.77 g of LDH was obtained. From this, 11.07 g was subjected to calcination in a muffle furnace at 600 °C for 2 h, yielding 6.21 g of calcined hydrotalcite. This corresponds to an overall mass reduction of approximately 45%, consistent with values reported in previous studies [8].

2.3 Experimental Setup

The removal efficiency of LDH was tested on both dyes under varying LDH dosages. For the experiment using methyl orange, a 25 mL solution was prepared with a dye concentration of 20 mg/L plus 0.125 g of calcined LDH (5 mg/ml). Then, a stir bar was introduced and the flask was placed on a magnetic stirrer at 200 rpm for 30 min, 60 min, and 24 h.

Subsequently, two additional 50 mL solutions were prepared, each containing 20 mg/L of methyl orange. One solution was dosed with 0.125 g of calcined LDH (2.5 mg/mL), while the other contained 0.0625 g (1.25 mg/mL). The same stirring procedure was applied as in the previous experiment, but in this case, samples were collected every 5 min over a 20-min period to capture the adsorption kinetics in the early stages.

Similar experiments were conducted using 50 mL of crystal violet solution at a concentration of 10 mg/L. Three different LDH dosages were tested: 5 mg/mL, 2.5 mg/mL, and 1.25 mg/mL. Aliquots were collected at 20, 40, and 60 min, with a final sample taken after 12 h to monitor long-term dye degradation.

At each sampling time, the absorbance of the solution was measured to determine dye removal efficiency at the dye's characteristic maximum wavelength (see Table 1). Measurements were conducted using a UV/VIS Spectrometer Lambda XLS (PerkinElmer©). Prior to analysis, all samples were filtered using a 0.45 μm syringe filter to halt LDH activity and ensure consistent and stable absorbance readings.

3 Results and Discussion

3.1 Methyl Orange Removal

After conducting the experiments, the results showed significant variations depending on the dye being removed.

In the case of methyl orange, the performance was highly efficient, as evidenced by the low final concentration, and the short time required. In fact, this dye showed rapid adsorption on LDH, with nearly 99% removal in 30 min at higher LDH doses (5 mg/ml). Therefore, a second much more specific experiment was carried out with a maximum duration of 20 min (taking measurements at 5 min intervals) and with two different concentrations (1.25 and 2.5 mg/mL) of calcined LDH. The results obtained with the 2.5 mg LDH/mL dosage are shown in Table 2. It can be observed that an almost complete removal was attained after 20 min. When the LDH dose was reduced to 1.25 mg/mL, a 96% of color removal was obtained after 20 min.

The literature reports a broad spectrum of removal efficiencies for various types of LDH. According to Yadav & Dasgupta [9], Mg–Al LDH achieved near-complete removal of methyl orange even at low doses. Likewise, doping LDH with polyethylene glycol-400 led to a decolorization of 99%, while Mg–Fe LDH exhibited an efficiency ranging from 78 to 97% [10]. Slightly lower removal efficiencies (79%) were attained with polyvinylidene fluoride-intercalated Cu–Fe LDH membranes [11].

Figure 1 shows the adsorption rate of methyl orange over time. Since the dosages of LDH used did not reach saturation, in both cases a complete removal of the dye was achieved, with almost no differences in their performance.

Table 2 Temporal evolution of the removal and concentration of Methyl Orange treated with 2.5 mg/mL and 1.25 mg/mL of LDH

Time (min)	Absorbance (nm)		Removal (%)		Final concentration (mg/L)	
	2.5 mg/mL	1.25 mg/mL	2.5 mg/mL	1.25 mg/mL	2.5 mg/mL	1.25 mg/mL
0	1,346	1,369	0	0	20	20
5	0,211	0,441	84	68	3,13	6,83
10	0,137	0,165	90	88	1,91	2,37
15	0,120	0,107	92	93	1,63	1,40
20	0,013	0,046	99	97	0,13	0,40

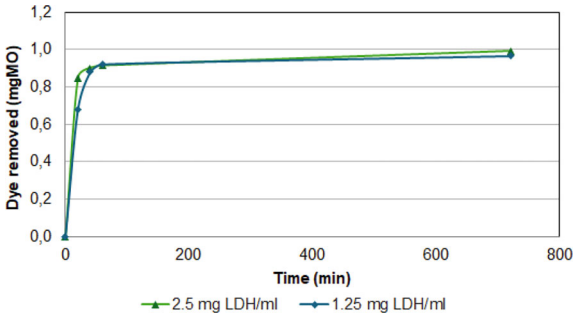


Fig. 1 Adsorption rate of methyl orange (MO) over time

3.2 Crystal Violet Removal

In the case of crystal violet, the results were not as effective as those for methyl orange. The dye removal was slower, requiring up to 12 h for 99% removal at the higher LDH dose (5 mg/mL). After 1 h, only a 30% fade was achieved. Similarly, to the case of methyl orange, decreasing the LDH dosage resulted in lower removal efficiencies. As shown in Table 3, after 12 h in contact with 2.5 mg LDH/mL, 94% of crystal violet was removed, while this percentage was reduced to 87% in the case of 1.25 mg LDH/mL.

Few studies have investigated the removal of crystal violet with LDH. Using Zn–Mg–Al LDH at dosages similar to those in this study resulted in a 70–80% removal of crystal violet [12]. In contrast, polyvinylidene fluoride-intercalated Cu–Fe LDH membranes demonstrated superior performance, achieving 100% removal of the dye [11]. Other strategies, based on advanced oxidation processes, such as UV/H_2O_2 or Fenton's reagent have shown very good efficiencies in removing crystal violet [13]. Thus, future studies could combine these strategies with LDH to find a treatment technique valid for a wide spectrum of textile dyes.

Figure 2 confirms that the adsorption rate of crystal violet is almost uninfluenced by the LDH dosage, so the slower adsorption rate in comparison with methyl orange (Fig. 1) should be caused by differences in their adsorption mechanism.

Table 3 Temporal evolution of the removal and concentration of Crystal Violet treated with 2.5 mg/mL and 1.25 mg/mL of calcined LDH

Time (min)	Absorbance (nm)		Removal (%)		Final concentration (mg/L)	
	2.5 mg/mL	1.25 mg/mL	2.5 mg/mL	1.25 mg/mL	2.5 mg/mL	1.25 mg/mL
0	2,235	2,152	0	0	10	10
20	1,738	1,725	22	20	8,24	8,17
40	1,624	1.627	27	24	7,71	7.72
60	1,458	1,498	35	30	6,94	7,12
12 h	0,121	0,271	94	87	0,72	1,42

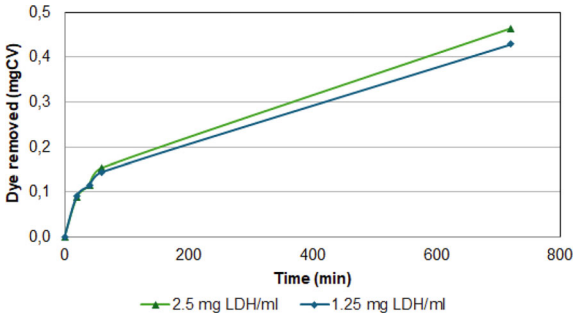


Fig. 2 Adsorption rate of crystal violet (CV) over time

4 Conclusion

For Methyl Orange, LDH proved to be highly efficient, achieving a dye removal of approximately 90% within just 10 min of application. In addition, increasing the LDH dosage only reduced the time required for dye removal without affecting the final amount eliminated.

In contrast, for crystal violet, LDH was ineffective at low exposure times, achieving only around 30% removal after 1 h. However, when the application time was extended to 12 h, the removal efficiency increased to approximately 90%.

Overall, the main conclusion of this study is that LDH has demonstrated itself to be a good alternative for removing anionic dyes such as methyl orange; while for cationic compounds, such as crystal violet, the process should be combined with other technologies to increase the removal rate and thus reduce the treatment time.

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Geological Factors Responsible for the Variations in Radon Exhalation Rates Across a Building

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Abstract Radon gas is produced from the disintegration of uranium found in various types of soils and rocks. The World Health Organization (WHO) states that inhalation of radon gas is the second leading cause of lung cancer after smoking. European and Spanish legislation establishes an average annual threshold of 300 Bq/m^3 for dwellings, public buildings, and workplaces. To implement effective mitigation strategies, it is essential to identify the main sources of radon entry into buildings. In this study, a thorough investigation was conducted in a large ground-floor office building located in northwest Spain to identify the main geological determinants and sources of radon that may explain the variability of radon concentration throughout the building. The methodology followed involved characterizing (physical, mechanical, chemical, and mineralogical) various granite samples taken from the soil and applying Ground-Penetrating Radar (GPR) techniques across the building floor. The analysis of the rocks revealed two types of granite present in the building, and clear relationships were found between indoor radon levels and geological characteristics. Furthermore, the GPR images also showed different disintegration and weathering of the soil rocks, proving the existence of two types of granites.

Keywords Radon concentration · Granite · Ground-Penetrating radar · Geological characteristics

1 Introduction

Radon (^{222}Rn) is a radioactive noble gas with a half-life of 3.8 days, produced by the decay of uranium (^{238}U) present in various soils and rocks [1]. Due to its high ionization capacity, radon, and its decay products pose significant health risks when

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accumulated in indoor environments, where people spend approximately 90% of their time. The World Health Organization (WHO) identifies radon as the second leading cause of lung cancer, and the primary cause among non-smokers, recommending indoor concentrations below 100 Bq/m^3 [2]. However, both European Directive 2013/59/EURATOM and Spanish legislation set an annual average threshold of 300 Bq/m^3 for dwellings, public buildings, and workplaces.

Indoor radon levels exhibit considerable variability across buildings, within different rooms of the same structure, and throughout the day and year [3]. Factors influencing radon accumulation include geological composition, building materials, ventilation, foundation type, and meteorological conditions [4–6]. Granite-rich areas tend to have higher radon exhalation rates due to elevated uranium and radium content than clayey or calcareous areas [4]. However, while the influence of the underlying rock type is well-established, other soil properties such as rock permeability, ground structure, topography of the terrain, soil porosity, moisture and composition, are less studied [4–6] and their impact on indoor radon levels remains unclear.

The objective of this study is to identify the primary geological factors influencing radon entry and concentration variability in a large ground-floor office building in northwest Spain. To achieve this, granite samples from the subsoil were analyzed for their physical, mechanical, chemical, and mineralogical properties, while Ground-Penetrating Radar (GPR) was used to assess subsurface conditions beneath the building. With this integrated approach, we seek to provide insights into the geological determinants of radon infiltration in buildings, aiding in the development of targeted mitigation measures.

2 Materials and Methods

2.1 Description of the Building and the Geological Area

The study focuses on a building situated in Marín, a municipality located in Galicia in the northwestern of Spain. This region of Spain is dominated by igneous rocks, mostly granitoids and granites, intruded during the Variscan orogeny [7]. Among the different types of rock substrates on which a building can be built, granite is the rock with the highest natural ^{238}U content. For this reason, as indicated by the Nuclear Safety Council's radon potential map [8], this region is among the areas in Spain with the highest radon levels (Fig. 1). Consequently, without the implementation of mitigation measures, there is a considerable likelihood that buildings in this municipality may exceed radon concentrations of 300 Bq/m^3.

The two-floor building used for this study is located less than 50 m from the coast, serves as an office space and has a rectangular footprint of 50 m × 15 m. The ground floor comprises 14 offices and 2 seminar rooms symmetrically arranged along a central corridor, ending in a larger meeting room. The first floor has a similar design, accommodating 19 offices, a meeting room, a boardroom, and restroom facilities.

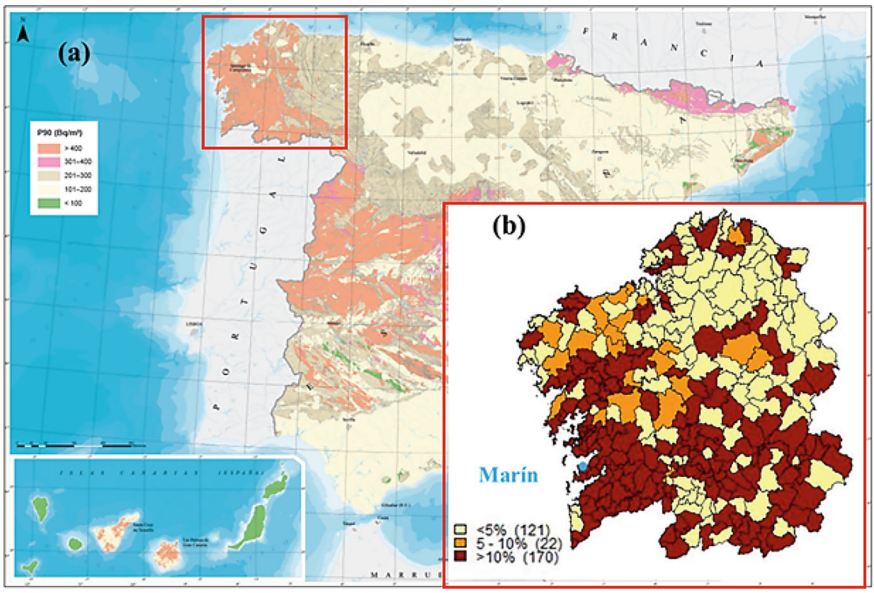

Fig. 1 (**a**) Map of radon potential in Spain and (**b**) percentage of measurements over 300 Bq/m^3 per municipality in Galicia. Adapted from [8, 9]

Windows are evenly distributed across the façades, ensuring that every office and room has at least one large window. Additionally, each floor is equipped with an independent HVAC system to maintain comfortable working conditions.

2.2 Assessment of Radon Risk in the Building Under Study. Preliminary Diagnosis.

In order to carry out a preliminary diagnosis about the risk of exposure to radon gas, the concentration of this gas was continuously monitored in various rooms of the building using twenty RadonEye RD200 active radon detectors. These devices were installed in accordance with the guidelines set by the Spanish Technical Building Code (CTE) to ensure accurate and standardized measurements. The detectors were strategically placed to achieve a uniform spatial distribution, allowing for comprehensive mapping of radon concentrations throughout the building. Data collected during August 2023 were analyzed, a period when the offices remained unoccupied and the building's ventilation system was deactivated, minimizing external influences on radon levels.

To investigate the subsoil structure beneath the building GPR data were collected along the building's longitudinal axis (central corridor of the ground floor). While

GPR is widely applied in geological studies, only a few studies used GPR to identify radon-prone zones [10]. A continuous profile, approximately 40 m in length, was acquired using a ProEx system equipped with 500 MHz and 800 MHz center-frequency antennas. An odometer wheel was attached to the rear of the antenna to accurately measure profile lengths and maintain consistent trace spacing. The data were recorded with a trace interval of 8 mm, while the total time window was set to 95 ns (656 samples per trace) for the 500 MHz antenna and 43 ns (632 samples per trace) for the 800 MHz antenna. The collected GPR signals were subsequently processed using ReflexW software to enhance data interpretation and visualization.

2.3 Characterization of Granite Samples

An initial examination of the key macroscopic features of the building's subsoil suggests the existence of two main zones with two different granites, one granite named GS1, located on the eastern side, and another named GS2, located on the western side. Based on this division, a total of 12 granite samples were gathered from soil depths of 0.5 m–1 m of both zones, at an approximate distance of 1 m from the southern façade of the building, for their physical, mechanical, chemical, and mineralogical characterization. In particular, the following analyses were conducted:

- Water accessible porosity and real and bulk densities were determined in accordance with UNE-EN 1936:2007 standard. Capillarity water absorption coefficient was determined according to UNE-EN 1925:1999 standard, and capillary porosity were determined following [11].
- Ultrasonic pulse velocity was measured using PUNDIT CNS electronics instruments and transducers of 1 MHz frequency and 11.82 mm diameter.
- Radioactivity analysis. The radioactivity levels of the two granite types, specifically for ^{238}U, ^{226}Ra, ^{232}Th, ^{40} K, and ^{137}Cs, were measured using gamma-ray spectrometry at the Laboratory of Radioactivity of the University of Cantabria, following the ISO 18589–3 standard procedures.
- Petrographic characterization was performed on thin sections by transmitted polarized light optical microscopy using a Carl Zeiss Jenapol polarized light optical microscope.

3 Results and Discussion

3.1 Variation of Radon Concentration Inside the Building

Radon measurements taken in August 2023 indicated that over 80% of the analyzed rooms on the ground floor significantly exceeded the 300 Bq/m^3 threshold set by the EU Directive 2013/59/EURATOM and Spanish regulation RD 1029/2022 (Fig. 2). As

expected, lower concentrations were observed on the first floor, since radon is rapidly diluted as it rises from its source, in this case the foundation [6]. A detailed analysis of radon concentration on the ground floor reveals a heterogeneous distribution (Fig. 2), reaching higher concentrations in the northwestern half of the floor (on average 700 Bq/m^3) than those achieved in the northeastern side (355 Bq/m^3).

Since all radon measurements were conducted simultaneously, meteorological factors are expected to have minimal influence on the recorded concentrations. Likewise, the building materials are consistent throughout, and no cracks were detected in the enclosure materials, indicating that construction factors neither account for the significant differences observed between the building's sides. Therefore, these differences in the radon level could be attributed to the existence of a fracture that facilitates the circulation of gas, to a different degree of weathering or to the existence of differences in the composition of the bedrock. All these factors have influences on the radon exhalation rates and consequently on the differences in radon concentrations.

The radargram survey conducted in the ground-floor corridor using the GPR reveals differences in the type of bedrock that can be attributed either to a different degree of weathering or to a different granite rock (Fig. 3), which confirms one of the above-mentioned hypotheses. The strong reflections on the right margin (orange lines) suggest denser or less weathered rock, whereas the weaker reflections in the left-central area indicate greater disintegration and signs of weathering (green lines).

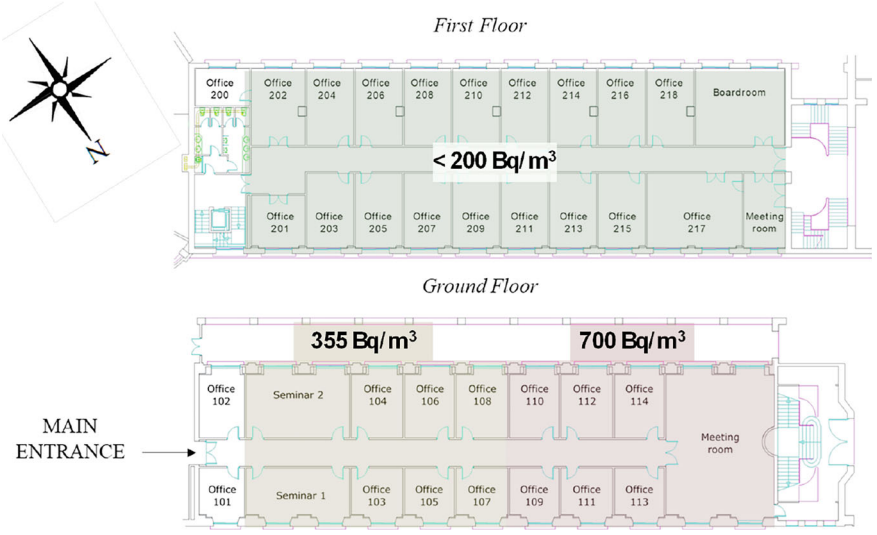

Fig. 2 Floor plan of the two-story building showing the distribution of average radon concentrations across different building areas

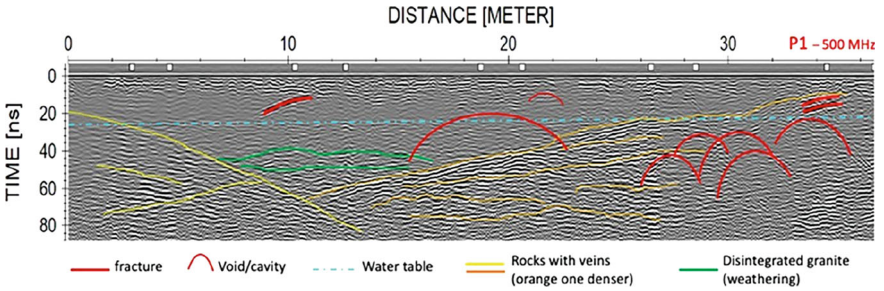

Fig. 3 Radargrams at 500 MHz generated along the building's longitudinal section. Colored lines indicate different geological elements

3.2 Granite Characterization

Table 1 shows the data on the main physical properties of the two kinds of granites found in the foundation of the building assessed in this study, namely GS1 extracted from the east side and GS2 from the west side. The data show that both granites have very similar water-accessible porosity, around 3.2%, as well as very similar real and apparent density. This fact rules out the possibility that this property influences the greater or lesser radon exhalation capacity shown by both materials and also indicates that the differences detected in the radargrams obtained by GPR are due to a different mineralogical composition.

The main difference detected refers to capillary behavior, the main mechanism of fluid transport through porous materials, as well as ultrasonic pulse transmission, which is influenced by: (1) the presence of fractures or voids in the material and (2) the proportion and composition of the minerals present in the rock.

The GS1 granite has a higher capillary absorption capacity and a higher absorption coefficient by this mechanism, which indicates that water absorption occurs more rapidly in this granite. This fact suggests that the GS1 granite has a larger pore volume within the range of capillary access between 0.1 and 100 μm. Therefore, it is

Table 1 Main physical properties of both granites

Physical property	Granite sample	
	GS1	GS2
Water accessible porosity (% v/ v)	3.21 ± 0.15	3.24 ± 0.14
Bulk density (kg/m^3)	2543 ± 9	2563 ± 8
Real density (kg/m^3)	2629 ± 12	2649 ± 9
Capillary coefficient (kg/m^2 s$^{0.5}$)	0.013	0.008
Capillary porosity (% v/ v)	1.48	1.23
Ultrasonic pulse velocity (average, m/s) [Maximum, Minimum]	3552 [3892, 2561]	3981 [4127, 2978]

expected, considering the porosity accessible to water, that much of the porosity of GS2 is due to the presence of pores and fissures with a larger size than that found in GS1. This hypothesis is supported by ultrasonic propagation measurements, which are always higher in the GS2 granite.

Regarding the radioactivity concentrations, the GS2 granite exhibited significantly higher activity levels of ^{238}U (223 ± 40 Bq/kg) and ^{226}Ra (149 ± 13 Bq/kg), nearly doubling those of GS1, which indicates that at the compositional level there must also be differences between the two stony materials. Additionally, ^{232}Th was only detected at notable levels in GS2 (35.8 ± 9.0 Bq/kg). The ^{226}Ra concentrations in both granite types were relatively high compared to samples from nearby regions [12]. While ^{238}U and ^{226}Ra activity is generally correlated with Rn exhalation [1], the emanation coefficient is also influenced by other properties of the rock, such as mineral composition [12].

Petrographically, the polarized light images shown in Fig. 4 indicate that GS1 granite is composed mainly of partially kaolinized potassium feldspar of the orthoclase type (400 × 580 μm), sodium plagioclase of the albite type, forming polysynthetic twins (280 × 300 μm) and quartz with corroded edges (400 × 350 μm). Among the accessory minerals, there is a high content of micas forming large sheets of muscovite (1.4 × 1.7 mm) and biotite (160 × 700 μm), a low proportion of tiny zircon crystals (18-57μm) developing pleochroic halos, arranged as inclusions in the micas or scattered randomly along the surface. Titanium and calcium silicates of the titanite (sphene) type also appear, occupying the intergranular space (73 × 200 μm). In some of the sodium and potassium feldspars, fracturing is locally observed. On the other hand, GS2 is a granite of brecciated appearance composed mainly of quartz, alkali feldspar of the sodium and potassium type and micaceous minerals. The feldspars develop a perthitic texture due to the intergrowth of kaolinized orthoclase and sodium plagioclase of albite type, which present a high degree of fracturing. Quartz occurs in two sizes: larger crystals (approximately 1.3 × 1.7 mm) with corroded edges and internal microcracks, and smaller mosaic crystals (25 × 25 μm). Around the quartz edges and fissures, there are micaceous mineral laminae with thicknesses between 16 and 26 μm. Locally there are zones where smaller quartz crystals are embedded between the mica layers. Zircon accumulations and fractured pyramidal crystals of titanite are identified.

Comparatively, sample GS2 has a higher zircon content than sample GS1 and contains fractured titanite. The GS2 granite shows a high degree of fissuring, which favors the migration and accumulation of zircon and increases the accumulation of radioactive elements in certain zones of the rock. There is also evidence of crushing of the material in the GS2 zone by a local cataclastic effect with the formation of quartz in small-sized mosaics. All this indicates that the western region, where the GS2 granite is located, is much more diaclased than the eastern region, which agrees with the red lines of the radargram obtained in the GPR radargram. Both minerals have a higher radioactivity than the rest of the typical minerals that make up a granite, because they have a higher uranium content. This content in the zircon is 1330 ppm and in the titanite or sphene is 280 ppm. The rest of the minerals have a lower content,

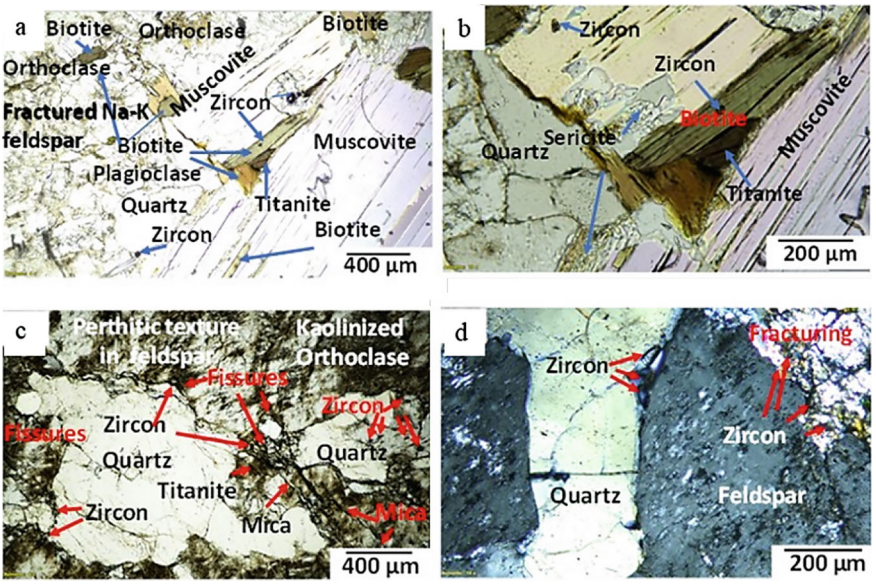

Fig. 4 Micrographs taken under petrographic microscopy of GS1 granite (**a** and **b**) and GS2 granite (**c** and **d**) taken in parallel at different magnifications. Annotations indicate mineral types and features such as fissures

with micas in the range between 8 and 12 ppm, feldspars around 2.7 ppm and quartz at 1.7 ppm [13].

4 Conclusions

The main conclusions to be drawn from this study are summarized below:

- GPR surveys offer valuable insights into a building's subsurface structure, aiding in the understanding of radon distribution, especially when non-intrusive techniques are required.
- Among the studied granites, porosity alone does not appear to significantly affect radon gas emanation, as the most porous granite (GS1) emits the least radon.
- Pore size and the presence of fractures or diaclases may influence gas migration, as seen in GS2, which has a larger pore volume and is located in a highly diaclastic region.
- The presence and degree of cracking in radioactive minerals, such as zircon and titanite, appear to be key factors in radon gas emanation.

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Image-Enabled Characterisation and Oxidation Assessment of Microplastics in Soil-Like Materials Obtained During Mining of Dumpsite

Avinash Sajwan, Deepak Kumar Haritwal, and G. V. Ramana

Abstract The widespread occurrence of microplastics (MPs) in soil-like materials (SLMs) at landfill sites presents significant geoenvironmental concerns, necessitating reliable evaluation techniques. This study employs a combination of image-based classification and chemical analysis to assess MPs concerning their dimensions, morphology, abundance and oxidation levels, thereby addressing the ecological risks posed by their degradation. Initially, manually retrieved plastic samples from SLMs were processed using image analysis to establish reference metrics. MPs were extracted via dual-density separation using NaCl and $ZnCl_2$ solutions, following organic matter removal. High-resolution images obtained with a Nikon DSLR camera were converted to 8-bit greyscale and subjected to thresholding to evaluate parameters such as size, aspect ratio, circularity, and solidity. Further oxidation level assessment was conducted through FTIR spectroscopy by computing indices such as the Carbonyl Index (CI) and Hydroxyl Index (HI). The automated image-based characterisation demonstrated an efficiency of 80–95% compared to manual analysis, thereby improving the precision and consistency of MP quantification. FTIR spectra identified polyethylene, and polypropylene as dominant polymer types, highlighting their oxidative deterioration. These results underscore the effectiveness of an integrated approach to MP assessment and its potential to mitigate environmental hazards associated with oxidised MPs in landfill settings.

Keywords Microplastics · Landfill mining · Image processing · Soil-like Material · FTIR

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1 Introduction

The rising global demand for plastic materials has driven production to approximately 390 million tonnes in 2024 [1]. However, plastics' durability and inadequate waste disposal have led to the widespread formation of microplastics (MPs). These MPs, typically ranging from 5 mm to 1 μm, are well-documented in scientific literature [2–4]. Due to their large surface area and strong affinity for contaminants, MPs can act as carriers for heavy metals and organic pollutants, posing significant risks to aquatic and terrestrial environments [5–7]. Their pervasive presence across marine, freshwater and soil ecosystems and their recent detection in humans [8, 9] has raised growing concerns about potential health implications.

MPs are traditionally classified based on physical properties such as size, shape, colour and concentration. Identification often relies on manual visual assessment, which, while effective, is labour-intensive and time-consuming [10, 11]. Advanced microscopic techniques offer automated characterisation through built-in analytical software, but high costs and maintenance limit accessibility. A practical alternative is image processing software like ImageJ/Fiji, an open-source tool widely used in biological and environmental research for image analysis, quantification and visualisation [1, 7, 12]. With features such as image segmentation and particle analysis, ImageJ enables efficient characterisation of MPs, including their size and shape. This study explores the potential of image-based classification for characterising MPs from manually collected plastic samples and applies the method to MPs extracted from soil-like material (SLM) obtained through landfill mining. The findings serve as a reference for researchers seeking to adopt image-based classification in MP characterisation.

Landfill conditions significantly influence the oxidation and degradation of plastics such as polyethylene (PE) and polypropylene (PP) [2, 13]. Oxygen levels, microbial activity and moisture content determine degradation rates. In aerobic conditions, oxidation occurs more rapidly due to atmospheric oxygen, leading to increased carbonyl formation [14, 15]. In contrast, anaerobic burial slows degradation, particularly for PE, which undergoes oxidative chain scission. While synthetic polymers like PE and PP resist microbial degradation, certain microorganisms can contribute to surface-level oxidation.

Additionally, high moisture content accelerates hydrolytic degradation, especially in PP, resulting in elevated hydroxyl and carbonyl index values. These environmental factors affect the carbonyl index (CI) and hydroxyl index (HI), influencing the long-term stability of polymeric waste. This study computes the oxidation indices of PE and PP due to their significant contribution to MPs in SLM and aims to evaluate MPs' oxidation levels using FTIR-based indices.

2 Material and Methodology

2.1 Site Description and Sample Collection

The SLM samples were obtained from the dumpsite in Delhi. Operational since 1996, this landfill was declared operationally inoperable in 2010 and permanently closed in 2018. It currently stands at a height of 55 m, exceeding its permissible limit threefold. The legacy waste, per the directives of the National Green Tribunal (NGT), was categorised into three size fractions: > 30 mm, 6 mm–30 mm and < 6 mm. The fraction below 6 mm was retrieved and subsequently sieved using a 4.75 mm sieve to obtain SLM for analysis.

2.2 Procedure for MPs Extraction and Image-Enabled Classification

To analyse the size distribution of MPs in SLM, an initial 100 g sample underwent wet sieving, followed by drying at 60 °C for 24 h. The dried sample was then fractionated using sieves with mesh sizes of 75 µm, 0.425 mm, 1 mm and 4.75 mm, separating it into four distinct particle size categories. MPs are extracted using a widely adopted methodology described in Haritwal et al. [7], Goli et al. [11] and Guo et al. [13]. The image-based classification was applied to MPs within the 4.75 mm–1 mm and 1 mm–0.425 mm size ranges. High-quality digital images of MPs were captured using a DSLR camera under optimal lighting and contrast conditions. These images were then calibrated in ImageJ using a scale bar or a known reference measurement to ensure accurate size determination of MPs. Following calibration, individual MP pieces were marked using the freehand selection tool for visual analysis and classified as film, fibre, fragment, and foam. Overlay options were used to organise these groups effectively. A threshold image was required for automated MP counting. The images were first converted to 8-bit grayscale, followed by image thresholding, with threshold values between 0 and 255. After segmentation, particle analysis was conducted to measure key parameters, including size, shape, aspect ratio, solidity and roundness. The extracted data was then analysed using statistical methods to derive meaningful insights.

2.3 Oxidation Level Assessment of MPs in SLM Using FTIR

Polypropylene (PP) and Polyethylene (PE) are widely used thermoplastic polymers that undergo oxidative degradation when exposed to environmental factors such as ultraviolet (UV) radiation, oxygen and moisture. The extent of degradation is

quantified using the Carbonyl Index (CI) and Hydroxyl Index (HI), both derived from Fourier Transform Infrared (FTIR) spectroscopy.

The CI measures the formation of carbonyl-containing degradation products (e.g. ketones, aldehydes, carboxylic acids), while the HI indicates the presence of hydroperoxides and hydroxyl groups, which are intermediates in polymer oxidation. These indices are calculated using Eq. 1 [16]. FTIR data for PE and PP samples were analysed, and CI and HI values were determined through numerical integration of absorbance within specific wavenumber ranges. The results were compared to reference values from fresh polymer samples documented in established literature.

$$CI = \frac{A_{C=O}}{A_{CH2}}; HI = \frac{A_{OH}}{A_{CH2}} \quad (1)$$

where $A_{C=O}$ is the integrated absorbance (area under the curve) within the 1600–1780 cm-1 range (carbonyl region), A_{CH2} is the integrated absorbance in the 1420–1500 cm-1 region (methylene bending reference peak) and A_{OH} is the integrated absorbance in the 3060–3600 cm-1 range (hydroxyl stretching region).

3 Results and Discussion

3.1 Image-Enabled Characterisation of Manually Collected Plastic Pieces

Plastic pieces were collected directly from the soil and various plastic products, then stored in a petri dish. High-resolution images were captured and calibrated using a scale bar of known measurement. Individual plastic pieces were marked and colour-coded as follows: fibre (green), film (red), fragment (blue) and foam (black), as shown in Fig. 1a. The image was converted to 8-bit greyscale to enable automated counting and then processed into a threshold image. However, white and transparent plastic pieces were not detected in 8-bit final threshold images, as shown in Fig. 1b and c.

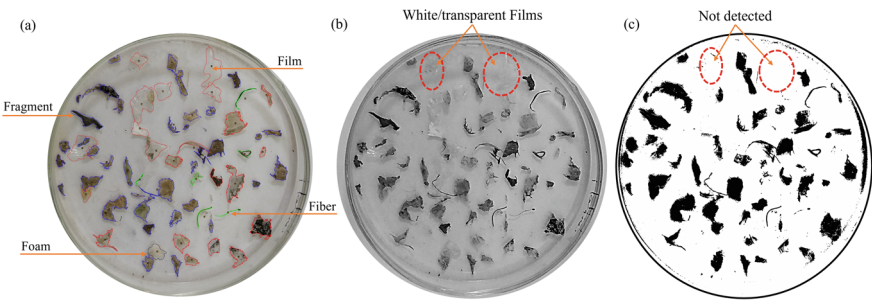

Fig. 1 **a** Visualisation of MPs based on **a** shapes, **b** 8-bit image and **c** threshold image

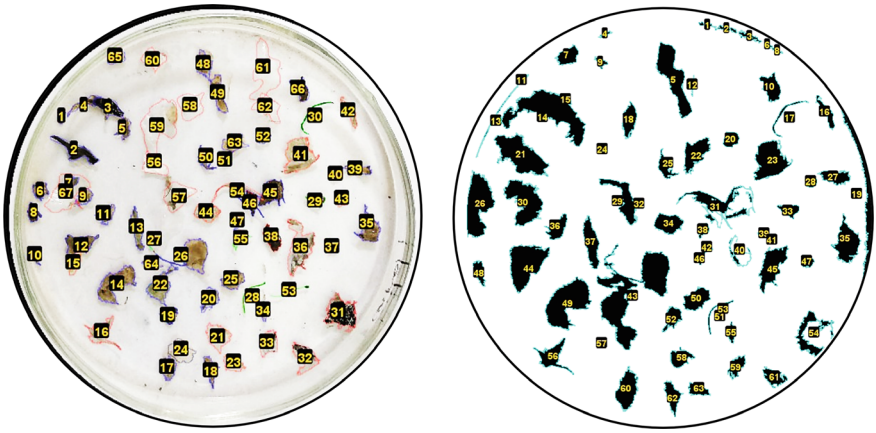

Fig. 2 MPs using manual counting (67 pieces) and automated counting (63 pieces)

Manual counting identified 67 discrete MP pieces, whereas automated counting identified 63 discrete MP pieces, achieving an overall efficiency of 94%. Figure 2 presents a comparison between image-enabled manual and automated counting. It is important to note that some MP pieces positioned closely together in the raw image were merged into single entities during thresholding, affecting the count. Additionally, white and transparent MP films were undetectable against the white background during 8-bit conversion, leading to their exclusion from the automated count.

3.2 SLM Obtained from Landfill Mining

The MPs extracted from SLM (1000 μm–4750 μm) were categorised by shape and colour-coded as follows: fibre (green), fragment (blue), film (red) and foam (black). Figure 3a shows the raw image, Fig. 3b presents the marked MPs by shape and Fig. 3c displays the processed image. Visual analysis identified 16 discrete MP pieces, while automated counting detected 11, achieving an efficiency of 68.75%. Excluding white or transparent MPs from automated counting could improve efficiency by approximately 80%. Additional geometric parameters, including aspect ratio, roundness, sphericity and solidity, were measured for each MP, summarising results in Table 1. Notably, fibre-shaped MPs exhibited a higher aspect ratio than other shapes. When combined with larger datasets, these geometric insights can further enhance MP classification. This similar technique was further applied to SLM within the 425 μm–1000 μm size range (Fig. 4). Visual analysis identified 83 discrete MPs, whereas automated counting detected 76, achieving an efficiency of 92%.

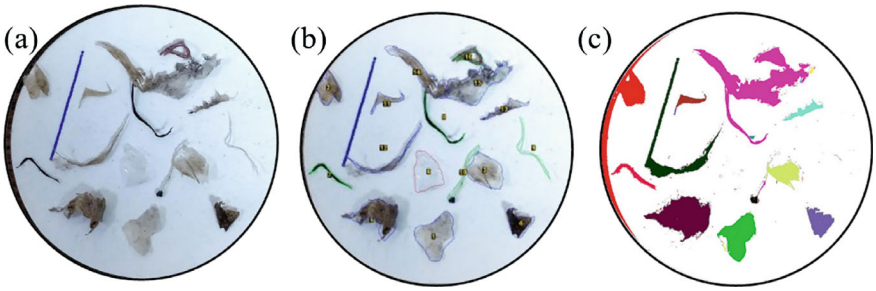

Fig. 3 **a** Raw image, **b** classification of MPs based on shapes, c processed image for automated counting of MPs extracted from SLM (1000 μm–4750 μm)

Table 1 Geometric features of individual MPs for size range (1000 μm–4750 μm)

Sr.no	Area	Mean	Circ	AR	Round	Solidity	Shape
1	39.338	158.409	0.695	1.351	0.74	0.884	Fragment
2	16.885	123.744	0.518	1.308	0.764	0.851	Fragment
3	29.899	167.211	0.503	1.374	0.728	0.951	Fragment
4	13.704	70.295	0.573	1.219	0.821	0.789	Fragment
5	30.833	91.707	0.435	1.436	0.696	0.713	Fragment
6	28.786	195.606	0.754	1.39	0.72	0.935	Film
7	5.111	81.57	0.154	4.136	0.242	0.423	Fibre
8	6.917	79.929	0.092	3.642	0.275	0.202	Fibre
9	5.776	211.454	0.08	3.248	0.308	0.254	Fibre
10	7.77	179.014	0.164	10.047	0.1	0.57	Fibre
11	5.827	147.86	0.194	1.611	0.383	0.435	Fragment
12	8.784	132.845	0.072	1.082	0.325	0.16	Fragment
13	7.684	145.42	0.195	1.867	0.205	0.591	Fragment
14	25.578	136.383	0.232	1.633	0.178	0.657	Fragment
15	44.834	145.109	0.367	1.422	0.413	0.773	Fragment
16	5.819	113.402	0.128	3.752	0.571	0.524	Fibre

3.3 Polymer Composition and Oxidation Assessment of MPs from SLM

The FTIR spectra was obtained using attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR). To identify the composition of the samples, the obtained spectra were compared with reference databases, including the “Specta Polymers and Plasticizers” by ATR and the “Comprehensive Forensic FT-IR Collection”.

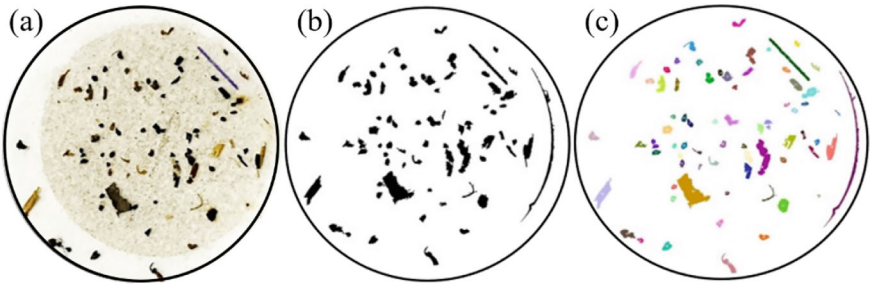

Fig. 4 **a**Raw image, **b** threshold image, and **c** processed image for automated counting of MPs extracted from SLM (425 μm–1000 μm)

The analysis confirmed that the predominant MPs were polyethylene (PE) and polypropylene (PP). Characteristic absorption bands for PP were observed at 2953–2845 cm^{-1}, 1459 cm^{-1}, 1377 cm^{-1}, 1156 cm^{-1} and 971 cm^{-1}, corresponding to its main chain structure (Fig. 5a). Similarly, PE exhibited distinctive absorption peaks at 2915 cm^{-1}, 2846.5 cm^{-1}, 1468 cm^{-1} and 716 cm^{-1}, associated with C–H stretching and bending vibrations (Fig. 5b). These findings confirm that the excavated plastic waste samples primarily consisted of PE and PP polymers.

The CI value for PP (1.50) is significantly higher than that of fresh PP (0.5–1.2) [15], indicating substantial carbonyl formation due to prolonged oxidation. The elevated HI value (4.67) further reflects extensive hydroperoxide accumulation and moisture-driven degradation. PP continuously increases CI and HI over time as carbonyl groups accumulate without significant volatilisation. For PE, the CI value (0.96) exceeds that of fresh PE (0.1–1.1) [15], while the HI value (2.56) indicates hydroperoxide presence. Initially, PE follows an increasing CI trend, but it declines due to chain scission and volatilisation of carbonyl compounds. The higher-than-expected CI suggests that the sample is in a mid-stage degradation phase, where oxidation remains active.

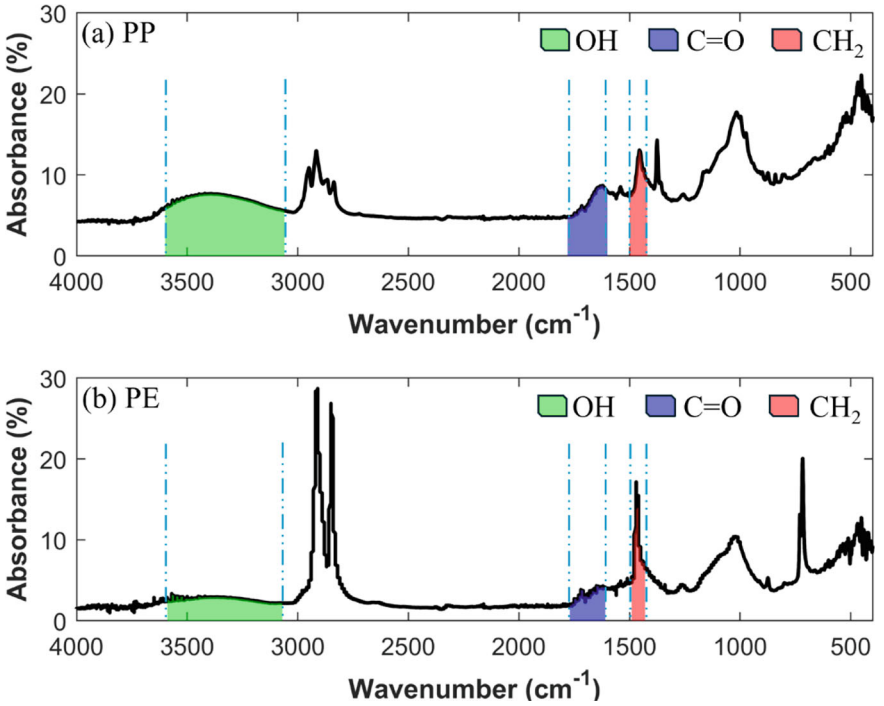

Fig. 5 FTIR spectra of MPs from the SLM identifying **a** Polypropylene and **b** Polyethylene

4 Conclusion

This study confirms that image-based classification is an efficient and practical method for characterising MPs. It provides detailed insights into MP size, morphology and distribution while offering additional geometric parameters such as aspect ratio and circularity. Automated particle counting achieved 80–95% accuracy compared to manual methods, demonstrating its reliability for large-scale applications. However, limitations persist, as white or transparent MPs are not easily detected in threshold images, potentially leading to underestimation in automated assessments.

Additionally, landfill environments significantly influence the degradation of PE and PP. Under anaerobic conditions, PE degrades more slowly due to limited oxygen availability, while PP undergoes progressive hydrolytic degradation due to moisture exposure. FTIR analysis revealed substantial oxidative changes, with PP exhibiting continuous carbonyl accumulation and PE displaying unexpectedly high CI values, suggesting ongoing oxidation. These findings contribute valuable insights into polymer degradation and landfill waste management. Further research incorporating longitudinal ageing experiments and environmental exposure simulations is

recommended to refine our understanding of polyethylene degradation pathways and confirm long-term oxidation trends.

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Influence of Electrolyte Concentration on Electroosmotic Permeability of Variable Charge Soil

Yang Min, Yang Xiaojuan, Sun Honglei, Ángel Yustres, and Rubén López-Vizcaíno

Abstract Electroosmosis is an efficient process that can facilitate the dewatering of high-water content clays and the decontamination of soils. Addition of background electrolyte is commonly used particularly in electrokinetic remediation. However, some experiments indicate that electroosmotic drainage rates increase in electrolyte solutions with higher concentration, while others suggest that there is an optimal concentration, below or above which the dewatering flow gets slower. Besides, it is also found that the electroosmotic permeability can be taken as independent of electrolyte concentration. The classical Helmholtz–Smoluchowski model fails to explicit the relationship of electroosmotic permeability coefficient and electrolyte concentration. The recent ion migration model interpreted the optimal salt content of electroosmotic permeability, but in a qualitative way. To further clarify the influence of electrolyte concentration, electroosmotic tests were conducted with variable charge soil in series of initial NaCl concentrations. The results showed that the electroosmotic permeability reached a maximum value at NaCl concentration of 0.12 mol/L. The authors tried to substitute the cation current density and cation electro-migration with the apparent electric current and total electro-migration flux of cations

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and anions in ion migration model, for the purpose of easier accessibility of parameters required. It turned out that the calculated electroosmotic permeability coefficient was in better agreement with the experimental values.

Keywords Electroosmotic permeability · Electrolyte concentration · Variable charge soil · Ion migration

1 Introduction

Electroosmosis is a phenomenon that utilizes an applied electric field to drive the migration of water and ions in soil. During electrokinetic restoration, background electrolytes are usually added to improve the electroosmotic efficiency. However, there is no consistent conclusion on whether the electrolyte can enhance the electroosmotic efficiency and how the electrolyte concentration affects the electroosmotic efficiency. Shang [1] and Gargano [2] concluded that the electroosmotic coefficient is maximum when the NaCl concentration is 8 g/L. Zhou [3] found that electroosmotic drainage was optimal when the background solution conductivity was 0.201 S/m. However, Citeau [4] obtained faster water removal at lower electrolyte concentrations of about 0.001 M concentration. Tao [5] stated that higher salinity leads to a lower rate of dehydration. Therefore, the electroosmotic behavior in response to electrolyte concentration are unclear.

The conventional Helmholtz–Smoluchowski electroosmotic model is widely used due to its wide applicability, and the nonlinear prediction of the electroosmotic coefficient relies on the relationship between parameters such as pore fluid pH and zeta potential. Yustres [6] proposed an ionic hydration electroosmosis model based on the Donnan equilibrium, which indicated that the electroosmosis coefficient is related to the total pore ratio of the soil body and the specific surface area. Zhou [7] proposed an electroosmotic model based on cation migration by using the capillary theory in the H–S model in combination with the Nernst–Planck equation, which was able to better reflect the correlation between electroosmotic drainage and the electric and chemical fields.

The electroosmotic behavior and surface electrochemical properties of variable charge clays distributed in tropical and subtropical climatic zones vary are yet to be identified. Therefore, this paper intends to explore the electroosmotic behavior of variable charge under different NaCl concentrations, and compare with predicted results based on modified the total ion migration electroosmosis model.

Table 1 Physicochemical properties of soil samples

Specific gravity	Hydraulic conductivity(cm/s)	Liquid limit (%)	Plastic limit (%)	Plasticity index	pH
2.61	7.722×10^{-6}	44.6	17.1	27.5	6.2

Table 2 Grain size distribution

Particle size	Yunnan soil (%)
Clay (<0.002mm)	11.79
Silt (0.002–0.05mm)	55.05
Sand (0.05–2mm)	33.15

2 Materials and Methods

2.1 Soil Materials

The variable charge soils are from Yunnan Province, China, where the soil is located in an area with a predominantly tropical climate, and rich in iron oxides and dominated by variable charge clay minerals. The soil is dominated by silt with a water content of 25.7%, and the basic physical properties are shown in Tables 1 and 2.

2.2 Experimental Setup

The one-dimensional electroosmosis experimental setup used in this study is shown in Fig. 1, a DC power supply model SPPS-C305 was used in the experiments, and its output voltage was set to 30 V. Probes were set at 1 cm from the cathode and anode electrode plates for measuring the effective voltage of the soil. In addition, the current and voltage data were monitored in real time by means of sensors, and these data were recorded and analyzed using a computerized data acquisition system.

2.3 Test Procedure

The soil used in the experiment was initially dried, crushed into fine particles and passed through a 0.25 mm sieve. NaCl was chosen as the electrolyte for the experiment, and the soil was mixed with deionized water and the corresponding concentration of electrolyte to prepare a soil–water mixture with 70% water content. The mixture was left to stand for 12 h. Subsequently, the mixture was filled into the soil chamber in 4 layers and compacted as it was filled to eliminate macropores in the soil and to provide a uniform pore distribution. A potential gradient of 1 V/cm was used

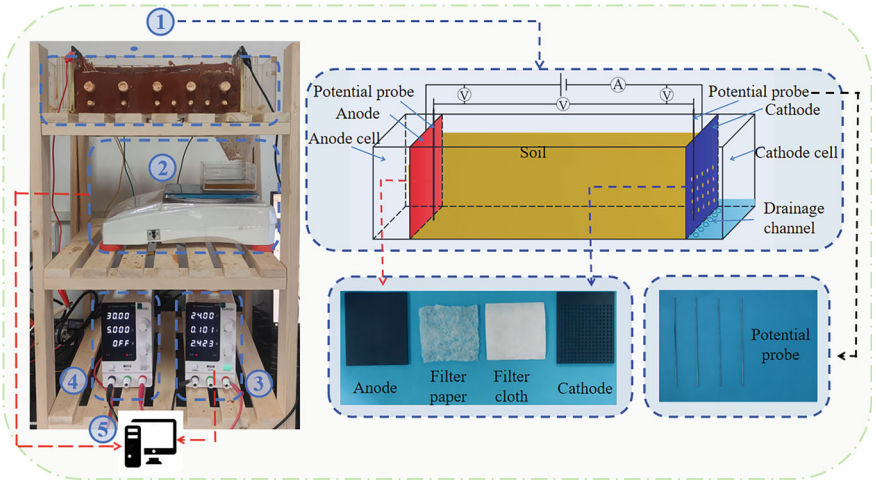

Fig. 1 One-dimensional electroosmotic experimental device. (1) Test box; (2) Drainage collector; (3) Data acquisition sensor power supply; (4) DC power supply; (5) Computer and data acquisition software

for all experiments. Seven sets of electrolyte solutions with different concentrations (0.06, 0.08, 0.1, 0.12, 0.15, 0.18, and 0.2 mol/L) were designed. The soil chamber was divided into three zones: anode, middle, and cathode. Changes in conductivity, pH, and zeta potential were measured during the experiment; pH was measured in the ratio of 1:2.5 [8], and electroosmosis was considered to terminate once the cathode drainage rate was less than 1 ml/h.

2.4 Formulation of Total Ion Flux Migration Model

Ion migration electroosmosis model proposed by Zhou [7] is based on the capillary theory in the classical H–S electroosmosis model assuming that the pore liquid current (I_s) is mainly carried by the cation migration flux, thus the pore liquid current can be regarded as the cation current (I_c). The cation migration flux is applied to calculate the permeability coefficient as seen in Eq. 1.

$$K_{ec} = \frac{\sigma \delta}{c_i(x) u_i^* \mu} \frac{1}{F \sum z_i} \frac{I_c}{A \cdot E_f} \tag{1}$$

However, the I_c is difficult to be obtained. Therefore, in the present study the apparent current calculation based on the total anion and cation fluxes is proposed.

First, the experimentally measured apparent current I of the soil is caused by the migration of anionic and cathodic charged ions in the pore fluid [9–11], the electrical conductivity of the soil particles is neglected. The relationship between the apparent

current and the anion and cation fluxes is built as follows neglecting the ion fluxes carried by diffusion and convection:

$$(c_i^+(x)u_i^{+*} + c_i^-(x)u_i^{-*})\frac{v\mu}{\sigma\delta} = \frac{I}{F\sum z_i} \tag{2}$$

In addition, pH and the electrolyte concentration affects the surface charge density of the variable charge soil. Herein, the calculation method proposed in Yang [12] based on the Stern–Gouy double model in the study of the zeta potential of variable charge soil is applied.

In summary, the equation for calculating the electroosmotic coefficient based on the total ion mobilization flux in this paper can be obtained:

$$K_e = \frac{\sigma\delta}{(c_i^+(x)u_i^{+*} + c_i^-(x)u_i^{-*})u_i^*\mu}\frac{1}{F\sum z_i}\frac{I}{A\cdot E_f} \tag{3}$$

where σ is the surface charge density of soil particles; δ is the thickness of the sliding layer in the bilayer, according to the Stern–Gouy model of the bilayer, this paper adopts 0.6 nm; $c_i^+(x)$ is the cation concentration in the pore fluid, u_i^{+*} is the effective migration rate of cations, $c_i^+(x)$ is the anion concentration in the pore fluid, u_i^{-*} is the effective migration rate of anions; μ is the viscosity of the pore water, at room temperature for the 1×10^{-3}Ns/m^2; F is the Faraday constant of 96485C/mol; z_i is the number of charges carried by ions; E_f is the effective potential gradient at both ends of the soil body, V/cm; A is the cross-sectional area of the soil body through the current, and A is a constant value of 0.0064m^2 in this experiment.

In this study, u_i^{+*} is the Na^+ effective migration rate, calculated as

$$u_i^{+*} = \frac{D_i^{+*}}{RT} \tag{4}$$

$$D_i^{+*} = \phi S_r \tau D_i^+ \tag{5}$$

where D_i^{+*} is the diffusion coefficient of cations in the soil; ϕ is the porosity of the soil, calculated as 0.652; S_r is the degree of saturation, which is calculated as 1; D_i^+ is the diffusion coefficient of cations in dilute solution; τ is the tortuosity of the soil, which adopted the same value as porosity from Liu [13]; u_i^{-*} is the effective Cl^- migration rate.

3 Results and Discussion

3.1 Electroosmotic Behavior on Various Salinity

Seven groups of electroosmotic experiments basically ended at about 50 h. The cumulative flow of electroosmotic drainage over 48 h was plotted in Fig. 2. The total amount of drainage from EK1 to EK7 for 48 h was 219.85 ml, 249.4 ml, 311.96 ml, 352.94 ml, 301.5 ml, 335.7 ml, and 258.87 ml, respectively. The total amount of drainage from EK1 to EK4 increased with the increase of electrolyte concentration, and the amount of drainage in EK5, EK6, and EK7 was lower than that of EK4. The maximum drainage in EK4 was 60.54% higher than the minimum drainage in EK1, indicating that electrolyte concentration has significant effects on electroosmotic dewatering. Besides, the proportion of the drainage volume within the first 24 h to the total drainage volume from EK1 to EK7 was 72.78%, 70.44%, 87.28%, 91.76%, 85.54%, 84.98%, and 78.71%, respectively. From EK1 to EK4, the percentage of drainage in the initial 24 h increases with the increase of electrolyte concentration; while it is relatively lower than that of EK4 for EK5, EK6, and EK7. It shows that a higher electrolyte concentration can promote the electroosmotic drainage process. However, its effect on the drainage is weakened after reaching a potential optimal concentration.

The variation curves of electroosmotic rate are decreasing gradually with time as shown in Fig. 3. The average rate of EK1-EK7 in the first 24 h was 2.68, 3.86, 6.83, 11.15, 5.91, 5.66, and 3.69 times of that during the last 24 h, respectively. The EK4 group had the highest drainage rate of 13.49 ml/h during the first 24 h, showing the greatest difference from the drainage rate of 1.21 ml/h during the second stage of 24 h.

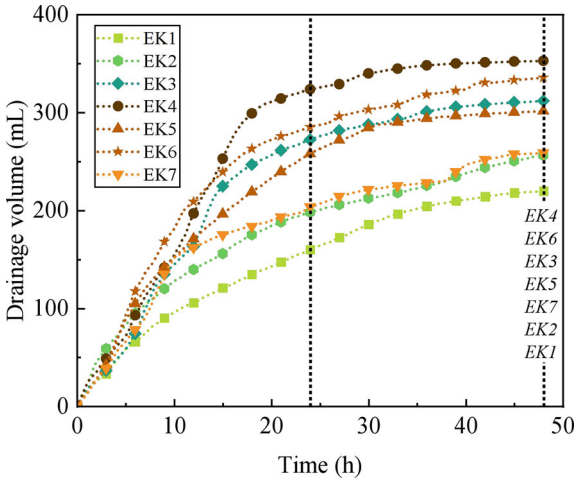

Fig. 2 Variation of drainage volume over time

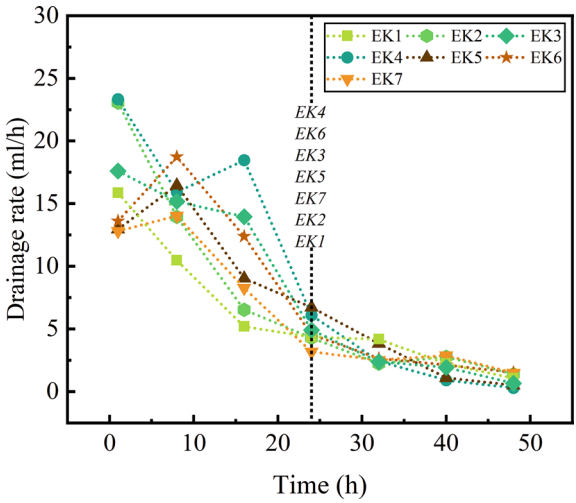

Fig. 3 Variation of drainage rate over time

The electric current was declining continuously during the electroosmotic process as seen in Fig. 4a. A positive correlation was observed between initial current values and salinity. Linear regression analysis of current decay rates and electrolyte concentrations during initial 24 h revealed a positive correlation in groups EK1–EK4 (R^2 = 99.17), attributed to linear increase in the number of mobile ions with electrolyte concentration under low-concentration conditions. In contrast, nonlinear behavior in current decay rates was observed in high-concentration groups EK5–EK6 (R^2 = 53.56), likely due to enhanced inter-ionic interactions at higher concentrations.

The electric current in EK5–-EK7 exhibited an atypical trend with an initial rise followed by a decline within the first 3 h (Fig. 5). The anode pH decreased rapidly to 2.5 ± 0.2 within 3 h and stabilized, while the cathode pH increased to 11.5 ± 0.3

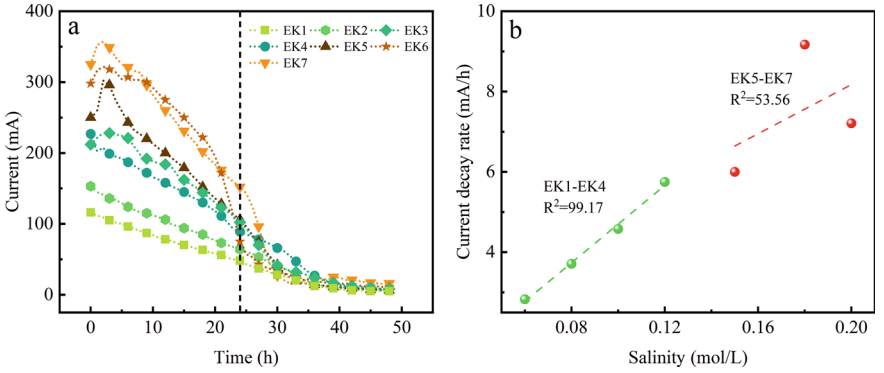

Fig. 4 The behavior of the electrical current **a** Current evolution **b** Fitted relationship between current decay rate and salinity

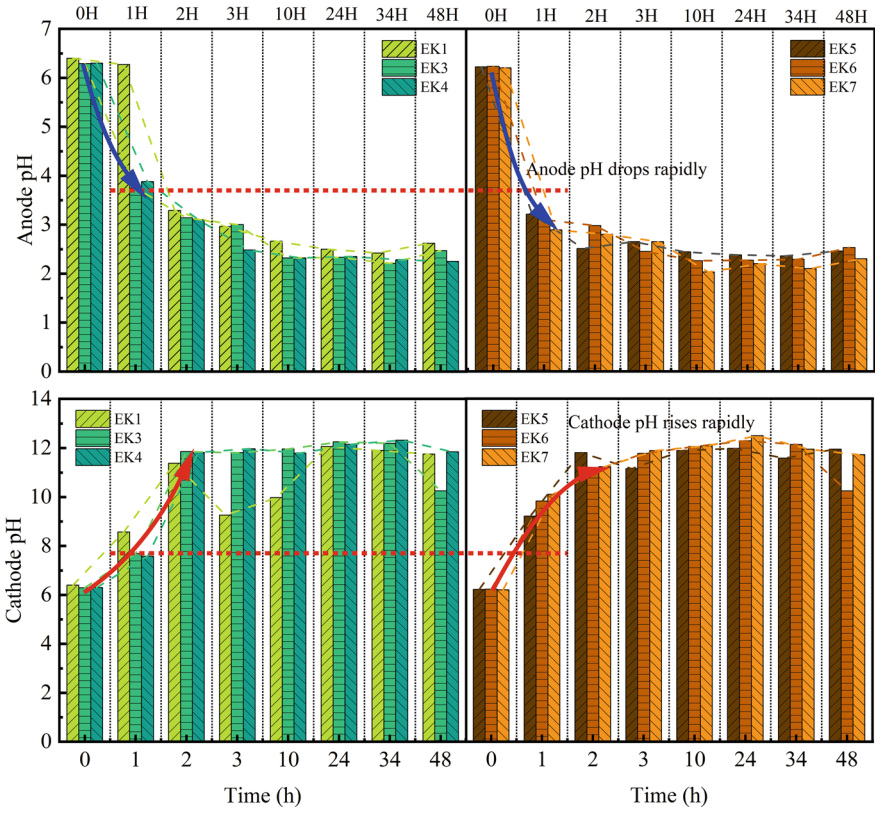

Fig. 5 The pH variation of the soil sample near the anode and cathode

as shown in Fig. 5. The pH variation rates in EK5–EK7 were significantly higher than those in EK1–EK4, driven by increased fluxes of H^+ and OH^- ions generated through water electrolysis at higher salinities. This phenomenon accelerated pH polarization at both electrodes, inducing transient current elevation. Conversely, low-salinity groups maintained a monotonic current decline, resulting from the synergistic effects of sustained ion migration and progressive soil matrix resistance.

3.2 Calculation of Electroosmotic Permeability Coefficient

According to the calculation method of the electroosmotic coefficient presented in Sect. 2.4, the experimental values of the electroosmotic coefficient (K_{eo}), the cation flux electroosmotic coefficient (K_{ec}), and the total flux electroosmotic coefficient (Ke) were determined at 48 h using Eqs. 1 and 3, respectively. The surface charge density of soil particles at each electrolyte concentration was calculated based on the

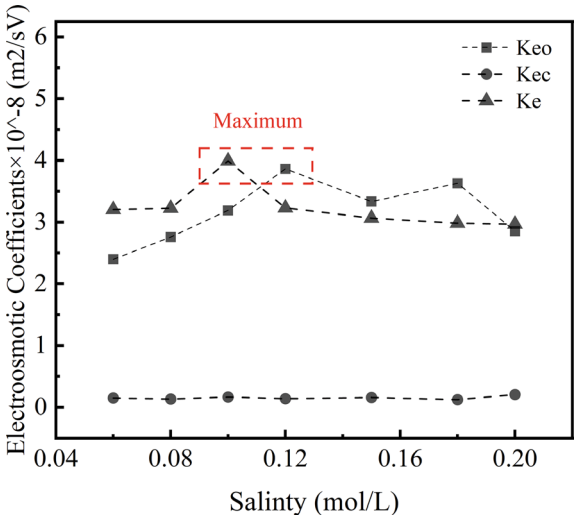

Fig. 6 Comparison of electroosmotic coefficients

method described by Yang [12] in Sect. 2.4. In the calculation of K_{ec}, the proportion of cation current (I_c) to solution current (I_s), denoted as $\alpha = 0.016$ [7], where $Ic = \alpha I_s \approx \alpha I$. The comparisons of these results are presented in Fig. 6.

An optimal value of the experimental electroosmotic coefficient (K_{eo}) exists also in EK4: at concentrations lower than 0.12 mol/L, K_{eo} increases with the increase of electrolyte concentration; however, when the concentration exceeds 0.12 mol/L, K_{eo}decreases with further increases in electrolyte concentration. The electroosmotic coefficient (K_e) calculated using the total flux model is closer to the average value of experimentally measured K_{eo}, and an optimal concentration of 0.1 mol/L was identified. The electroosmotic coefficient (K_{ec}) calculated using Eq. 1 [7] showed discrepancy from the experimental K_{eo} values and was not unaffected by changes in electrolyte concentration. This might be caused by the inaccurate estimation for the cation current (I_c) and the neglection of the variation in surface charge density of variable charge soil particles with the concentration of the background solution.

3.3 Interpretation on Optimal Electrolyte Concentration

The ion migration rate is considered to be constant, and the ratio of current to effective voltage represents the conductivity of the soil–water system. Therefore, the electroosmotic coefficient is related to the ratio of soil conductivity to ion concentration:

$$K_e \propto \frac{\sigma_{soil}}{c_i(x)} \tag{6}$$

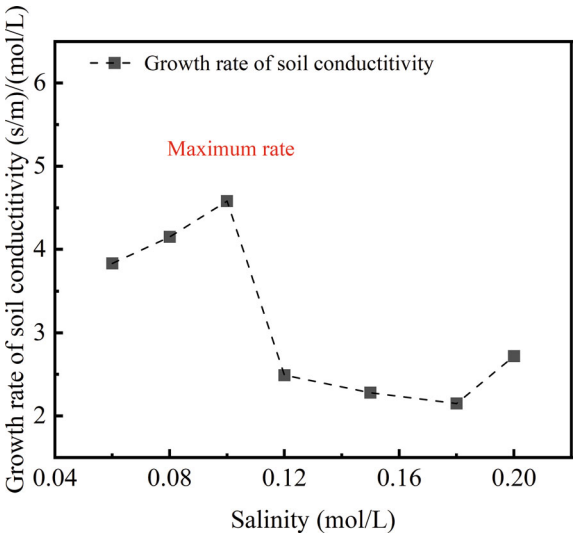

Fig. 7 Growth rate of EC with salinity

The variation rate of electric conductivity with respect to electrolyte concentration is then calculated based on the salinity of the background solution, as shown in Fig. 7. The growth rate of soil conductivity increased with the increase of electrolyte concentration before reaching 0.1 mol/L. However, the growth rate of soil conductivity decreased with further increase in concentration exceeding 0.1 mol/L. This resulted in the maximum value of the electroosmotic coefficient calculated by the total flux ion migration model at 0.1 mol/L. This may be due to the significant enhancement of ion interactions in the solution when the concentration exceeded 0.1 mol/L, which led to a slowdown in the increase of solution conductivity. Consequently, the growth rate of soil conductivity also decreased. The results could provide guidance on salinity selection of electrolyte solution for filed applications.

4 Conclusions

The experiments demonstrated that electrolyte concentration significantly influenced the electroosmotic effect. The 48-h electroosmotic drainage volume and rate was highest at a concentration of 0.12 mol/L. The volume was 60.54% higher and the rate was 1.6 times higher than the minimum group.

The calculated values of the total ion flux electroosmotic coefficients showed good agreement with the experimental values, applying the apparent current to be obtained more easily. This model is capable of predicting the optimal concentration of the background solution, and interpreted the electrochemical mechanism of optimal electrolyte concentration.

Surface charge density affected by pH and water content can also alter the relationship between the apparent current of the soil and current carried by the pore solution. Therefore, the ability to calculate the electroosmotic coefficient of the total ion flux electroosmosis model under different initial pH and water content conditions requires further investigation.

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Contaminant Transport Through Sand–Bentonite Composite Landfill Liner: Effect of Sand Gradation

R. Mahender Reddy and Bande Giridhar Rajesh

Abstract Engineered landfills are designed to protect the soil and groundwater by containing the municipal solid waste disposal sites that prioritize environmental protection. These landfills typically contain the soil and groundwater by containing the municipal solid waste which decomposes and generates leachate. To mitigate these risks, low-permeability landfill liners act as barriers to prevent contaminant transport. However, a few studies have focused on simulating contaminant transport to evaluate the thickness of liner. Therefore, this study investigates the effectiveness of sand–bentonite mixtures as liner materials by simulating contaminant transport using a one-dimensional finite difference method. The model incorporates the coupled process of advection–dispersion–adsorption–diffusion. Furthermore, the study examines the effect of sand gradation (fine sand, medium sand, and coarse sand), and composite sand–bentonite liner thickness on the leaching of contaminants such as chloride, zinc, and lead. The results indicate that reducing the bentonite content from 30 to 10% increases the permeability (3.14 X 10^{-11} m/s) by 51%, 73%, and 98% for fine, medium, and coarse sand–bentonite mixtures, respectively, leading to higher contaminant concentration at the exit of liner. Conversely, increasing bentonite content enhances contaminant retention by lowering the permeability and increasing cohesion by 1.66, 3.39, and 5.95 times for fine, medium and coarse sand respectively. These findings highlight the importance of optimizing sand–bentonite mixtures to improve the effectiveness of landfill liners.

Keywords Contaminant transport · Sand gradation · Composite Sand–Bentonite liners · Finite difference method · Partial distribution coefficients

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1 Introduction

Landfill liners are low-permeability barriers that prevent leachate from contaminating soil and groundwater. The primary characteristic influencing the effectiveness of the liners is their low permeability (1.0×10^{-9} m/s). Common liner materials include compacted clay liners, sand–bentonite mixtures, and geosynthetic clay liners. Bentonite clay is widely used due to its low permeability, high swell potential, and adsorption capacity, but its susceptibility to desiccation cracking and low shear strength necessitates mixing with sand for improved performance. When hydrated, sand–bentonite mixtures swell, reducing permeability by filling pore spaces and enhancing their function as engineered barriers.

Several studies have investigated optimal sand–bentonite mix proportions to achieve a required permeability of less than 1.0×10^{-9} m/s. Mollins [2], Hoeks et al. [3] and Gueddouda et al. [4] found that 5%, 5–15%, and 12% bentonite in the sand meets the hydraulic conductivity requirement. These studies primarily considered sand as a filler material for its volumetric stability, but did not account for the effect of sand gradation. Further, the influence of sand and silt mixtures on liquid limit and shrinkage limit [5, 6]. The authors also concluded that fine sand mixtures exhibit a higher liquid limit than medium sand mixtures for same bentonite content. Additionally, Srikanth and Mishra investigated the impact of sand particle size on the hydraulic conductivity of the sand–bentonite mixtures through oedometer tests. Their findings reveal that bentonite content below 20% is insufficient to fill the voids, regardless of sand particle size.

To effectively assess the performance of landfill liners and to evaluate thickness, numerical simulations of contaminant transport through liner system are conducted. Chen et al. [7] developed a 1D analytical solution for contaminant diffusion through multilayered media and concluded that untreated clay liners require a thickness of 2.9m–5m, whereas HDTMA treated clay liners require on 0.6m for same level of protection. Further, Xie et al. [8, 9] and Feng et al. [10] developed an analytical model to evaluate the performance of a triple liner system comprising of geomembrane, geosynthetic clay liner, and attenuation layer.

In this study, an attempt has been made to evaluate the feasibility of fine, medium, and coarse sand mixtures with bentonite for liner applications. The objectives include determining the optimal sand–bentonite mix proportion based on sand gradation and assessing the required liner thickness under various contaminants. A combined experimental and numerical modeling approach was adopted. Laboratory tests were conducted to assess the hydraulic conductivity and the shear strength parameters, while numerical simulations were performed to evaluate the thickness of the liner for major contaminants such as chloride, zinc, and lead.

Table 1 Properties of bentonite and sand used in this study

Property	Value	Property	Value
Bentonite properties		Sand properties	
Clay	66.1%	D_{60}	0.91 mm
Silt	33.9%	D_{30}	0.44 mm
Sand	0	D_{10}	0.248 mm
Liquid limit	350.2%	C_u	3.67
Plastic limit	45.1%	C_c	0.857
Shrinkage limit	12.7%	Specific gravity	2.63
Specific gravity	2.72	Electrical conductivity	1.104S/cm
pH	9.4		

2 Materials

This study examines the effect of sand gradation on the behavior of sand–bentonite mixtures using locally available river sand and bentonite. The sand is classified into fine (0.075–0.425 mm), medium (0.425–2 mm), and coarse sand (2–4.36 mm). Particle size distribution of both bentonite and river sand was determined using hydrometer and sieve analysis, following ASTM D 7928 and ASTM D 6913, respectively. Index properties, including shrinkage limit, plastic limit, and liquid limit were evaluated in accordance with ASTM D 4318, while specific gravity was measured as per ASTM D 854. The pH and electrical conductivity of the bentonite were assessed by mixing dry bentonite with deionized water in a 1:25 ratio. Various properties of bentonite and sand used in this study are listed in Table 1.

3 Methodology

Fine sand (FS), Medium sand (MS) and Coarse sand (CS) are mixed with bentonite in the proportions of 70–90% by dry weight. These mixtures were tested for compaction characteristics, permeability and unconsolidated undrained triaxial strength tests. The compaction characteristics including optimum moisture content and maximum dry density were determined using the standard proctor compaction method as per ASTM D 698. Table 2 presents the MDD and OMC values for various sand–bentonite mixtures.

Falling head permeability tests, following the ASTM D 5856, were conducted out using a rigid wall permeameter to determine the hydraulic conductivity of the sand–bentonite mixtures. The unconsolidated undrained (UU) triaxial test was performed as per ASTM D 2850 to determine the shear strength parameters.

Table 2 Compaction characteristics for sand–bentonite mix proportions

	Fine sand			Medium sand			Coarse sand		
Sand (%)	70	80	90	70	80	90	70	80	90
Bentonite (%)	30	20	10	30	20	10	30	20	10
OMC (%)	16.9	15.14	16.24	15.8	12.96	15.91	13.52	11.31	9.85
MDD (g/cc)	1.67	1.70	1.709	1.81	1.87	1.74	1.778	1.768	1.61
Porosity (%)	41.4	37.62	33.09	31.6	31.06	30.22	32.3	31.32	30.8

4 Numerical Simulation

To evaluate the effectiveness and long-term performance of the landfill liner, contaminant transport through liners. Contaminant transport through compacted soil is influenced by physical, chemical, and biological mechanisms. The physical mechanisms include diffusion, advection, and dispersion which govern the contaminant transport within the soil. Chemical process incorporates the sorption, dissolution, complexation, hydrolysis, and oxidation play a critical role in contaminant transport. Biological mechanisms include degradation or transformation of contaminants further influencing their behavior. The governing equation for 1D contaminant transport combines with these mechanisms can be expressed as shown below [11]:

$$\frac{\partial C}{\partial t} = \frac{D_h}{R}\frac{\partial^2 C}{dz^2} - \frac{v}{R}\frac{\partial C}{\partial z} \tag{1}$$

where C denotes contaminant concentration at any time and any space, D_h, R, and v denotes the hydrodynamic dispersion coefficient, retardation factor, and seepage velocity, respectively. The hydrodynamic dispersion coefficient and the retardation factor can be expressed as [11].

$$D_h = D_0 \times (n)^e; R = 1 + \frac{K_d \rho_d}{n} \tag{2}$$

where D_0, n, K_d, and v_d denote are the free diffusion coefficient, porosity, partial distribution coefficient, and dry density, respectively.

The initial and bottom boundary condition of the liner system can be expressed as

$$C(z, 0) = 0; \frac{\partial C(\infty, t)}{dz} = 0 \tag{3}$$

Finite difference method is employed to solve the above governing Eq. (1), and the final expression can be expressed as.

$$C(m, n+1) = C(m, n)(1 - 2\alpha - \beta) + C(m+1, n) \cdot \alpha + C(m-1, n) \cdot (\alpha + \beta) \tag{4}$$

Table 3 Parameters and concentration of contaminants used in this study

Contaminant	Chloride	Lead	Zinc
Initial concentration (mg/L)	4000	19.4	20

$$\alpha = \frac{D_h \cdot \Delta t}{R \cdot \Delta z^2}; \beta = \frac{v \cdot \Delta t}{2R \cdot \Delta z}$$

The step size in both space and time should be opted such that the following stability equation is fulfilled ($\alpha + \beta \leq 0.5$).

Equation (4), which is discrete form of governing equation, has been solved using MATLAB2024b for the simulation of contaminant transport. The initial concentration in the liner systems is assumed to be zero and the inflow concentrations of different contaminants such as chloride, lead, and zinc are taken based on the available studies, which are reported in Table 3.

5 Model Validation

A 1D contaminant transport through the landfill liners has been developed by Garg et al. [11] is used to validate the contaminant transport in the current study. The comparison is carried out for contaminants iron and chloride were considered. Figure 1 shows the comparison of contaminant concentrations along the thickness of the liner in a period of 100 years. The comparison shows that the proposed contaminant transport model agrees well with Garg et al. [11].

6 Results and Discussions

Permeability tests: To evaluate the optimum mix proportion of sand–bentonite with varying the gradation of the sand was carried out using the falling head permeability tests. Figure 2 shows the variation of permeability with varying the bentonite content and gradation of the sand.

From Fig. 2, it is evident that all the sand-bentonite mixtures considered satisfies the hydraulic conductivity criteria of ($k \leq 1.0 \times 10^{-9}$ m/s), except for coarse sand mixed with bentonite in 9:1 proportion. It is also clear that with increasing the bentonite content, reduces the permeability and gradation of sand also has significant effect. For a sand–bentonite mix proportion of 9:1, fine sand and medium sand–bentonite mixtures have satisfied the hydraulic conductivity criteria and is in the range of ($3.14 - 18.95 \times 10^{-11}$ m/s). But for coarse sand mixed in the proportion of 9:1 has resulted in an increase in hydraulic conductivity up to 7.68×10^{-9} m/s. This increase in hydraulic conductivity is due to 10% bentonite not being enough to block the void spaces in the coarse sand mixtures.

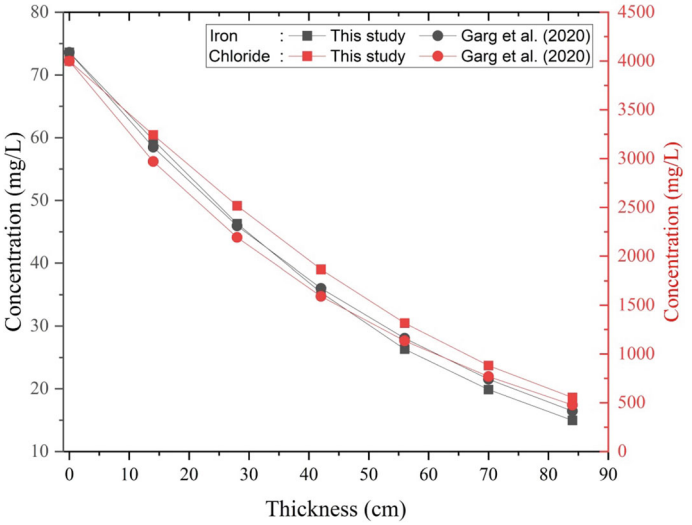

Fig. 1 Comparison of contaminant concentration at various depths in the current study and Garg et al. (2020)

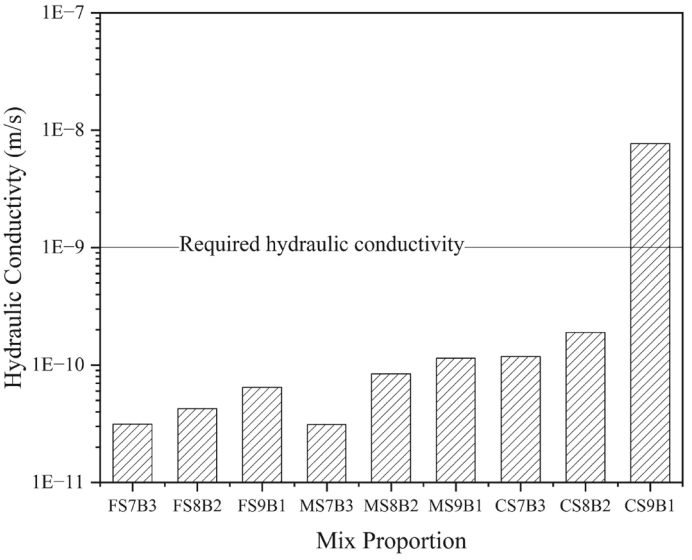

Fig. 2 Variation of hydraulic conductivity for varying sand gradation and bentonite content

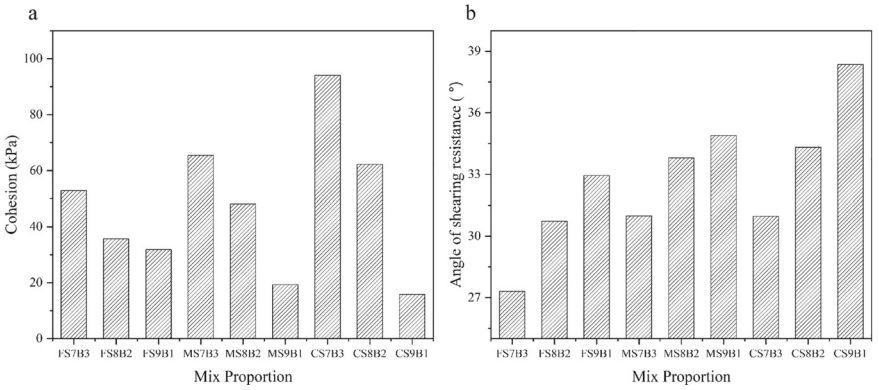

Fig. 3 Variation of cohesion and angle of shearing resistance with gradation of sand and bentonite content

Unconsolidated undrained triaxial tests: To evaluate the shear strength parameters of the sand–bentonite mixtures, unconsolidated undrained triaxial tests were performed. Figure 3 shows the variation of cohesion and angle of shearing resistance with varying the gradation and the bentonite content in the mix proportion.

From Fig. 3a, for fine sand with bentonite content from 30 to 10%, cohesion is in the range of 52.89–31.82 kPa, while with increasing the particle size to medium sand and coarse sand, the cohesion is in the range of 65.47–19.28 kPa and 94.04–15.82 kPa, respectively. This increase in cohesion when bentonite content is 30% is majorly due to the gradation of sand and bentonite filled in the pore spaces of the sand. It is observed that with increasing the particle size of the sand reducing the bentonite content resulted in an increase in the angle of shearing resistance. This is due to the insufficient bentonite in the pores of the mixtures.

Contaminant transport: Contaminant transport of three ionic contaminants through the landfill liner is simulated. The hydrodynamic dispersion coefficients are calculated using the free diffusion coefficient and the tortuosity factor and the retardation factor have been calculated using partial distribution coefficient, porosity, and dry density of the sand-bentonite mixtures. Figure 4 shows the variation of contaminant concentration along the depth for sand–bentonite mixtures varying the gradation of sand and bentonite content at 50 years.

From Fig. 4, it is observed that for CS9B1 mix, the contaminant concentration along the depth is significantly greater when compared with remaining mixes. This is due to the higher hydraulic conductivity (7.68×10^{-9} m/s). For CS9B1 mix, the concentration of chloride observed is 1421.9 mg/L. This indicates that, in case of CS9B1 mix, liner with thickness of 1.8 does not resist the containment of these contaminants. In remaining mix proportions, fine sands with bentonite content of 30% and 20% were observed to have higher contaminant concentration along the depth at $t = 50$ years. This is observed due to the higher porosity and less dry density achieved.

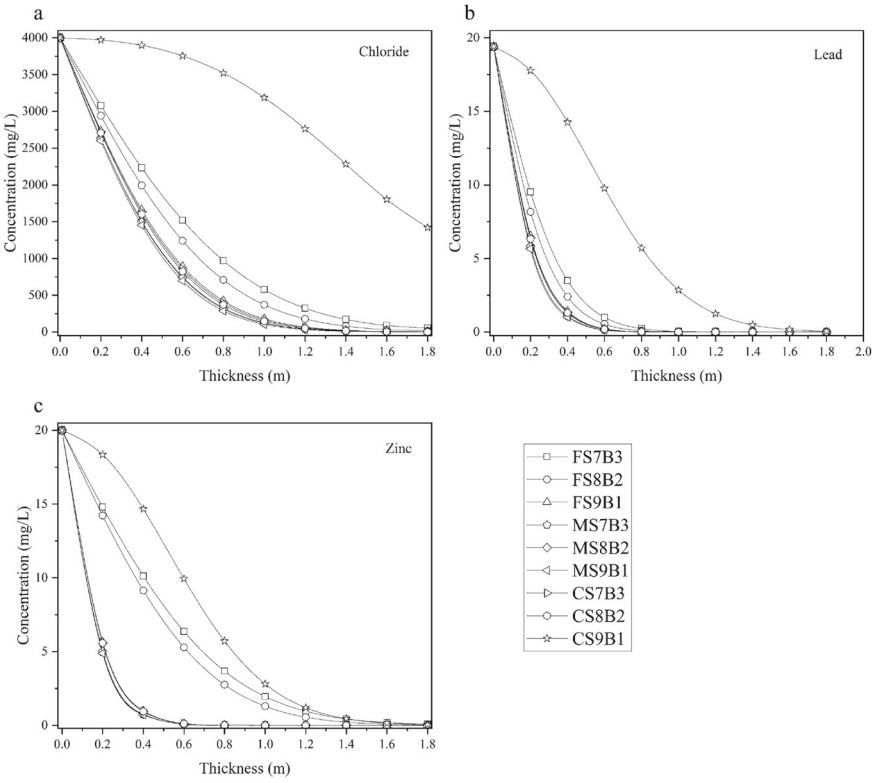

Fig. 4 Variation of contaminant concentration with depth at t = 50 years

7 Conclusions

In the present study, the effect of sand gradation and the bentonite content for the application of landfill liner is evaluated using experimental investigations. Further, a finite difference model is employed to simulate contaminant transport through landfill liner. In this study, simulation of three different contaminants were carried out which includes chloride, lead, and zinc. From hydraulic conductivity tests, with decreasing the bentonite content from 30 to 10% in the bentonite content in fine sand, medium sand and coarse sand, the hydraulic conductivity of the mix was increased by 51%, 73%, and 98%, respectively. While with increasing the bentonite content, the cohesion is increased by 1.66, 3.39, and 5.95 times when fine sand, medium sand, and coarse sand are used. Simulation of contaminant transport demonstrated that the gradation of sand and bentonite content has significant effect. Higher bentonite content effectively reduces contaminant transport due to lower hydraulic conductivity. But in case of coarser sand and low bentonite content exhibit higher hydraulic conductivity and resulted in higher concentrations at the exit of the liner. These findings highlight the importance of optimizing the sand–bentonite mixtures to achieve

a balance between permeability and contaminant transport for effective performance of landfill liners.

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Assessment of Electrokinetic Treatment and Associated Health Risks of Metal-Contaminated Sites

Sonam Taneja, Oznur Karaca, and Anil Kumar Haritash

Abstract The effectiveness of the Electrokinetic remediation (EKR) technique for treating Cr (VI), Pb, and Cd-contaminated soils was investigated. Further, a Human Health Risk Assessment (HHRA) was performed using the USEPA methodology to investigate the reduction in health risks post-EKR. The best removal efficiency was achieved for Cr (VI) (77%), followed by Pb (65%) and Cd (30%). The synergistic interplay of CrO_4^- and $Cr[EDTA]^-$ complexes towards the anode resulted in higher removal than Pb, where the antagonistic action of Pb^{2+}/Cd^{2+} and $Pb[EDTA]^-/Cd[EDTA]^-$ complexes counteracted the movement, resulting in low net electromigration. The results from the HHRA indicated a significant reduction in the non-carcinogenic adverse effects from very high category to low and medium in adults and children, respectively. Carcinogenic risk, on the other hand, could not be reduced below the acceptable range of 10^{-6} after treatment but reduced from one in a hundred to one in thousand (10^{-3}). The carcinogenic risks are expected to decrease further by increasing treatment time. The findings from this study concluded that EKR has the potential to reduce metal toxicity in soils. The results from risk assessment can be useful in forming risk-based management strategies for contaminated sites.

Keywords Soil pollution · Electrokinetic remediation · Chromium · Lead · Cadmium · Health risk assessment

1 Introduction

Soil is one of the most critical natural resources available to human beings, acting as the pillar for agriculture and food security around the world. However, population growth at a very high rate, urbanization, and industrialization have resulted in an

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increased use of resources and a higher generation of waste, much of which finds its way onto land. This has led to the introduction of contaminants in soil, such as toxic organic pollutants and heavy metals, hence hindering the natural attenuation potential of soil and leading to cumulative soil pollution. High levels of heavy metals greatly alter soil microbiology through interference with microbial diversity, abundance, and growth, resulting in diminishing soil fertility, hindering plant growth, and, in extreme cases, leading to necrosis of plants [1]. In addition, the affected soils can serve as secondary sources of contamination by releasing the contaminants into neighboring aquatic systems and hence enhancing groundwater contamination risk.

Heavy metal toxicity also presents very serious hazards to the human body. Exposure for long periods of time can negatively impact the central and peripheral nervous system, cardiovascular diseases, trigger kidney failure, and gastrointestinal injury [2, 3]. Metals may enter the human body through oral ingestion or skin absorption and cause acute poisoning, dizziness, weakness, headaches, and loss of memory [4]. Hence, heavy metal pollution is an important environmental and public health issue, emphasizing the need for efficient soil treatment techniques. However, surface and subsurface remediation is still a challenge due to its labor-intensive nature, high economic burden, and time-consuming needs.

Electrokinetic remediation (EKR) is a potential method that has been gaining significant attention in the recent past for its versatility to treat wastewater [5], sludge [6], groundwater [7], sediments [8], acid mine tailings [9], and contaminated soil [10]. EKR utilizes the strength of an applied electric field to cause contaminants to migrate towards oppositely charged electrodes. The technique is highly effective for fine-grained soils, which tend to be resistant to traditional remediation methods as a result of low permeability [11]. The success of EKR in different matrices lies mainly in the fact that it relies on path length rather than soil permeability [12]. The process generally involves three major steps: (i) the creation of an electric field by means of an externally supplied power source, (ii) movement and transport of contaminant ions towards the electrodes with an opposite charge, and (iii) elimination of the contaminants built up in electrode reservoirs by post-treatment operations like electrodeposition or chemical precipitation [13].

Based on the numerous benefits of EKR, specifically the removal of metals with less disruption to soil structure, the current study evaluates its use for the remediation of soil polluted with three priority metals of public health concern, namely hexavalent chromium (Cr^{6+}), cadmium (Cd^{2+}), and lead (Pb^{2+}). To further understand the migration process of these metals, sequential extraction methods were used. Lastly, an extensive health risk assessment was made to determine the efficacy of EKR in reducing metal-related risks to human health.

2 Materials and Methods

2.1 Materials

The analytical grade potassium dichromate (99.9%) and cadmium metal (99%) were purchased from SRL Chemicals, India. The lead nitrate salt (99.5%) was purchased from Sarabhai Chemicals Ltd., under Merck. (Germany) and Synthetic solutions of metals were prepared using ultrapure Type-I water. Ethylenediaminetetraacetic acid disodium salt (98%), sodium hydroxide (NaOH), and sulfuric acid (H_2SO_4) were purchased from Central Drug House (CDH) (P) Ltd., India.

2.2 Methods

Soil Collection and Preparation. The soil sample was collected from the campus of Delhi Technological University, Delhi, at 0–20 cm depth after removing the surface cover, with the help of soil auger, and sun-dried for 2 days. The dried soil was then crushed using a mortar and pestle and sieved through a 2 mm mesh to remove coarse particles. The sieved soil was dry-spiked with the finely crushed salts of Cr, Pb, and Cd with concentrations 1000 mg/kg, 1000 mg/kg, and 250 mg/kg, respectively. These concentrations were selected based on the range of industrial soil contamination and previous literature. For Cr^{6+}, 4.23 g of potassium dichromate salt was thoroughly mixed with 1.5 kg of the sieved soil to achieve 1000 mg/kg of Cr^{6+} concentration. Similarly, for Pb^{2+}, 1.59 g of lead nitrate salt was used to achieve 1000 mg/kg of Pb^{2+} in soil. For Cd^{2+}, 0.25 g of Cd metal with 99% purity, dissolved in nitric acid, was added to the soil to achieve 250 mg/kg of Cd concentration. Further, to wet the soil, deionized water was added to the dry spiked soil specimen till the moisture content of 15% was achieved. The soil was left to dry naturally with intermittent mixing at regular time intervals and this wetting–drying cycle was repeated 04 times. This was carried out to ensure that metals are evenly distributed (through diffusion and dispersion) [9] and to maintain a moisture content of 15%. The spiked soil was stored in a plastic bag for a week before the experiment to avoid moisture loss.

EKR setup. The experimental setup for EKR, as shown in Fig. 1, consisted of soil cell of 19.5 cm length, 5.5 cm width, and 4.5 cm height, a dual-output regulated variable DC power supply (0-60 V, 0-5A), electrodes, and electrode wells. For this study, 0.1 M EDTA solution (37.2 g in 1L ultrapure water) was used as an enhancing solution as it improves metal solubilization and is easily available [3]. The spiked soil was saturated with 0.1 M EDTA solution to achieve a moisture content of 25% and then compactly filled in the soil cell. The electrode wells were filled with 120 ml of EDTA solution before the start of the experiment. This study employed two commercially available electrodes, graphite, and stainless steel (SS 304) of size 10 mm diameter × 150 mm length placed 15 cm apart. The experiments were performed for 10 h

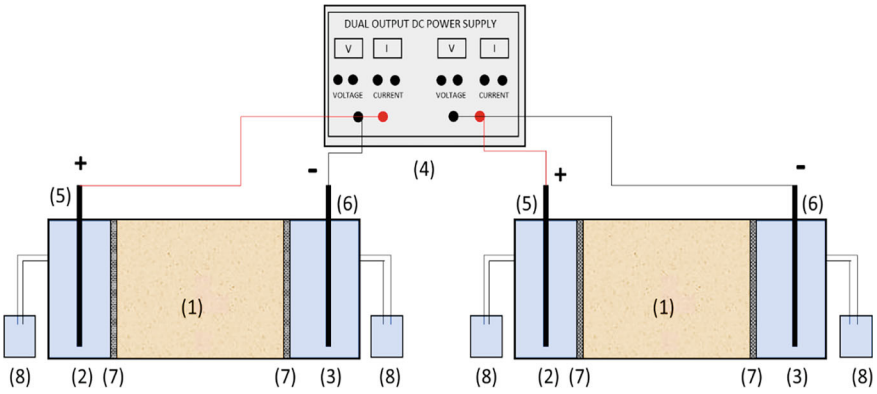

Fig. 1 Schematic of Electrokinetic setup and its components labeled as (1) soil cell (2) anode well (30 cathode well (4) DC power supply (5) anode (6) cathode (7) nylon mesh (8) electrode reservoirs

over 2 days (5 h/day). Two sets of experiments were performed for each metal, one with graphite electrode and one with stainless-steel electrode at a constant 2.5 V/cm voltage gradient.

Analytical Procedure. The soil was sectioned into equal divisions from the anode to the cathode (S1, S2, S3, and S4) and extruded into the containers after the treatment. Three samples were taken from each section for analysis. The electrolytes were analyzed for pH and conductivity using a bench-top multiparameter kit. The metal concentrations were analyzed using Atomic Absorption Spectrophotometer (AAS) (Analytik-Jena novAA 350). The electrolytes were centrifuged at 5000 rpm prior to metal analysis for removal of any solid particles of soil. The soil was oven-dried at 105ºC for 24 h in a hot-air oven for moisture content analysis. Next, 5 g of oven-dried soil was thermally oxidized in a muffle furnace at 550ºC for 2 h to remove organic compounds (standard ASTM D6357). The inorganic soil residue was shaken with a dilute nitric acid solution (5%) for 2 h for extraction. The supernatant was then collected and centrifuged at 5000 rpm for 10 min and used for analysis of metals. Before every experiment, washing of the soil cell and electrode wells was done using diluted acid (0.1 M sulfuric acid) to prevent cross-contamination. The electrodes were thoroughly cleaned to eliminate deposition and new nylon cloth was employed for every test.

Sequential Extraction Procedure. A widely used procedure given by [14], originally developed by [15], was used to determine the speciation of metals in five fractions, exchangeable and soluble form, carbonate bound, Fe–Mn oxide bound, organic matter bound, and residuals.

Human Health Risk Assessment (HHRA). A human health risk assessment was performed to assess the overall health impacts in terms of carcinogenic, non-carcinogenic, and radiological risks. Non-carcinogenic risk is estimated according to the USEPA 1989 method by calculating the Average Daily Dose (ADD, mg/kg/day)

of a contaminant through all three routes of exposure: oral, dermal, and inhalation, which is then used to calculate the Hazard Index (HI) (Table 1). Carcinogenic risks, however, are estimated through the product of ADD with the slope factor (SF). Total Carcinogenic Risk (TCR) is estimated from the sum of risks from the three pathways (Table 2). The normal value of TCR ranges from 10^{-6} to 10^{-4}, above which, the associated contaminant poses adverse carcinogenic risk [16]. The following equations were used to calculate the risk assessment.

$$ADDing = \frac{CX\ IngR\,X\ EF\ X\ ED\,X\ EF}{BW\ X\ AT} X\ 10^{-6} \tag{1}$$

$$ADDinh = \frac{CX\ Ingh\,X\ ED\,X\ EF}{BW\ X\ AT\ X\ PEF} \tag{2}$$

$$ADDder = \frac{CX\ SAF\ X\ ESA\,X\ PEF\ X\ EF\ X\ ED}{BW\ X\ AT} X\ 10^{-6} \tag{3}$$

Table 1 Selected values for the variables used in HHRA [16, 17]

Variables	Physical significance and units	Values	
		Adults	Children
IngR	Ingestion rate of soil (mg/day)	100	200
InhR	Inhalation rate of soil (m^3/day)	12.3	7.6
EF	Exposure frequency (days/year)	365	365
ED	Exposure duration (years)	30	6
BW	Average body weight (kg)	70	20
AT	Average exposure time (non-carcinogenic, days)	8760	2190
ESA	Exposed skin surface area (cm^2)	4359	1600
SAF	Soil-to-skin adherence factor (mg/cm^2)	0.07	0.2
PEF	Particle emission factor (m^3/kg)	1.36×10^6	1.36×10^6

Table 2 Reference dose and slope factor of the selected metals [16]

Exposure pathway		Cr	Pb	Cd
R*f*D	Ingestion	3×10^{-3}	3.5×10^{-3}	4×10^{-2}
	Dermal absorption	6×10^{-5}	5×10^{-4}	1.2×10^{-2}
	Inhalation	2.86×10^{-5}	3.5×10^{-3}	1×10^{-3}
SF	Ingestion	0.5	8.5×10^{-3}	0.38
	Dermal absorption	20	-	0.38
	Inhalation	42	4.2×10^{-2}	6.3

$$HQ = \frac{ADDi}{RfDi} \tag{4}$$

$$HI = \sum HQ \tag{5}$$

$$CRing = ADDing\ X\ SFing \tag{6}$$

$$CRinh = ADDinh\ X\ SFinh \tag{7}$$

$$CRinh = ADDder\ X\ SFder \tag{8}$$

$$TCR = \sum (CRing + CRinh + CRder) \tag{9}$$

3 Results

3.1 Variations in pH

The variations in pH of electrolyte and soil sections after EK treatment are shown in Fig. 2. The pH at anode varied between 1.5 and 3 for all tests due to the oxidation of water and generation of H^+ ions. The pH at cathode was strongly alkaline due to the generation of OH^- ions. The pH of soil varied from 6.5 to 8.5. In all tests, the pH after EKR increased from the initial value, except for exp. Cr–G of Cr (VI) with graphite, where the soil pH was lower than the initial pH. It was observed that graphite electrode led to slightly lower pH possibly due to more acidic migration front and stronger H^+ ion transport. The higher pH in soil in Pb and Cd systems could lead to the formation and precipitation of metal hydroxides.

3.2 Metal Removal Through EKR

As shown in Fig. 3, the average removal rate of metals was in the order Cr (VI) > Pb > Cd. While the removal was comparable for Cr (VI) and Pb, Cd showed significantly low removal. The Cr (VI) ions exist as CrO_4^{2-} in aqueous form, which tend to migrate towards anode under the applied electric field. In the presence of EDTA, negatively charged soluble Cr-EDTA complexes are formed over a wide range of pH that tend to migrate towards anode as well. This synergistic movement of highly soluble chromate ions and EDTA complexes towards the anode resulted in high net removal of the metal. In case of Pb^{2+}, the metal ion exhibits low solubility and high

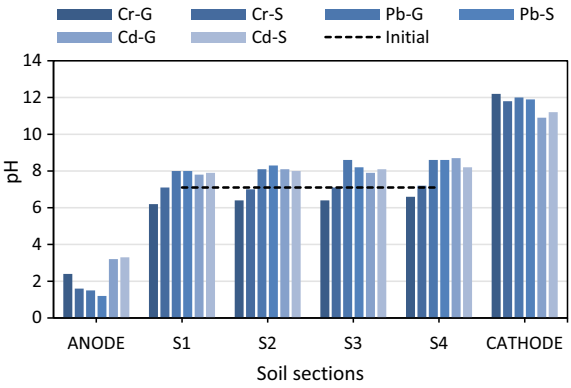

Fig. 2 Variations in pH of soil and electrolytes after EK treatment

affinity towards the soil particles. Due to the neutral to slightly alkaline pH of soil in this study, the Pb ions precipitated as $Pb(OH)_2$ near the cathode. In the presence of EDTA, negatively charged Pb-EDTA complexes were formed. However, in addition to the precipitation, the counteractive migration of Pb^{2+} towards the cathode and $[Pb\text{-}EDTA]^{2-}$ towards the anode resulted in a lower net removal of Pb. A similar trend was observed for Cd^{2+} ions. The low initial concentration of Cd in the soil also contributed towards its less removal as compared to Pb. The removal was maximum for Cr (VI), followed by Pb and Cd when graphite electrodes were used. However, when stainless steel was used as electrodes, Pb showed higher removal (65.1%) as compared to Cr (VI). This was because higher oxidation and subsequent corrosion were observed at the graphite anode and higher electrodeposition at the graphite cathode which caused high resistance and reduced flow of current in the case of Pb as compared to Cr (VI). The stainless-steel electrode had relatively less corrosion/or deposition than graphite in Pb-contaminated soils. Though Cd removal was less than the other two metals, comparable results were obtained for both electrodes. Graphite electrode was better suited for Cr (VI) removal while stainless steel was better was Pb and Cd removal.

3.3 Mobilization of Metals

The distribution of Cr (VI), Pb, and Cd in soil both before and after EK treatment was determined using the sequential extraction procedure analysis is summarized in Fig. 4. With the help of sequential extraction, the different forms of metals in soil can be determined which serves as a useful tool to regulate soil–metal chemistry. In the case of Cr (VI), most of the fraction was in the exchangeable and soluble fraction. Due to this, the removal was relatively higher and faster. Upon EKR, the pH of the cathode increased which led to the partial reduction of Cr (VI) to Cr (III) and the precipitation of Cr as hydroxides or carbonates, preventing further migration. In the case of Pb, a major portion of the metal initially was present in the residual fraction

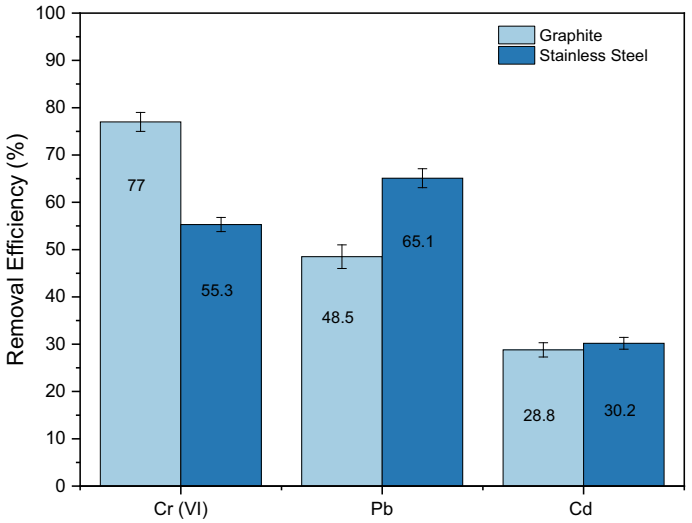

Fig. 3 Average removal efficiency (%) of Cr (VI), Pb, and Cd in experiments with graphite electrode and stainless-steel electrode

resulting in the immobilization of metals. Upon EK treatment, the metals shifted to the exchangeable fraction which could be due to the development of an acidic front in the anolyte that led to solubilization of metal precipitates or hydroxides. The Cd distribution in the soil is similar to Cr (VI) with a major portion in the exchangeable form and soluble form, which migrated readily upon the induced electric field. The findings suggest that the metals when present in exchangeable and soluble form cause higher migration during EK treatment.

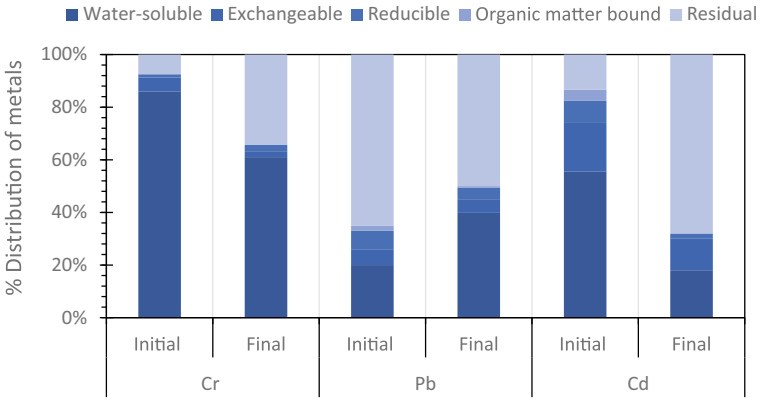

Fig. 4 Percentage distribution of metals in different fractions from the sequential extraction procedure

3.4 Human Health Risk Assessment (HHRA)

In order to assess the EKR-induced toxicity reduction, a Human Health Risk Assessment (HHRA) was performed on the soil before and after the treatment using Eqs. 1–9. The exposure to the metals in soil was assessed taking into account, accidental ingestion, dermal contact, and inhalation. The daily dose through these exposure routes was calculated, using which the Hazard Index (HI) was determined. Ideally, HI < 1 is considered to pose no adverse non-carcinogenic effects. However, in this study, the non-carcinogenic effects based on HI have been categorized as follows: From Table 3, it was found that the HI index for untreated soils was under the very high risk to medium risk category for children, while only Cr (VI) and Pb soils were under the very high-risk category for adults. Upon treatment with EKR, the HI index reduced to the negligible and medium category for Cr (VI), Pb, and Cd-contaminated soils in adults and children respectively. The results indicate that Cr (VI) posed the most adverse non-carcinogenic health effects, followed by Pb when left untreated in soil. The non-carcinogenic effects can effectively be reduced through treatment with EKR. As the metals selected for this study are considered carcinogens, the assessment of Total Cancer Risk was also performed as per the U.S. EPA framework. The accepted level of cancer risk is in the range of 10^{-4}–10^{-6}. TCR was calculated by multiplying the daily dose with the slope factor, given in Table 4. From the results, it was found that the maximum cancer risk was worst in Cr (VI)-contaminated soil without remediation, followed by cadmium. The children were at higher risk as compared to the adults, due to more exposure. However, the EK remediation of 10 h showed a decrease in LCR from 4.23 X 10^{-2} to 6.5 X 10^{-3}, and from 1.37×10^{-3} to 9.5×10^{-4} in cadmium-contaminated soils. EKR was successful in bringing down the TCR due to cadmium within the acceptable range, while the TCR due to chromium decreased but not in line with the permissible limits, which could be due to relatively higher initial concentrations of chromium in soil. In the case of Pb-contaminated soil, the exposure through dermal contact was not established, thus the TCR due to oral ingestion and inhalation was in line with the acceptable range, before and after the treatment.

4 Conclusion

Electrokinetic Remediation was performed to investigate the toxicity reduction due to the high concentration of metals in the soil. The results showed significant potential of EKR to remove metals from soils in a relatively short duration. The best results were achieved for Cr (VI) (77%), followed by Pb (65%) and Cd (30%). EKR can thus be used as a rapid remediation technique for highly contaminated sites. Apart from the significant metal removal from soils with high initial concentrations, it can also be employed for bringing down the overall toxicity and associated health risks. The results from the HHRA indicated a significant reduction in the non-carcinogenic

Table 3 Total daily intake (mg/kg/day) via ingestion, inhalation, and dermal contact pathways, and HI for Cr (VI), Pb, and Cd in adults and children

Metals	Conc. (mg/kg)	ADD_{ing}		ADD_{inh}		ADD_{der}		HQ_{ing}		HQ_{inh}		HQd_{er}		HI	
		Adult	Children	Adult	Children	Adult	Children	Adult	Children	Adult	Children	Adult	Children	Adult	Children
Cd untreated	246.5	0.0004	0.0025	0.0000	0.0001	0.0000	0.0000	0.44	2.4	0.039	0.069	0.0003	0.0002	0.4	2.5
Cd treated	171.9	0.0003	0.0017	0.0000	0.0000	0.0000	0.0000	0.30	1.7	0.027	0.048	0.0002	0.0002	0.3	1.7
Pb untreated	778.6	0.0014	0.0078	0.0001	0.0002	0.0000	0.0000	1.39	7.7	0.125	0.218	0.0010	0.0008	1.5	8.0
Pb treated	271.7	0.0005	0.0027	0.0000	0.0001	0.0000	0.0000	0.48	2.7	0.043	0.076	0.0003	0.0003	0.5	2.7
Cr (VI) untreated	2519.3	0.0045	0.0252	0.0004	0.0007	0.0000	0.0000	4.49	25.1	0.406	0.706	0.0031	0.0025	4.9	25.9
Cr (VI) treated	387.3	0.0007	0.0039	0.0001	0.0001	0.0000	0.0000	0.69	3.8	0.062	0.108	0.0005	0.0004	0.7	3.9

Table 4 Carcinogenic risks of adults and children under different exposure pathways

	CR_{ing}		CR_{inh}		CR_{der}		TCR	
	Adult	Children	Adult	Children	Adult	Children	Adult	Children
Cd untreated	1.67×10^{-4}	9.37×10^{-4}	2.51×10^{-4}	4.36×10^{-3}	1.17×10^{-9}	9.37×10^{-10}	4.18×10^{-4}	1.37×10^{-3}
Cd treated	1.17×10^{-4}	6.53×10^{-4}	1.75×10^{-4}	3.04×10^{-4}	8.16×10^{-10}	6.53×10^{-10}	2.92×10^{-4}	9.57×10^{-4}
Pb untreated	1.18×10^{-5}	6.62×10^{-5}	5.28×10^{-6}	9.17×10^{-6}	–	–	1.71×10^{-5}	7.54×10^{-5}
Pb treated	4.12×10^{-6}	2.31×10^{-5}	1.84×10^{-6}	3.20×10^{-6}	–	–	5.97×10^{-6}	2.63×10^{-5}
Cr (VI) untreated	2.25×10^{-3}	1.26×10^{-2}	1.71×10^{-2}	2.97×10^{-2}	6.30×10^{-7}	5.04×10^{-10}	1.93×10^{-2}	4.23×10^{-2}
Cr (VI) treated	3.46×10^{-4}	1.94×10^{-3}	2.63×10^{-3}	4.56×10^{-3}	9.68×10^{-8}	$7.75E \times 10^{-8}$	2.97×10^{-3}	6.50×10^{-3}

adverse effects from very high category to low and medium in adults and children, respectively. Carcinogenic risk, on the other hand, could not be reduced below the acceptable range of 10^{-6} after treatment but reduced from one in a hundred to one in thousand (10^{-3}). The results from risk assessment can be useful in forming risk-based management strategies for contaminated sites**.**

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Subsurface Permeability Profiling at an MSW Dumpsite: Insights from CPT and HPT

Pranjal Singh, D. K. Haritwal, G. V. Ramana, and Manoj Datta

Abstract Permeability estimation in heterogeneous subsurface environments remains a significant challenge, particularly at municipal solid waste (MSW) dumpsites where material properties vary widely. This study compares permeability profiles obtained using cone penetration testing (CPT) and hydraulic profiling tool (HPT) methods at the Bhalswa landfill in Delhi, India. Six locations were investigated using both methods to evaluate permeability in waste, unsaturated, and saturated zones. Four different approaches were compared: correlation of permeability with soil behavior index (Ic) correlation and normalized cone parameters (Qt-Fr), and two HPT-based methods (in 2010 and 2021). Results showed that HPT methods provided more consistent permeability estimates in waste materials (10^{-5}–10^{-4} m/s) where CPT measurements were erratic due to debris. However, in low-permeability clay layers, HPT methods overestimated permeability compared to Ic-based correlations (10^{-9}–10^{-8} m/s). The Qt-Fr method and HPT-2021 correlation generally provided intermediate values with reduced scatter across different materials. The study demonstrates that combining CPT and HPT methods provides more reliable permeability assessment in heterogeneous materials, with method selection depending on material type and expected permeability range. These findings have important implications for contamination assessment and monitoring at waste disposal sites.

Keywords Permeability · Cone penetration test · Hydraulic profiling tool · Municipal solid waste · Heterogeneous soils

1 Introduction

The accurate estimation of soil permeability at contaminated sites is crucial for assessing contaminant migration, designing remediation strategies, and evaluating groundwater vulnerability. However, obtaining reliable permeability measurements

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at contaminated sites presents significant challenges due to several factors. First, the subsurface often exhibits high spatial heterogeneity, with permeability varying by several orders of magnitude over short distances. Second, traditional sampling methods can disturb the soil structure, potentially altering its in situ permeability characteristics. Third, the presence of different porous media—ranging from natural soils to anthropogenic materials like municipal solid waste (MSW)—creates additional complexity in measurement and interpretation. These challenges are particularly pronounced at MSW dumpsites, where the heterogeneous nature of waste materials, coupled with varying degrees of decomposition and settlement, makes conventional permeability assessment methods less reliable [1].

Both laboratory and field methods have been developed to measure soil permeability, each with its own advantages and limitations. Laboratory methods such as constant and falling head permeameter tests offer controlled testing conditions but suffer from sample disturbance effects and may not represent the true field conditions, especially in heterogeneous deposits. Traditional field methods like pump tests and slug tests provide more representative values but only yield averaged estimates over large volumes of soil, missing critical small-scale variations that could act as preferential flow paths [2]. More recent field methods include indirect estimation using cone penetration tests (CPT), and direct push hydroprobes tools like permeameter (DPP) and injection logger (DPIL) [3–6]. CPTu can provide continuous soil classification and estimate permeability through empirical correlations with the soil behavior index or through dissipation tests at discrete depths. However, its accuracy can be compromised in waste materials or highly heterogeneous soils where cone resistance readings become erratic due to presence of foreign materials like construction and demolition waste [7].

Hydraulic Profiling Tool (HPT) is a form of DPIL that can work well in such heterogenous porous media due to its direct approach of measurement [8, 9]. HPT typically operates by injecting water at 200–400 ml/min through a downhole screen on the probe while measuring the injection pressure, with data logged every 1–2 cm during probe advancement. A higher injection pressure generally indicates lower permeability soil, while lower pressure suggests higher permeability materials. Permeability can also be obtained by correlation with the flow-pressure ratio [5, 10]. HPT offers several advantages over traditional methods, including rapid data acquisition, minimal site disturbance, and the ability to provide continuous permeability profiles in both saturated and unsaturated zones [11]. Unlike conventional methods that require separate boreholes for different depths, HPT can profile an entire site in a single push, significantly reducing investigation time and cost [2]. However, the interpretation of HPT data requires careful consideration of factors such as probe advancement rate, flow rate variations, and the validity of empirical correlations used to convert pressure readings to permeability values.

Based on the capabilities of HPT, an attempt was made in this study to compare the permeability values obtained by CPT and HPT for waste and soil layers at a MSW dumpsite. The results from this study can help evaluate the reliability of HPT, which will help in designing the most ideal framework to measure permeability at contaminated sites with heterogeneous layers.

2 Materials and Methods

2.1 Site Description

This study was conducted at the Bhalswa landfill, a non-engineered municipal solid waste dumpsite located in North Delhi, India. The dumpsite lacks engineered containment systems such as bottom liners or leachate collection systems, making the underlying natural soil layers the only barrier against contaminant migration to groundwater. The subsurface geology primarily consists of alluvial deposits overlying quartzite bedrock. The depth of alluvium at the site averages around 80 m, characterized by layers of fine sand (locally known as Yamuna sand) interspersed with lenses of clay and silt. The groundwater table fluctuates between 5 and 10 m below ground level, with a seasonal variation of $\pm$ 2 m during monsoon periods. The regional groundwater flow is generally northeastward toward the Yamuna River, located approximately 10 km east of the landfill, though local variations exist due to the influence of leachate mounding beneath the waste mass.

2.2 Site Investigation

Site investigations were conducted at six locations around the landfill perimeter during February–April 2023, utilizing a combination of CPT and HPT measurements. The investigation points were strategically selected to capture different hydrogeological conditions: two points on the western boundary (P01–P02) where clay layers were previously identified, two points on the northern side (P03–P04) characterized by sandy deposits, and two points on the southern boundary (P05–P06) where soil-like material from waste mining operations had been deposited. A direct-push drill rig (Geoprobe 7822DT) was used to advance both the CPT and HPT probes. To achieve sufficient penetration force in dense soils, the rig was anchored using two helical augers, increasing its pushing capacity from 4 to 16 tons. This enhanced capacity enabled probe advancement to depths of 15–20 m below the waste surface. Prior to testing, pre-boring of 1.5–3 m was carried out at each location to avoid any surface debris that could damage the probes. The testing program was supplemented by continuous soil sampling using a dual-tube sampling system to validate the probe readings and collect samples for laboratory analysis. The soil profile used in this study is derived from the continuous soil samples.

CPT was first prepared by saturating the brass filter ring by applying keeping the rings dipped in silicone oil in vacuum for 24 h. The ring was then placed between the cone tip and sleeve friction and the void was filled with silicone oil. The depth potentiometer, data acquisition system were connected to the laptop. Before starting the investigation, the rig was anchored using two solid helical augers with a flight length 8 feet. This generated an additional capacity of 10–15 tons to carry out the test with CPTu with a total of 15–20 tons (the weight of the rig was 4 tons). The

initial 3 m of the investigation was carried using a solid cone of the same diameter as the CPTu due to the presence of C&D waste that could damage the cone. The investigation was carried out to qc reaching 30 MPa or the inclination reaching up to 15° or if the inclination was greater than 1° in one meter.

HPT was prepared by cleaning the injection screen from any debris. The injection logger was then checked for calibration by inserting the probe in a standing pipe filled with water up to the top, which is above the level of the injection screen. The pressure difference is measured when the water is filled up to the top and when it is at the level of the screen. This difference should be equal to the hydrostatic pressure of the water column above the screen up to top of the tube. After obtaining satisfactory calibration check results, the probe was connected with data acquisition system, pump, flow controller, depth potentiometer, and a laptop. Similar to CPTu, the initial 3 m of the subsurface was drilled using a solid cone to prevent any damage to the probe.

2.3 Permeability Estimation

The permeability profiles were developed using four different methods—two based on CPT data and two utilizing HPT measurements. Each method has specific theoretical foundations and limitations that make them suitable for different soil conditions.

2.4 CPT-Based Methods

Robertson [6] proposed a correlation between soil behavior type index (Ic) and permeability based on the observation that soil permeability generally decreases as soils become more fine-grained. The relationship is defined in two parts:

$$1.0 < Ic \leq 3.27 : k = 10^{0.952-3.04\times Ic} m/s \tag{1}$$

$$3.27 < Ic < 4.0 : k = 10^{-4.52-1.37\times Ic} m/s \tag{2}$$

where Ic is calculated as

$$Ic = \left[(3.47 - logQ_{tn})^2 + (logF_r + 1.22)^2\right]^{0.5} \tag{3}$$

The normalized parameters Q_{tn} and F_r are defined as

$$Q_{tn} = \left[\frac{q_t - \sigma_v}{p_a}\right]\left(\frac{p_a}{\sigma_{v0}^{'}}\right)^n \tag{4}$$

$$Fr = [f_s/(q_t - \sigma_{v0})] \times 100\% \tag{5}$$

where q_t is the corrected cone resistance, f_s is sleeve friction, σ_v and σ'_{v0} are total and effective overburden stresses, respectively, p_a is atmospheric pressure, and n is a stress exponent that varies with soil type.

Elsworth and Lee [12] developed a theoretical framework relating permeability (k) to normalized CPT parameters through:

$$K_D = \frac{4k\sigma'_{v0}}{Ua\gamma_w} = \frac{1}{Q_t\left[1 + \frac{1}{Q_t} - \frac{F_r}{tan\phi}\right]} \tag{6}$$

where K is a dimensionless permeability parameter, U is penetration rate (typically 2 cm/s), a is cone radius, γ_w is unit weight of water, Q_t is the normalized cone resistance, and ϕ is the angle of shearing resistance.

2.5 HPT-Based Methods

McCall and Christy [5] proposed an empirical correlation between permeability (k) and the HPT flow–pressure ratio:

$$k = 21.14ln\left(\frac{Q}{P_c}\right) - 41.71 \tag{7}$$

where Q is the injection flow rate (mL/min) and P_c is the corrected injection pressure (psi) after subtracting hydrostatic effects. This relationship was developed through comparison with slug tests in alluvial deposits and is valid for permeabilities between approximately ~ 10^{-6}–10^{-4} m/s.

Borden, et al. [10] later developed a more comprehensive physically based approach:

$$k = \frac{E\left(0.1235VD^2 + 0.119Q\right)}{P_c^{1.017}} \tag{8}$$

where V is probe advancement rate (cm/s), D is the probe diameter (cm), and E is a hydraulic efficiency factor that accounts for soil disturbance around the probe. Based on field validation at multiple sites, E typically equals 2. This method explicitly incorporates probe geometry and advancement rate, making it more theoretically robust than purely empirical correlations.

For this study, all four methods were applied where appropriate data existed. The results from each method were compared to evaluate consistency and identify

potential limitations in different materials. The pore pressure dissipation tests were also carried out using CPT but was not considered in this study due to poor results.

3 Results

3.1 CPT and HPT Profiles

The soil behavior index (Ic) profiles shown in Fig. 1 revealed distinct layering across the six test locations (P01–P06). The top layer (consisting of MSW, top soil, and SLM) showed highly variable Ic values (ranging from 1 to 4) due to its heterogeneous nature. Below this at P01 and P03, a relatively uniform clay layer (Clay1) was identified with Ic values between 2.95 and 3.60. This layer could potentially attenuate the contaminant migration to groundwater. However, this clay layer was not present at other locations, but consisted of major sand that could allow significant contaminant flux. The sand and silt layers consisted of Ic values between 1.31 and 2.95. A second clay layer (Clay2) was observed at 3–8 m below groundwater table at all locations. The Ic profiles effectively delineate the different material types except for the waste layers and show good consistency in layer boundaries across adjacent boreholes.

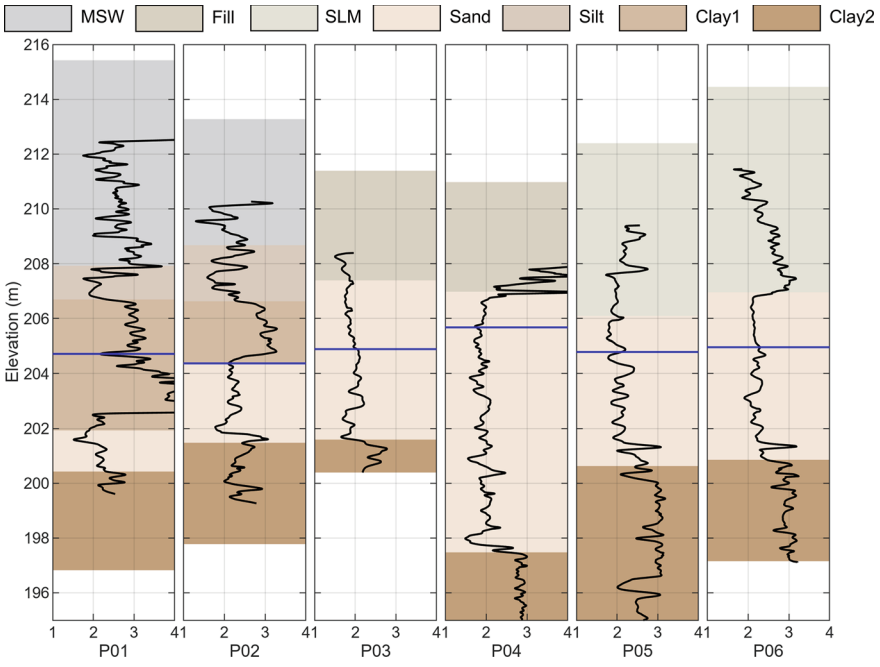

Fig. 1 Depth profile of soil behavior index (Ic)

The excess pore pressure (Δu2) profiles in Fig. 2 clearly delineated the existence of Clay2 below the groundwater table. Both positive and negative excess pore pressures were observed in the layer indicating possible dilative and contractive behavior, which could be due to over consolidated clays. However, this could also be due to desaturation of the filter in the probe, which may give erroneous results [13]. Due to this limitation and lower confidence in data, Δu2 and corresponding dissipation tests was not used for permeability calculation. Δu2 was nearly zero in the unsaturated zone, except at P02, indicating a higher water content than other locations.

The HPT flow–pressure ratio (Q/P) profiles in Fig. 3 demonstrate clear contrasts between the soil and waste layer. The waste layer showed high and variable Q/P ratios (1–100 ml/min/kPa), indicating its generally high but heterogeneous permeability. The clay layers exhibit consistently low Q/P ratios (<1 ml/min/kPa). The sand layers showed high Q/P values that varied between 10 and 100 ml/min/kPa. The Q/P profiles effectively complement the CPTu data by providing direct measurements of flow characteristics, particularly useful in the highly variable MSW layer where CPTu parameters may be less reliable.

The heterogeneity of the landfill environment, particularly the presence of highly variable MSW and SLM, plays a crucial role in permeability estimation. The large scatter observed in Q/P ratios for Ic values between 2 and 3 indicates significant heterogeneity, which traditional CPT correlations may not fully capture. While HPT measurements provide direct indications of flow characteristics, the disturbed nature

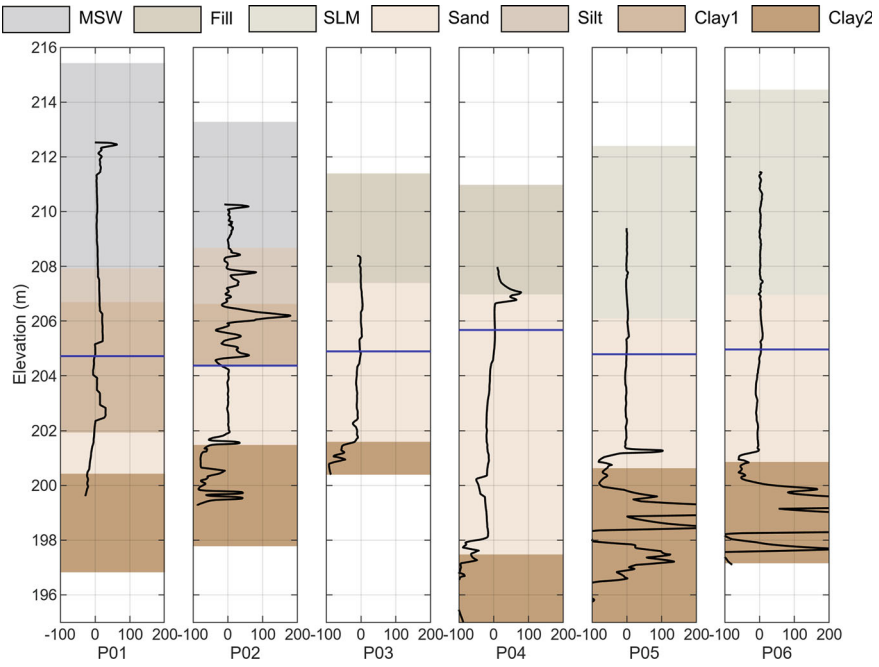

Fig. 2 Depth profile of excess pore pressure (Δu_2)

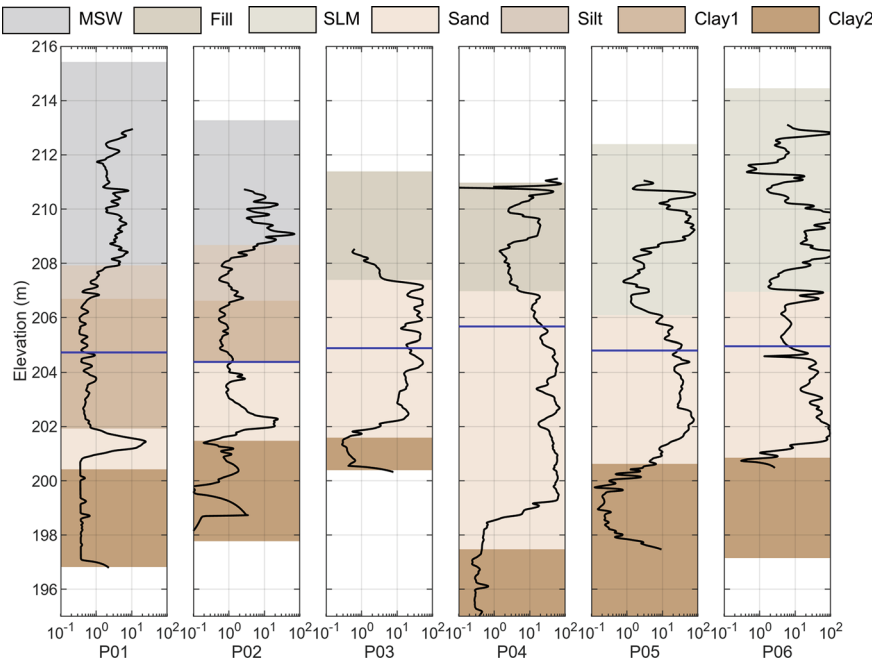

Fig. 3 Depth profile of flow-pressure ratio (Q/P) from HPT

of waste deposits and potential preferential flow paths could lead to localized overestimation of permeability. A spatial statistical approach, such as variogram analysis or geostatistical interpolation (e.g., kriging), could help map permeability variations more accurately across the study area. This would allow for a more refined assessment of contamination risk based on spatial patterns in permeability rather than relying solely on pointwise estimates.

3.2 Correlation Between HPT and CPT Readings

The relationship between the flow–pressure ratio (Q/P) from HPT testing and both soil behavior type index (Ic) and excess pore pressure (Δu2) from CPTu is shown in Fig. 4. The Q/P ratio exhibits a generally decreasing trend with increasing Ic, indicating lower permeability in finer-grained soils with higher Ic values. The Spearman correlation between the two is -0.49 with $p < 0.01$. However, considerable scatter is observed in this relationship, particularly for Ic values between 2 and 3, where Q/P varies over two orders of magnitude (0.1–100 ml/min/kPa). This scatter likely reflects the complex nature of the heterogeneous waste material. Δu2 correlation with Ic indicates that excess pore pressure were only generated with Q/P <1 ml/min/

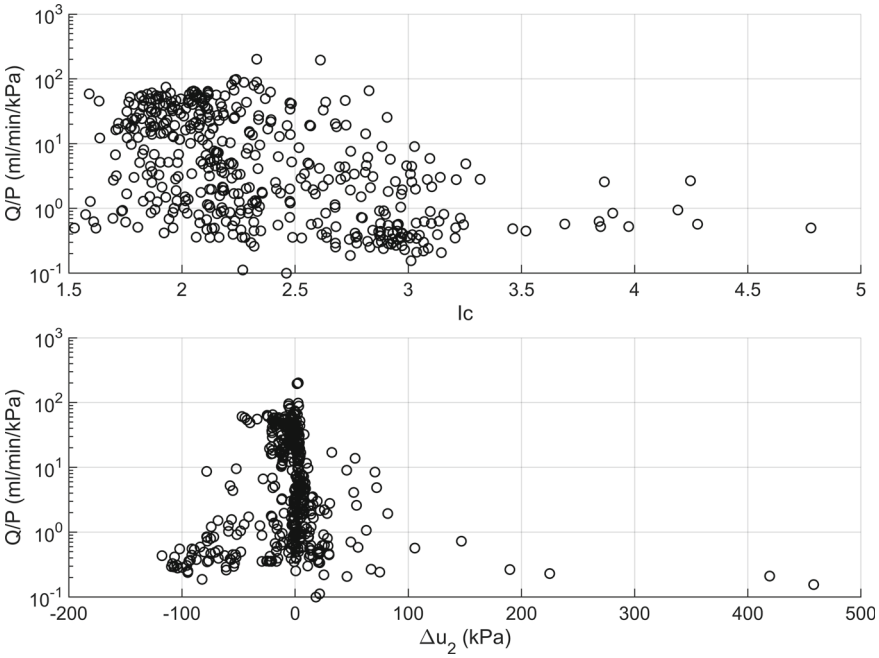

Fig. 4 Correlation of flow–pressure ratio (Q/P) from HPT with soil behavior index (Ic) and excess pore pressure (Δu_2)

kPa. However, no information about the positive and negative excess pore pressure could be derived.

3.3 Comparison of Permeability

The comparison of permeability (k) estimates shows distinct patterns across different materials using four methods: Ic-based correlation, Qt–Fr correlation, and two HPT correlations (2021 and 2010) [5, 6, 10, 12]. For waste materials (MSW) and soil-like material (SLM), the methods show interesting trends. In MSW, the Ic method shows the widest range and lowest median values (10^{-7} m/s), while both HPT correlations predict consistently higher permeabilities (10^{-5}–10^{-4} m/s). The Qt–Fr method provides intermediate estimates. Similar patterns were also observed in SLM. For the topsoil layer, all methods show relatively high permeability estimates, but with considerable scatter, particularly in the Ic-based method. HPT-2010 consistently provided higher permeability estimates than the rest of the methods.

For natural soils, particularly sand, all methods show substantial variation in estimates. The Ic-based method shows the widest range (10^{-6}–10^{-4} m/s) with significant outliers. The Qt–Fr method provides more conservative estimates around 10^{-6} m/

s, while both HPT correlations predict higher permeabilities, with the 2010 method giving the highest estimates (10^{-4} m/s). For silts, the spread between methods is smaller, with estimates generally ranging between 10^{-6} and 10^{-5} m/s, though the HPT-2010 method again predicts slightly higher values.

In clay layers (Clay1 and Clay2), there are notable differences between methods. The Ic-based correlation predicts significantly lower permeabilities (10^{-9}–10^{-8} m/s) compared to other methods. The Qt–Fr method shows considerable scatter but generally estimates higher values (10^{-6}–10^{-4} m/s). Both HPT correlations provide relatively consistent estimates around 10^{-5} m/s, with the 2010 method showing slightly higher values than the 2021 method. This suggests that the Ic-based method may be overly conservative for clay layers, while HPT methods may overestimate permeability in low-permeability materials.

A consistent pattern across all materials is that the Ic-based method generally shows the largest variability and tends to predict lower permeabilities, while the HPT-2010 method typically provides the highest estimates. The Qt–Fr method and HPT-2021 correlation often provide intermediate values with less scatter, suggesting they may offer more reliable estimates across different material types. This systematic difference between methods highlights the importance of using multiple approaches when estimating soil permeability, particularly in heterogeneous materials like MSW. While Ic-based method is widely used, its tendency to be overly conservative could limit its applicability for certain soil types, especially when designing remediation strategies that depend on accurate permeability assessments (Figs. 5, 6).

4 Conclusion

This study compared permeability profiles obtained using CPT and HPT methods at an MSW dumpsite, leading to several important findings:

1. The CPT and HPT methods showed complementary strengths in different materials. While CPT-based methods provided more reliable permeability estimates in natural soils, HPT methods proved more suitable for the heterogeneous waste layers where CPT measurements were erratic due to debris.
2. HPT methods tended to overestimate permeability in low-permeability clay layers. CPT-based Ic correlations generally predicted lower permeabilities with higher variability across all materials. The Qt–Fr method and HPT-2021 correlation typically provided intermediate values with less scatter, suggesting better reliability across different material types.
3. In natural soils (sands and silts), all methods showed reasonable agreement within 1–2 orders of magnitude. For clay layers, significant differences were observed between methods, with Ic-based correlations being potentially overconservative. In waste materials, HPT methods provided more consistent and realistic permeability estimates compared to CPT-based approaches.

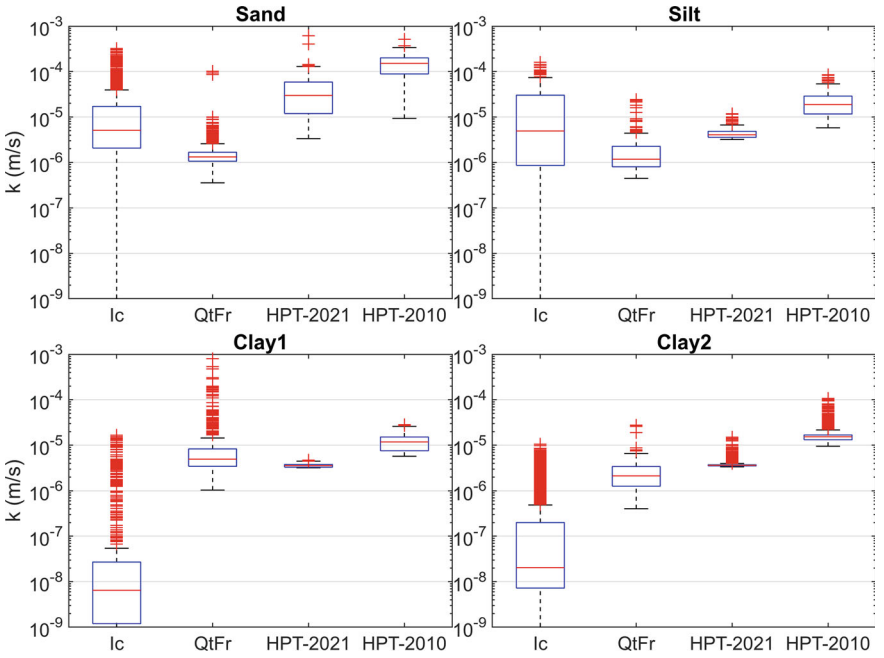

Fig. 5 Variation of permeability (in m/s) by CPT and HPT in different waste layers

The findings suggest that a combined approach using both CPT and HPT methods provides the most comprehensive permeability assessment at contaminated sites. The selection of appropriate method(s) should consider the type of material being tested, the expected permeability range, the required resolution of the permeability profile, and the presence of heterogeneities or artificial materials.

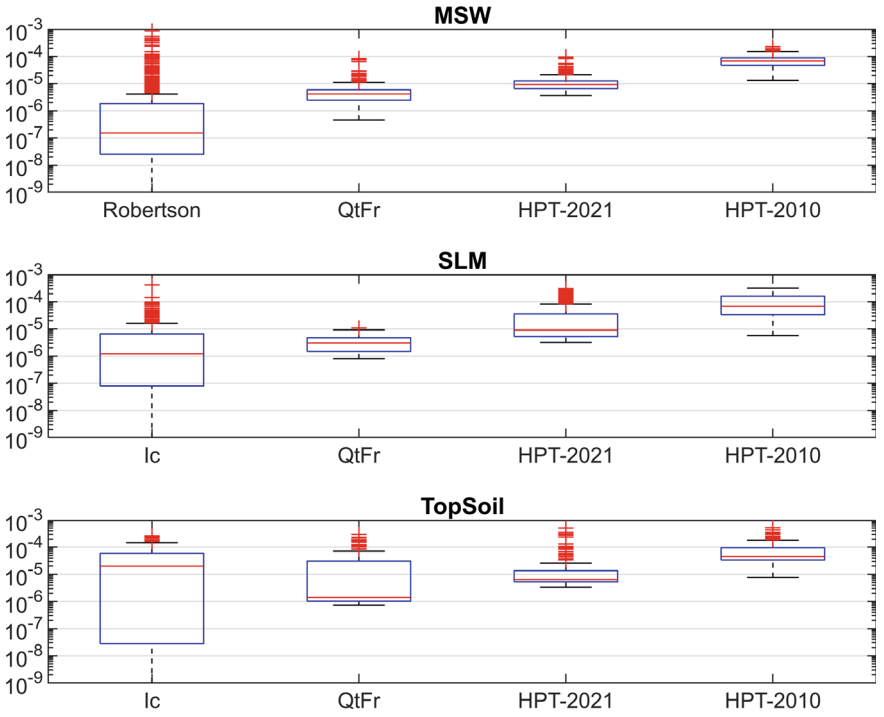

Fig. 6 Variation of permeability by CPT and HPT in different soil layers

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Hydraulic Conductivity of Conventional Sodium Geosynthetic Clay Liners to Synthetic Coal Combustion Product Leachate

Asena İrem Demir Sürer, Nilhan Demirci, and A. Hakan Ören

Abstract In this study, the hydraulic conductivity of conventional sodium geosynthetic clay liner (Na-GCL) to coal combustion product leachate (CCP leachate) was investigated. The hydraulic conductivity of Na-GCL was then evaluated in terms of bentonite mass per unit area (μ_{bA}) of Na-GCL. The hydraulic conductivity of GCL with 3.0 kg/m^2 μ_{bA} was high throughout the test and was finally measured as 3.7×10^{-8} m/s. The bundles of fibers on the GCL preferentially conduit the flow across the GCL, resulting in high hydraulic conductivity. The existence of bundles of fibers was approved by performing a dye test on the GCL. When the μ_{bA} of GCL increased to 4.0 kg/m^2, the hydraulic conductivity of the GCL decreased to 2.4×10^{-11} m/s, which is several orders of magnitude lower than that of GCL with μ_{bA} of 3.0 kg/m^2. The hydraulic conductivity stayed low when the μ_{bA} was further increased to 5.0 kg/m^2. This result showed that the critical μ_{bA} for the conventional Na-GCL was 4.0 kg/m^2, where the hydraulic conductivity was well below the allowable limit suggested for the barriers (i.e., 10^{-9} m/s).

Keywords Coal combustion · Geosynthetic clay liner · Mass per unit area · Hydraulic conductivity

1 Introduction

Coal combustion producta (CCPs) are the by-products of burning coal in power plants or industrial boilers. These products mainly include fly ash, bottom ash, boiler slag, and flue gas desulfurization residuals, which can pose significant environmental

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hazards if not managed properly. One primary concern is the leachate from CCPs, which contains various harmful cations and anions. When these leachates come into contact with water, they can seep into the groundwater and lead to contamination, threatening soil and water quality. A commonly used method to reduce this risk is implementing GCLs as barriers in waste containment systems [1, 2].

GCLs are composite materials consisting of a thin layer of bentonite sandwiched between woven and non-woven geotextiles. GCLs generally contain sodium bentonite and have low hydraulic conductivity to water ($<2.0 \times 10^{-11}$ m/s), which helps prevent the migration of contaminants [3, 4].

CCP leachates contain divalent ions ($Ca2^+$ and $Mg2^+$). When Na-GCL encounters these leachates, it can undergo a cation exchange and significantly lose its impermeability [4–6]. Therefore, it is vital to investigate the hydraulic conductivity behavior of GCLs to CCP leachates. Researchers have reported that aggressive leachates with high ionic strength and multivalent cations suppress bentonite swelling, creating large voids between particles, which can result in high hydraulic conductivity for the GCLs [1, 2, 4, 7, 8].

Research on GCLs indicates that the mass per unit area (μ_A) of a GCL significantly affects hydraulic conductivity [9]. When deionized water is used as the permeant, hydraulic conductivity decreases as the μ_A of GCL increases. However, the effect of μ_A on hydraulic conductivity when the permeant is different from deionized water has not been investigated in detail. While previous studies have examined polymer-enhanced GCLs with various permeants, limited work has focused on the hydraulic conductivity of Na-GCLs with CCP leachates under the influence of μ_A. Demir Sürer et al. [10] demonstrated that the impact of bentonite mass per unit area (μ_{bA}) is significant for GCLs permeated with trona ash leachate, a CCP leachate with high ionic strength. They reported that the hydraulic conductivities of Na-GCL with μ_{bA} of 3.3 and 4.1 kg/m2 were 2.6×10^{-6} m/s and 7.6×10^{-7} m/s, respectively. However, the hydraulic conductivity of the same GCL with 5.0 kg/m2 μ_{bA} was 4.1×10^{-11} m/s. As the μ_{bA} of GCL decreased, hydraulic conductivity increased.

This study aims to investigate the hydraulic conductivity of conventional Na-GCLs using a CCP leachate in terms of μ_{bA}. Coal combustion leachate (CCL) produced synthetically in the laboratory was used as the permeant. The hydraulic conductivity results were supported by chemical analyses performed on samples obtained from the inflow and outflow of the GCLs during the tests.

2 Materials and Methods

2.1 Materials

Needle-punched Na-GCL was used in this study. The liquid and plastic limits of sodium bentonite were determined as 231% and 61%, respectively, following ASTM D4318–17 [11]. The μ_{bA} of the GCL was between 3.0 and 5.0 kg/m^2 according to ASTM D5993–99 [12].

Coal combustion leachate (CCL) was used as the permeant in the hydraulic conductivity tests. CCL was prepared synthetically in the laboratory by dissolving 7.133 mM $CaSO_4$, 5.216 mM Na_2SO_4, 1.2 mM $MgSO_4$, 0.95 mM K_2SO_4, 0.667 mM NaCl, and 0.667 mM $CaCl_2$ salts in deionized water. CCL is prepared as a 5-L stock solution.

The ionic strength of CCL was 55.4 mM and the relative abundance of monovalent to divalent cations of CCL was 0.13 M$^{1/2}$. The pH and electrical conductivity of CCL were 8.5 and 0.2 S/m, respectively. Detailed information about the preparation of CCL was given by Demir Sürer [13].

2.2 Methods

Hydraulic conductivity tests. Hydraulic conductivity tests were conducted according to ASTM D6766-14 [14]. GCLs were cut 100 mm in diameter from a roll and weighed. The thickness and diameter of the samples were measured. Then, GCL was placed between the top and base pedestals of a flexible-wall permeameter. Non-woven geotextiles with the same diameter as GCL were placed on top and bottom of the GCL. The perimeter of GCL was sealed with bentonite paste to prevent sidewall leakage. Then, GCL was covered with a latex membrane, which was secured to the top and base pedestals using O-rings. 35 kPa cell pressure and an average hydraulic gradient 45 were applied to simulate the field conditions. Hydraulic conductivity tests were conducted using the falling head and constant tail water method.

Chemical Analysis. Samples were collected from inflow and outflow during hydraulic conductivity tests. The pH and electrical conductivity of liquid samples were measured using Accumet XL50 bench-top pH and Electrical Conductivity meters. Cation and anion analyses were conducted on the influent and effluent samples using inductively coupled plasma optical emission spectrometry (ICP-OES) to determine the exchangeable ions released from the bentonite [10, 13].

3 Results

The influence of μ_{bA} was investigated as a factor affecting the hydraulic conductivity of Na-GCL. The μ_{bA} of Na-GCL was 3.0, 4.0, and 5.0 kg/m2, referred to as μ_{bA3}, μ_{bA4}, and μ_{bA5}, respectively, in the paper henceforth. Table 1 presents the final hydraulic conductivities of Na-GCLs with CCL. Each test was conducted only once without replication.

The hydraulic conductivity behavior of Na-GCL with 3.0 kg/m^2 μ_{bA} (μ_{bA3}) obtained with CCL is given in Fig. 1a. The hydraulic conductivity of μ_{bA3} was high throughout the test. Since the hydraulic conductivity of μ_{bA3} was above the allowable limit suggested for the liners (1.0×10^{-9} m/s), chemical analysis was not performed to determine exchangeable cations and anions in the influent and effluent samples.

The hydraulic conductivity, pH, and electrical conductivity (EC) measurements, along with the concentrations of Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-, and SO_4^{2-} in influent and effluent of μ_{bA4}, are plotted in Fig. 2a-d as a function of the pore volume of flow. The hydraulic conductivity of μ_{bA4} gradually decreased from 4.1×10^{-9} m/s to 3.9×10^{-11} m/s at 4.5 pore volume of flow (Fig. 2a) and remained approximately constant until the end of the test. The final hydraulic conductivity was measured as 2.5×10^{-11} m/s.

Figure 2b shows the pH and EC values of influent and effluent samples of μ_{bA4}. The pH of the effluent decreased from 9.4 to 8.8, while the pH of the influent was 8.2 (Fig. 2b). The initial EC of the effluent was 12.1 mS/cm, and it decreased to 4.2 mS/

Table 1 Final hydraulic conductivities of Na-GCLs with CCL

Test No	μ_{bA} (kg/m^2)	μ_{bA}	Pore volume of flow	Test duration (Days)	Hydraulic conductivity (m/s)
1	3.04	μ_{bA3}	17.4	~1	3.7×10^{-8}
2	4.01	μ_{bA4}	20.4	400	2.4×10^{-11}
3	4.98	μ_{bA5}	15.2	385	1.5×10^{-11}

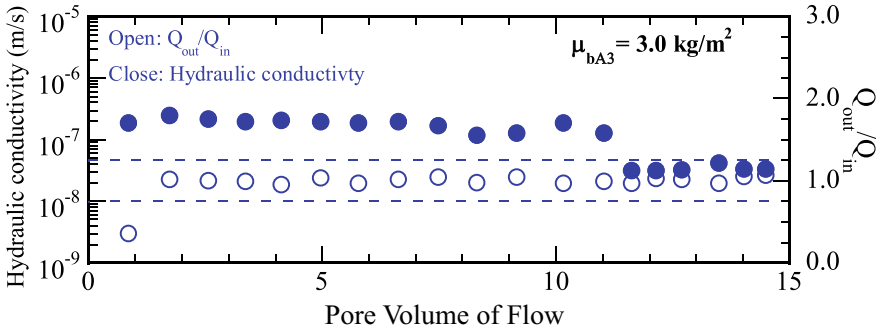

Fig. 1 The hydraulic conductivity of μ_{bA3} with coal combustion leachate (CCL)

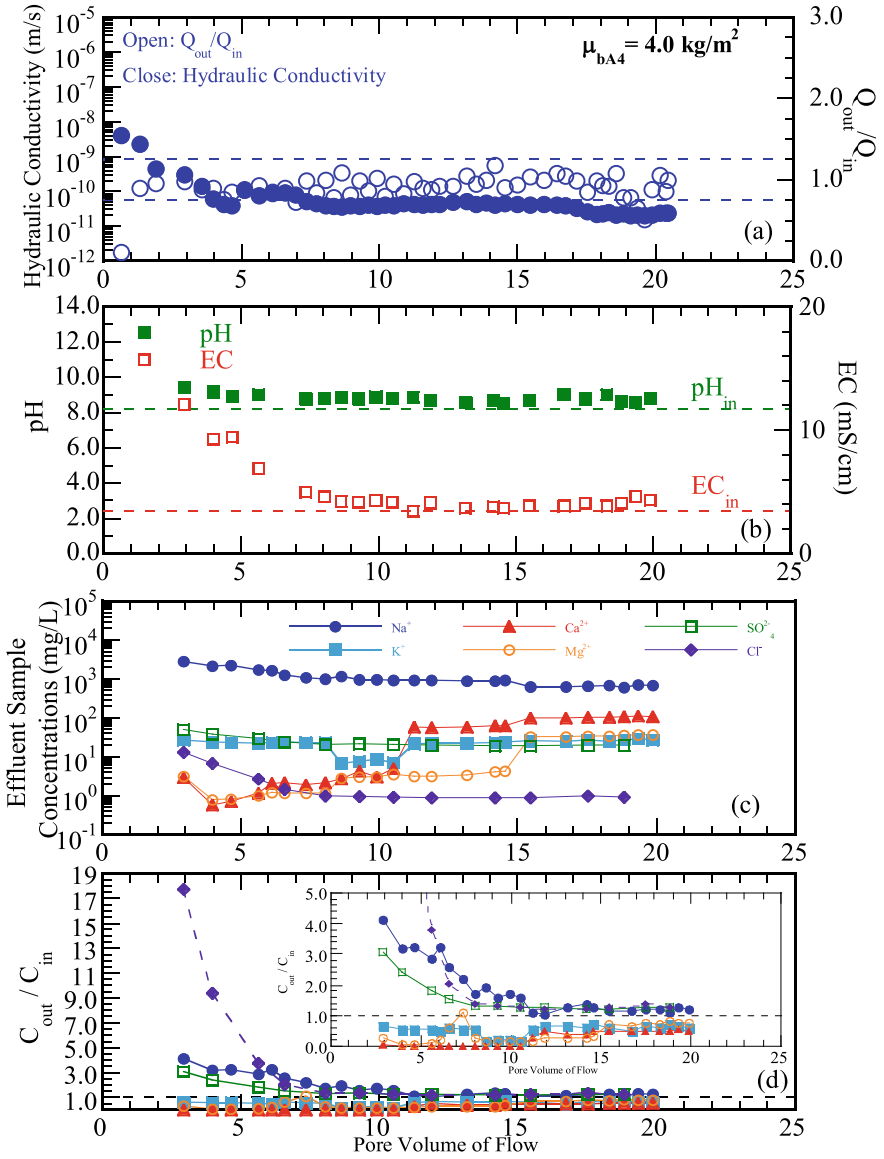

Fig. 2 Results of μ_{bA4} with CCL as a function of the pore volume of flow (PVF): **a** hydraulic conductivity, **b** pH and EC of influent and effluent, **c** Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^- and SO_4^{2-} concentrations in the effluent, **d** effluent-to-influent ratio of Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^- and SO_4^2

cm at 10 pore volume of flow. The EC did not change until the end of the test, and the final electrical conductivity value was measured at 4.4 mS/cm, which is close to the EC of the influent.

The concentrations of exchangeable cations (Na^+, K^+, Ca^{2+}, and Mg^{2+}) and anions (Cl^- and SO_4^{2-}) detected in samples taken from the effluent of μ_{bA4} are shown in Fig. 2c. The cation and anion concentrations in the effluent were divided by the concentrations of the influent (C_{out}/C_{in}) to better represent the cation exchange (Fig. 2d). Since the C_{out}/C_{in} of the Cl^- anion was initially high (~18), the second graph is embedded in Fig. 2d to show the initial part of the C_{out}/C_{in} in detail. The equilibrium was reached ($C_{out}/C_{in} = 1.0 \pm 0.1$) at 12 pore volume of flow for Na^+, Cl^-, and SO_4^{2-} and at 15.5 pore volume of flow for K^+, Ca^{2+}, and Mg^{2+}. The C_{out}/C_{in} of anions and cations are close to one, indicating that μ_{bA4} reached chemical equilibrium by the end of the test (Fig. 2d).

The hydraulic conductivity of μ_{bA5} was 6.6×10^{-9} m/s initially. Then, at 10 pore volume of flow, the hydraulic conductivity decreased to 2.0×10^{-11} m/s, two times lower than the initial value (Fig. 3a). The hydraulic conductivity of μ_{bA5} was also slightly lower than that of μ_{bA4} (2.0×10^{-11} m/s versus 2.5×10^{-11} m/s).

The pH and EC values of the samples taken from the effluent of μ_{bA5} with CCL are shown in Fig. 3b as a function of the pore volume of flow. The initial pH of the effluent is approximately 9.5. After 5 pore volume of flow, the pH nearly reached the pH of the influent (i.e., 8.2). In μ_{bA4} and μ_{bA5}, pH stabilized as the test progressed, indicating less change in the bentonite composition, which correlated with low hydraulic conductivity values ($2.4 \times 10^{-}11$ m/s and $1.5 \times 10^{-}11$ m/s, respectively). While the EC of the effluent was 16.8 mS/cm during the initial stage of permeation, it subsequently decreased to 4.3 mS/cm at 9 pore volume of flow. This value is also close to the influent EC (4.2 mS/cm – 3.9 mS/cm). In the μ_{bA4} and μ_{bA5}, the EC of the effluent gradually approached that of the influent, indicating chemical equilibrium and reduced permeability.

In Fig. 3c, the concentrations of exchangeable cations and anions determined in the effluent samples are plotted. While it was expected that the Na^+ cation would decrease as the pore volume of flow increased due to the cation exchange occurring between the permeant and Na-GCL, there was a slight decrease in the Na^+ concentration in the samples taken from the effluent (~1000 mg/L). This concentration is high, as in μ_{bA4}. Its elevated level is primarily a result of the abundance of Na^+ ions in the CCL. The equilibrium condition was obtained at 7 pore volume of flow for anions (Cl^- and SO_4^{2-}), while the equilibrium was reached at a higher pore volume of flow for the exchangeable cations (Fig. 3d).

One of the main factors controlling the hydraulic conductivity of Na-GCLs is the swelling of bentonite. Swell index tests were performed with CCL on bentonite collected from GCL that was not subjected to the hydraulic conductivity test (virgin) and on bentonite collected from μ_{bA3}, μ_{bA4}, and μ_{bA5} after being removed from the permeameter at the end of the test. At the end of 24h, the swell index of virgin bentonite was found to be 16.7 mL/2g. The swell index of μ_{bA3}, μ_{bA4}, and μ_{bA5} was measured at 16.5 mL/2g, 6.5 mL/2g, and 7.4 mL/2g, respectively.

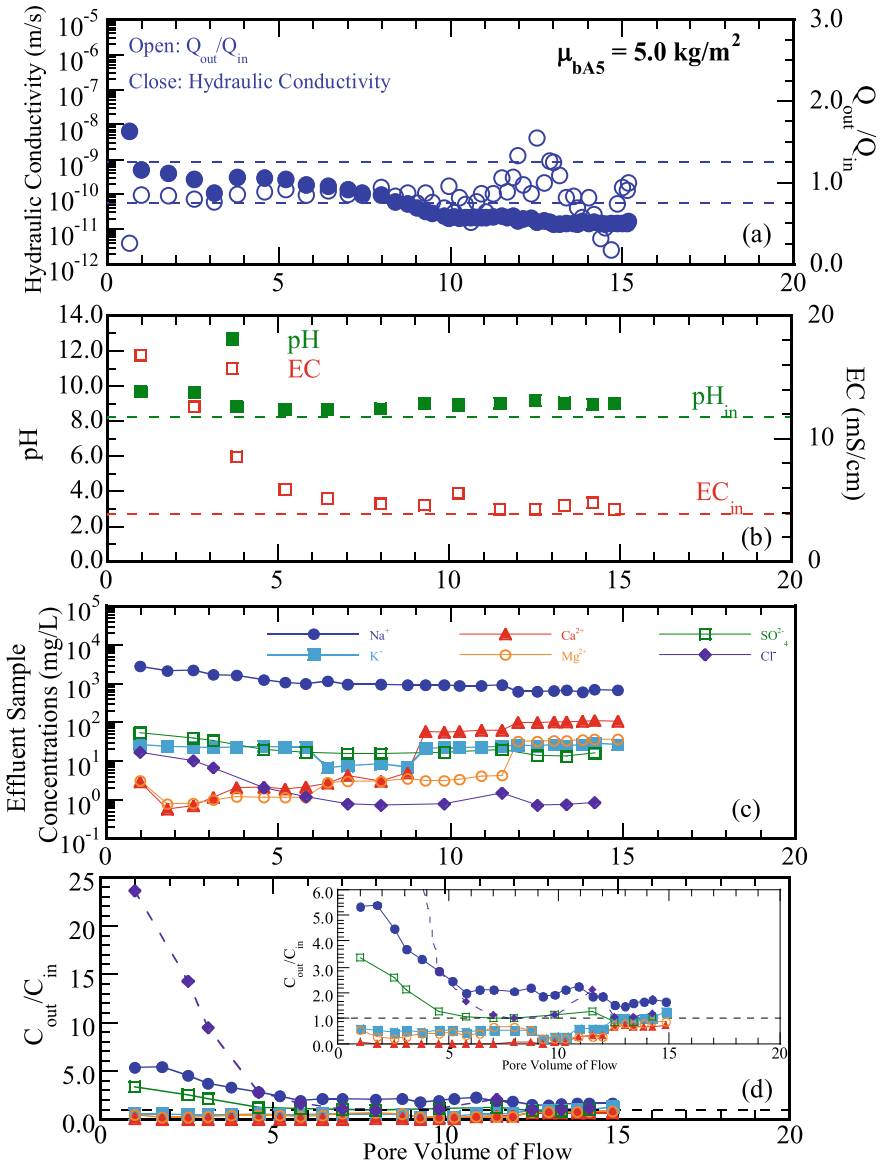

Fig. 3 Results of μ_{bA5} with CCL as a function of the pore volume of flow (PVF): **a** hydraulic conductivity, **b** pH and EC of influent and effluent, **c** Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^- and SO_4^{2-} concentrations in the effluent, **d** effluent-to-influent ratio of Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^- and SO_4^{2}

4 Conclusion

The hydraulic conductivity of Na-GCL with CCL was investigated and discussed in terms of μ_{bA}. For this purpose, Na-GCL with μ_{bA} of 3.0 kg/m^2, 4.0 kg/m^2, and 5.0 kg/m^2 were subjected to hydraulic conductivity tests. The effect of μ_{bA} was significant when μ_{bA} of Na-GCL was 3.0 kg/m^2. This study demonstrated that the μ_{bA} of GCL significantly influences the hydraulic conductivity of Na-GCL using a CCP leachate as the permeant. The hydraulic conductivity of μ_{bA3} permeated with CCL was measured as 3.7×10^{-8} m/s, while those of μ_{bA4} and μ_{bA5} were measured as 2.4×10^{-11} m/s and 1.5×10^{-11} m/s, respectively. The higher hydraulic conductivity of μ_{bA3} compared to μ_{bA4} and μ_{bA5} is due to the flow over the bundles of fibers. The hydraulic conductivity of Na-GCLs started to decrease at 4.0 kg/m^2 μ_{bA} when the Na-GCL was permeated with CCL. The hydraulic conductivity of Na-GCLs with μ_{bA} of 4.0 and 5.0 kg/m^2 was low. The abundance of Na^+ in the CCL was responsible for the low hydraulic conductivity. The evidence of this phenomenon can be seen from the swell index of bentonite in CCL (i.e., 16.7 mg/L) and the chemical analyses conducted on the influent and effluent.

From a practical point of view, the results obtained herein indicate that a minimum μ_{bA} of 4.0 kg/m2 for Na-GCL is required to maintain hydraulic conductivity suggested for the barriers (i.e., 1.0×10^{-9} m/s) when using CCL as a permeant. However, the transition μ_{bA} (i.e., 4.0 kg/m^2), where hydraulic conductivity is low, can change depending on GCL and permeant type. Therefore, performing hydraulic conductivity tests on different kinds of GCLs and CCP leachates is strongly recommended to increase the database.

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Hydraulic Conductivity of P-GCLs with Different Mass per Unit Area Values against Bauxite Liquor

H. Efe Kayrak, Taylan Cem Aytekin, and A. Hakan Ören

Abstract Red mud is a solid waste classified as hazardous obtained from bauxite ore. The red mud produced from bauxite ore processing for aluminum production is highly alkaline. In addition to this, red mud and its leachate contain heavy metals such as aluminum and iron. Red mud leachate can infiltrate soil or groundwater when not correctly stored. Due to its active chemicals, red mud poses a significant threat to aquatic ecosystems. Despite being a less-known material in Civil Engineering practice, red mud has become the subject of numerous studies in recent years. Particularly, hydraulic conductivity tests conducted with red mud leachate (also referred to as bauxite liquor in the literature) constitute the main focus of these studies. This study focuses on performing hydraulic conductivity tests using geosynthetic clay liners (GCLs) containing polymerized bentonite (P-GCLs). The aim is to determine the resistance of P-GCL against bauxite liquor with varying mass per unit area of bentonite (μ_{bA}).

Keywords Bauxite liquor · Geosynthetic clay liner · Hydraulic conductivity · Mass per unit area · Red mud

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1 Introduction

Geosynthetic clay liners (GCLs) are composite materials mostly used in landfills, manufactured by laying bentonite between two geotextiles and needle-punching the geotextiles together. In conventional GCL, sodium bentonite is used; however, the long-term performance of Na-GCL against aggressive leachates is generally poor. Thus, the use of polymer bentonite in GCL has increased over the years [1–3].

Scalia et al. [4] compared the hydraulic performances of bentonite-polymer composite GCLs (BPC GCLs) and Na-GCLs. Under the influence of aggressive inorganic solutions (1 M NaOH, 1 M HNO, and $CaCl_2$ concentration $\geq$ 50 mM), the barrier performance of BPC GCL was better than Na-GCL in terms of hydraulic conductivity. Chen et al. [5] also compared the hydraulic conductivities of BPC GCLs and Na-GCL to coal combustion leachates. At similar bentonite mass per unit area (μ_{bA}), the hydraulic conductivity of BPC GCLs was lower than that of Na-GCLs. When the polymer content of BPC GCLs increased, the hydraulic conductivity decreased further.

Red mud is a by-product of aluminum production, commonly known as bauxite residue. Bauxite residue is characterized by very fine particle size and high alkalinity [6] which is due to the presence of heavy metals [7].

Li et al. [8] permeated Na-GCL and BPC GCLs with bauxite liquors at two different ionic strengths. One conventional Na-GCL and six BPC GCLs were used in their study. BPC GCLs were manufactured by dry mixing granular proprietary polymers with granular sodium bentonite. Polymer loadings were in the range of 6.3 ± 0.3–12.1 ± 0.4%. At any ionic strength, it was found that the hydraulic conductivity of BPC GCLs to bauxite liquor was lower than that of NaB-GCL. When the polymer loading was less than 7.5%, the hydraulic conductivity ranged from 10^{-13} m/s to 10^{-9} m/s. When polymer loading exceeded %7.5, the hydraulic conductivity of BPC-GCLs was mostly lower than 10^{-10} m/s. In contrast, the hydraulic conductivity of NaB GCL was ~10^{-8} m/s.

Chen et al. [9] performed hydraulic conductivity tests with two bauxite liquors, one of which was synthetic (BLS) and the other was actual (BLA). The BLA was collected from red mud disposal site in China, Chongqing. Both bauxite liquors were hyperalkaline and the pH of BLS and BLA was 13.4 and 12.8, respectively. Conventional (GA) and polymer-enhanced bentonite GCLs (PB-GCLs) were used in tests. Initial polymer loadings of PB-GCLs were %1.0, %2.5, %5.0, %10.0, and %15.0. The hydraulic conductivity of PB-GCLs to both bauxite liquors was lower than that of conventional GCL (10^{-7}–10^{-12} m/s versus 10^{-8} m/s). In addition, increasing the initial polymer loading of PB-GCLs also improved the chemical compatibility.

Mass per unit area of GCL (μ_A) is also an important parameter affecting the hydraulic conductivity of GCLs. Regardless of μ_{bA}, the hydraulic conductivity of GCLs to distilled water (DIW) is around 2.0×10^{-11} m/s. However, when the permeants were inorganic solutions, the hydraulic conductivity performance of GCLs can change. Ören et al. [10] reported that Na-GCL preserved its low hydraulic conductivity even at low μ_{bA} (the hydraulic conductivity of Na-GCL to DIW was 1.8 $\times$

10^{-11} m/s at 1.8 kg/m^2). Demir Sürer et al. [11] also reported similar results for another Na-GCL and with μ_{bA} ranged between 3.0 kg/m^2 and 5.0 kg/m^2. That is, the hydraulic conductivity of Na-GCL to DIW was low. However, the μ_{bA} was important regarding hydraulic conductivity when the permeant was changed to trona ash leachate. For example, the hydraulic conductivity of Na-GCL was 7.6×10^{-7} m/s and 4.1×10^{-11} m/s, when the μ_{bA} was 4.1 kg/m^2 and ~5.0 kg/m^2, respectively.

The studies conducted so far have been on determining the hydraulic conductivity of GCLs against coal combustion permeant (CCP) leachates by increasing the polymer content. In recent years, it has been observed that similar studies have been conducted using bauxite liquor as the permeant. However, in these studies the μ_A was greater than 3.5 kg/m^2. This study aims to investigate and discuss the hydraulic performance of GCL, containing polymer bentonite (P-GCL) with low μ_{bA} (i.e., between 3.0 and 3.5 kg/m^2), to bauxite liquor. By this means, geotechnical engineers can get an insight into the hydraulic conductivity performance of P-GCLs with relatively low μ_{bA} against bauxite liquor.

2 Materials and Methods

2.1 Materials

In this study, a geosynthetic clay liner that contains polymerized bentonite, which had been applied in the field, was used. The type of polymer was unknown. Three GCL samples with a diameter of 100 mm were cut from the roll, and GCL masses and heights were measured. The μ_{bA} of GCLs was calculated according to ASTM D5993–18 [12]. The GCL heights ranged between 0.42 and 0.56 mm. The μ_{bA} was 3.08, 3.35 and 3.45 kg/m^2, which are denoted as $\mu_{bA3.1}$, $\mu_{bA3.4}$, $\mu_{bA3.5}$, respectively.

Red mud was supplied from Seydişehir Aluminum Plant, Konya/Türkiye. Red mud was in cobble size when obtained (Fig. 1). The cobbles of red mud were grounded using a small-scale jaw crusher and passed through No. 10 U.S. standard sieve (2.0 mm). Then, the bauxite liquor was obtained by washing the grounded red mud with DIW.

Bauxite liquor obtained from red mud was used as the permeant. To prepare bauxite liquor, solid to liquid ratio of 0.1 was used. Since high volumes of bauxite liquor are required during hydraulic conductivity tests, 2.0 kilogram of grounded red mud was placed in a 20-liter plastic drum and 20 liter of DIW was added into the drum. Then, the suspension was vigorously shaken several times so that the cations in the red mud could desorb into the water. After the shaking process was completed, the suspension was left to sedimentation. After sedimentation, the supernatant water above the sediment was transferred to another clean drum. This liquid was used as the bauxite liquor during the hydraulic conductivity tests.

Fig. 1 Cobble sized red mud used in the study

Table 1 Geotechnical index properties of the red mud

w_L (%)	w_p (%)	I_p (%)	G
52.6	38	15	3.03

2.2 Methods

Geotechnical Index Properties. Consistency limits [13] and specific gravity [14] tests were conducted on the red mud to determine geotechnical index properties. The summary of the geotechnical index properties of red mud is given in Table 1. The specific gravity of red mud was 3.03, which is greater than the specific gravity of natural soils (i.e. 2.60–2.70). This is possibly because of the iron present in red mud.

Hydraulic Conductivity Test. Hydraulic conductivity tests were conducted using flexible wall permeameters and following ASTM D6766–12 [15]. Before placing in the permeameter, the perimeter of all three GCLs were sealed with bentonite using fingers to prevent possible side-wall leakage. Then, the GCLs were placed on the base pedestals of the permeameters. Other details about the placement of GCL on the permeameter cell can be found in [16–18]. Cell pressure was kept at 35 kPa throughout the tests. Falling head and constant tailwater methods were applied. The flow direction was from top to bottom of the GCL to simulate in-situ conditions. Outflow was collected in a 100 ml graduated cylinder.

3 Results

The first test was conducted on $\mu_{bA3.1}$. Figure 2 shows the pictures of $\mu_{bA3.1}$ before performing the hydraulic conductivity test. As seen in Fig. 2a-b, the dark areas on the GCL indicate bentonite-filled parts, while the yellow/white areas indicate bentonite-free and polymer-filled parts through which light passes easily. Figure 2 also proves the uneven distribution of bentonite and polymer in GCL.

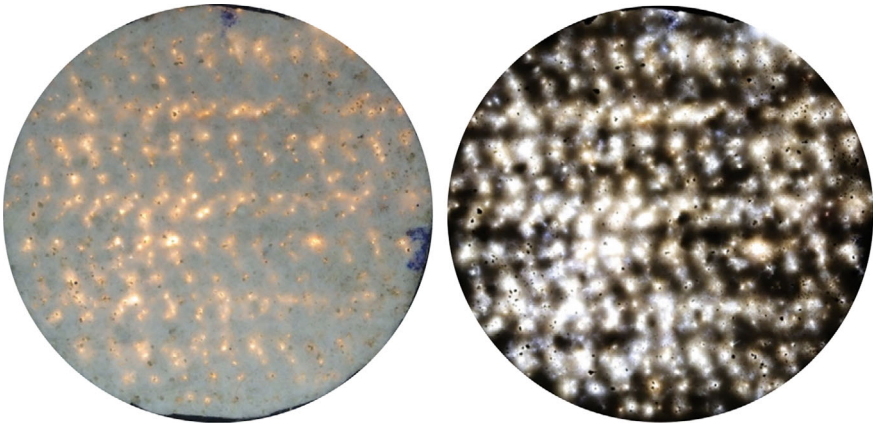

Fig. 2 Photographs of μbA3.1 taken light-on (left) and light-off (dark room) conditions before performing hydraulic conductivity test

Prior to permeation, the GCL was hydrated for 48 hours with bauxite liquor. The hydraulic conductivity was measured as 1.2×10^{-10} m/s at the beginning, and the hydraulic conductivity values fluctuated as the permeation progressed (Fig. 3). This fluctuation in the hydraulic conductivity may be due to the free polymers, which might have broken away from the GCL and blocked the flow paths (Fig. 2). At specific time intervals, the test was stopped, and the influent tubing was flushed with the bauxite liquor. After flushing, the permeation began, and hydraulic conductivity readings were recorded. The test has been continuing for 90 days. To terminate the test, the last four consecutive readings should be 1.0 ± 0.25. Therefore, the test is still in progress. The last hydraulic conductivity measured so far was 4.4×10^{-11} m/s, which is 2.7 times lower than the initial value of 1.2×10^{-10} m/s (Fig. 3).

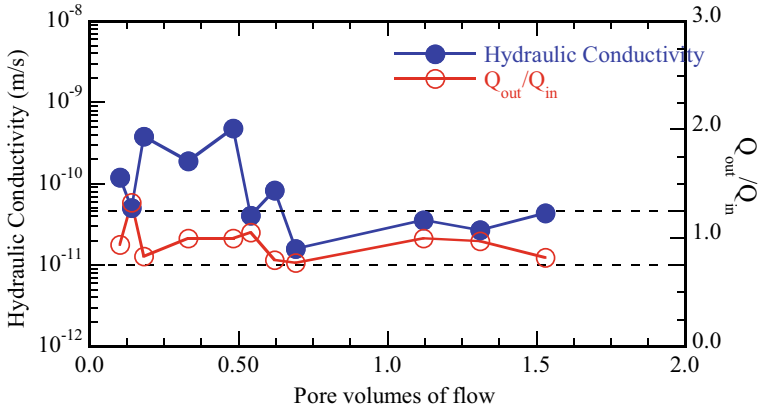

Fig. 3 Hydraulic conductivity of $\mu_{bA3.1}$ to bauxite liquor

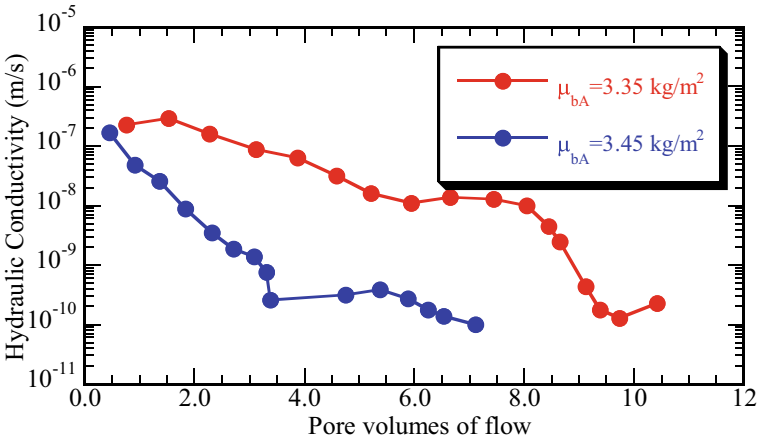

Fig. 4 Hydraulic conductivities of $\mu_{bA3.35}$ and $\mu_{bA3.45}$ to bauxite liquor

The μ_{bA} of the P-GCL was increased to 3.35 and 3.45 kg/m^2 for the next two tests. Both GCLs were pre-hydrated with bauxite liquor for 48 hours. The initial hydraulic conductivity of $\mu_{bA3.35}$ was 2.3×10^{-7} m/s and that of $\mu_{bA3.45}$ was 1.7×10^{-7} m/s (Fig. 4). Although μ_{bA} increased, no significant difference was observed between the initial hydraulic conductivity values. However, in the later stages of the permeation, the hydraulic conductivity of $\mu_{bA3.35}$ decreased to 2.3×10^{-10} m/s at 10.5 PVF, while $\mu_{bA3.45}$ decreased to 1.0×10^{-10} m/s at 7.0 PVF (Fig. 4). Note that the tests have not terminated yet.

The final hydraulic conductivity of P-GCL to bauxite liquor is shown in Fig. 5 as a function of μ_{bA}. Since all three tests are still in progress, the averages of the last three consecutive readings were considered as the final hydraulic conductivity. The final hydraulic conductivity of P-GCL within 3.1 and 3.5 kg/m^2 μ_{bA} ranged between 3.6×10^{-11} m/s and 1.4×10^{-10} m/s, indicating the negligible influence of μ_{bA} on the hydraulic conductivity of P-GCL to bauxite liquor. Although the hydraulic conductivity of $\mu_{bA3.1}$ was five times lower than those of $\mu_{bA3.4}$ and $\mu_{bA3.5}$, the hydraulic conductivity of P-GCL was significantly lower than 1.0×10^{-9} m/s, which is the allowable limit suggested for the liners. This difference can be attributed to the uneven distribution of bentonite within the GCL and the size and amount of bundles of fibers that existed on the GCL. The use of polymerized bentonite within GCL masked the negative influence of μ_{bA} on hydraulic conductivity. At least, this is valid when bauxite liquor was used as the permeant.

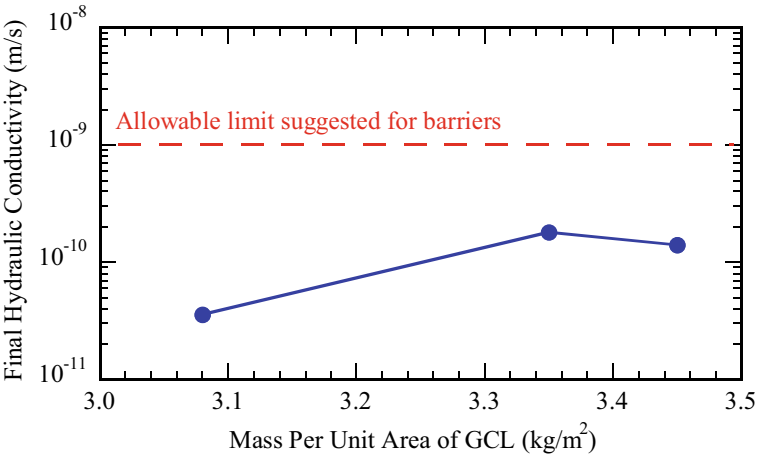

Fig. 5 The final hydraulic conductivity of P-GCL as a function of bentonite mass per unit area

4 Conclusion

In this study, the hydraulic conductivities of P-GCL against bauxite liquor were examined as a function of the μ_{bA} of the GCL.

P-GCLs with μ_{bA} ranged between 3.1 and 3.5 kg/m^2 were pre-hydrated with bauxite liquor for 48 hours. When permeation was started, the initial hydraulic conductivity of $\mu_{bA3.1}$ was recorded as 1.2×10^{-10} m/s. The hydraulic conductivity fluctuated throughout the test duration for this GCL, which was attributed to the clogging of flow paths by the free polymers. The last hydraulic conductivity reading for $\mu_{bA3.1}$ was 4.4×10^{-11} m/s.

Two more hydraulic conductivity tests with greater μ_{bA} were conducted using bauxite liquor (3.35 and 3.45 kg/m^2). The initial hydraulic conductivities were 2.3×10^{-7} m/s and 1.7×10^{-7} m/s for $\mu_{bA3.35}$ and $\mu_{bA3.45}$, respectively. However, the hydraulic conductivity of $\mu_{bA3.35}$ and $\mu_{bA3.45}$ decreased to 2.3×10^{-10} m/s and 1.0×10^{-10} m/s at the subsequent permeation stages, respectively.

According to the results obtained from these three tests, the increase in the μ_{bA} did not cause a significant difference in the final hydraulic conductivity of P-GCL. The final hydraulic conductivities were still significantly lower than the allowable limits suggested for the liners (1.0×10^{-9} m/s). However, all three tests have not been terminated yet. The hydraulic conductivity tests will be lasted until stable hydraulic conductivity values will be obtained. That is, last four consecutive hydraulic conductivity readings will be 1.0 ± 0.25. Therefore, the final hydraulic conductivity values can change over time.

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An Alternative Approach to Reusing Mine Waste in Underground Mines: CPB

Okan Yüksel, Oznur Karaca, and Tugay Aktaş

Abstract Improper mining activities cause water, soil, and air pollution. They may cause problems affecting all living things in the area where the activity is carried out and in the nearby geography. Therefore, the economic development to be obtained from mining should be evaluated together with the environmental problems it will cause. The contemporary approach in this regard should be to take environmental protection measures by predicting the problems that may occur with good planning, monitoring, and environmental technologies, and to ensure that the development is continuous and sustainable. In this context, samples were taken, and initial analyses were performed to examine the environmental impacts of the waste ponds in the mining operation in Balıkesir (Türkiye). Accordingly, it was observed that lead and zinc were in high concentrations in the region. Sulfide-Sulfur (SO_3^{2-}—S^{2-}) tests were performed and it was determined that the area's waste and mining waste produced acid and were harmful to the environment. In this study, the cemented paste backfill (CPB) method was applied to the samples to intertie waste and prevent acid production. Leaching tests were performed on the sample that completed the 28-day curing period from the CPB mixtures prepared using mining waste. The leaching test was applied to the waste samples to achieve this objective, with results remaining below the regulatory limit values. Acid mine drainage (AMD) tests showed that CPB samples which were cured for 28 days had no acid generation potential. Pyrite was determined to be the primary mineral responsible for acid formation, while neutralizing minerals were found to belong to carbonate and silicate groups. CPB application decreased acid-forming minerals by increasing the neutralization capacity.

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1 Introduction

Although mining is essential in many respects, it also brings environmental problems. Mining operations generate huge amounts of solid and liquid waste and impact on the environment. Furthermore, the main effect of mining activities is related to acid mine drainage (AMD), a metal-containing acid effluent generated in mines by the oxidation of buried minerals when exposed to atmospheric oxygen [1–3]. The amelioration of a contaminated area with metals is challenging. Remediation techniques involve mostly expensive and time-consuming tasks [4].

Paste fill is a high-density, fluid, pumpable material composed of a mixture of mining waste, water, and cement, used to fill gaps in underground mines [5]. It is widely used in mines worldwide because it can effectively solve the safety problems arising from mining areas and waste reservoirs [6]. In the kinetic test moisture cell study conducted by Verburg (2001) on Cemented Paste Backfill (CPB) samples, it was stated that increasing the cement dosage improved the geochemical properties from an environmental perspective and could contribute to sulfur waste management. It was emphasized that CPB provides sufficient strength in dry environments, but its structural and geochemical properties may change over time in high-humidity conditions. In a study conducted to investigate the ensuring geochemical properties of CPB, it was observed that the curing stage had an effect on the geochemical properties and that the fragmentation of the samples could change their chemical properties [7]. The use of CPB to fortify underground production voids aims to reduce the surface footprint of the waste generated after the process while also ensuring the efficient use of resources by considering the protection-use balance [8].

This study aims to prevent the contact of sulfide minerals with oxygen using the CPB method in waste materials containing sulfide and to prevent acid formation. In this context, samples were taken from the flotation wastes of the Balya (Balıkesir-Türkiye) lead and zinc mining facility and were subjected to analyses at national and global standards to determine the acid production potential and metal release of the prepared mixtures. The analyses were prepared by considering the 28-day cure periods in the materials to be used in the application and the mixture ratios. Tests regarding the acid production potential of the materials to be used as CPB and metal leaching analyses were performed, and characterization was carried out.

2 Materials and Methods

The following experimental studies were conducted on the sample from the underground mine pit flotation waste to indicate all standards that have been applied.

1. Preparation and curing of cement paste filling mixtures
2. Conducting sulfur tests and static tests
3. Conducting mineralogical tests
4. Conducting monolithic leaching tests of CPB samples according to EN 15863 Standard
5. Evaluation of the results

Internationally accepted standard methods were used in the conduct of experimental studies.

2.1 Material Characterization

A 6 kg waste sample was collected from the field for the study. 1.5 kg of the sample was separated by the quartering method in the laboratory and used for this study. For acid mine drainage tests, samples were first dried in atmospheric conditions (room temperature) to remove moisture. After the drying process, all samples were passed through a 6.3 mm sieve and gradual crushing was performed to release acid-producing and acid-consuming minerals. Grinding was performed and passed through a 150 µm sieve for sulfur and static tests. 250-gram samples were prepared for static tests. The remaining samples were prepared to be used for leaching tests.

For the preparation of CPB samples, flotation waste, cement, and DI water were mixed with a mixer for 15 min (Table 1). They were transferred to cylindrical sample molds made of polyurethane material with a diameter of 10 cm and a height of 10 cm, following EN 12390–1 standard. Then, it was subjected to waiting for 28 days for curing in an 80% humidity and 25 °C temperature environment [9]. For the Monolithic Leaching Tests carried out according to the EN 15863 Standard, cubes with a geometric surface area of 4x4 in the standard were obtained from the samples that completed the 28-day curing period.

To classify a sample as potential acid-producing or non-acid-producing, evaluations were made based on Acid Base Accounting (ABA) analysis test using different criteria. The most common approach is based on the use of the Neutralization Potential Ratio (NPR). NPR criteria were evaluated in the assessment of acid rock drainage potential.

Table 1 Sample preparation definitions

Study No	Prepared mixture flotation waste %	Recipe (Gram)			
		Waste	Cement	Water	Total
1	5.5	1440	88	75	1603.0
2	7.5	1440	124	90	1654.0

Sulfur Analysis. In the samples, sulfide-sulfur contents were determined using the Eltra CS-2000 Carbon-Sulfur analyzer by Gemar Environmental Laboratory (Çanakkale, Türkiye) according to ASTM E 1915 "Standard Test Methods for Analysis of Metal Bearing Ores and Related Materials for Carbon, Sulfur, and Acid-Base Characteristics."

pH and electrical conductivity of flotation waste. pH measurements were conducted on samples prepared in a 2:1 volume ratio (waste: DI water) according to the EPA 600.2 standard. For electrical conductivity measurements, the sample was prepared with a 1:5 (m/V) extraction ratio according to the TS ISO 11265 standard. The measurements were performed using a Gach Lange branded multimeter.

Static Tests. Sulfide-containing mine waste has the potential to generate AMD. The potential for a waste to produce AMD is determined by calculating the neutralization potential ratio (NPR) using its acid potential (AP) and neutralization potential (NP) (Fig. 5). To determine the acid generation potential (AP), firstly the iron-bound sulfur content of the waste ($\%S^{2-}$) is determined. The obtained value is multiplied by a constant coefficient (31.25) [10]. NPR calculations are expressed as H^+ content in mol/kg for both AP and NP. The conversion factor is expressed as carbonate equivalents ($CaCO_3$) in kg/t. AP is calculated by determining the sulfur content. The acid potential was calculated from the data obtained from analyses conducted according to the ASTM D1915 standard (Fig. 5).

In this calculation, it is assumed that all the sulfur in a sample is potentially reactive and, therefore, can produce acid. However, this assumption overlooks the fact that not all sulfur contributes to acidity (such as sulfur in minerals like gypsum, barite, etc.). The acid potential (AP) is expressed as kilograms of calcium carbonate equivalent per ton of material (kg $CaCO_3$/t). A specific amount of 0.1 N HCl (hydrochloric acid) is added to the prepared sample, and the NP is determined by back-titrating the diluted acid amount with sodium hydroxide. The amount of acid added is of great importance for the test. Therefore, the carbonate rating (sometimes referred to as inorganic carbon in certain standards) is first calculated. When using the test described in this standard, it is assumed that the entire neutralization capacity is determined within the 24-hour test period. However, some minerals respond faster, while others respond more slowly.

3 Results

3.1 Material Characterization

In order to characterize the mineralogical properties of flotation waste samples from the project site and paste filling mixtures that have completed a 28-day curing period, and to identify the minerals that will form acidity and neutralization, XRD (X-ray Diffraction) analyses were performed on the samples (Fig. 1). In the flotation waste,

the presence of carbonate and silicate minerals was detected, and an increase in these minerals was observed after the CPB application (Table 2). Considering these results, it is anticipated that the neutralizing effect on acidic minerals will increase. Based on this, the presence of basic minerals can prevent the dissolution of acidic minerals, potentially aiding in the accumulation or stabilization of certain minerals.

As a result of the sulfide-sulfur (SO_3^{2-}–S^{2-}) analysis, the value for the raw flotation tailings was found to be 1.81% (Fig. 2). After the CPB mixture, it was observed that the sulfide-sulfur (SO_3^{2-}–S^{2-}) content in the flotation tailings decreased to 1.67% and 1.61%, respectively (Fig. 2). Sulfide (SO_3^{2-}) minerals can transform into sulfate (SO_4^{2-}) minerals during the oxidation process. Accordingly, the decrease in the sulfide-sulfur ratio after CPB application may lead to a reduction in acidic conditions and an increase in the pH level of the drainage water (Fig. 3). This could contribute to reducing the negative impacts on the ecosystem.

An increase in pH levels in acid rock drainage indicates the formation of a more neutral environment (Fig. 3). This is a positive indicator in terms of mitigating environmental impacts and improving water quality. The increase in sulfate ions affects electrical conductivity. As shown in Fig. 3, electrical conductivity values range from 586.0 to 3910 µS/cm.

The NPR values of the samples examined in the study are determined by the NP/AP ratio. As shown in Fig. 4, the NP values marked in red are higher than the AP values, indicating that the risk of the rocks producing acid drainage is low. In this case, it is expected that the environmental impact will be lesser. The observed increase in NPR values after CPB suggests that the environmental impacts may be lower, and preventive measures may be less required.

Dynamic Monolithic Leaching Test with Periodic Renewal under Fixed Conditions: TS EN 15863.

This analysis is a method used to determine the leaching (dissolution) behavior of monolithic wastes under dynamic conditions. The release of inorganic components from CPB monolithic wastes in contact with DI water over time has been determined (Fig. 5). In this method, the monolithic test sample is placed in a container and completely covered with liquid. The liquid is renewed at predefined intervals in the experiment conducted within a specific temperature range (Table 3). The type of liquid, temperature, liquid/surface area, test duration, and the number of samples to be taken at specific time intervals are predetermined. The obtained sample liquid is physically and chemically analyzed according to current standards.

In the first stage of the sampling process, the volume of the liquid to be used is calculated using the following equation:

$V_1 = (8 \pm 2)$ x A; V_1 = liquid volume; A = Surface area of monolithic sample.

In the first sampling stage, after the liquid is added, it is sealed and left to stand for 6 h ±15 min. At the end of this period, the solid particles are left in the container, and the filtered liquid is taken, after which the second stage begins immediately.The process is repeated in stages 2–8 of the sampling process. The expected duration for each stage is provided in Table 3.

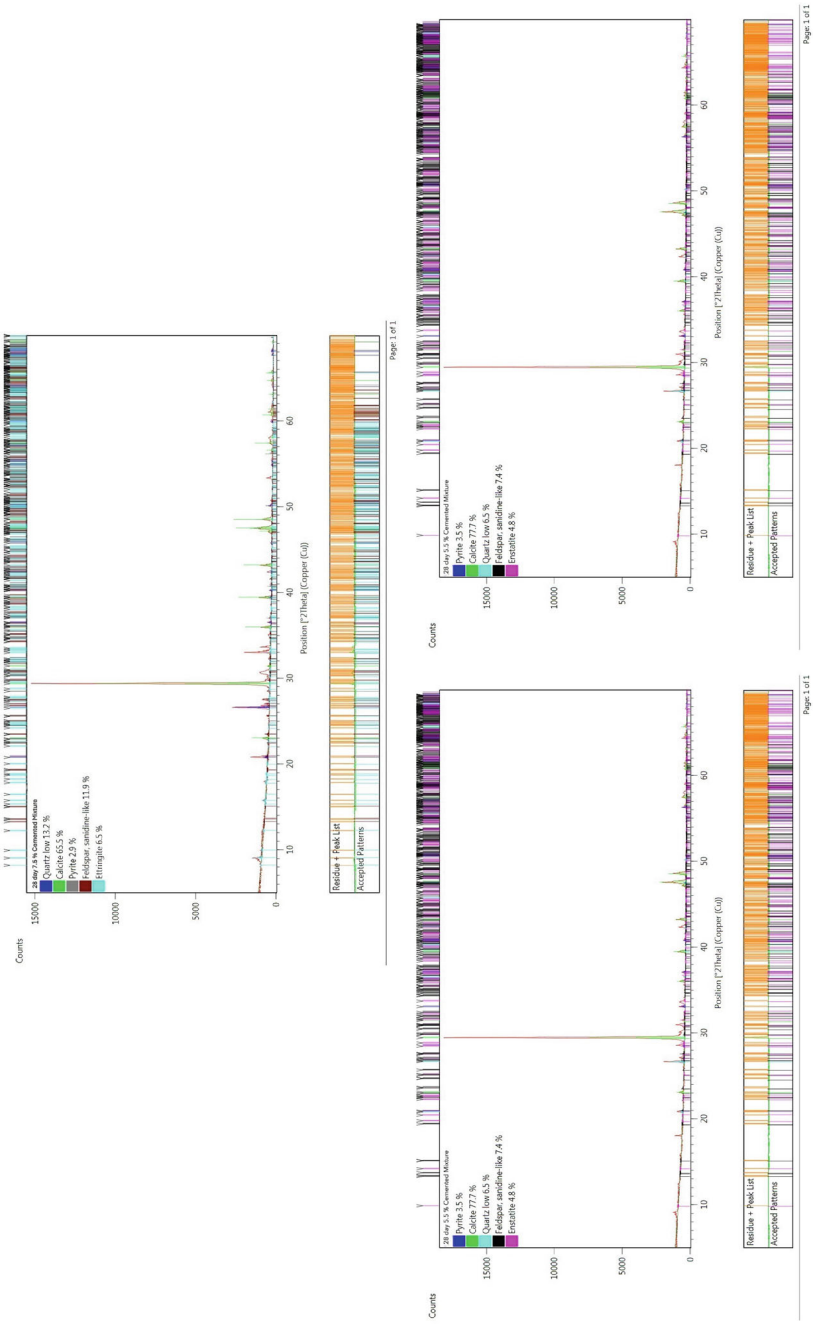


Fig. 1 XRD mineralogical analysis results

Table 2 Mineral quantification

Sample type	Quartz (%)	Calcite (%)	Pyrite (%)	Feldspar (%)	Dolomite (%)	Illite (%)	Gypsum (%)	Enstatite (%)	Ettringite(%)
Raw flotation waste	10.8	64.8	2.8	11.8	4.7	4.2	0.7	–	–
5.5% cemented mixture	6.5	77.7	3.5	7.4	–	–	–	4.8	–
7.5% cemented mixture	13.2	65.5	2.9	11.9	–	–	–	–	6.5

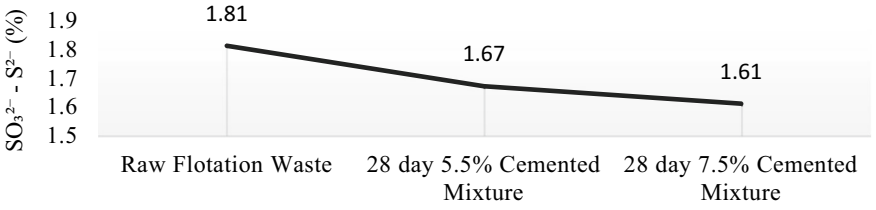


Fig. 2 Sulfur analyses results

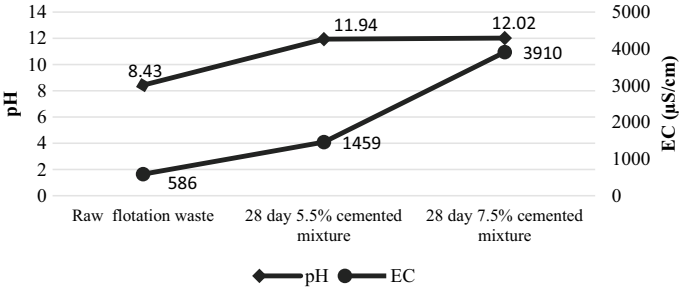


Fig. 3 pH and EC values of the samples before and after the tests

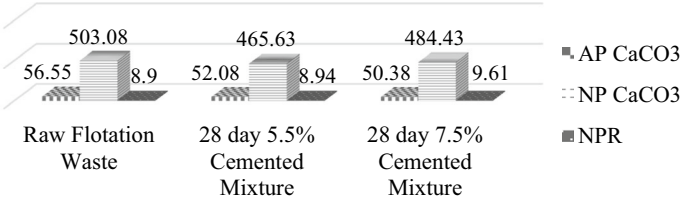


Fig. 4 AP, NP, and NPR of the flotation waste before and after CPB

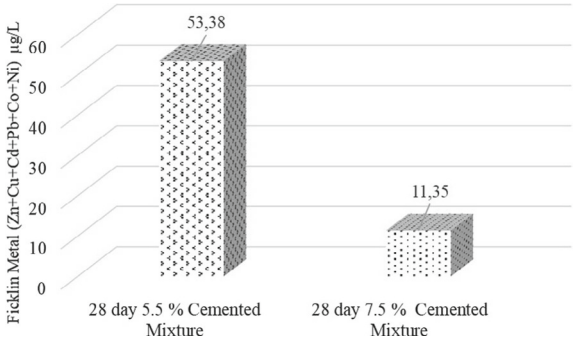


Fig. 5 Ficklin metals release from flotation waste following CPB application

Table 3 Leaching test time intervals

Step	Time interval	Elapsed time since the start of the test
1	6 h ± 15 m	6 h ± 15 min
2	18 h ± 45 min	1 day ± 45 min
3	1 day 6 h ± 1 h	2 days 6 h ± 2 h
4	1 day 18 h ± 2 h	4 days ± 4 h
5	5 days ± ± 6 h	9 days ± 18 h
6	7 days ± ± 8 h	16 days ± 18 h
7	20 days ± ± 24 h	36 days 42 h
8	28 days ± 25 h	64 days ± 0.25 day

Ficklin Metal Values. Ficklin Metals generally play an important role in water pollution assessments. Ficklin metals refer to metal ions that are present in leachate or wastewater samples and can have environmental impacts. Measuring these metals is important for evaluating water quality, understanding the effects on ecosystems, and identifying potential risks to human health [11]. The Ficklin metal value is calculated by considering the total of elements such as Cd, Co, Cu, Zn, Pb, and Ni, which are mobilized under low pH conditions in the filtered leachate waters (Fig. 5). Based on the Ficklin diagram in Fig. 6, when comparing the results of metals with pH, it shows lower metal release. Flotation tailings with a 7.5% cemented mixture show less release compared to those with a 5.5% cemented mixture.

pH. After the preparation of the CPB mixture and the 28-day curing period, the analyses conducted indicated a pH value greater than 9, characterizing the mixture in the basic range (Fig. 7). Basic waters facilitate the neutralization of acidic pollutants. This reduces the effects of pollution that may be caused by acidic conditions and improves water quality.

Dissolved Solids and Conductivity. The values ranged from 1144 to 1778 µS/cm (Fig. 8). Total Dissolved Solids (TDS) concentrations, expected to parallel conductivity, decreased after the CPB process. Reduced conductivity generally indicates

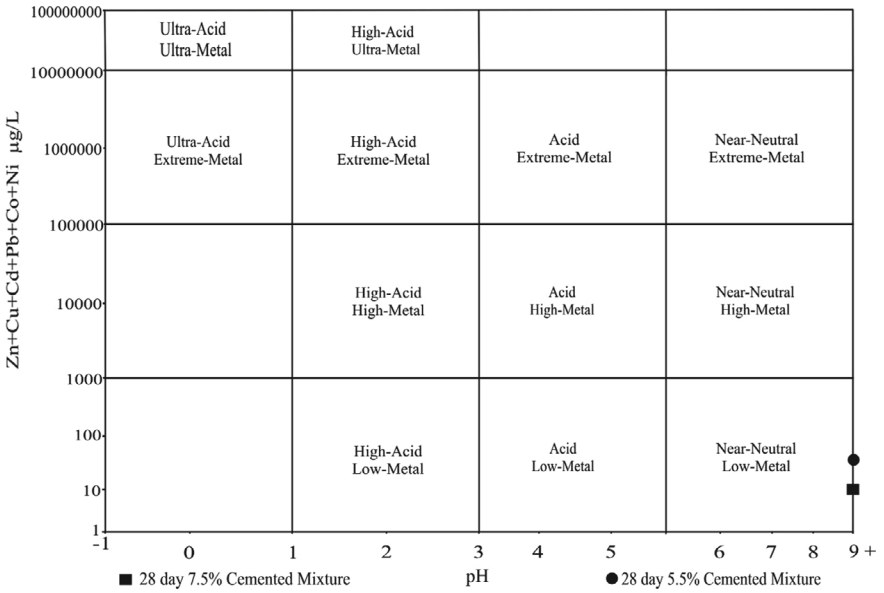
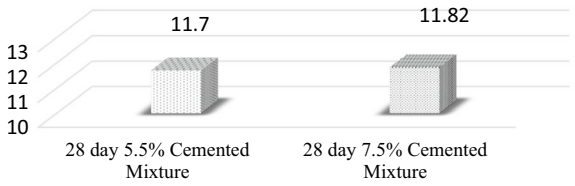


Fig. 6 Ficklin diagram [11]

Fig. 7 Leachate pH graph



water purification, lower pollutant levels, and healthier ecosystems. A reduction in TDS indicates decreased dissolved pollutants in the water, especially salts, heavy metals, and organic pollutants.

Table 4 shows the data obtained from all tests for raw flotation waste and samples mixed with waste and two different ratios of cement.

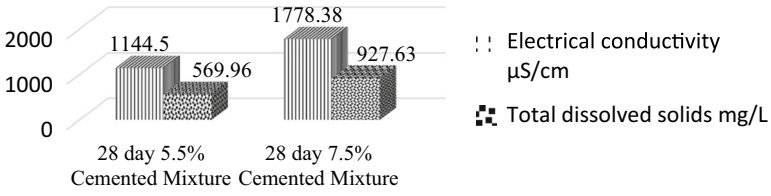


Fig. 8 Total dissolved solids (TDS) and conductivity

Table 4 Data obtained from all tests for raw flotation waste and two different cemented mixtures

Parameter	Raw flotation waste	28-day 5.5% cemented mixture	28-day 7.5% cemented mixture	Effect/observation
pH	8.43	11.94	12.02	pH increase observed
Sulfide-sulfur (SO_3^{2-} –S^{2-}) (%)	1.81	1.67	1.61	Decrease observed
Electrical conductivity (µS/cm)	586.00	1459.00	3910.00	Affected by sulfate ions
AP value ($CaCO_3$ mg/kg)	56.55	52.08	50.38	Decrease observed
NP value ($CaCO_3$ mg/kg)	503.08	465.63	484.43	Decrease and increase observed
NPR (%)	8.9	8.94	9.61	Increase observed
CPB Ficklin metal graph	–	761.6	142.6	Change in metal content
CPB leachate pH	–	11.70	11.82	pH increase observed
CPB dissolved solids (mg/L)	–	569.96	927.63	Increase observed
CPB electrical conductivity (µS/cm)	–	1144.5	1778.38	Change observed

4 Conclusion

The acid mine drainage (AMD) tests were conducted on flotation tailing samples within the scope of CPB studies, with samples cured for 28 days. As a result of the tests, it was determined that none of the samples exhibited 'acid production' potential and that the paste fill application increased neutralization, thereby reducing the potential for acid production. When the results of the mineralogical tests were examined, it was found that the major mineral responsible for acid formation is pyrite, while the major minerals with neutralizing properties belong to the carbonate and silicate groups. Mineralogical analyses of samples from cemented paste backfill (CPB) mixtures revealed an increase in neutralizing minerals and a decrease in the proportion of minerals with acid-forming potential. When the pH values of the samples were examined, no acid formation was observed. It can be stated that the presence of easily soluble neutralizing minerals (Mg + Ca + Na), along with sulfate releases, which are one of the indicators of acid formation, contributed to the consumption of the produced hydrogen ions, thereby creating alkalinity. Ficklin metal values: The Ficklin metal value in the leachate was calculated for all the samples, and it was determined to have low metal release. It was also observed that the pH values,

electrical conductivity, and total dissolved solids (TDS) values had decreased. The decrease in pH values, along with electrical conductivity and total dissolved solids (TDS), indicates that the CPB is not acidic and contains low levels of contaminants. This is a positive environmental indicator and contributes to the protection of water resources. Considering the findings, it has been observed that the use of waste in CPB applications will prevent the formation of acid mine drainage, reduce metal leaching, and limit the impact on groundwater in the project area.

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